



# GumNaaM\_Helpers



## Mega File for Final Term

## All Files in ONE FILE

(Handouts\_waqar\_moaz\_junaaid\_vu toper\_orange monkey)

الحمد لله گمنام ہیلپرز ان تمام کتابوں میں مدد کرتے ہیں جن کے لنکس نیچے دیئے گئے ہیں  
لیذا ہم سے صرف انہی کتابوں کی مدد کا سوال کیا جائے جن کے لنکس نیچے موجود ہیں

**Groups Join Krty hi Description Read krna na bhooolyn**

**جزاک اللہ خیرا و احسن الجزاء**

### GumNaaM\_Helpers All What's App Group Links

- |                                                                                                                           |                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 📞Cs101📞 <a href="https://chat.whatsapp.com/KCQ3jK6FcoYEvDYTF1gnBS">https://chat.whatsapp.com/KCQ3jK6FcoYEvDYTF1gnBS</a>   | 📞Eng101📞 <a href="https://chat.whatsapp.com/Cwp4Ag9TG6ZKxyRgThqM6S">https://chat.whatsapp.com/Cwp4Ag9TG6ZKxyRgThqM6S</a>   |
| 📞Cs201📞 <a href="https://chat.whatsapp.com/DrYV1FAXmPpImVZFw6V2R">https://chat.whatsapp.com/DrYV1FAXmPpImVZFw6V2R</a>     | 📞Eng201📞 <a href="https://chat.whatsapp.com/EvvxcPtmrM22VTZh5NCTw">https://chat.whatsapp.com/EvvxcPtmrM22VTZh5NCTw</a>     |
| 📞Cs301📞 <a href="https://chat.whatsapp.com/EPLvH2f8EB7aH8e1AtidV">https://chat.whatsapp.com/EPLvH2f8EB7aH8e1AtidV</a>     | 📞Eco401📞 <a href="https://chat.whatsapp.com/K3NLvyue6BdK0sVMQDFoU2">https://chat.whatsapp.com/K3NLvyue6BdK0sVMQDFoU2</a>   |
| 📞Cs302📞 <a href="https://chat.whatsapp.com/BaYfW0P67vNIsI2n36otZM">https://chat.whatsapp.com/BaYfW0P67vNIsI2n36otZM</a>   | 📞Mcm301📞 <a href="https://chat.whatsapp.com/GUyzfdHzu8r43TW16vhZWF">https://chat.whatsapp.com/GUyzfdHzu8r43TW16vhZWF</a>   |
| 📞Cs304📞 <a href="https://chat.whatsapp.com/C0mk0wdo0BU46dlhc67IQ">https://chat.whatsapp.com/C0mk0wdo0BU46dlhc67IQ</a>     | 📞Mgt101📞 <a href="https://chat.whatsapp.com/BmrYaQYulBOKguHOIT6t2Y">https://chat.whatsapp.com/BmrYaQYulBOKguHOIT6t2Y</a>   |
| 📞Cs401📞 <a href="https://chat.whatsapp.com/Jz9WDdHX84IKgkrMPAV0Mf">https://chat.whatsapp.com/Jz9WDdHX84IKgkrMPAV0Mf</a>   | 📞Mgt201📞 <a href="https://chat.whatsapp.com/DZwztUyxV50D6AshnvWCD">https://chat.whatsapp.com/DZwztUyxV50D6AshnvWCD</a>     |
| 📞Cs402📞 <a href="https://chat.whatsapp.com/DW9AZLb4i1P9la4XmcCsRz">https://chat.whatsapp.com/DW9AZLb4i1P9la4XmcCsRz</a>   | 📞Mgt211📞 <a href="https://chat.whatsapp.com/EmX9xloZnbl12pX9m2DZwa">https://chat.whatsapp.com/EmX9xloZnbl12pX9m2DZwa</a>   |
| 📞Cs403📞 <a href="https://chat.whatsapp.com/HXV25Lc6MnLKZgrA7xOqjF">https://chat.whatsapp.com/HXV25Lc6MnLKZgrA7xOqjF</a>   | 📞Mgt301📞 <a href="https://chat.whatsapp.com/G62QTipycOXFrP0ZxfDIBK">https://chat.whatsapp.com/G62QTipycOXFrP0ZxfDIBK</a>   |
| 📞Cs408📞 <a href="https://chat.whatsapp.com/EFy2mrYJodx5d85W1jtII7">https://chat.whatsapp.com/EFy2mrYJodx5d85W1jtII7</a>   | 📞Mgt501📞 <a href="https://chat.whatsapp.com/H0dTUINbesw0GS4LLSt0W4">https://chat.whatsapp.com/H0dTUINbesw0GS4LLSt0W4</a>   |
| 📞Cs411📞 <a href="https://chat.whatsapp.com/lrOaF8Q3tkJKzeVTHapHR9">https://chat.whatsapp.com/lrOaF8Q3tkJKzeVTHapHR9</a>   | 📞Mgt502📞 <a href="https://chat.whatsapp.com/L3GSMFgwDl7DAsu9r0vAxl">https://chat.whatsapp.com/L3GSMFgwDl7DAsu9r0vAxl</a>   |
| 📞Cs501📞 <a href="https://chat.whatsapp.com/Eggx3X0SH2LhOysM57SII">https://chat.whatsapp.com/Eggx3X0SH2LhOysM57SII</a>     | 📞Mgt503📞 <a href="https://chat.whatsapp.com/G7BNxqyLYwP48jPofkbj">https://chat.whatsapp.com/G7BNxqyLYwP48jPofkbj</a>       |
| 📞Cs502📞 <a href="https://chat.whatsapp.com/FbLU96Xhmr2F199fnb5Yw">https://chat.whatsapp.com/FbLU96Xhmr2F199fnb5Yw</a>     | 📞Mgt602📞 <a href="https://chat.whatsapp.com/HXRpOrEL18vD7lb91nLHTB">https://chat.whatsapp.com/HXRpOrEL18vD7lb91nLHTB</a>   |
| 📞Cs504📞 <a href="https://chat.whatsapp.com/Gn7w0UzNCQ2FzRK7Zy4Io">https://chat.whatsapp.com/Gn7w0UzNCQ2FzRK7Zy4Io</a>     | 📞Mgt610📞 <a href="https://chat.whatsapp.com/BxMcR1ouiao0wuqLLlJ5dB">https://chat.whatsapp.com/BxMcR1ouiao0wuqLLlJ5dB</a>   |
| 📞Cs506📞 <a href="https://chat.whatsapp.com/BL4FFmDKf2S3WIoLSAD2JK">https://chat.whatsapp.com/BL4FFmDKf2S3WIoLSAD2JK</a>   | 📞Mth101📞 <a href="https://chat.whatsapp.com/LHsbl65FbwGH2SNFF9ipSb">https://chat.whatsapp.com/LHsbl65FbwGH2SNFF9ipSb</a>   |
| 📞Cs507📞 <a href="https://chat.whatsapp.com/EINDZKv4e4zGLGg8ImozR">https://chat.whatsapp.com/EINDZKv4e4zGLGg8ImozR</a>     | 📞Mth202📞 <a href="https://chat.whatsapp.com/GevRebusRaxFh6dgnzRJuM">https://chat.whatsapp.com/GevRebusRaxFh6dgnzRJuM</a>   |
| 📞Cs508📞 <a href="https://chat.whatsapp.com/HTMAwnJ4Bic2ky8MbT3PuW">https://chat.whatsapp.com/HTMAwnJ4Bic2ky8MbT3PuW</a>   | 📞Mth301📞 <a href="https://chat.whatsapp.com/EFm472nc2woDProEgnY1j">https://chat.whatsapp.com/EFm472nc2woDProEgnY1j</a>     |
| 📞Cs601📞 <a href="https://chat.whatsapp.com/KYoaAutdXRINDS0Mm5XfzP">https://chat.whatsapp.com/KYoaAutdXRINDS0Mm5XfzP</a>   | 📞Mth401📞 <a href="https://chat.whatsapp.com/EP7tlcc17TzLtaNaOgD4AcA">https://chat.whatsapp.com/EP7tlcc17TzLtaNaOgD4AcA</a> |
| 📞Cs602📞 <a href="https://chat.whatsapp.com/FpJkLAFNqpZHRyi6vhl5n8">https://chat.whatsapp.com/FpJkLAFNqpZHRyi6vhl5n8</a>   | 📞Mth501📞 <a href="https://chat.whatsapp.com/Dy7XUUVqqD5LLw4i5X4Xzi">https://chat.whatsapp.com/Dy7XUUVqqD5LLw4i5X4Xzi</a>   |
| 📞Cs604📞 <a href="https://chat.whatsapp.com/KjDhHjxawtU43Veo9Eawa4">https://chat.whatsapp.com/KjDhHjxawtU43Veo9Eawa4</a>   | 📞Mth601📞 <a href="https://chat.whatsapp.com/BoD3akuXGK25MFB4UEZcRr">https://chat.whatsapp.com/BoD3akuXGK25MFB4UEZcRr</a>   |
| 📞Cs605📞 <a href="https://chat.whatsapp.com/BoF34LtlMIYK1L4nQaEO2Ce">https://chat.whatsapp.com/BoF34LtlMIYK1L4nQaEO2Ce</a> | 📞isL201📞 <a href="https://chat.whatsapp.com/BtoqaMbbELW99MLLTQMgWc">https://chat.whatsapp.com/BtoqaMbbELW99MLLTQMgWc</a>   |
| 📞Cs610📞 <a href="https://chat.whatsapp.com/FcNcztKyYfd9szsJElino2">https://chat.whatsapp.com/FcNcztKyYfd9szsJElino2</a>   | 📞Pak301📞 <a href="https://chat.whatsapp.com/KuR9Y7YWeA32enRwKeZJaR">https://chat.whatsapp.com/KuR9Y7YWeA32enRwKeZJaR</a>   |
| 📞Cs615📞 <a href="https://chat.whatsapp.com/HLVjBNcorFN9bwZhGekF3">https://chat.whatsapp.com/HLVjBNcorFN9bwZhGekF3</a>     | 📞Phy101📞 <a href="https://chat.whatsapp.com/GjC6e47vF1zHwRs5iOc13j">https://chat.whatsapp.com/GjC6e47vF1zHwRs5iOc13j</a>   |
| 📞Cs619📞 <a href="https://chat.whatsapp.com/FEklYr700ui0ihxqnk55A6">https://chat.whatsapp.com/FEklYr700ui0ihxqnk55A6</a>   | 📞Phy301📞 <a href="https://chat.whatsapp.com/lBrOdznEHDi92fGa76Dqmr">https://chat.whatsapp.com/lBrOdznEHDi92fGa76Dqmr</a>   |
|                                                                                                                           | 📞Sta301📞 <a href="https://chat.whatsapp.com/BchodAnjbydKXLuxFfkGxR">https://chat.whatsapp.com/BchodAnjbydKXLuxFfkGxR</a>   |

**...Paid Service Available for LMS Activities...**

**Only for those students who are so busy in office or work or job that they can't watch their LMS due to their Busyness**

**FOR MORE INFO. CONTACT US ONLY WHAT'S APP 0302-1400084**

D\_C\_Chapter\_01\_Topic 1 to 17.pptx

D\_C\_Chapter\_02\_Topic 18 to 28.pptx

D\_C\_Chapter\_03\_Topic 29 to 53 (Complete).pptx

D\_C\_Chapter\_04\_1Topic 54 to 60.pptx

D\_C\_Chapter\_04\_2\_Topic 61 to 75.pptx

D\_C\_Chapter\_05\_Topic 76 to 88.pptx

D\_C\_Chapter\_06\_Topic 89 to 100.pptx

D\_C\_Chapter\_07\_Topic 101 to 113.pptx

D\_C\_Chapter\_08\_Topic 114 to 123.pptx

D\_C\_Chapter\_09\_Topic 124 to 134.pptx

D\_C\_Chapter\_10\_Topic 135 to 148.pptx

D\_C\_Chapter\_11\_Topic 149 to 165.pptx

D\_C\_Chapter\_12\_Topic 166 to 179.pptx

D\_C\_Chapter\_13\_Topic 180 to 190.pptx

D\_C\_Chapter\_14\_1\_Topic 191 to 196.pptx

D\_C\_Chapter\_14\_2\_Topic 197 to 204.pptx

D\_C\_Chapter\_15\_1\_Topic 205 to 210.pptx

D\_C\_Chapter\_15\_2\_Topic 211 to 212.pptx

D\_C\_Chapter\_16\_Topic 213 to 220.pptx

# Introduction to the Course

**The Need for  
Communication**

**Trends and Advancements**

**What is taught in this  
course?**

**What is NOT taught in this  
course?**

**Tips and Tricks to do well**

**Text and References**

# Introduction to Data Communications

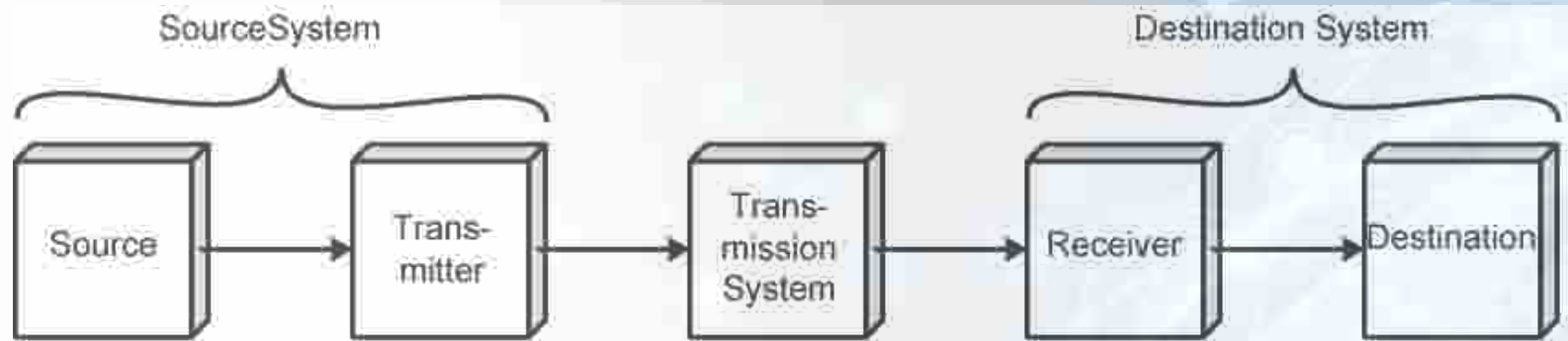
**Communication:** Sharing of Information (Local or remote)

**Telecommunications:** Communication at a Distance (includes telephony, telegraph, and television etc.)

**Data communications:** Exchange of data between two devices via some form of transmission media



# A Simple Communication Model



(a) General block diagram



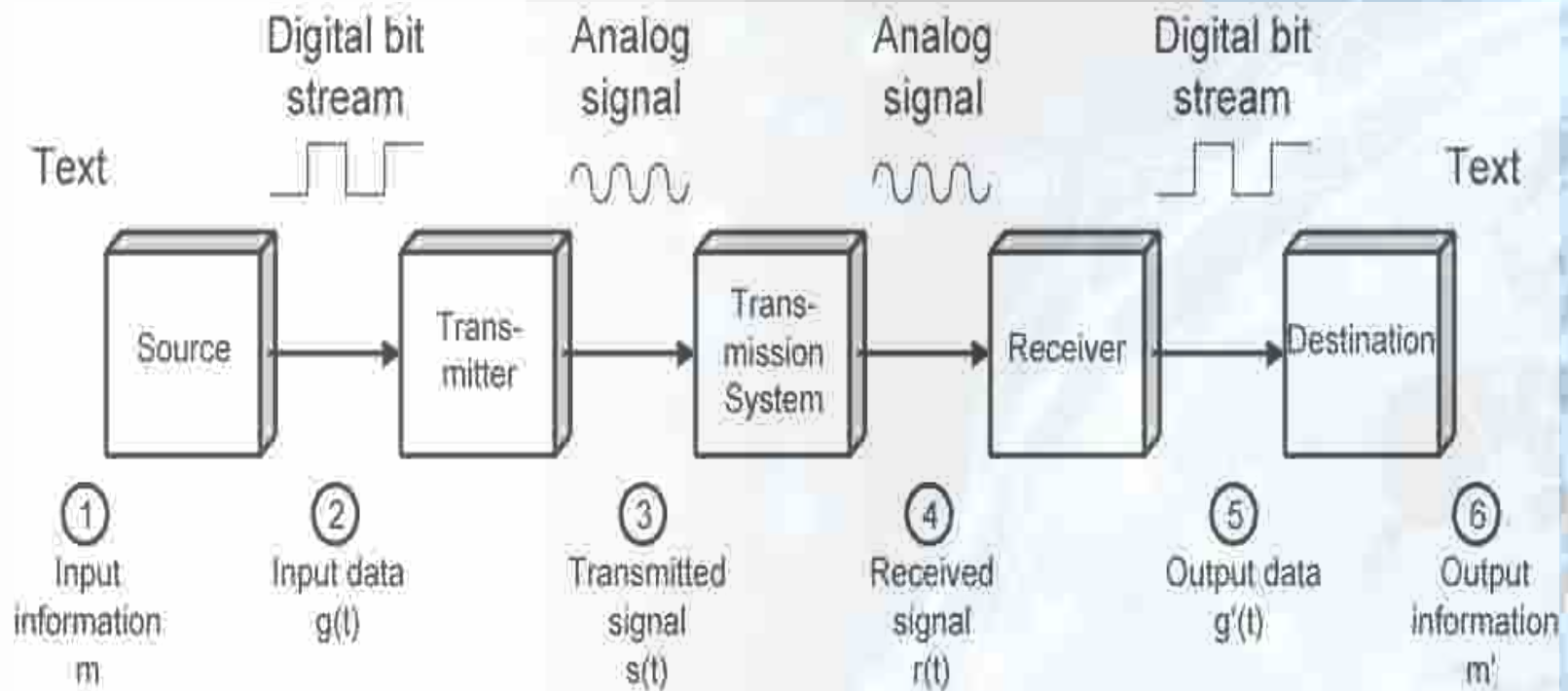
(b) Example

# Characteristics of a Data Communication System

## Effectiveness of a Data Communication System:

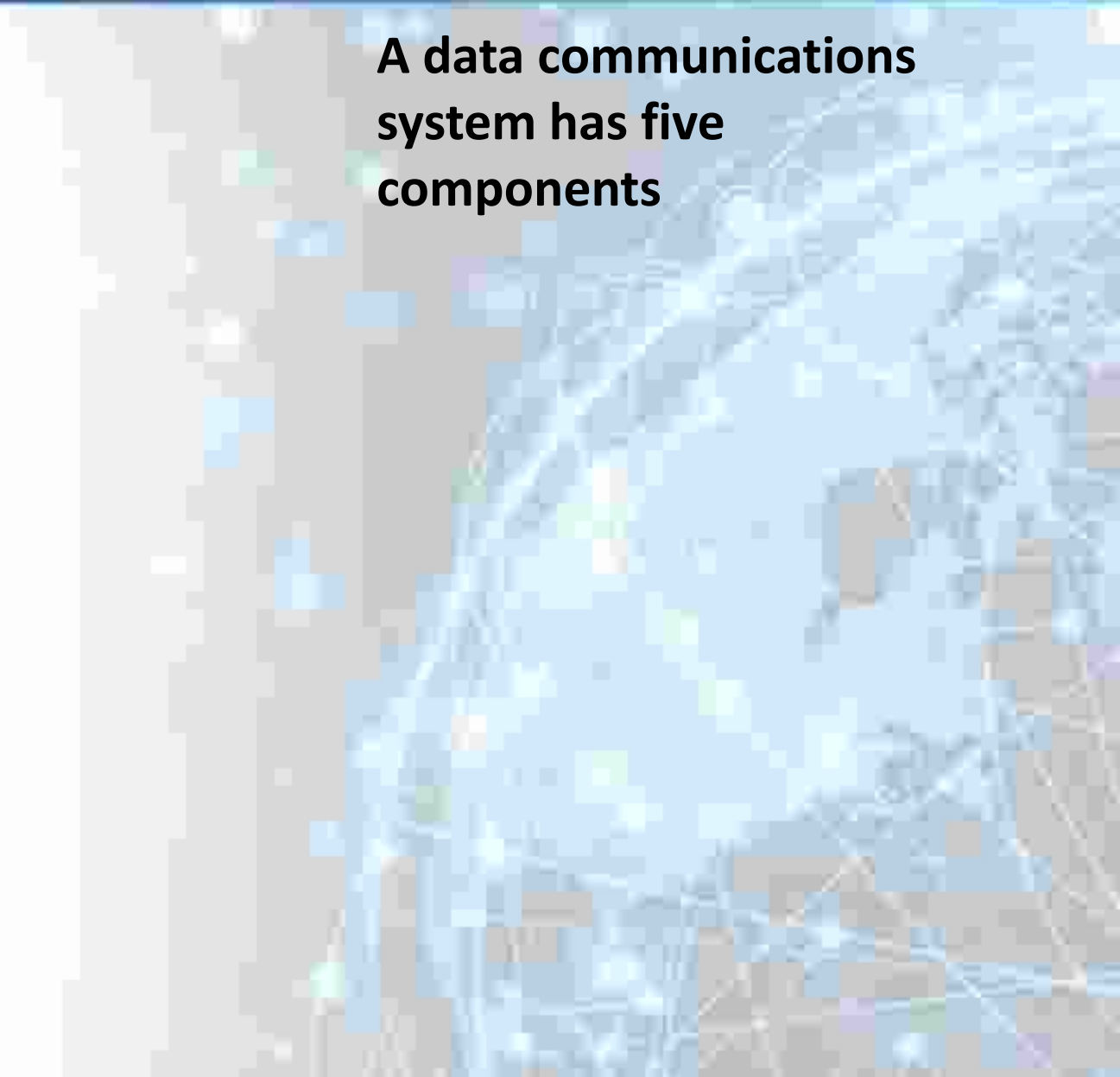
- **Delivery**
- **Accuracy**
- **Timeliness**
- **Jitter**

# Characteristics of a Data Communication System

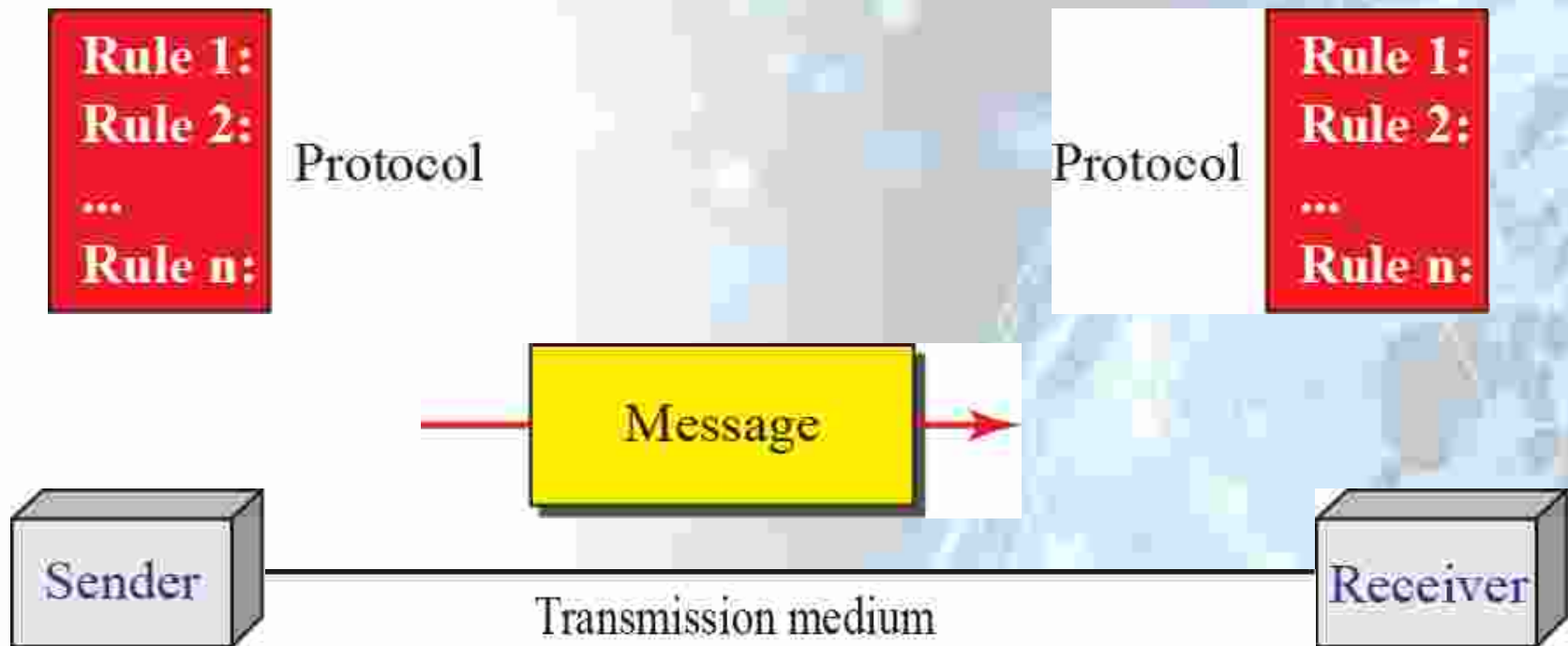


# Components of a Data Communication system

**A data communications system has five components**



# Components of Data Communication system



# Data Representation and Data Flow

## Forms of Information

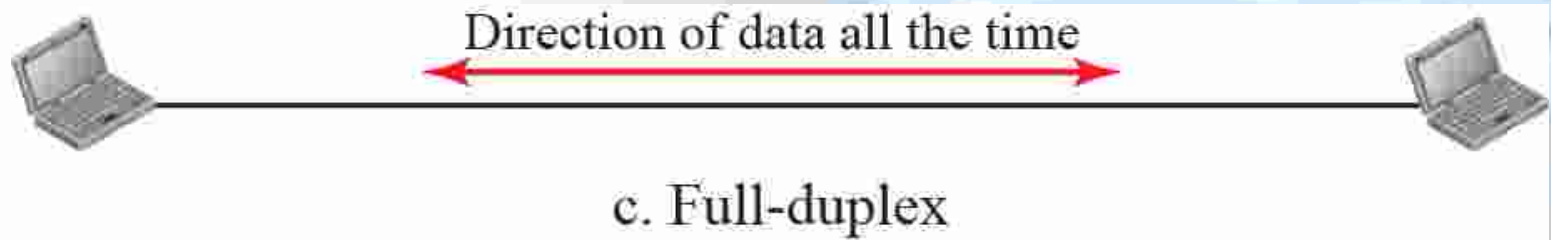
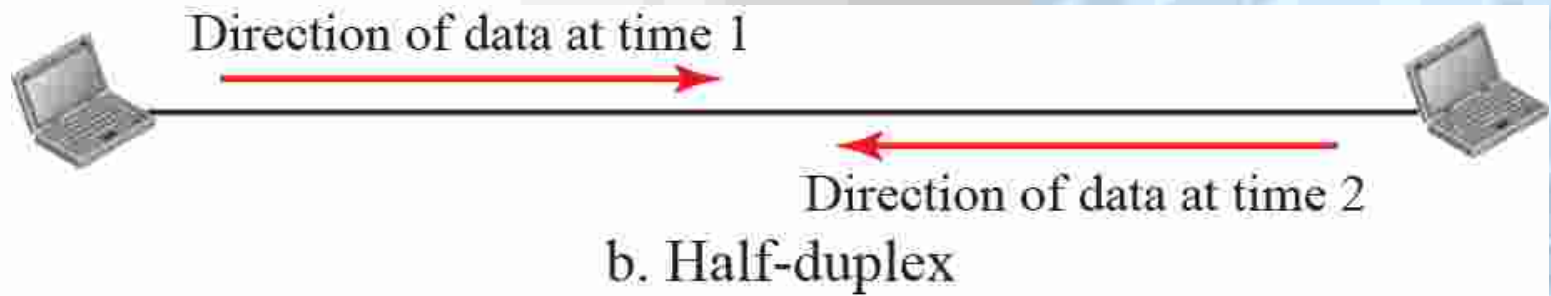
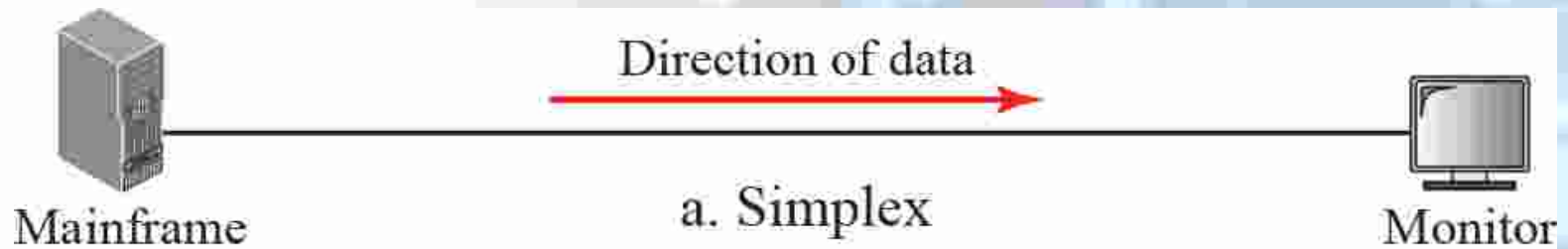
- Text
- Numbers
- Images
- Audio
- Video

# Data Representation and Data Flow

**Data Flow between two devices:**

- **Simplex**
- **Half-Duplex**
- **Full-Duplex**

# Data Flow





# Networks

- **Network:**  
Interconnection of a set of devices capable of communication
- **Host**
- **Connecting Device**

# Network Criteria

A network must be able to meet a certain number of criteria:

- **Performance**
  - ✓ Throughput
  - ✓ Delay
- **Reliability**
- **Security**

# Physical Structures

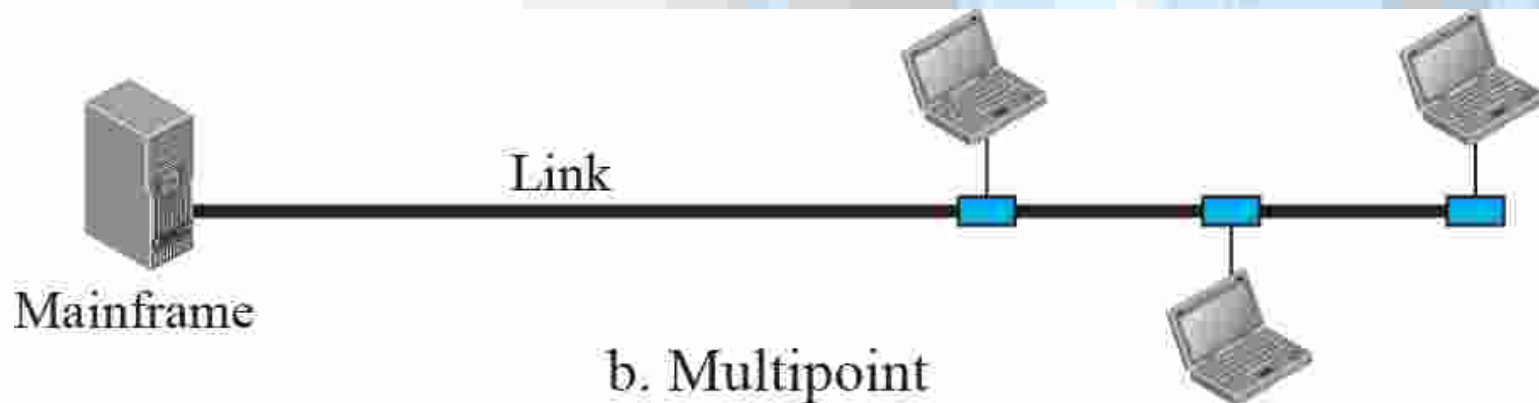
## Physical Network Attributes

- **Link**
- **Type of Connection**
  - ✓ **Point-to-Point**
  - ✓ **Multipoint**

# Physical Structures



a. Point-to-point

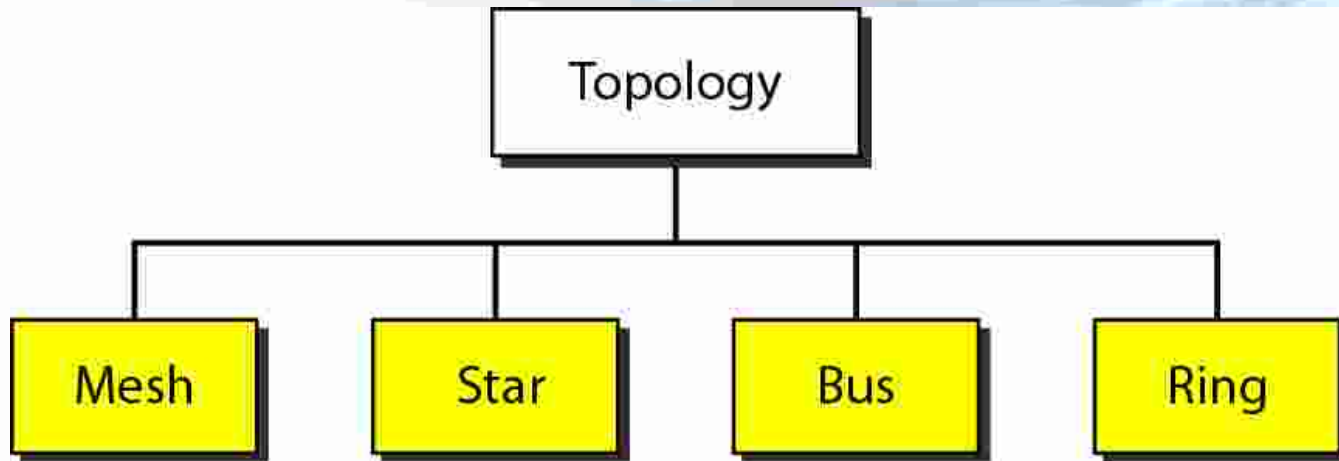


b. Multipoint

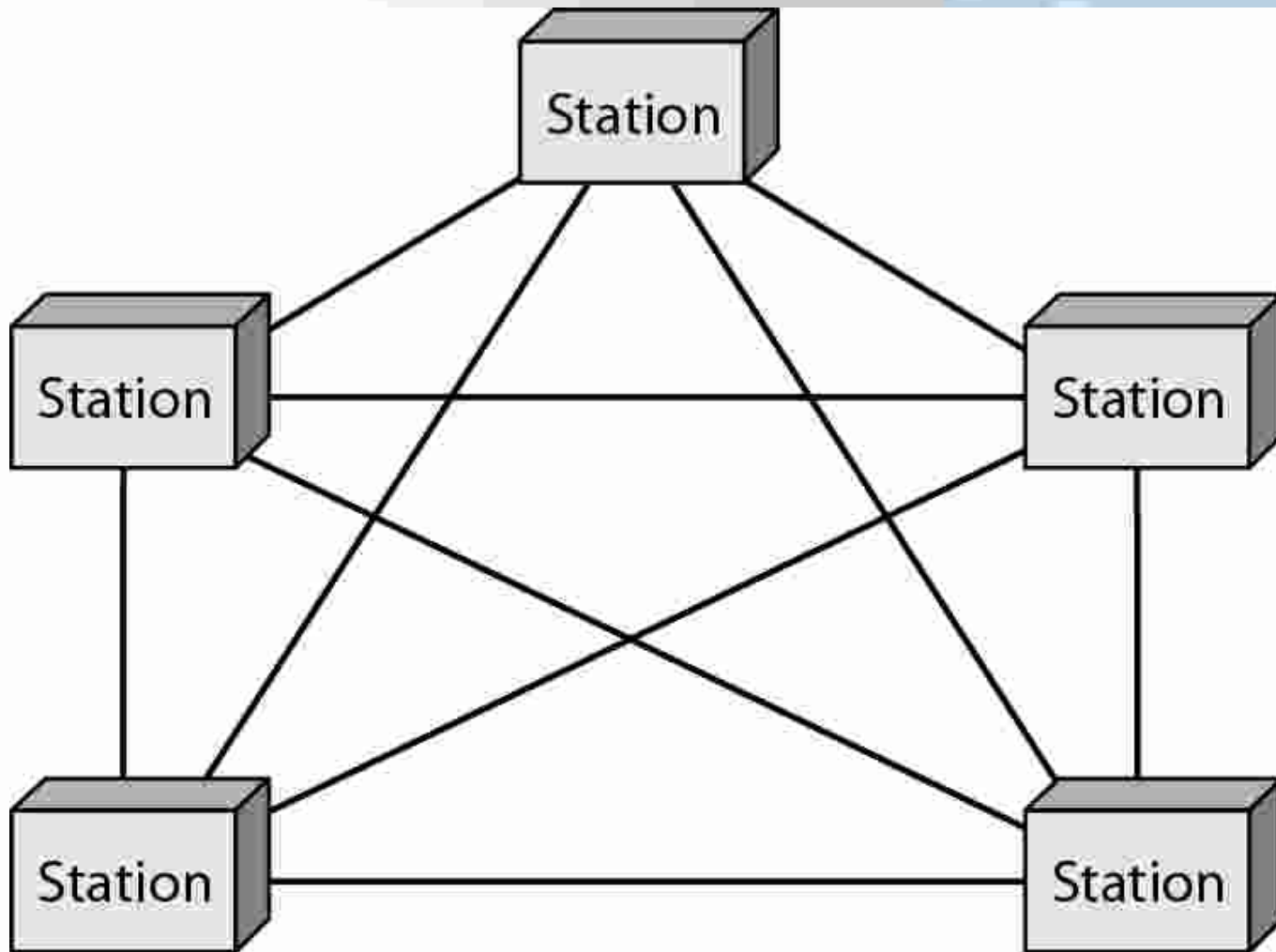
# Physical Topologies

- **Physical Layout of Network**
- **Links + Nodes = Topology**
- **Physical Topologies:**
  - ✓ **Mesh**
  - ✓ **Star**
  - ✓ **Bus**
  - ✓ **Ring**

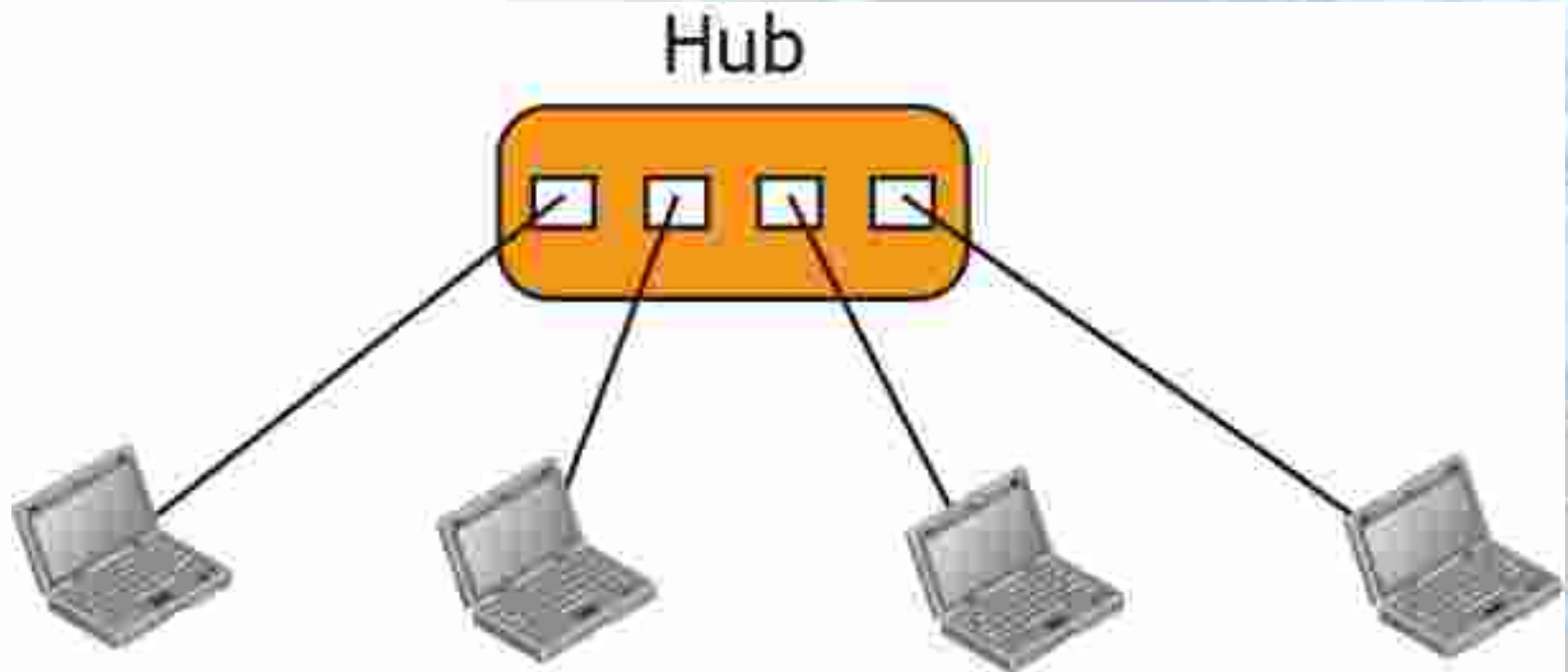
# Physical Topologies



# Mesh Topology

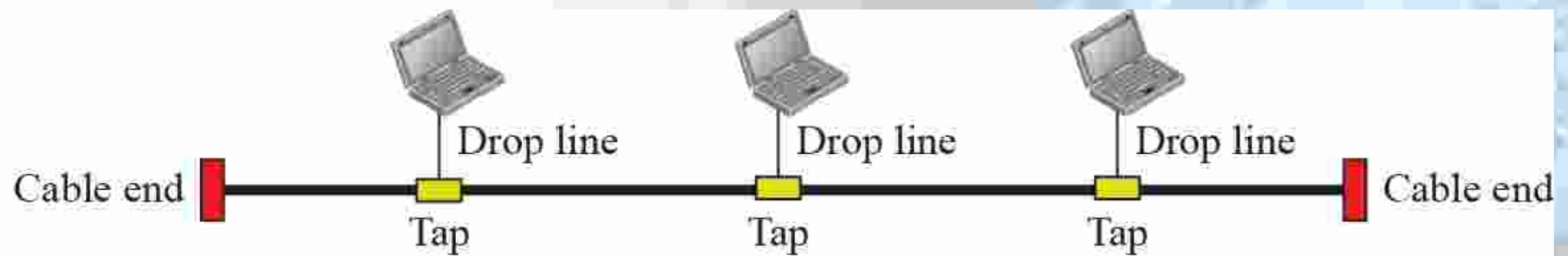


# Star Topology

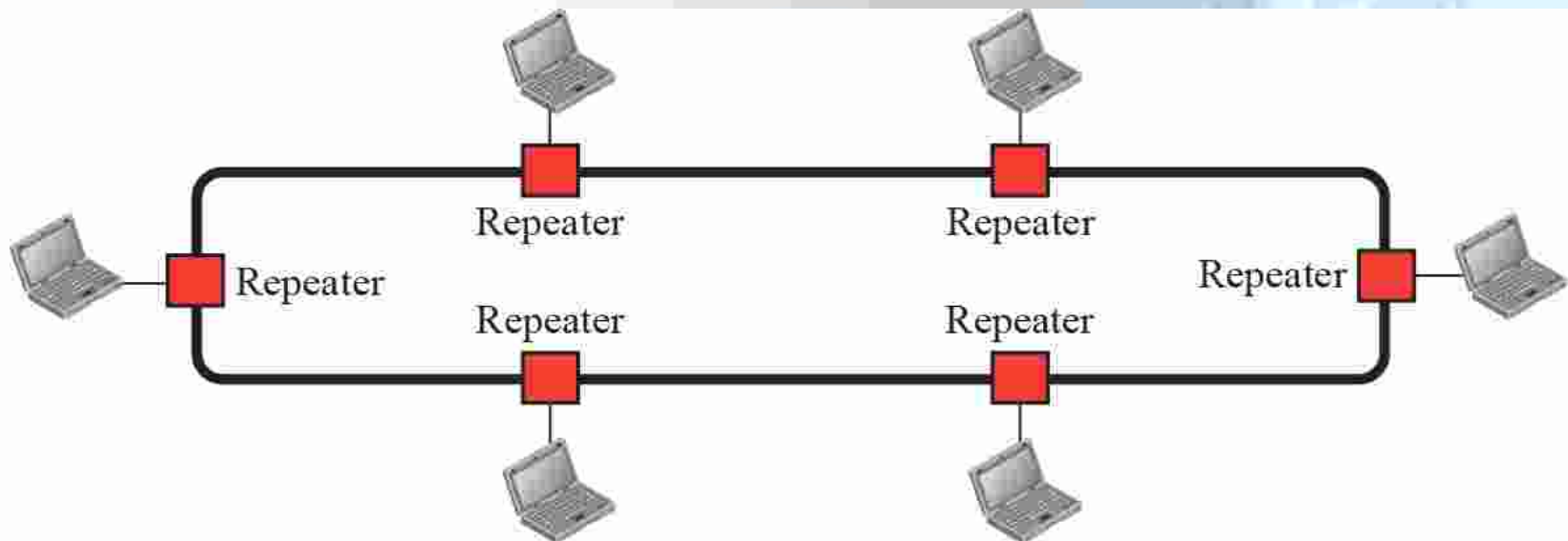




# Bus Topology



# Ring Topology



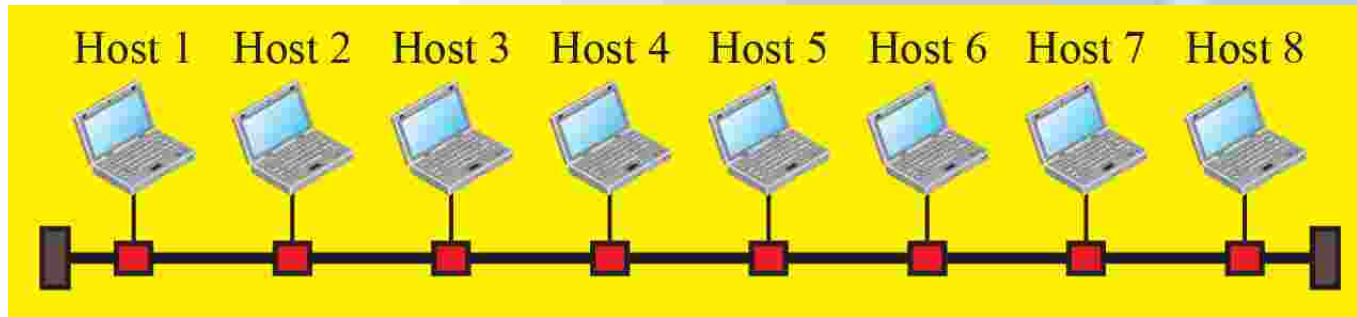
# Networks Types

- **Network classification:**
  - ✓ **Size**
  - ✓ **Geographical Coverage**
  - ✓ **Ownership**
- **Local Area Networks (LANs)**
- **Wide Area Networks (WANs)**

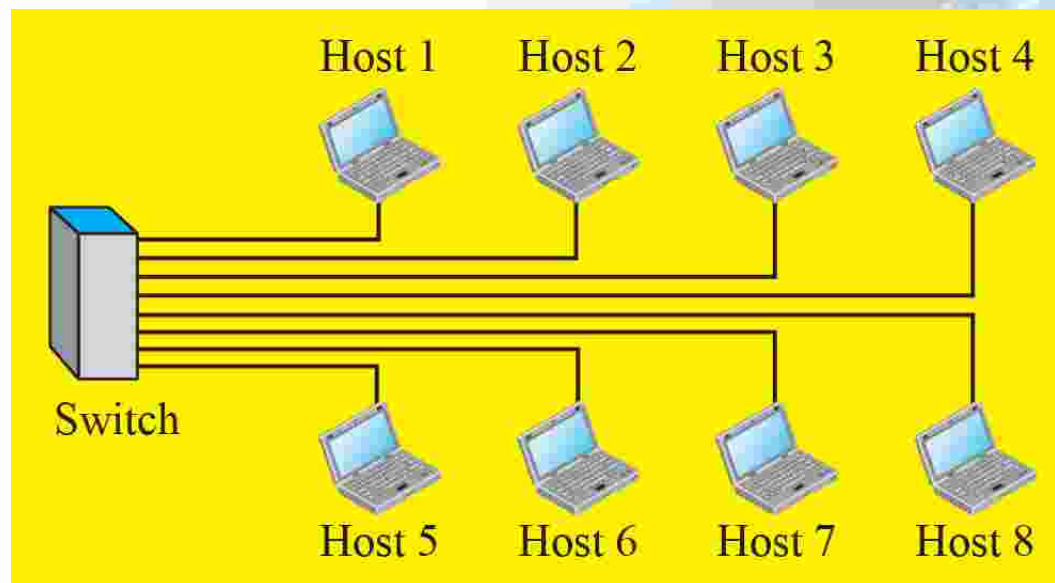
# Local Area Networks

- **Usually Privately owned**
- **Connects some hosts in a single office, building, or campus**
- **Can be as simple as two PCs and a printer in someone's home office**
- **Can extend throughout a company**
- **Host Address**

# Local Area Networks



a. LAN with a common cable (past)



b. LAN with a switch (today)

## Legend

-  A host (of any type)
-  A switch
-  A cable tap
-  A cable end
-  The common cable
-  A connection

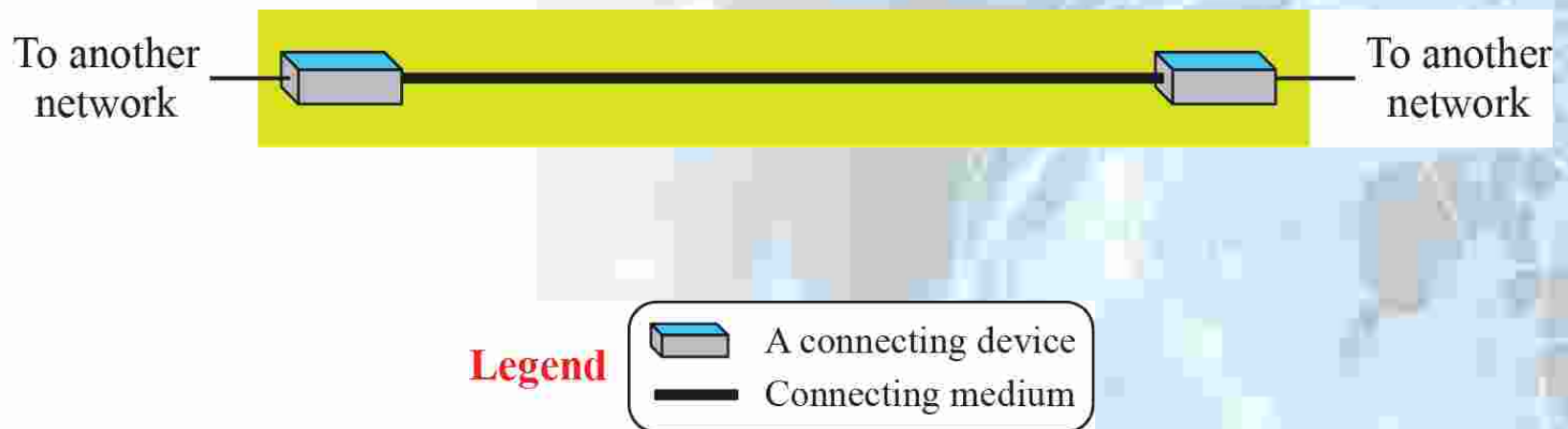
# Wide Area Network

- Wider geographical span than a LAN
- Spans a town, a state, a country, or even the world
- Interconnects connecting devices such as switches, routers, or modems
- Normally created and run by communication companies

# Wide Area Network

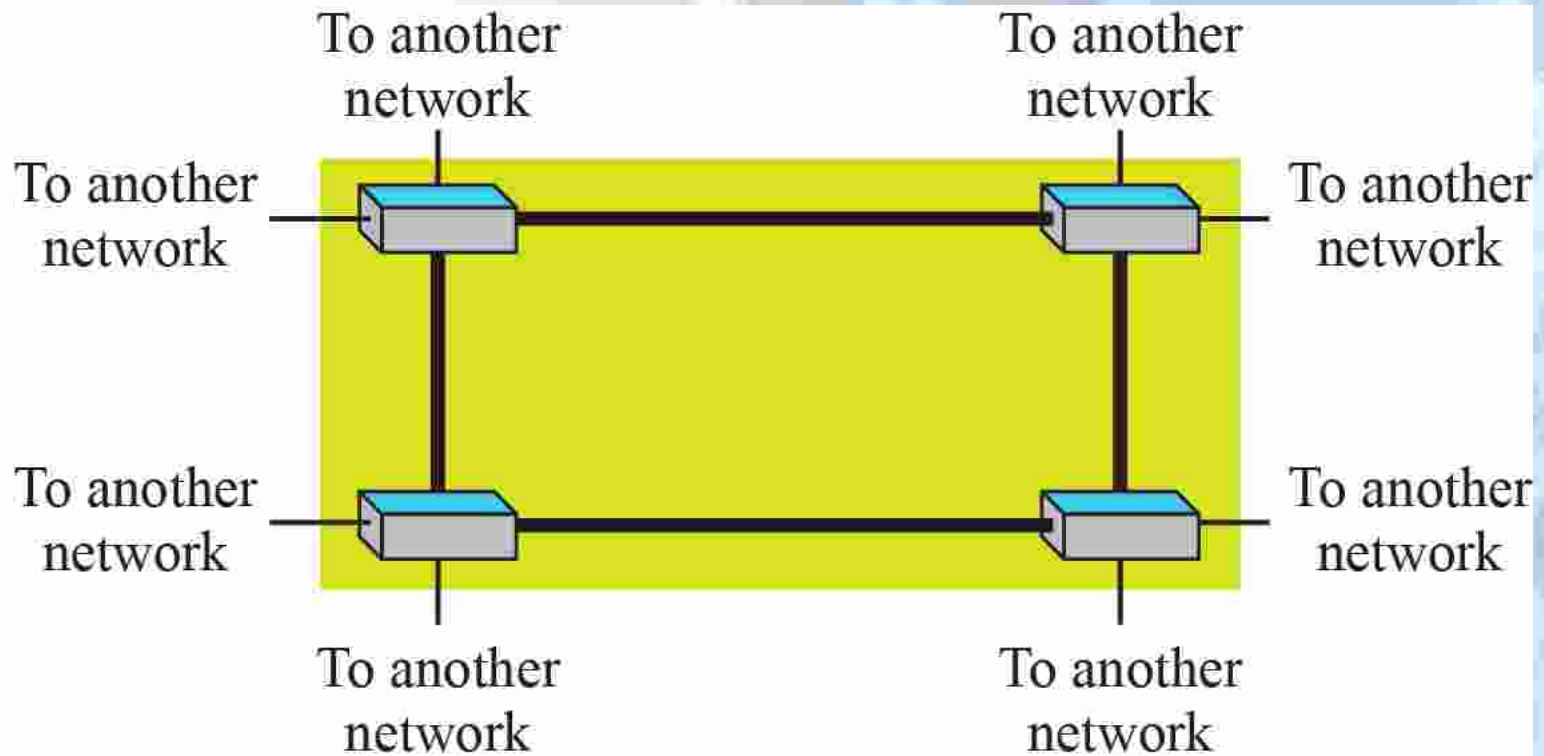
- **Point-to-Point WAN**
- **Switched WAN**
- **Internetwork**

# Point-to-Point WANs





# Switched WANs



## Legend

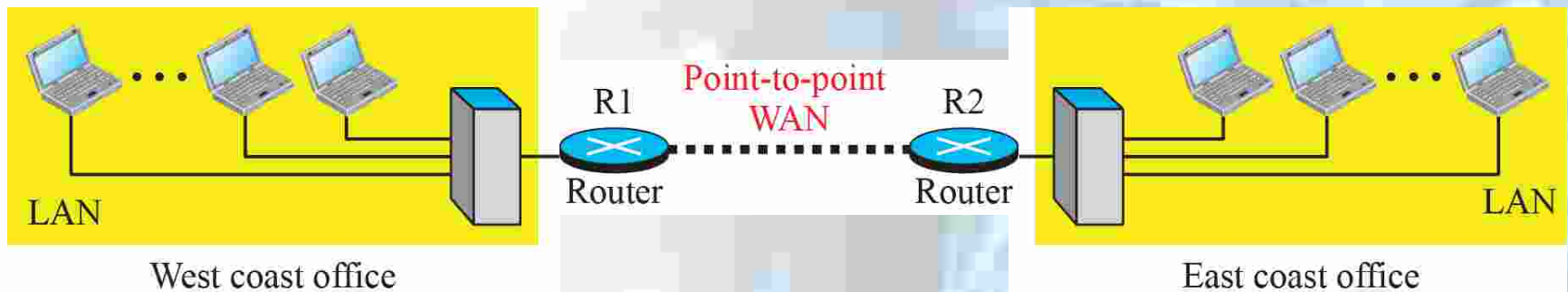


A switch



Connecting medium

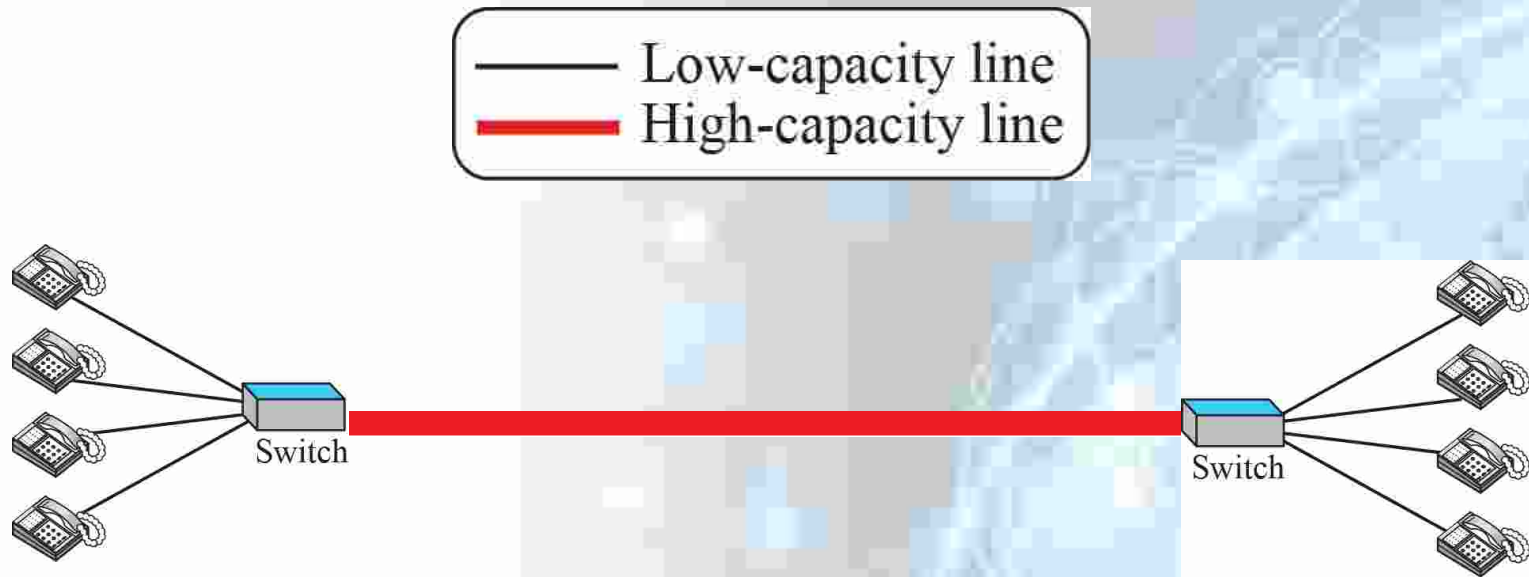
# Internetwork



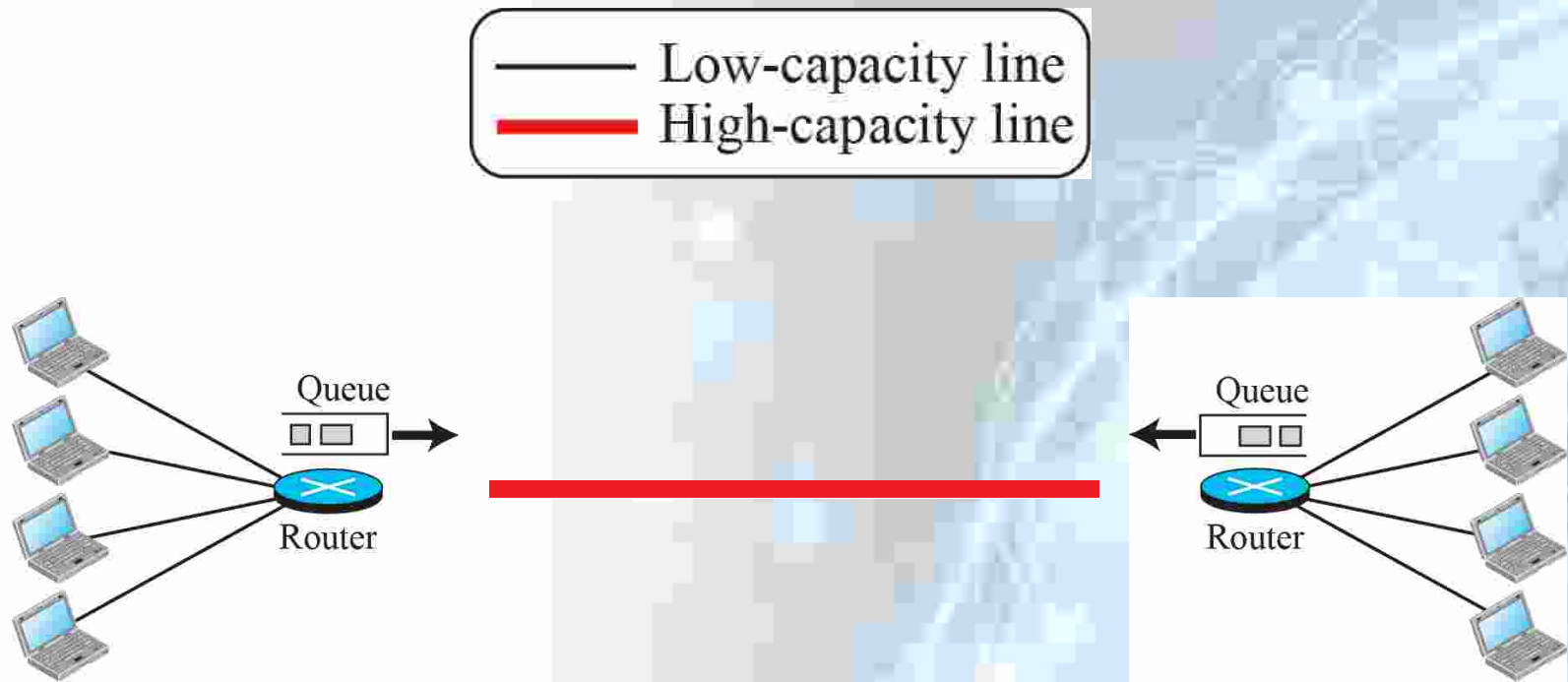
# Switching

- **Switching**
  - ✓ **Circuit-Switched Network**
  - ✓ **Packet-Switched Network**

# Circuit Switched Network



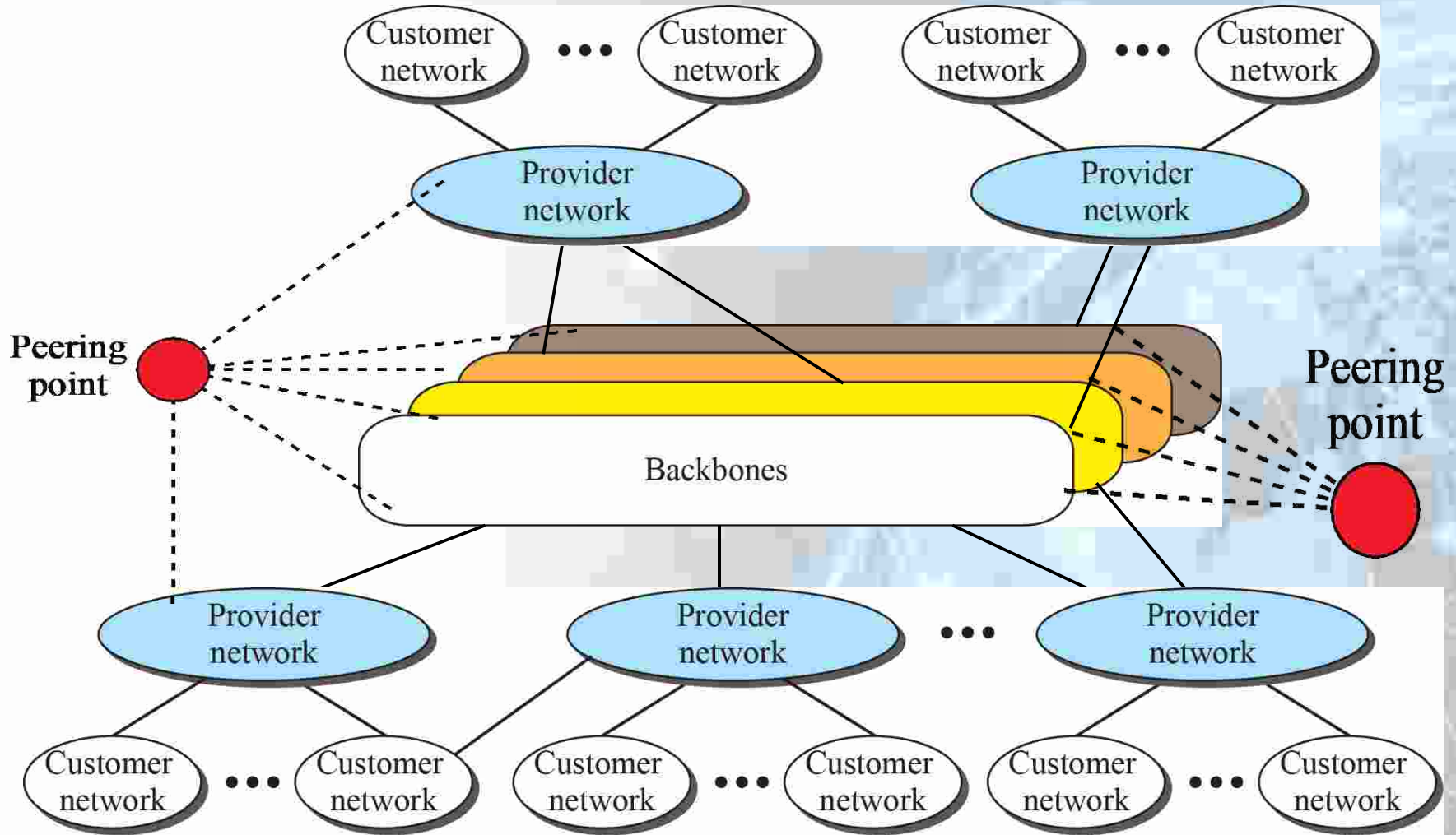
# Packet Switched Network



# The Internet

- An internet (note the lowercase i) is two or more networks that can communicate with each other
- The Internet (uppercase I ), and is composed of thousands of interconnected networks.
- Accessing the Internet

# The Internet



# Internet History

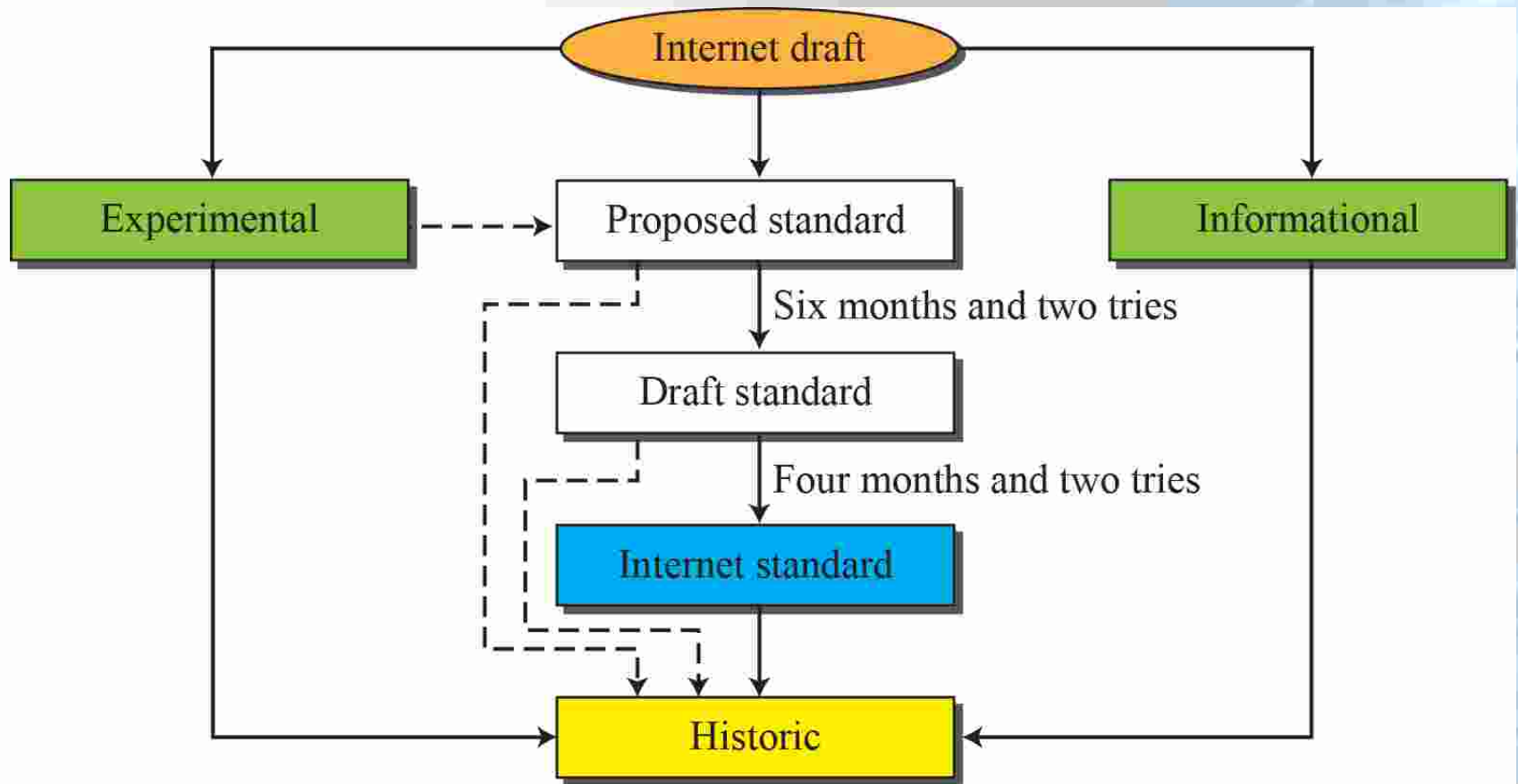
- **Telegraph and Telephone networks, before 1960:**
  - ✓ **Constant-rate communication only**
- **ARPANET- Packet Switched**
- **Birth of the Internet & TCP/IP**
- **MILNET**
- **CSNET**
- **NSFNET**
- **Internet Today**



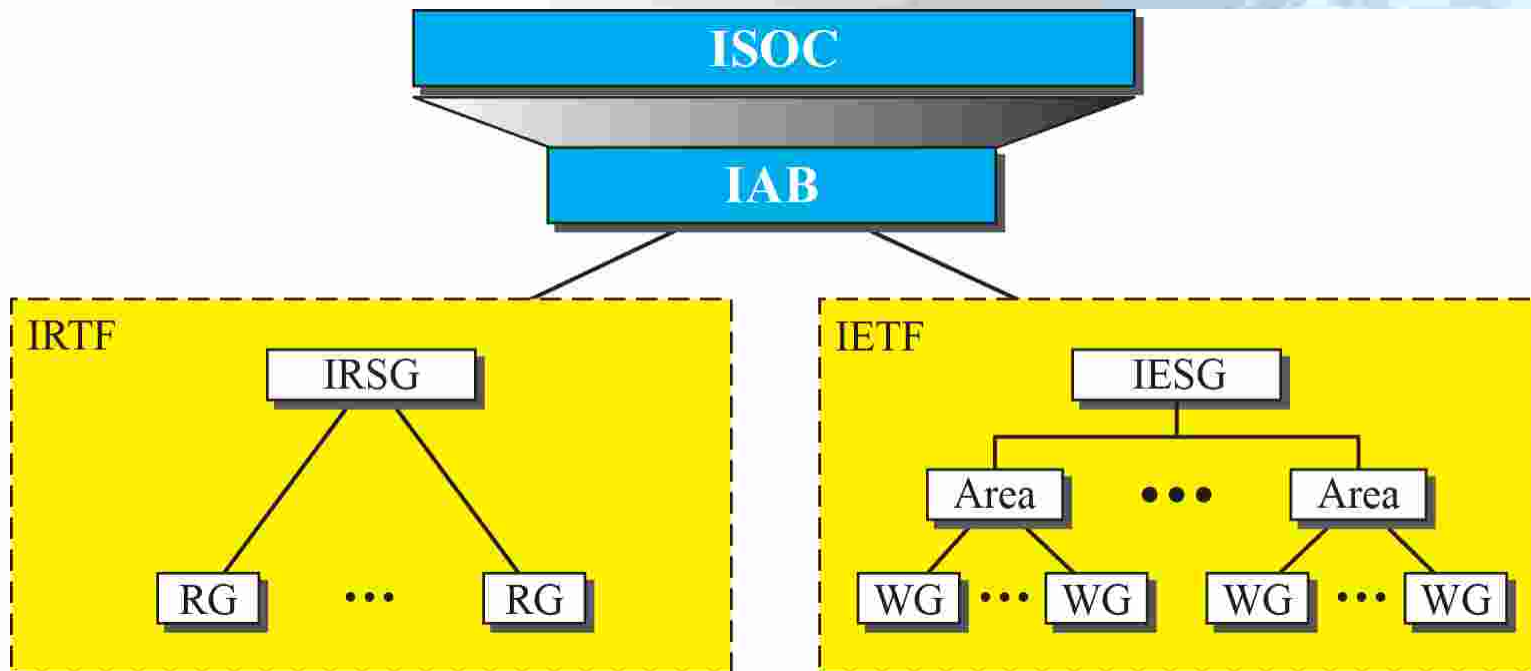
# Internet Standards and Administration

- Internet draft
- Request for Comments (RFC)
  - ✓ Proposed Standard
  - ✓ Draft Standard
  - ✓ Internet Standard
  - ✓ Historic
  - ✓ Experimental
  - ✓ Informational

# Internet Standards



# Internet Administration



# Protocol Layering - Introduction

- **Protocol**

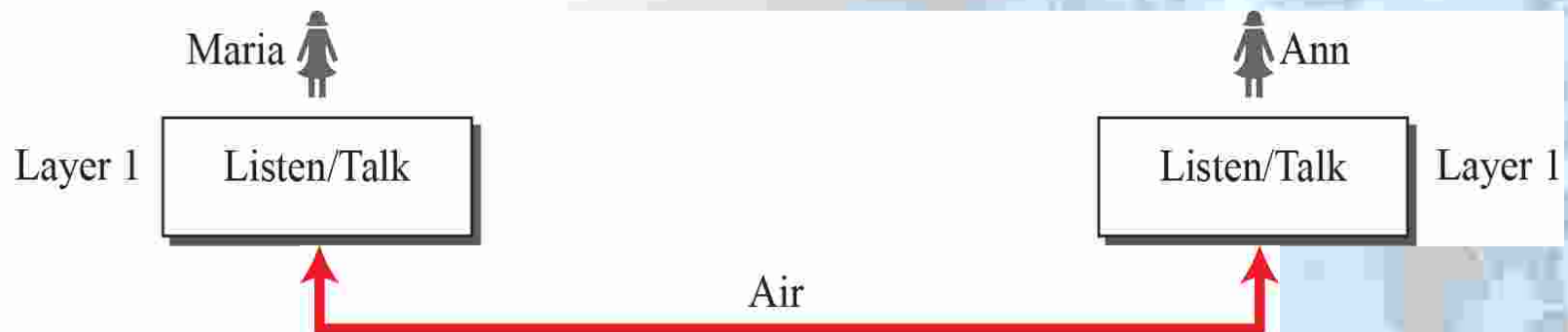
Rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively

- **Protocol Layering**

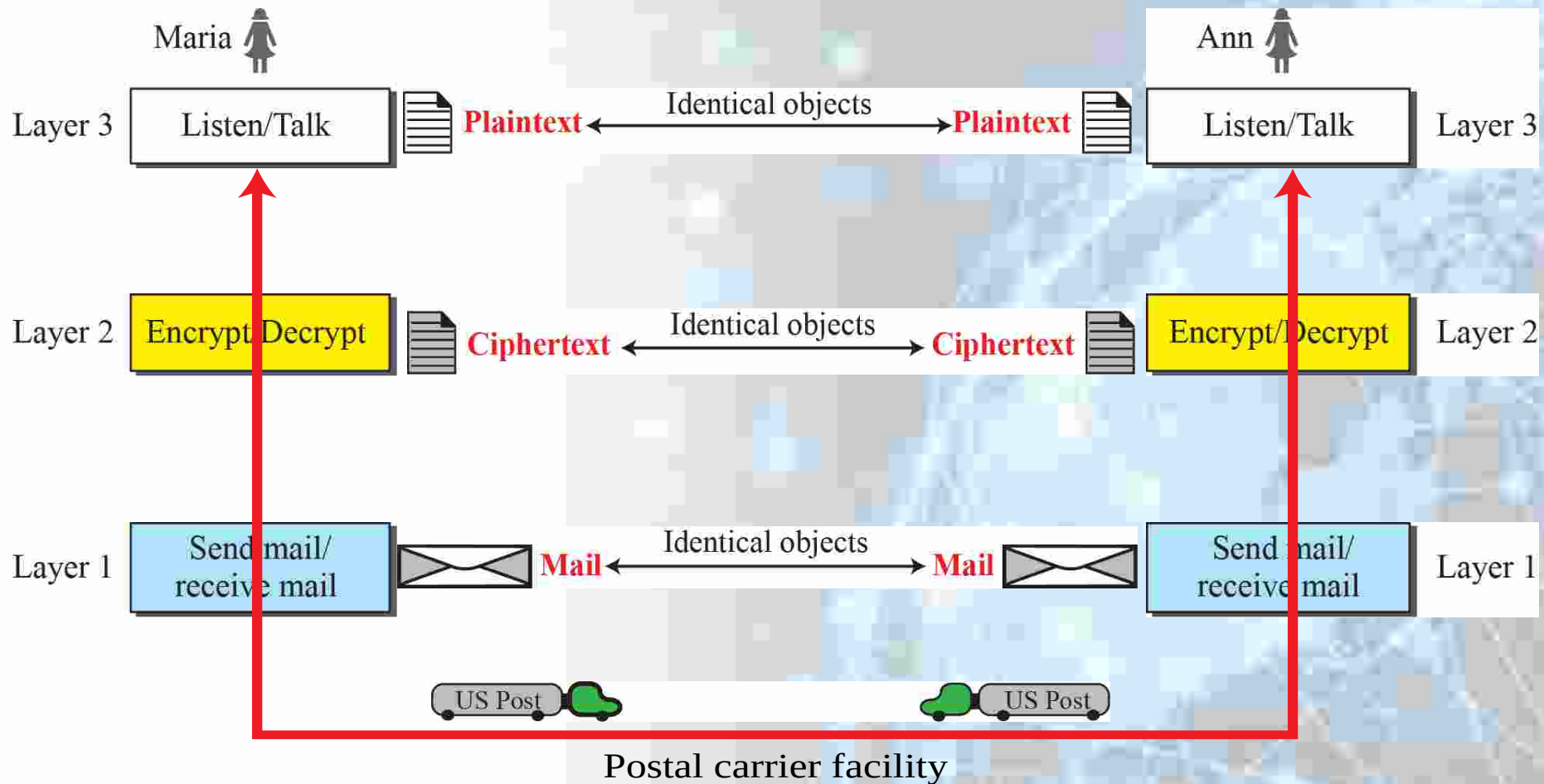
Simple Communication:  
only one simple protocol

Complex Communication,  
we need a protocol at each  
layer, or Protocol Layering

# Protocol Layering - Example Scenario 1



# Protocol Layering - Example Scenario 2



# Protocol Layering - Advantages and Disadvantages

- **Advantages**
  - ✓ Modularity
  - ✓ Separation of Service & Implementation
  - ✓ Reduced Complexity & Cost
- **Disadvantages**
  - ✓ None Really!

# Protocol Layering - Principles

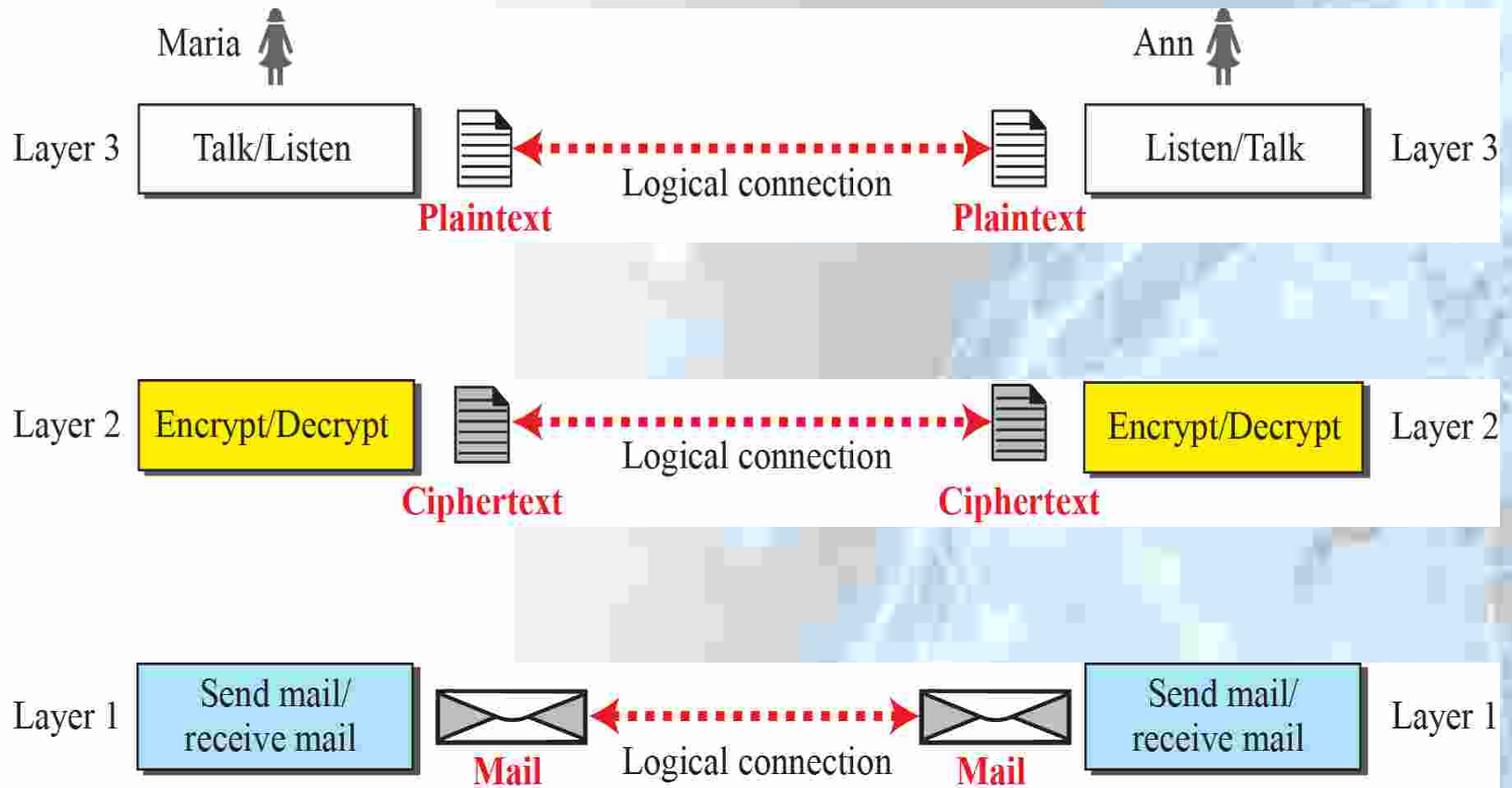
- **Two Principles**
  - ✓ Bidirectional Communication → Each Layer performs two opposite tasks in each direction
  - ✓ Two objects under each layer at both sites should be identical



# Protocol Layering - Logical Connections

- **Logical Connections**
  - ✓ Imaginary connection between each layer

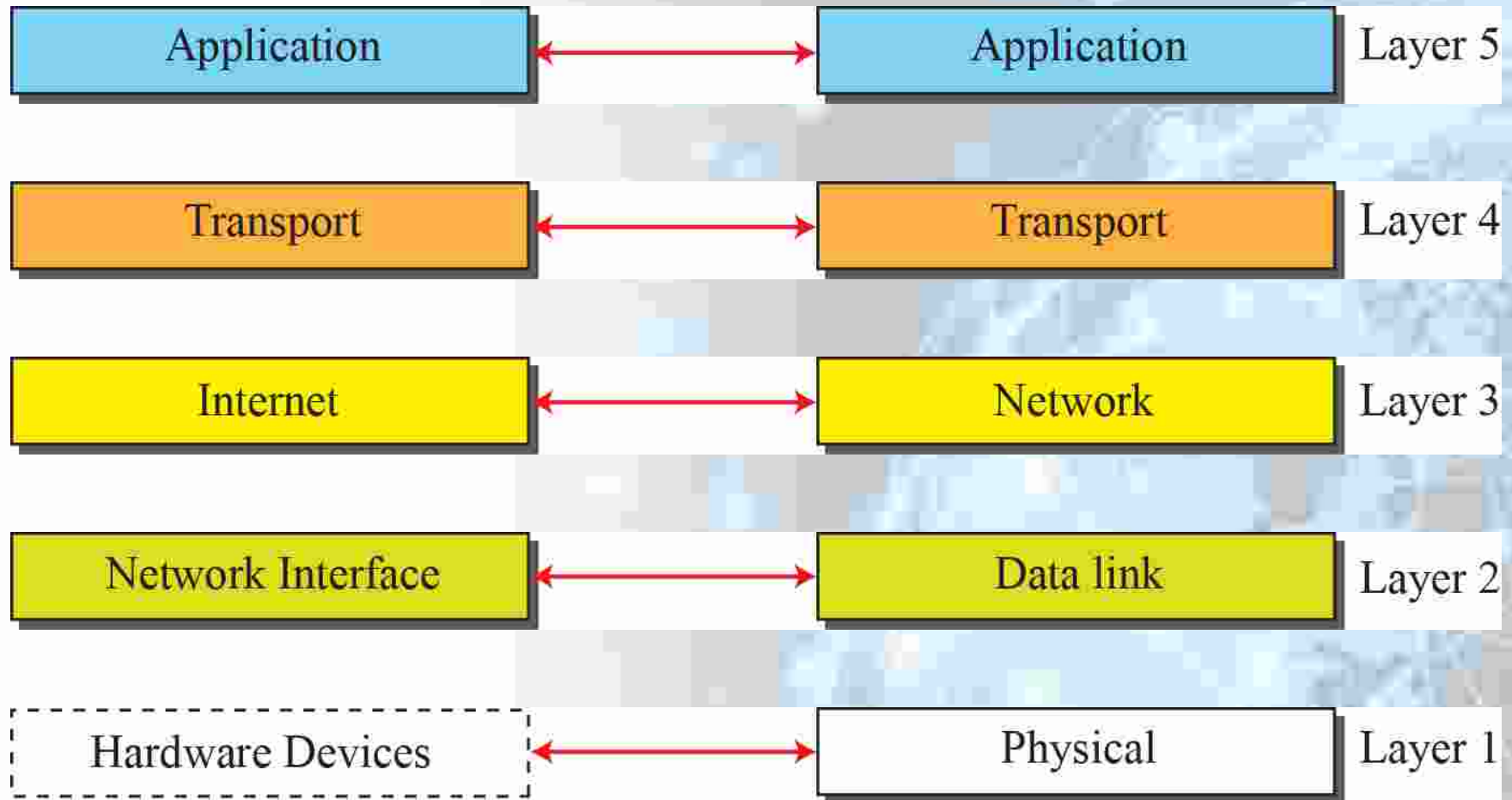
# Protocol Layering



# TCP/IP Protocol Suite

- **TCP/IP Protocol Suite**
  - ✓ Protocol suite used in Internet today
  - ✓ Each Layer provides specific functionality
  - ✓ Hierarchical Protocol
  - ✓ Presented in 1973 and chosen to be the official protocol of Internet in 1983

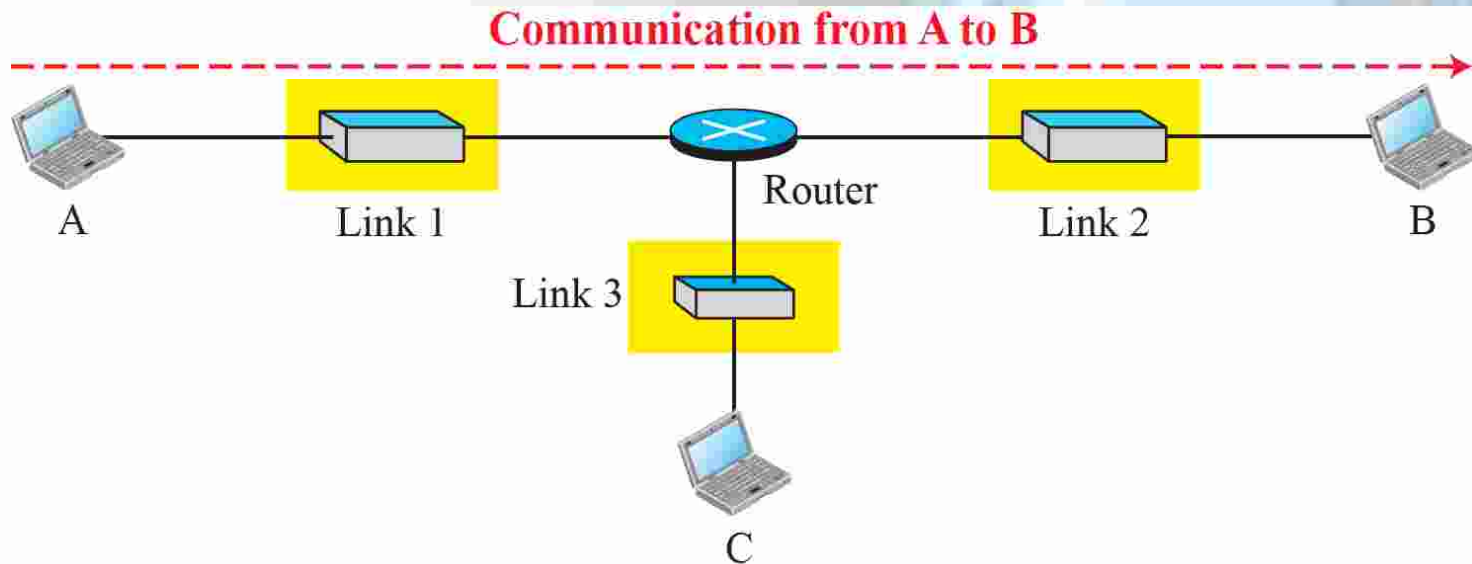
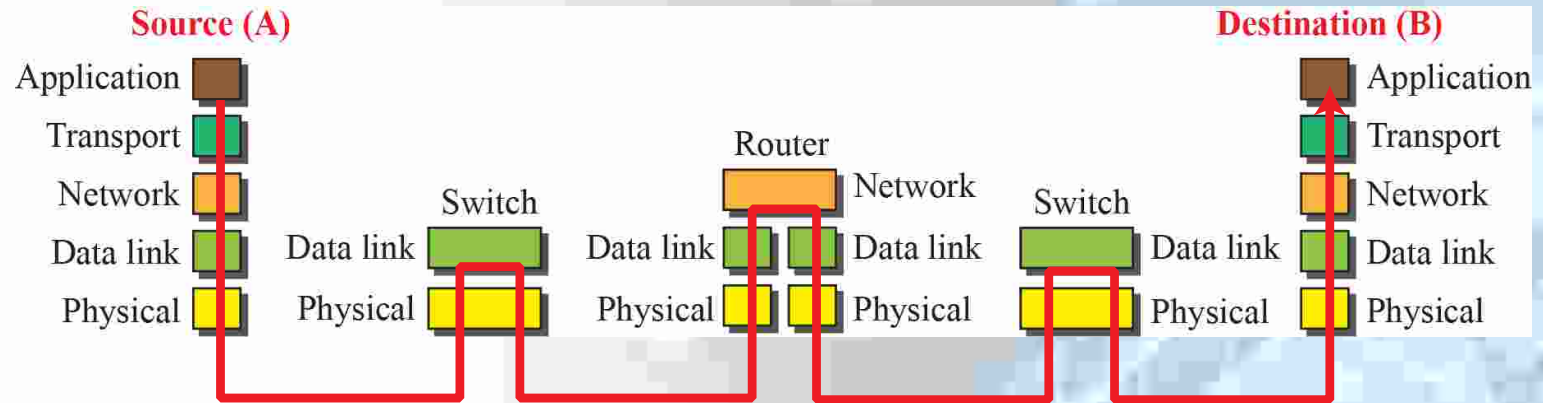
# TCP/IP Protocol Suite



a. Original layers

b. Layers used in this book

# TCP/IP Protocol Suite - Layered Architecture

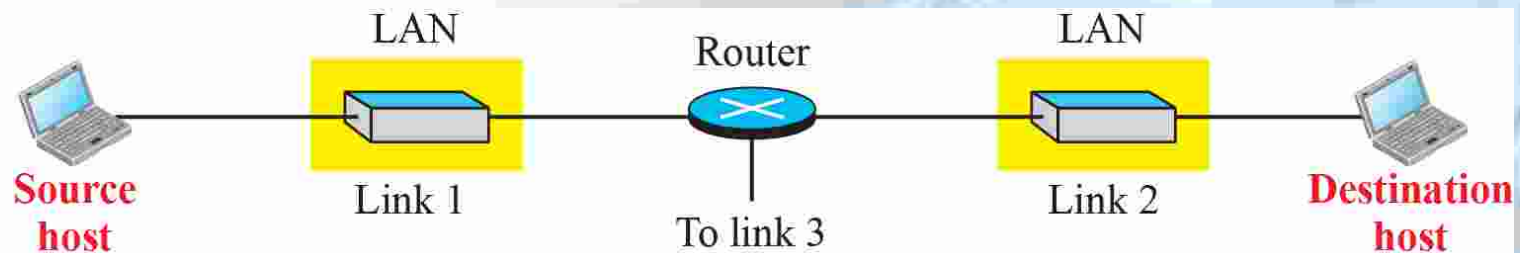
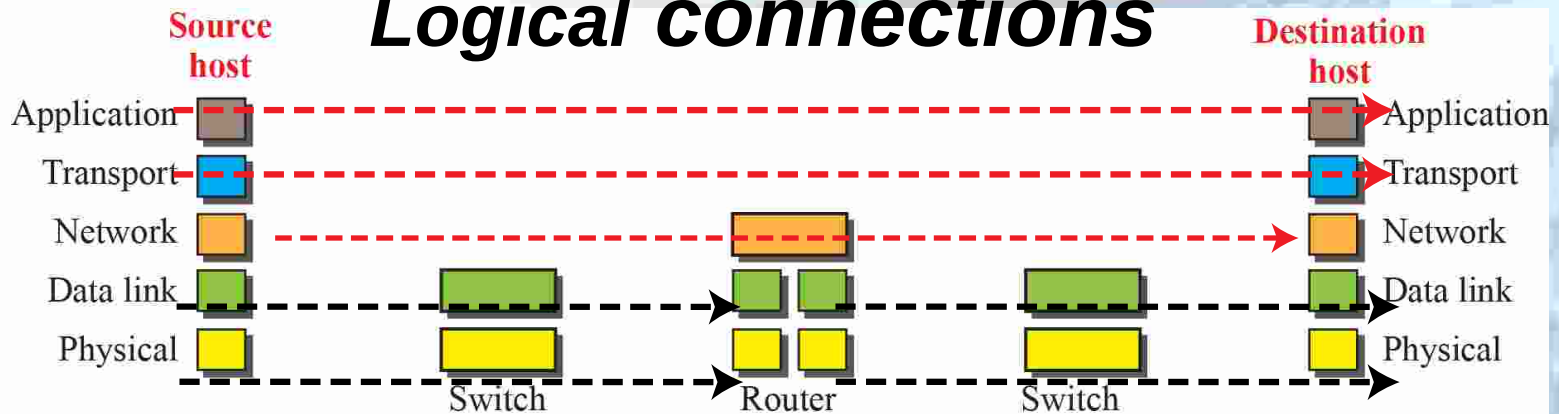


# TCP/IP Protocol Suite – Function of Layers

- **Logical Connections between TCP/IP Layers**

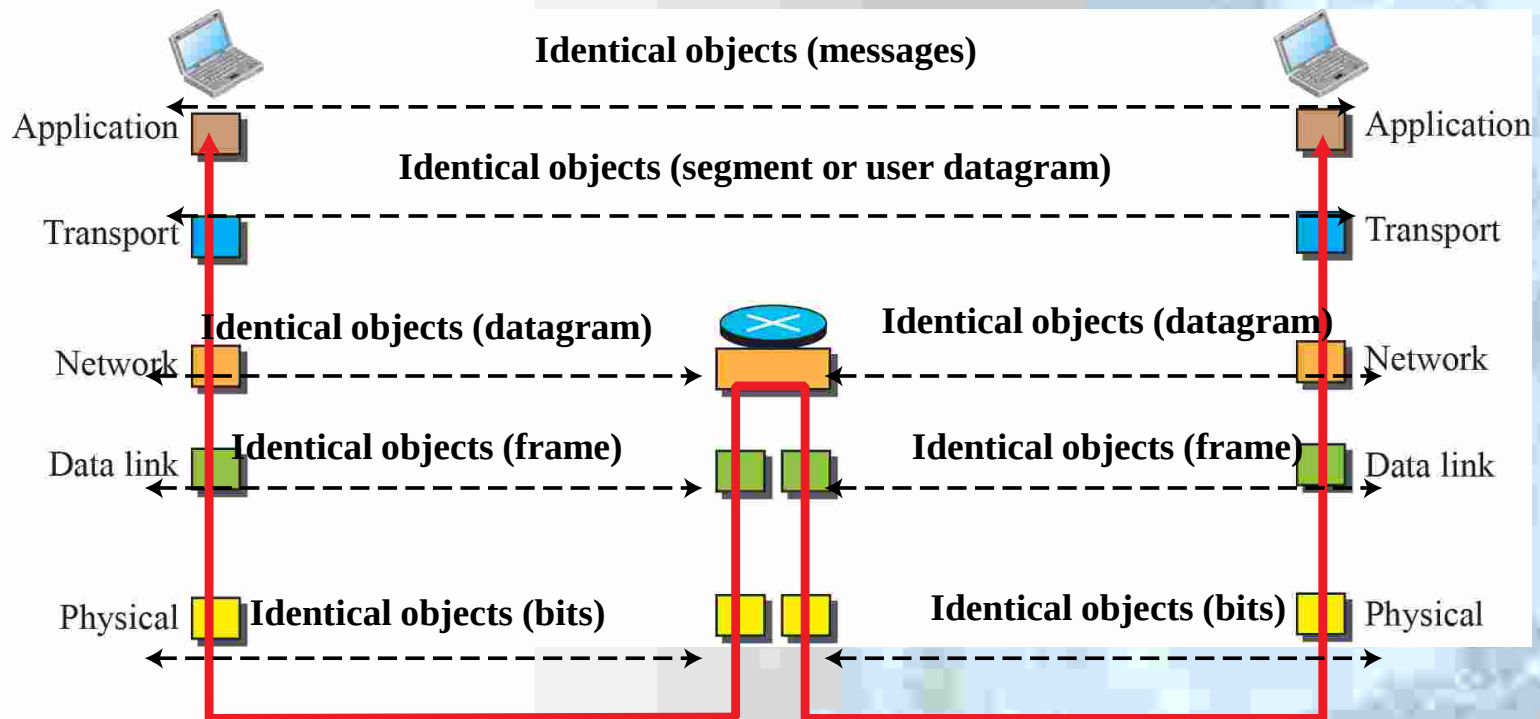
# TCP/IP Protocol Suite – Function of Layers

## *Logical connections*



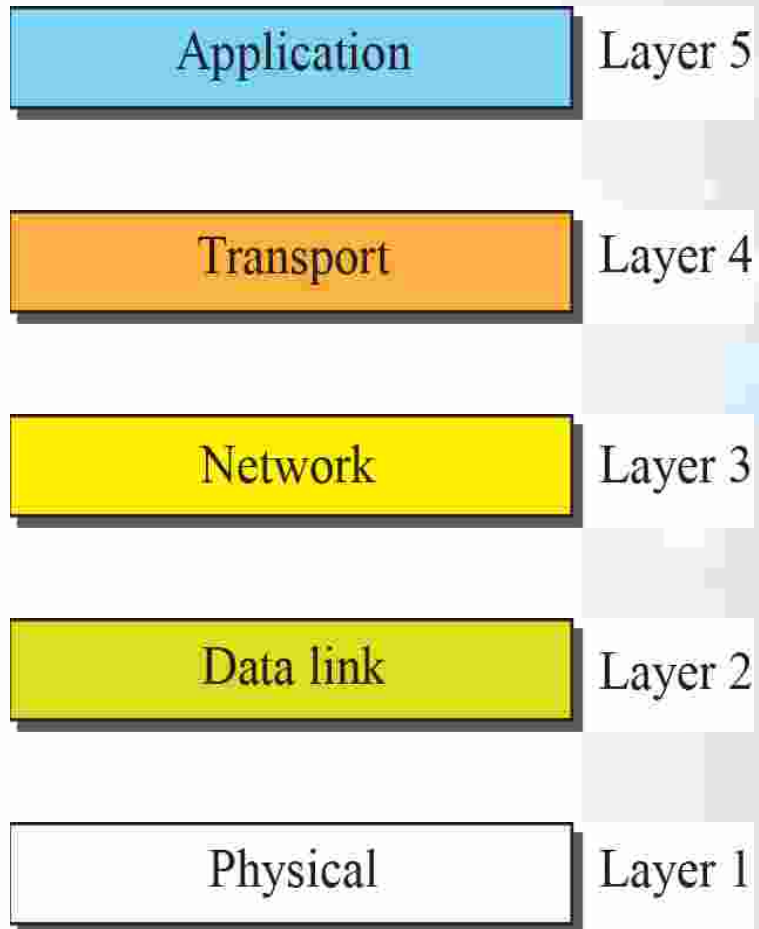
# TCP/IP Protocol Suite – Function of Layers

**Notes:** We have not shown switches because they don't change objects.

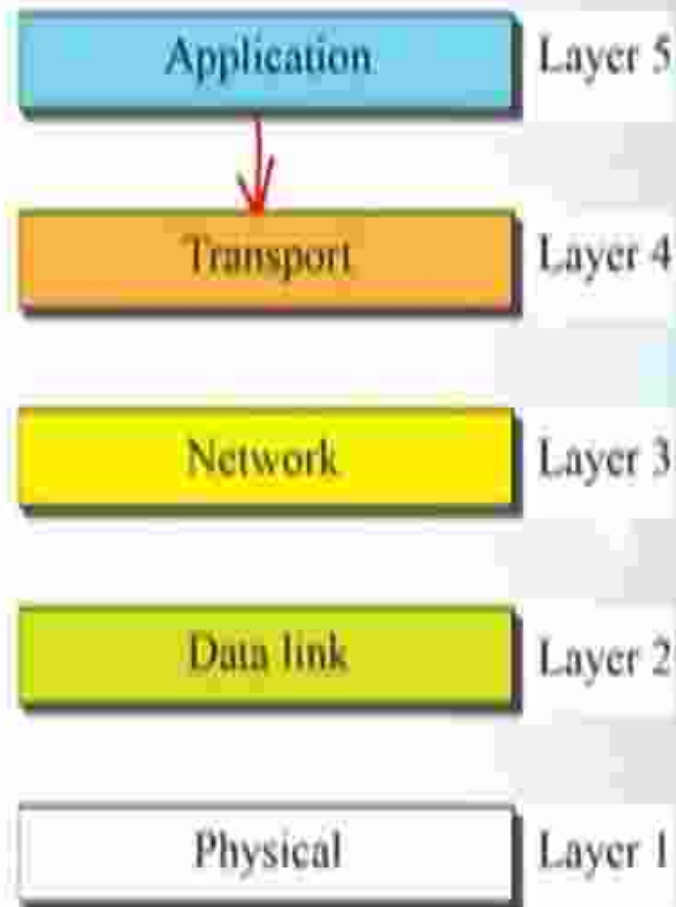




# TCP/IP Protocol Suite – Layer Description



# TCP/IP Protocol Suite – Layer Description

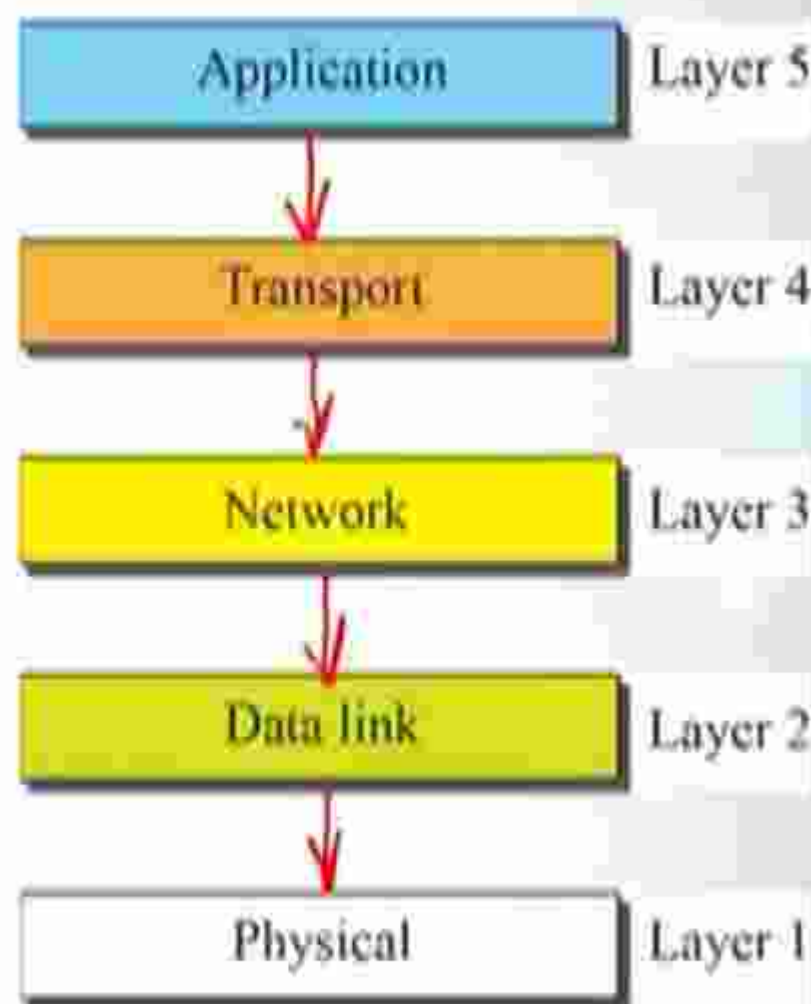


*Message (object) HTTP, SMTP, FTP, ...*

*Segment (User Datagram) TCP, UDP*

*Datagram → Routing → IP*

# TCP/IP Protocol Suite – Layer Description



*Message (object) HTTP, SMTP*

*Segment (User Datagram) TCP*

*Datagram → Routing →*

*Frame (object) Standard,*

*Bits (object)*



*TM*

# Encapsulation & Decapsulation

- Important Concept in Internet Protocol Layering
- Layer Header

# Encapsulation & Decapsulation

## Legend

4 Header at transport layer

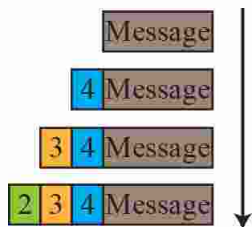
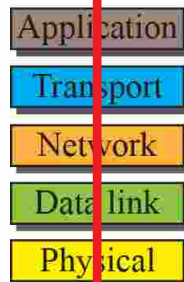
3 Header at network layer

2 Header at data-link layer

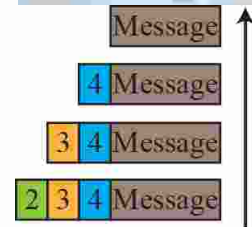
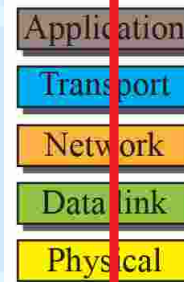
↓ Encapsulate

↑ Decapsulate

Source host



Destination host



# Addressing in TCP/IP Protocol Suite

- Every communication needs at least two addresses:
  - ✓ Source Address &
  - ✓ Destination Address
- Addressing by Layer
- Physical Layer is an exception

# Addressing in TCP/IP Protocol Suite

| Packet names            | Layers            | Addresses            |
|-------------------------|-------------------|----------------------|
| Message                 | Application layer | Names                |
| Segment / User datagram | Transport layer   | Port numbers         |
| Datagram                | Network layer     | Logical addresses    |
| Frame                   | Data-link layer   | Link-layer addresses |
| Bits                    | Physical layer    |                      |

# The Open System Interconnection (OSI) Model

- International Organization for Standardization (ISO)
- ISO established in 1947
- Close to three-fourths of countries represented
- Introduced OSI Model in late 1970s
- OSI: a 7-Layer Model



# The Open System Interconnection (OSI) Model

Layer 7

Application

Layer 6

Presentation

Layer 5

Session

Layer 4

Transport

Layer 3

Network

Layer 2

Data link

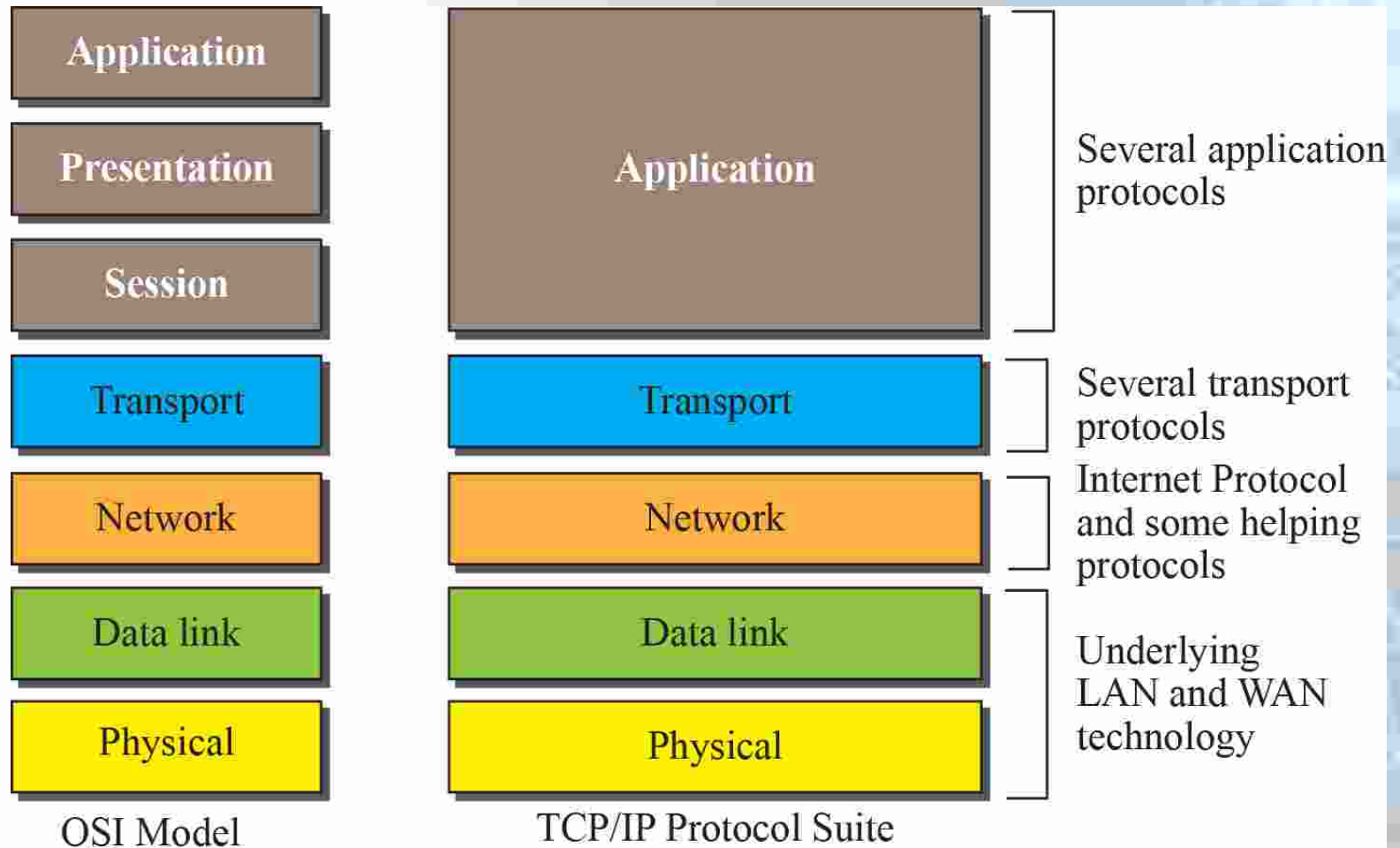
Layer 1

Physical

# OSI Model vs TCP/IP Protocol suite

- **Two Layers of OSI missing from TCP/IP**
- **Application (TCP/IP) = Application + Presentation + Session (OSI)**

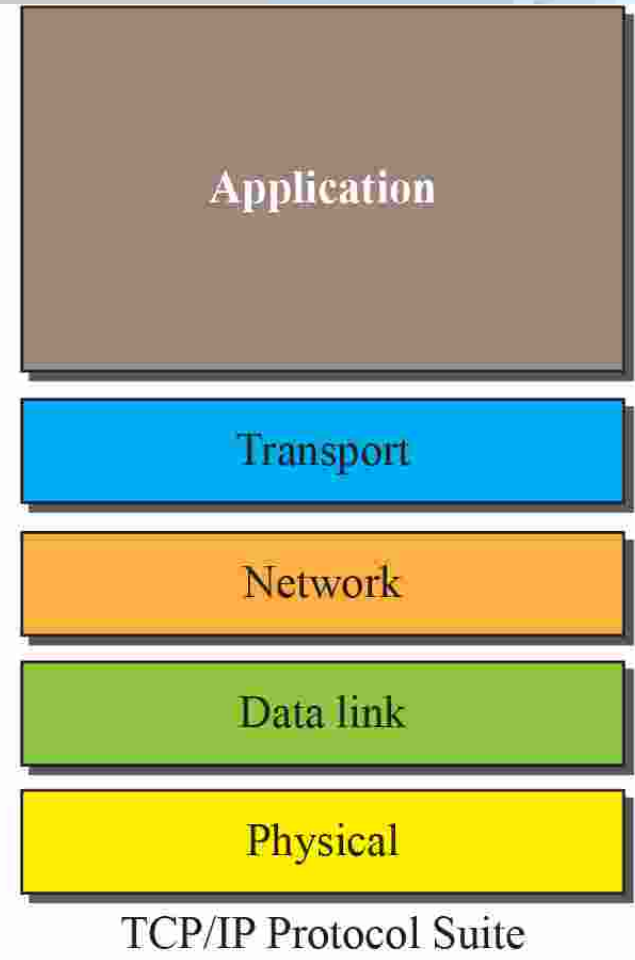
# OSI Model vs TCP/IP Protocol suite



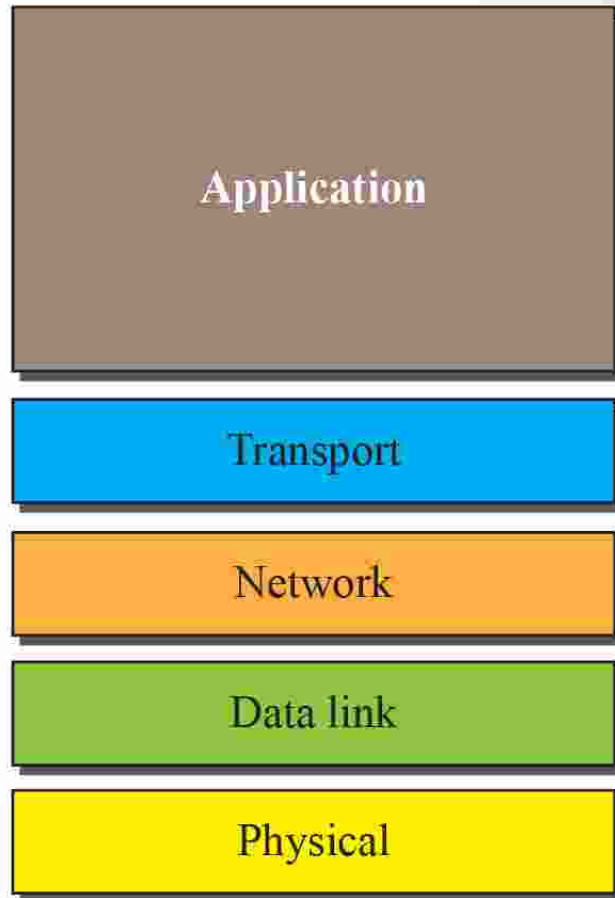
# Lack of OSI Model's Success

- Three reasons OSI did not replace TCP/IP:
  - ✓ OSI was completed when TCP/IP was fully in place
  - ✓ Some layers in OSI not fully defined
  - ✓ Performance of TCP/IP better than that of OSI

# Data Communication versus Computer Networks



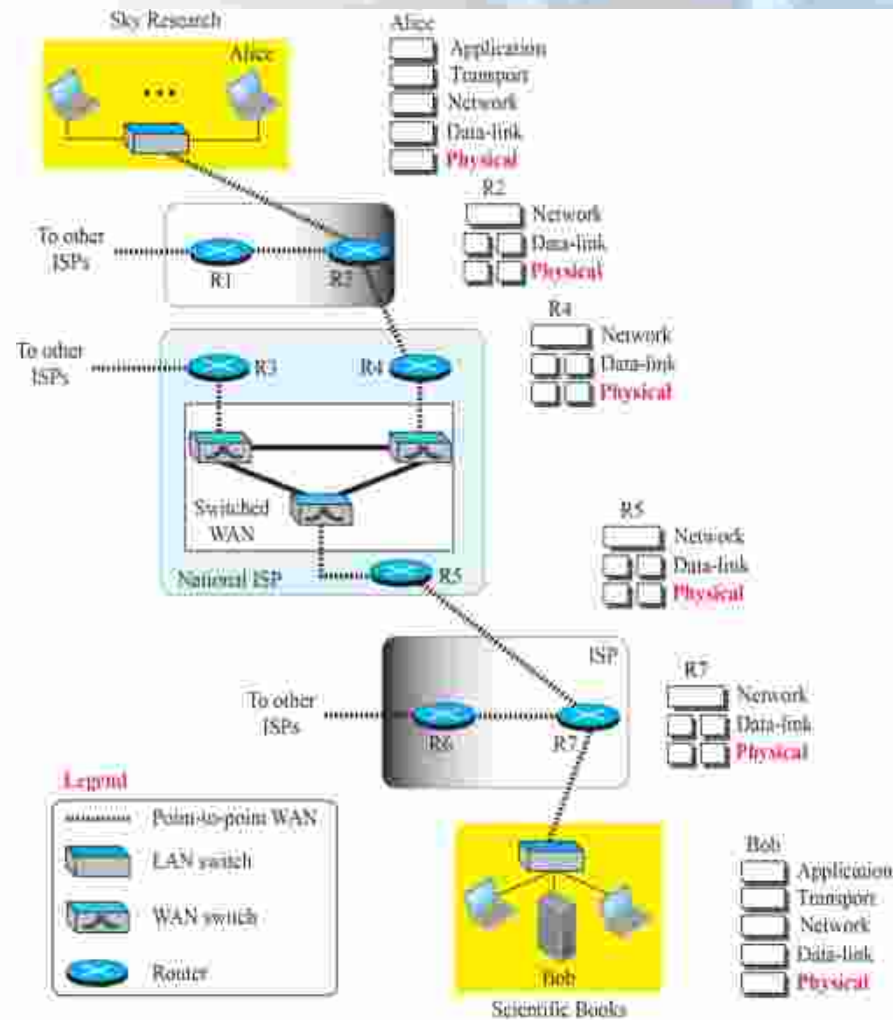
# Data Communication versus Computer Networks



TCP/IP Protocol Suite

- Analog & Digital Transmission
- Transmission Media
- Switching
- Error Detection and Correction
- Media Access and Data Link Control
- Wired and Wireless LANs

# Communication at Physical Layer



# Analog & Digital Data

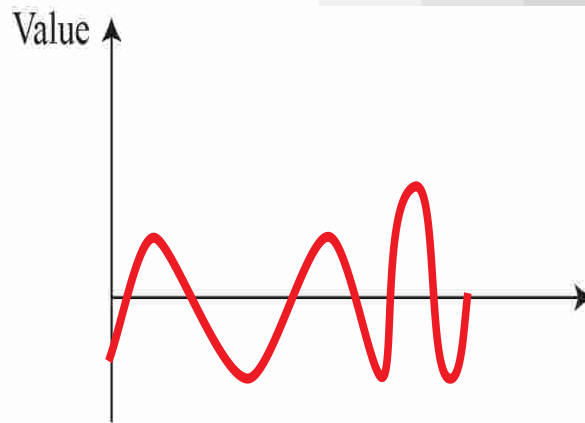
- **Data → Analog or Digital**
- **Analog Data → Continuous**
- **Digital Data → Discrete**
- **Examples: Analog Clock vs. Digital Clock**
- **Human voice vs. Data in Computer**



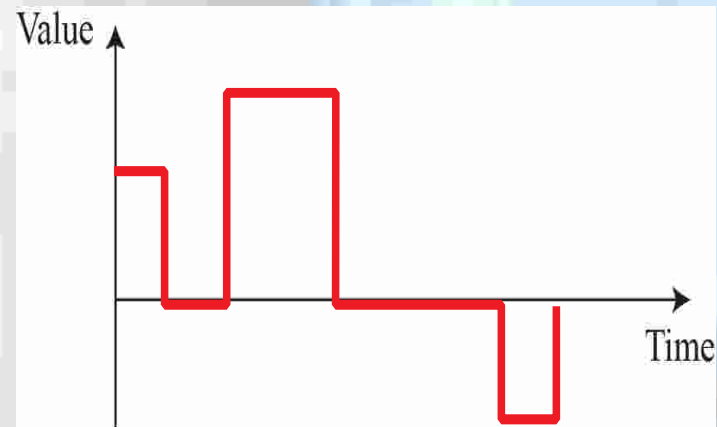
# Analog & Digital Signals

- **Signals represent Data**
- **Signals → Analog or Digital**
- **Analog Signal → Infinite Levels of Intensity over time**
- **Digital Signal → Limited number of defined values**

# Analog & Digital Signals



a. Analog signal



b. Digital signal

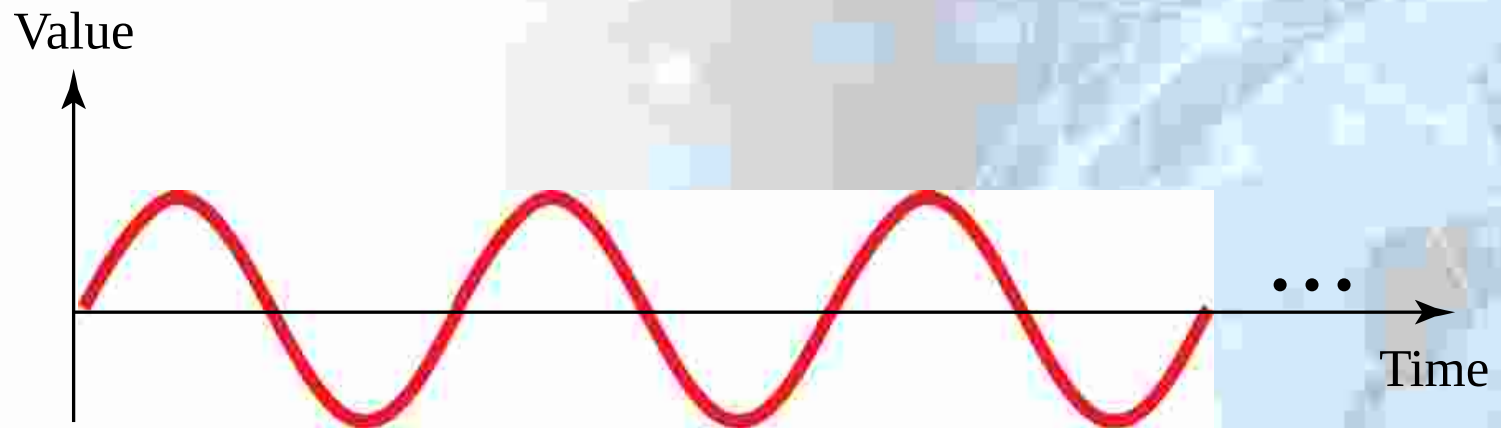
# Periodic & Non-periodic Signals

- **Analog/Digital Signal → Periodic or Non-periodic**
- **Periodic Signal → Pattern**
- **Period and Cycle**
- **Non-Periodic → No Pattern**
- **Periodic ANALOG Signals and Non-periodic DIGITAL Signals**

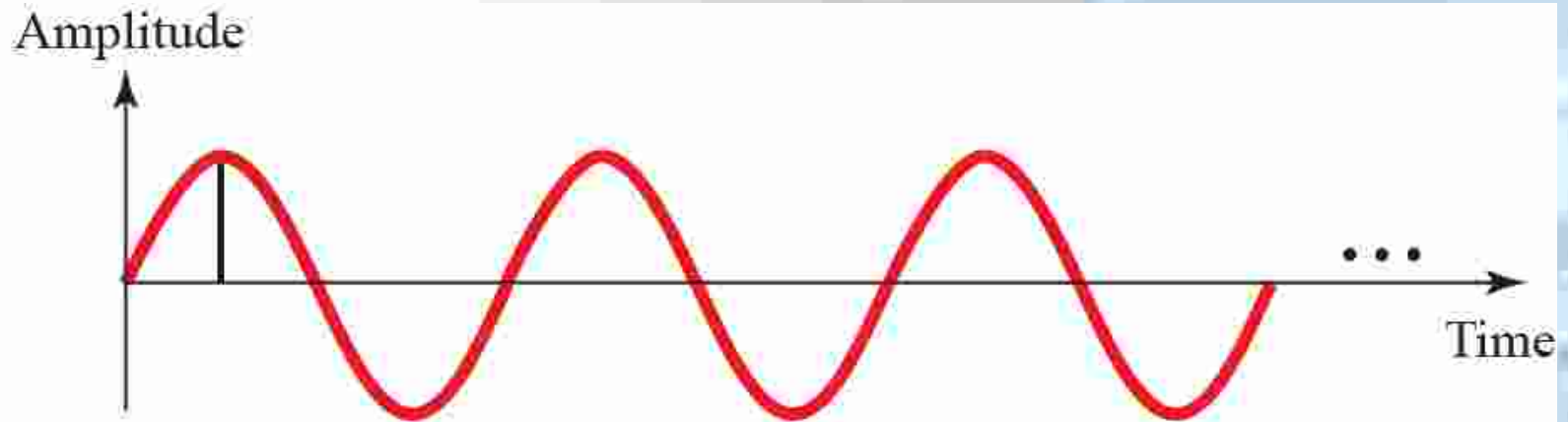
# Periodic Analog Signals

- **Periodic Analog Signals**  
→ Simple or Composite
- **Simple Periodic Analog signal** → Sine wave
- **Composite Periodic Analog signal** → Composed of multiple sine waves

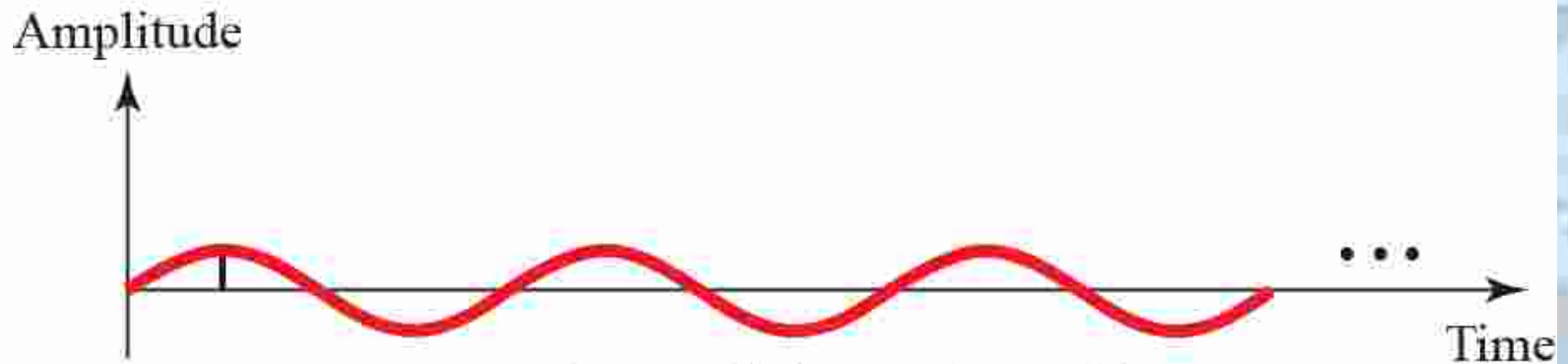
# Sine Wave



# Sine Wave – Peak Amplitude



a. A signal with high peak amplitude



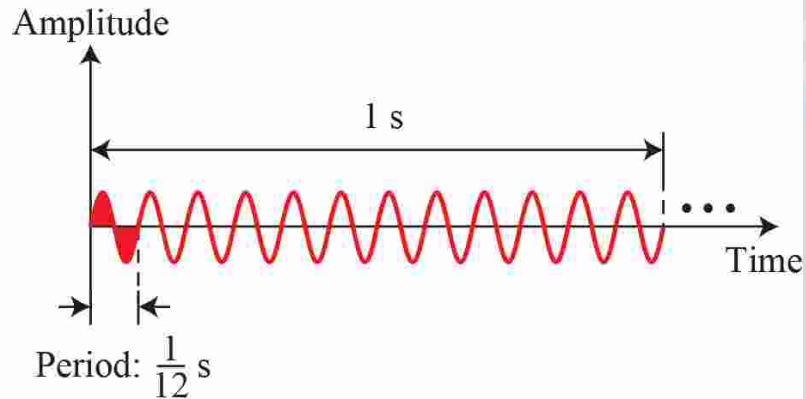
b. A signal with low peak amplitude

# Sine Wave –Frequency

- Period (T) → Amount of time required to complete 1 cycle
- Frequency (f) → No. of Periods in 1 sec
- $f = 1/T$  or  $T = 1/f$

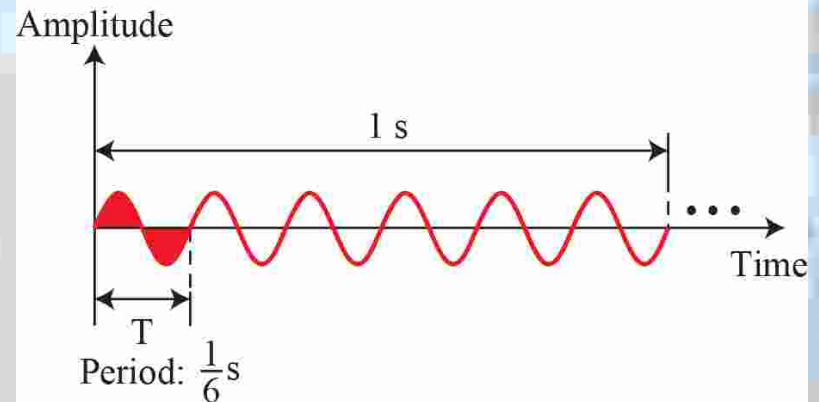
# Sine Wave – Frequency

12 periods in 1 s  $\rightarrow$  Frequency is 12 Hz



a. A signal with a frequency of 12 Hz

6 periods in 1 s  $\rightarrow$  Frequency is 6 Hz



b. A signal with a frequency of 6 Hz

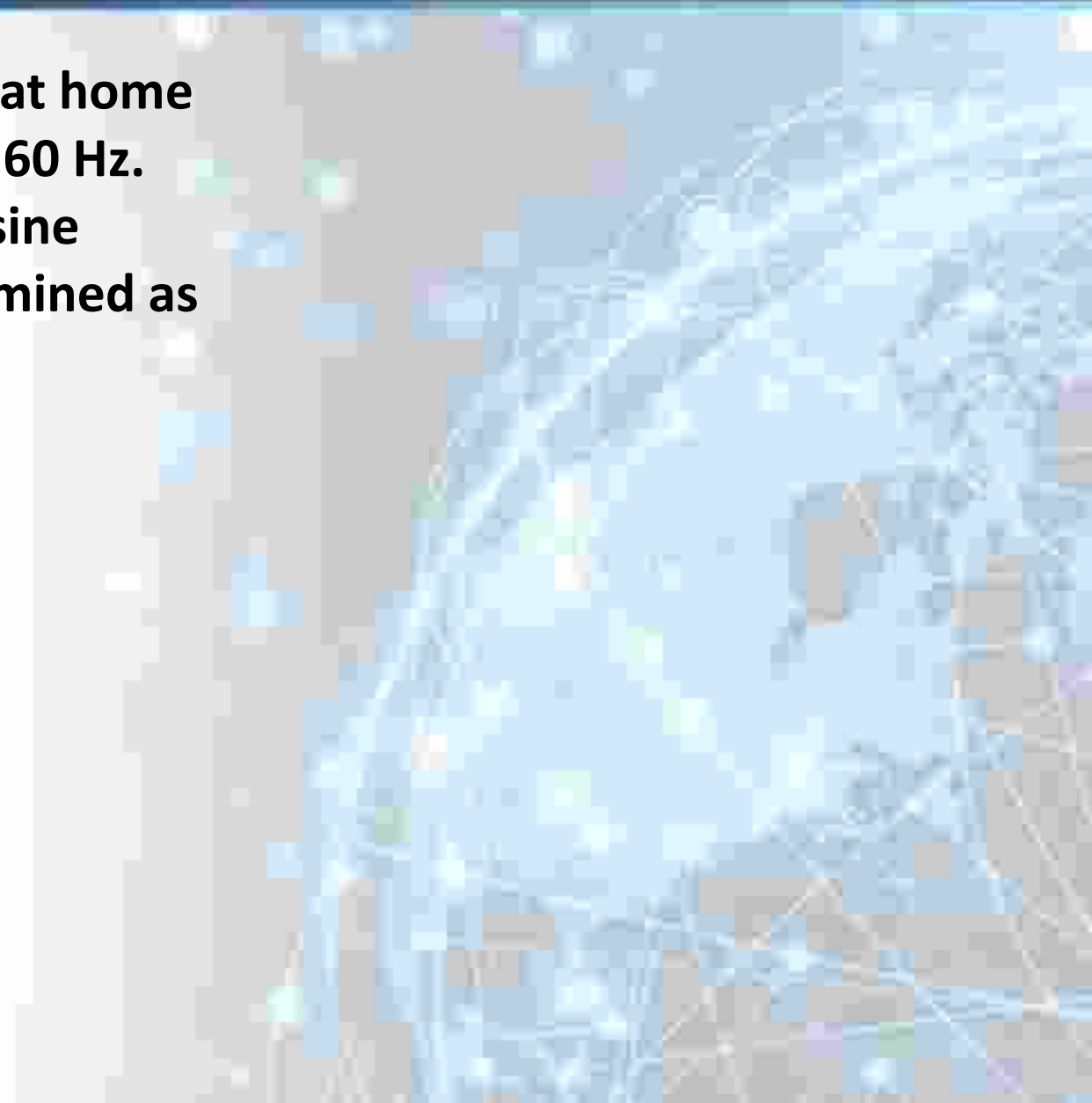


# Sine Wave –Frequency

| <i>Period</i>           |                   | <i>Frequency</i> |                   |
|-------------------------|-------------------|------------------|-------------------|
| <i>Unit</i>             | <i>Equivalent</i> | <i>Unit</i>      | <i>Equivalent</i> |
| Seconds (s)             | 1 s               | Hertz (Hz)       | 1 Hz              |
| Milliseconds (ms)       | $10^{-3}$ s       | Kilohertz (kHz)  | $10^3$ Hz         |
| Microseconds ( $\mu$ s) | $10^{-6}$ s       | Megahertz (MHz)  | $10^6$ Hz         |
| Nanoseconds (ns)        | $10^{-9}$ s       | Gigahertz (GHz)  | $10^9$ Hz         |
| Picoseconds (ps)        | $10^{-12}$ s      | Terahertz (THz)  | $10^{12}$ Hz      |

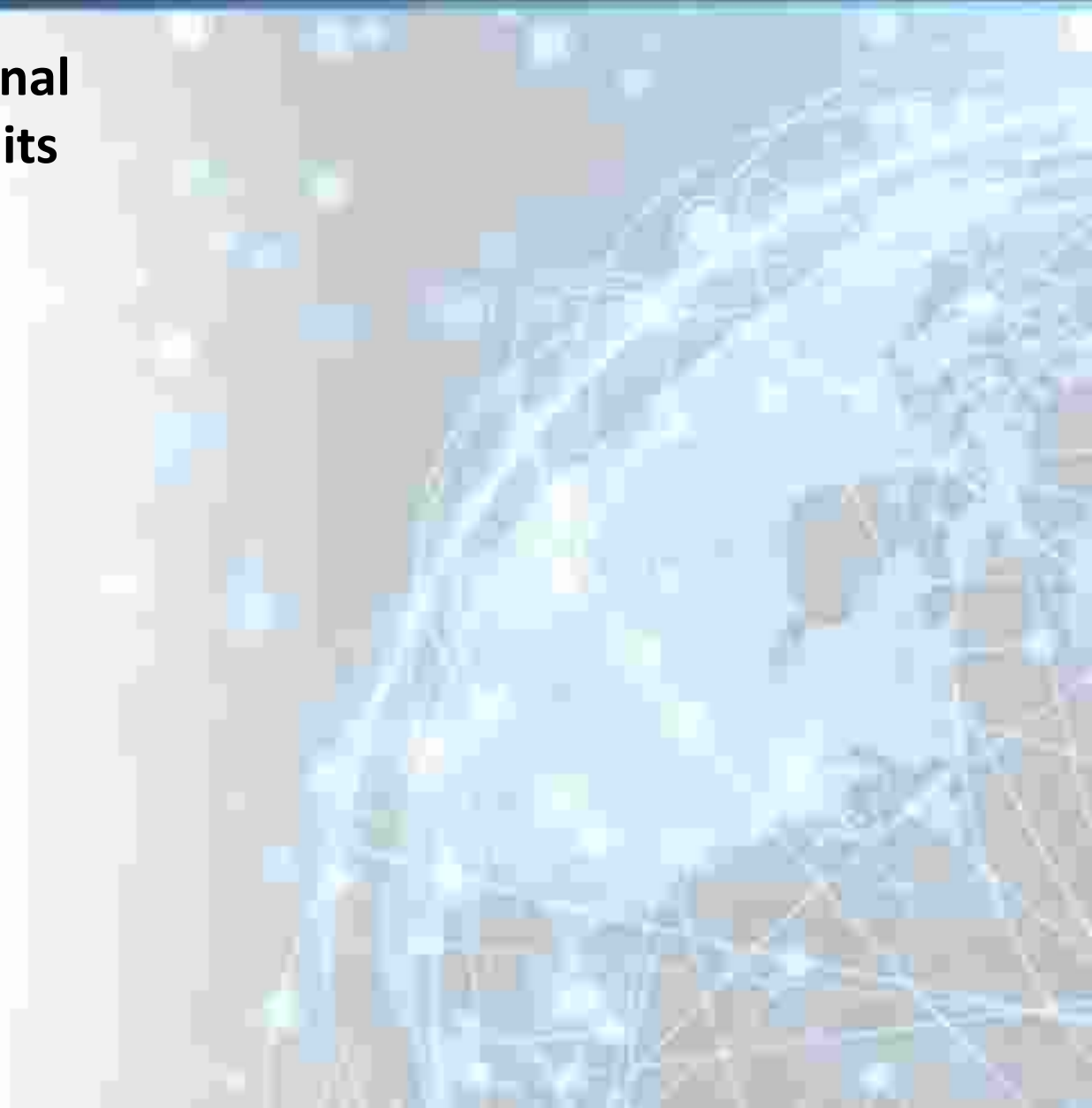
## Example

The power we use at home has a frequency of 60 Hz.  
The period of this sine wave can be determined as follows:



## Example

The period of a signal is 100 ms. What is its frequency in kilohertz?.



# Phase (or Phase Shift)

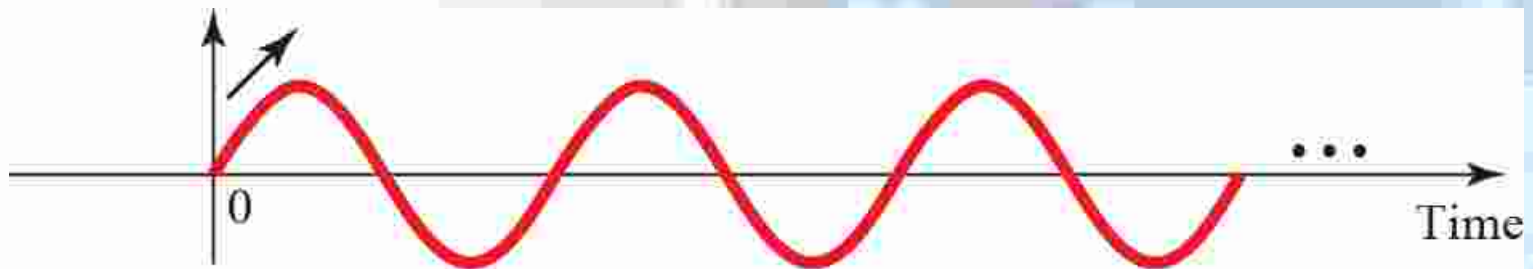
- Position of waveform relative to time 0
- Phase describes the amount of shift of the wave
- Indicates start of the first cycle

## Phase

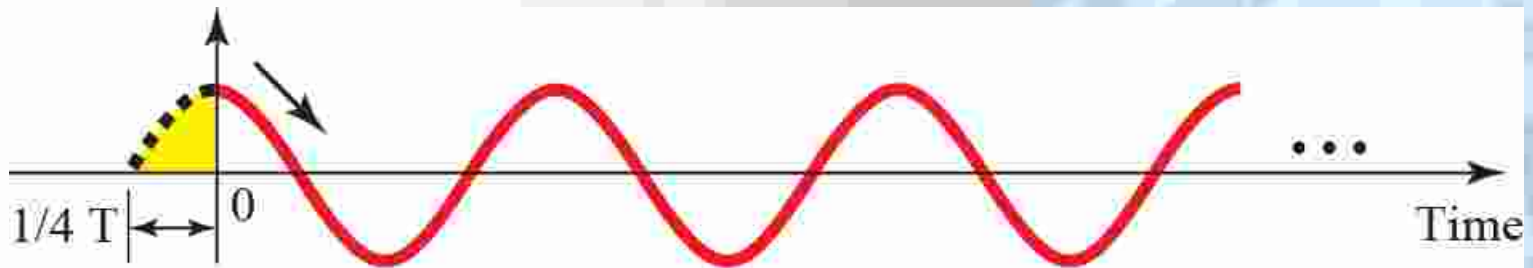
$$360^\circ = 2\pi \text{ radians}$$

$$1^\circ = \frac{2\pi}{360} \text{ rad}$$

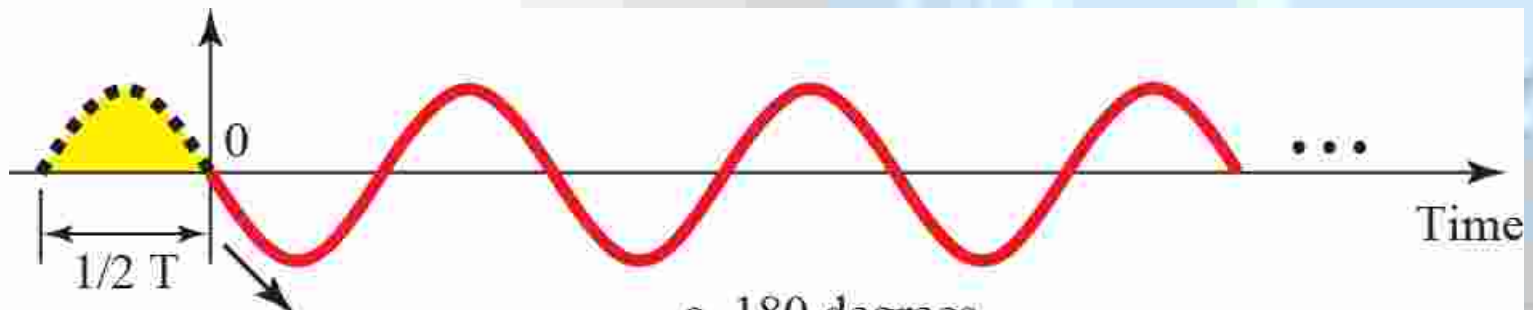
# Phase



a. 0 degrees



b. 90 degrees



c. 180 degrees

## Example

A sine wave is offset  $\frac{1}{6}$  cycle with respect to time 0. What is its phase in degrees and radians?

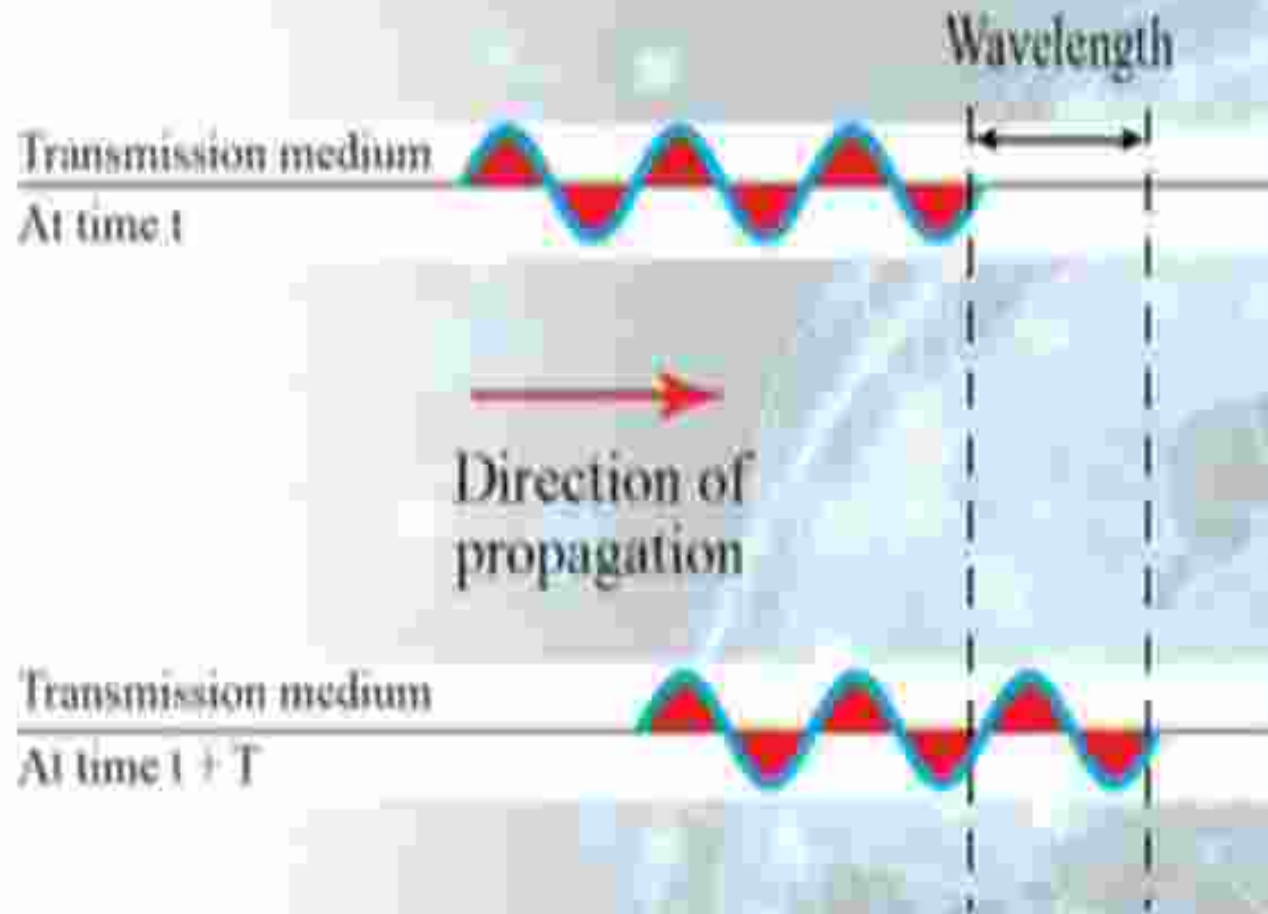
# Wavelength

**Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium (see Figure 3.7).**



# Wavelength

Figure 3.7



# Wavelength

Prop. Speed =  $c$  = Light (speed)

frequency =  $f$

Wavelength =  $\lambda$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/sec}}{f}$$

$$\lambda = \frac{3 \times 10^8}{4 \times 10^{14}}$$

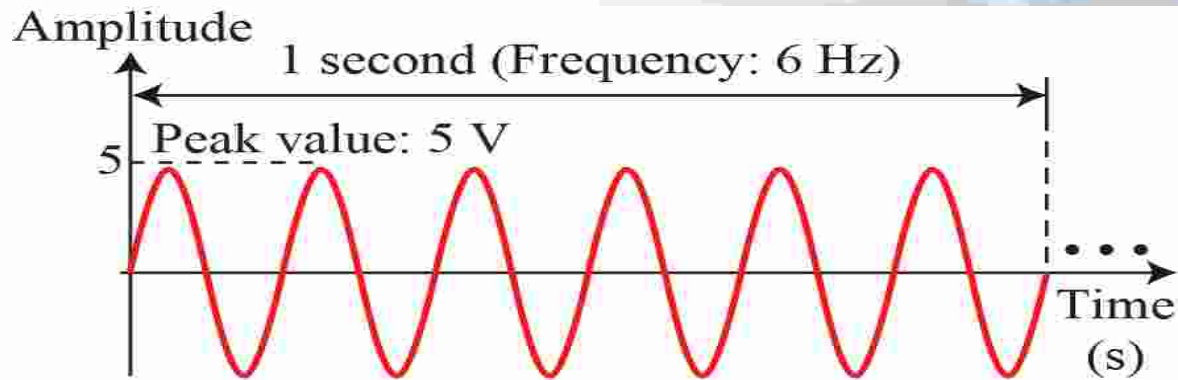
# Wavelength

$$\begin{aligned}\lambda &= 0.75 \times 10^{-6} \text{ m} \\ &= 0.75 \mu\text{m} \\ &= \underline{\underline{\quad\quad\quad}}\end{aligned}$$

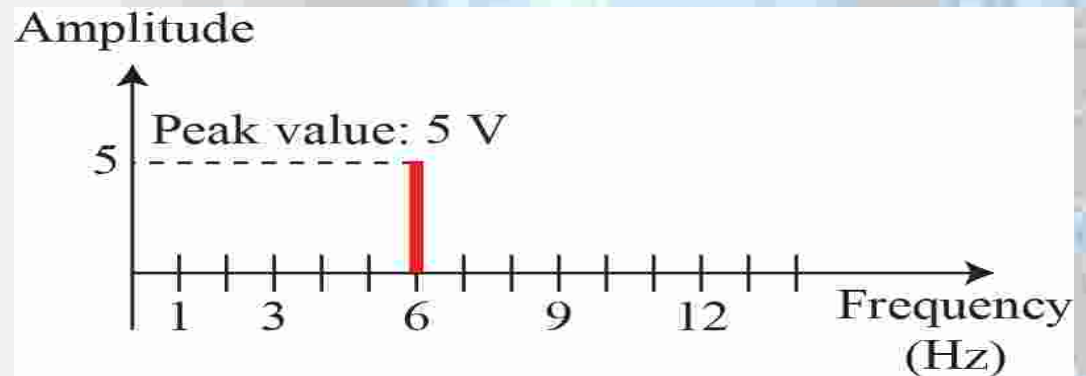
# Time & Frequency Domains

A sine wave is comprehensively defined by its amplitude, frequency, and phase. We have been showing a sine wave by using what is called a time domain plot. The time-domain plot shows changes in signal amplitude with respect to time (it is an amplitude-versus-time plot). Phase is not explicitly shown on a time-domain plot.

# Time & Frequency Domains



a. A sine wave in the time domain



b. The same sine wave in the frequency domain

## Example 3.7

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.9 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

# Time & Frequency Domains

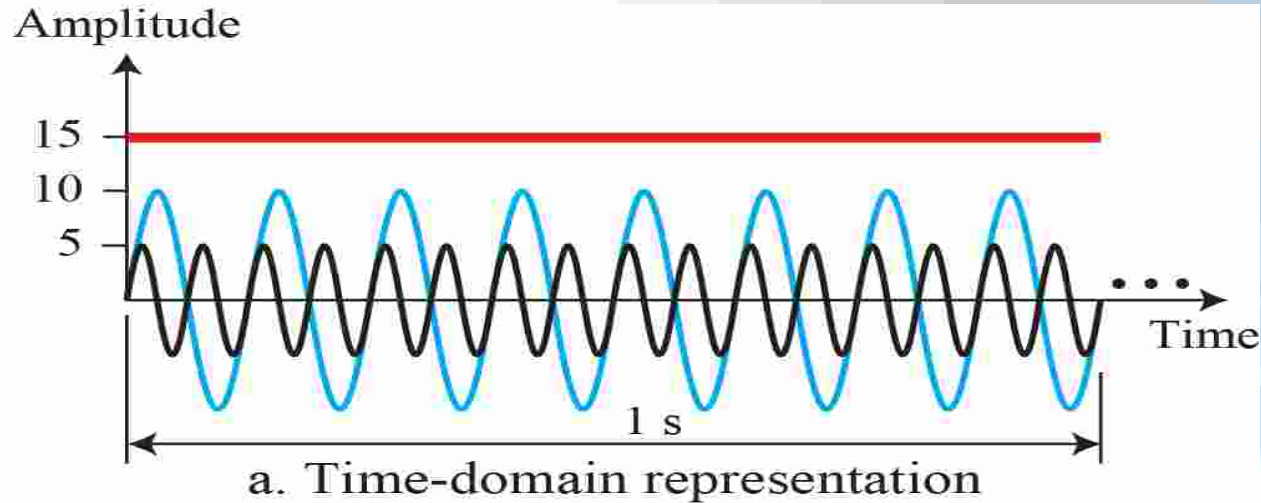
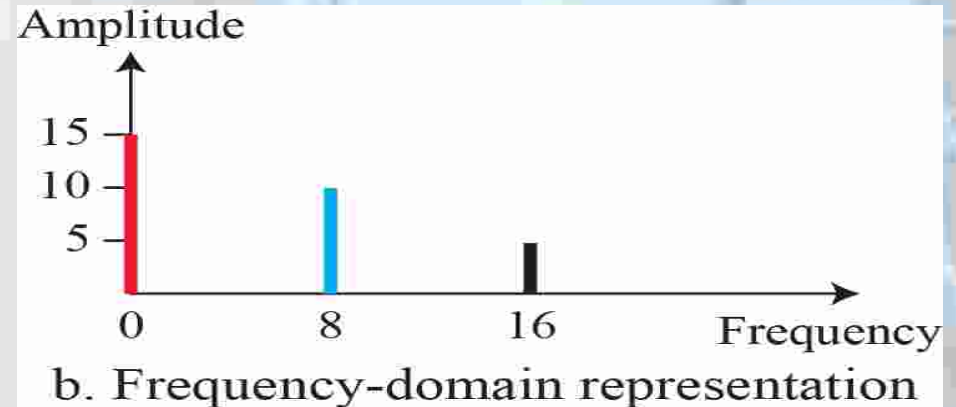


Figure 3.9



# Composite Signals

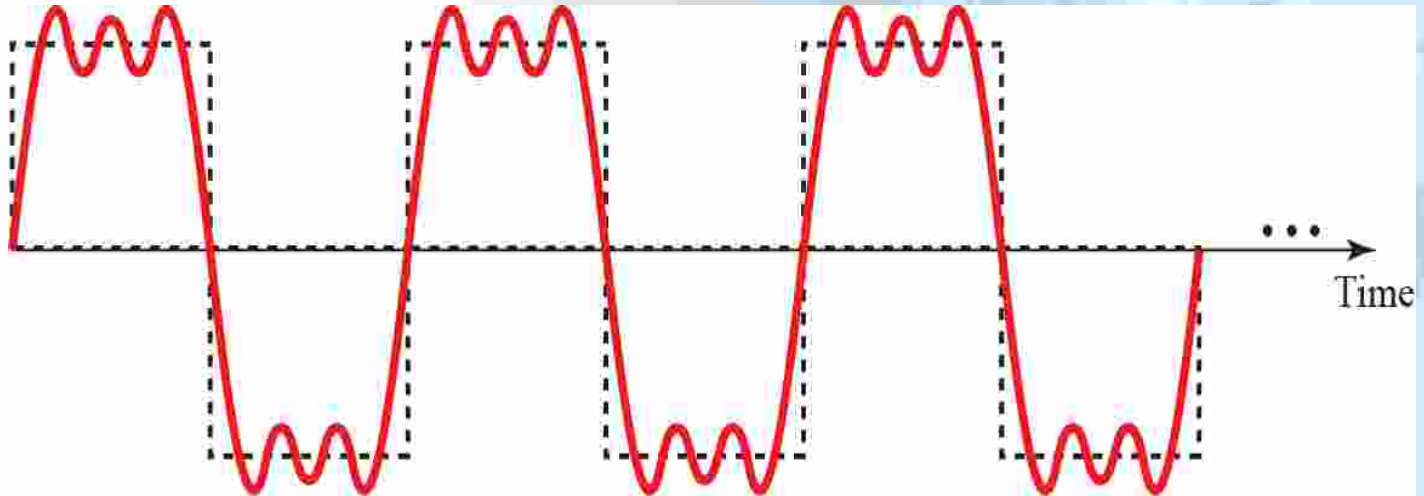
So far, we have focused on simple sine waves. Simple sine waves have many applications in daily life. We can send a single sine wave to carry electric energy from one place to another. For example, the power company sends a single sine wave with a frequency of 60 Hz to distribute electric energy to houses and businesses.



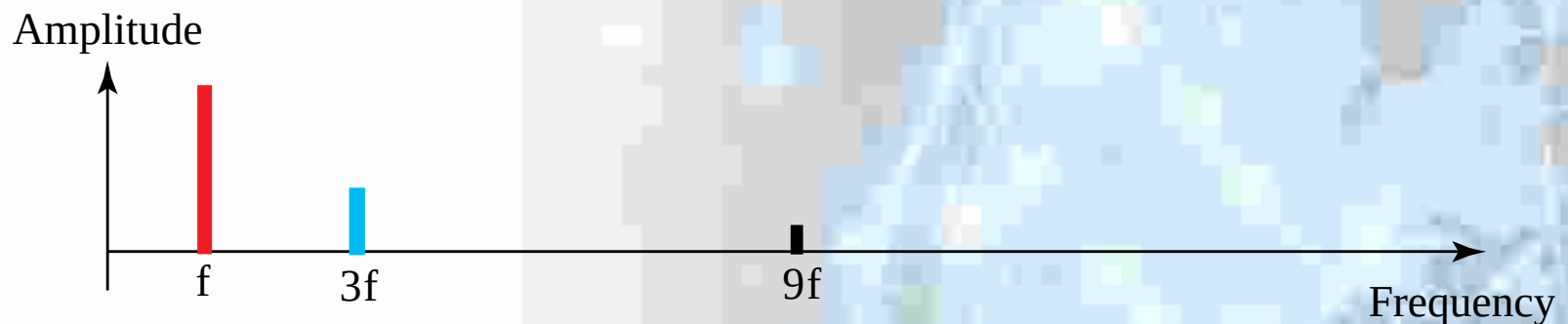
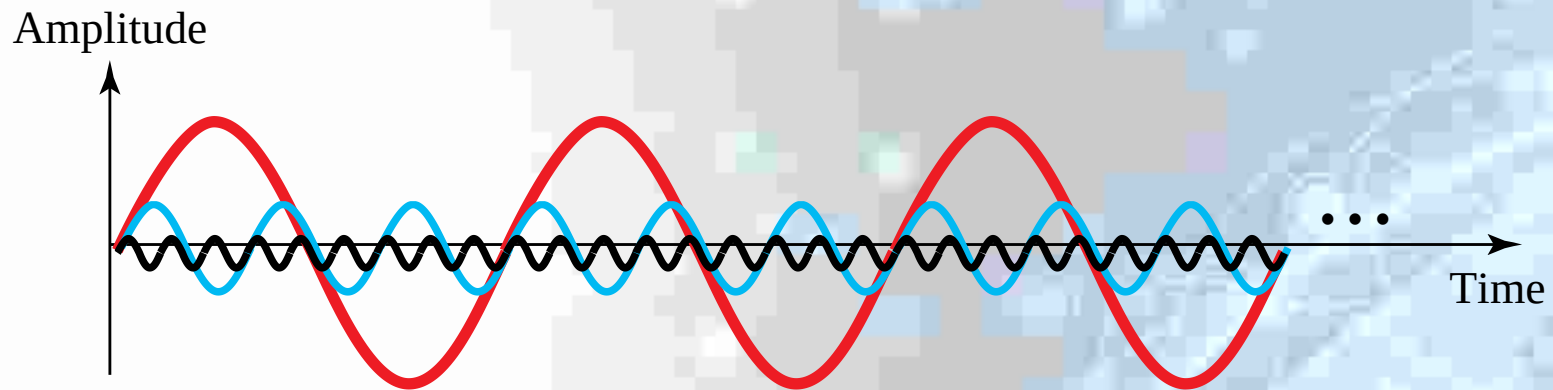
# Composite Signals

- **Single Sine Wave can only carry limited information**
- **Composite Signal is made up of multiple simple sine waves**
- **Can be periodic or non-periodic**

# A Composite Periodic Signal



# Decomposition of Composite Periodic Signal

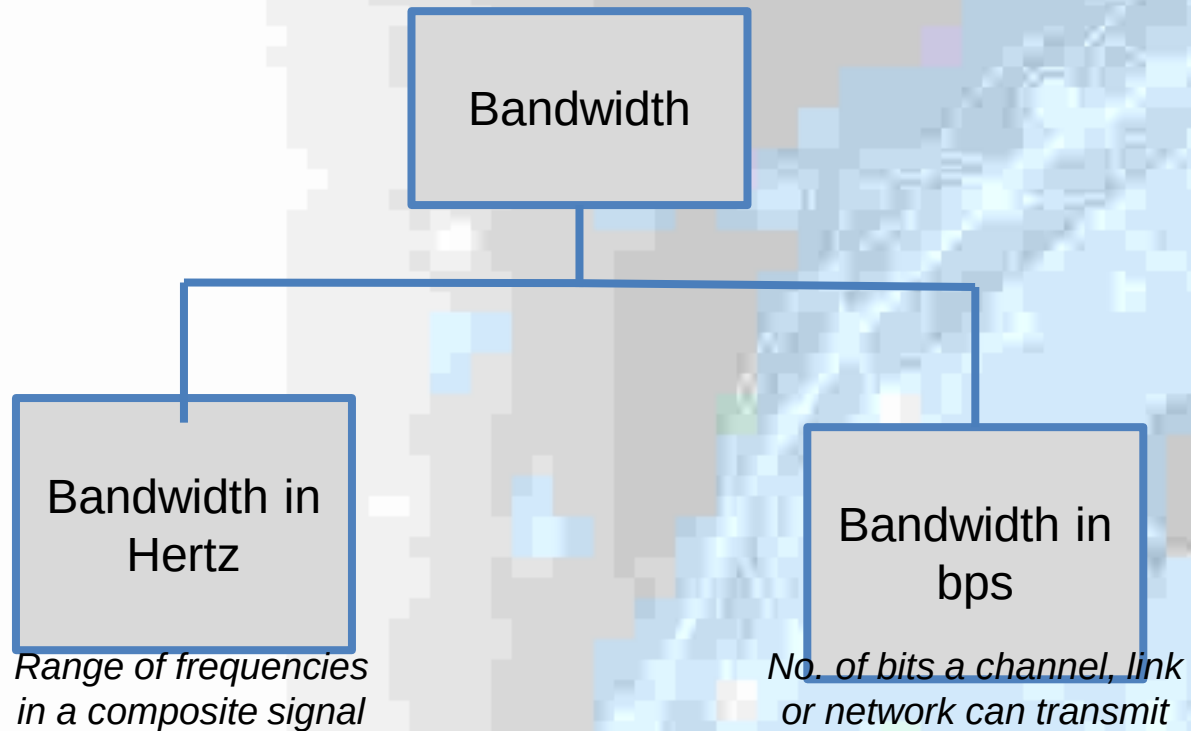


b. Frequency-domain decomposition of the composite signal

# Bandwidth

- **An important characteristic that measures Network Performance**
- **Bandwidth can be used in two different contexts with two different measuring values:**
  - **Bandwidth in Hertz**
  - **Bandwidth in bits per second**

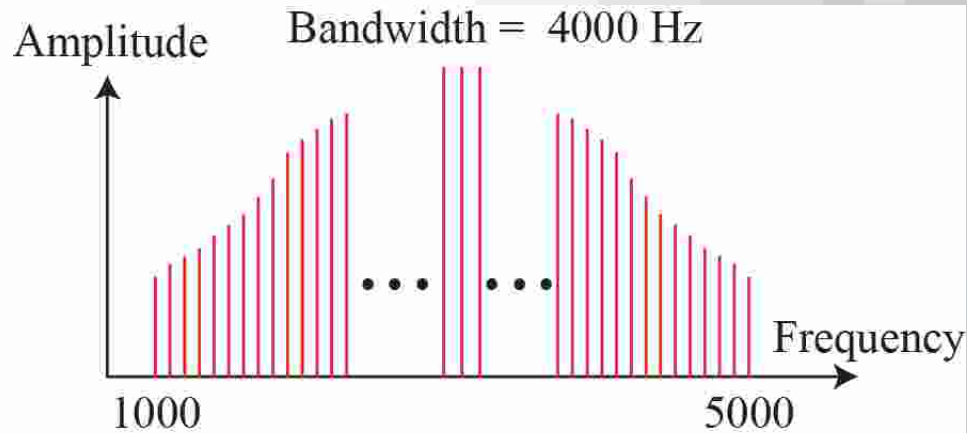
# Bandwidth



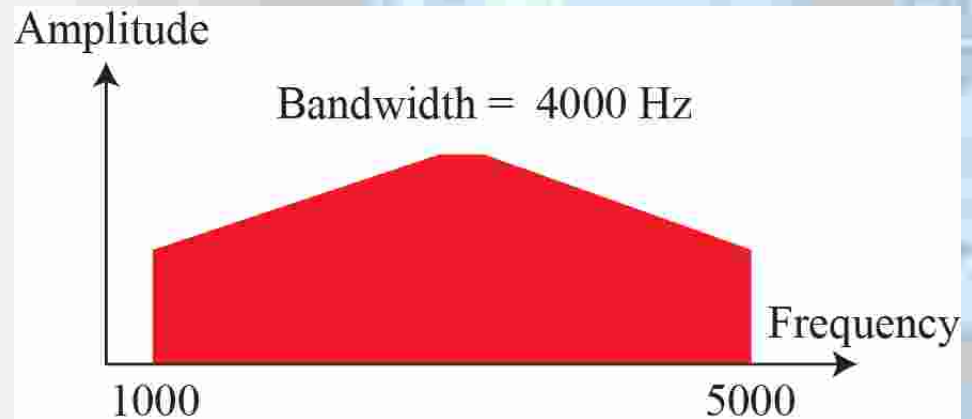
# Bandwidth

- **Range of frequencies contained in a Composite Signal**
- **The bandwidth is normally a difference between two frequencies (the highest and the lowest)**

# Bandwidth of a composite signal



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

## Example

**If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.**



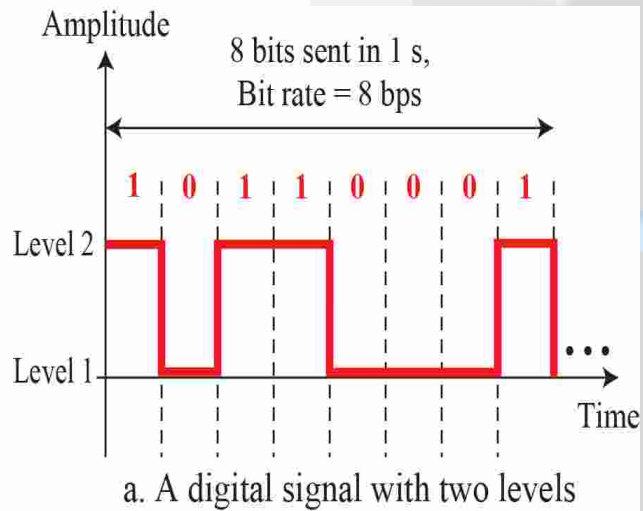
# Digital Signals

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels.

# Digital Signals

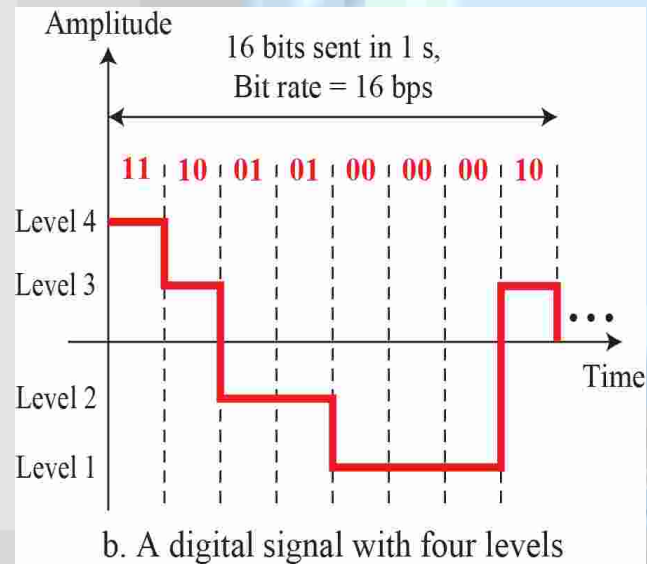
- Information can also be represented by a digital signal
- For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage
- A digital signal can have more than two levels so that we can send more than one bit for each level

# Digital Signals



In general, if a signal has  $L$  levels, each level needs  $\log_2 L$  bits. So, we can send  $\log_2 2 = 1$  bit per level

$$\log_2 4 = 2 \text{ bits}$$



# Digital Signals

In this case, we can send more than 1 bit for each level. Figure 3.17 shows two signals, one with two levels and the other with four.

## Example 3.16

A digital signal has eight levels. How many bits are needed per level?

We calculate the number of bits from the following formula. Each

Number of bits per level

bits. =  $\log_2 8 = 3$

# Example

A digital signal has eight levels. How many bits are needed per symbol?

$$\begin{aligned}\text{No of bits} &= \log_2 L \\ &= \log_2 8 \\ &= 3 \text{ bits}\end{aligned}$$

## Example 3.17

A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula. Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

# Example

A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula.

$$\log_2 L = \log_2 9 = \underline{\underline{3.17 \text{ bits}}}$$

↑ ↑  
integer power of 2

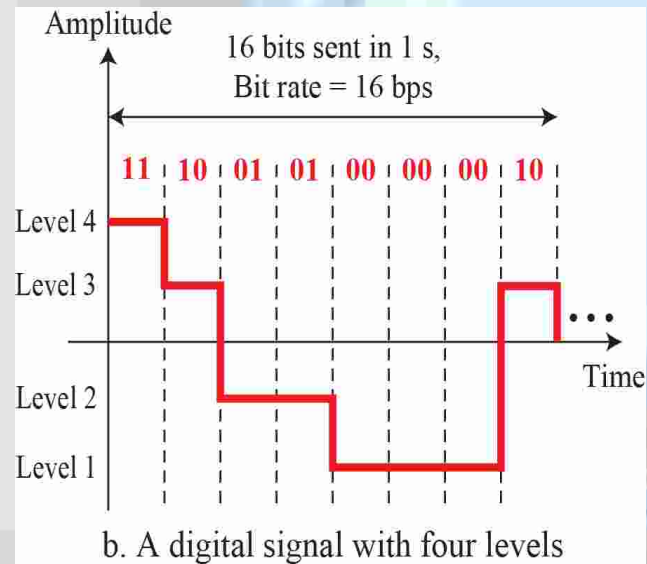
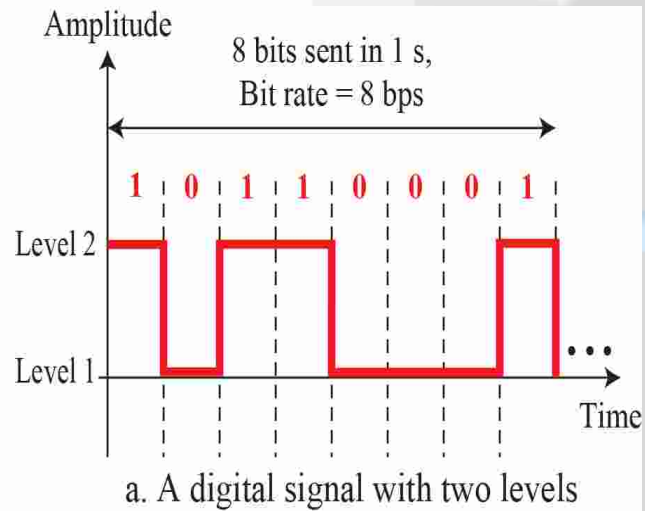
4 bits



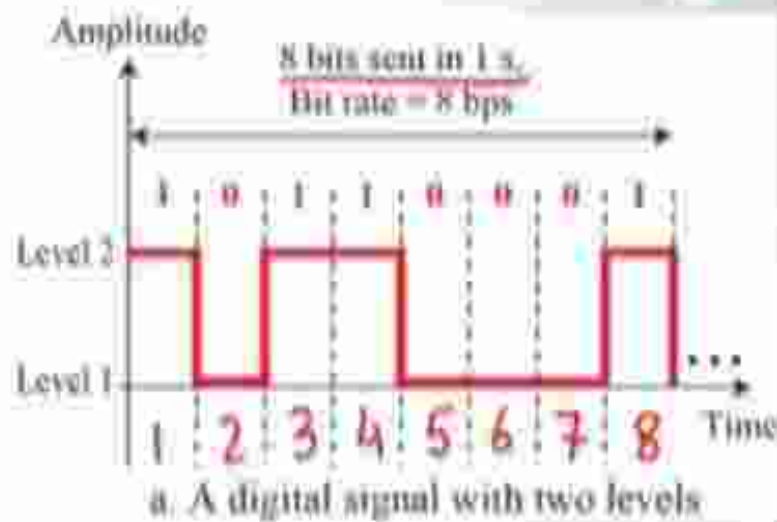
# Bit Rate

- Number of bits sent in 1 second
- Bit Rate is expressed in bits per second (bps)
- Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics

# Bit Rate

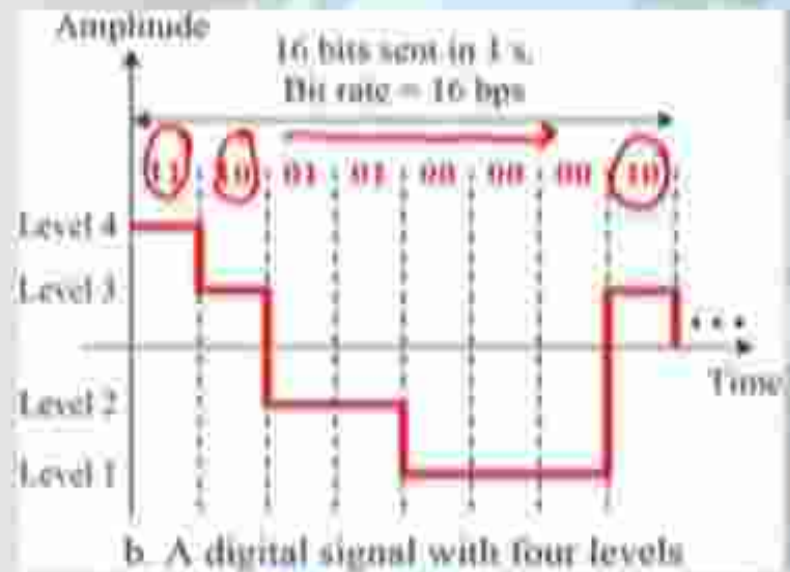


# Bit Rate



Bit Rate = 8 bps

BR = 1



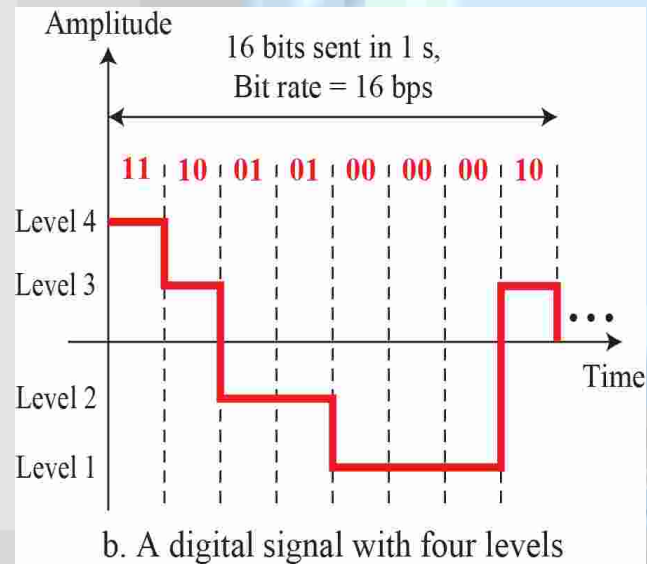
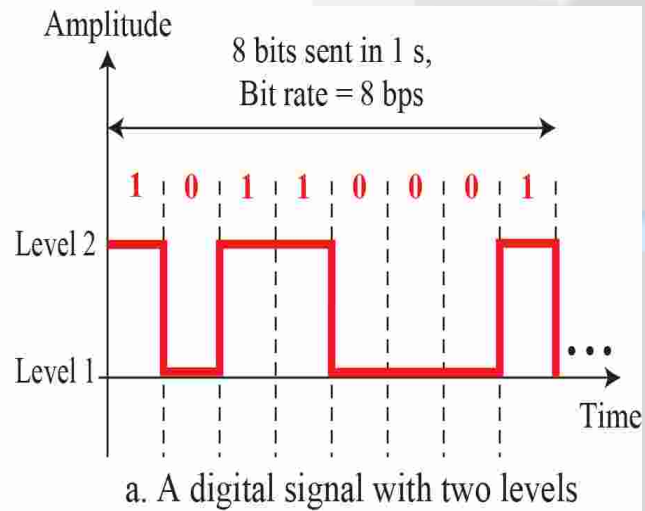
# Bit Rate

Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics. Another term—bit rate (instead of frequency)—is used to describe digital signals. The bit rate is the number of bits sent in 1s, expressed in bits per second (bps). Figure 3.17 shows the bit rate for two signals.

# Bit Rate

- Number of bits sent in 1 second
- Bit Rate is expressed in bits per second (bps)
- Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics

# Bit Rate



## Example

Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the

$$1 \text{ page} = 24 \text{ lines}$$

$$1 \text{ line} = 80 \text{ characters}$$

$$1 \text{ ch} = 8 \text{ bits}$$

$$\text{Bit Rate} = 100 \times 24 \times 80 \times 8$$

$$= \underline{\underline{1.536 \text{ Mbps}}}$$

## Example 3.18

### **Solution**

From Table 3.1 we find the equivalents of 1 ms (1 ms is  $10^{-3}$  s) and 1 s (1 s is  $10^6 \mu\text{s}$ ).

We make the

$$100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps}$$

$$\text{subst} = 1.536 \text{ Mbps} ;$$



## Example

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

$$\begin{aligned} &2 \times 4000 \times 8 \\ &= 64 \text{ kbps} \\ &= (64,000 \text{ bps}) \end{aligned}$$

## Example 3.19

### **Solution**

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires

$$2 \times 4000 \times 8 = 64,000 \text{ bps}$$

$$= 64 \text{ kbps}$$

## Example 3.20

What is the bit rate for high-definition TV (HDTV)?

### **Solution**

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9 (in contrast to 4 : 3 for regular TV), which means the screen is wider. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second.

## Example 3.20

### **Solution**

Twenty-four bits represents one color pixel. We can calculate the bit rate

$$\begin{aligned} & 1920 \times 1080 \times 30 \times 24 \\ & = 1,492,992,000 \approx 1.5 \text{ Gbps} \end{aligned}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

# Bit Length

We discussed the concept of the wavelength for an analog signal: the distance one cycle occupies on the transmission medium. We can define something similar for a digital signal: the bit length. The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

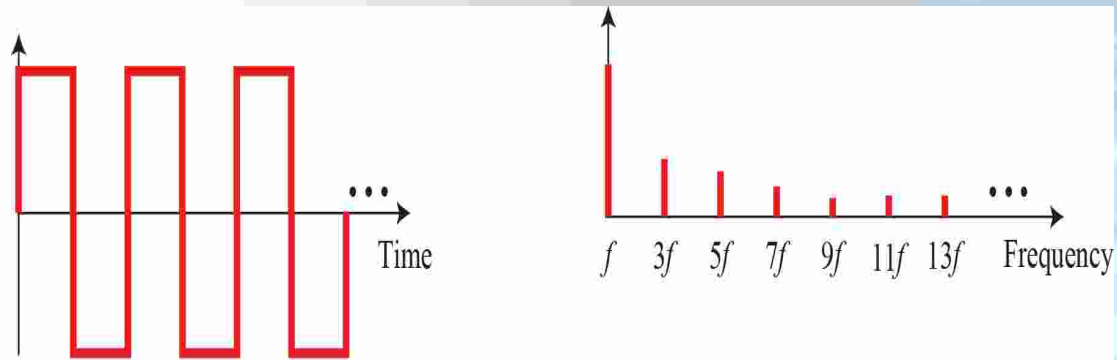
# Digital Signal as Composite Analog Signal

- Based on Fourier analysis, a digital signal is a composite analog signal
- A digital signal, in the time domain, comprises connected vertical and horizontal line segments
- Infinite Bandwidth

# Digital As Composite Analog

A vertical line in the time domain means a frequency of infinity: a horizontal line in the time domain means a frequency of zero. Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

# Digital Signal as Composite Analog Signal



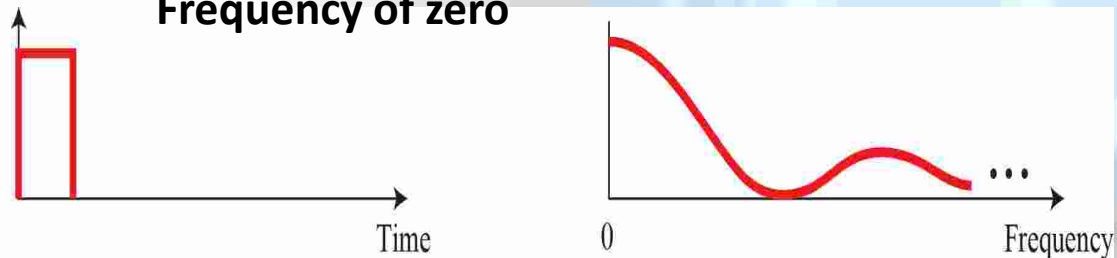
a. Time and frequency domains of periodic digital signal

**Vertical line in the time domain:**

**Frequency of infinity**

**Horizontal line in the time domain:**

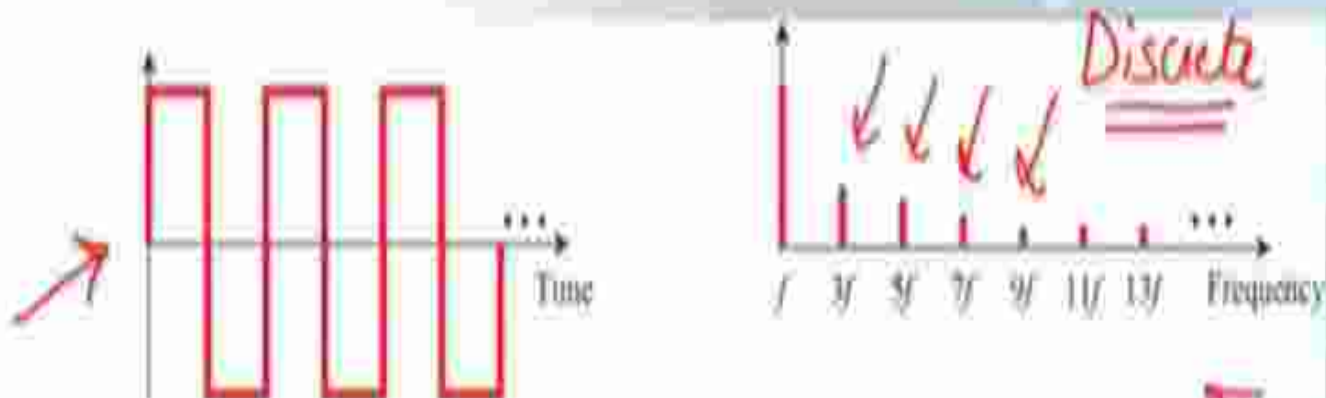
**Frequency of zero**



b. Time and frequency domains of nonperiodic digital signal



# Digital Signal as Composite Analog Signal

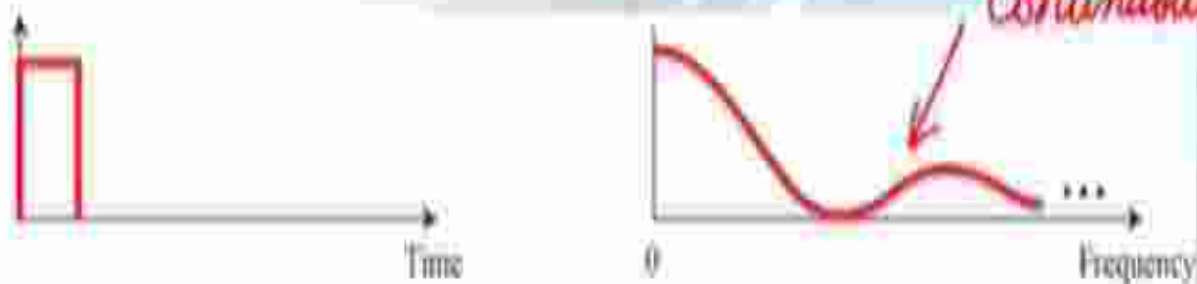


a. Time and frequency domains of periodic digital signal

→ Vertical line in the time domain: Frequency of infinity  
→ Horizontal line in the time domain: Frequency of zero

$f = \frac{1}{T}$

for  $f_0$



b. Time and frequency domains of nonperiodic digital signal

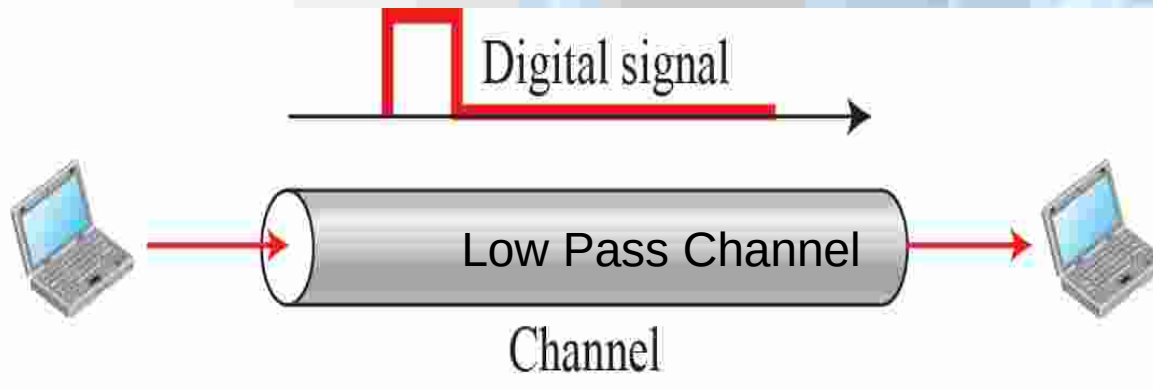
# Transmission of Digital Signals

- **Digital signal, periodic or non-periodic, is a composite analog signal with frequencies between zero and infinity (Infinite Bandwidth)**
- **Two approaches for transmission:**
  - ✓ **Baseband Transmission**
  - ✓ **Broadband Transmission**

# Transmission of Digital Signals

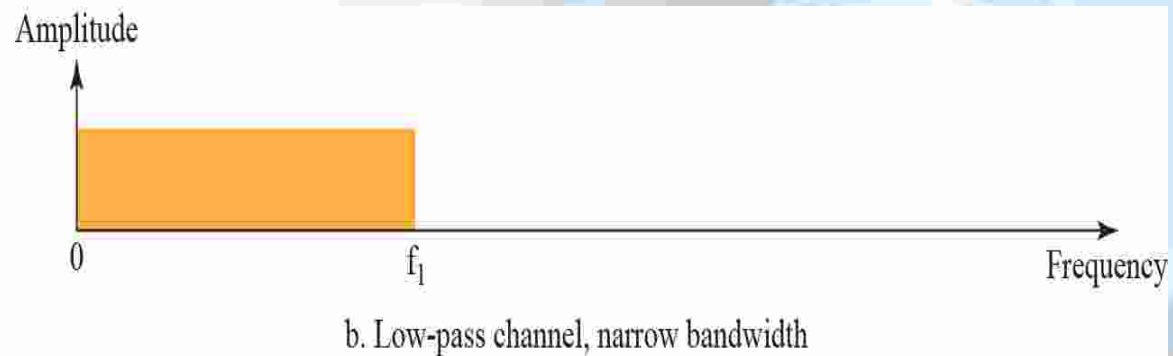
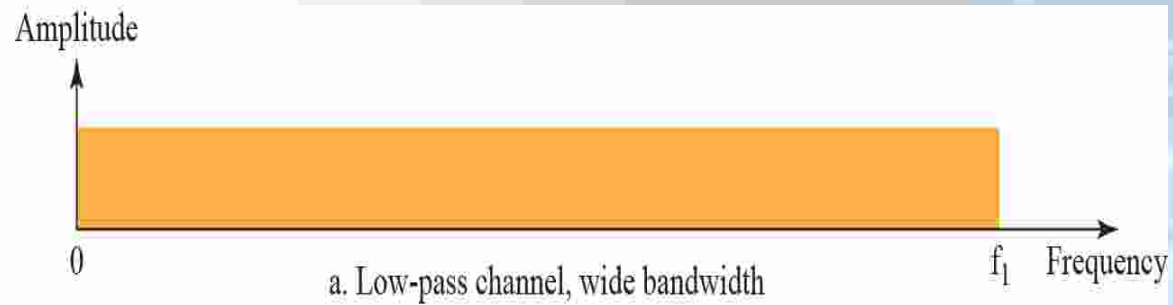
A vertical line in the time domain means a frequency of infinity: a horizontal line in the time domain means a frequency of zero. Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

# Baseband Transmission

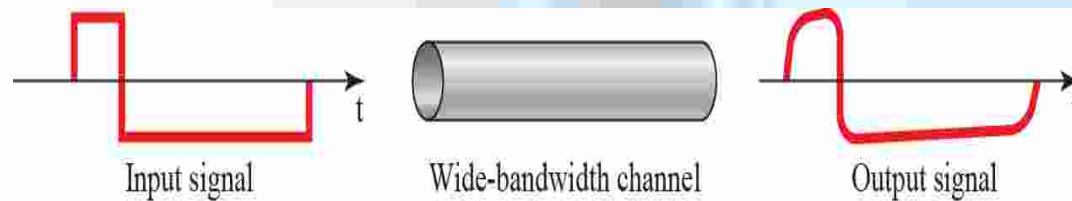
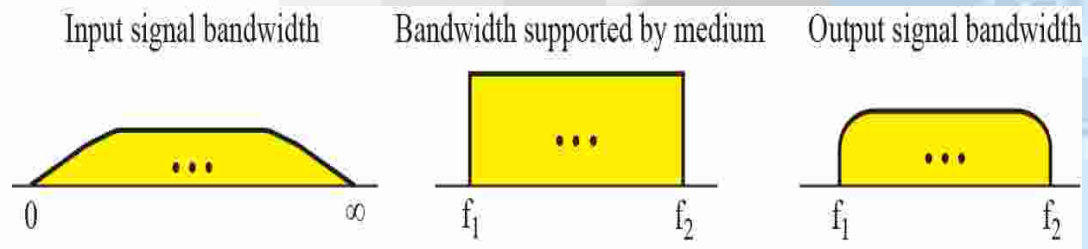


Sending a **Digital Signal** without changing it to an **Analog Signal**

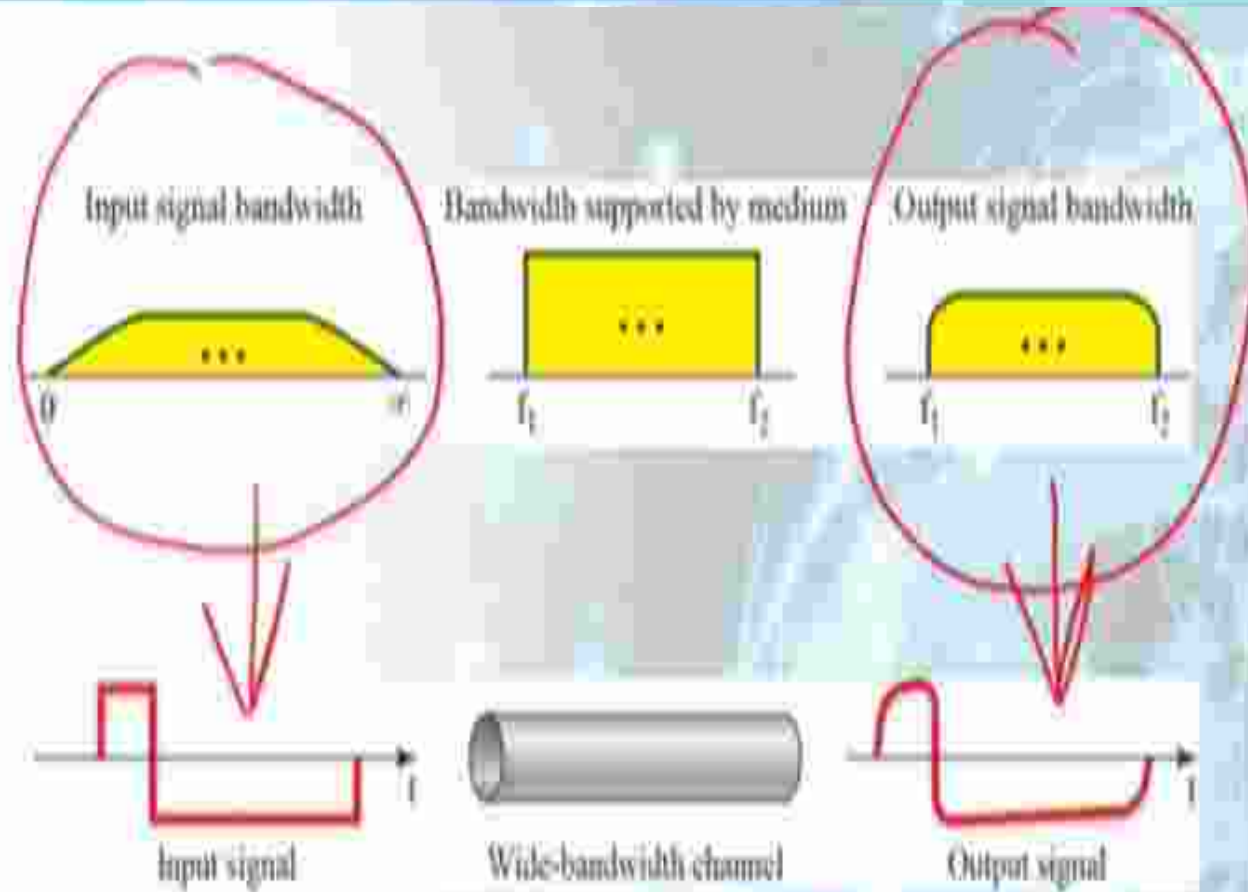
# Baseband Transmission



# Baseband Transmission



# Baseband Transmission

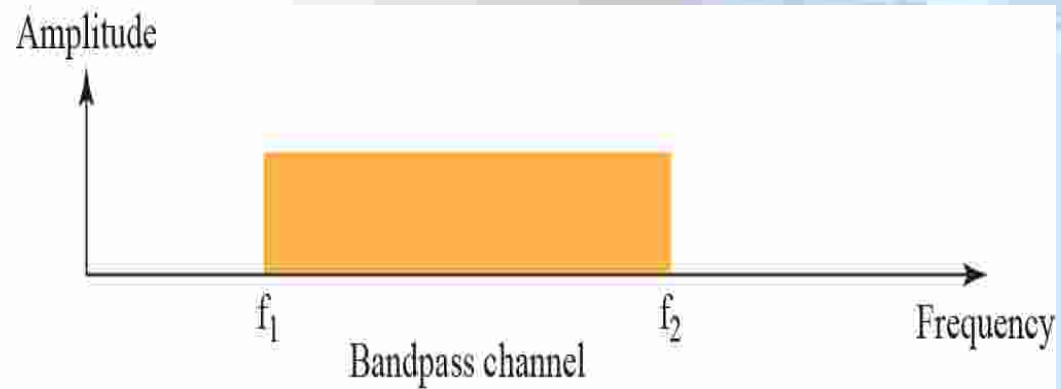


# Broadband Transmission (Modulation)

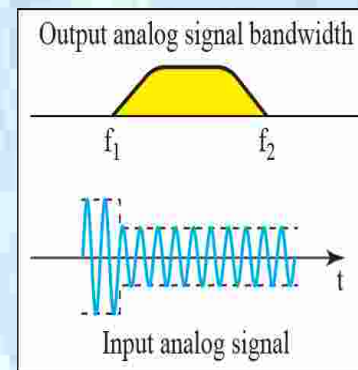
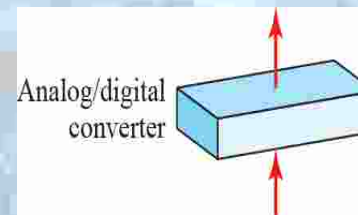
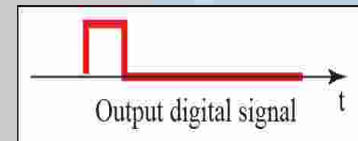
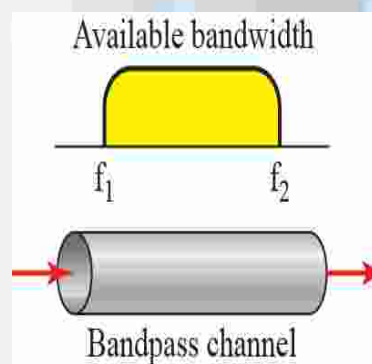
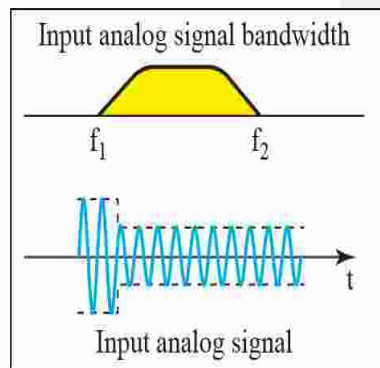
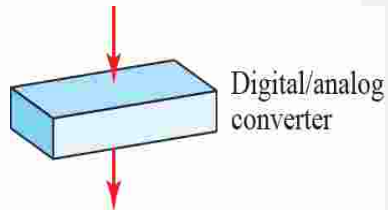
- Changing the Digital signal to an Analog signal for transmission
- Modulation allows us to use a bandpass channel—a channel with a bandwidth that does not start from zero
- More available than a low pass



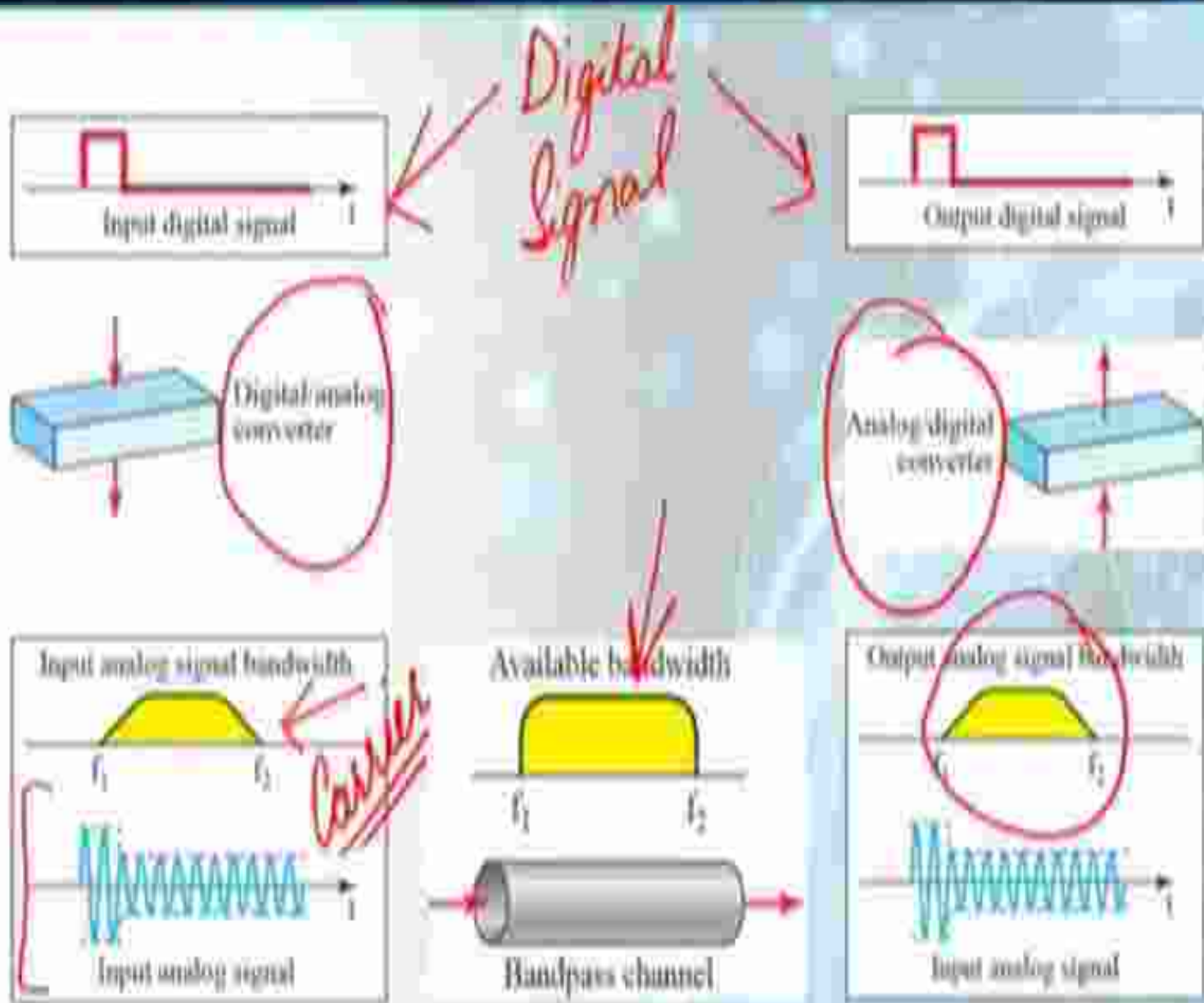
# Broadband Transmission (Modulation)



# Broadband Transmission (Modulation)



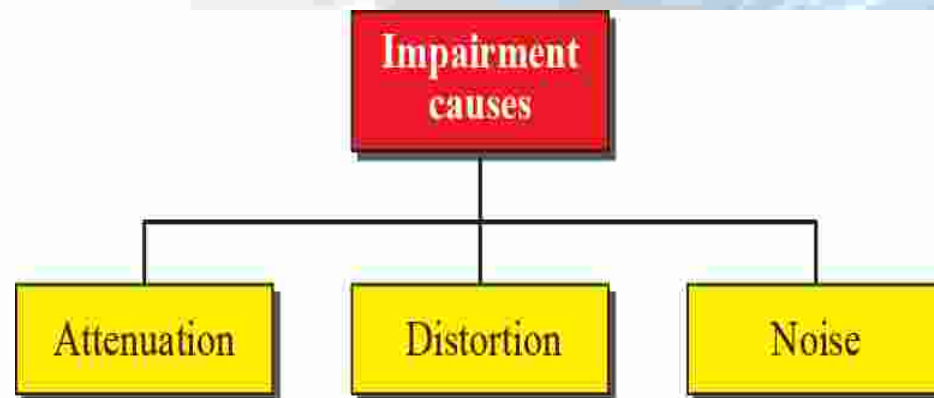
# Broadband Transmission (Modulation)



# Transmission Impairments

- **Transmission media are not perfect**
- **Cause Signal impairments**
- **Signal sent is not the same as the signal received**

# Causes of Transmission Impairment



# Transmission Impairment

**What is sent is not what is received.**

**Three causes of impairment are attenuation, distortion, and noise (see Figure 3.26).**

# Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.

# Attenuation

**Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat.**

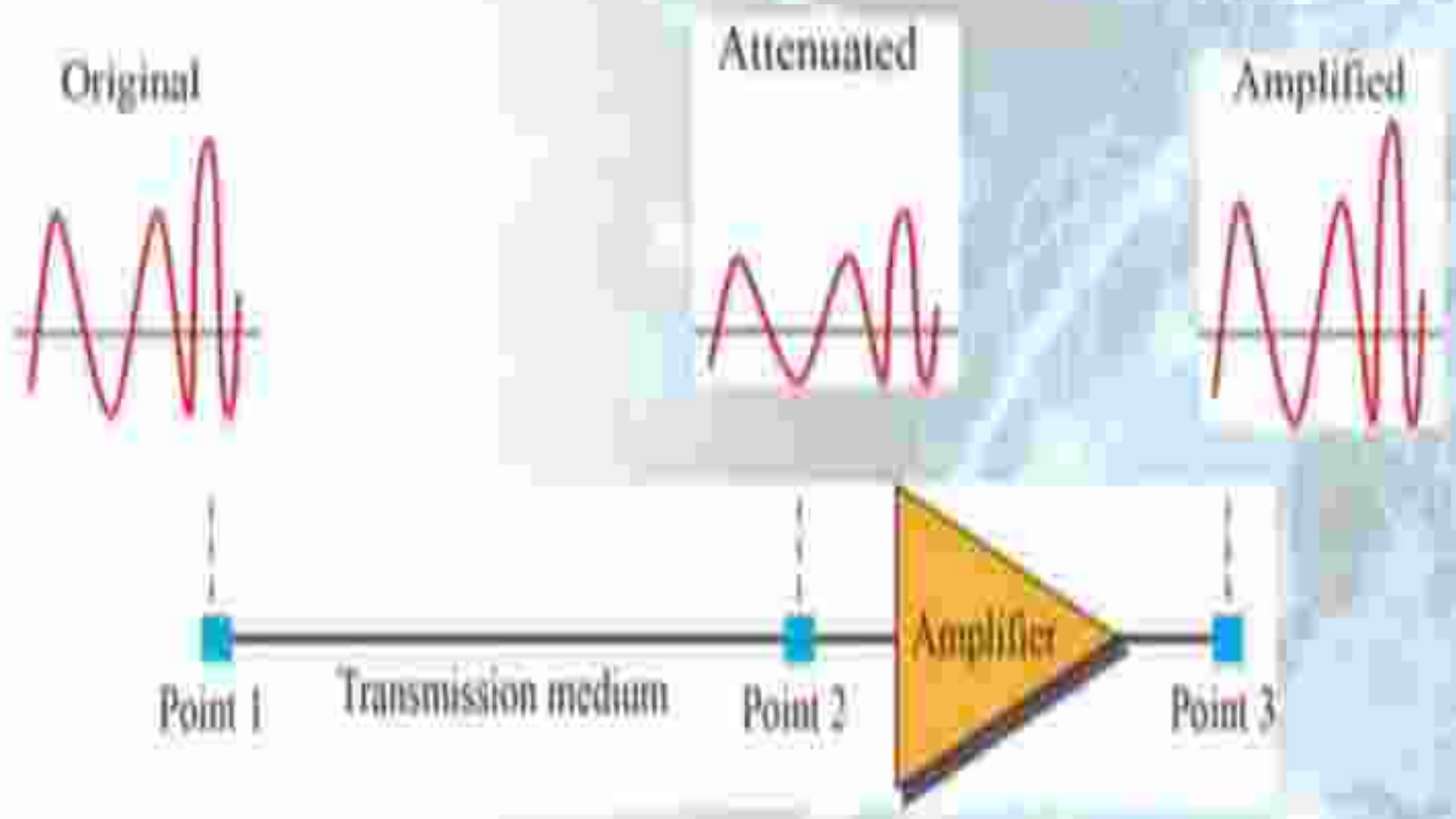


# Attenuation

To compensate for this loss, amplifiers are used to amplify the signal. Figure 3.27 shows the effect of attenuation and amplification..

# Attenuation and amplification

Figure 3.27



## Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that  $P_2 = 0.5 P_1$ . In this case, the attenuation (loss of power) can be calculated as

$$\begin{aligned} 10 \log_{10} P_2/P_1 &= 10 \log_{10} (0.5 P_1)/P_1 \\ &= 10 \log_{10} 0.5 = 10 \times (-0.3) = -3 \text{ dB.} \end{aligned}$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

## Example 3.27

A signal travels through an amplifier, and its power is increased 10 times. This means that  $P_2 = 10P_1$ . In this case, the amplification

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} =$$

$$10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

## Example 3.28

One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.28 a signal travels from point 1 to point 4. The signal is attenuated by the time it reaches point 2.

## Example 3.28

Between points 2 and 3, the signal is amplified. Again, between points 3 and 4, the signal is attenuated. We can find the resultant decibel value for the signal just by adding the decibel measurements between each set of points. In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

## Example 3.29

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as  $\text{dB}_m$  and is calculated as  $\text{dB}_m = 10 \log_{10} P_m$ , where  $P_m$  is the power in milliwatts.

Calculate the power of a signal if its  $\text{dB}_m = -30$ .

## Example 3.29

### Solution

We can calculate the power in the signal as

$$dB_m = 10 \log_{10} \longrightarrow dB_m = -30 \longrightarrow$$

$$\log_{10} P_m = -3 \longrightarrow P_m = 10^{-3} \text{ mW}$$



## Example 3.30

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as  $\text{dB}_m$  and is calculated as  $\text{dB}_m = 10 \log_{10} P_m$ , where  $P_m$  is the power in milliwatts.

Calculate the power of a signal if its  $\text{dB}_m = -30$ .

## Example 3.30

### Solution

The loss in the cable in decibels is  $5 \times (-0.3) = -1.5$  dB. We can

$$\text{dB} = 10 \log_{10} (P_2 / P_1) = -1.5 \text{ dB}$$

$$\rightarrow (P_2 / P_1) = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 \text{ mW} = 1.4 \text{ mW}$$

# Attenuation and Amplification - Decibel

- Unit of Signal strength is Decibel or dB
- Decibel (dB) measures the relative strengths of two signals or one signal at two different points

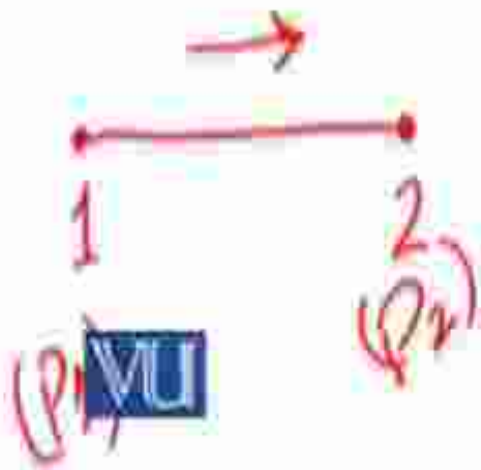
$$10 \log_{10} P_2/P_1$$

- Decibel is negative if a signal is attenuated and positive if signal is amplified

# Example

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that  $P_2 = 0.5 P_1$ . In this case, the

a  
r  
c



$P_2 \rightarrow$  Power at point 2

$P_1 \rightarrow$  Power at point 1

$$P_2 = 0.5 P_1$$

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1}$$

Signal has lost one half the power

$$= 10 \log_{10} 0.5$$
$$= \underline{\underline{-3 \text{ dB}}}$$

## Example

A signal travels through an amplifier, and its power is increased 10 times. This means that  $P_2 = 10P_1$ . In this case, the **amplification** (gain of power) can be calculated as

$$\begin{aligned} P_2 &= 10 P_1 \\ 10 \log_{10} \frac{P_2}{P_1} &= 10 \log_{10} \frac{10 P_1}{P_1} \\ &= 10 \text{ dB} \end{aligned}$$

# Distortion

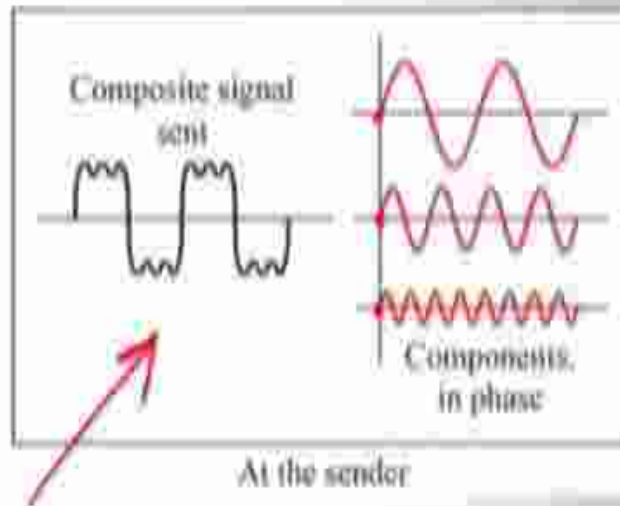
- **Distortion means that the signal changes its form or shape.**
- **Distortion can occur in a composite signal made of different frequencies.**
- **Each signal component has its own propagation speed (see the next section) through a medium and, therefore, its own delay in arriving at the final destination.**

# Distortion

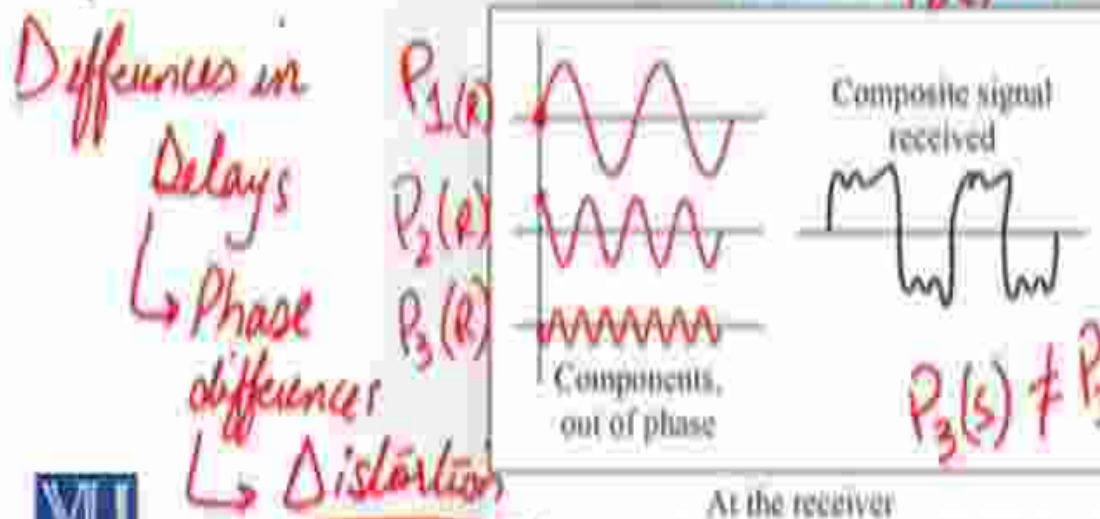
Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.

# Distortion

Figure 3.2



$$P_1(s) = P_1(r)$$
$$P_2(s) \neq P_2(r)$$



$$P_3(s) \neq P_3(r)$$



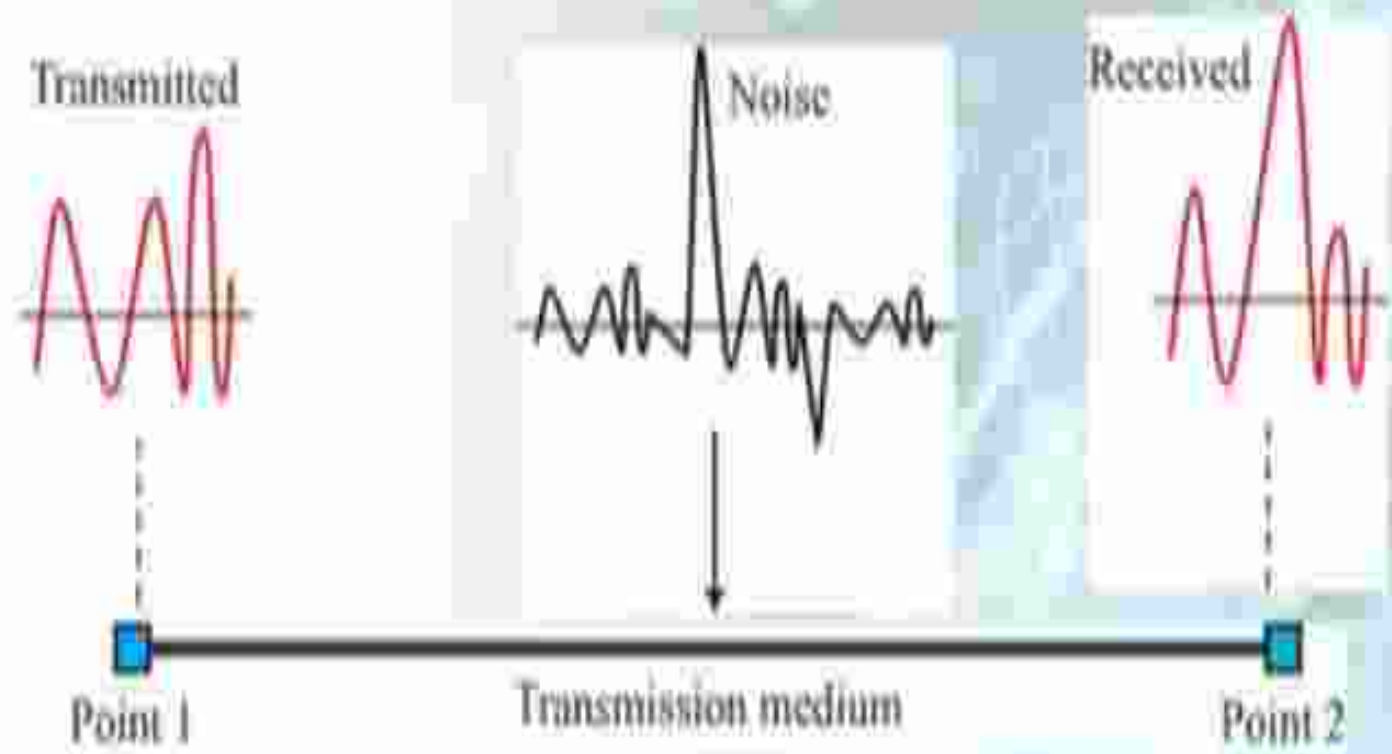
# Noise

- Noise is another cause of impairment.
- Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- Thermal noise is the random motion of electrons in a wire, which creates an extra signal not

# Noise

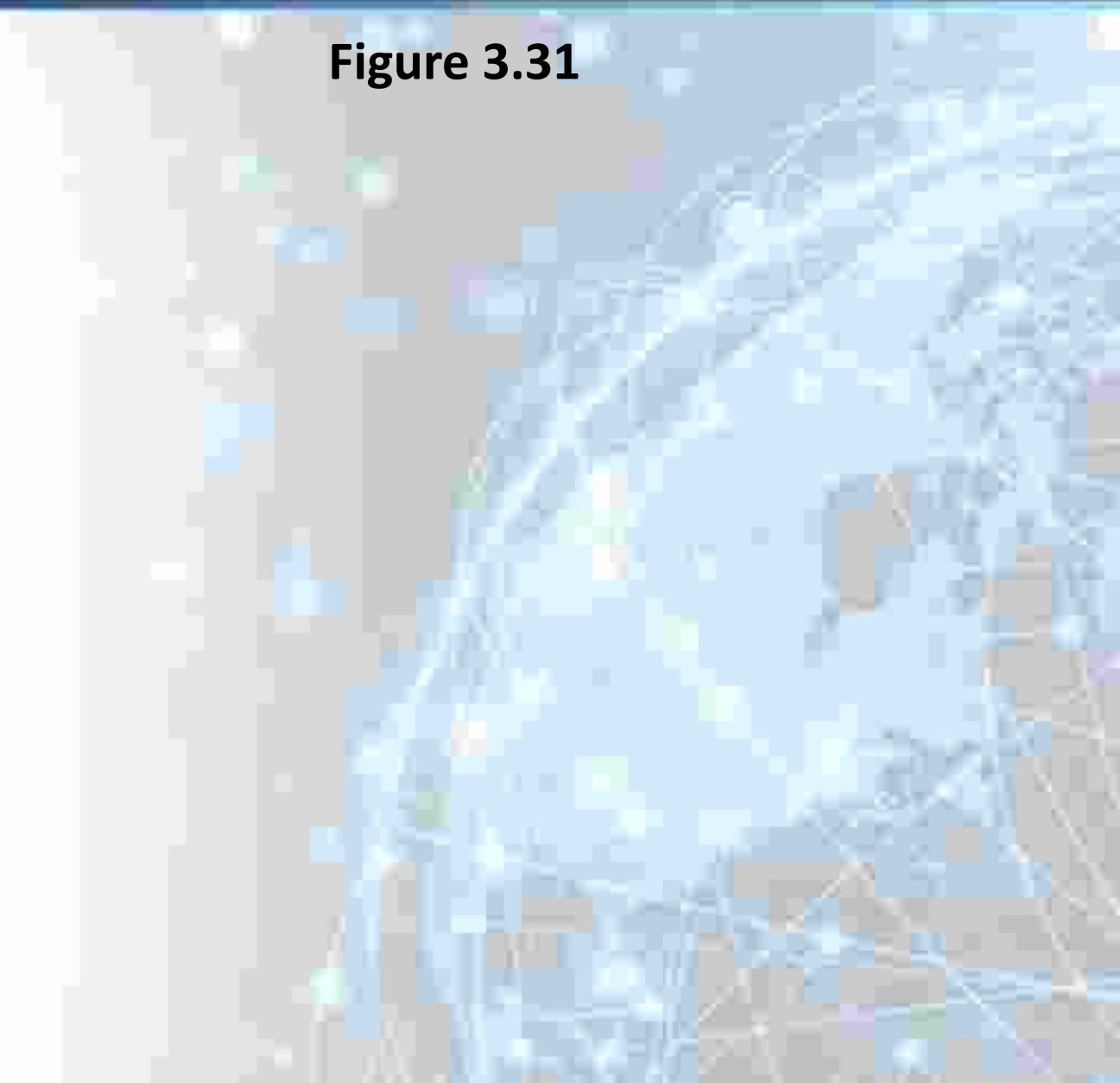
- **Induced noise comes from sources such as motors.**
- **Crosstalk is the effect of one wire on the other.**

# Noise



# Noise

**Figure 3.31**

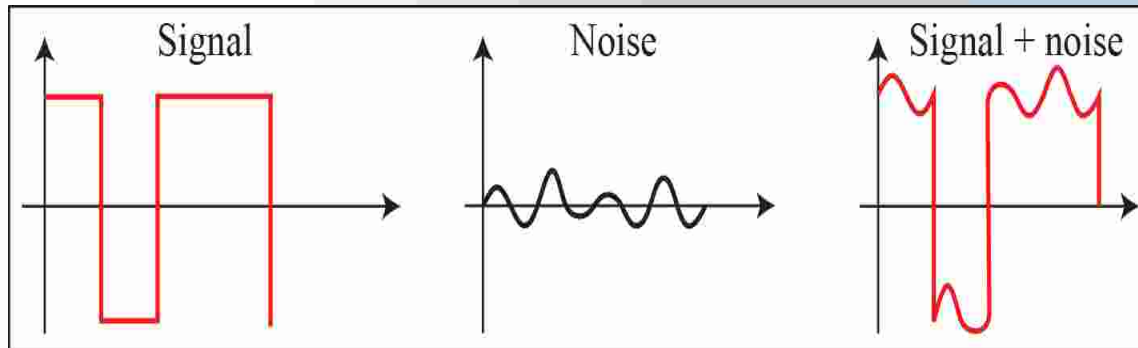


# Noise – Signal to Noise Ratio (SNR)

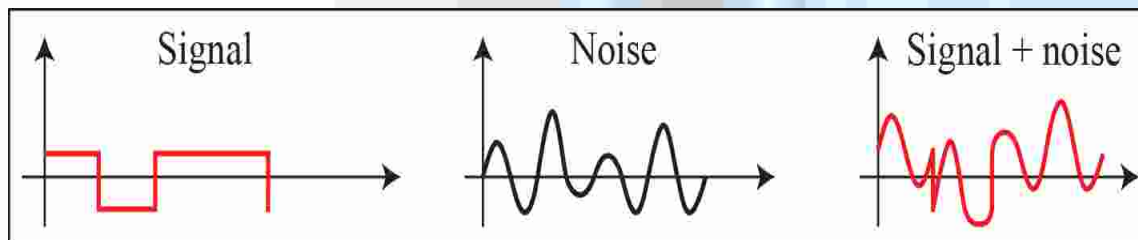
- Signal to Noise Ratio (SNR) is used to find the theoretical bit rate limit of a signal

$$\text{SNR} = \frac{\text{average signal power}}{\text{noise power}}$$

# Noise – Signal to Noise Ratio (SNR)



a. High SNR

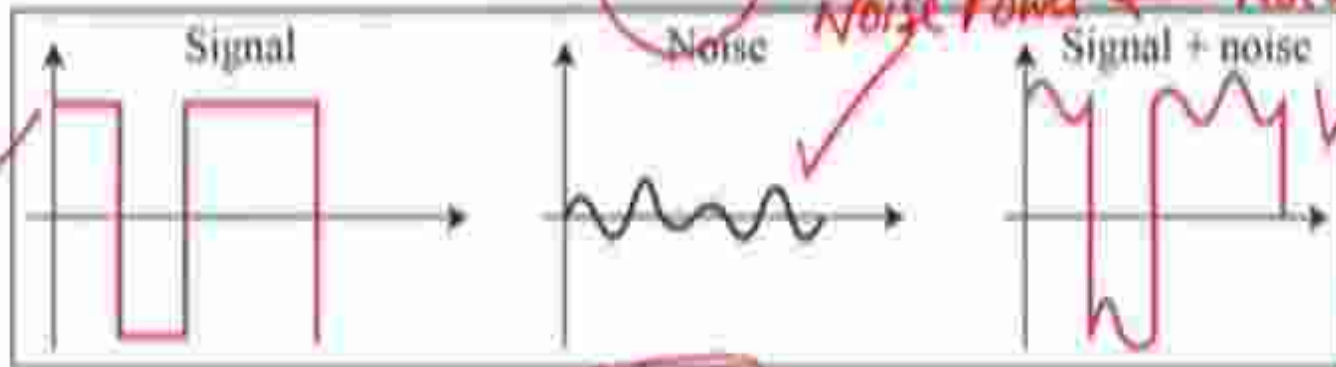


b. Low SNR

# Noise – Signal to Noise Ratio (SNR)

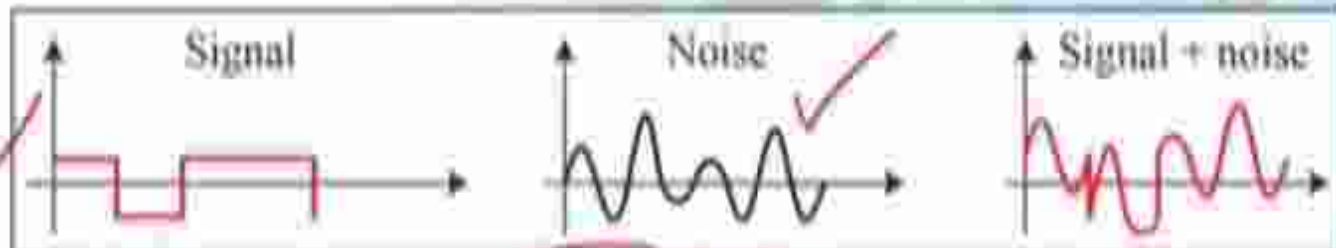
**SNR**

$\text{SNR} = \frac{\text{Signal Power}}{\text{Noise Power}}$   
← wanted  
← Not wanted



**a. High SNR**

less corrupted



**b. Low SNR**

more corrupted

$\text{SNR} = 10 \log_{10} \frac{P_s}{P_n}$

## Example

The power of a signal is 10 mW and the power of the noise is 1  $\mu$ W; what are the values of SNR and SN

$$SNR = \frac{10 \text{ mW}}{1 \mu\text{W}} = 10,000$$

$$SNR_{dB} = 10 \log_{10} 10,000 = \underline{\underline{40 \text{ dB}}}$$



# Example

The values of SNR and  $\text{SNR}_{\text{dB}}$  for a noiseless channel are calculated as

$$\underline{\text{Noise} = 0}$$

$\rightarrow$  NOT a real life scenario

$$\text{SNR} = \frac{(\text{Sig. Power})}{0} = \infty$$

$$= 10 \log_{10} \infty = \infty$$

Not Real

## Example 3.31

The power of a signal is 10 mW and the power of the noise is 1  $\mu$ W; what are the values of SNR and SNR<sub>dB</sub>?

### Solution

The values of SNR and SNR<sub>dB</sub> can be calculated as

$$\begin{aligned} \text{SNR} &= (10,000 \mu\text{W}) / (1 \mu\text{W}) = 10,000 & \text{SNR}_{\text{dB}} \\ &= 10 \log_{10} 10,000 = 10 \log_{10} 10^4 = 40 \end{aligned}$$

## Example 3.32

The values of SNR and  $\text{SNR}_{\text{dB}}$  for a noiseless channel are

### Solution

The values of SNR and  $\text{SNR}_{\text{dB}}$  for a noiseless channel are

$$\text{SNR} = (\text{signal power}) / 0 = \infty$$

$$\longrightarrow \text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

# Data Rate Limits

- How fast we can send data, in bits per second, over a channel?
- Data Rate depends on 3 factors:
  - ✓ The Bandwidth available
  - ✓ The level of the signals we use
  - ✓ The level of noise

# Data Rate Limits

- **Two theoretical formulas developed to calculate the data rate:**
  - ✓ **one by Nyquist for a noiseless channel**
  - ✓ **another by Shannon for a noisy channel**

# Noiseless Channel : Nyquist Rate

$$\text{Bit Rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

BW = BW of channel

L = No. of signal levels

BR = bps

Bit Rate  $\propto L$

$L \uparrow \Rightarrow \text{Bit Rate} \uparrow$

$(L \Rightarrow 0, 1) \quad L = 2 \text{ levels}$

- For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate
- Finding balance between Bit rate

## Example

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\begin{aligned}BR &= 2 \times 3000 \times \log_2 2 \\ &= \underline{6000 \text{ bps}}\end{aligned}$$

# Example

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The

n  
c  
a

$$\underline{BR \propto L}$$

$$BR = 2 \times 3000 \times \log_2 4$$
$$= 12,000 \text{ bps}$$

(Reliability)



# Noisy Channel : Shannon Capacity

- In reality, we cannot have a noiseless channel; the channel is always noisy
- In 1944, Claude Shannon introduced a formula, to determine the theoretical highest data rate for a noisy channel

$$\text{Capacity} = \text{Bandwidth}_{(\text{channel})} \times \log_2 \left( 1 + \text{SNR} \right)$$

Capacity  $\rightarrow$  Capacity of Channel  $\rightarrow$  Bit Rate(max)

SNR  $\rightarrow$  (Sig to Noise Ratio)

---

Levels  $\rightarrow L \times$

Max BR < Capacity of Channel

## Example

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity  $C$  is calculated as

$$\text{SNR} \approx \text{zero}$$

$$C = B \log_2(1 + \text{SNR})$$

$$= B \log_2 1$$

$$= B \times \text{zero}$$

$$C = \text{zero}$$

→ Cannot receive any data through channel.

## Example

Theoretical highest bit rate of a Telephone line with a Bandwidth of 3000 Hz assigned for data communication. SNR is usually 3162. The capacity is calculated as:

$$C = 3000 \times \log_2(1 + 3162)$$
$$= 34,860 \text{ bps}$$

Telephone line

# Using Both Limits

- In practice, we need to use both methods to find the limits and signal levels
- Shannon's formula gives us the upper limit while the Nyquist formula gives us the signal levels

## Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this

$$C = B \log_2(1 + \text{SNR}) = B \log_2(1 + 0)$$

$$= B \log_2 1 = B \times 0 = 0$$

capacity  $C$  is calculated as

## Example 3.37

**This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.**

## Example 3.38

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot

$$C = B \log_2 (1 + \text{SNR}) =$$

$$3000 \log_2 (1 + 3162) =$$

$$3000 \times 11.62 = 34,860 \text{ bps}$$

## Example 3.38

**This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.**



## Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that  $\text{SNR}_{\text{dB}} = 36$  and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

## Example 3.39

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \longrightarrow$$

$$\text{SNR} = 10^{\text{SNR}_{\text{dB}}/10} \longrightarrow$$

$$\text{SNR} = 10^{3.6} = 3981$$

$$C = B \log_2(1 + \text{SNR}) =$$

$$2 \times 10^6 \times \log_2 3982$$

$$= 24 \text{ Mbps}$$

## Example 3.40

When the SNR is very high, we can assume that  $\text{SNR} + 1$  is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to  $C = B \log_2 \text{SNR}_{\text{dB}}$ . For example, we can calculate the

$$C = 2 \text{ MHz} \times (36 / 3) = 24 \text{ Mbps}$$

of the previous example as

## Using Both Limits

**In practice, we need to use both methods to find the limits and signal levels. Let us show this with an example.**

## Example 3.41

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

### Solution

First, we use the

$$C = B \log_2(1 + \text{SNR}) = \text{a to bit}$$
$$10^6 \log_2(1 + 63) = 10^6 \log_2 64 =$$
$$6 \text{ Mbps}$$

## Example 3.41

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps. Then we use the Nyquist formula

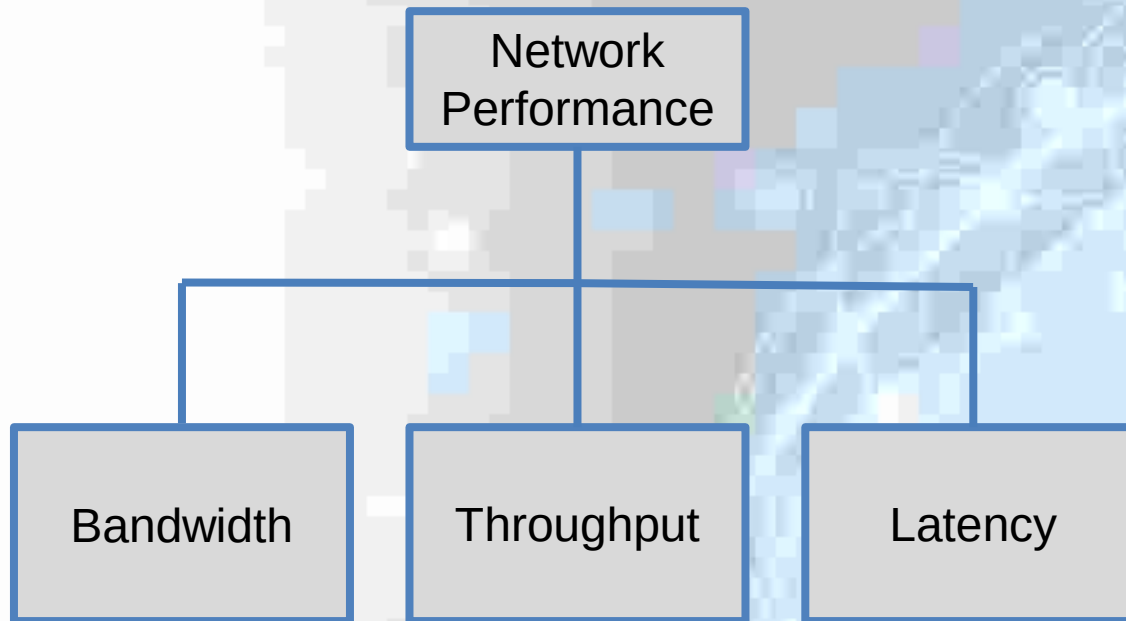
$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$$

number of levels  $\rightarrow L = 4$

# Network Performance

- Data transmission (in form of Signal) over a network and how network behaves is important
- More important is the performance of the network
- How good is our network?

# Network Performance



- **There are 3 characteristics of network performance**



# Bandwidth

- An important characteristic that measures Network Performance
- Bandwidth can be used in two different contexts with two different measuring values:
  - Bandwidth in Hertz
  - Bandwidth in

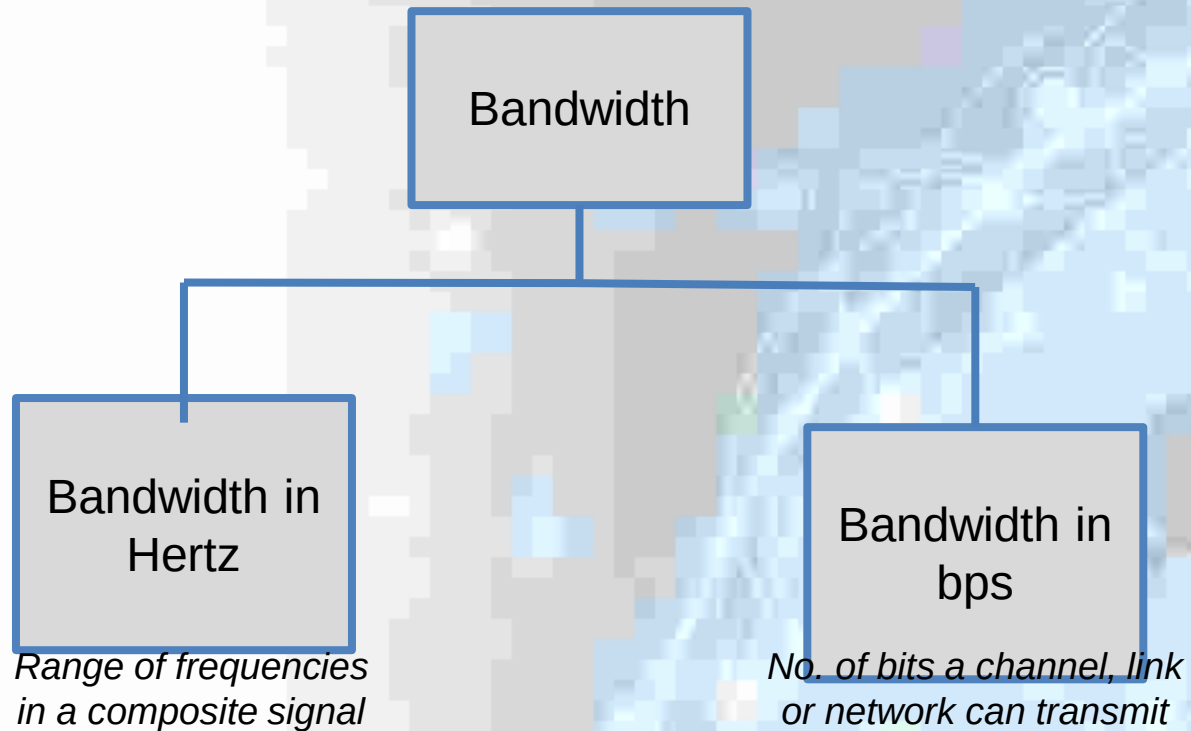
## Example 3.42

The bandwidth of a subscriber line is 4 kHz for voice or data. The bandwidth of this line for data transmission can be up to 56,000 bps using a sophisticated modem to change the digital signal to analog.

## Example 3.43

If the telephone company improves the quality of the line and increases the bandwidth to 8 kHz, we can send 112,000 bps by using the same technology as mentioned in Example 3.42.

# Bandwidth



# Throughput

- Measure of how fast we can actually send data through a network.
- Bandwidth is not the same as Throughput
- A link may have a bandwidth of  $B$  bps, but we can only send  $T$  bps through this link.

## Example

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of the network?

$$T = \frac{(12,000 \times 10,000)}{60}$$
$$= 2 \text{ Mbps}$$

$$T = 2 \text{ Mbps}$$
$$B = 10 \text{ Mbps}$$

# Throughput

The throughput is a measure of how fast we can actually send data through a network.

Although, at first glance, bandwidth in bits per second and throughput seem the same, they are different. A link may have a bandwidth of  $B$  bps, but we can only send  $T$  bps through this link with  $T$  always less than  $B$ .

# Throughput

**The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.**



# Throughput

We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.

Latency =  
propagation time +  
transmission time +  
queuing time +  
processing delay

## Example 3.44

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

## Example 3.44

### **Solution**

We can calculate the throughput as

Throughput =

$$(12,000 \times 10,000) / 60 = 2\text{Mbps}$$

The throughput is almost one-fifth of the bandwidth in this case.

## Example 3.45

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be  $2.4 \times 10^8$  m/s in cable.

### **Solution**

We can calculate the propagation time as

$$\text{Propagation time} = \frac{(12,000 \times 10,000)}{(2.4 \times 10^8)} =$$

## Example 3.45

The example shows that a bit can go over the Atlantic Ocean in only 50 ms if there is a direct cable between the source and the destination.

## Example 3.46

What are the propagation time and the transmission time for a 2.5-KB (kilobyte) message if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at  $2.4 \times 10^8$  m/s.

## Example 3.46

### **Solution**

We can calculate the propagation and transmission time as

$$\begin{aligned} \text{Propagation time} &= \\ &= (12,000 \times 1000) / \\ &= (2.4 \times 10^8) = 50 \text{ ms} \end{aligned}$$

$$\begin{aligned} \text{Transmission time} &= \\ &= (2500 \times 8) / 10^9 = \\ &= 0.020 \text{ ms} \end{aligned}$$

## Example 3.46

### **Solution**

Note that in this case, because the message is short and the bandwidth is high, the dominant factor is the propagation time, not the transmission time.



## Example 3.47

What are the propagation time and the transmission time for a 5-MB (megabyte) message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at  $2.4 \times 10^8$

## Example 3.47

**Solution**

**We can calculate the propagation and transmission times as**

$$\text{Propagation time} = (12,000 \times 1000) / (2.4 \times 10^8) = 50 \text{ ms}$$

$$\text{Transmission time} = (5,000,000 \times 8) / 10^6 = 40 \text{ s}$$

**We can calculate the propagation and transmission**

# Latency or Delay

- Latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source

$$\text{Latency} = \text{Propagation Time} + \text{Transmission Time} + \text{Queuing Time} + \text{Processing Delay}$$

$$PT = \frac{\text{Distance}}{\text{Prop Speed}}$$

$$TT = \frac{\text{Message Size}}{BW}$$

QT → Time the message is held

## Example

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be  $2.4 \times 10^8$  m/s in cable.

$$\begin{aligned} PT &= \frac{(12000 \times 1000)}{2.4 \times 10^8} \\ &= \underline{\underline{50 \text{ m sec.}}} \end{aligned}$$

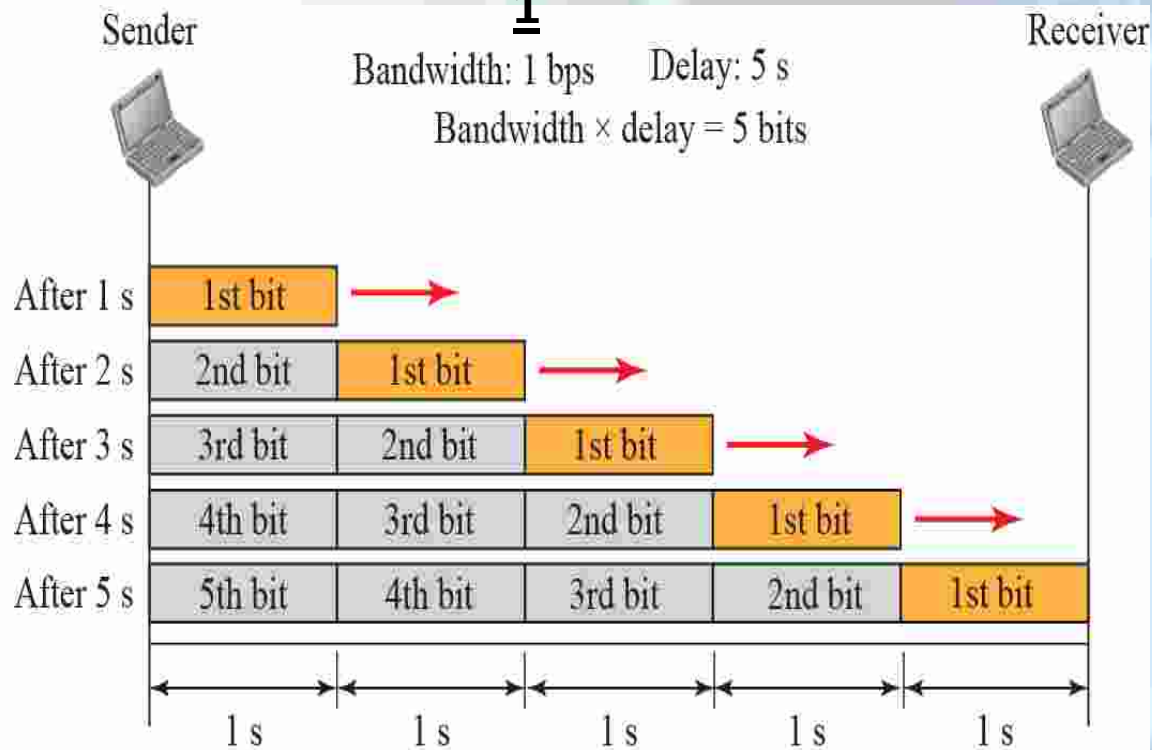
# Delay – Bandwidth Delay Product

- Bandwidth and delay are two performance metrics of a link
- Product of the two, The Bandwidth-Delay Product defines the number of bits that can fill a link

# Bandwidth-Delay Product

## Case

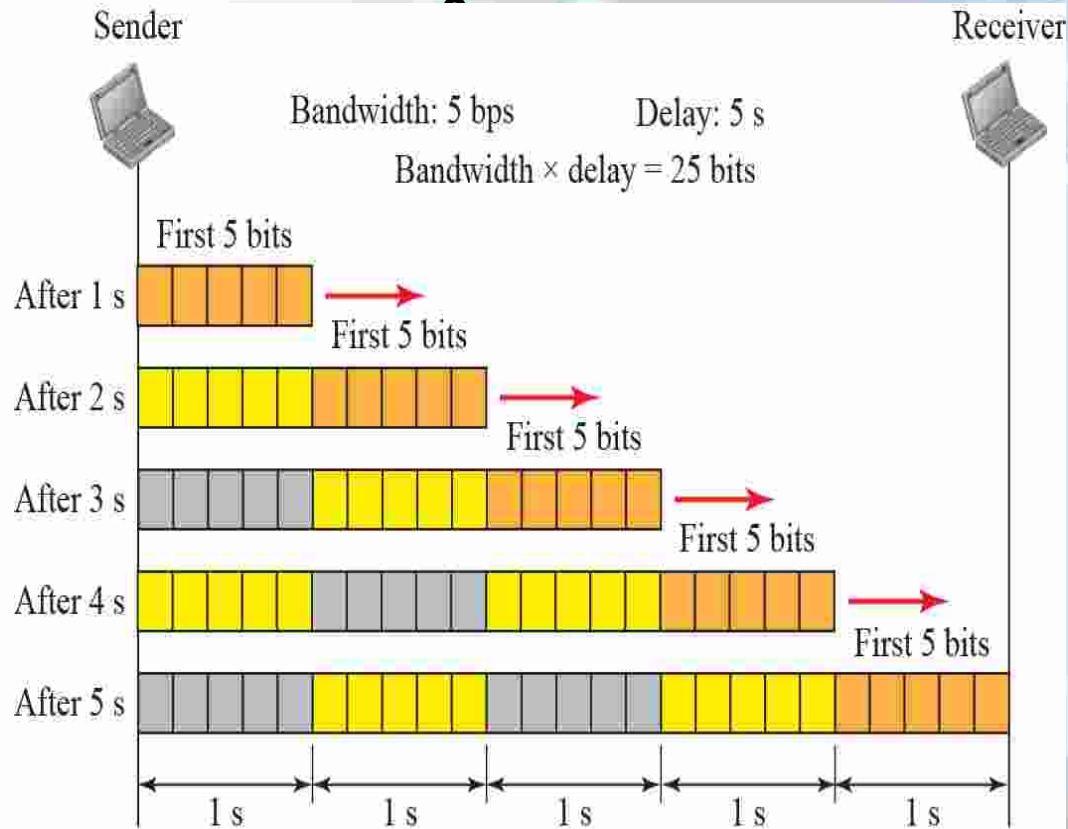
1



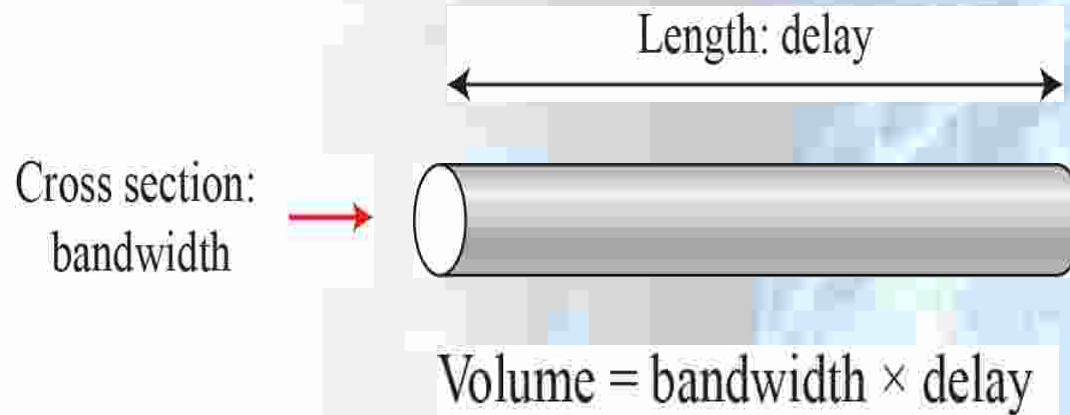
5 bit

# Bandwidth-Delay Product

## Case



# Bandwidth-Delay Product





## Example 3.48

We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product, as shown in Figure 3.34.

# Delay - Jitter

- Jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example)

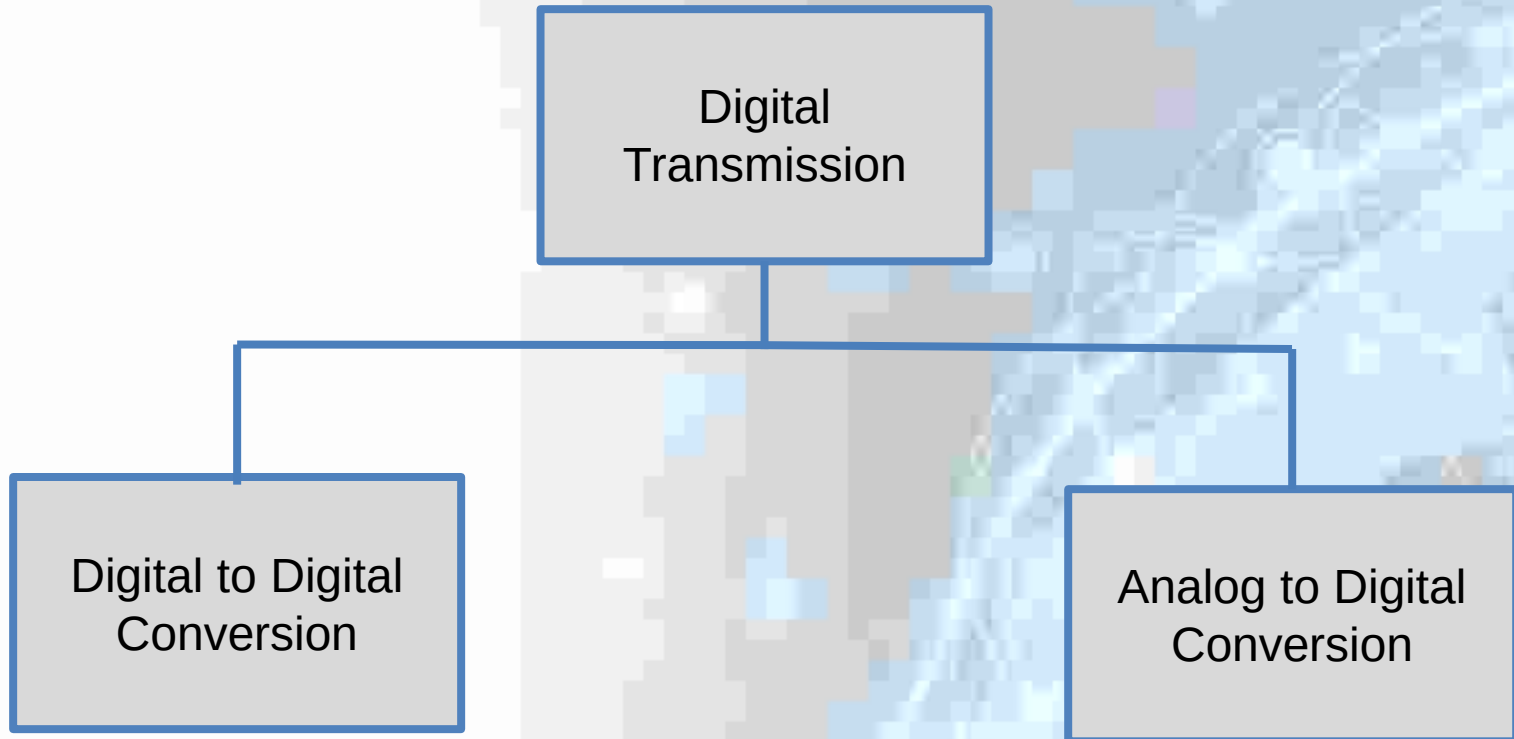
# Delay - Jitter

- Delay for first packet is 20 ms for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter

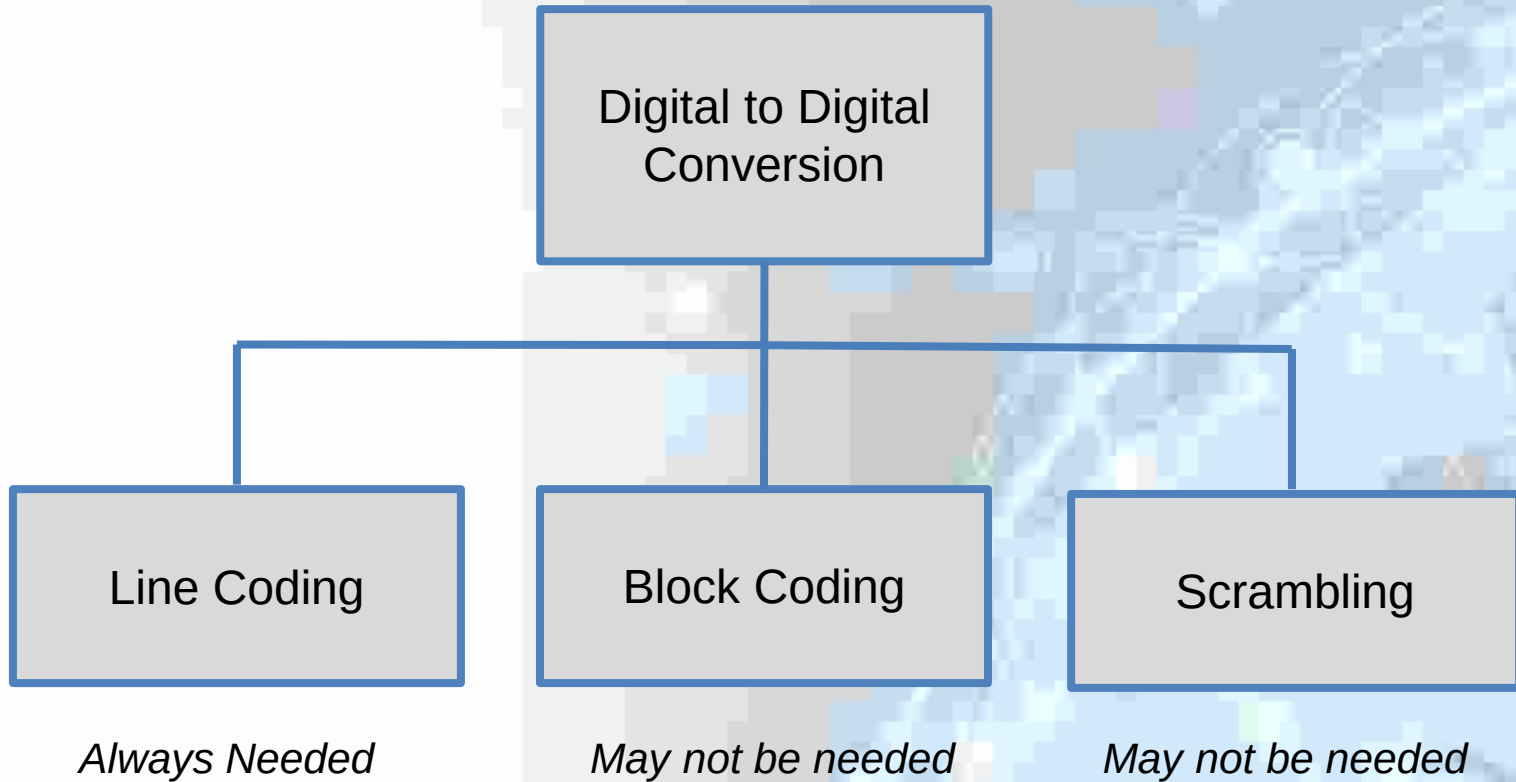
# Digital-to-digital Conversion

- **Data → Analog or Digital**
- **Signals → Analog or Digital**
- **Digital Transmission**
- **Analog Transmission**

# Digital Transmission



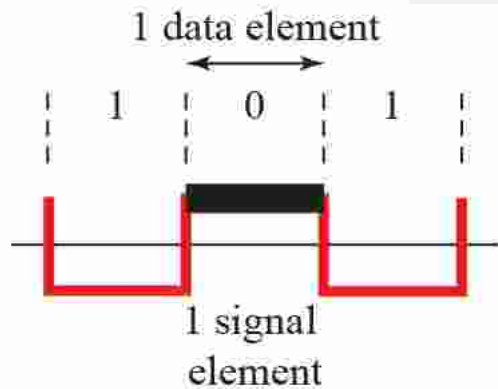
# Digital to Digital Conversion



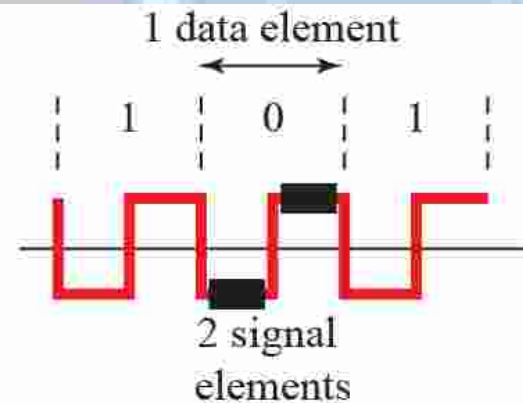
# Signal Element versus Data Element

- A Data element is the smallest entity that can represent a piece of information → Bit
- A Signal element is the shortest unit of a digital signal
- Data Elements: Carried
- Signal Elements: Carriers

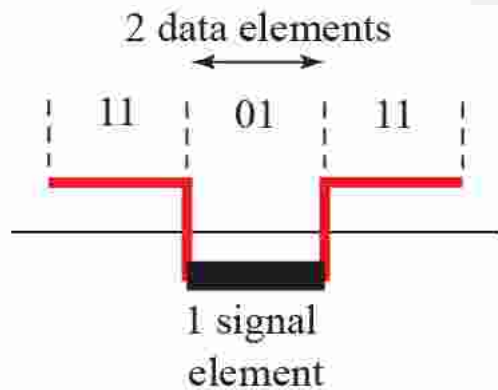
# Signal Element versus Data Element



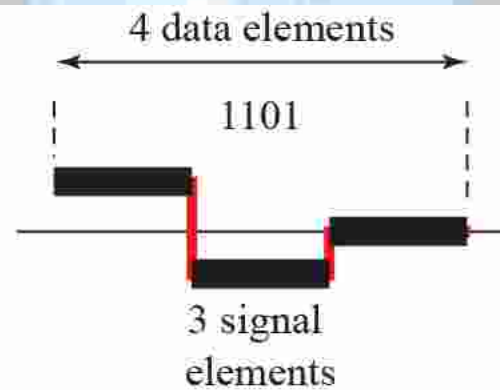
a. One data element per one signal element ( $r = 1$ )



b. One data element per two signal elements ( $r = \frac{1}{2}$ )



c. Two data elements per one signal element ( $r = 2$ )



d. Four data elements per three signal elements ( $r = \frac{4}{3}$ )



# Data Rate versus Signal Rate

- Data Rate is number of data elements sent in 1 sec (bps)
- Signal Rate is number of signal elements sent in 1 sec (baud)
- Data Rate → Bit Rate 
- Signal Rate → Pulse Rate, Modulation Rate or Baud Rate 

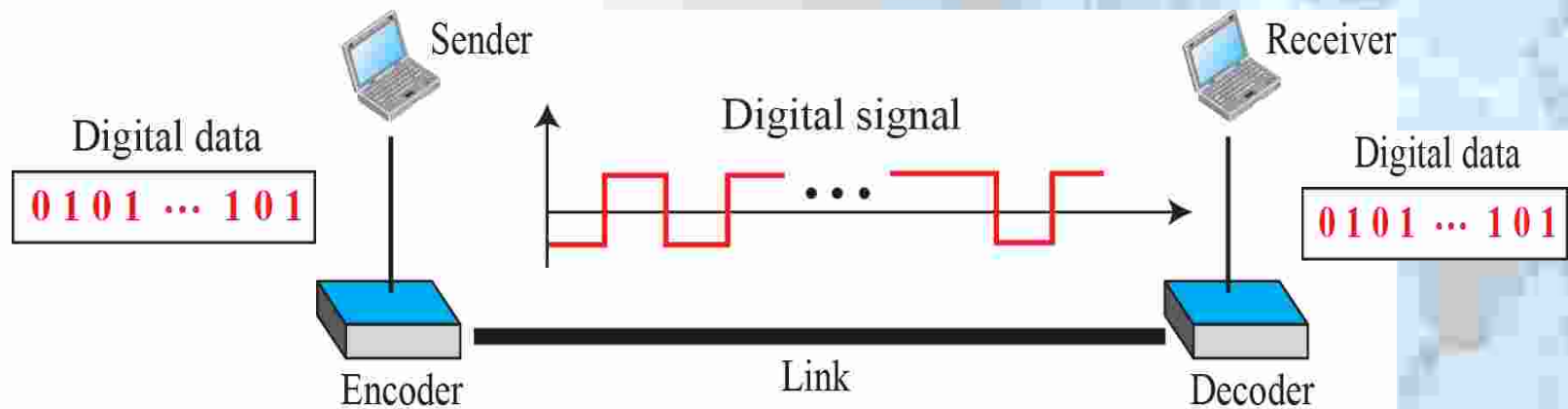
## Example

**A signal has a signal rate of 100 bauds. What is the Data rate if one data element is carried per signal element?**

# Line Coding

- **Digital data to Digital signals**
- **Data (Text, Numbers, Pictures, Audio, or Video) is stored in computer memory as sequences of bits**
- **Line coding converts a sequence of Bits to a Digital Signal**

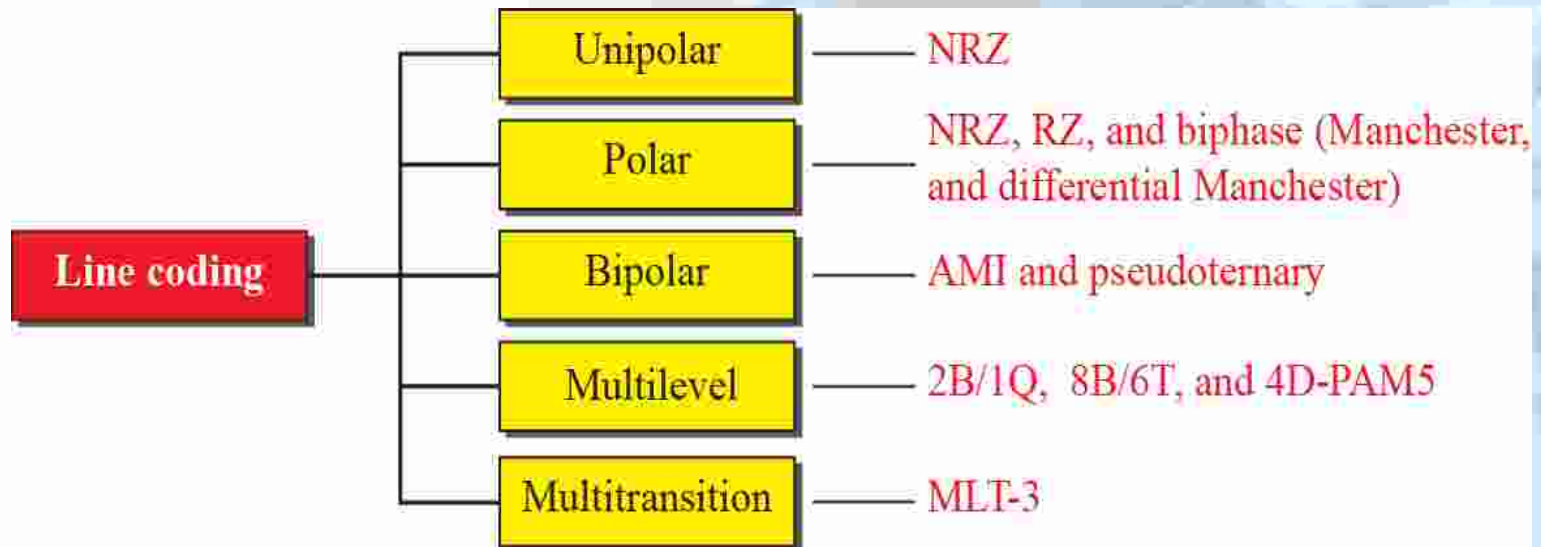
# Line Coding and Decoding



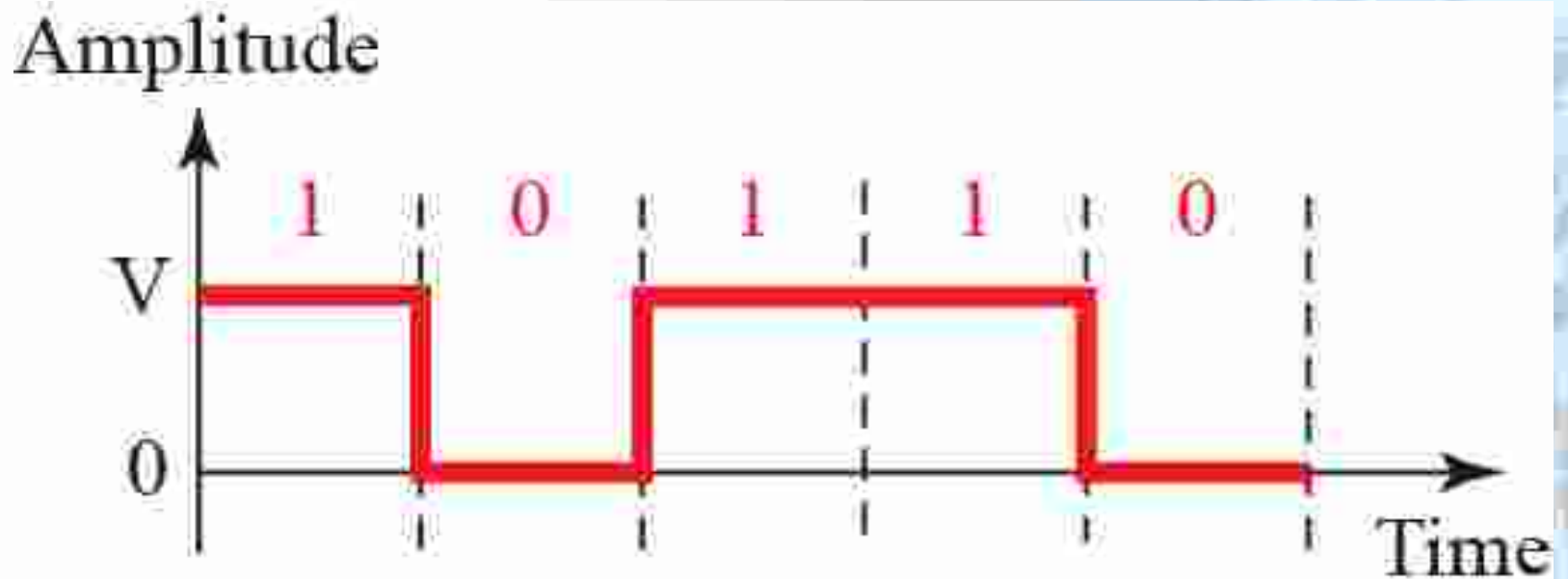
# Line Coding Schemes

- We can roughly divide line coding schemes into five broad categories

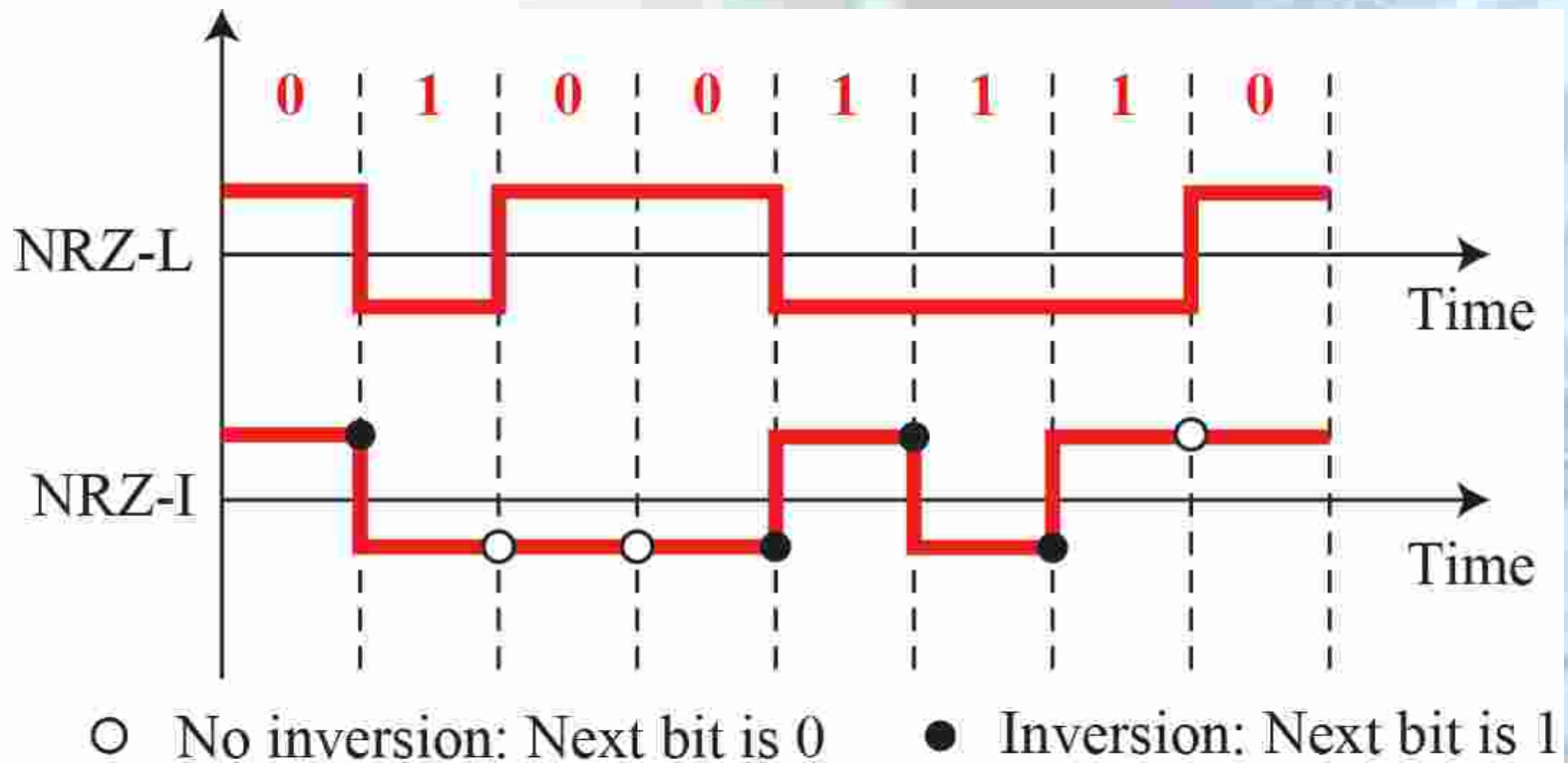
# Line Coding Schemes



# Unipolar NRZ scheme

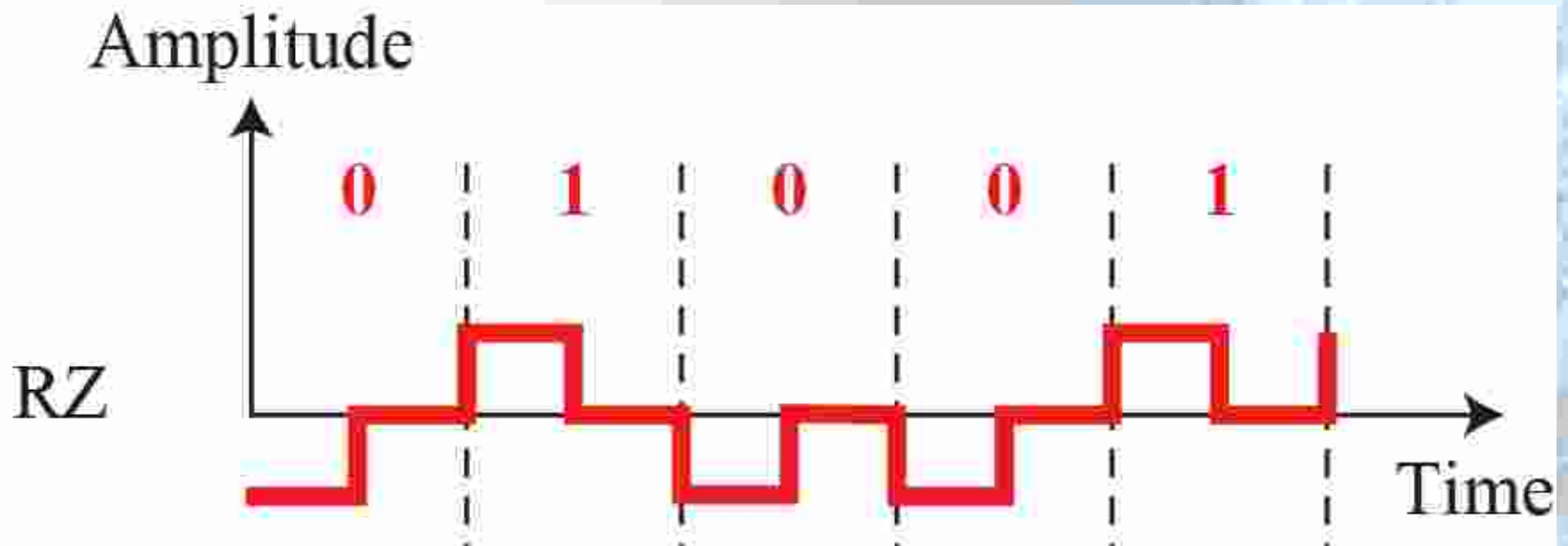


# Polar schemes (NRZ)





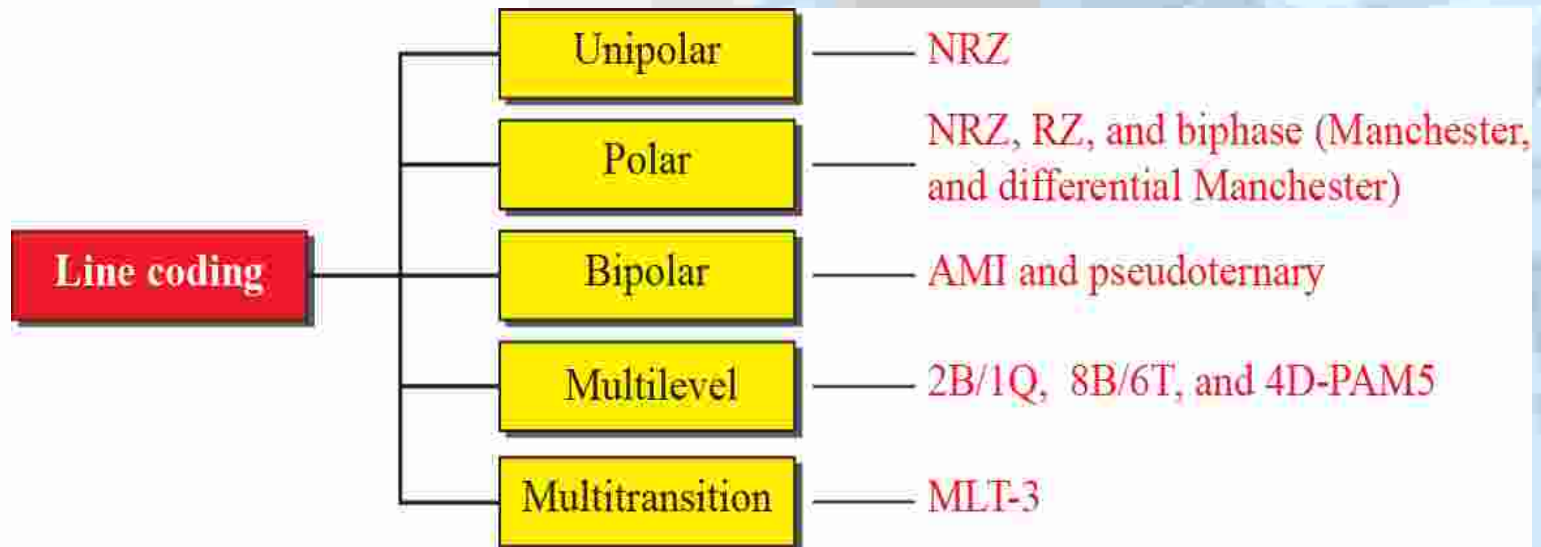
## Polar schemes (RZ)



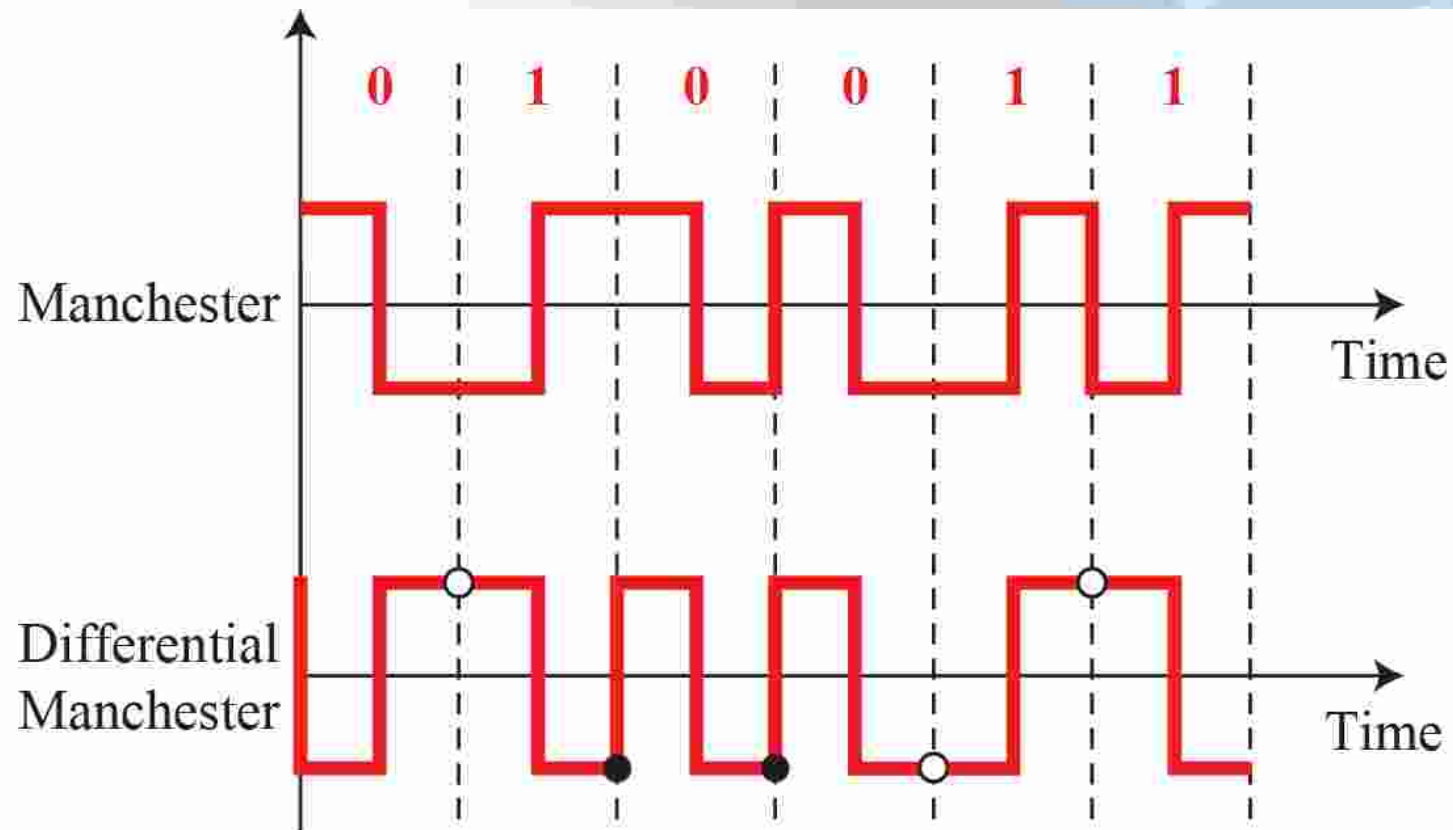
# Line Coding Schemes

- We can roughly divide line coding schemes into five broad categories

# Line Coding Schemes



# Polar Biphase

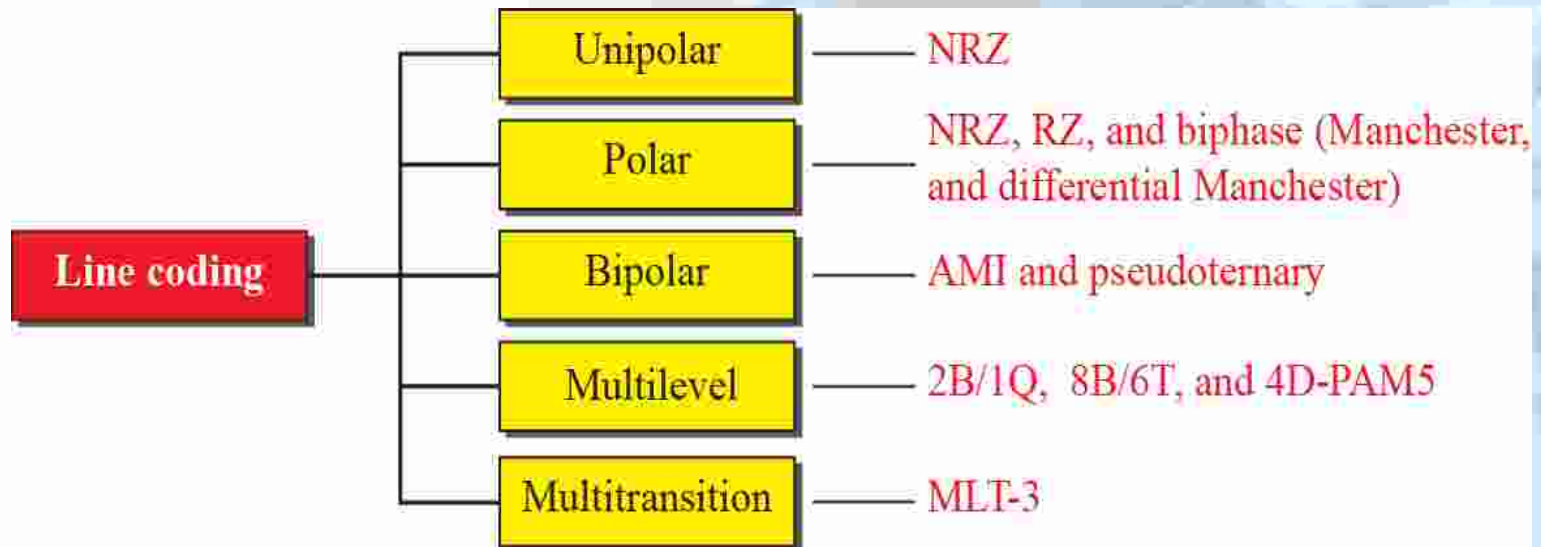


○ No inversion: Next bit is 1    ● Inversion: Next bit is 0

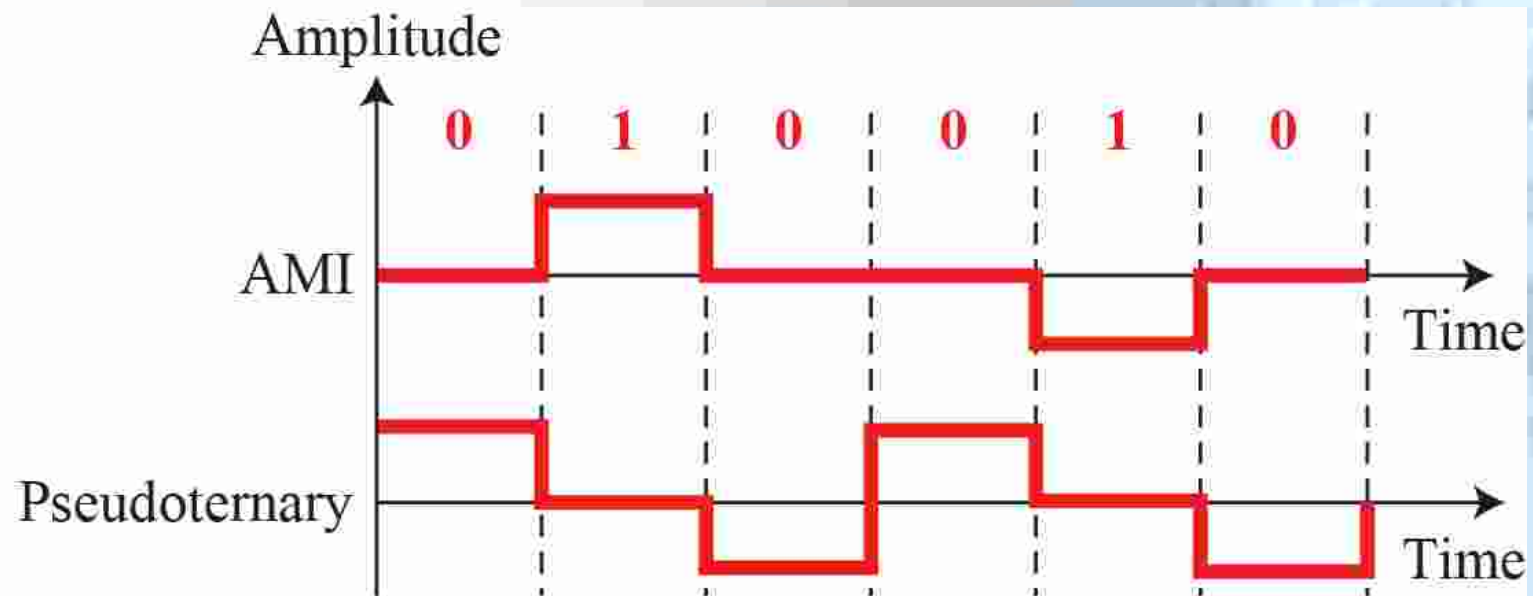
# Line Coding Schemes

- We can roughly divide line coding schemes into five broad categories

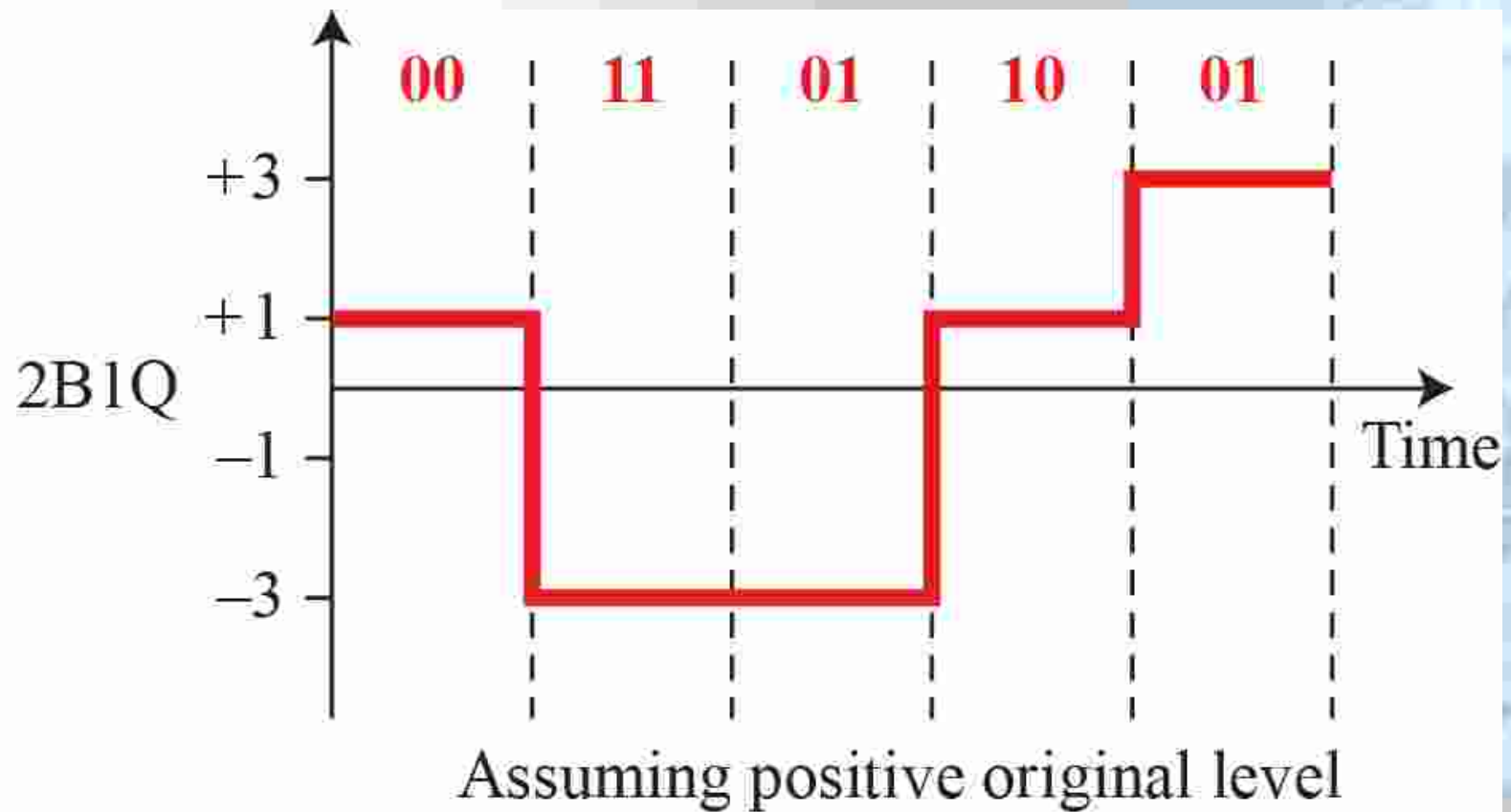
# Line Coding Schemes



# Bipolar schemes: AMI & Pseudoternary



## Multilevel: 2B1Q





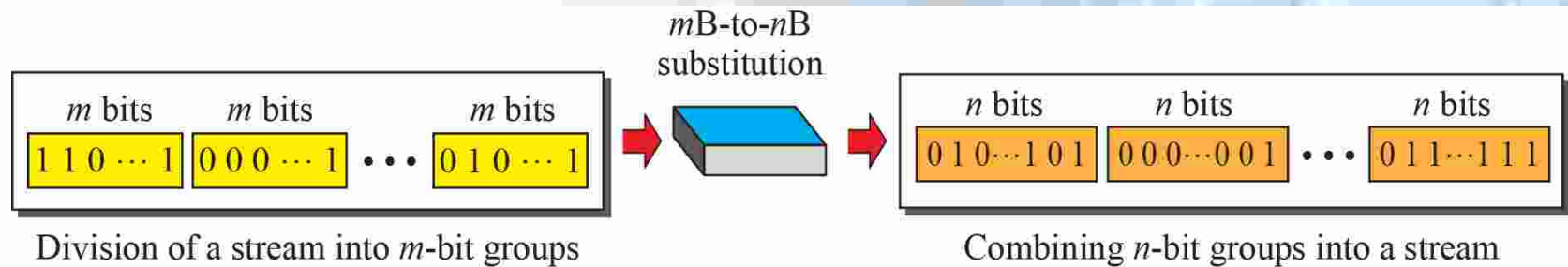
# Table 4.1 : Summary of line coding schemes

| <i>Category</i> | <i>Scheme</i> | <i>Bandwidth<br/>(average)</i> | <i>Characteristics</i>                               |
|-----------------|---------------|--------------------------------|------------------------------------------------------|
| Unipolar        | NRZ           | $B = N/2$                      | Costly, no self-synchronization if long 0s or 1s, DC |
| Polar           | NRZ-L         | $B = N/2$                      | No self-synchronization if long 0s or 1s, DC         |
|                 | NRZ-I         | $B = N/2$                      | No self-synchronization for long 0s, DC              |
|                 | Biphase       | $B = N$                        | Self-synchronization, no DC, high bandwidth          |
| Bipolar         | AMI           | $B = N/2$                      | No self-synchronization for long 0s, DC              |
| Multilevel      | 2B1Q          | $B = N/4$                      | No self-synchronization for long same double bits    |
|                 | 8B6T          | $B = 3N/4$                     | Self-synchronization, no DC                          |
|                 | 4D-PAM5       | $B = N/8$                      | Self-synchronization, no DC                          |
| Multitransition | MLT-3         | $B = N/3$                      | No self-synchronization for long 0s                  |

# Block Coding

- Block coding changes a block of 'm' bits into a block of 'n' bits ( $n > m$ )
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

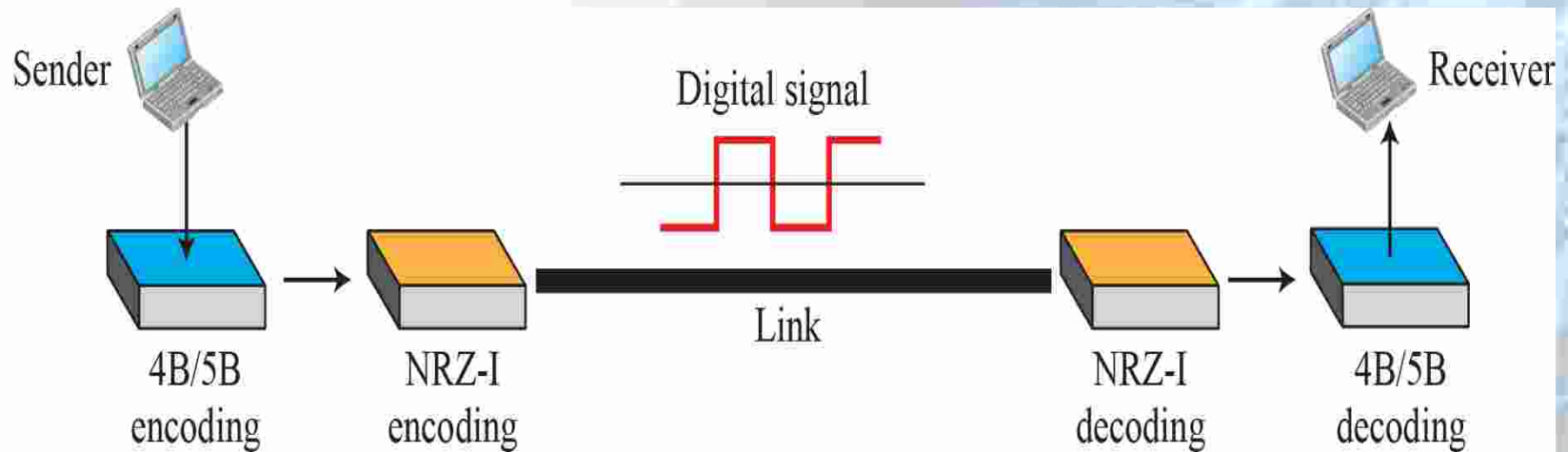
# Block coding concept



# Block Coding

- Block coding changes a block of 'm' bits into a block of 'n' bits ( $n > m$ )
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

# Using block coding 4B/5B with NRZ-I line coding



# Block Coding

- Block coding changes a block of 'm' bits into a block of 'n' bits ( $n > m$ )
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance

# 4B/5B mapping codes

| <i>Data Sequence</i> | <i>Encoded Sequence</i> | <i>Control Sequence</i> | <i>Encoded Sequence</i> |
|----------------------|-------------------------|-------------------------|-------------------------|
| 0000                 | 11110                   | Q (Quiet)               | 00000                   |
| 0001                 | 01001                   | I (Idle)                | 11111                   |
| 0010                 | 10100                   | H (Halt)                | 00100                   |
| 0011                 | 10101                   | J (Start delimiter)     | 11000                   |
| 0100                 | 01010                   | K (Start delimiter)     | 10001                   |
| 0101                 | 01011                   | T (End delimiter)       | 01101                   |
| 0110                 | 01110                   | S (Set)                 | 11001                   |
| 0111                 | 01111                   | R (Reset)               | 00111                   |
| 1000                 | 10010                   |                         |                         |
| 1001                 | 10011                   |                         |                         |
| 1010                 | 10110                   |                         |                         |
| 1011                 | 10111                   |                         |                         |
| 1100                 | 11010                   |                         |                         |
| 1101                 | 11011                   |                         |                         |
| 1110                 | 11100                   |                         |                         |
| 1111                 | 11101                   |                         |                         |

# Block Coding

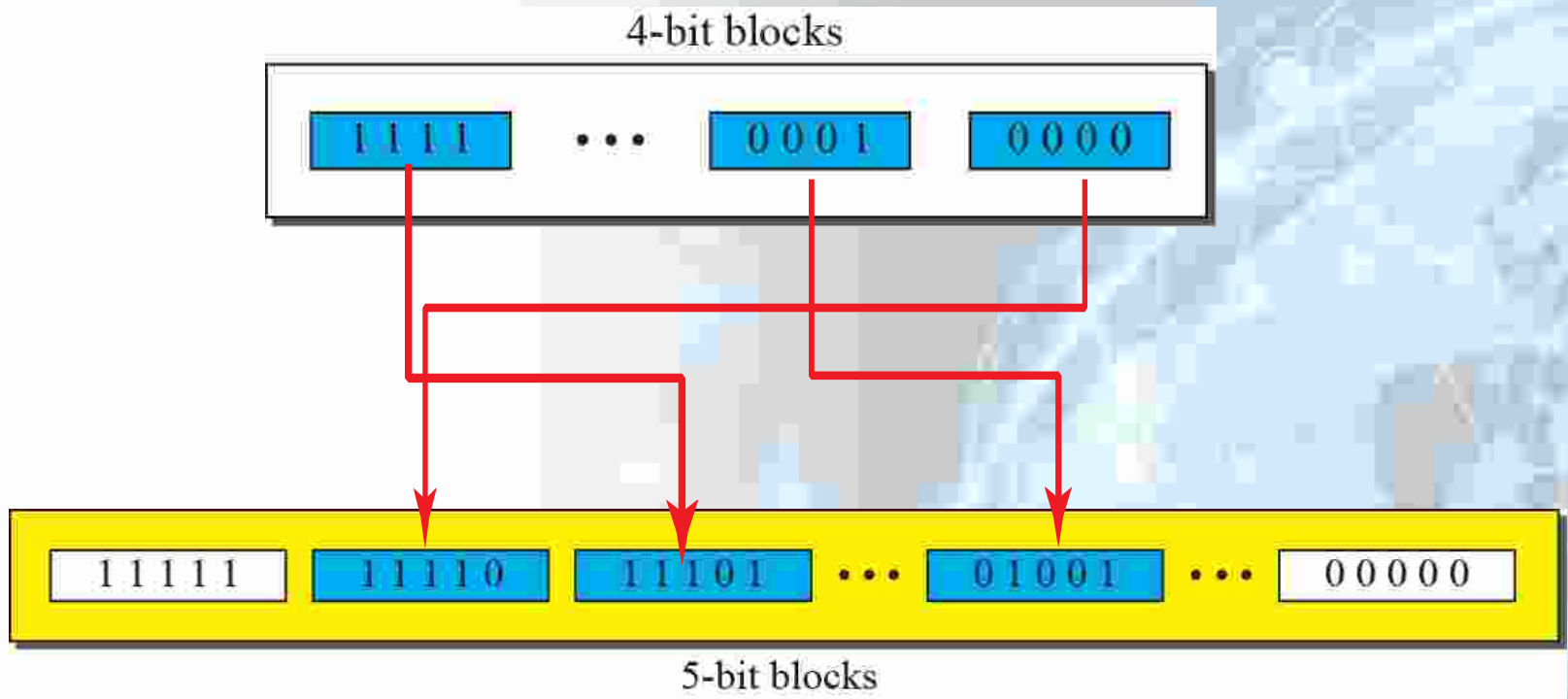
- Block coding changes a block of 'm' bits into a block of 'n' bits ( $n > m$ )
- mB/nB encoding technique
- We need Redundancy to ensure Synchronization
- Block coding gives us redundancy and improves line coding performance



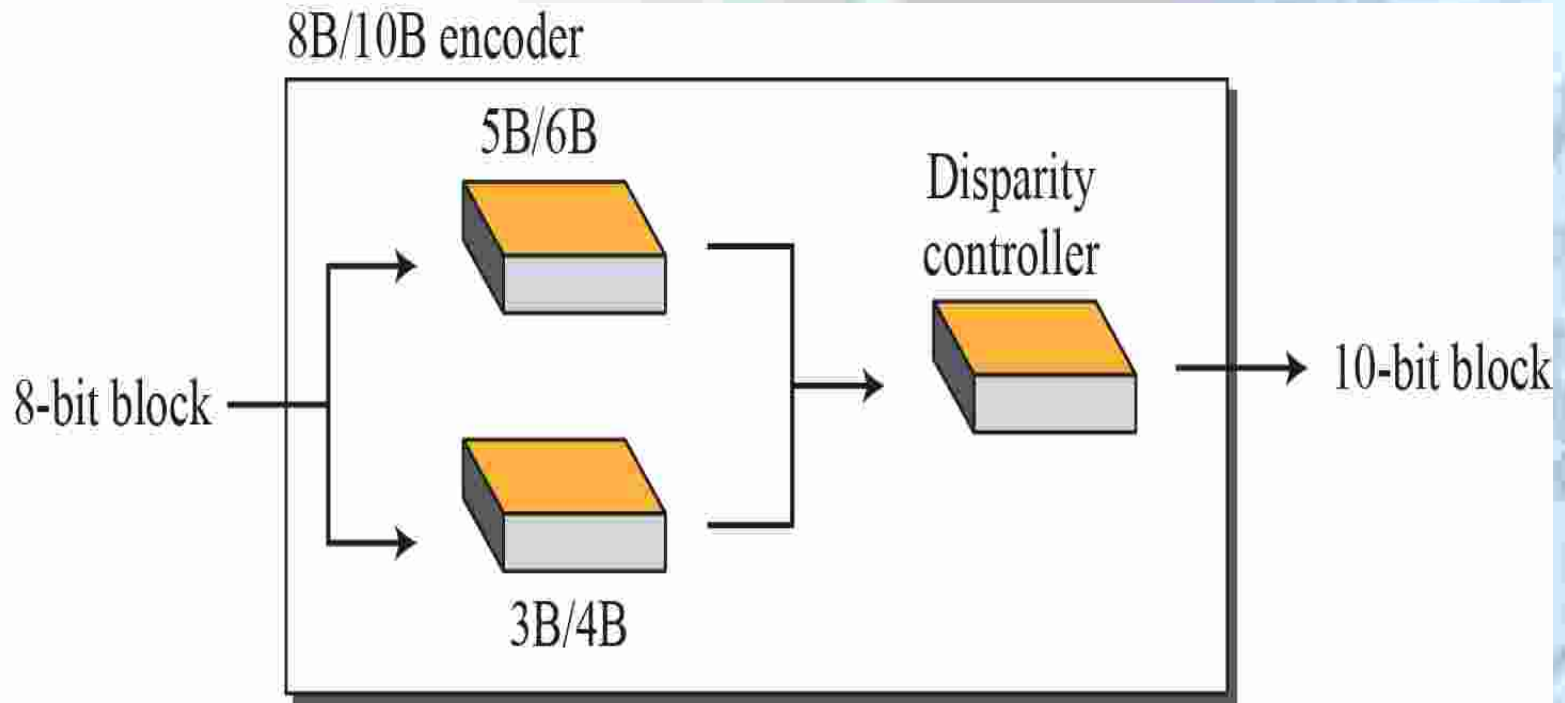
# Example

**We need to send data at a 1-Mbps rate. What is the minimum required bandwidth, using a combination of 4B/5B and NRZ-I or Manchester coding?**

# Example



# 8B/10B block encoding



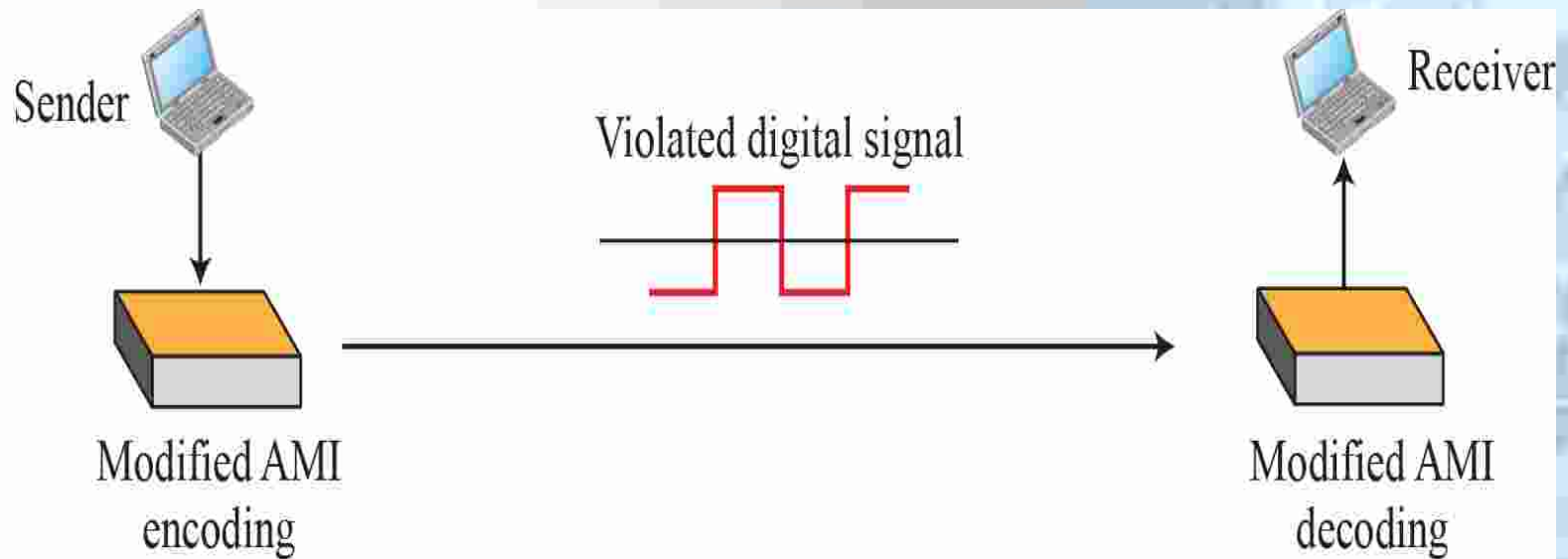
# Scrambling

- Biphase schemes suitable for LAN but not for Long Distance
- Block Coding + NRZ-I solves synch issue but has DC component
- Bipolar AMI has a narrow bandwidth (no DC Component) but synch issue (long series of 0s)

# Scrambling

- The system needs to insert the required pulses based on the defined scrambling rules

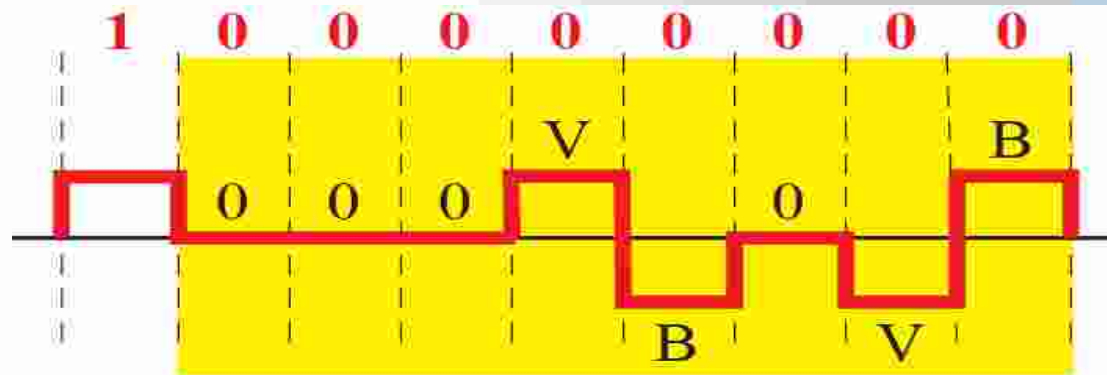
# AMI used with scrambling



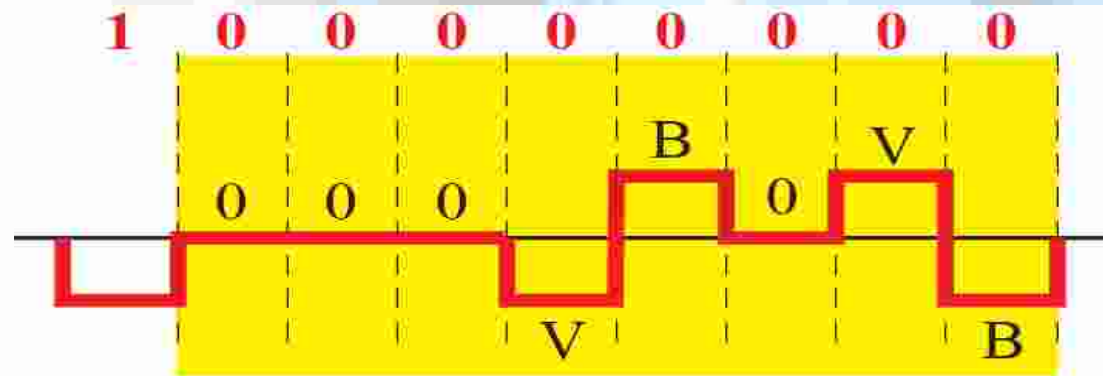
# Types of Scrambling Techniques

- Two common scrambling techniques are B8ZS and HDB3
- Bipolar with 8-Zero Substitution (B8ZS)
- High-density bipolar 3-zero (HDB3)

## Two cases of B8ZS scrambling technique



a. Previous level is positive.



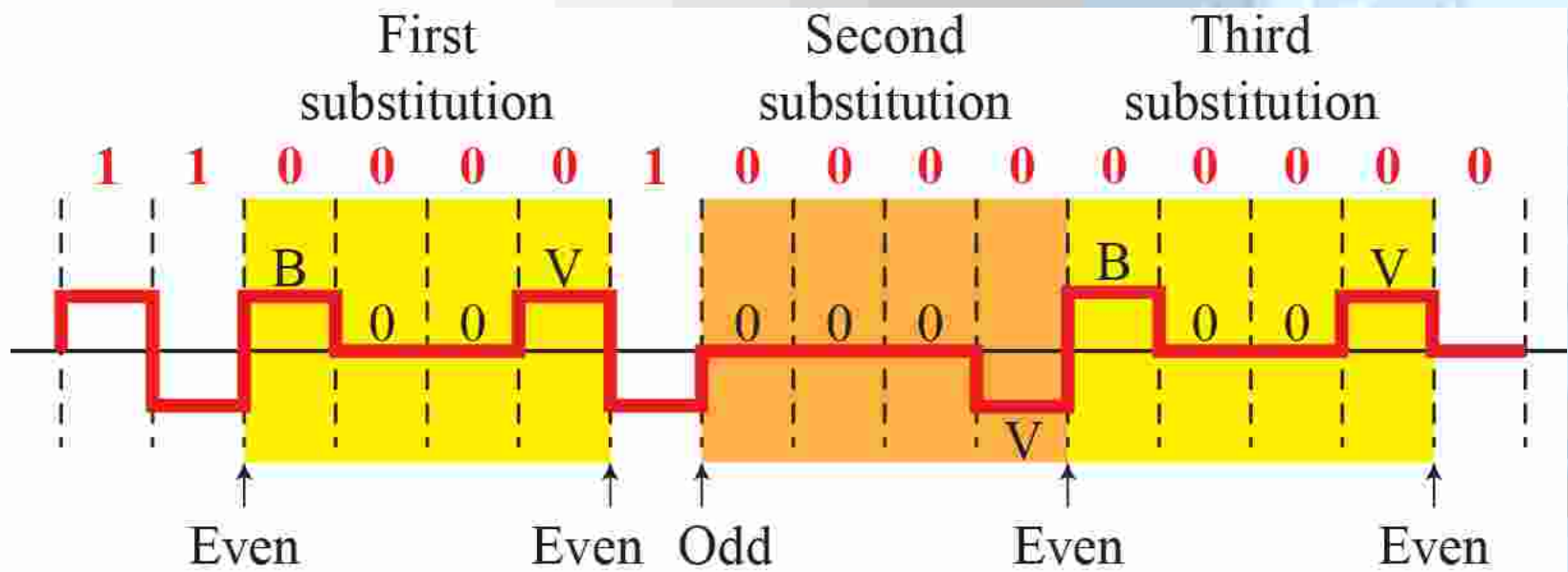
b. Previous level is negative.



# Types of Scrambling Techniques

- Two common scrambling techniques are B8ZS and HDB3
- Bipolar with 8-Zero Substitution (B8ZS)
- High-density bipolar 3-zero (HDB3)

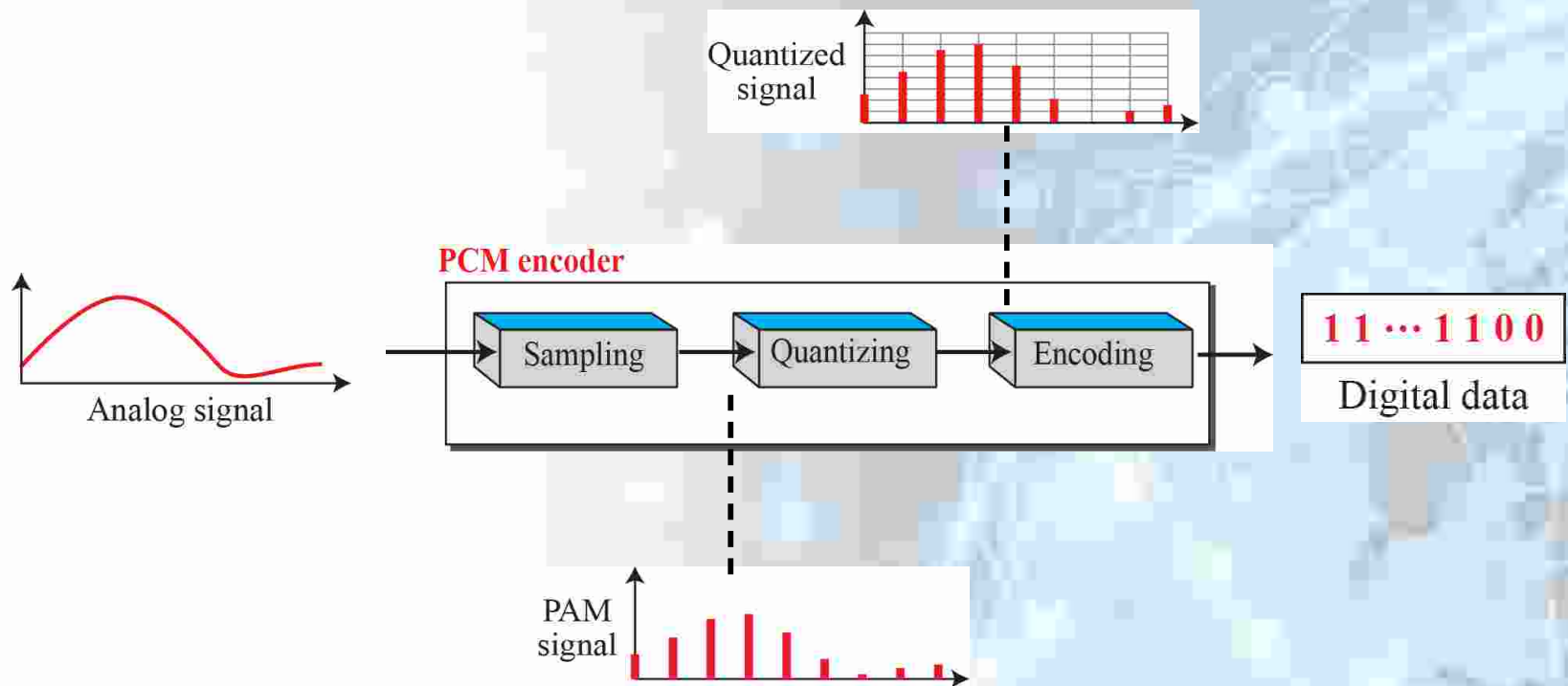
# Different situations in HDB3 scrambling technique



# Analog-to-digital Conversion

- **Analog Data to Digital Data**
- **Process of Digitization**
- **Two techniques:**
  - ✓ **Pulse Code Modulation (PCM)**
  - ✓ **Delta Modulation (DM)**

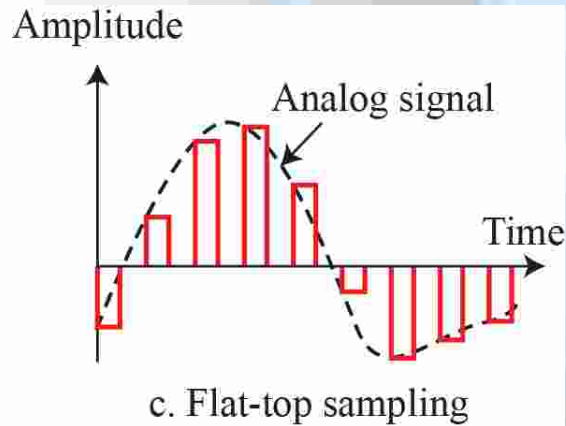
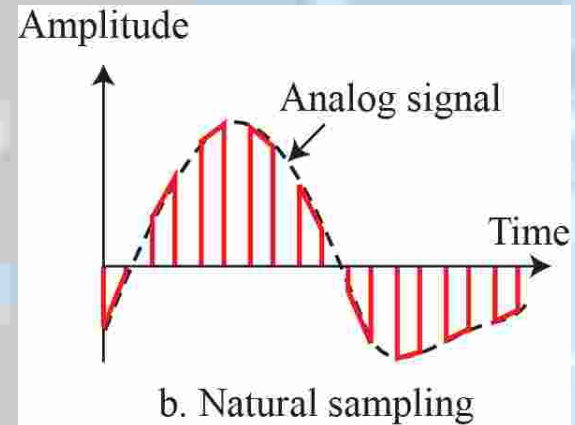
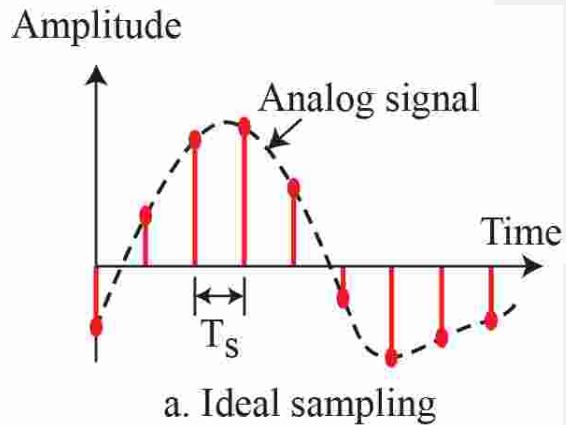
# Pulse Code Modulation (PCM)



# Pulse Code Modulation (PCM)

- Sampling
- Quantization
- Encoding

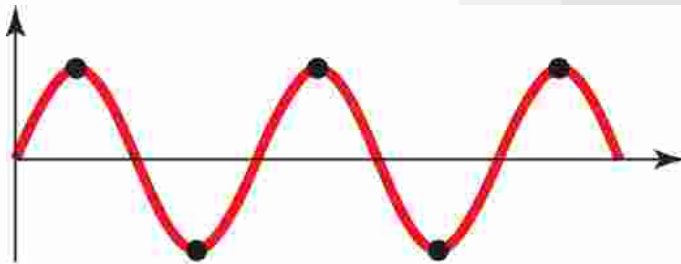
# Three different sampling methods for PCM



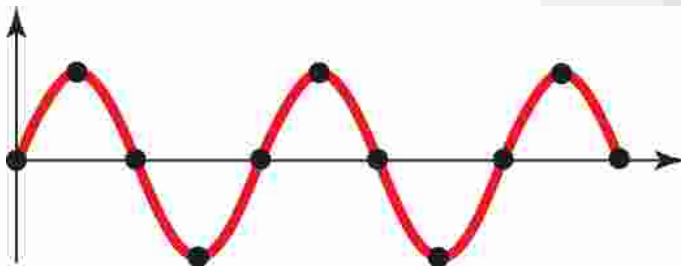
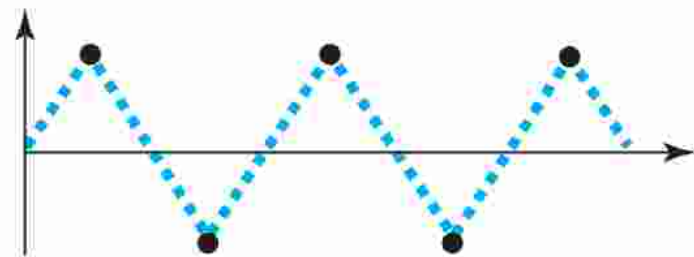
# Nyquist Sampling Rate

- Nyquist  $\rightarrow f_s = 2f_h$
- Sampling sine wave at three sampling rates:
  - ✓  $f_s = 4f$  (2 times the Nyquist rate)
  - ✓  $f_s = 2f$  (Nyquist rate)
  - ✓  $f_s = f$  (one-half the Nyquist rate)

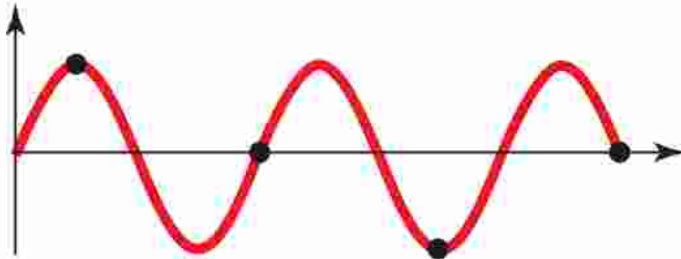
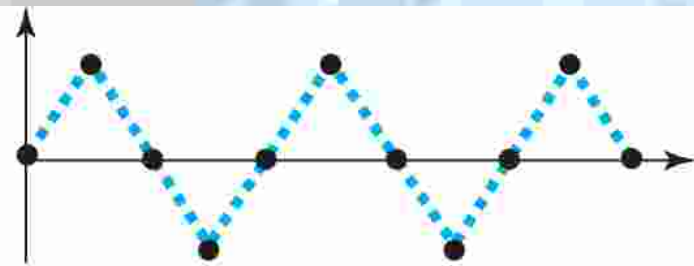
# Nyquist Sampling Rate



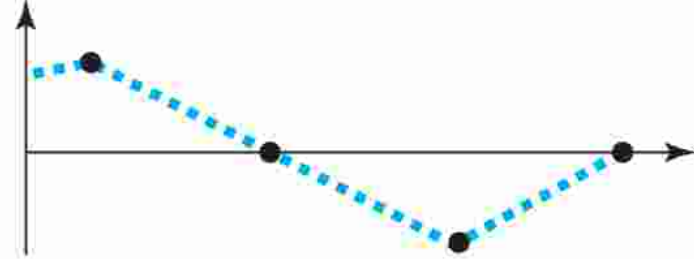
a. Nyquist rate sampling:  $f_s = 2 f$



b. Oversampling:  $f_s = 4 f$



c. Undersampling:  $f_s = f$

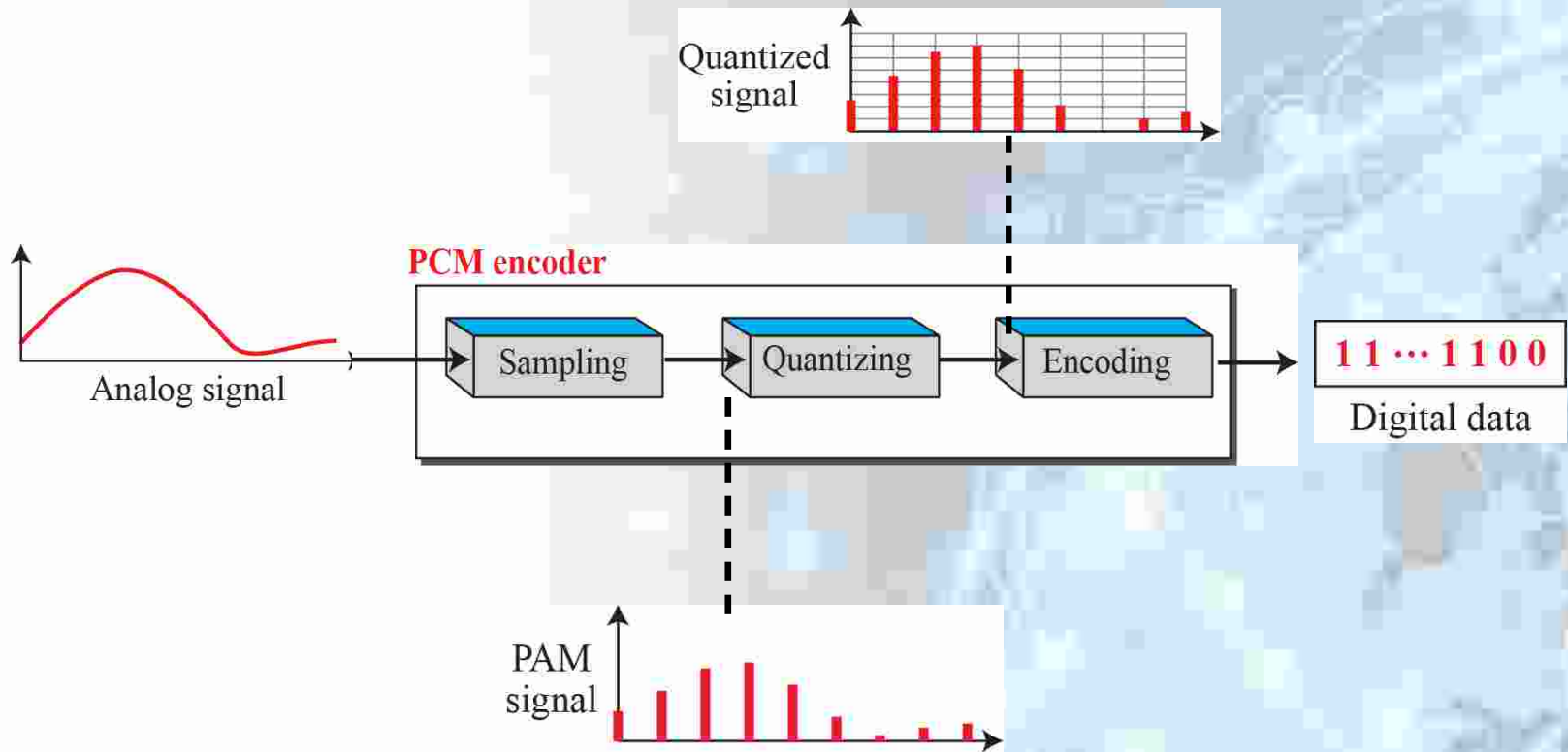




# Pulse Code Modulation (PCM)

- Most common technique
- Employs a PCM Encoder
- A PCM encoder has three processes:
  - ✓ Sampling
  - ✓ Quantization
  - ✓ Encoding

# Components of PCM encoder



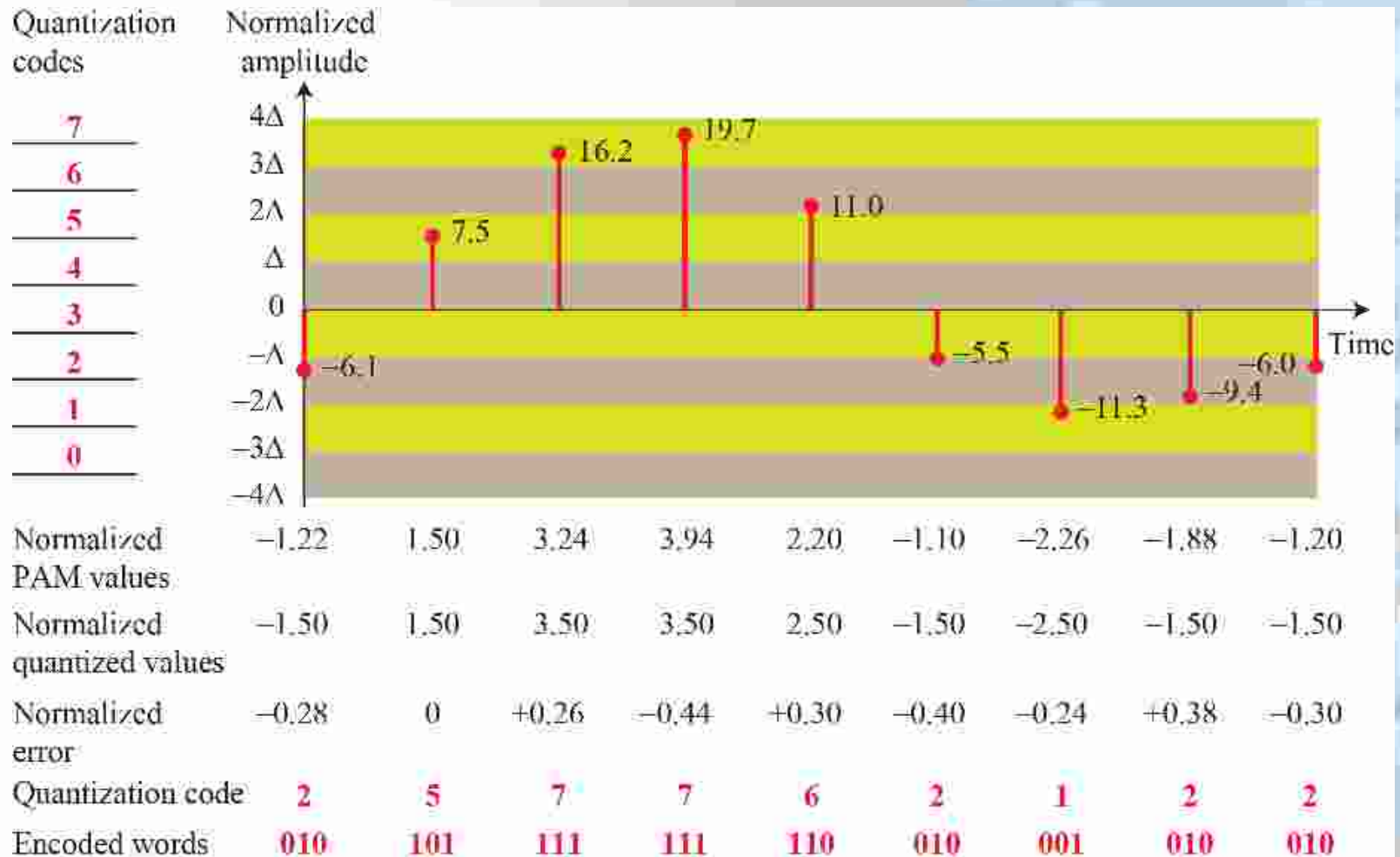
# Pulse Code Modulation (PCM)

- Sampling
- Quantization
- Encoding

# Quantization & encoding of a sampled signal

- Sampling → Series of pulses with amplitude values between min and max signal amplitude
- Infinite set with non-integral values not suitable for encoding
- We quantize the sampling output into certain levels based on range of amplitudes and how much accuracy is needed

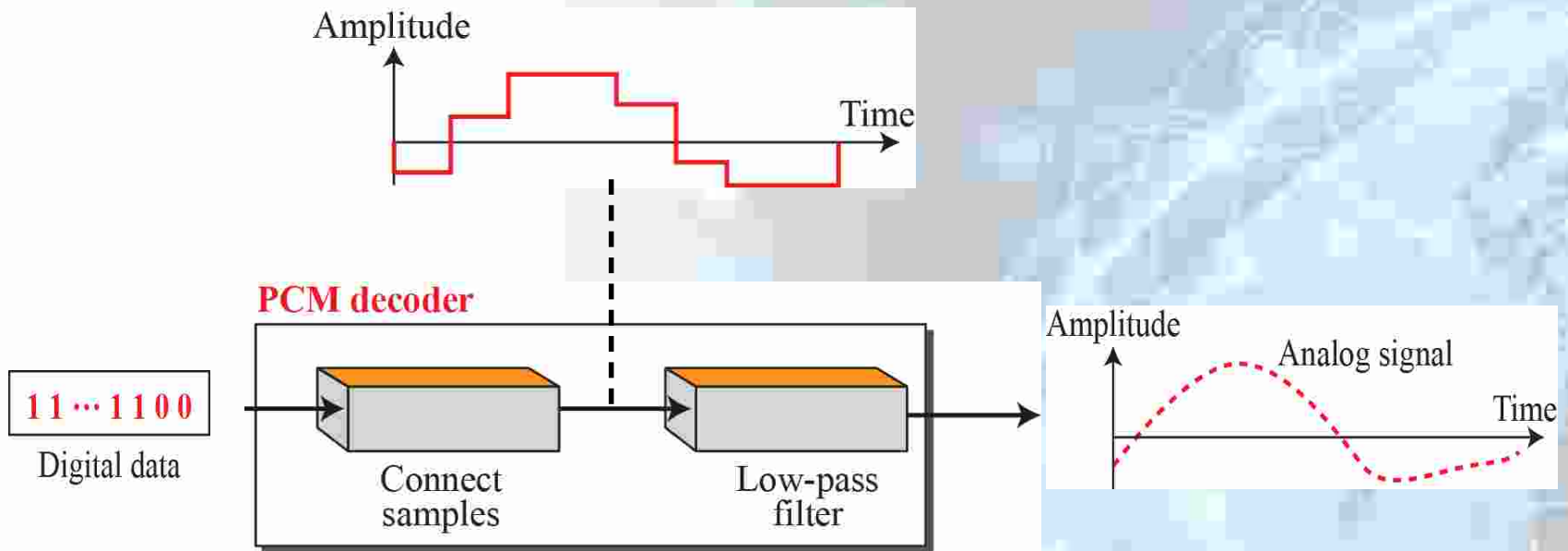
# Quantization & encoding of a sampled signal



# Pulse Code Modulation (PCM)

- Encoding
  - ✓ Sampling
  - ✓ Quantization
  - ✓ Encoding
- Decoding

# Original Signal Recovery- PCM Decoder



# Analog-to-digital Conversion

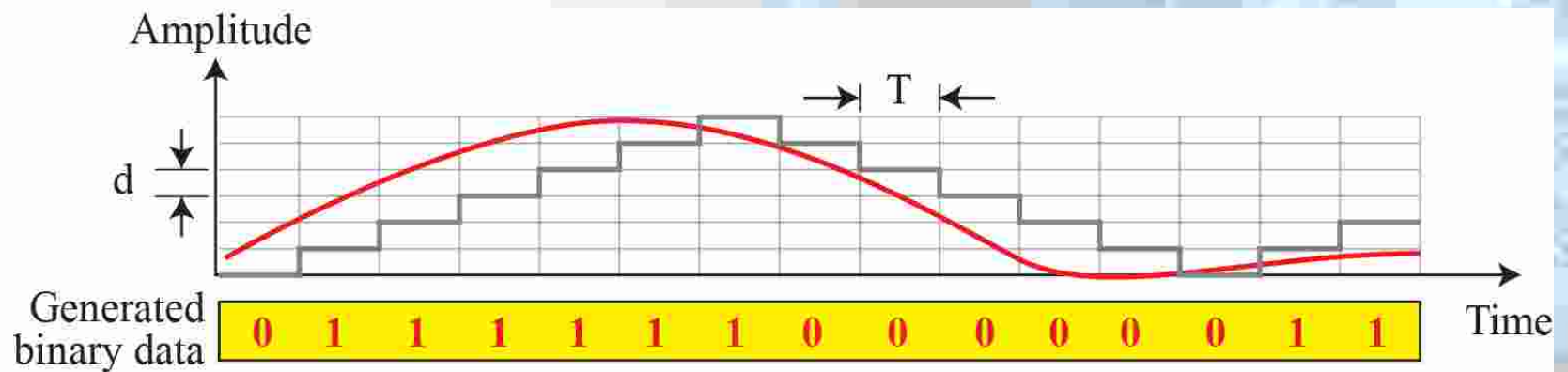
- **Analog Data to Digital Data**
- **Process of Digitization**
- **Two techniques:**
  - ✓ **Pulse Code Modulation (PCM)**
  - ✓ **Delta Modulation (DM)**



# Delta Modulation (DM)

- PCM is a very complex technique
- Delta modulation is a simpler technique
- PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample
- No code words

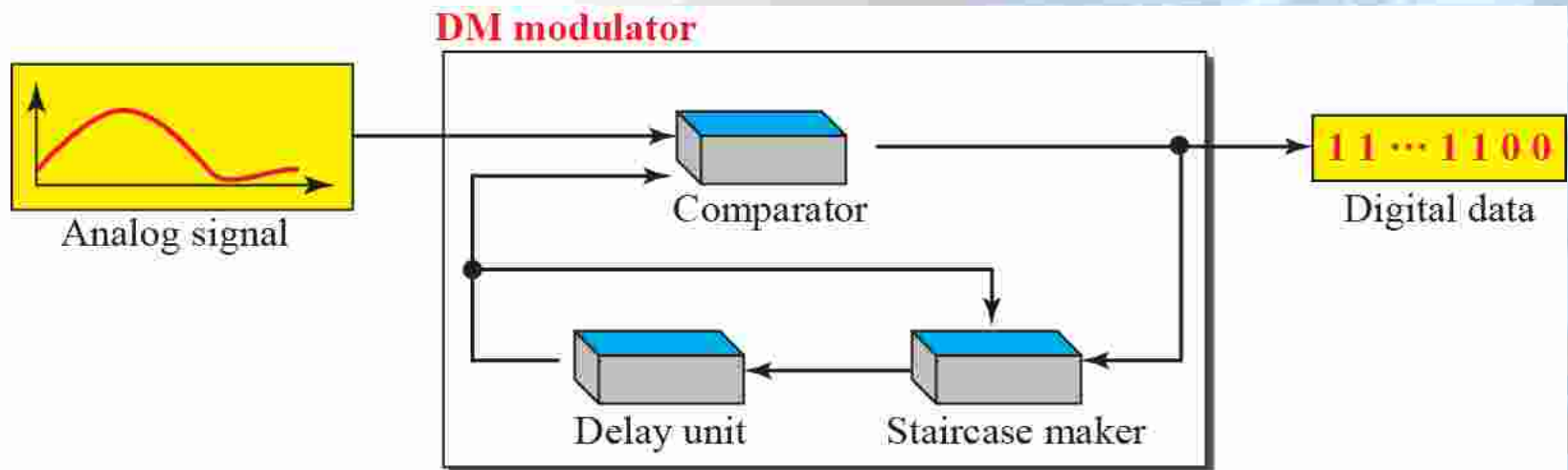
# The process of delta modulation



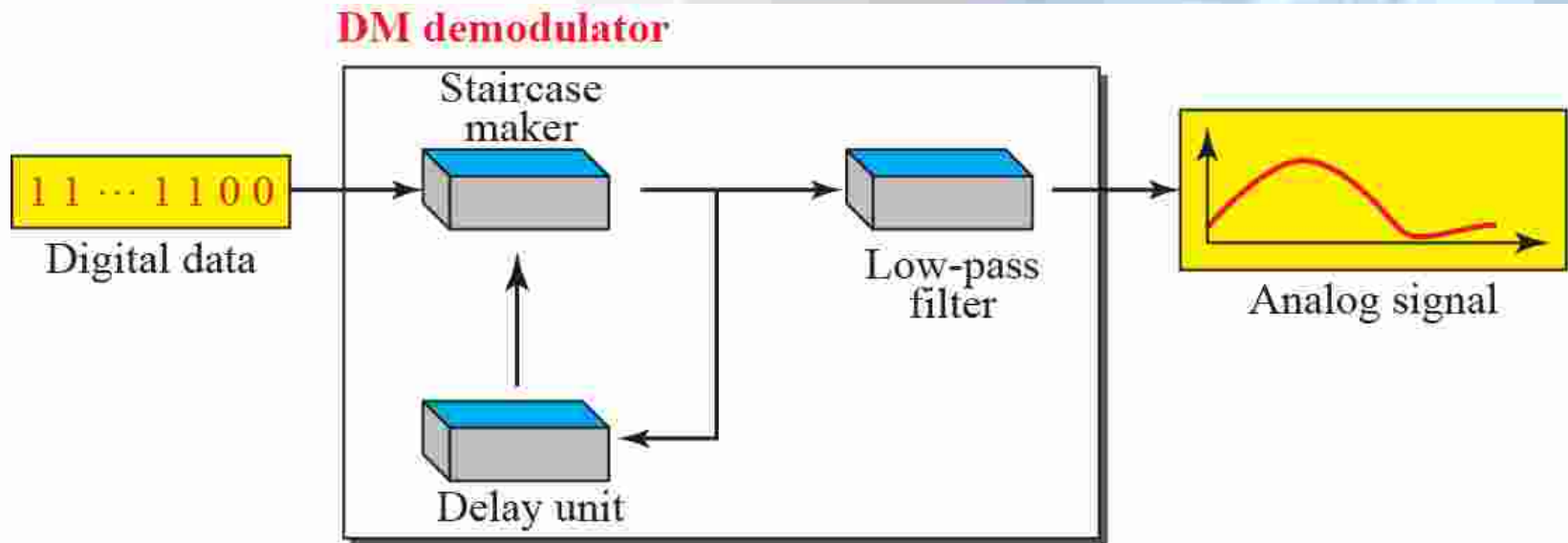
# Delta Modulation (DM)

- Delta modulation is a simpler technique
- DM finds the change from the previous sample
- No code words

# Delta Modulation Components



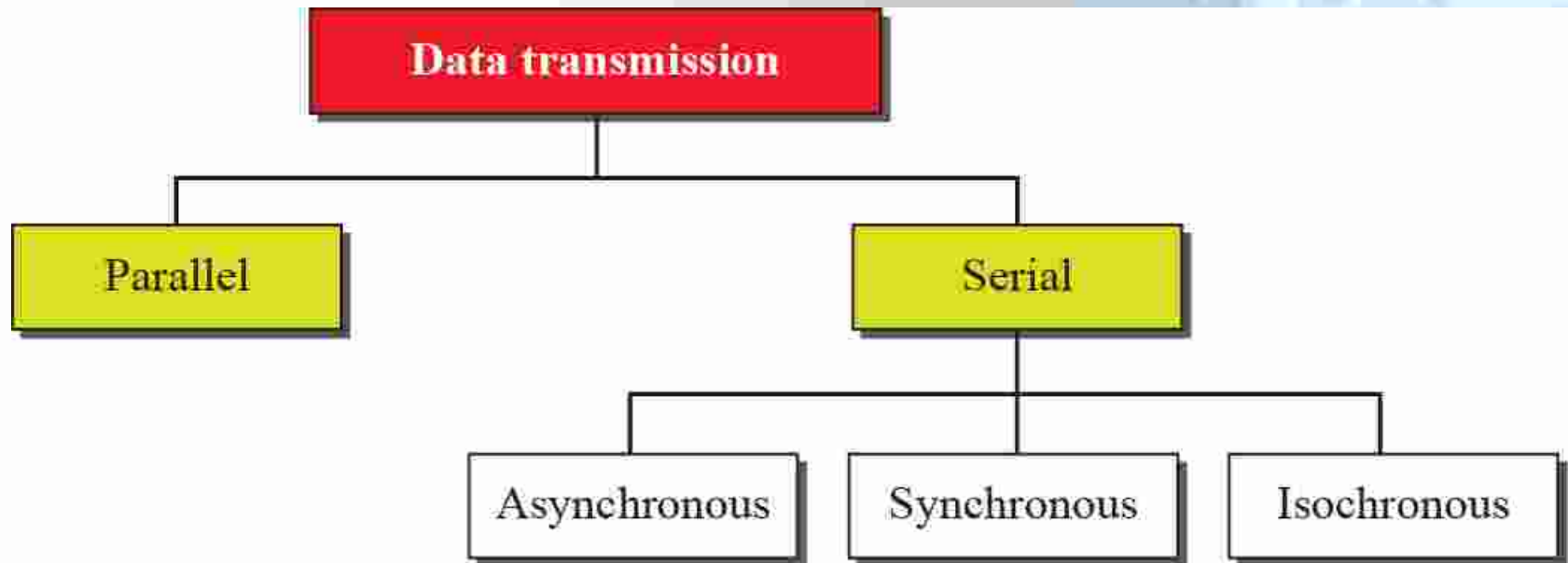
# Delta Demodulation Components



# Transmission Modes

- **Transmission of Data:**
  - ✓ **Wiring**
    - **Data Stream**
- **Do we send 1 bit at a time; or do we group bits into larger groups and, if so, how?**
- **Parallel or Serial Transmission**

# Data transmission modes

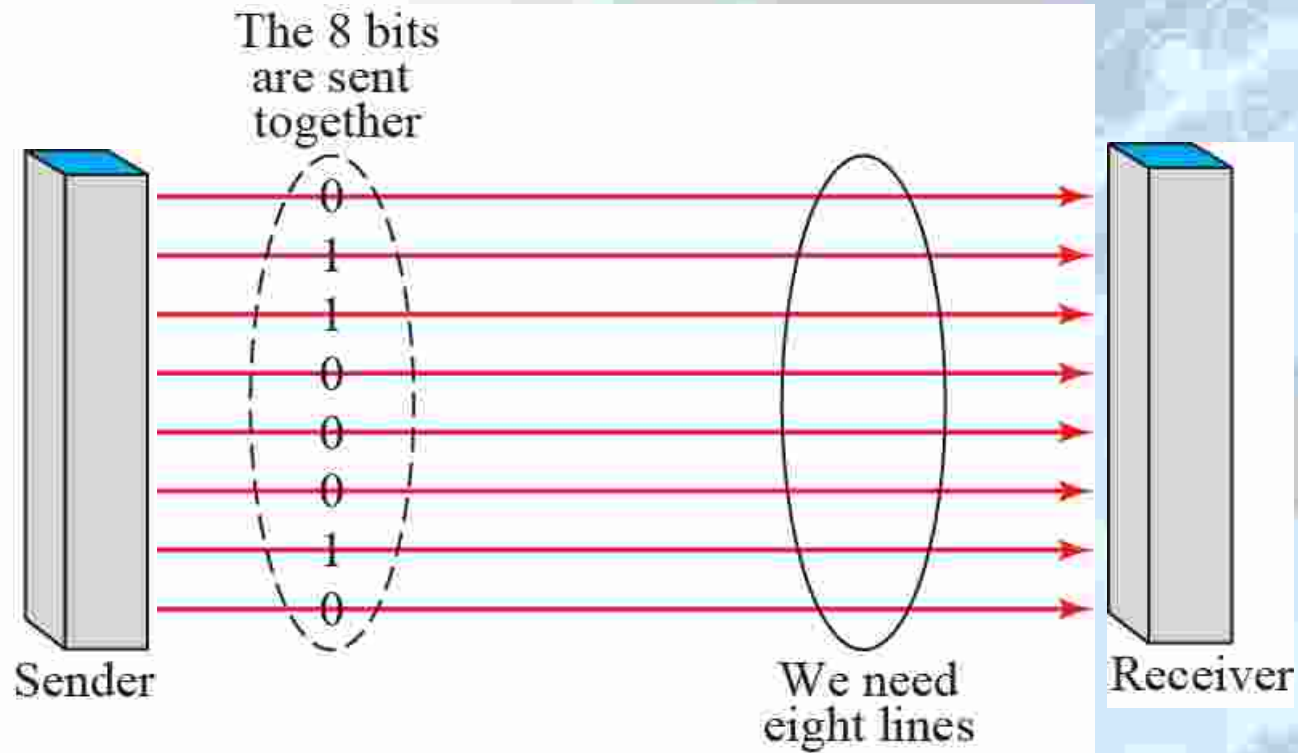


# Parallel Transmission

- Binary data (1s and 0s) organized in groups of 'n' bits
- We send 'n' bits at a time instead of just one
- 'n' wires required to send 'n' bits at one time



# Parallel Transmission

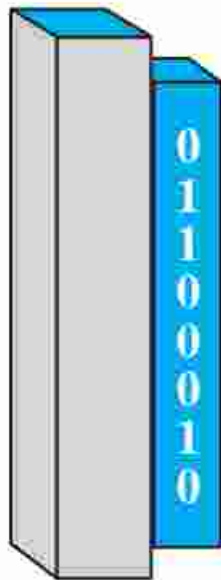


# Serial Transmission

- In serial transmission one bit follows another
- Only one communication channel rather than 'n' to transmit data

# Serial Transmission

Parallel/serial  
converter



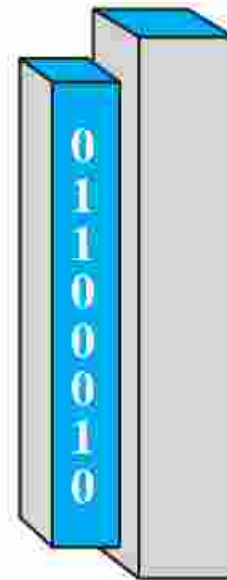
Sender

The 8 bits are sent  
one after another.

0 1 1 0 0 0 1 0

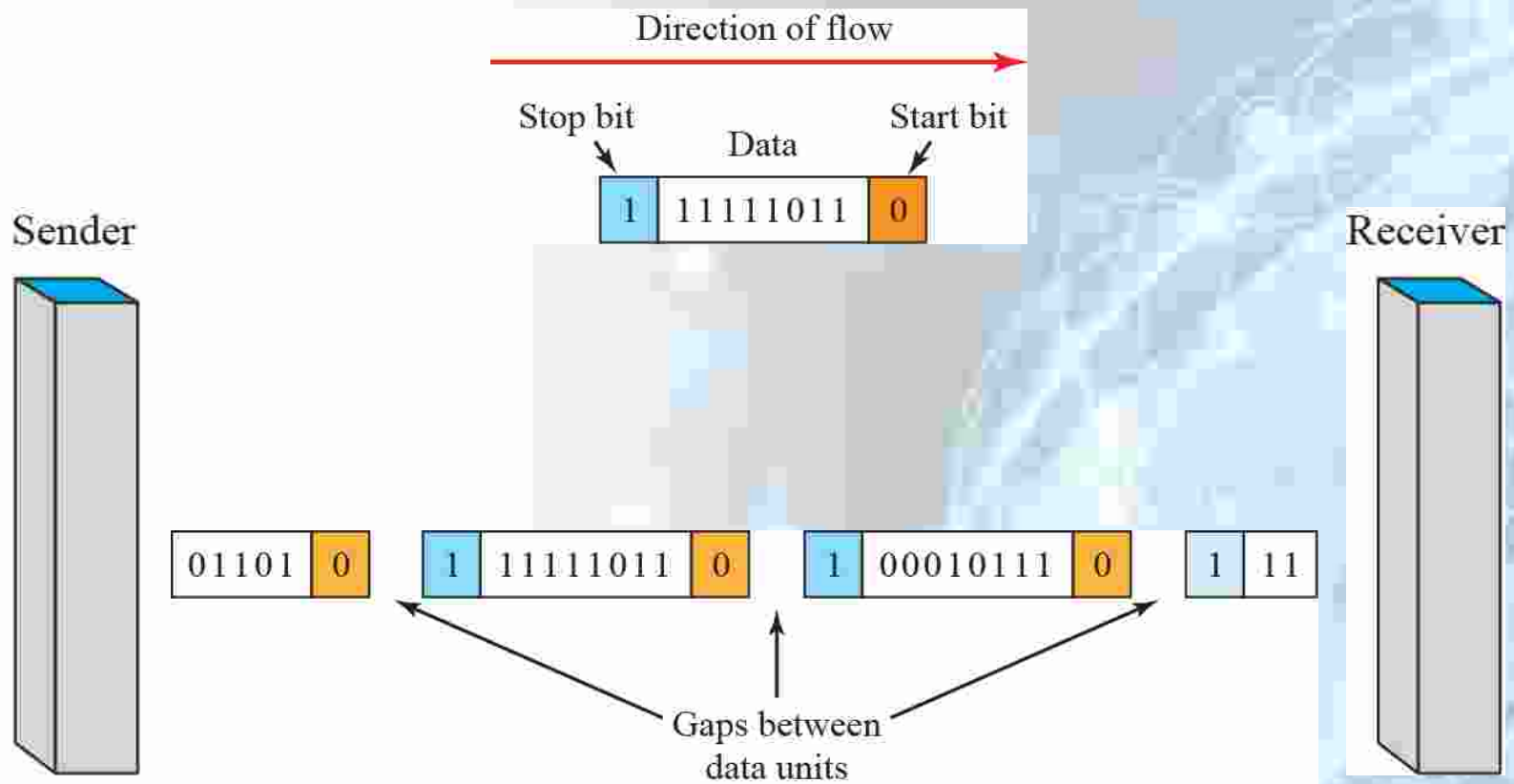
We need only  
one line (wire).

Serial/parallel  
converter



Receiver

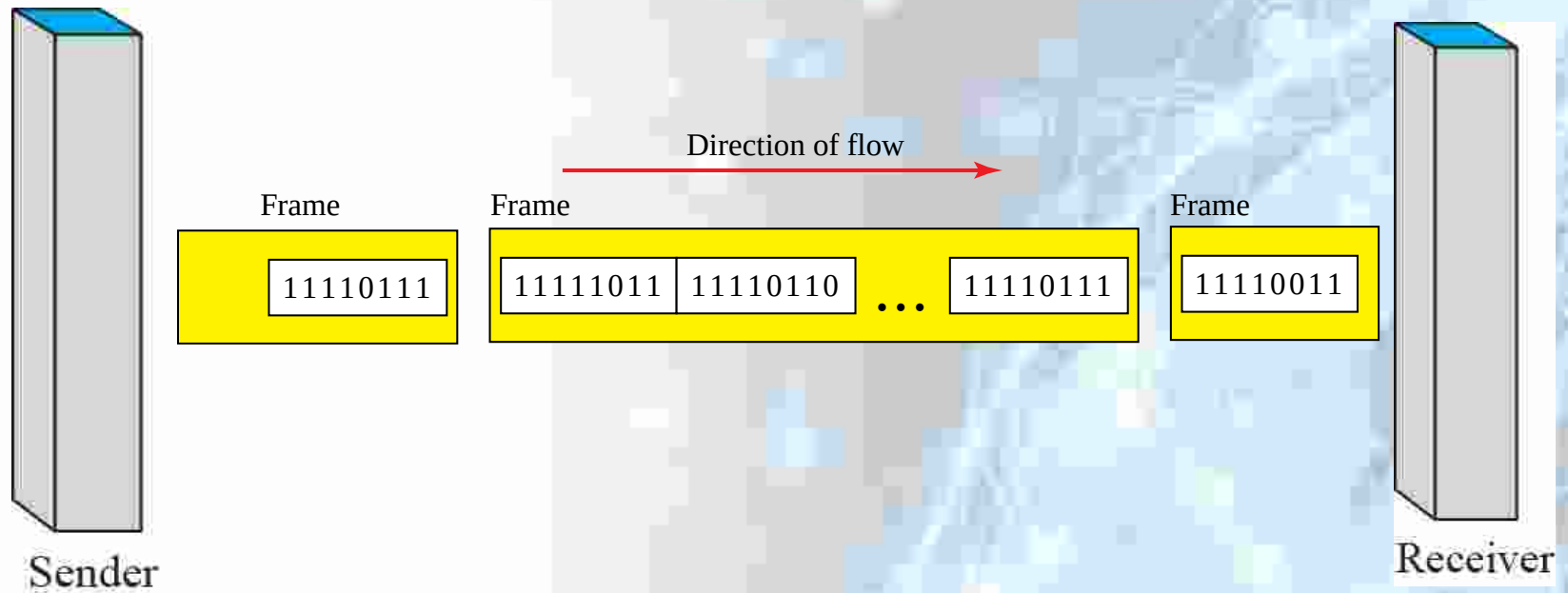
# Asynchronous Transmission



# Serial Transmission

- In serial transmission one bit follows another
- Only one communication channel rather than 'n' to transmit data

# Synchronous Transmission



# Isochronous Transmission

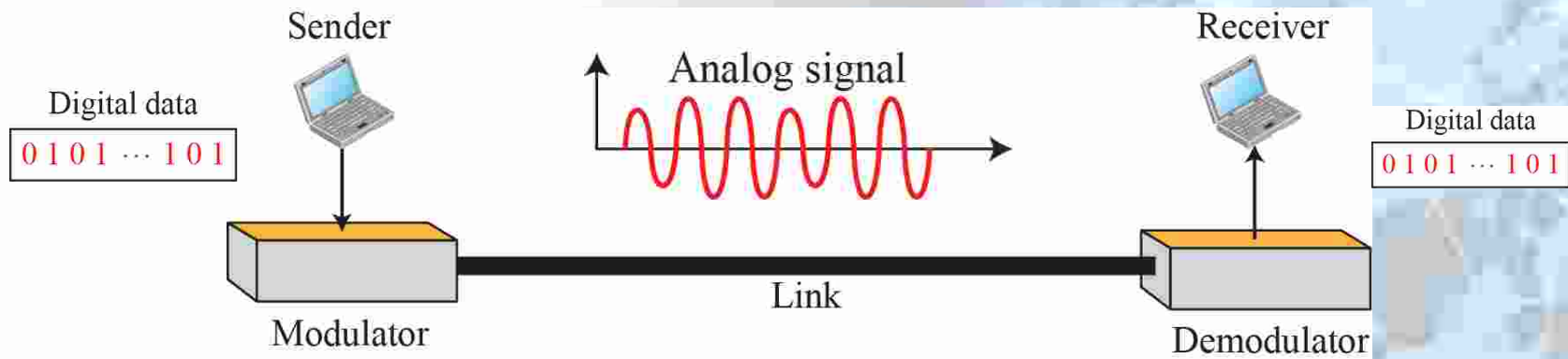
- Real time Audio and Video
- Synchronization between characters is not enough
- Entire stream should be synchronized
- Isochronous guarantees fixed rate data

# Digital-to-Analog Conversion

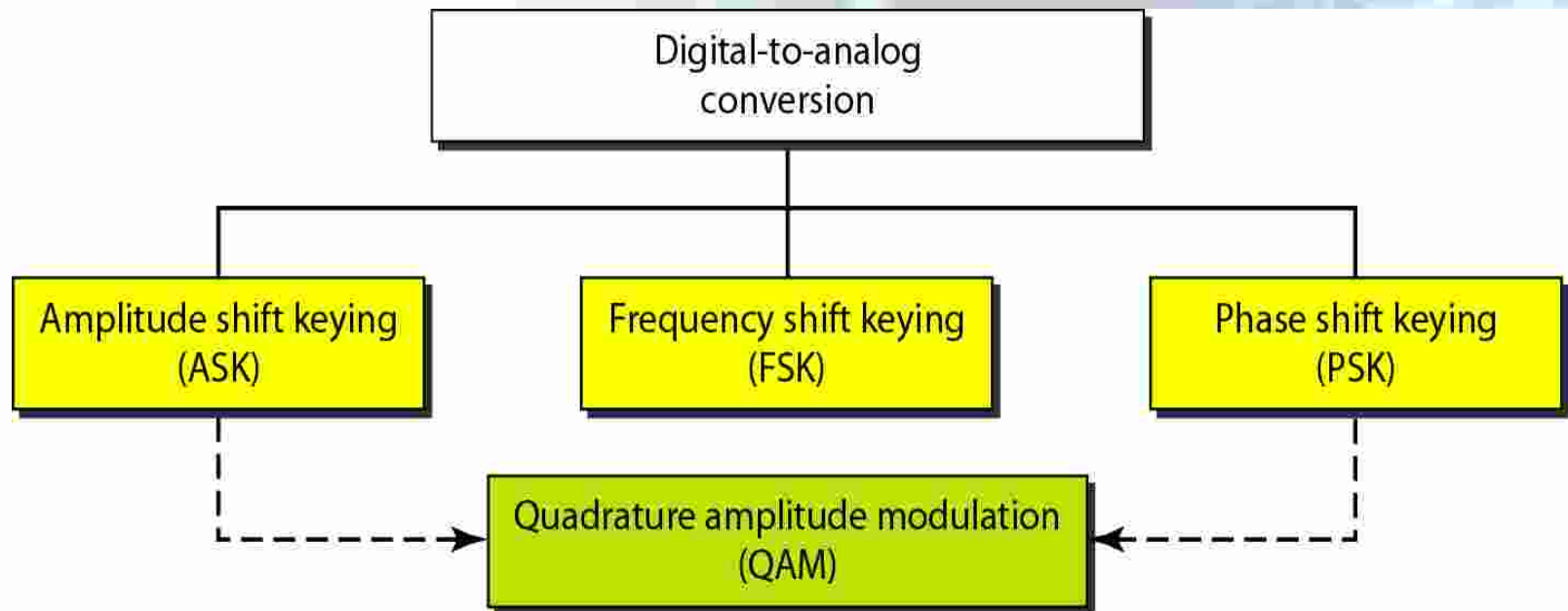
- Process of changing one of the characteristics of analog signal based on the information in digital data
- A sine wave is defined by 3 characteristics:
  - ✓ Amplitude
  - ✓ Frequency
  - ✓ Phase
- By changing one of these characteristics, we can use it to represent a digital signal



# Digital-to-Analog Conversion



# Types of Digital to Analog Conversion



# Aspects of Digital to Analog Conversion

- Before we discuss specific methods of digital-to-analog modulation, two basic issues must be reviewed:
  - ✓ Bit and Baud rates and
  - ✓ The Carrier Signal

# Aspects of Digital to Analog Conversion

- In Analog Transmission of Digital Data, Baud Rate is less than or equal to the Bit Rate
  - ✓ Data Element vs. Signal Element
  - ✓ Data Rate vs. Signal Rate
- Bandwidth Required  $\propto$  Signal Rate (except FSK)
- Carrier Signal

# Aspects of Digital to Analog Conversion

- Before we discuss specific methods of digital-to-analog modulation, two basic issues must be reviewed:
  - ✓ Bit and Baud rates and
  - ✓ The Carrier Signal

## Example

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate?

# Example

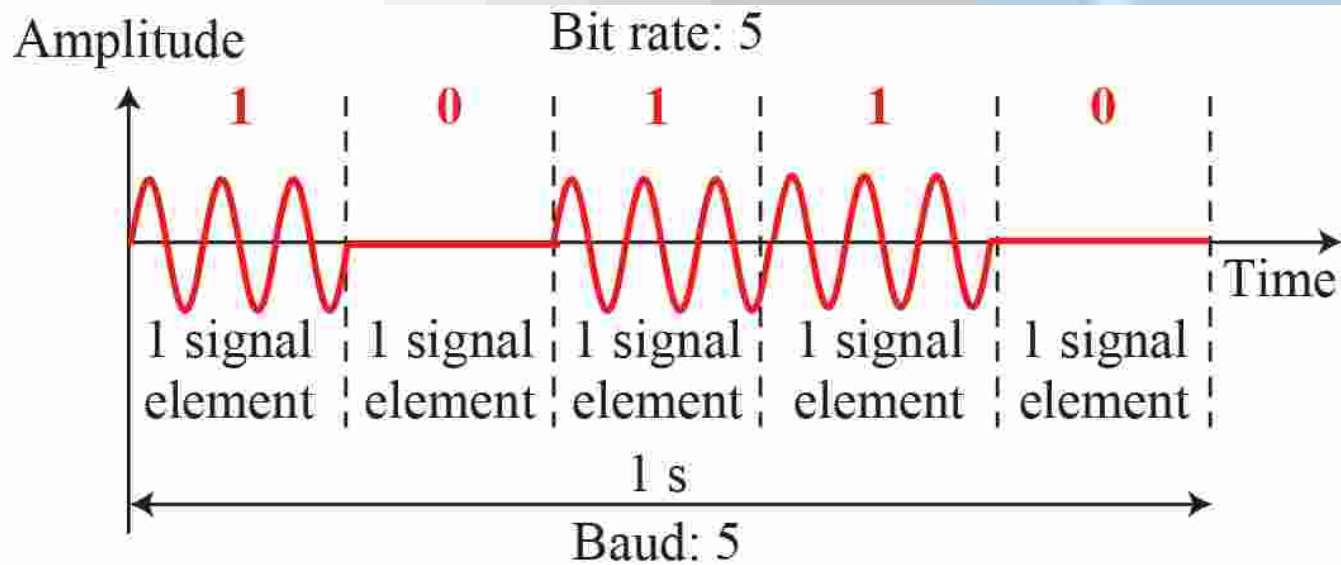
**An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?**

# Amplitude Shift Keying (ASK)

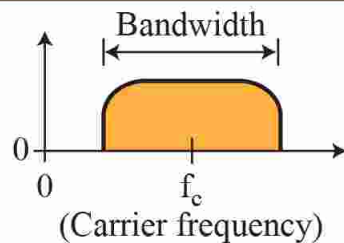
- The amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant while the amplitude changes
- Binary ASK or On-Off Keying (OOK)



# Binary Amplitude Shift Keying (Binary ASK)



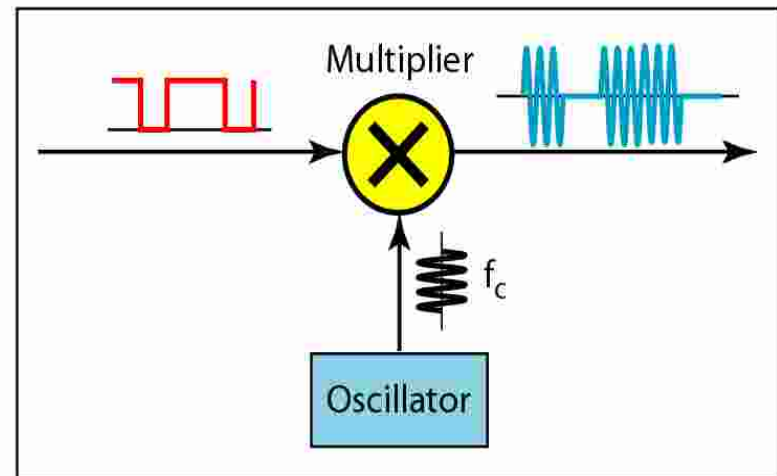
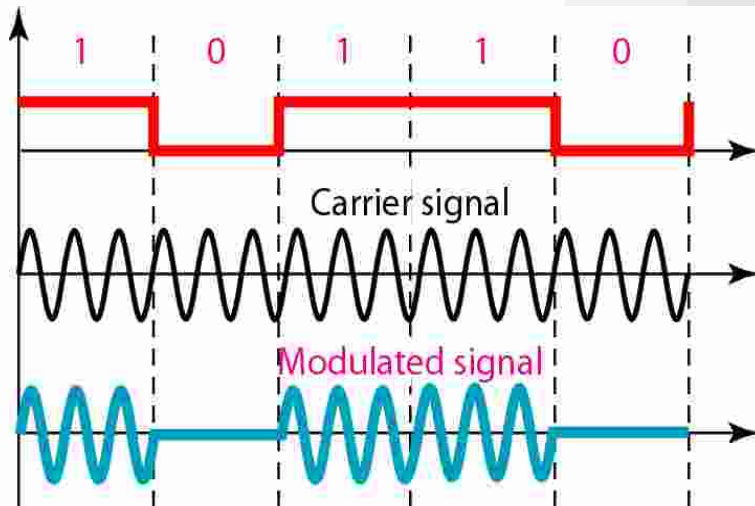
$$r = 1 \quad S = N \quad B = (1 + d)S$$



# Amplitude Shift Keying (ASK)

- The amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant while the amplitude changes
- Binary ASK or On-Off Keying (OOK)

# Implementation of Binary ASK



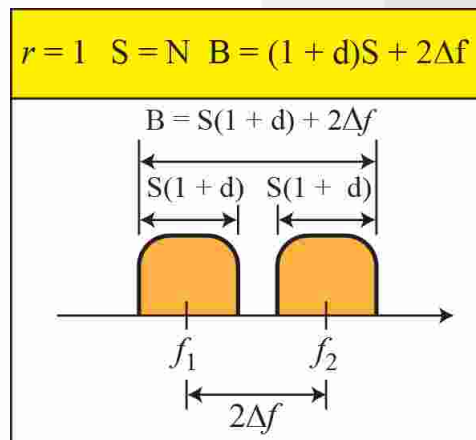
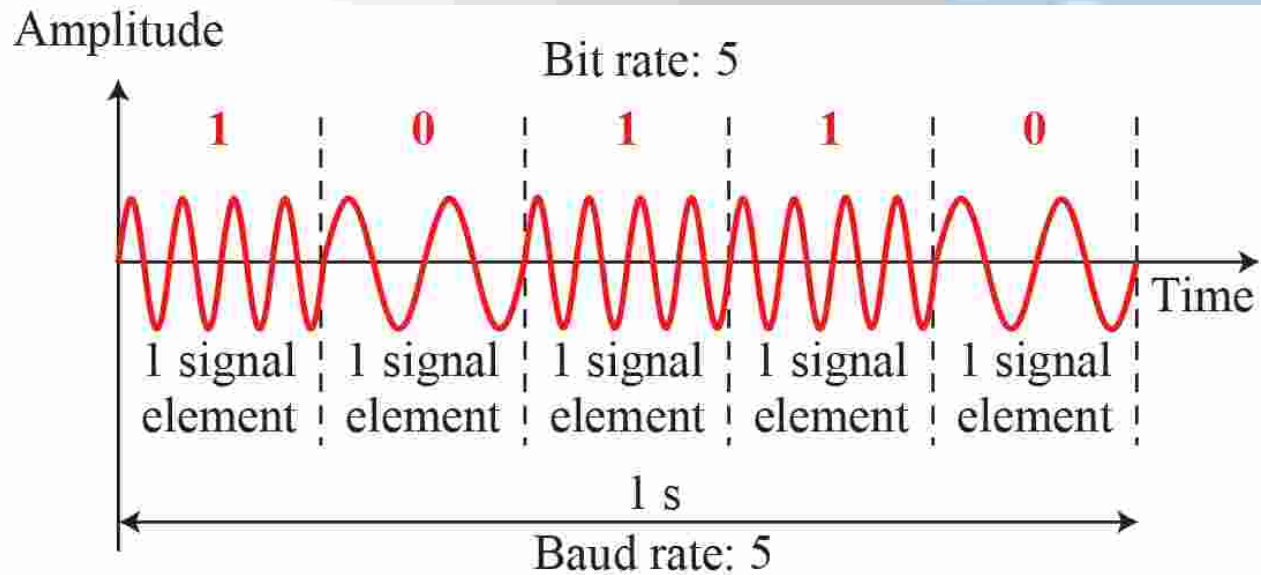
# Example

**We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with  $d = 1$ ?**

# Frequency Shift Keying (FSK)

- The frequency of the carrier signal is varied to represent data
- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes
- Both peak amplitude and phase remain constant

# Binary Frequency Shift Keying



# Frequency Shift Keying (FSK)

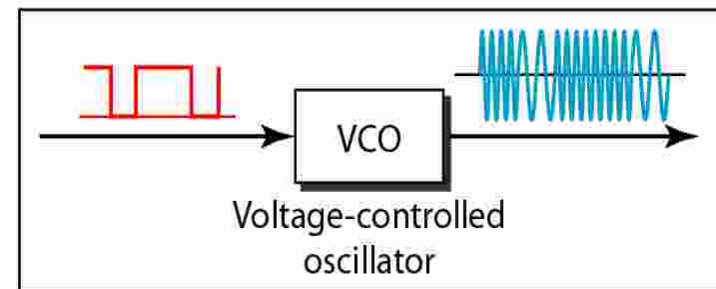
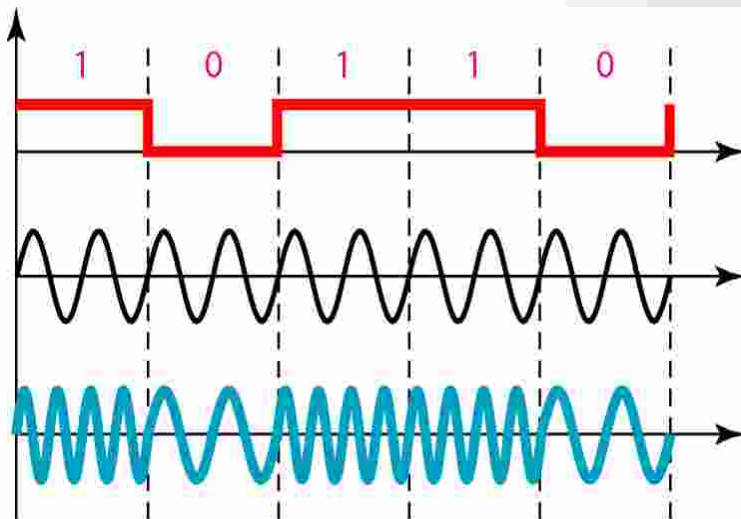
- The frequency of the carrier signal is varied to represent data
- Both peak amplitude and phase remain constant

## Example

We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with  $d = 1$ ?



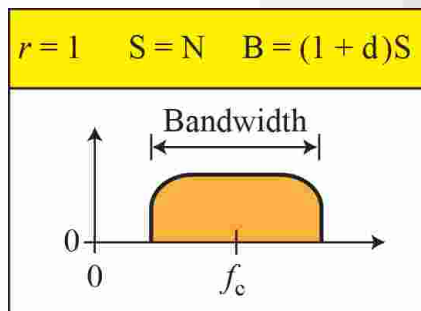
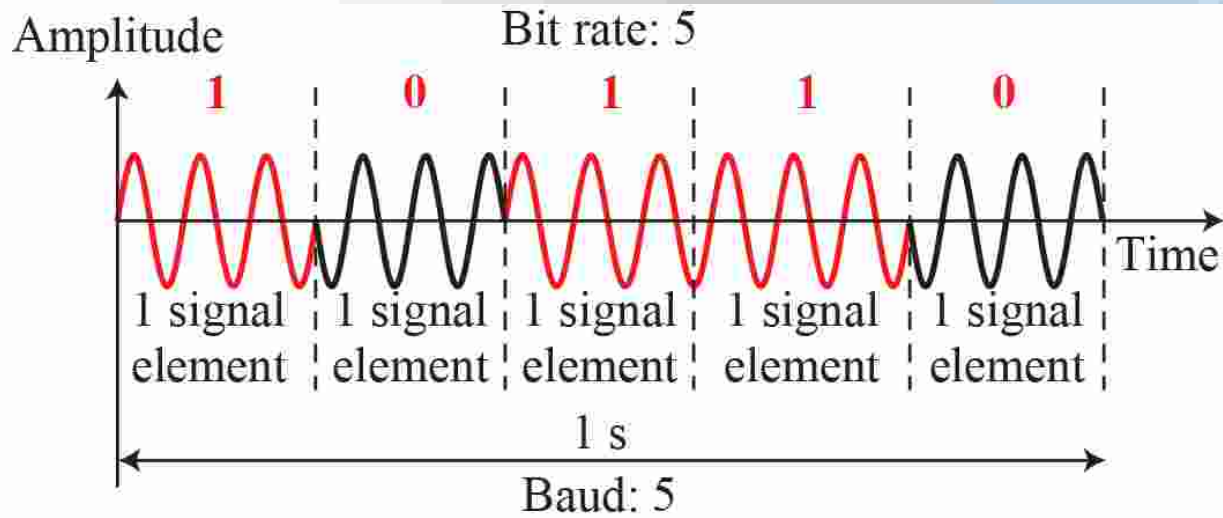
# Implementation of BFSK



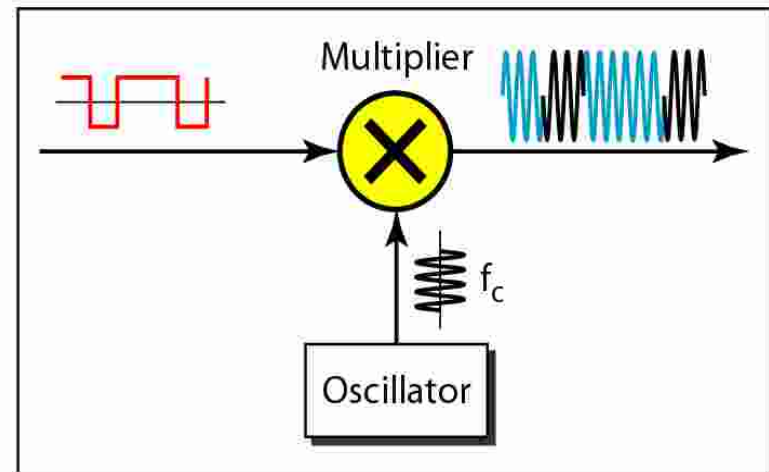
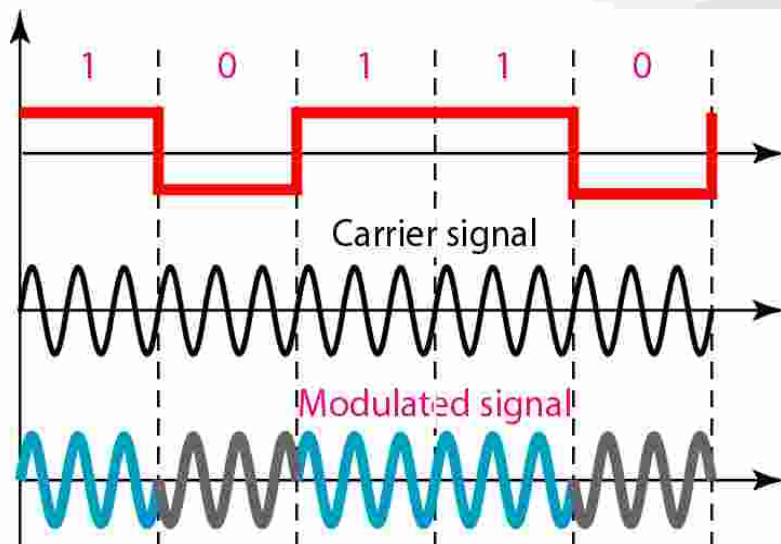
# Phase Shift Keying (PSK)

- The phase of the carrier is varied to represent two or more different signal elements
- Both peak amplitude and frequency remain constant
- PSK is relatively common than ASK or FSK

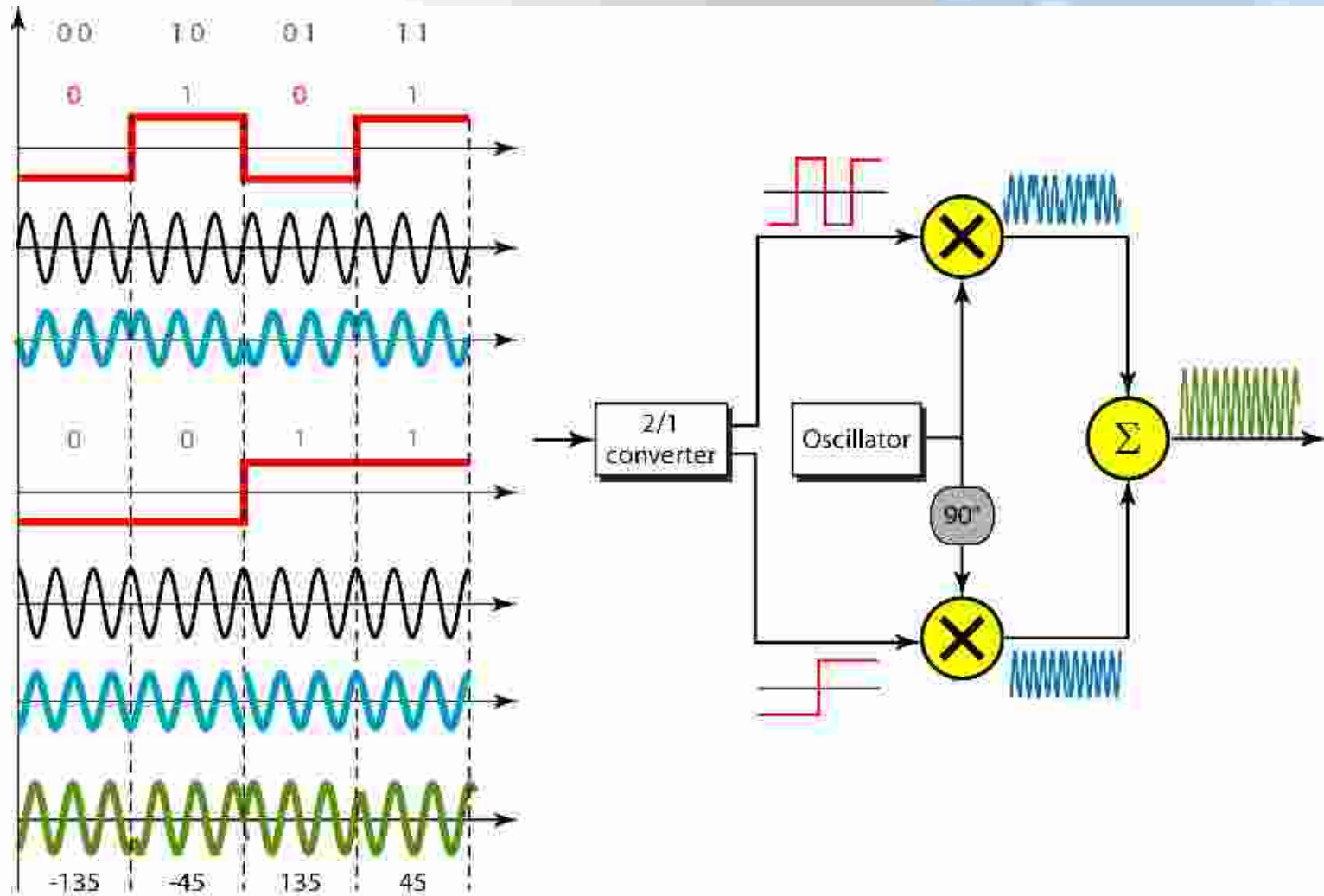
# Binary Phase Shift Keying



# Implementation of BPSK



# QPSK and its Implementation



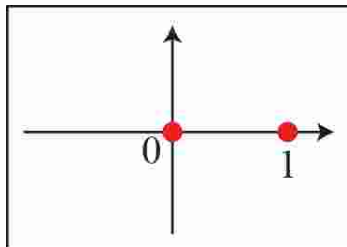
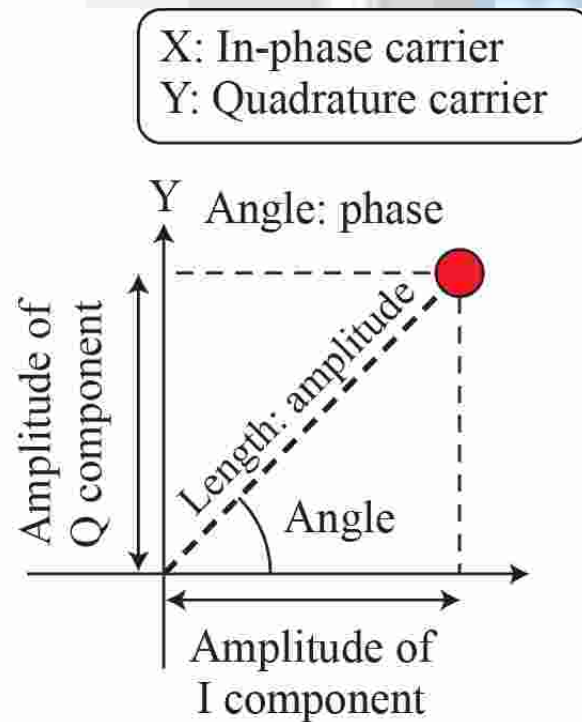
# Example

Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of  $d = 0$ .

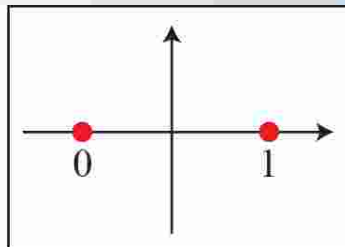
# Constellation Diagram

- Helps us define the phase and amplitude of a signal element when we are using two carriers (one in phase and other in quadrature)
- Signal element is represented as a dot

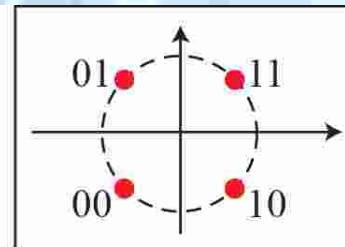
# Constellation Diagram



a. BASK (OOK)



b. BPSK



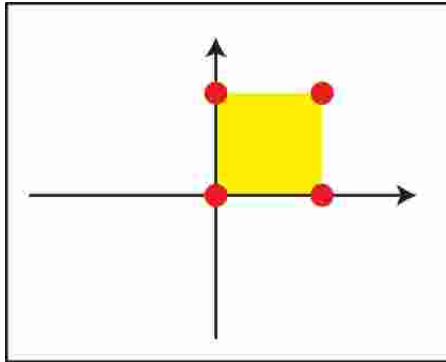
c. QPSK



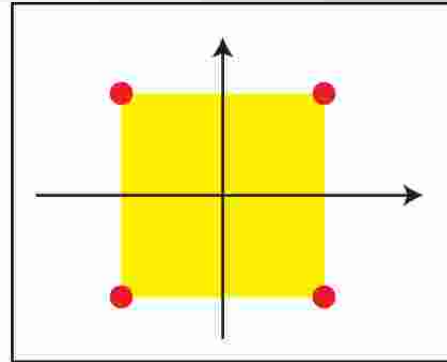
# Quadrature Amplitude Modulation (QAM)

- PSK is limited by the ability of the equipment to distinguish small differences in phase which limits its potential bit rate
- We have been altering only one of the three characteristics of a sine wave at a time; but what if we alter two?
- Why not combine ASK and PSK?

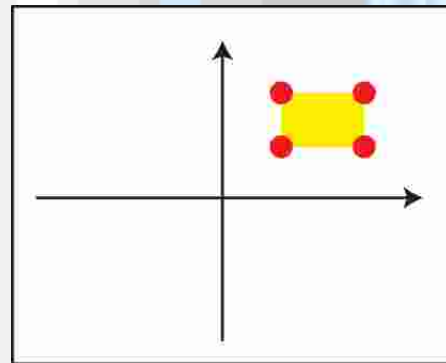
# Constellation diagrams for some QAMs



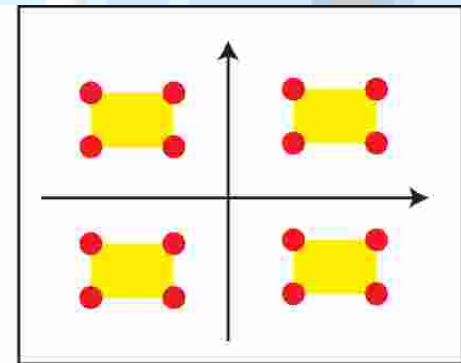
a. 4-QAM



b. 4-QAM



c. 4-QAM

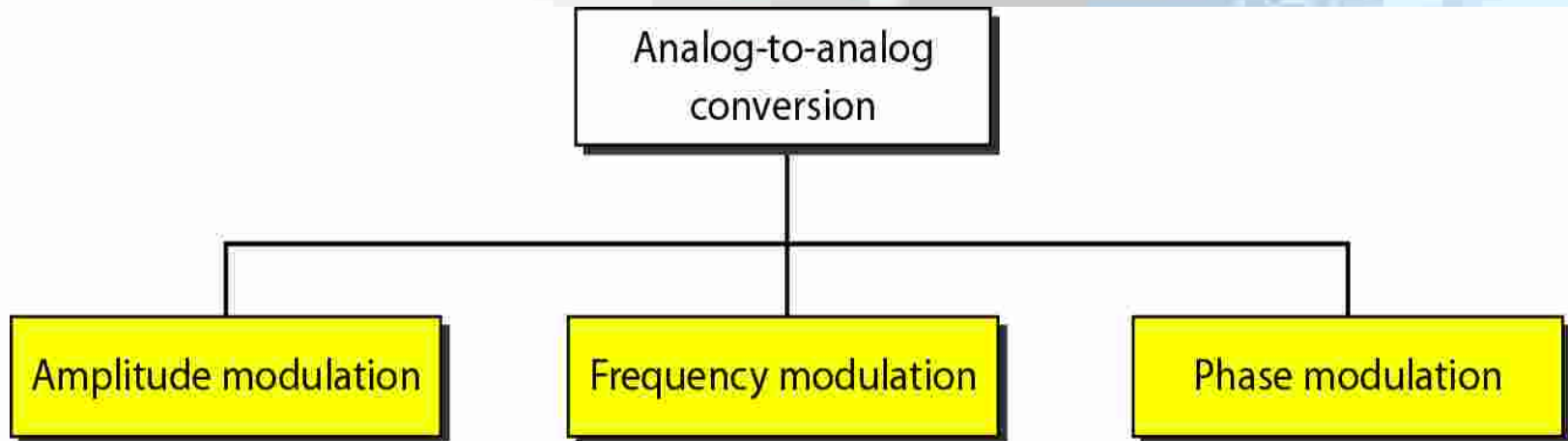


d. 16-QAM

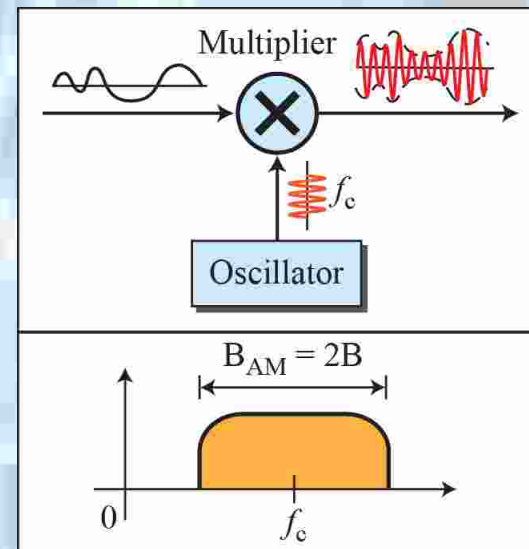
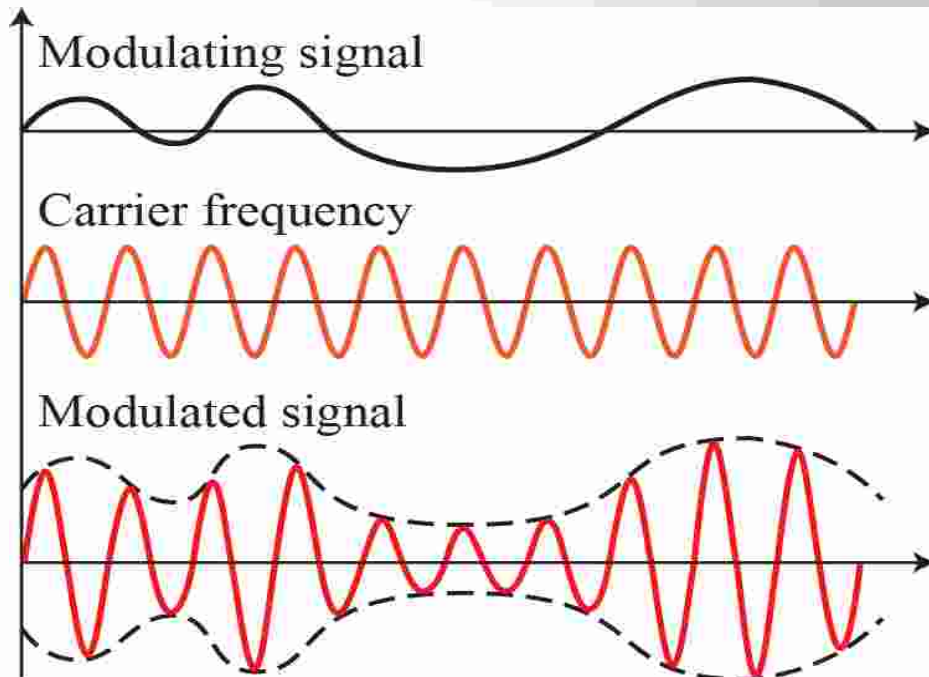
# Analog-to-Analog Conversion

- Representation of Analog information by an Analog signal
- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

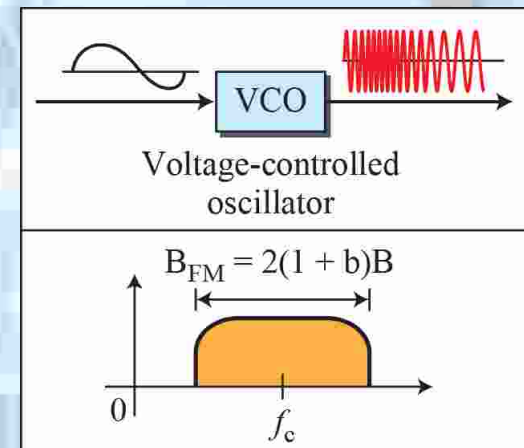
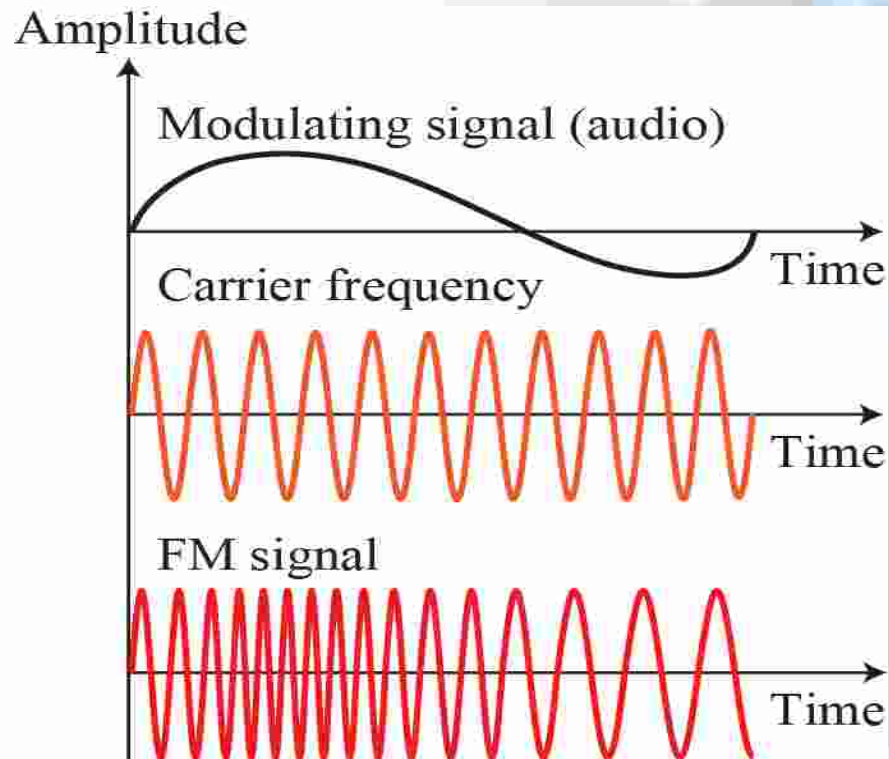
# Types of Analog-to-Analog Modulation



# Amplitude modulation



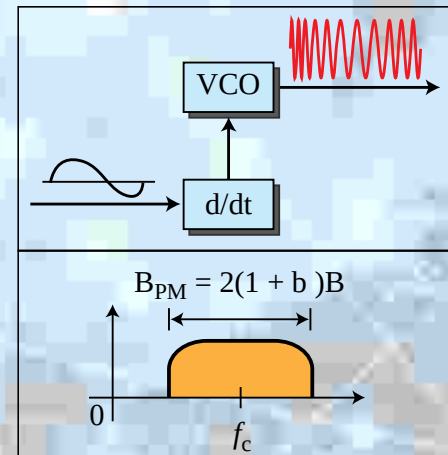
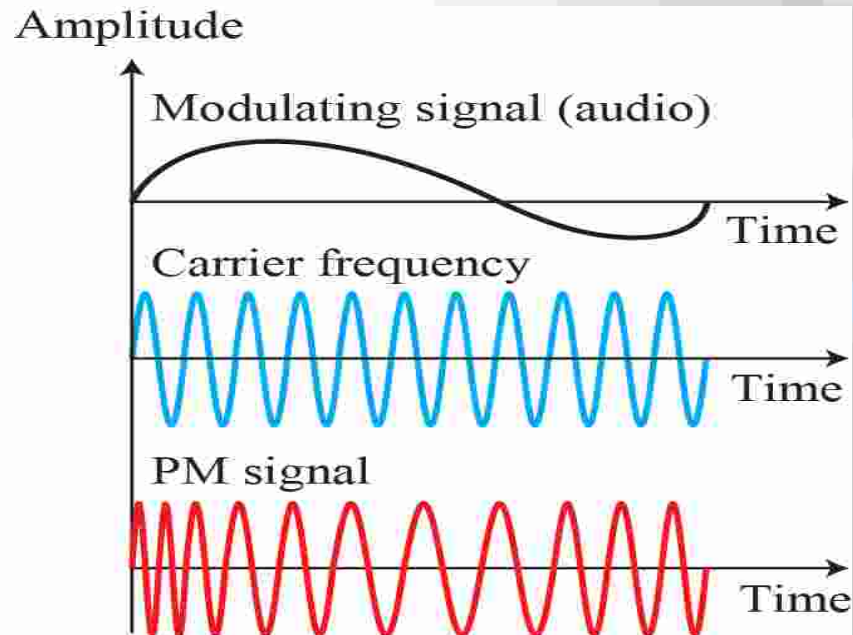
# Frequency Modulation



# Analog-to-Analog Conversion

- Representation of Analog information by an Analog signal
- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

# Phase Modulation

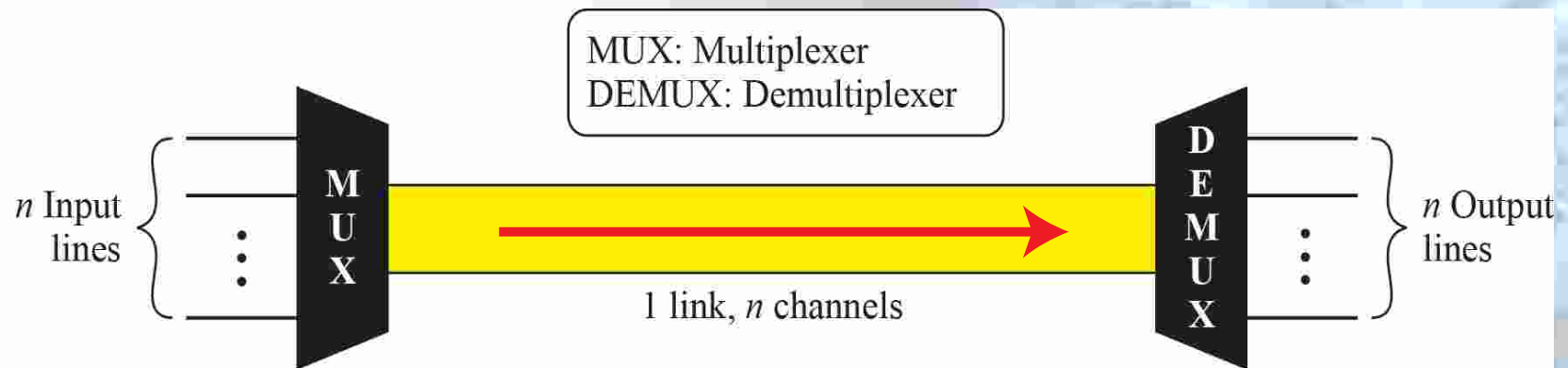




# Multiplexing

- Simultaneous transmission of multiple signals across a single data link
- As data & telecomm use increases, so does traffic
  - ✓ Add individual links each time a new channel is needed
  - ✓ Install higher-bandwidth links and use each to carry multiple signals

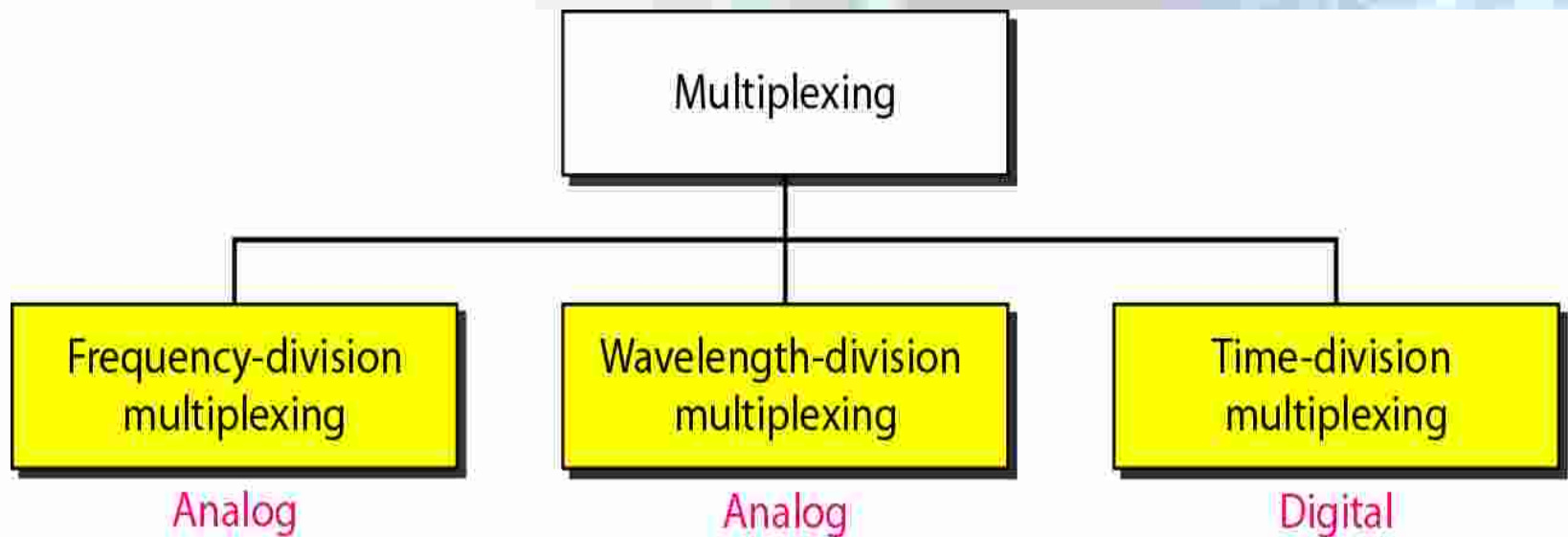
# Dividing a Link into Channels



# Multiplexing

- **Simultaneous transmission of multiple signals across a single data link**

# Categories of Multiplexing



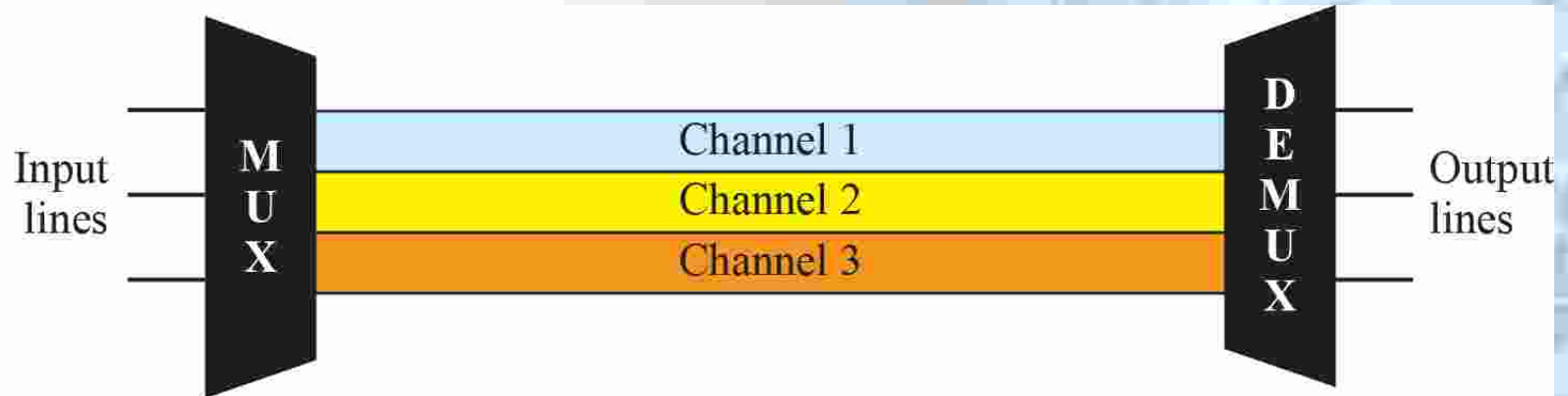
# Frequency-Division Multiplexing

- An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted
- Signals generated by each sending device modulate different carrier frequencies

# Frequency-Division Multiplexing

**These modulated signals are then combined into a single composite signal that can be transported by the link**

# Frequency-Division multiplexing

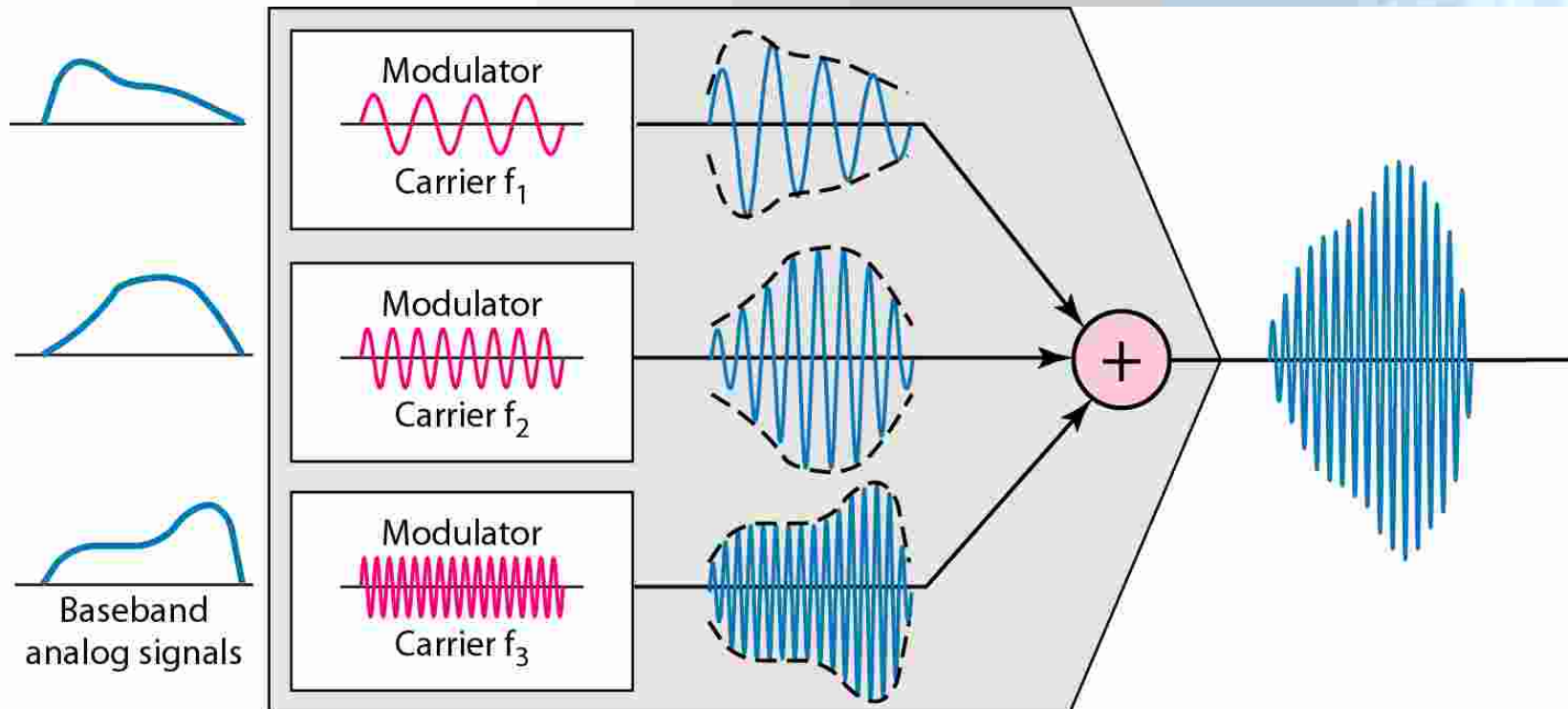


# Frequency-Division Multiplexing

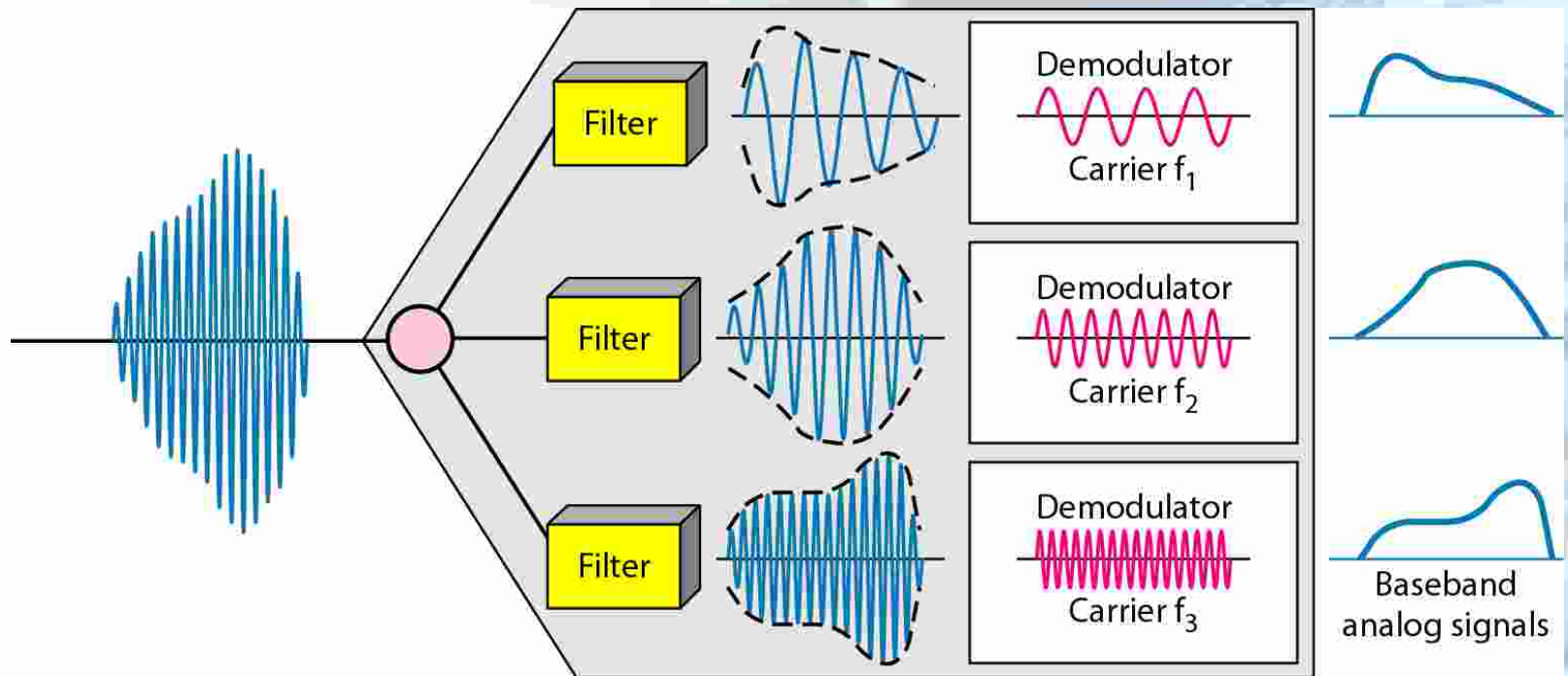
- An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted



# FDM Multiplexing

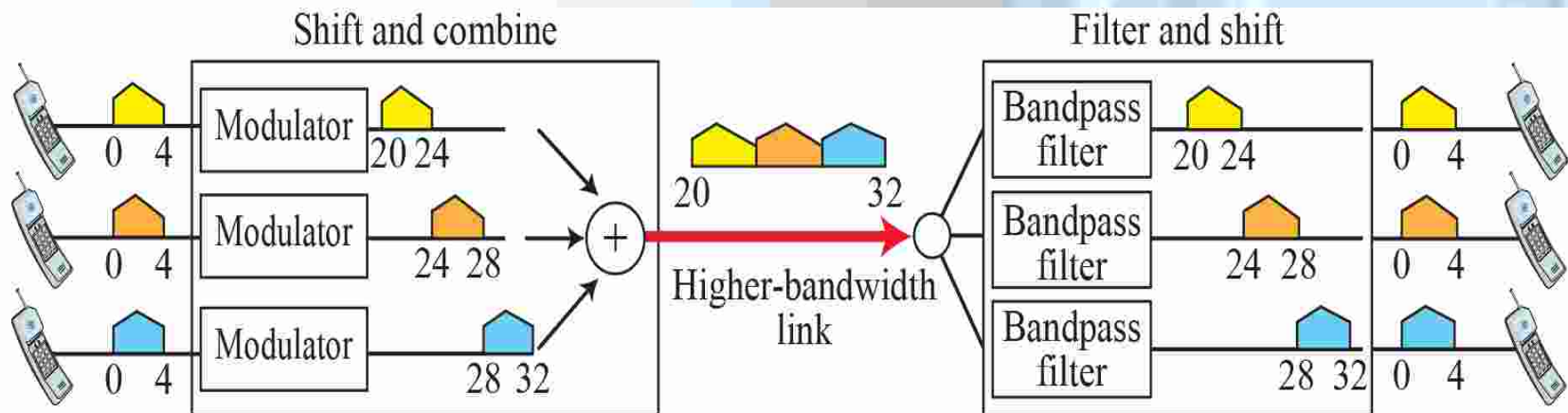


# FDM De-Multiplexing



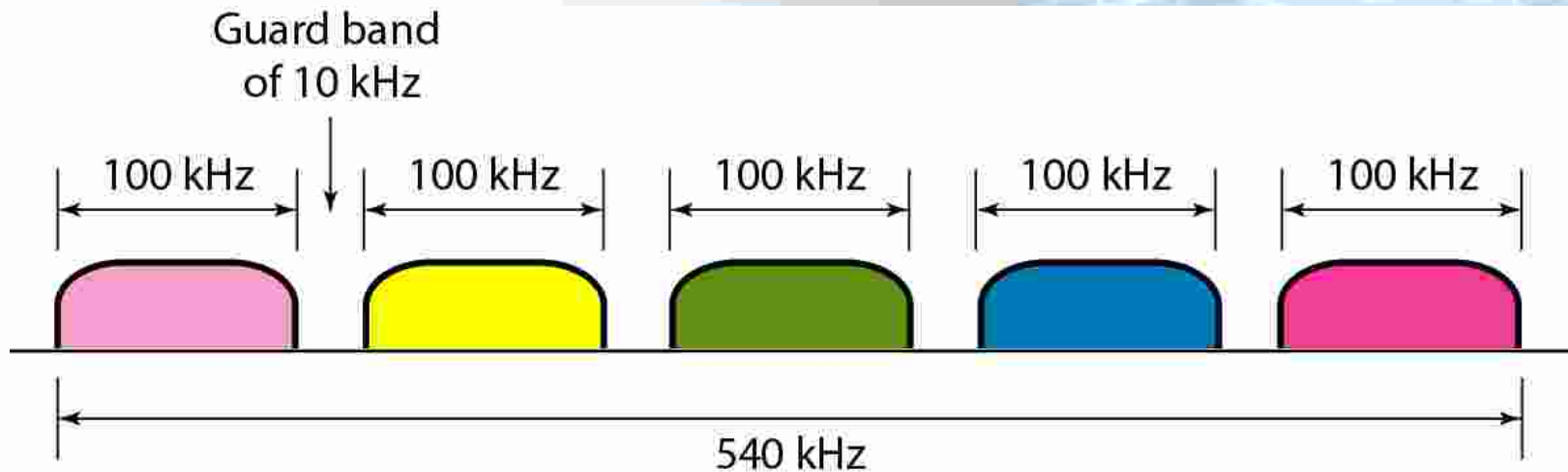
# Example

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.



# Example

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

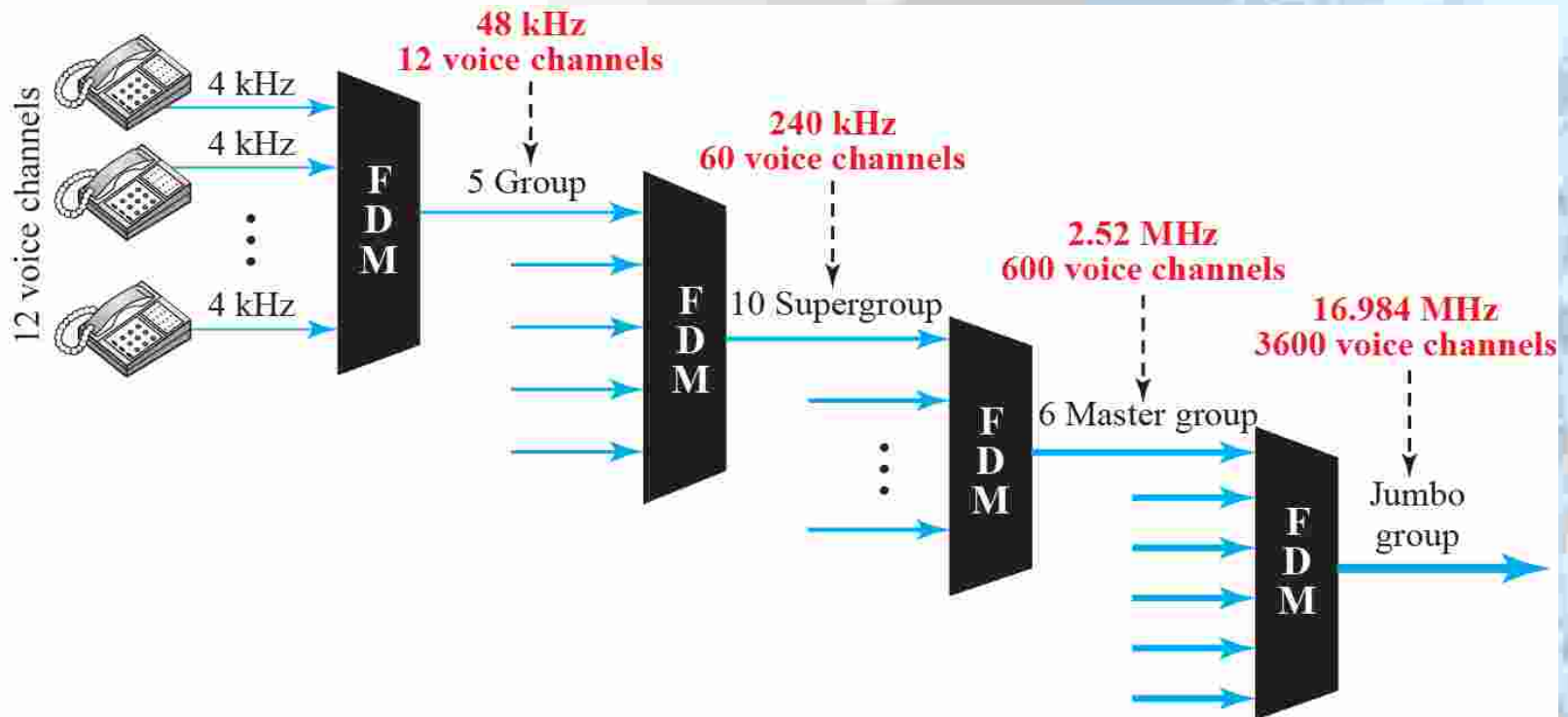


**= 540 KHZ**

# The Analog Carrier System

- Telephone companies multiplex signals from lower-bandwidth lines on to higher-bandwidth lines
- For Analog, FDM is used

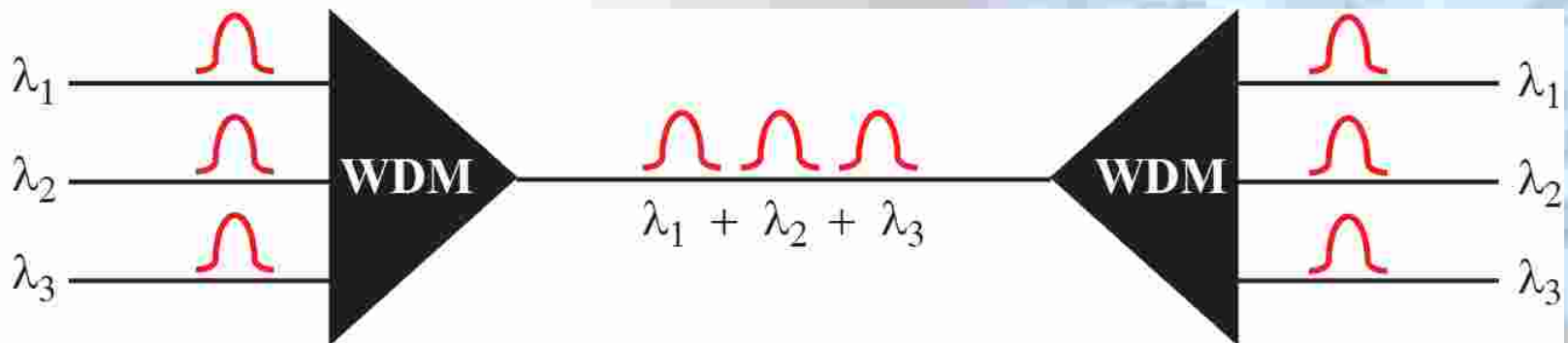
# Analog Hierarchy



# Wavelength-Division Multiplexing

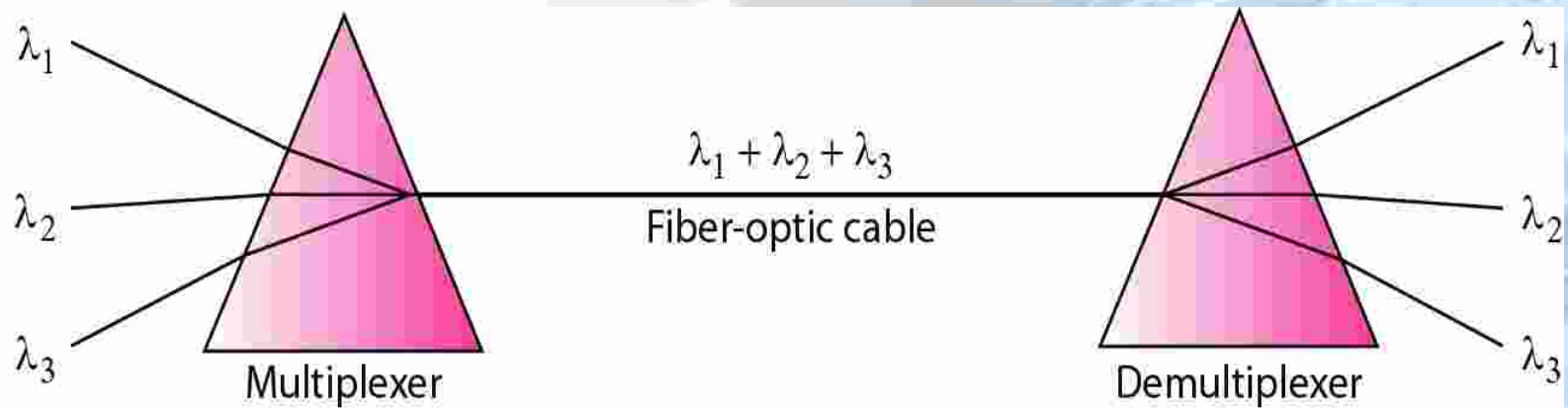
- Designed to use the high-data-rate capability of fiber-optic cable
- Fiber data rate is higher than the data rate of metallic transmission cable
- Using a fiber-optic cable for a single line wastes the available bandwidth
- Multiplexing allows us to combine several lines into one

# Wavelength-Division Multiplexing (WDM)





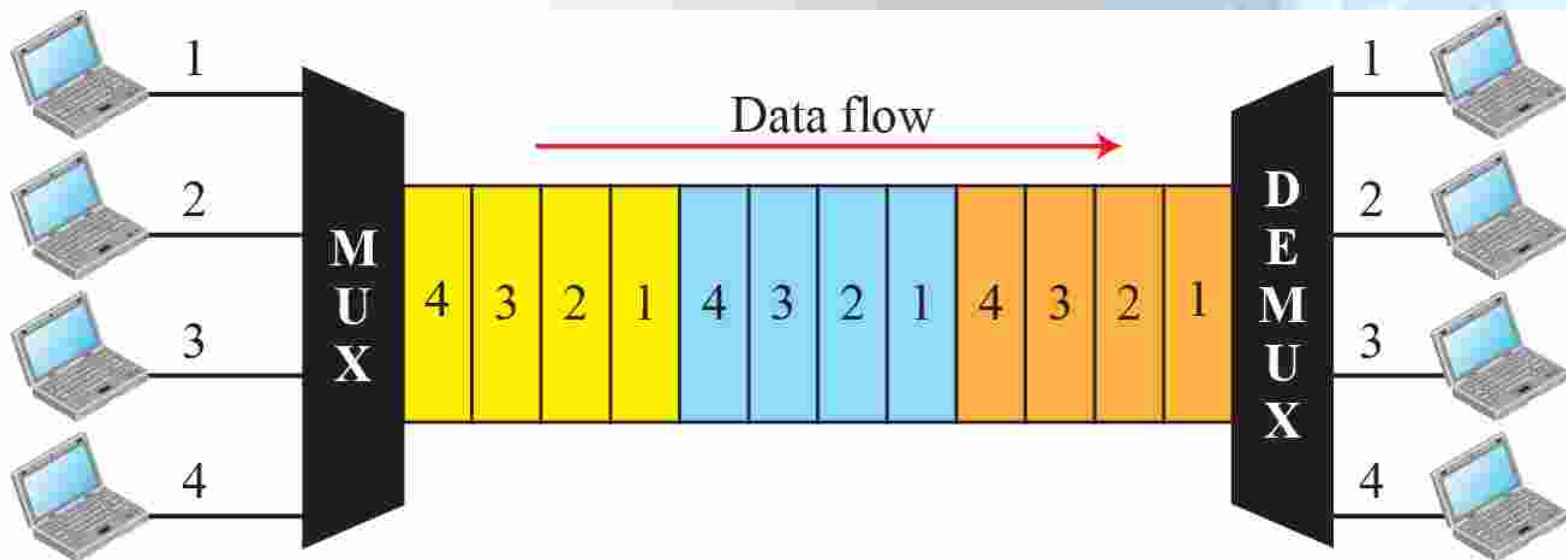
# Prisms in Wave-Length Division Multiplexing



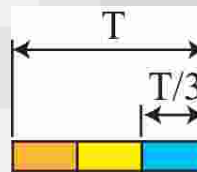
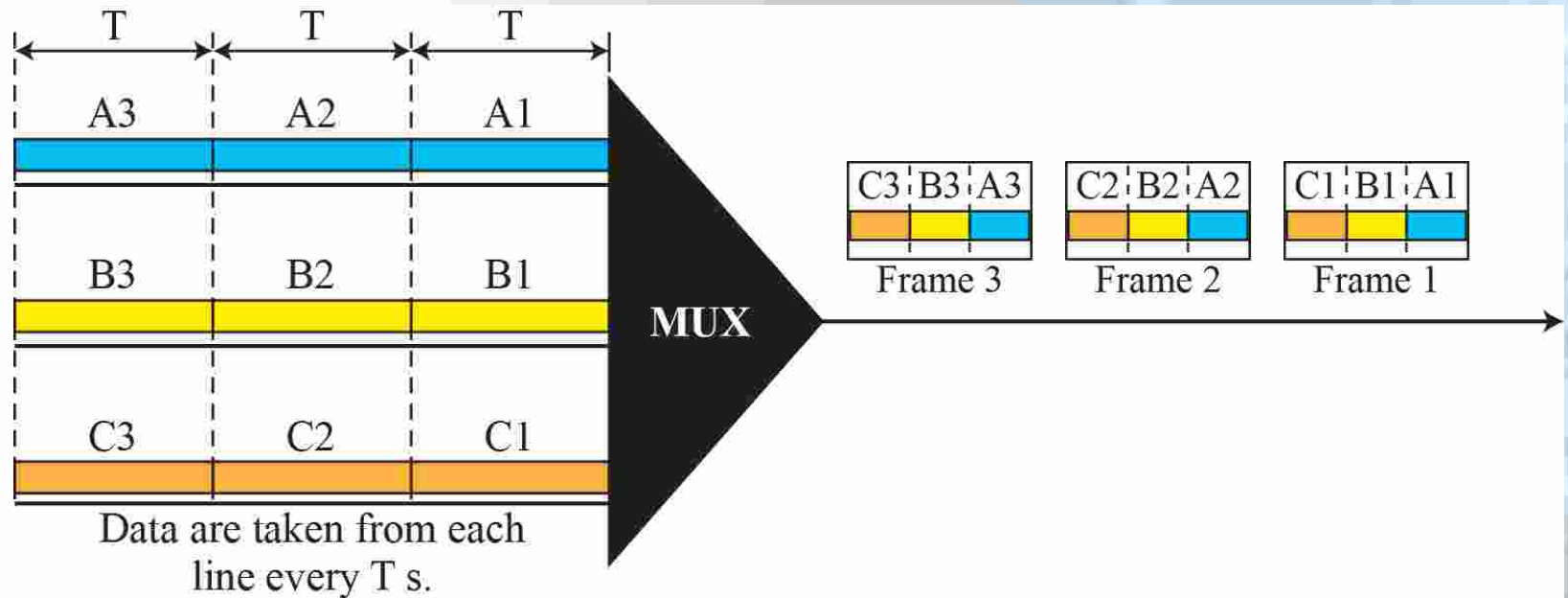
# Time-Division Multiplexing

- Digital process that allows several connections to share the high bandwidth of a link
- Time is shared i.e. each connection occupies a portion of time in the link

# TDM



# Synchronous Time-Division Multiplexing



Each frame is 3 time slots.  
Each time slot duration is  $T/3$  s.

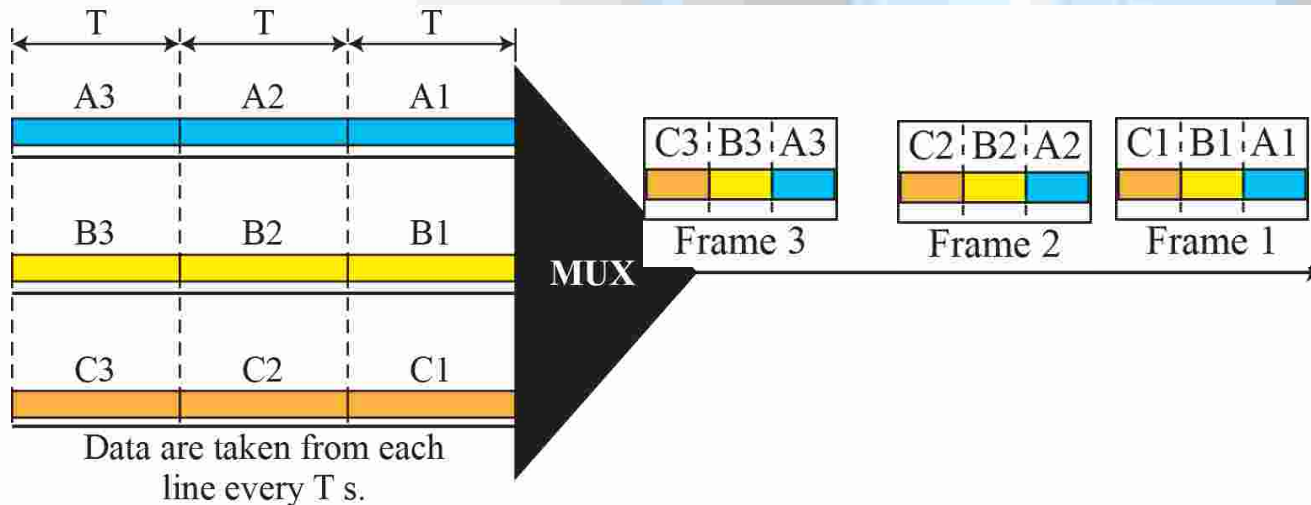
# Time-Division Multiplexing

- Digital process that allows several connections to share the high bandwidth of a link
- Time is shared i.e. each connection occupies a portion of time in the link

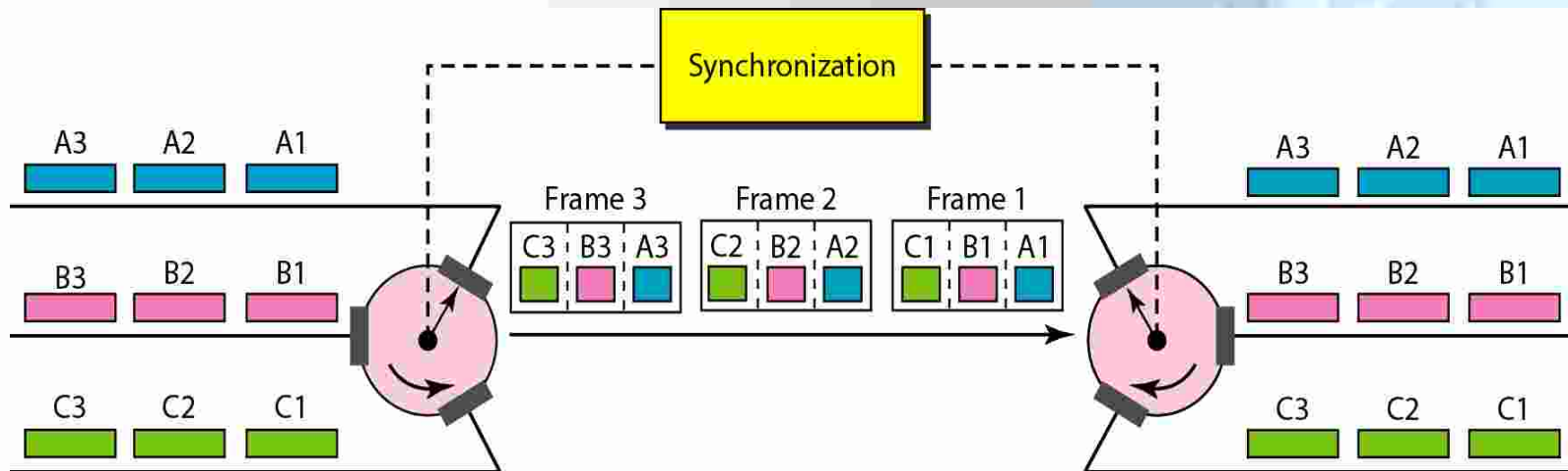
# Example

In Figure the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of

- each input slot,
- each output slot, and
- each frame?



# Interleaving

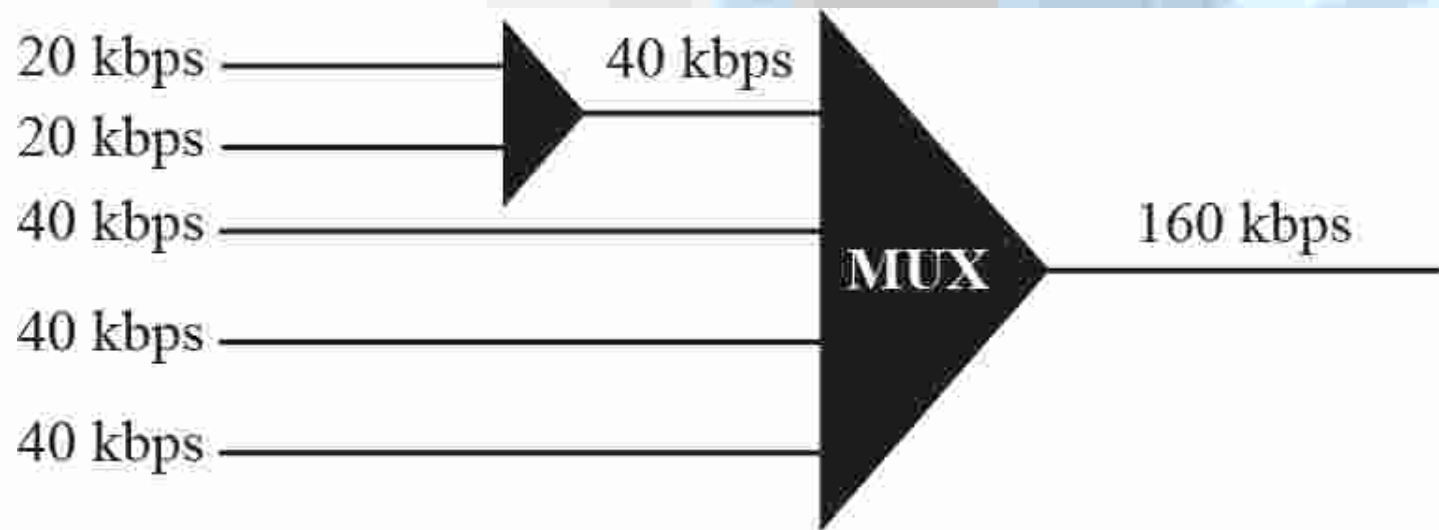


# Time-Division Multiplexing

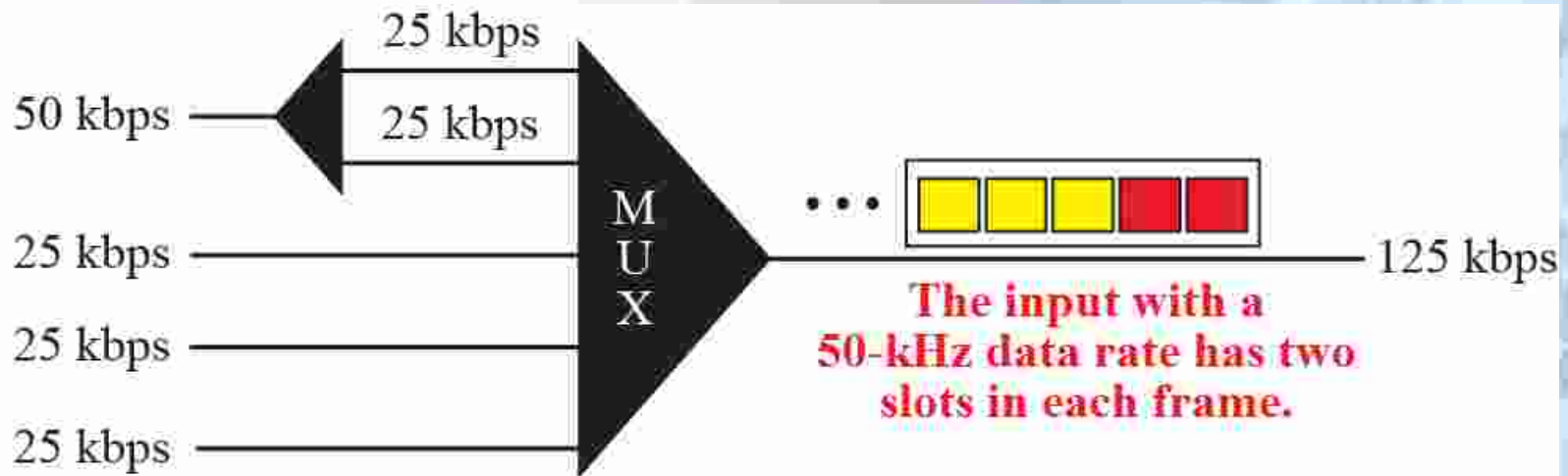
- Digital process that allows several connections to share the high bandwidth of a link
- Time is shared i.e. each connection occupies a portion of time in the link



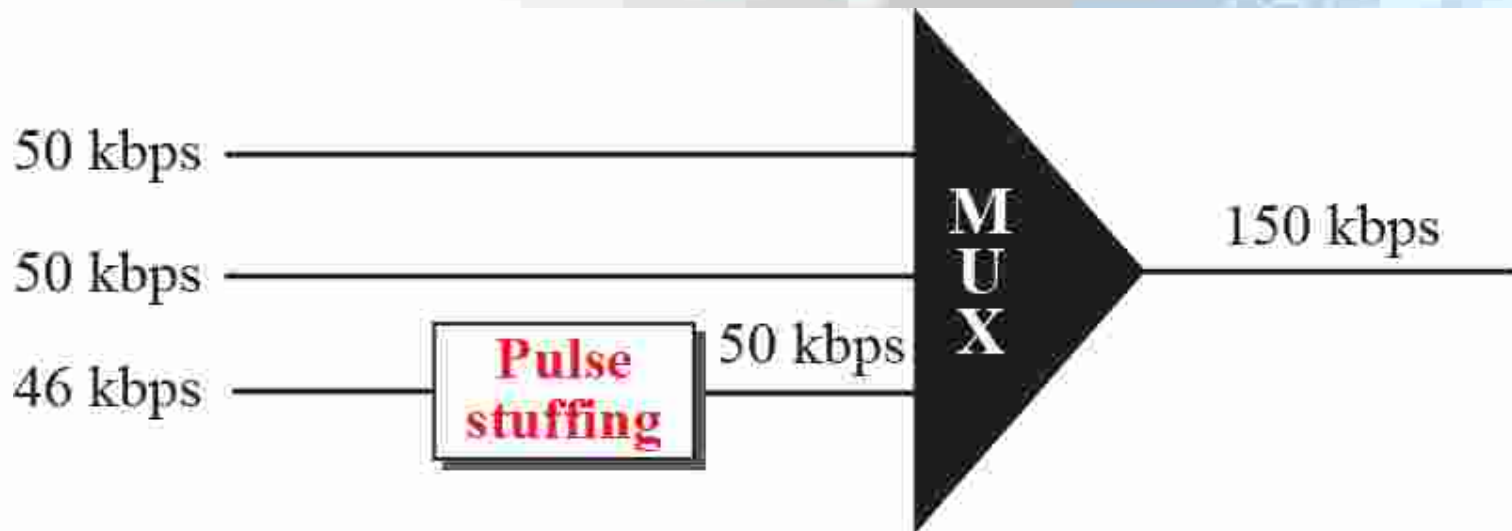
# Multilevel Multiplexing



# Multiple-Slot Multiplexing



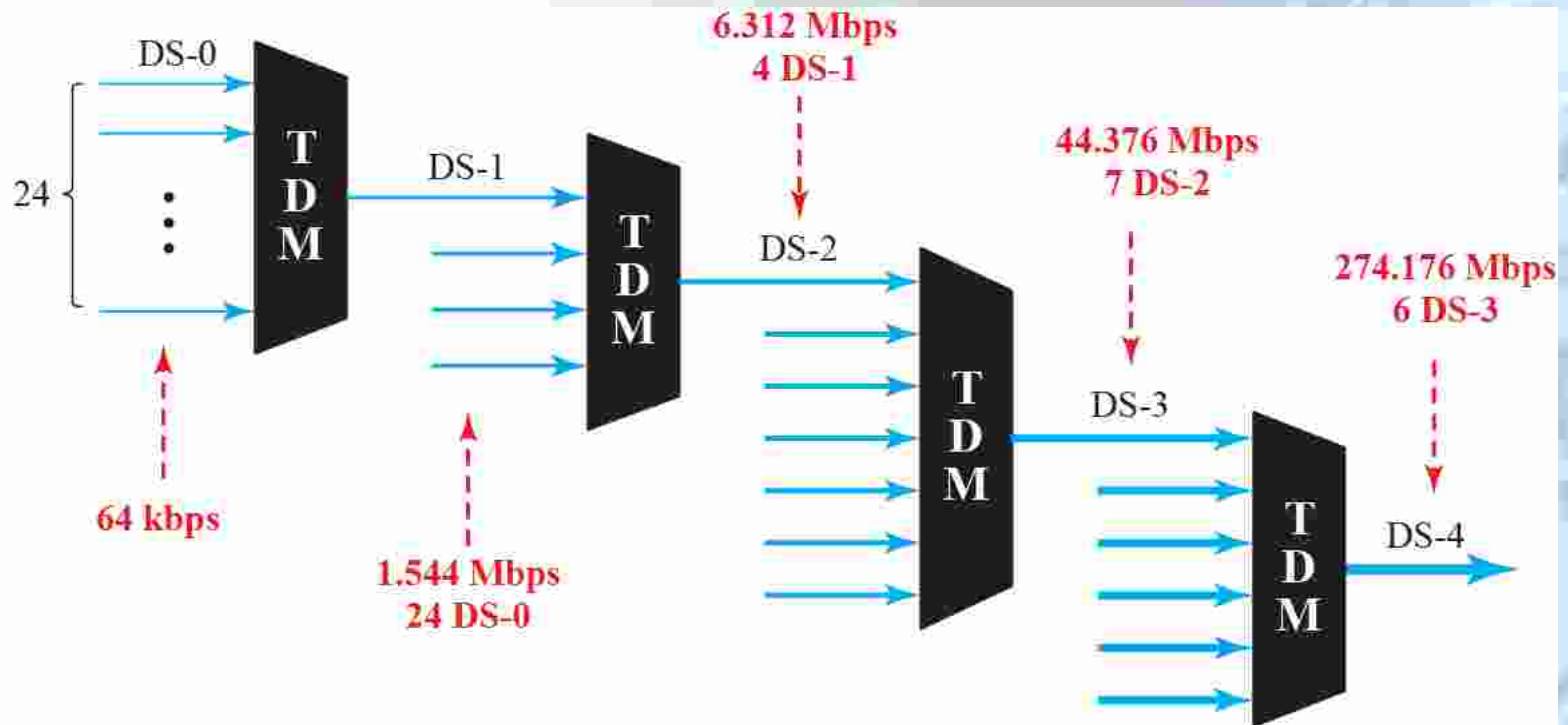
# Pulse Stuffing



# Time-Division Multiplexing

- Digital process that allows several connections to share the high bandwidth of a link
- Time is shared i.e. each connection occupies a portion of time in the link

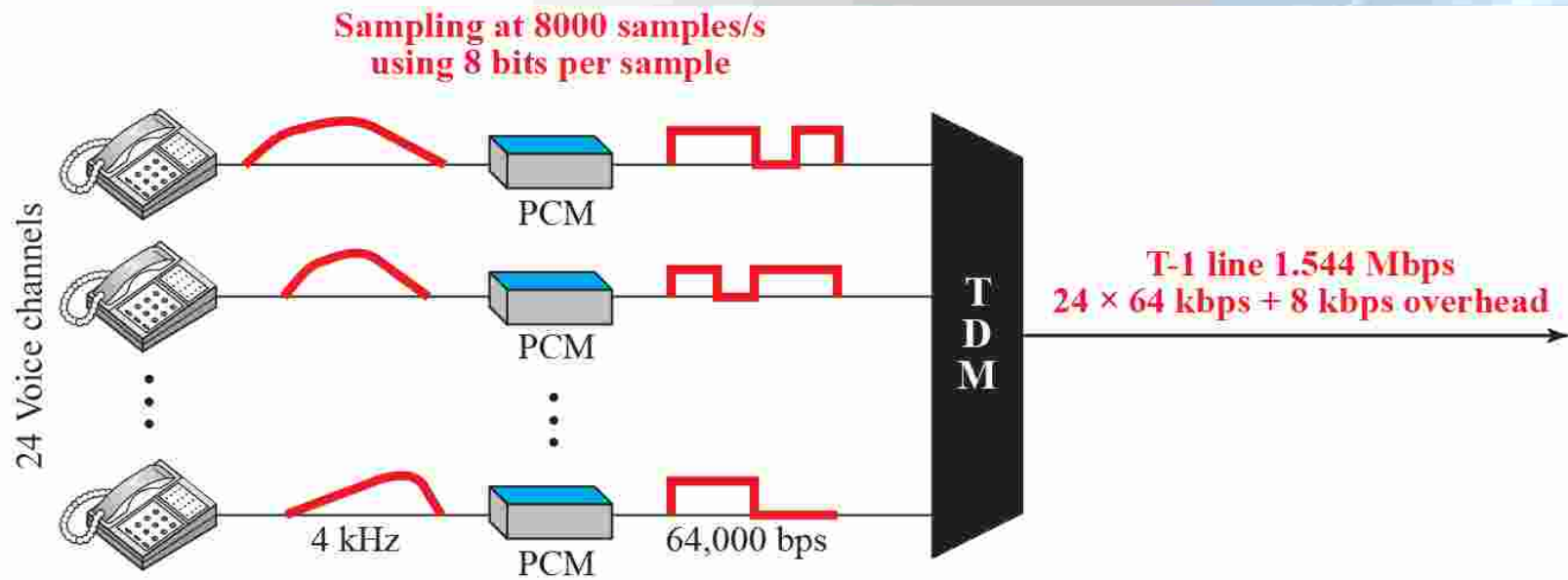
# Digital Hierarchy



## DS and T Line Rates

| <i>Service</i> | <i>Line</i> | <i>Rate (Mbps)</i> | <i>Voice Channels</i> |
|----------------|-------------|--------------------|-----------------------|
| DS-1           | T-1         | 1.544              | 24                    |
| DS-2           | T-2         | 6.312              | 96                    |
| DS-3           | T-3         | 44.736             | 672                   |
| DS-4           | T-4         | 274.176            | 4032                  |

# T-1 Line



## E Line Rates

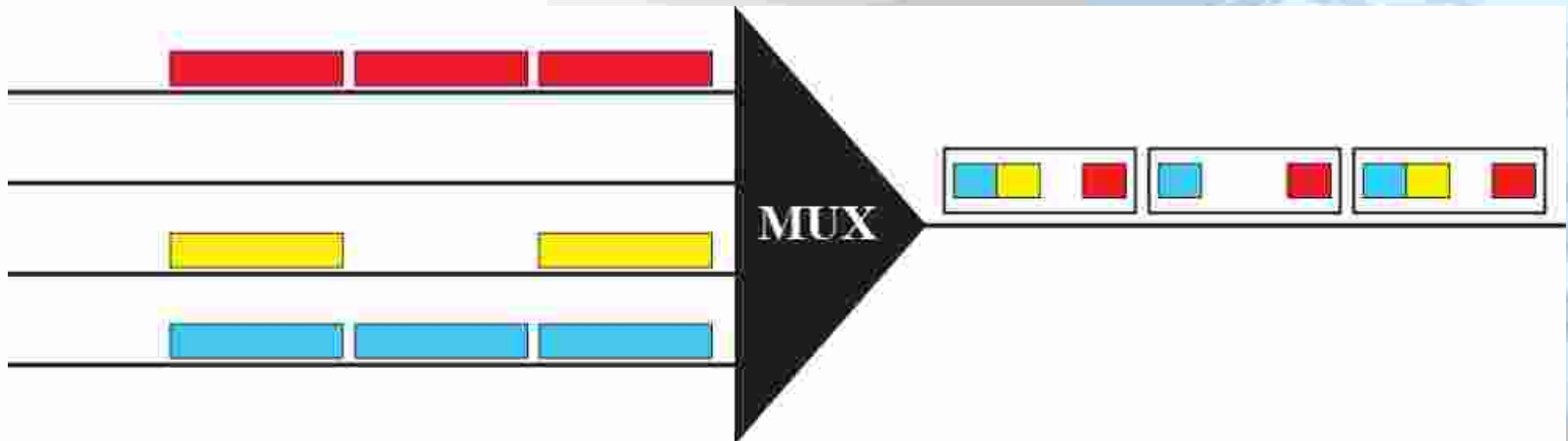
| <i>Line</i> | <i>Rate (Mbps)</i> | <i>Voice Channels</i> |
|-------------|--------------------|-----------------------|
| E-1         | 2.048              | 30                    |
| E-2         | 8.448              | 120                   |
| E-3         | 34.368             | 480                   |
| E-4         | 139.264            | 1920                  |



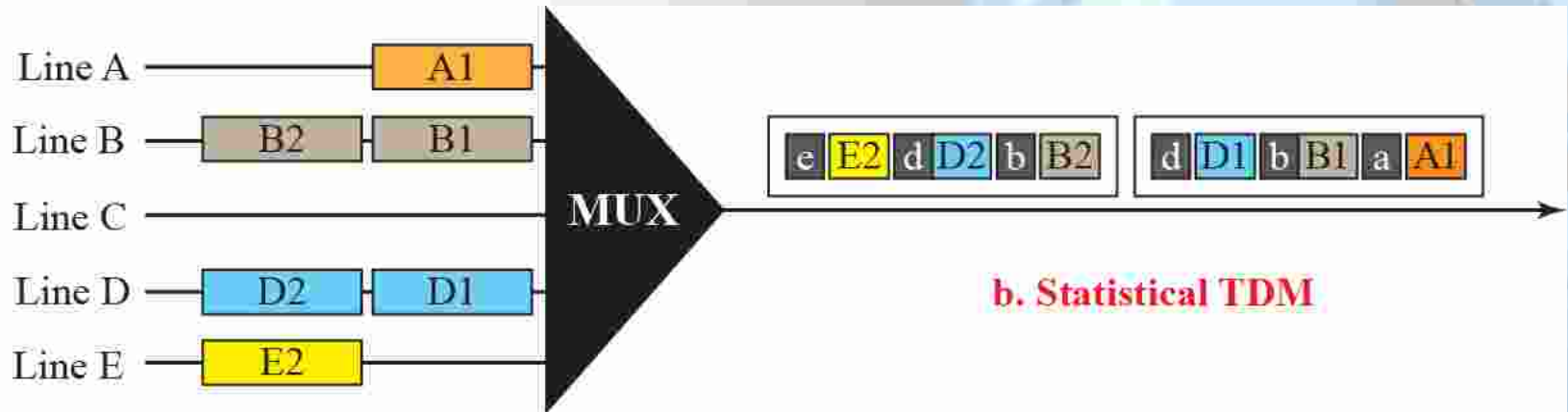
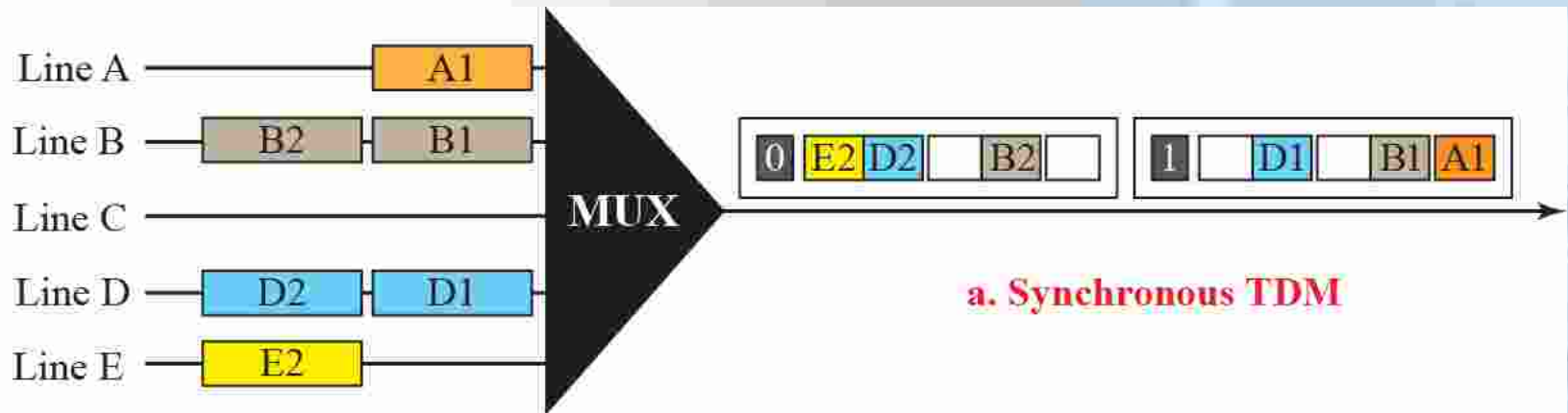
# Time-Division Multiplexing

- Synchronous TDM
- Statistical TDM

# Empty slots



# Statistical TDM



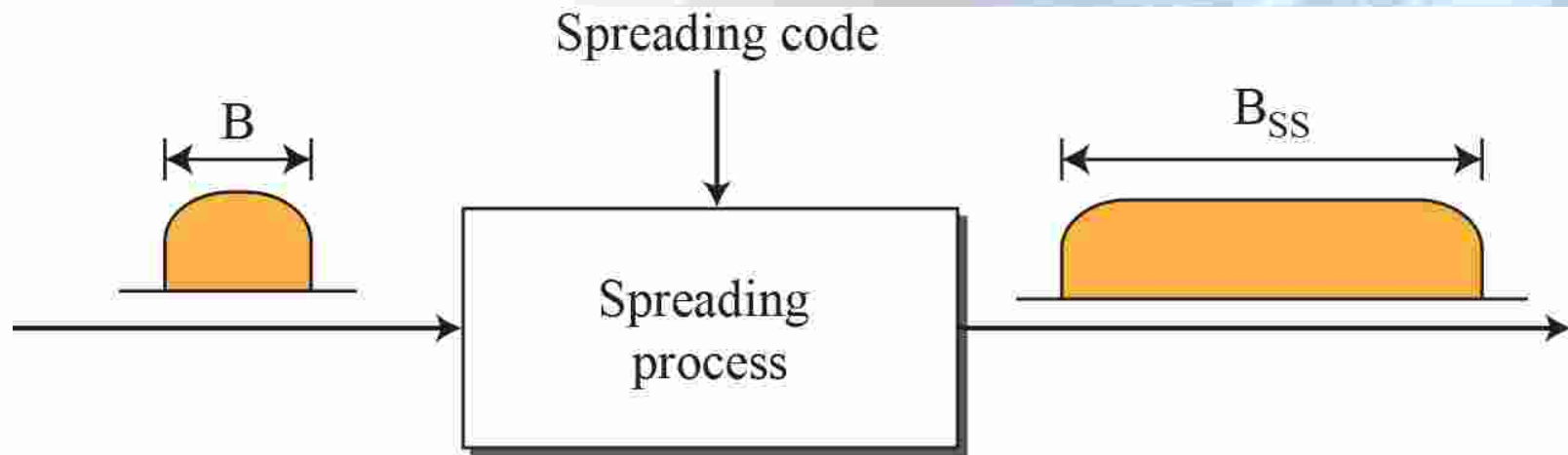
# SPREAD SPECTRUM

- In wireless applications, stations must be able to share the medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder
- To achieve these goals, spread spectrum adds redundancy and spread original spectrum needed for each station

# SPREAD SPECTRUM - Principles

- **Bandwidth allocated to each station needs to be larger than what is needed to allow Redundancy**
- **Spreading process should be independent of the original signal**

# Spread Spectrum



# SPREAD SPECTRUM TECHNIQUES

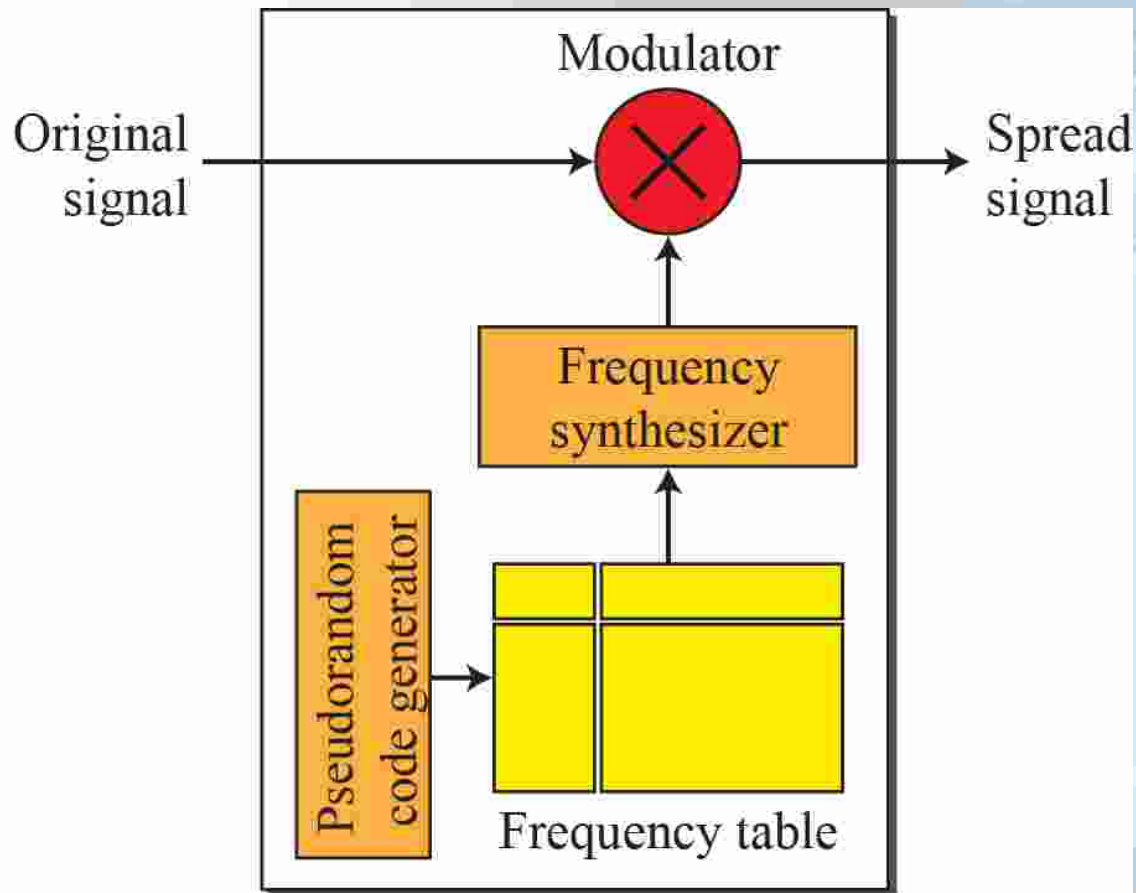
- **Frequency Hopping Spread Spectrum (FHSS)**
- **Direct Sequence Spread Spectrum (DSSS)**

# Frequency Hopping Spread Spectrum (FHSS)

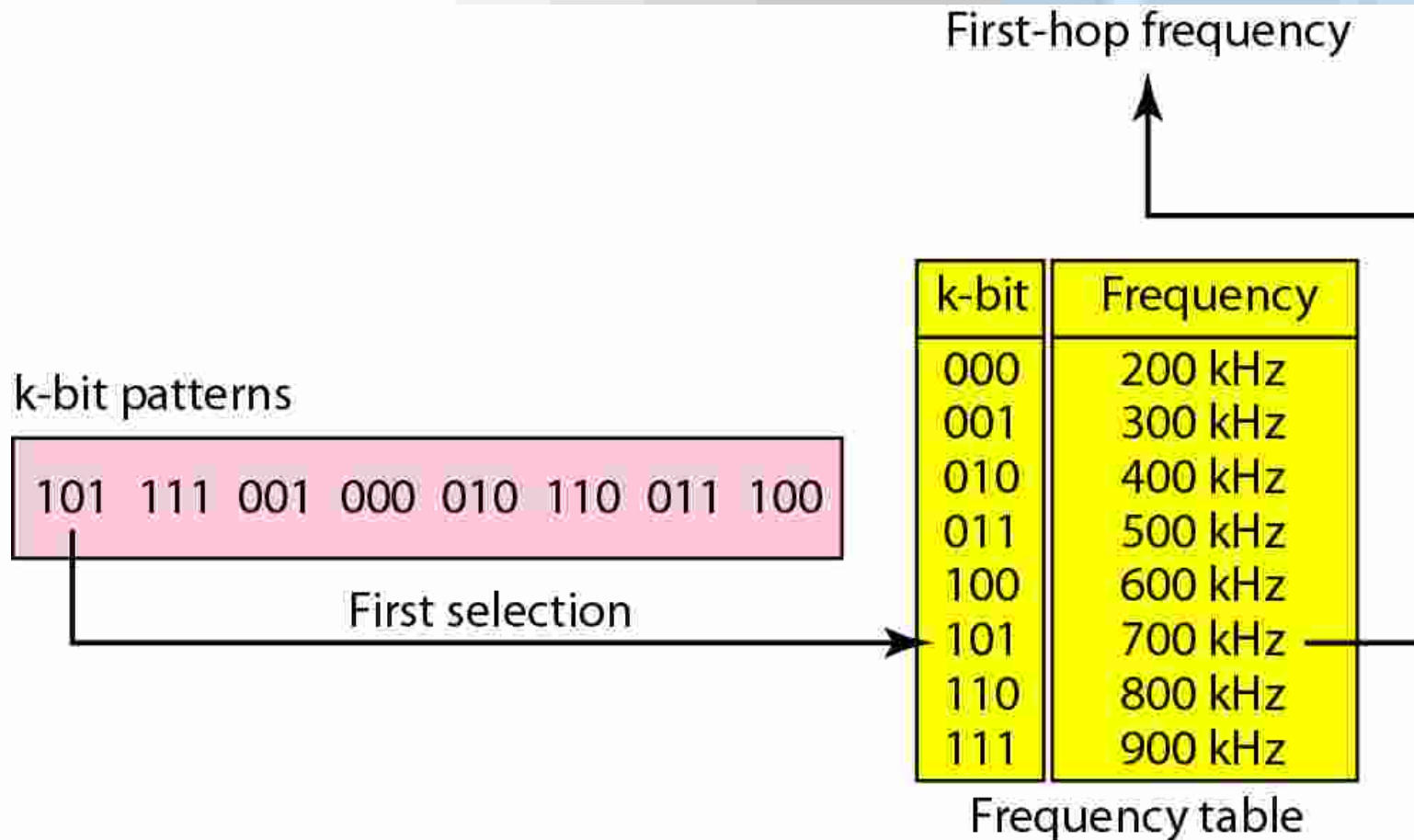
- ‘M’ different carrier frequencies that are modulated by the source signal
- At one moment, signal modulates one carrier frequency and at next moment, it modulates another



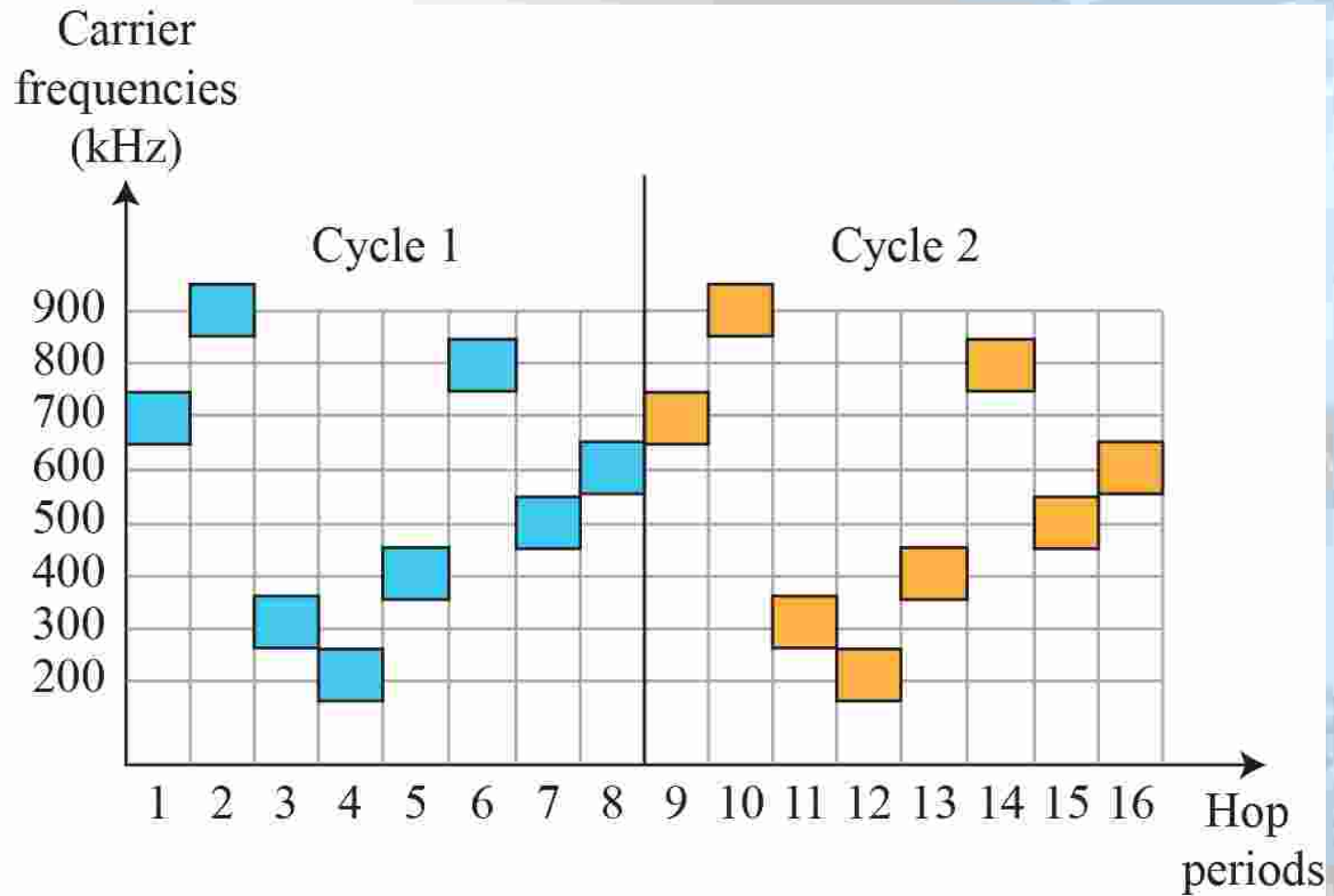
# Frequency Hopping Spread Spectrum (FHSS)



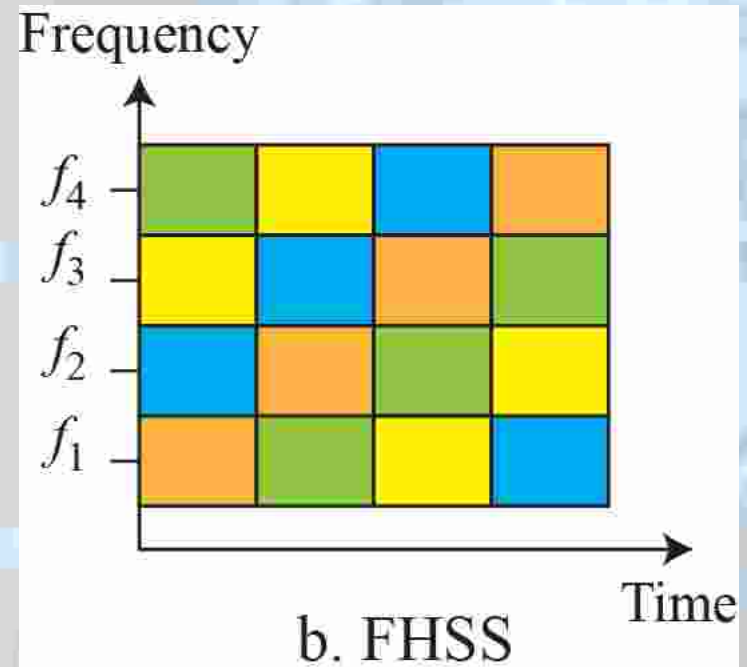
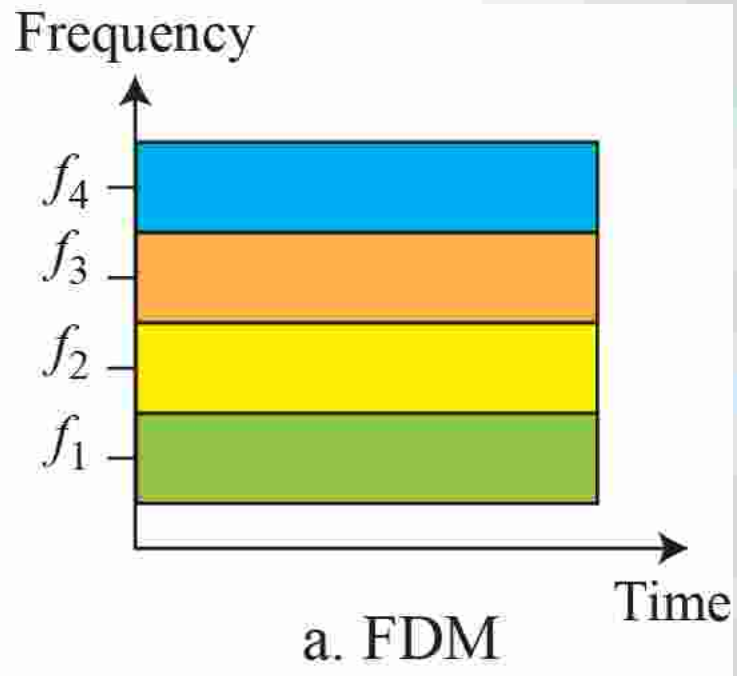
# Frequency Selection in FHSS



# FHSS Cycles



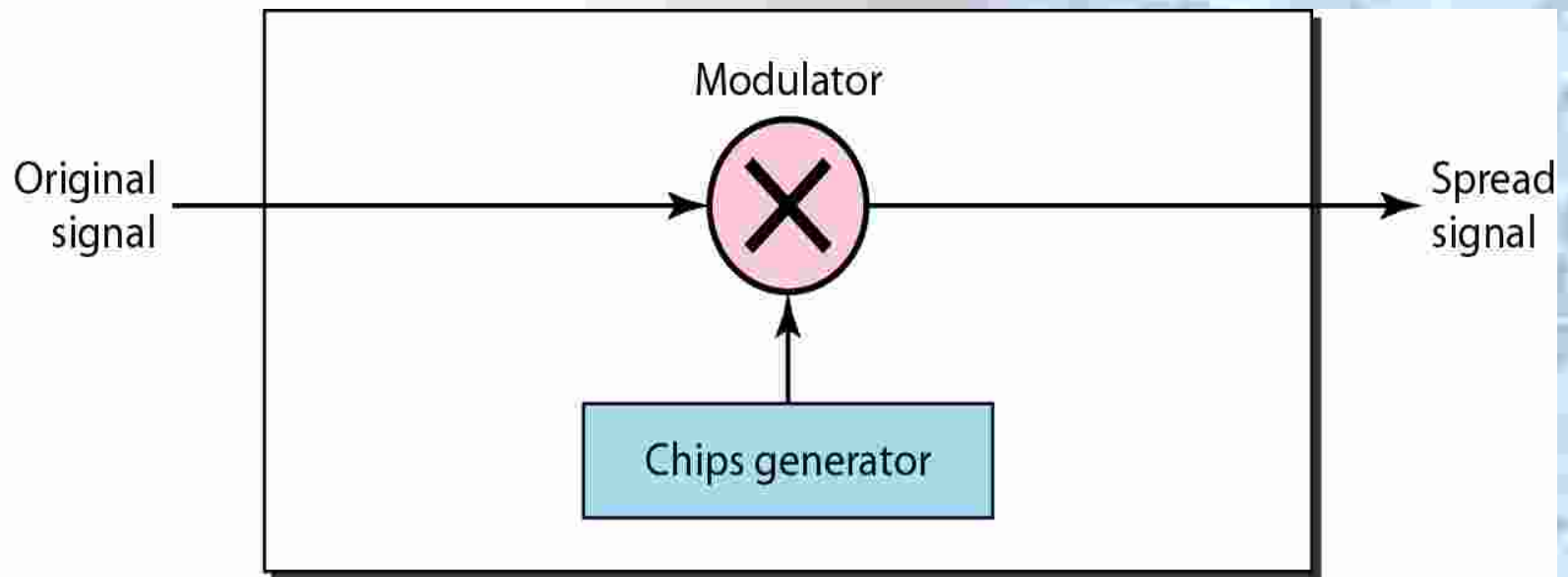
# Bandwidth Sharing



# DSSS

- DSSS also expands the bandwidth of the original signal, but the process is different
- We replace each data bit with 'n' bits using a spreading code
- Each bit is assigned a code of 'n' bits, called chips, where the chip rate is 'n' times that of the data bit

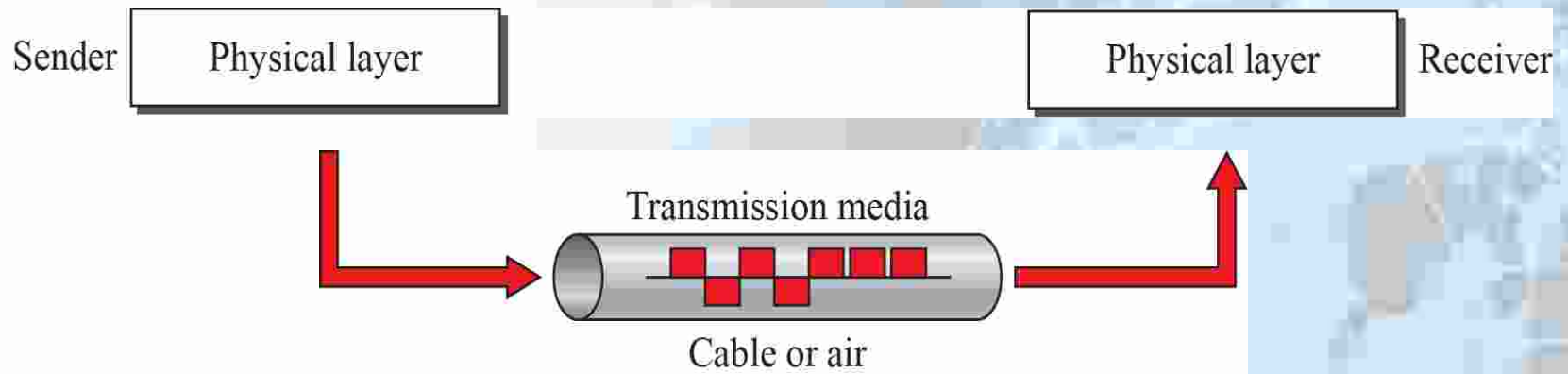
# DSSS



# Transmission Media

- Located below the physical layer and are directly controlled by the physical layer
- Belong to layer zero
- Metallic Media i.e. Twisted pair and Coaxial Cable
- Optical Fiber Cable
- Free Space i.e. Air, Vacuum

# Transmission Media & Physical Payer

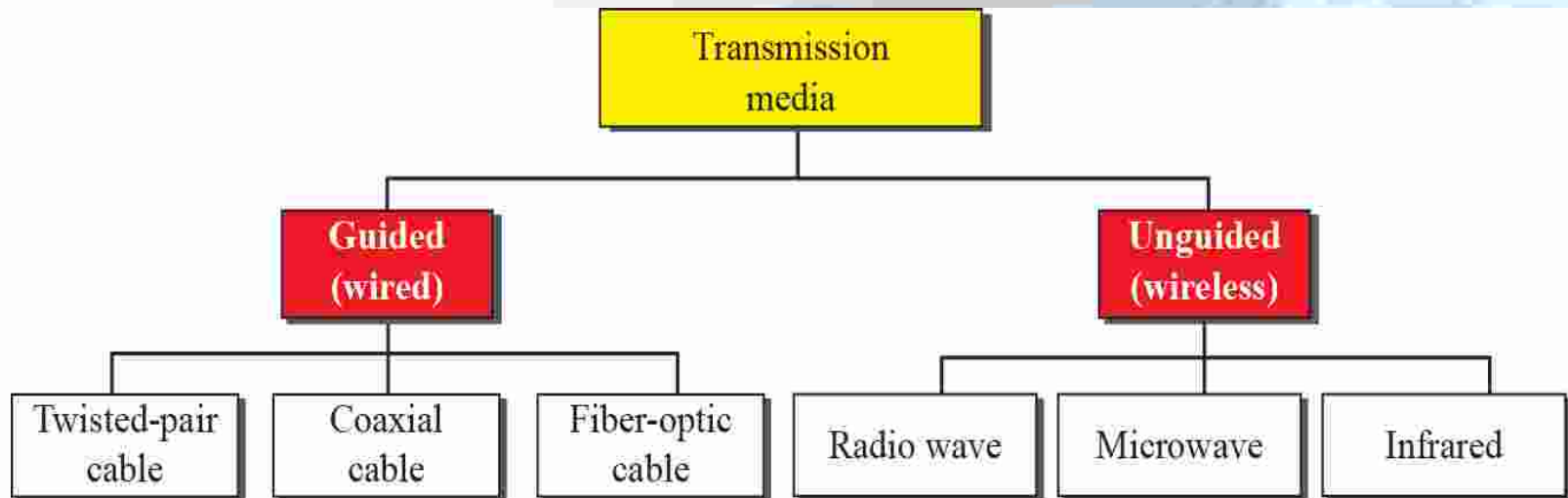




# Transmission Media

- Located below the physical layer and are directly controlled by the physical layer
- Belong to layer zero
- Metallic Media i.e. Twisted pair and Coaxial Cable
- Optical Fiber Cable
- Free Space i.e. Air, Vacuum

# Classes of Transmission Media



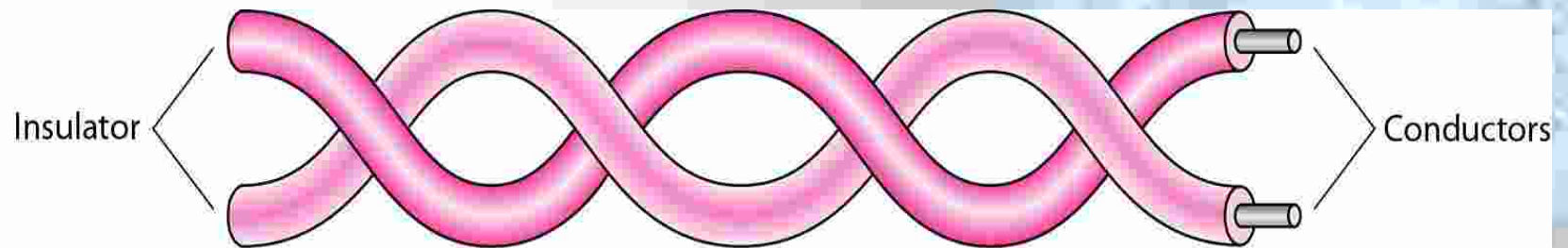
# Guided Media

- Media that provides a conduit from one device to another
- Twisted-pair cable, coaxial cable, and fiber-optic cable
- Signal traveling along any of these media is directed and contained by the physical limits of the medium

# Twisted-Pair Cable

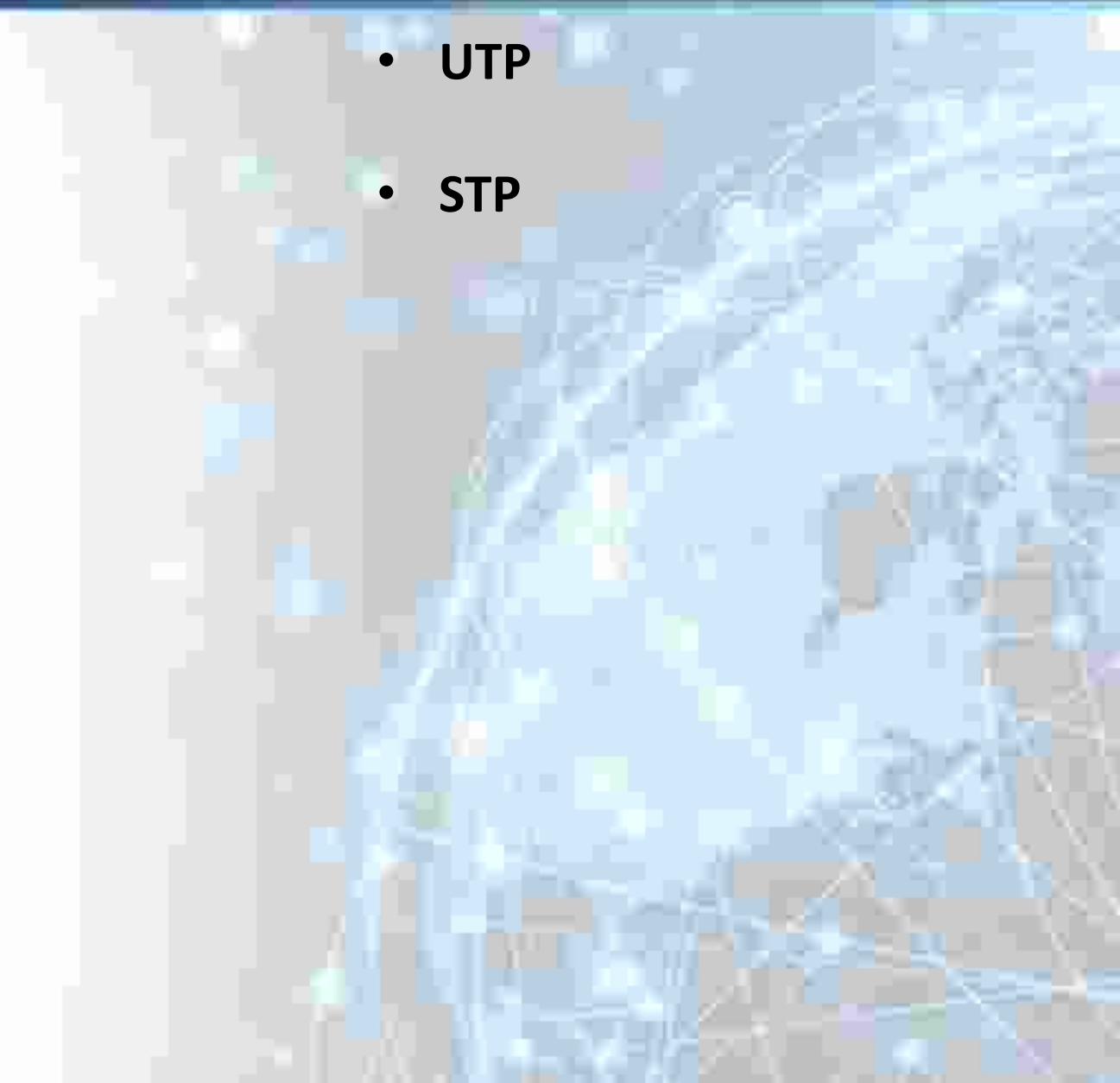
- Consists of 2 copper conductors, each with its own plastic insulation, twisted together
- One wire carries signals and other is ground reference
- Receiver uses difference between the two
- Interference (Noise) & Crosstalk

# Twisted-Pair Cable

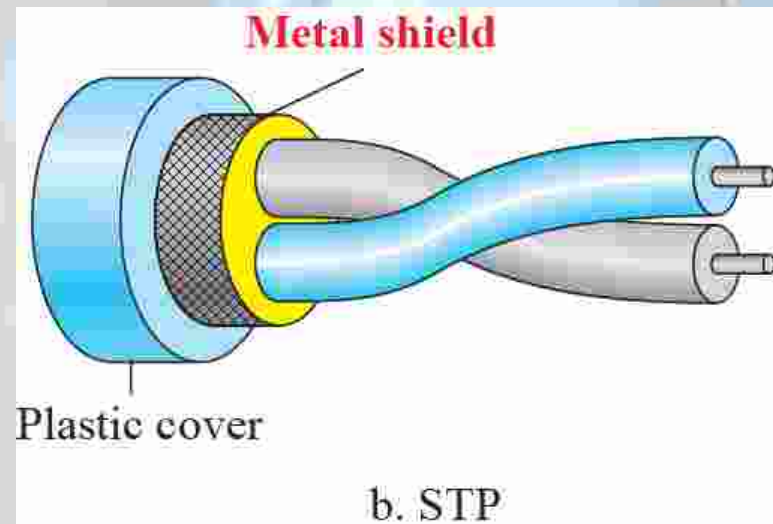
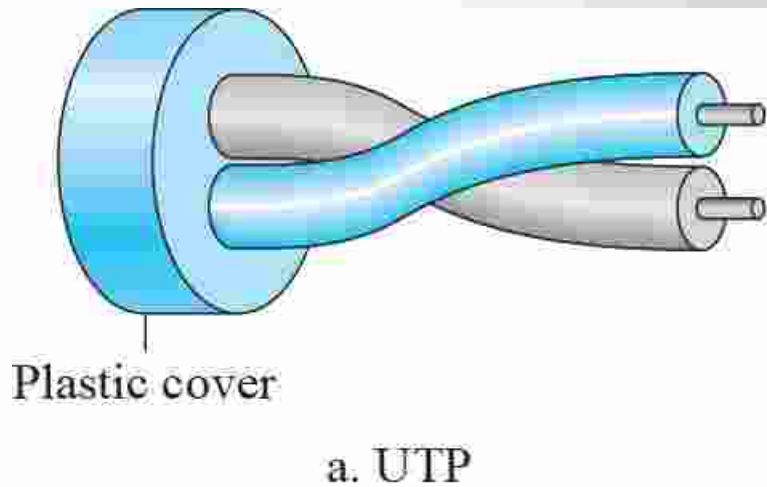


# Twisted-Pair Cable

- UTP
- STP



# Unshielded vs. Shielded Twisted Pair Cable



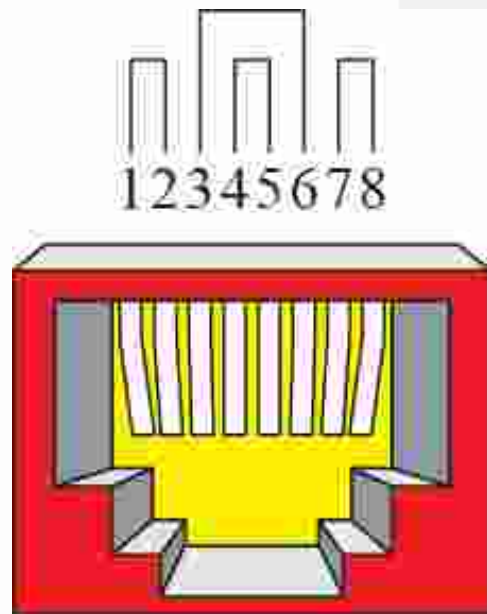
# Categories of Unshielded Twisted-Pair Cables

| <i>Category</i> | <i>Specification</i>                                           | <i>Data Rate (Mbps)</i> | <i>Use</i> |
|-----------------|----------------------------------------------------------------|-------------------------|------------|
| 1               | Unshielded twisted-pair used in telephone                      | < 0.1                   | Telephone  |
| 2               | Unshielded twisted-pair originally used in T lines             | 2                       | T-1 lines  |
| 3               | Improved CAT 2 used in LANs                                    | 10                      | LANs       |
| 4               | Improved CAT 3 used in Token Ring networks                     | 20                      | LANs       |
| 5               | Cable wire is normally 24 AWG with a jacket and outside sheath | 100                     | LANs       |

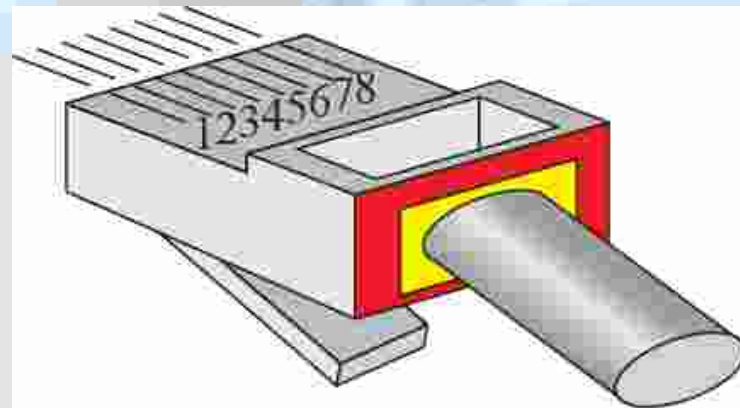
|    |                                                                                                                                                                                                                                                                            |     |      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|
| 5E | An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference                                                                                                                                                         | 125 | LANs |
| 6  | A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.                                                                                                                                                | 200 | LANs |
| 7  | Sometimes called <i>SSTP (shielded screen twisted-pair)</i> . Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate. | 600 | LANs |



# UTP Connectors

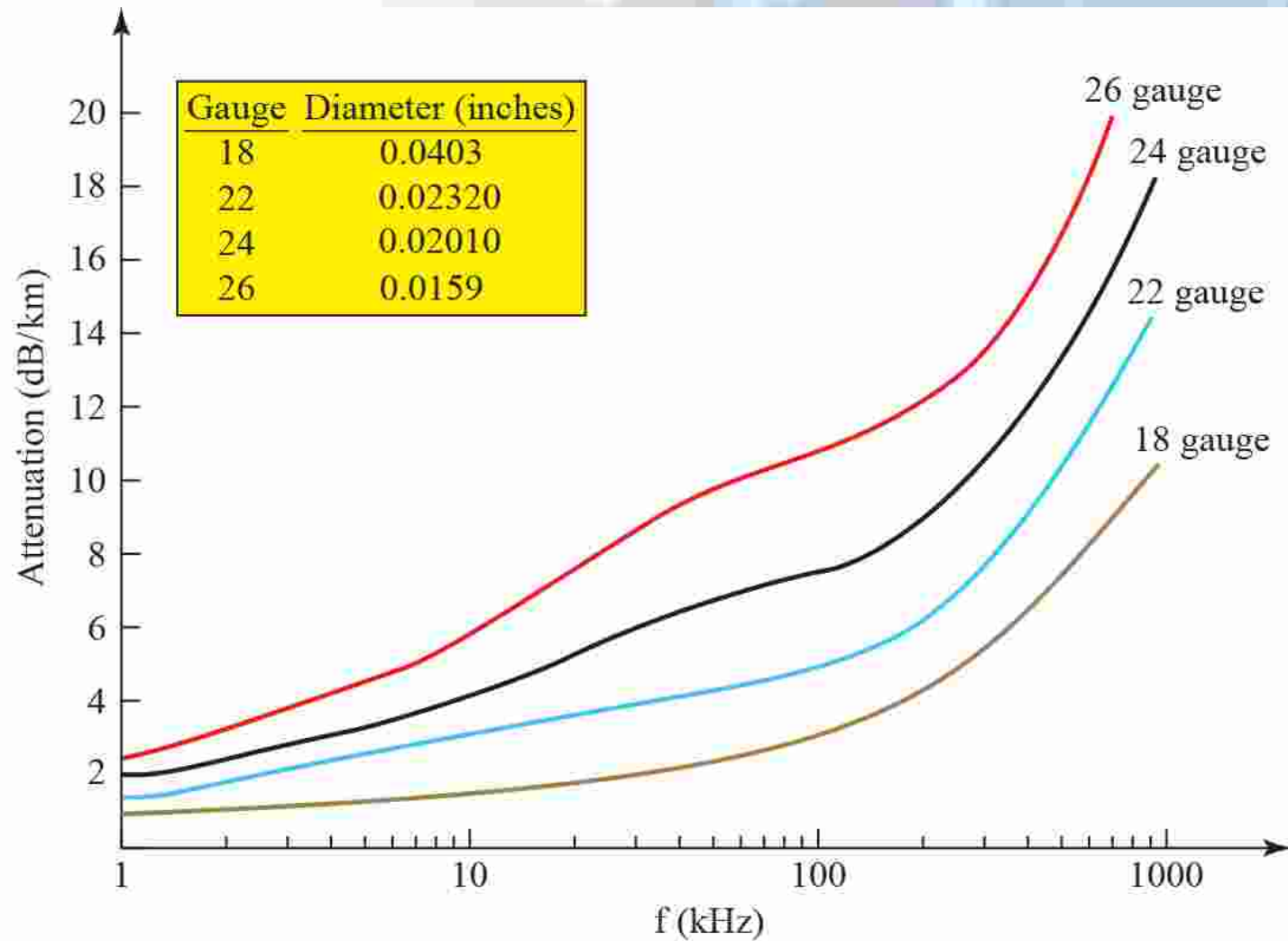


RJ-45 Female



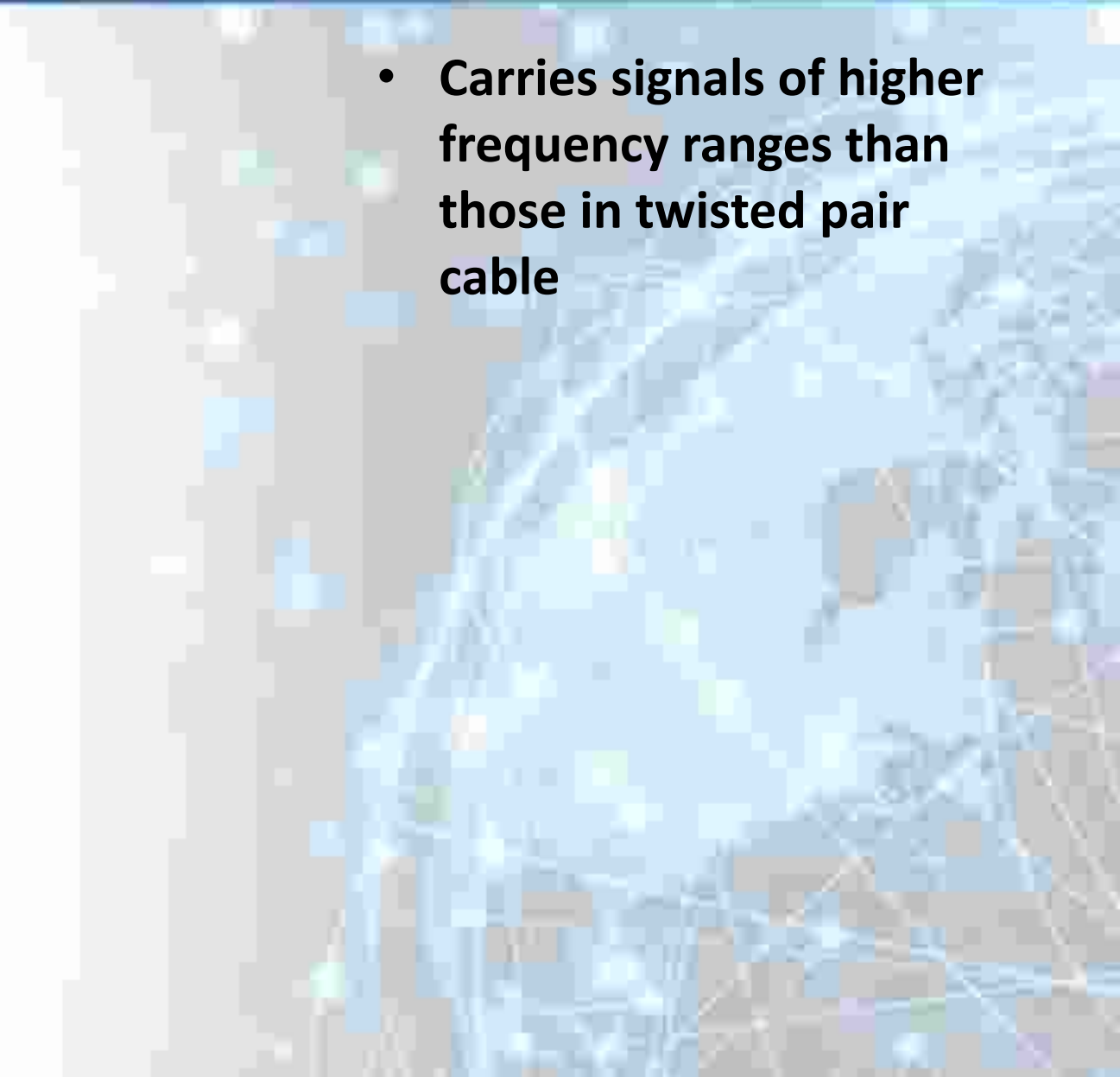
RJ-45 Male

# UTP Performance

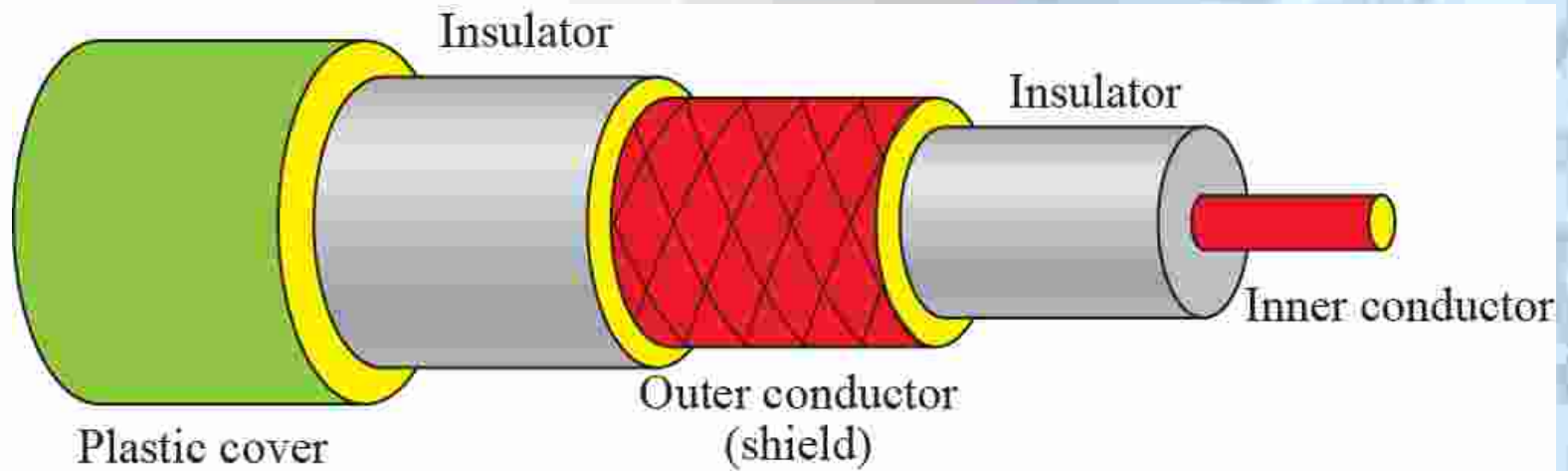


# Coaxial Cable

- Carries signals of higher frequency ranges than those in twisted pair cable



# Coaxial Cable

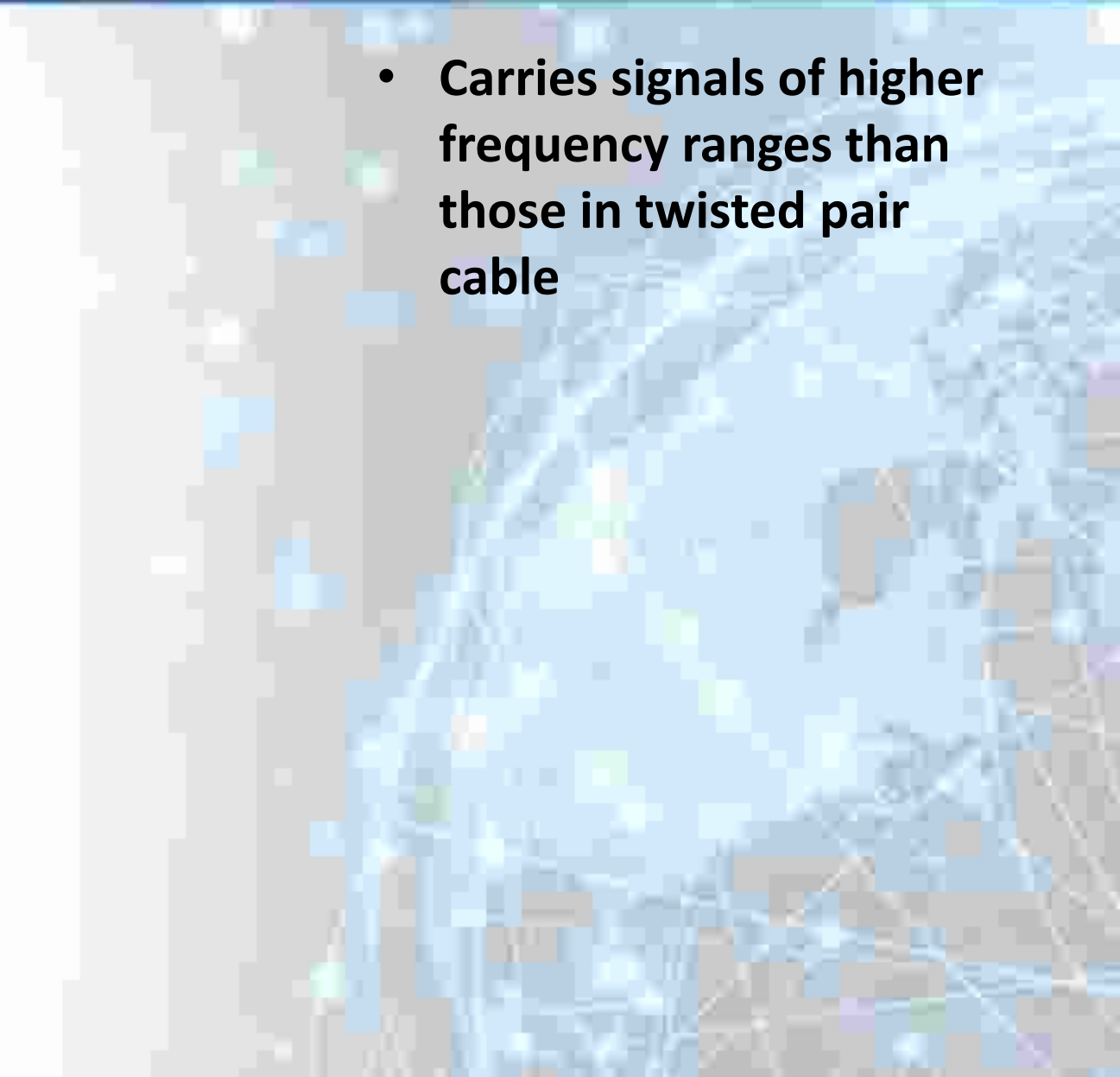


# Categories of Coaxial Cables

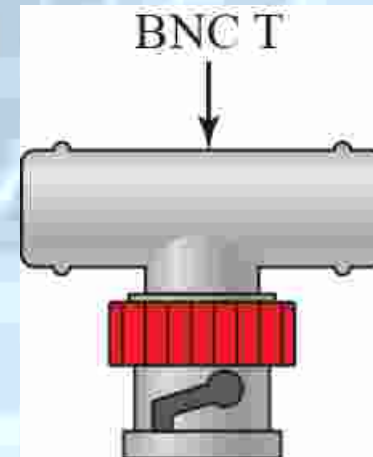
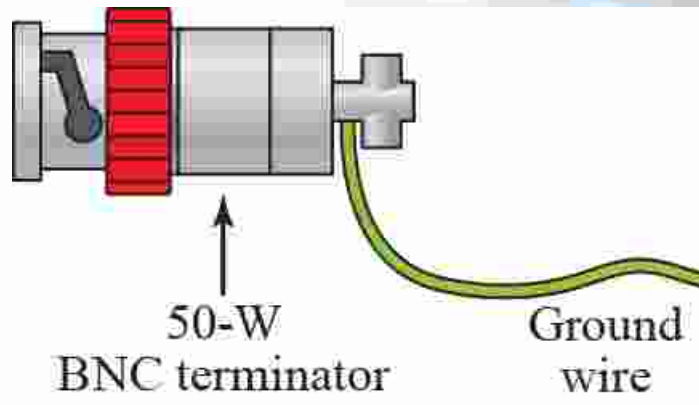
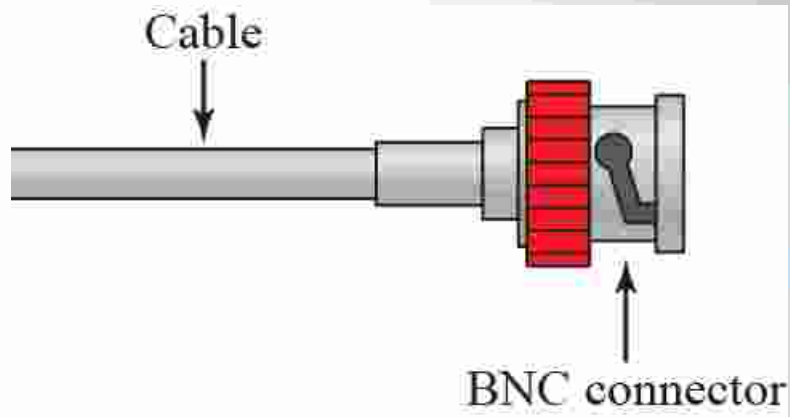
| <i>Category</i> | <i>Impedance</i> | <i>Use</i>     |
|-----------------|------------------|----------------|
| RG-59           | 75 $\Omega$      | Cable TV       |
| RG-58           | 50 $\Omega$      | Thin Ethernet  |
| RG-11           | 50 $\Omega$      | Thick Ethernet |

# Coaxial Cable

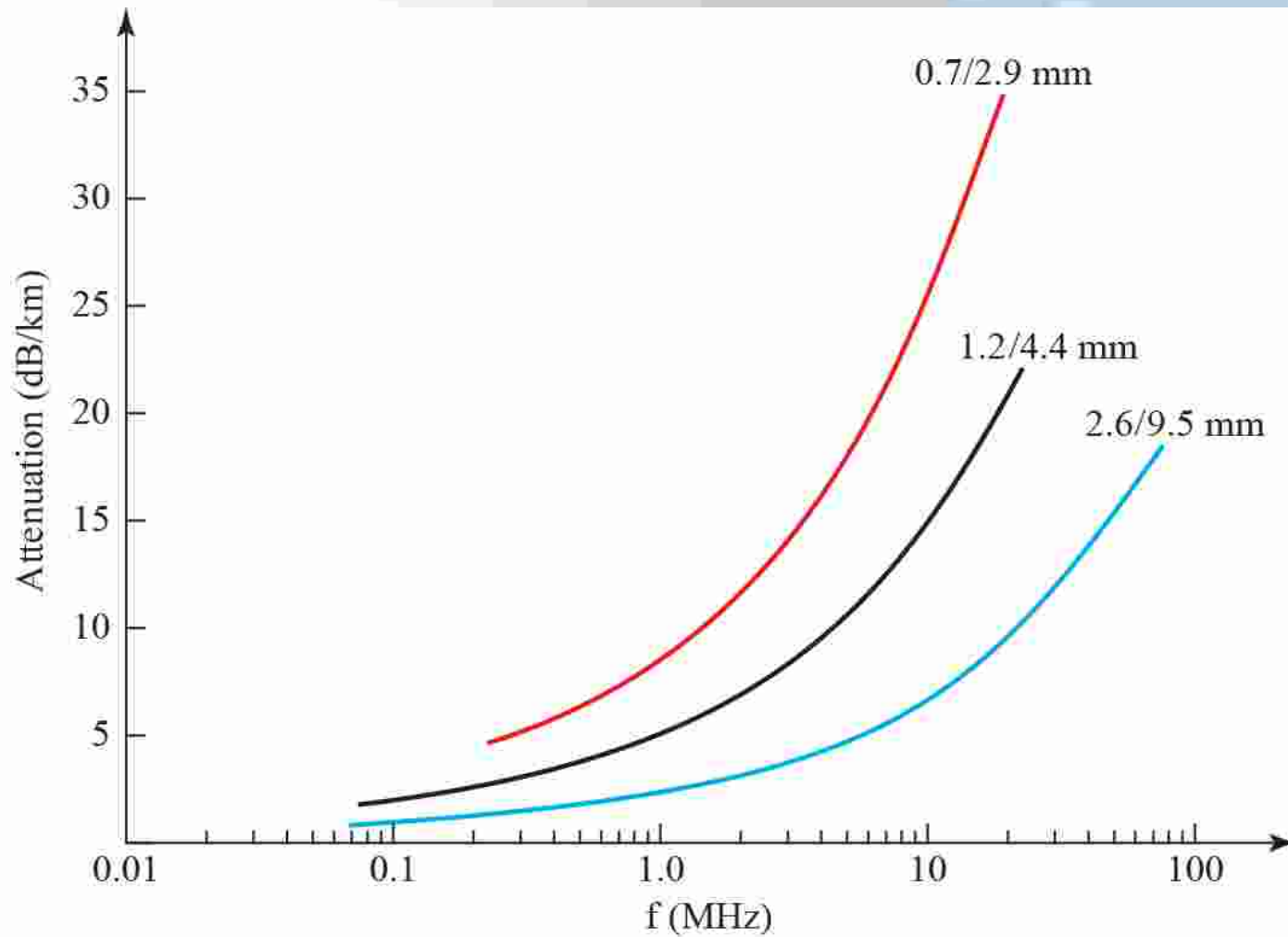
- Carries signals of higher frequency ranges than those in twisted pair cable



# BNC Connectors



# Coaxial Cable Performance

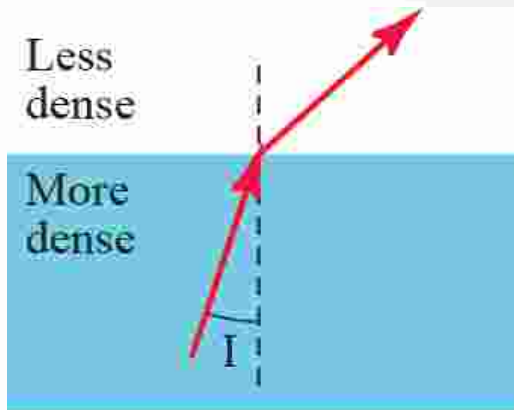




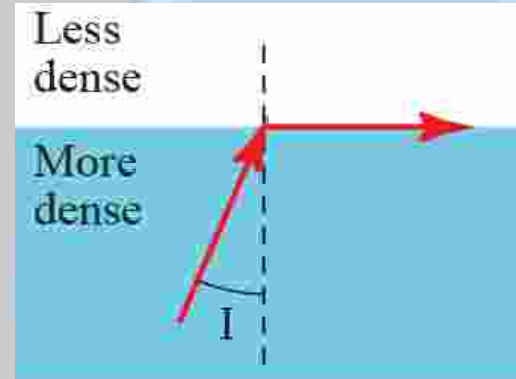
# Fiber-Optic Cable

- Made of glass or plastic and transmits signals in the form of light
- Light travels in a straight line as long as it is moving through a single uniform substance
- If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction

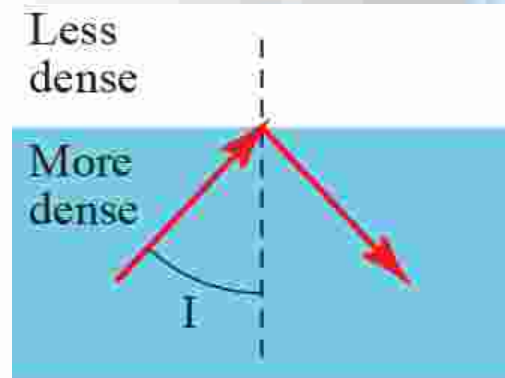
# Bending of Light Ray



$I < \text{critical angle,}$   
refraction



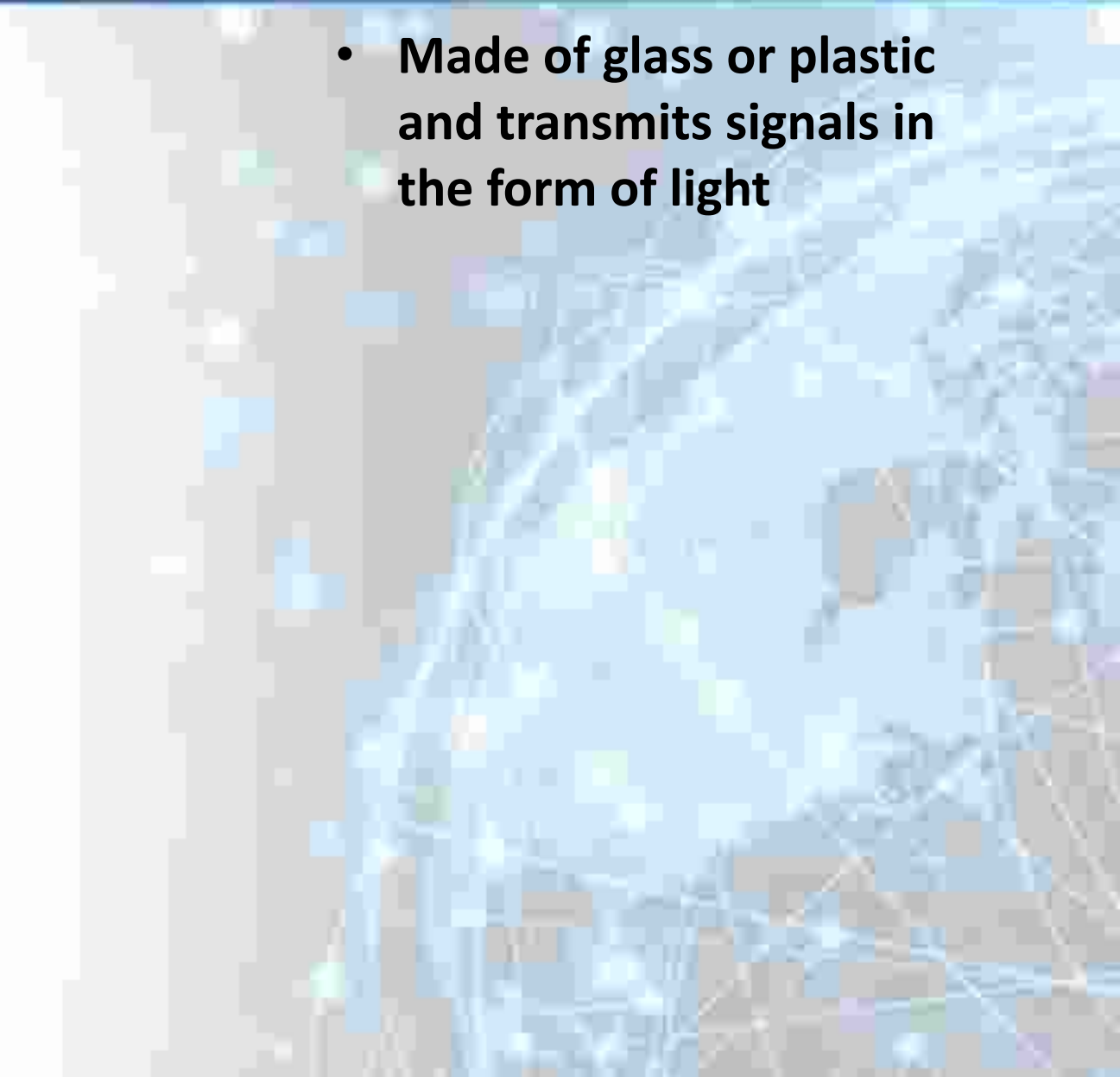
$I = \text{critical angle,}$   
refraction



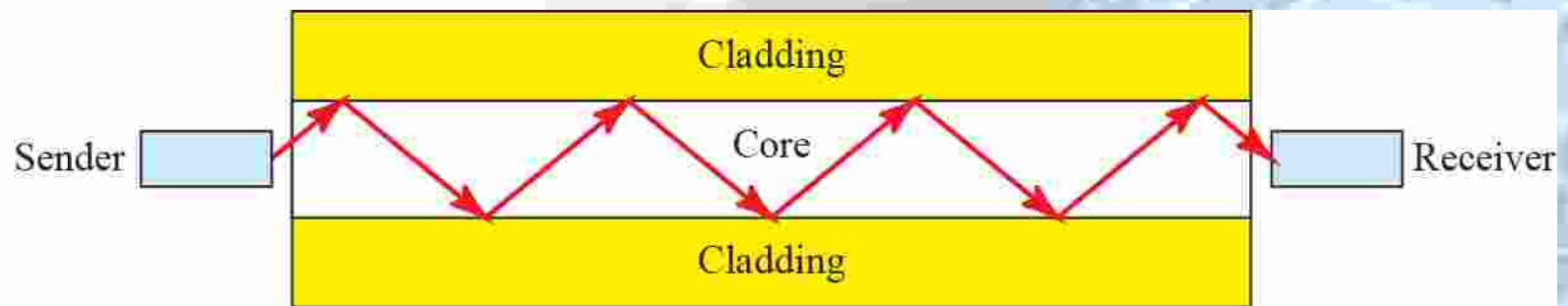
$I > \text{critical angle,}$   
reflection

# Fiber-Optic Cable

- Made of glass or plastic and transmits signals in the form of light

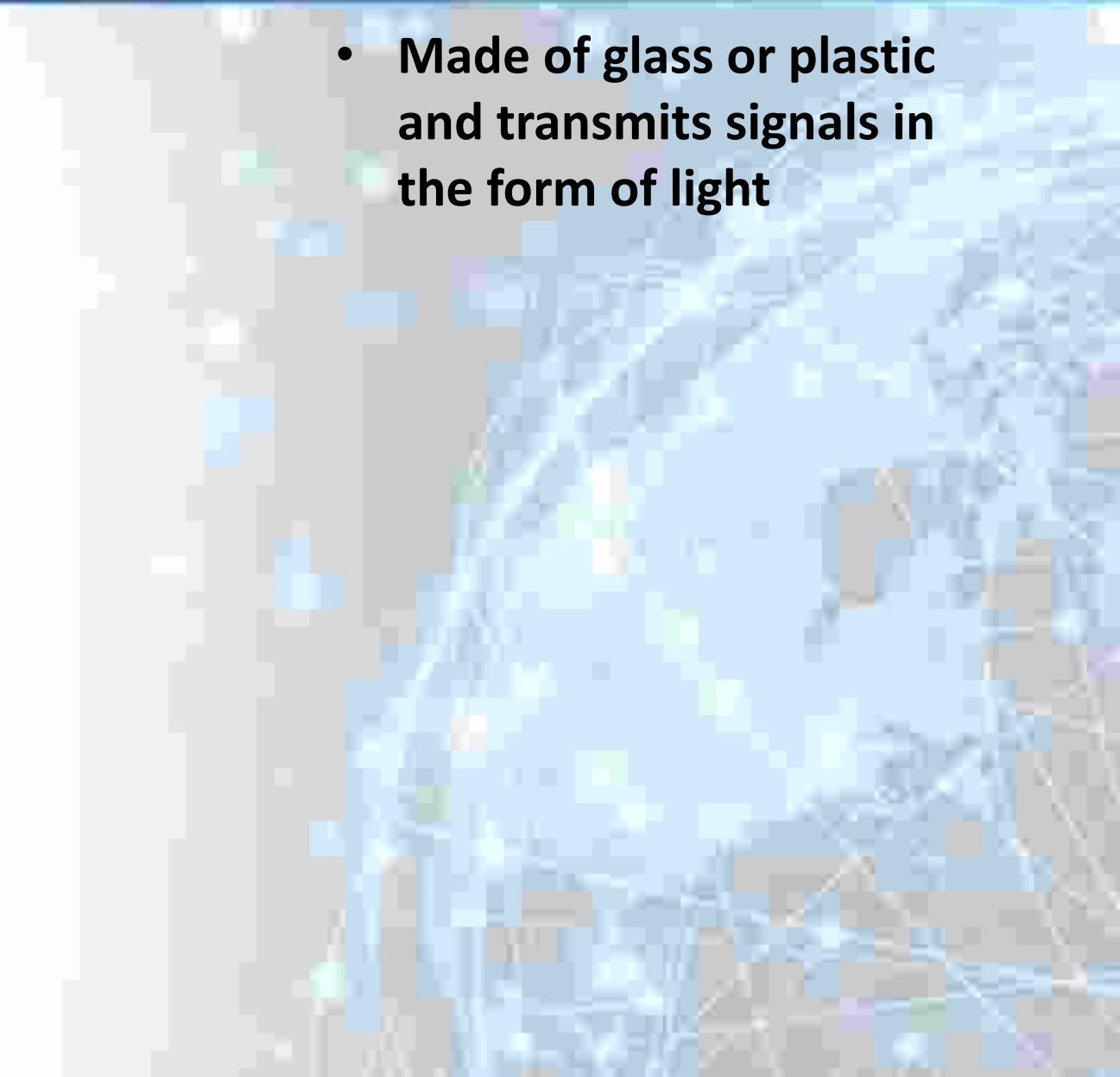


# Optical Fiber

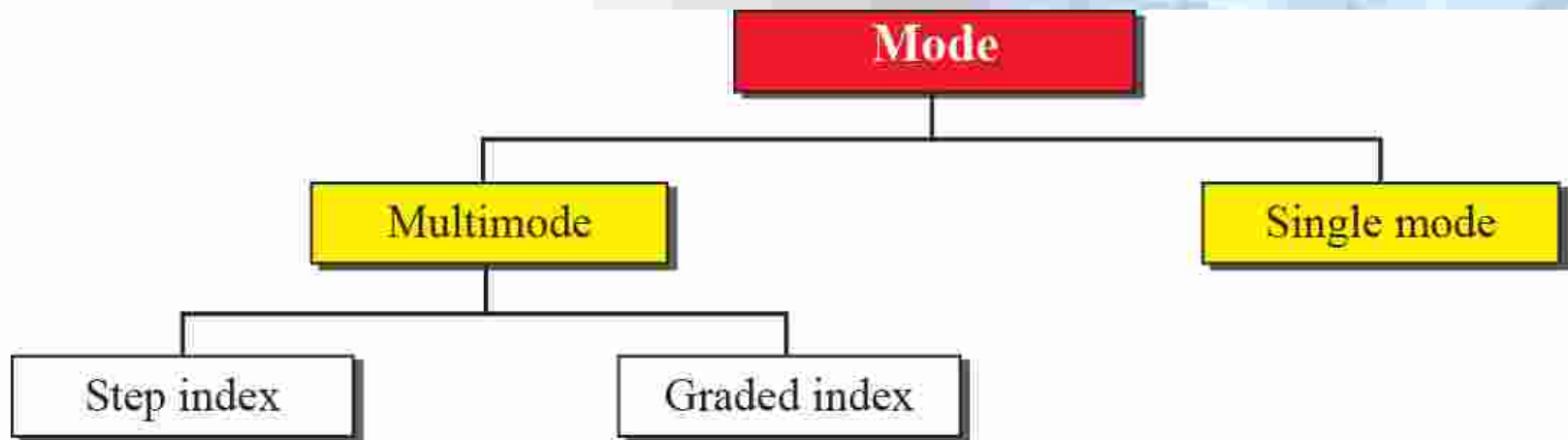


# Fiber-Optic Cable

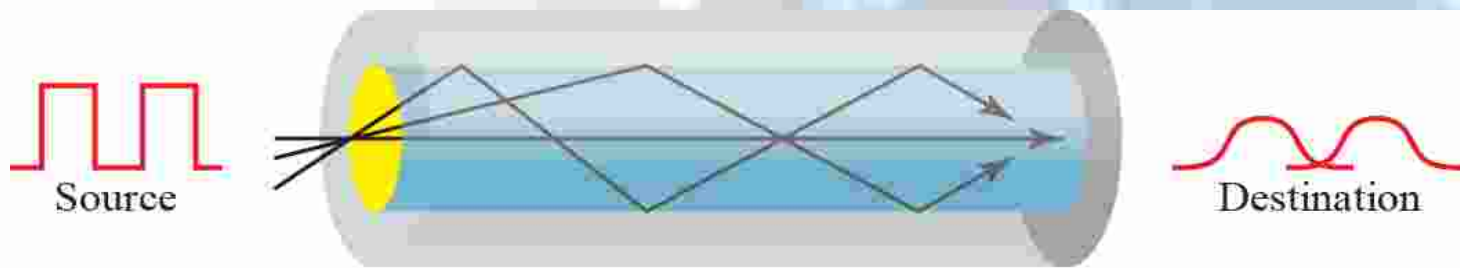
- Made of glass or plastic and transmits signals in the form of light



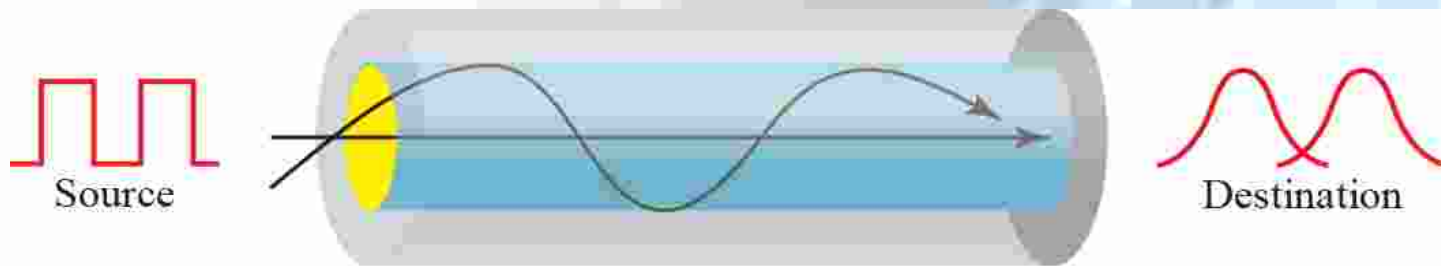
# Propagation Modes



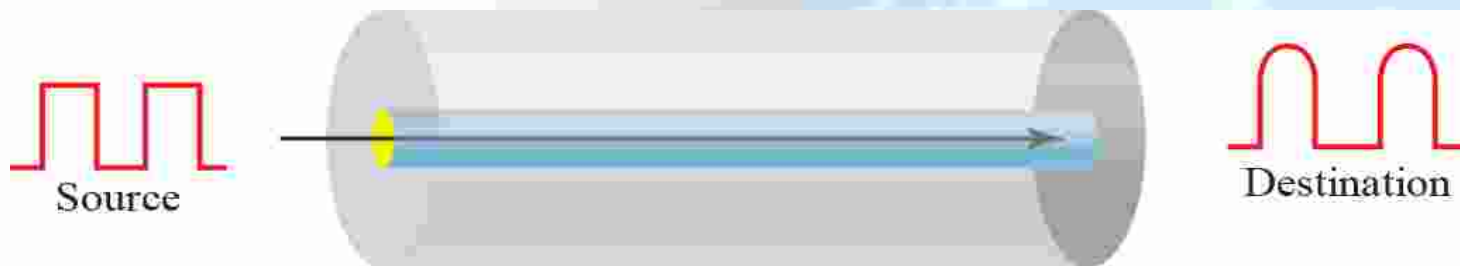
# Modes



a. Multimode, step index



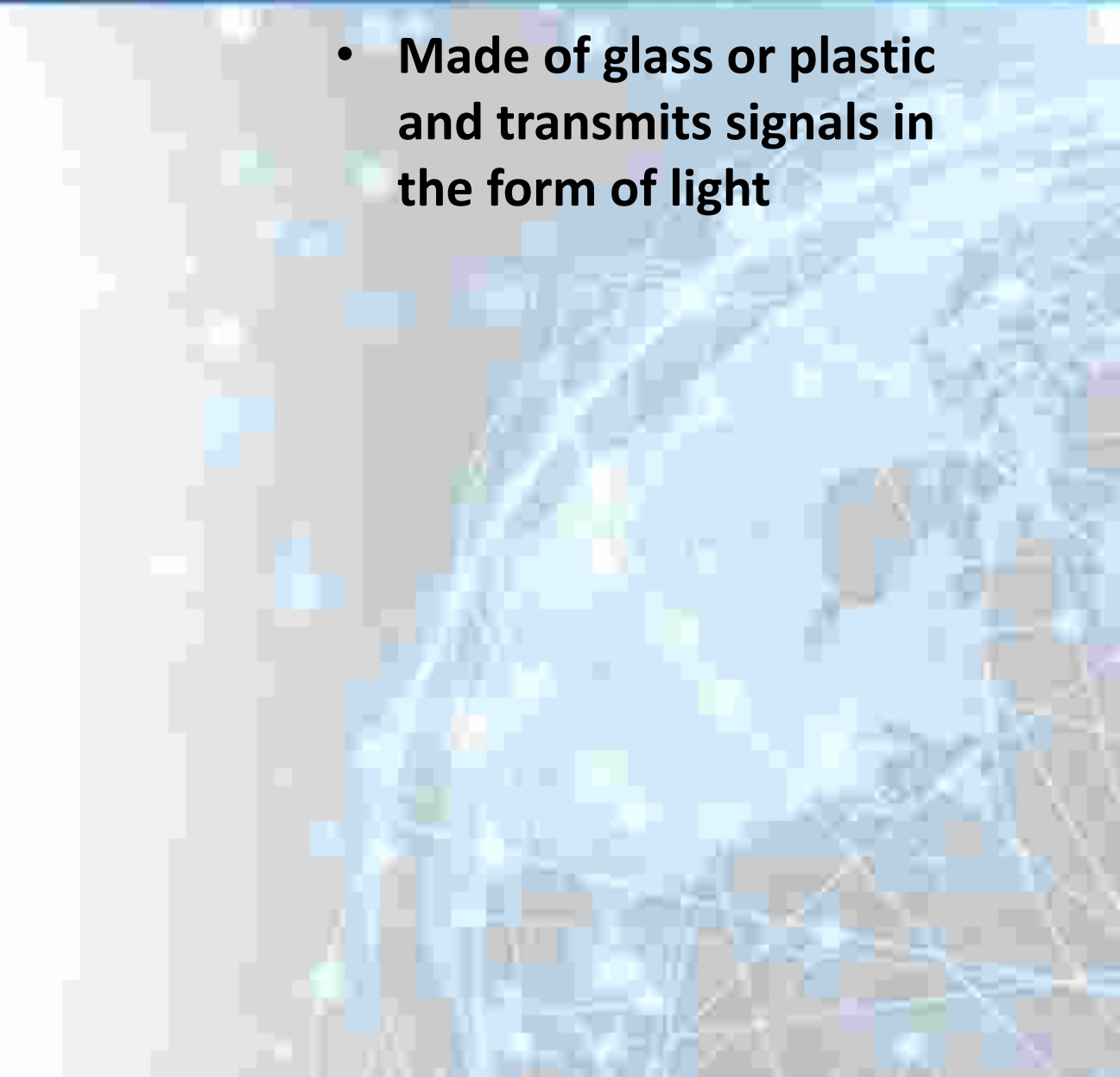
b. Multimode, graded index



c. Single mode

# Fiber-Optic Cable

- Made of glass or plastic and transmits signals in the form of light

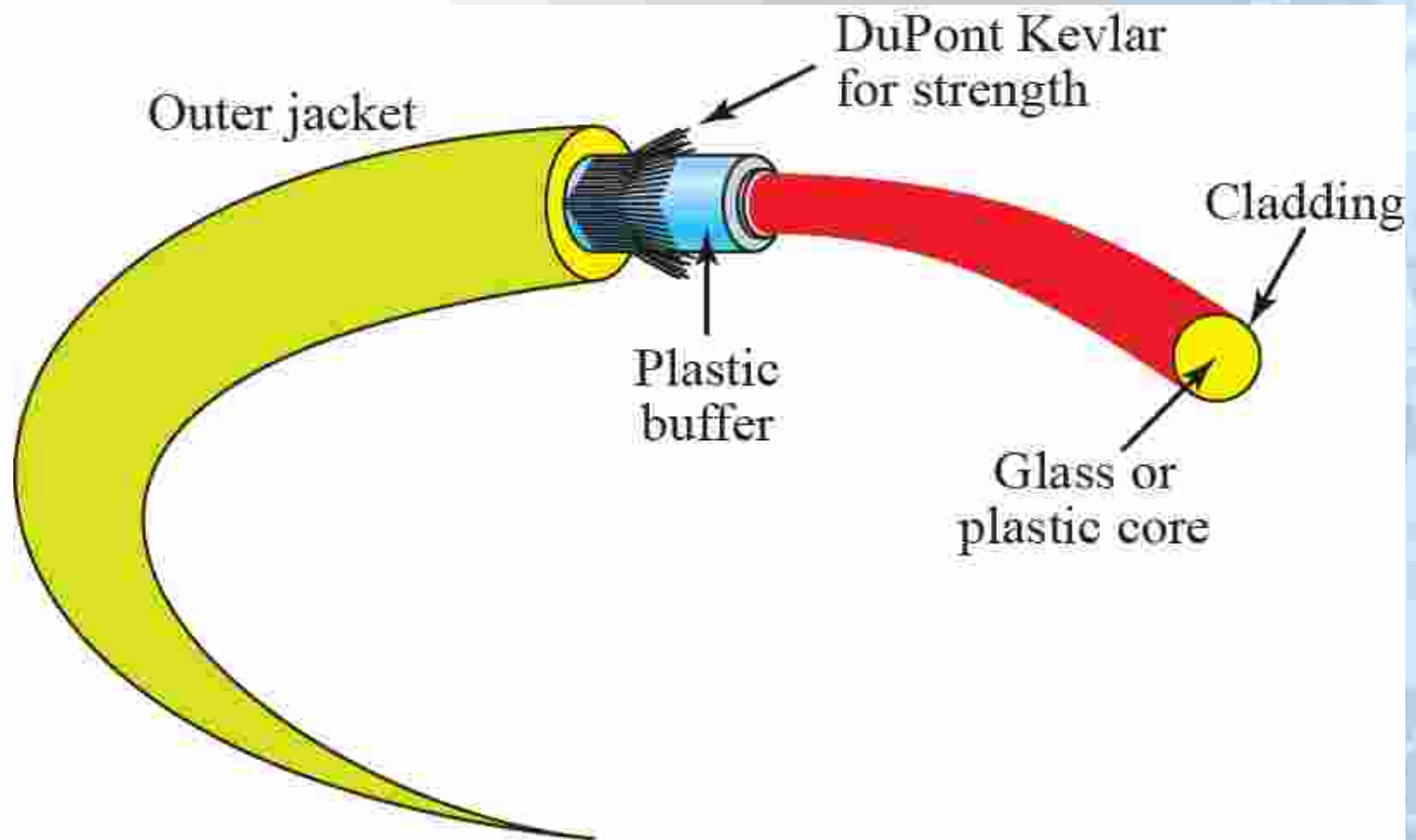




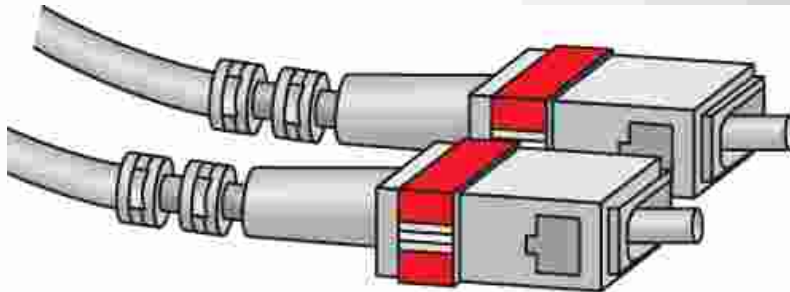
# Fiber Types

| <i>Type</i> | <i>Core (<math>\mu\text{m}</math>)</i> | <i>Cladding (<math>\mu\text{m}</math>)</i> | <i>Mode</i>             |
|-------------|----------------------------------------|--------------------------------------------|-------------------------|
| 50/125      | 50.0                                   | 125                                        | Multimode, graded index |
| 62.5/125    | 62.5                                   | 125                                        | Multimode, graded index |
| 100/125     | 100.0                                  | 125                                        | Multimode, graded index |
| 7/125       | 7.0                                    | 125                                        | Single mode             |

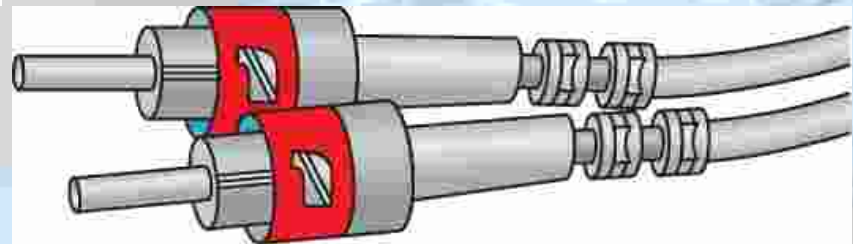
# Fiber Composition



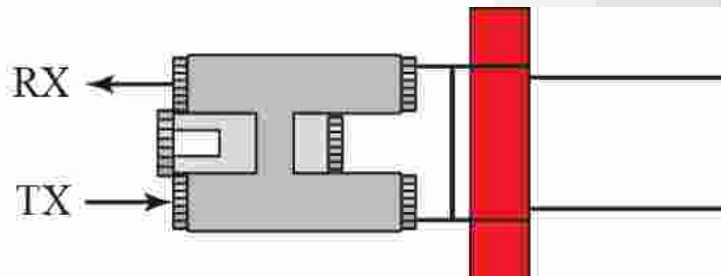
# Fiber-Optic Cable Connector



SC connector

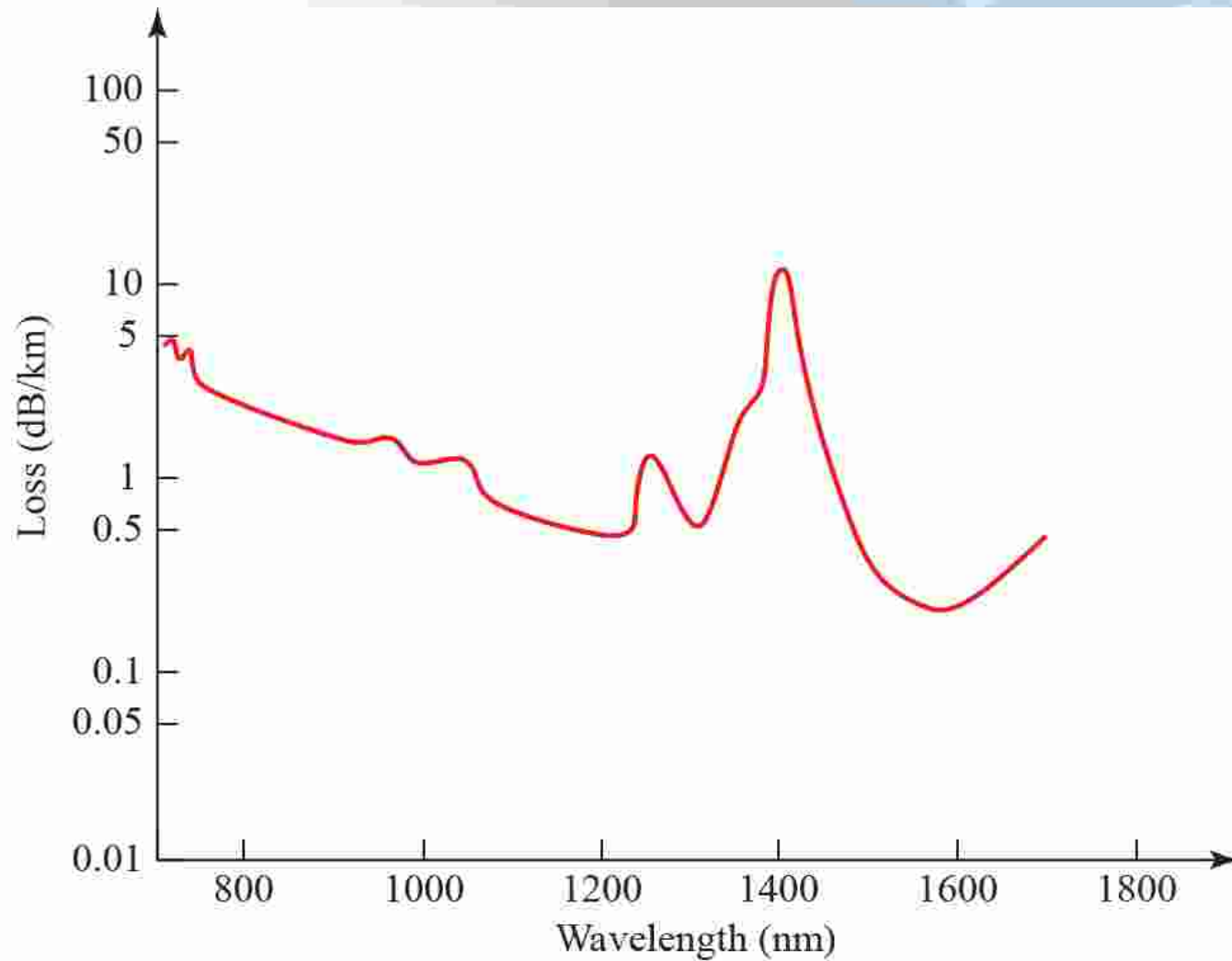


ST connector



MT-RJ connector

# Optical Fiber Performance



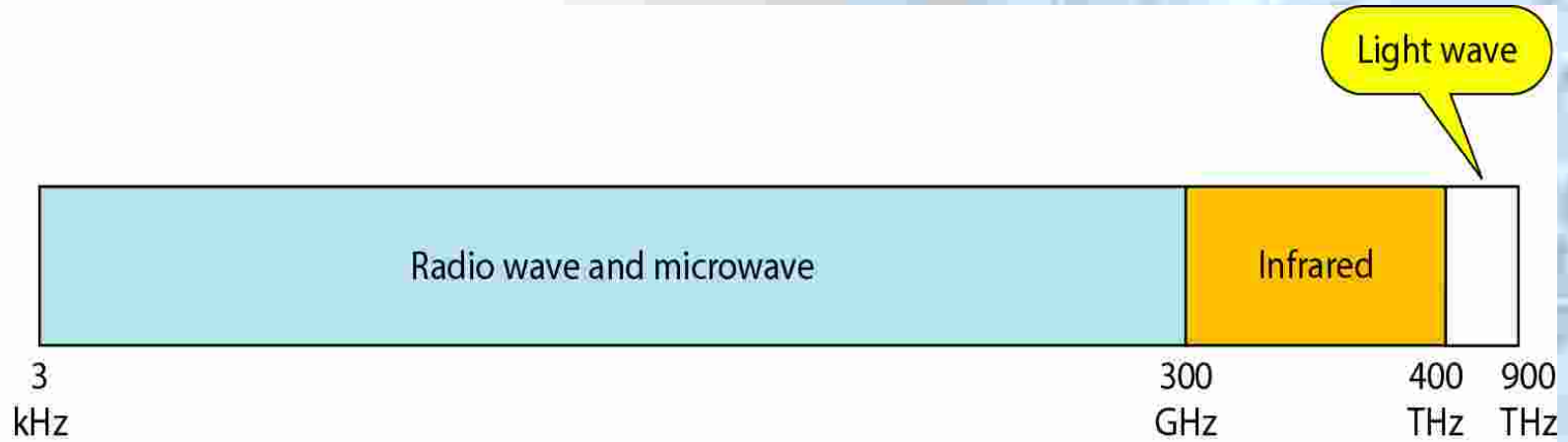
# Advantages & Disadvantages

- Higher Bandwidth
- Less Attenuation
- Less EM Interference
- Light Weight
- Less corrosive than copper
- Installation/Maintenance
- Unidirectional
- Cost

# Unguided Media

- Unguided medium transport waves without using a physical conductor
- Often referred to wireless communication
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them

# Electromagnetic Spectrum



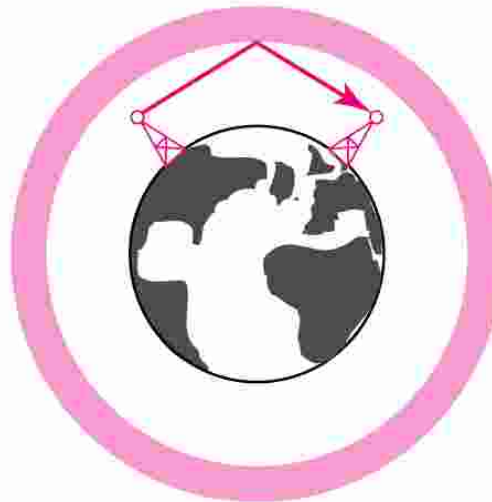
# Propagation Methods

Ionosphere



Ground propagation  
(below 2 MHz)

Ionosphere



Sky propagation  
(2–30 MHz)

Ionosphere



Line-of-sight propagation  
(above 30 MHz)



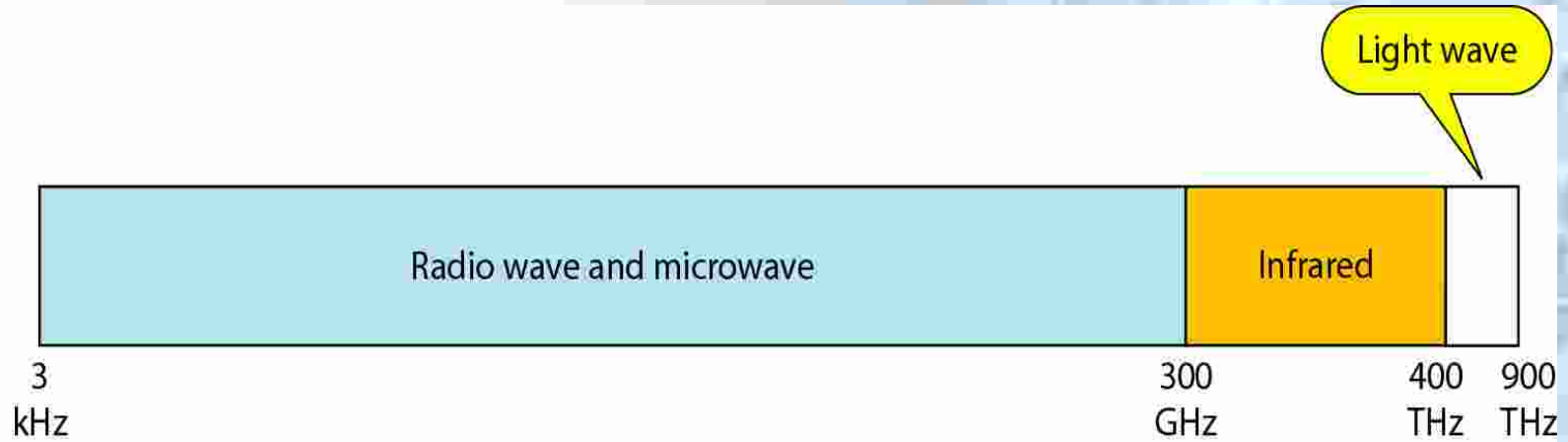
# Bands

| <i>Band</i>                    | <i>Range</i>  | <i>Propagation</i>    | <i>Application</i>                         |
|--------------------------------|---------------|-----------------------|--------------------------------------------|
| very low frequency (VLF)       | 3–30 kHz      | Ground                | Long-range radio navigation                |
| low frequency (LF)             | 30–300 kHz    | Ground                | Radio beacons and navigational locators    |
| middle frequency (MF)          | 300 kHz–3 MHz | Sky                   | AM radio                                   |
| high frequency (HF)            | 3–30 MHz      | Sky                   | Citizens band (CB), ship/aircraft          |
| very high frequency (VHF)      | 30–300 MHz    | Sky and line-of-sight | VHF TV, FM radio                           |
| ultrahigh frequency (UHF)      | 300 MHz–3 GHz | Line-of-sight         | UHF TV, cellular phones, paging, satellite |
| superhigh frequency (SHF)      | 3–30 GHz      | Line-of-sight         | Satellite                                  |
| extremely high frequency (EHF) | 30–300 GHz    | Line-of-sight         | Radar, satellite                           |

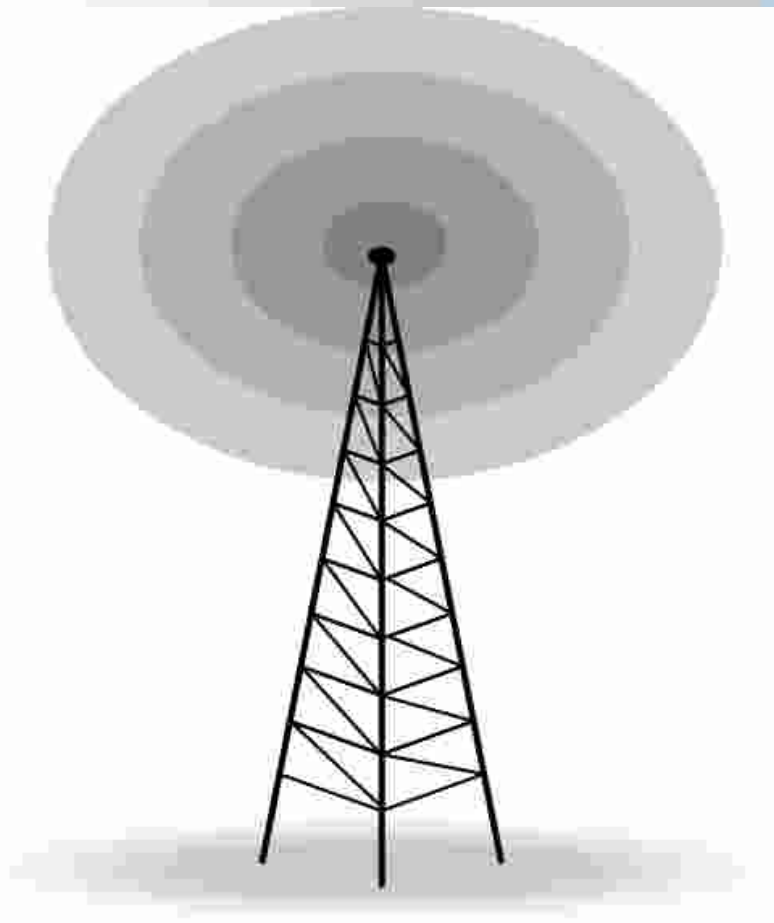
# Radio Waves

- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves
- Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called microwaves

# Electromagnetic Spectrum



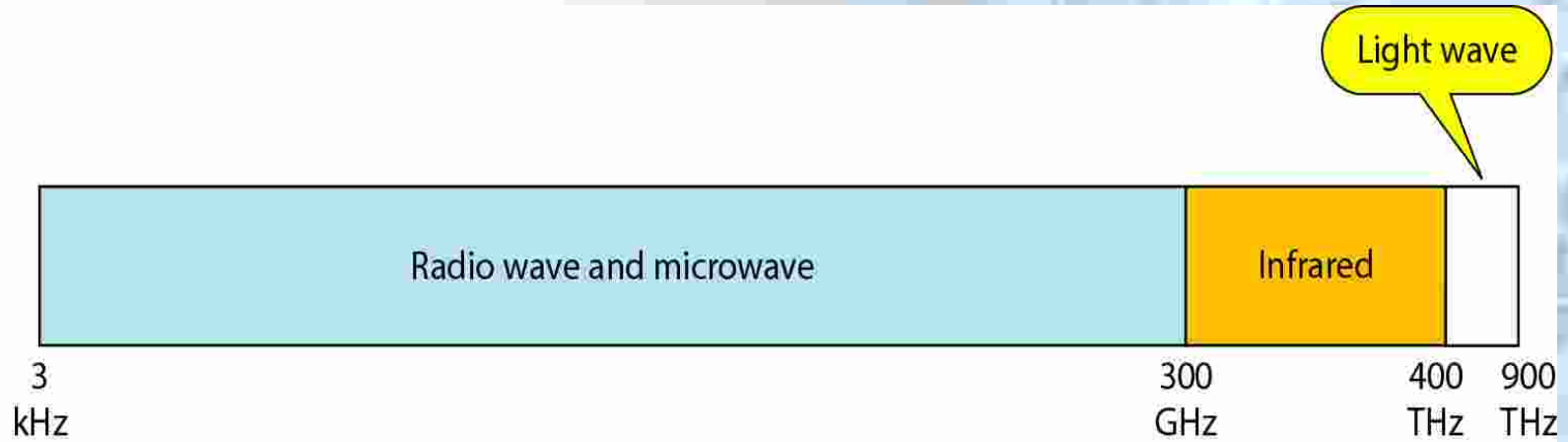
# Omnidirectional Antenna



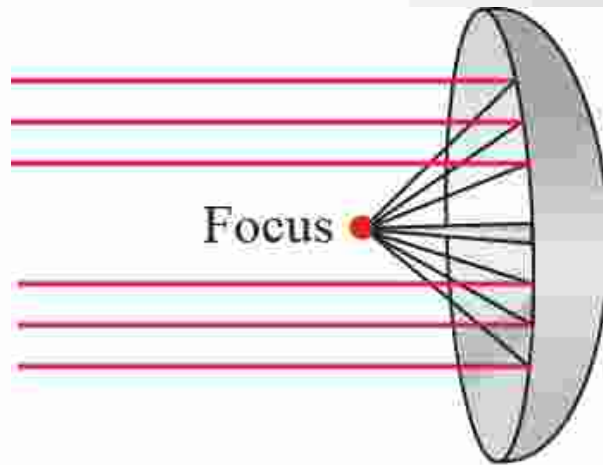
# Microwaves

- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves
- Microwaves are unidirectional
- When an antenna transmits microwaves, they can be narrowly focused

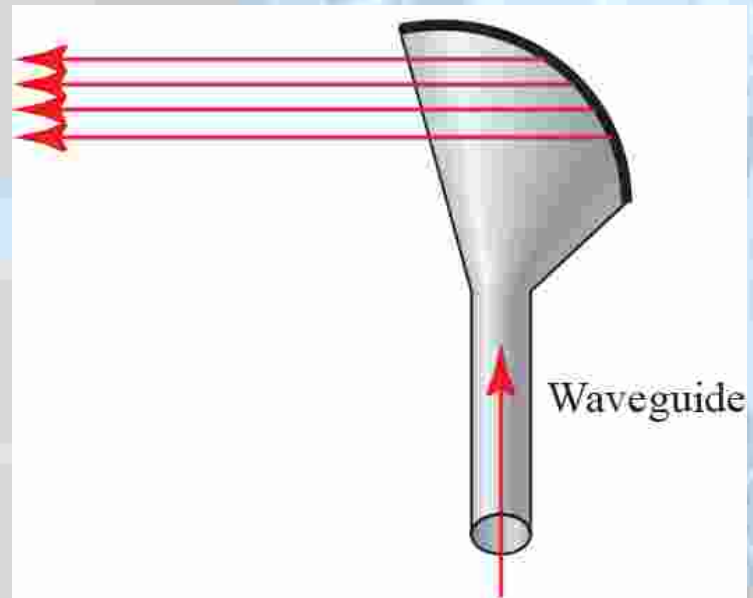
# Electromagnetic Spectrum



# Unidirectional Antennas



a. Parabolic dish antenna



b. Horn antenna

# Infrared

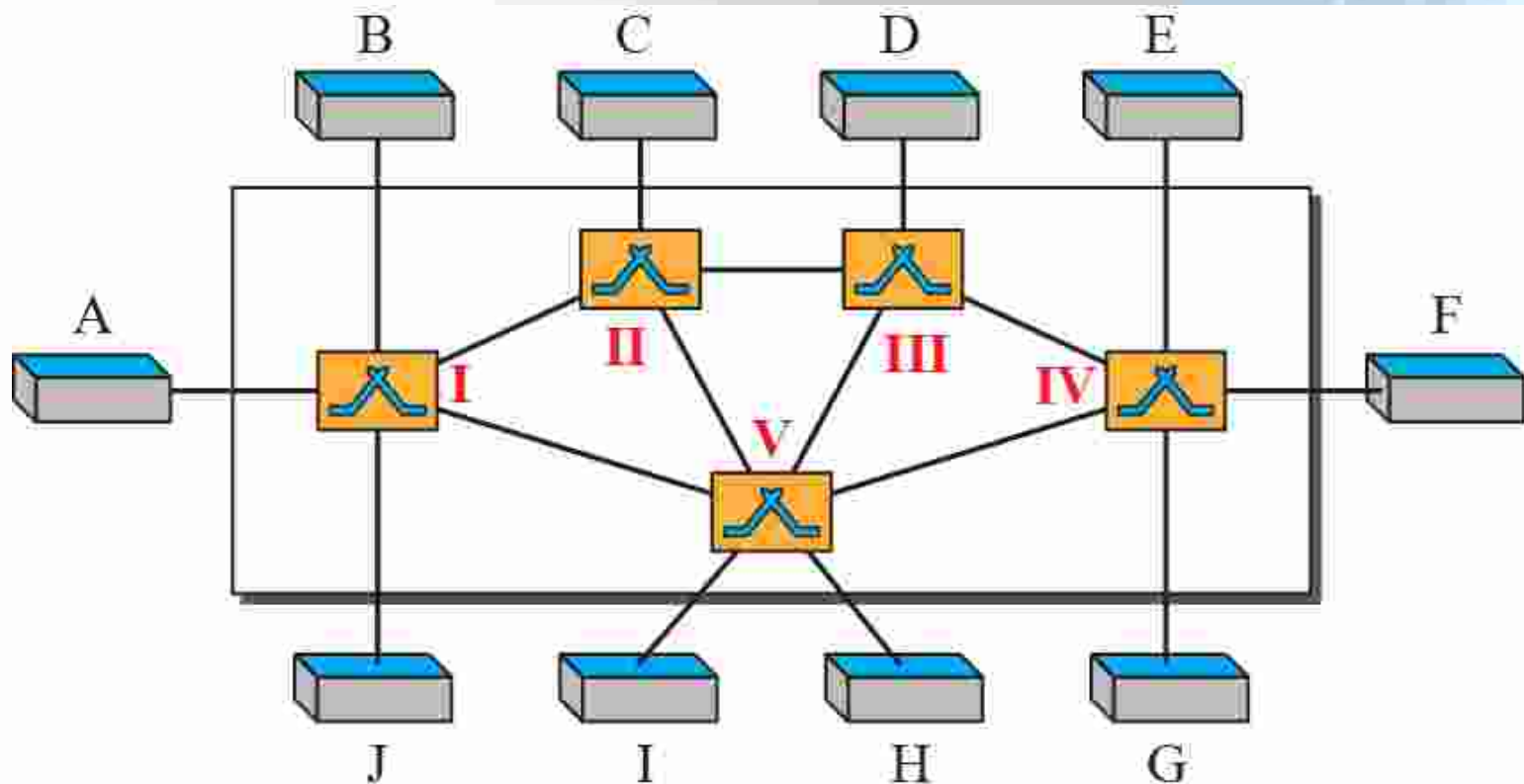
- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication
- Infrared waves, having high frequencies, cannot penetrate walls
- Prevents interference between one system and another



# Switching

- A network is a set of connected devices
- Problem of how to connect multiple devices to make one-to-one communication possible
- The solution is Switching
- Switched network consists of a series of switches

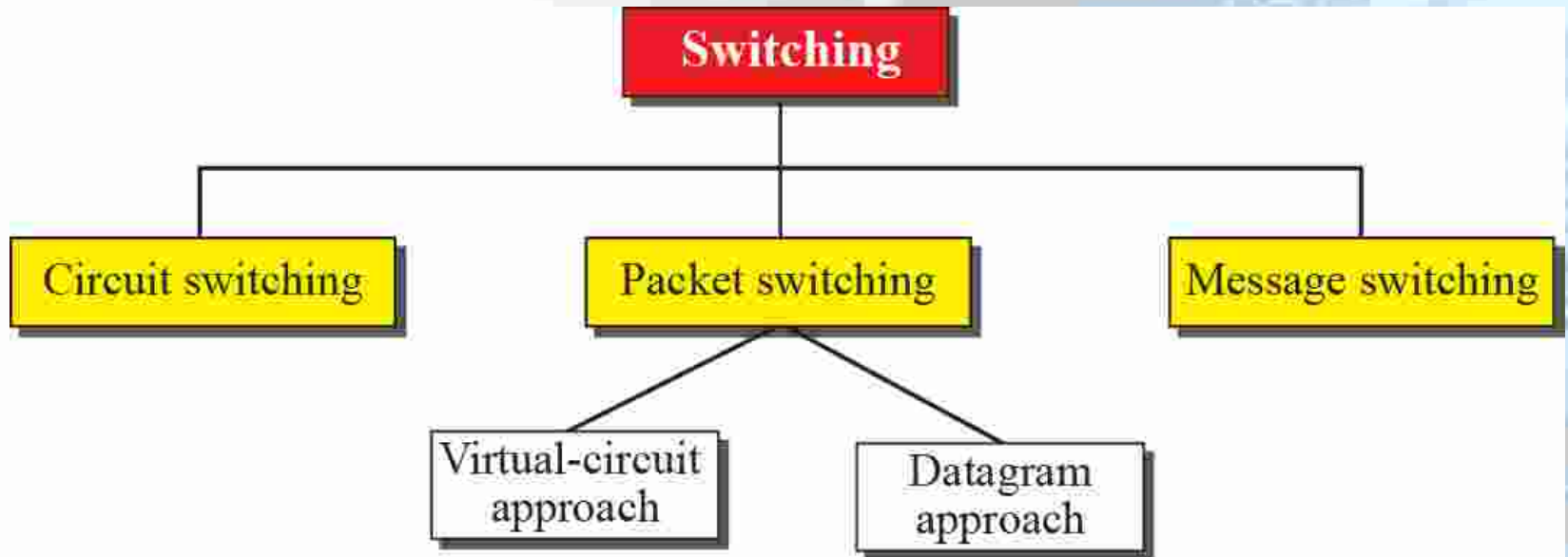
# Switched Network



# Three Methods of Switching

- **Three Methods:**
  - ✓ **Circuit Switching**
  - ✓ **Packet Switching**
  - ✓ **Message switching**
- **The first two are commonly used today**
- **The third has been phased out in general communications**

# Taxonomy of Switched Networks

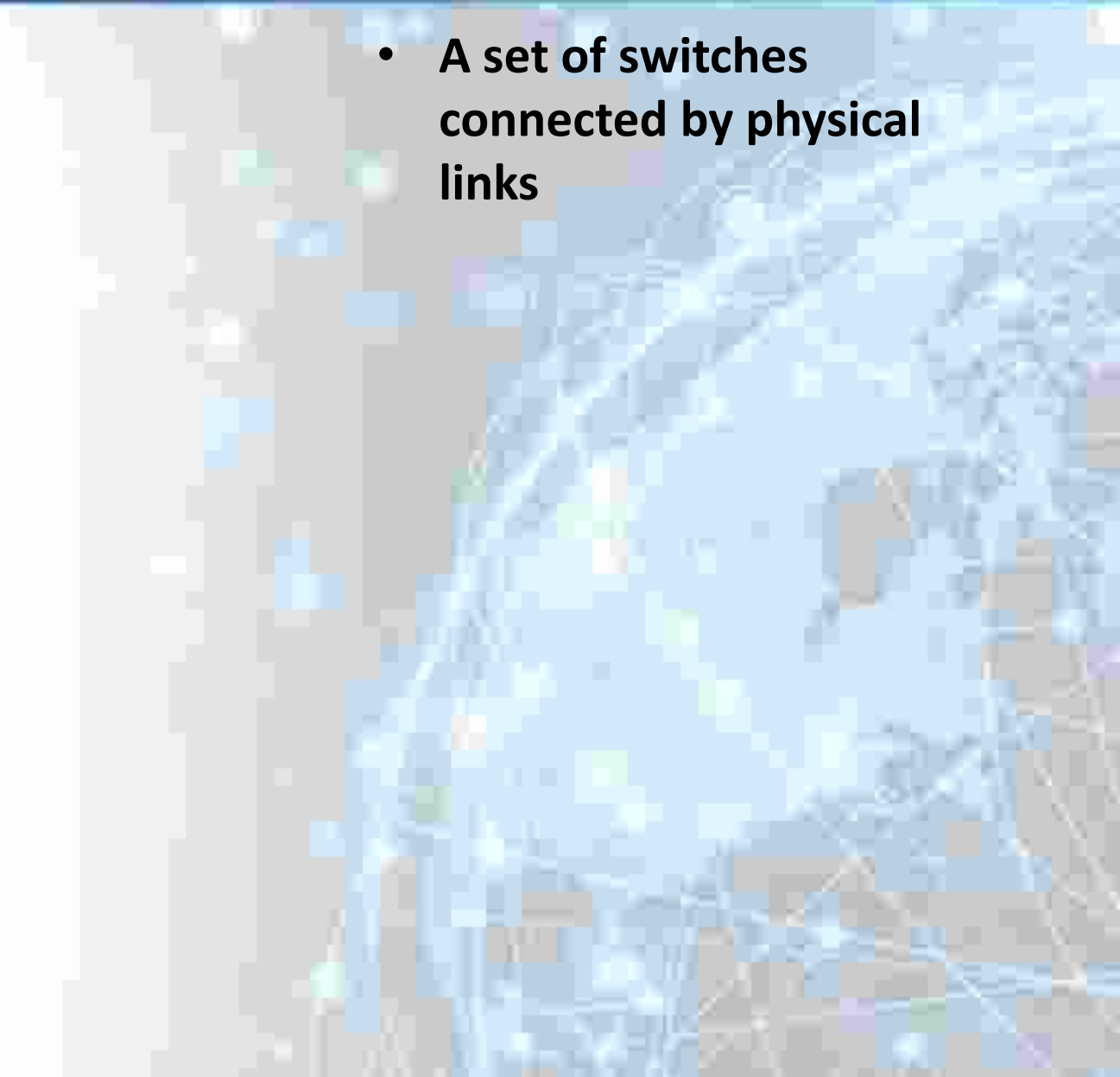


# Circuit-switched Networks

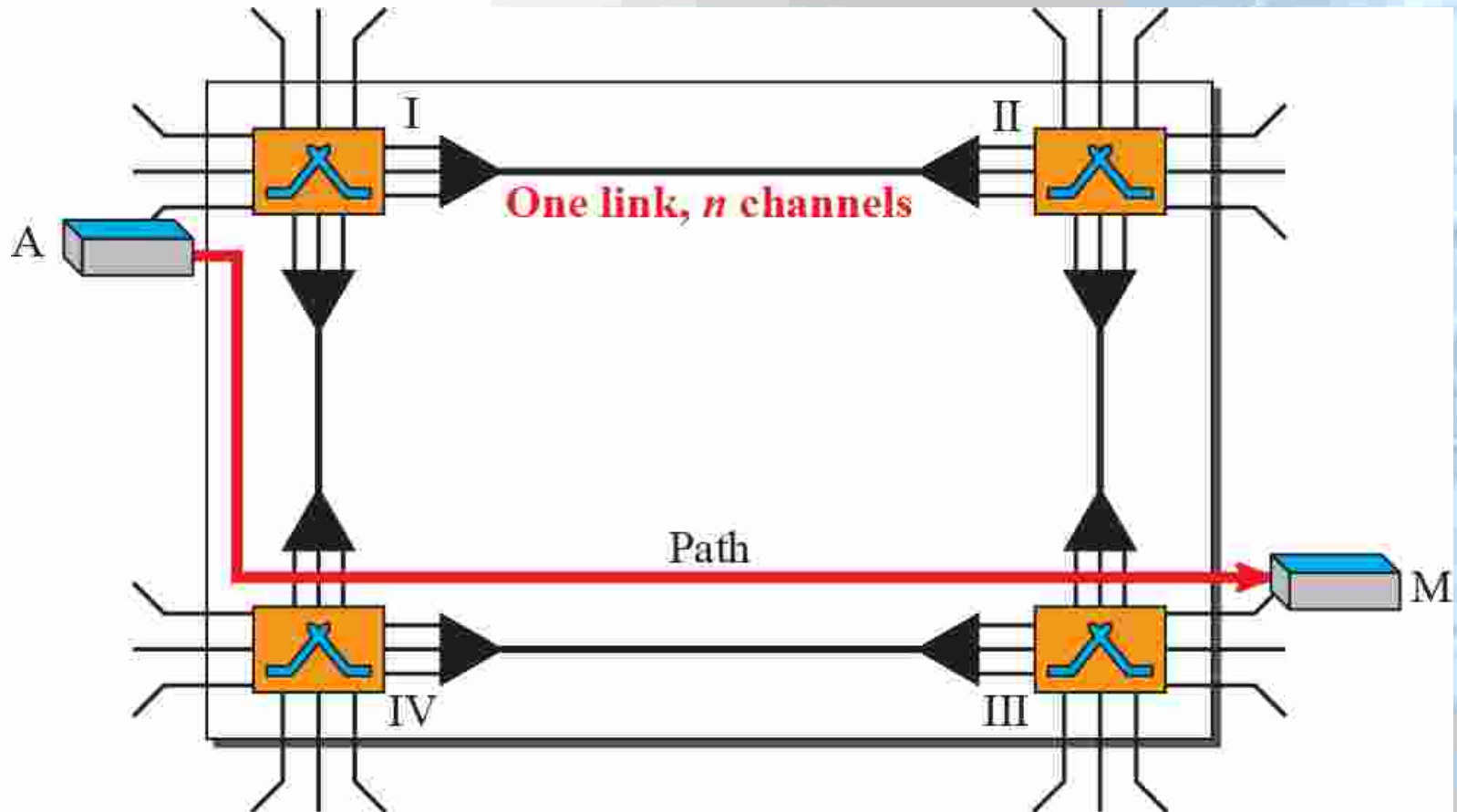
- A set of switches connected by physical links
- A connection between two stations is a dedicated path made of one or more links
- Each connection uses only one dedicated channel on each link
- Each link is normally divided into  $n$  channels by using FDM or TDM

# Circuit-switched Networks

- A set of switches connected by physical links

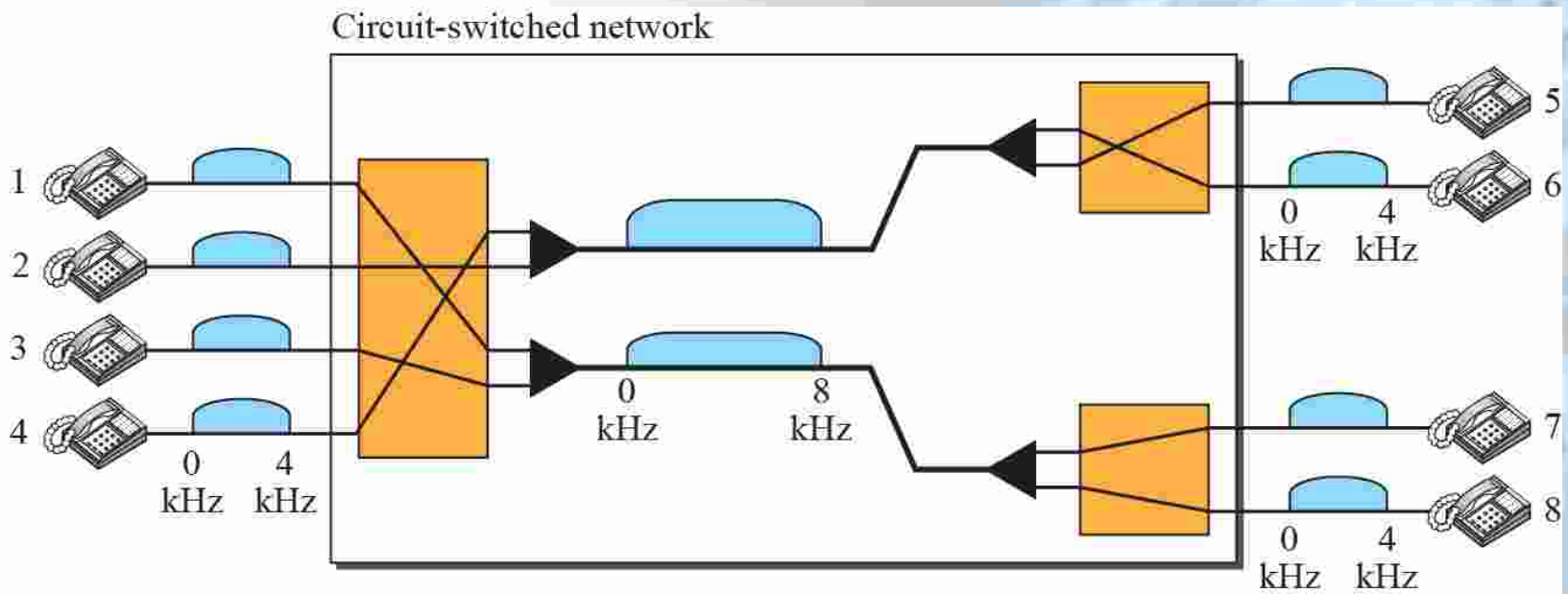


# A Circuit-Switched Network



# Example

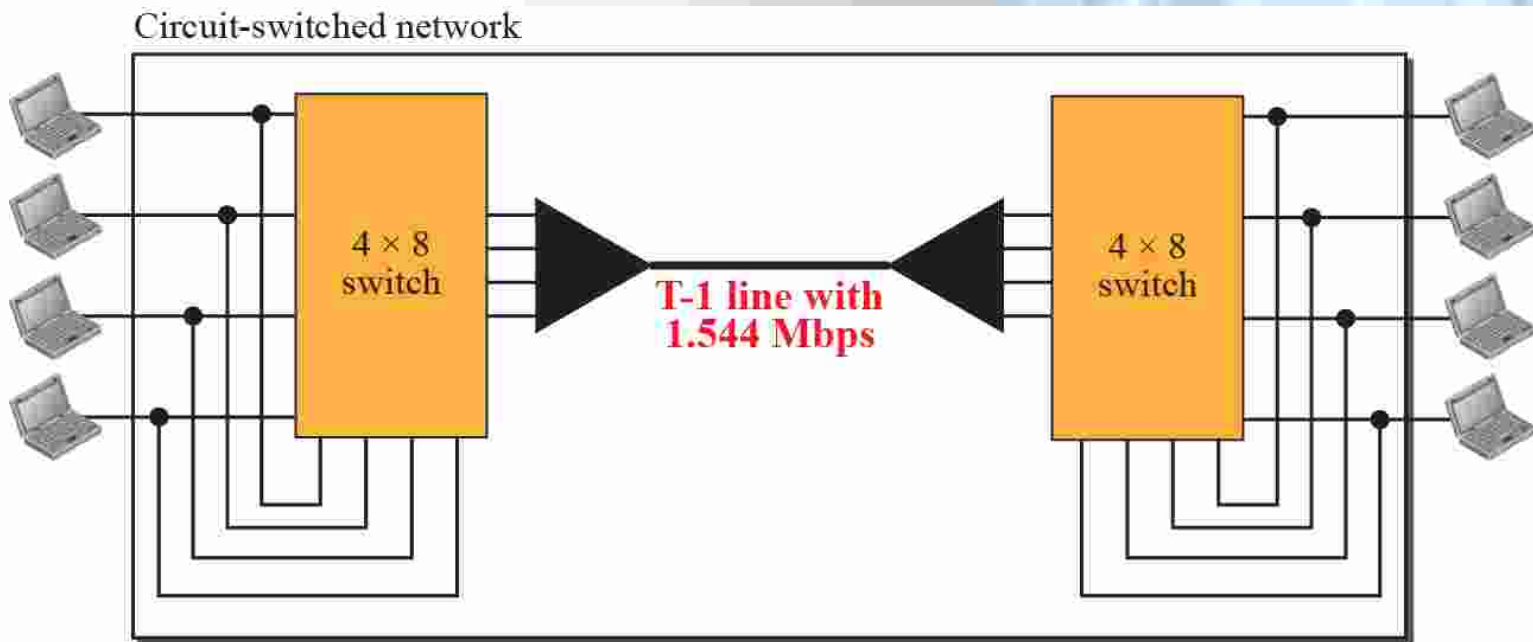
As a trivial example, let us use a circuit-switched network to connect eight telephones in a small area. Communication is through 4-kHz voice channels. We assume that each link uses FDM to connect a maximum of two voice channels. The bandwidth of each link is then 8 kHz.





# Example

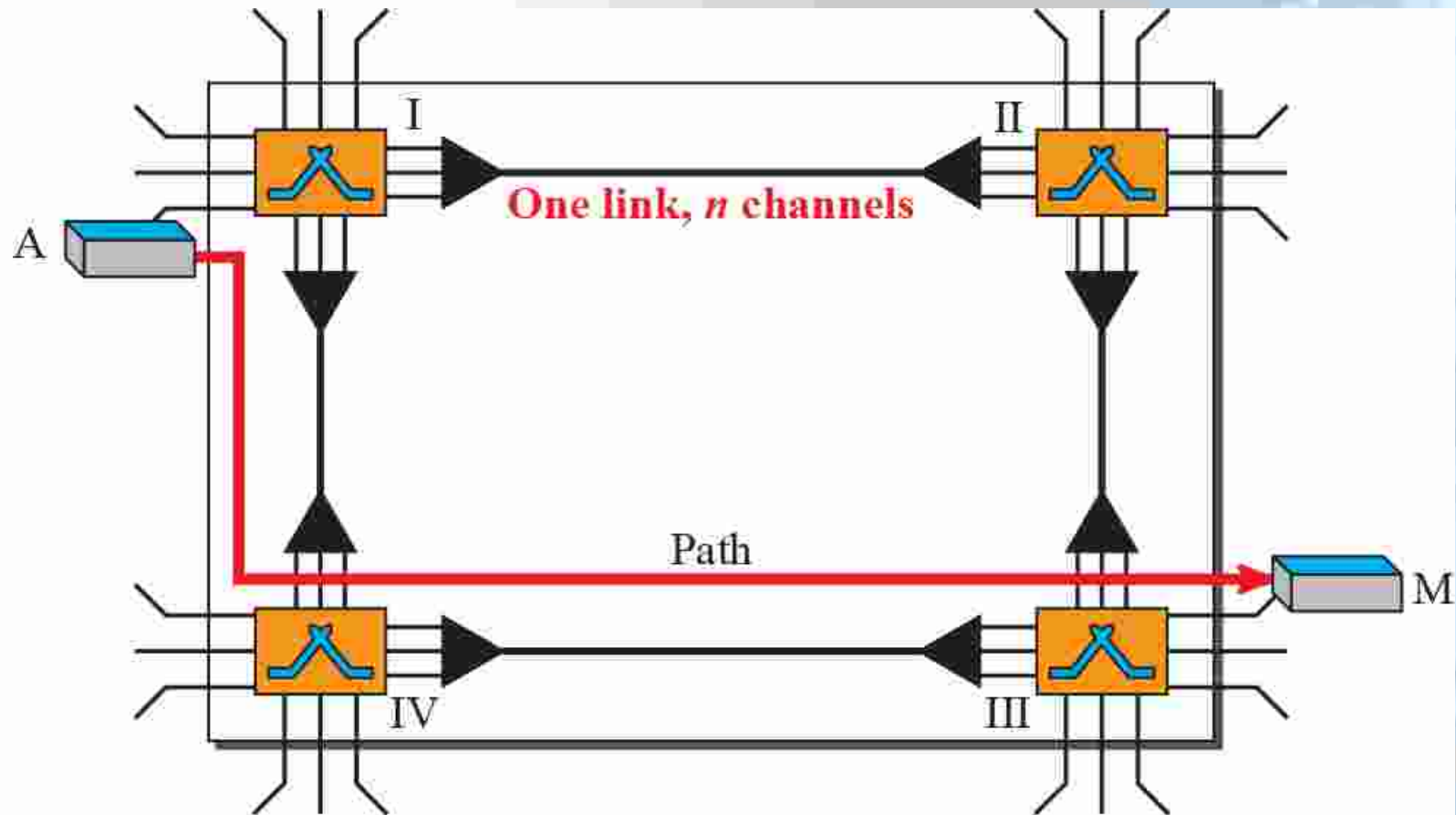
As another example, consider a circuit-switched network that connects computers in two remote offices of a private company. The offices are connected using a T-1 line leased from a communication service provider. There are two  $4 \times 8$  (4 inputs and 8 outputs) switches in this network.



# Three Phases in a Circuit Switched Network

- The actual communication in a circuit-switched network requires 3 phases:
  - ✓ Connection Setup
  - ✓ Data Transfer
  - ✓ Connection Teardown

# Three Phases in a Circuit Switched Network



# Efficiency of a Circuit-Switched Network

- Not as efficient as packet switching because resources are allocated during the entire duration of the connection and these resources are unavailable to other connections

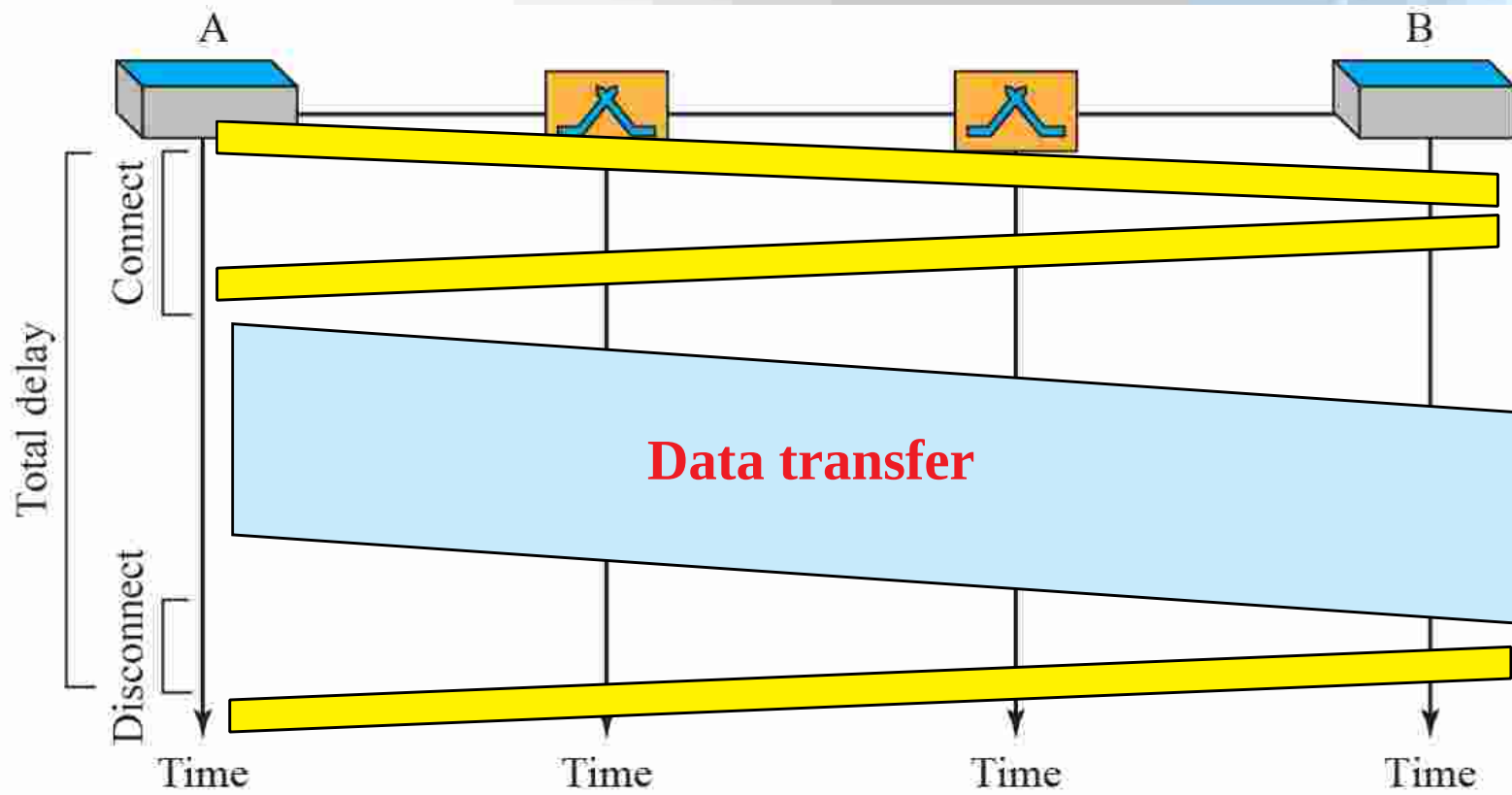
# Efficiency of a Circuit-Switched Network

- In a telephone network, people normally terminate the communication when they have finished their conversation
- Data Network is an issue

# Delay in a Circuit-Switched Network

- **Circuit switched networks have low efficiency but minimal delay**
- **Data is not delayed at each switch; the resources are allocated for the duration of the connection**

# Delay in a Circuit-Switched Network



# Packet Switching

- If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size
- The size of the packet is determined by the network and the governing protocol



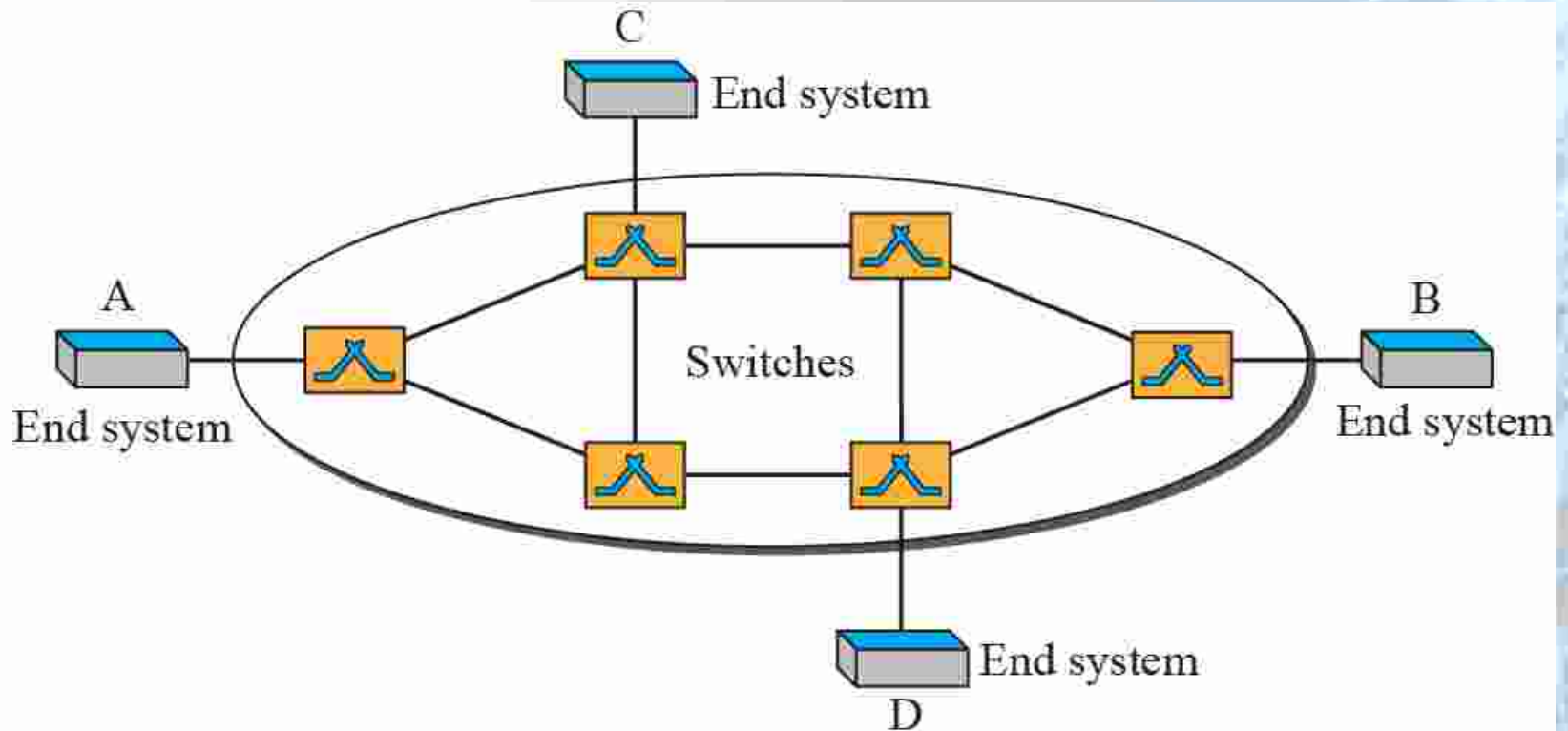
# Datagram Networks

- Each packet is treated independently of all others.
- Even if a packet is part of a multi-packet transmission, the network treats it as though it existed alone
- Packets are referred to as datagrams

# Virtual-Circuit Networks

- A virtual-circuit network is a cross between a circuit-switched network and a datagram network

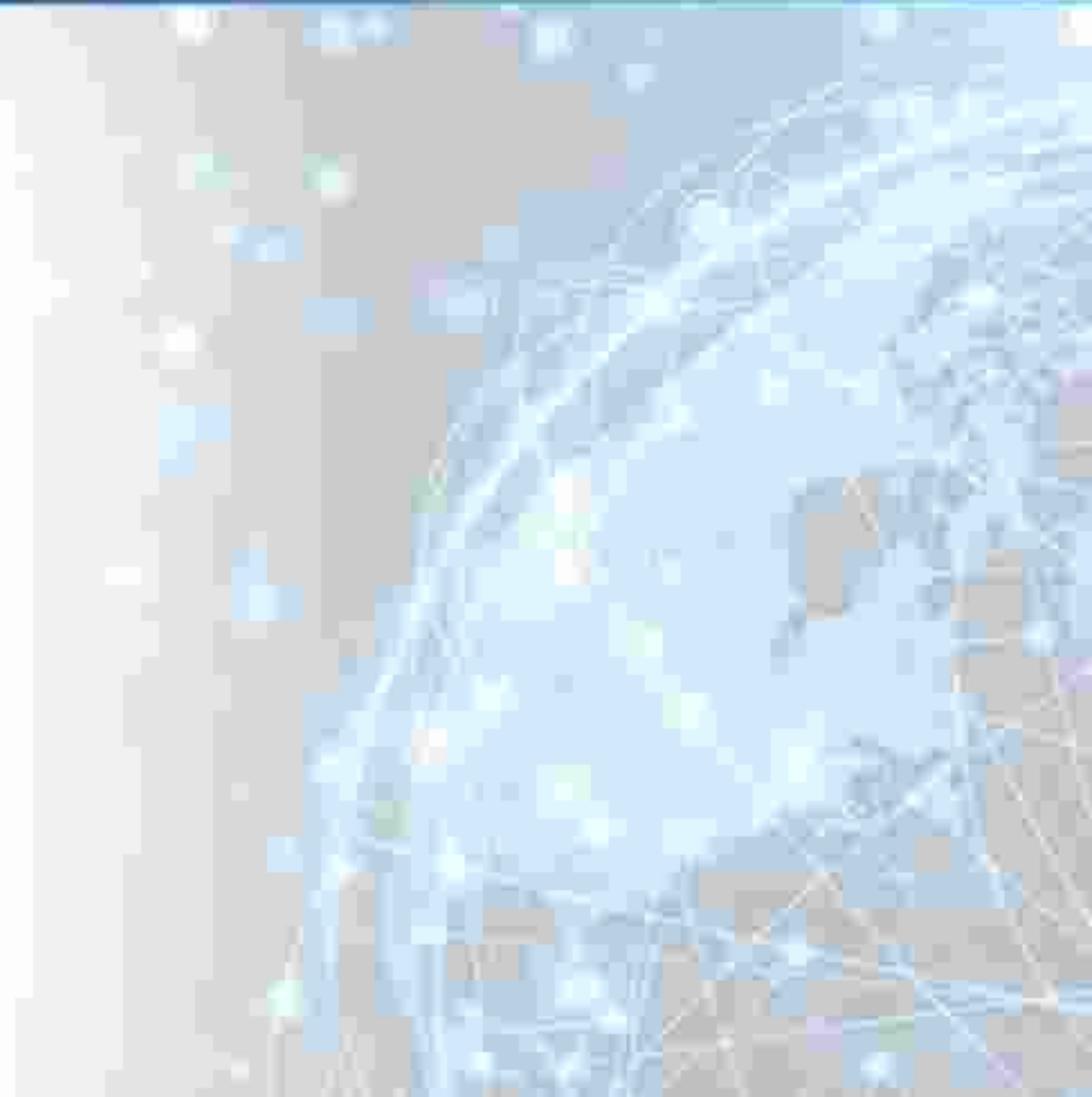
# Virtual-circuit network



# Virtual-Circuit Networks

- A virtual-circuit network is a cross between a circuit-switched network and a datagram network

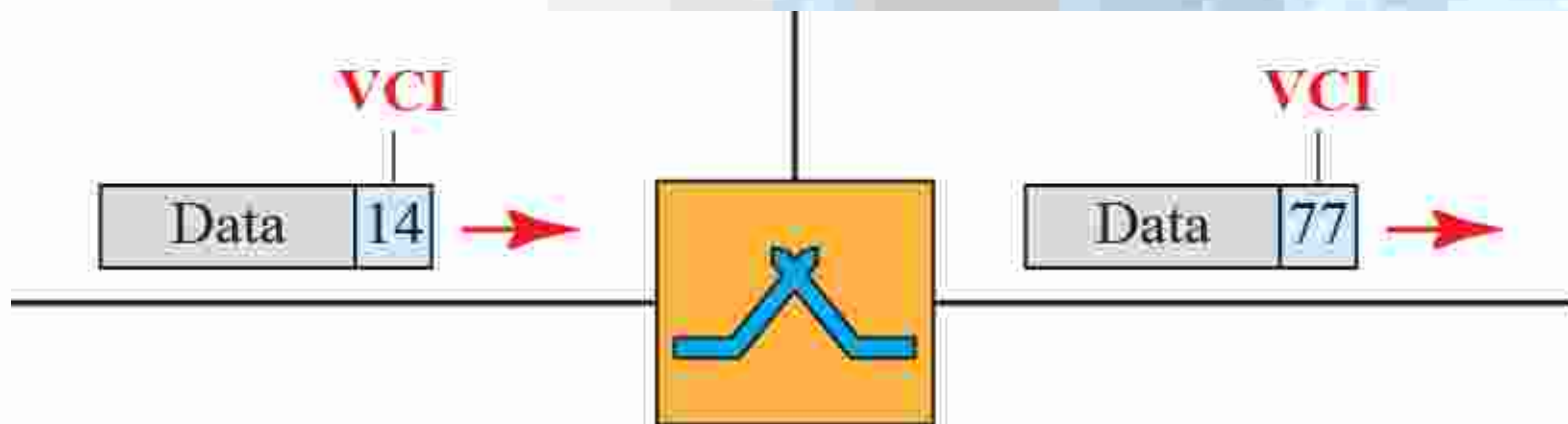
# Virtual-Circuit Networks



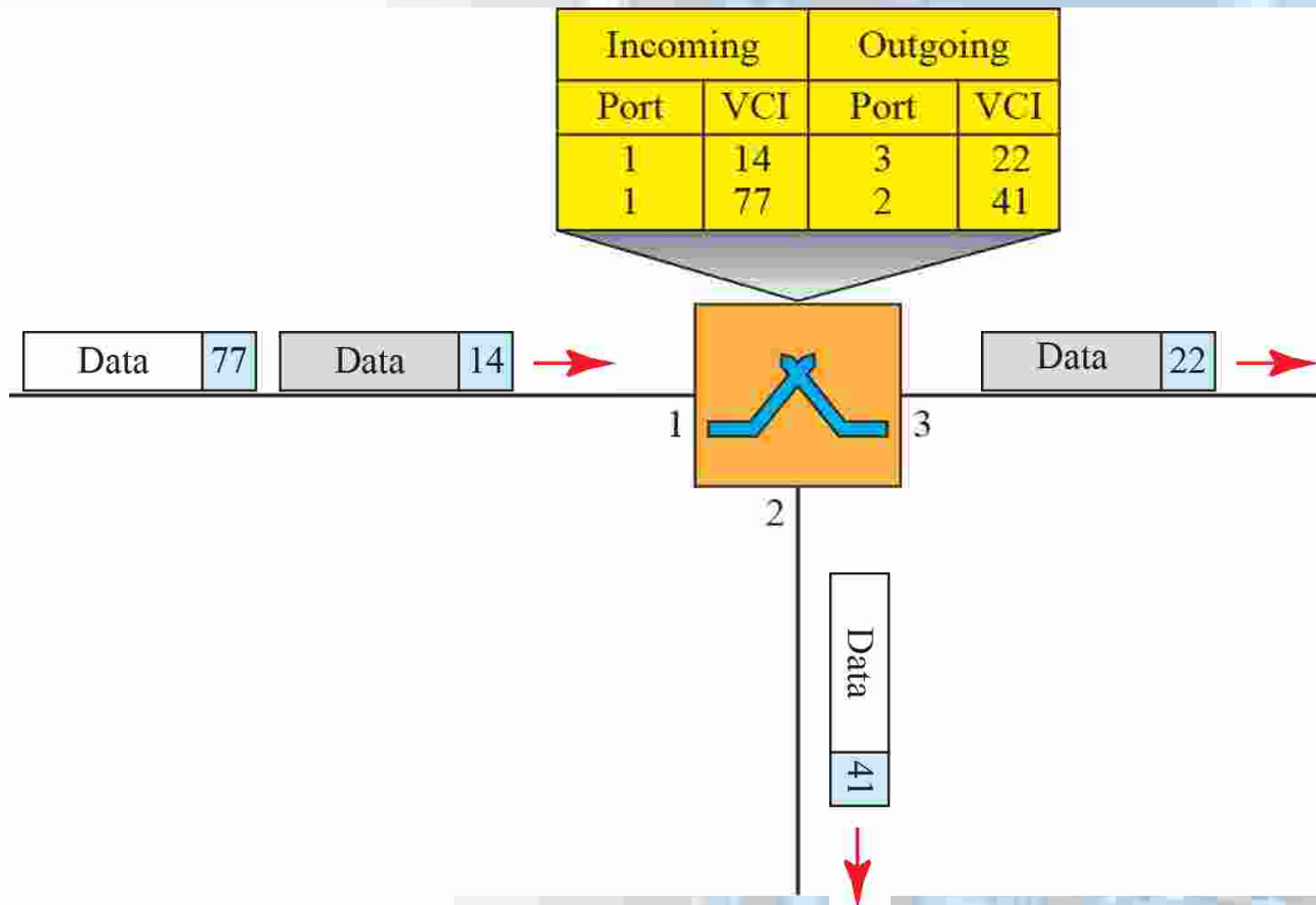
# Virtual-Circuit Networks

- A virtual-circuit network is a cross between a circuit-switched network and a datagram network

# Virtual-Circuit Identifier



# Switch & table for a virtual-circuit network

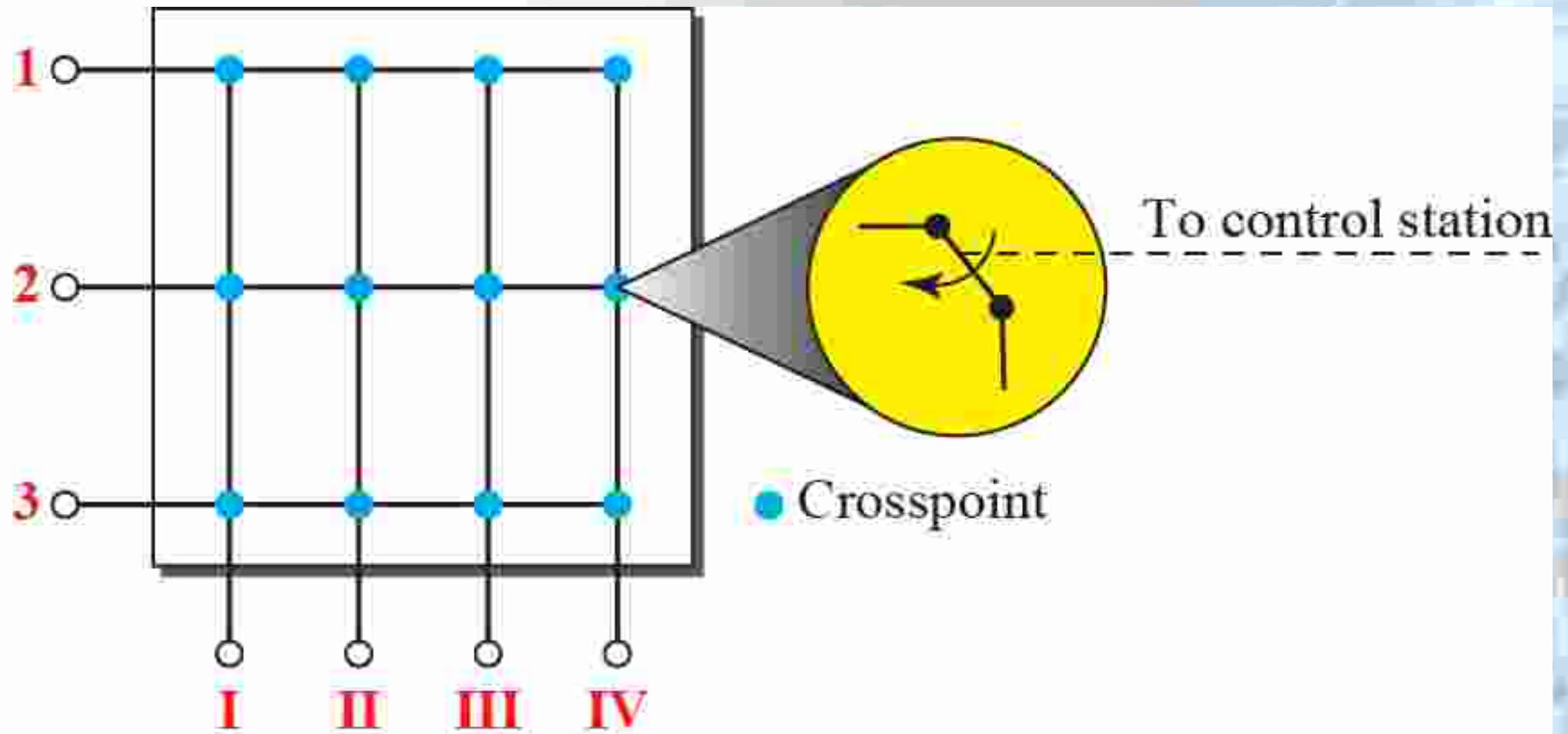




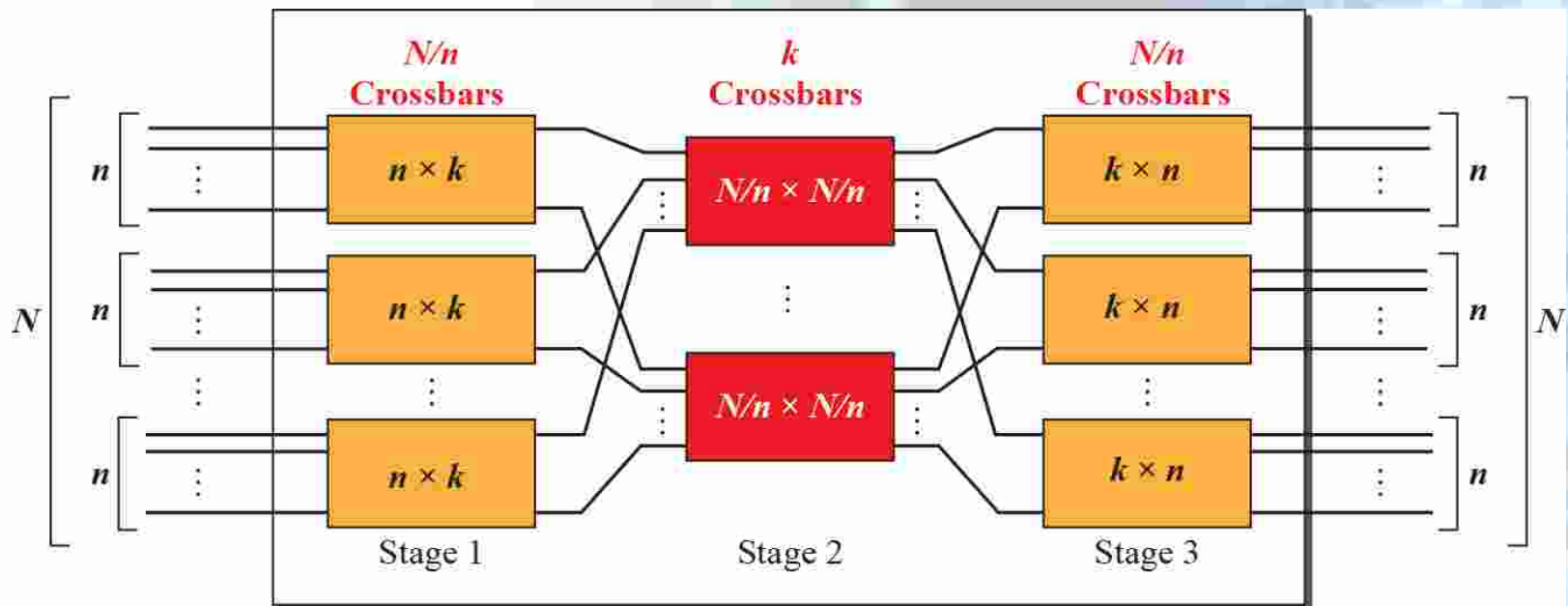
# Structure Of A Circuit Switch

- **Circuit switching today can use either of two technologies:**
  - ✓ **The Space-Division switch**
  - ✓ **The Time-Division switch**

# Crossbar switch with 3 inputs & 4 outputs



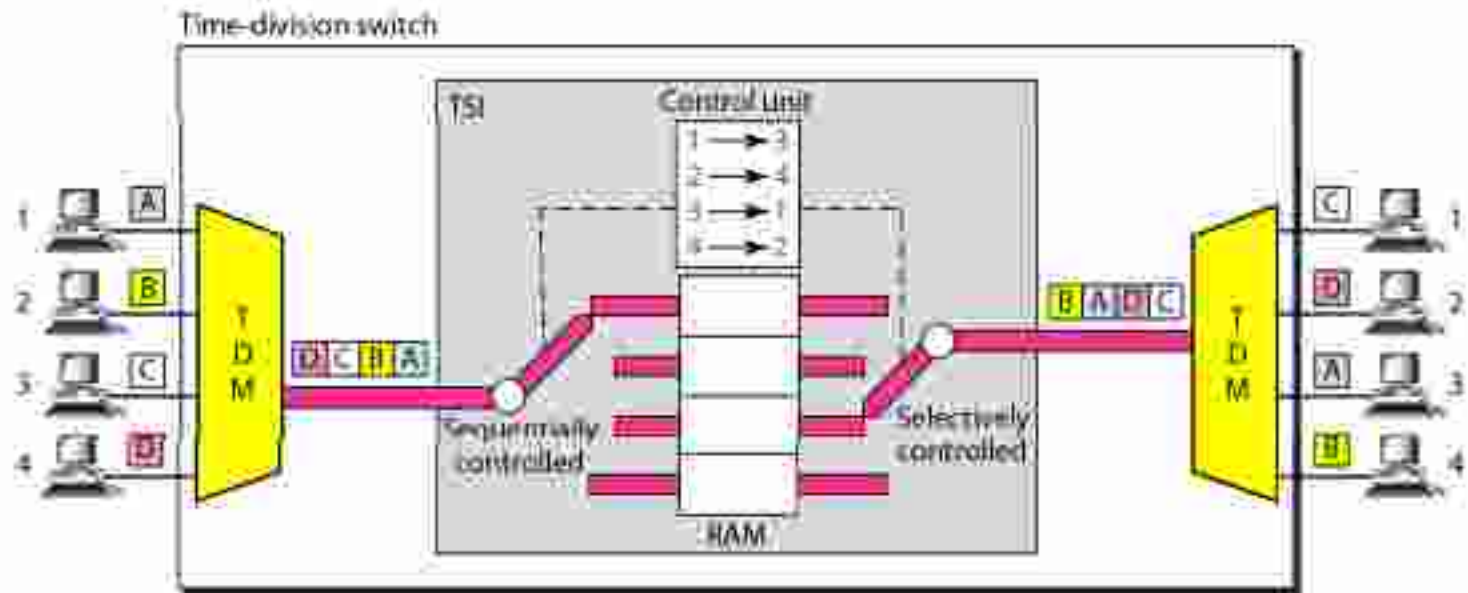
# Multistage Switch



# Time-Division Switch

- Uses TDM inside a switch
- Most popular technology is Time-Slot Interchange (TSI)

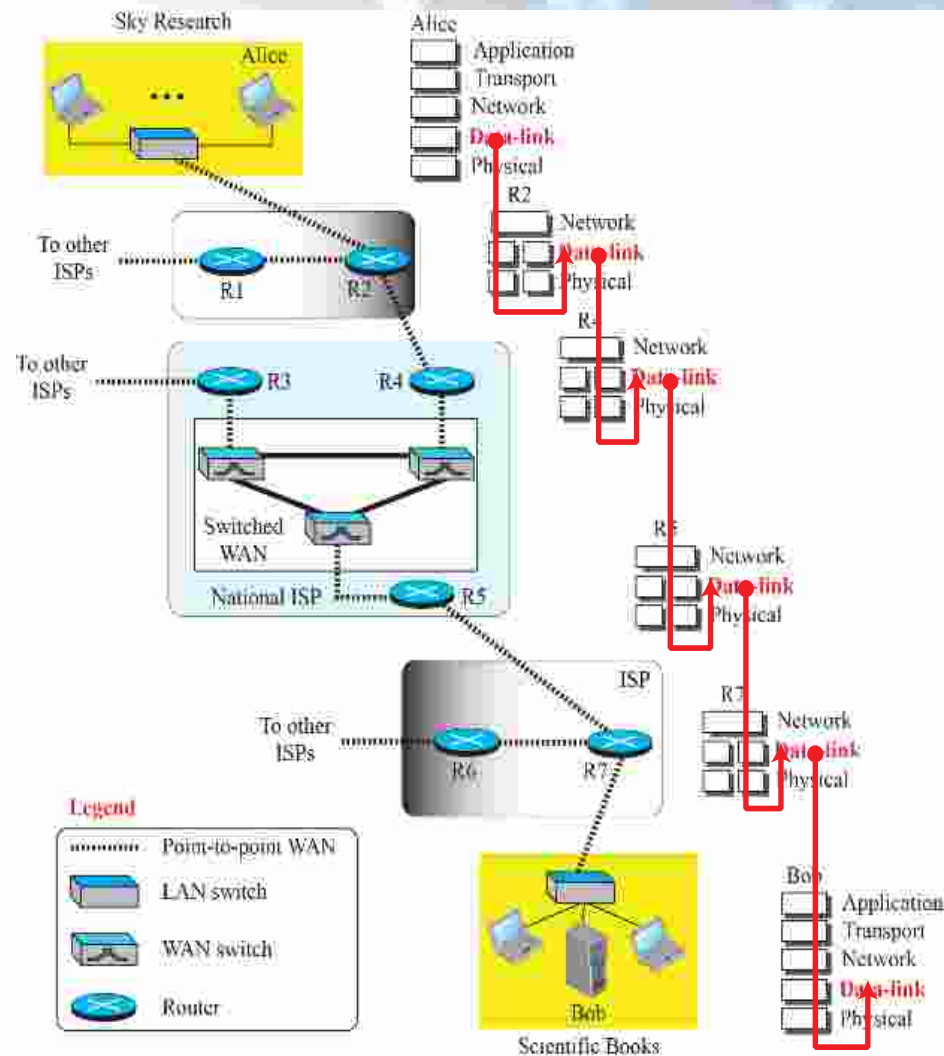
# Time-Division Switch



# Data-Link Layer

- The Internet is a combination of networks glued together by connecting devices (routers or switches)
- If a packet is to travel from a host to another host, it needs to pass through these networks
- Data Link layer controls node-to-node communication

# Communication at the Data-Link Layer

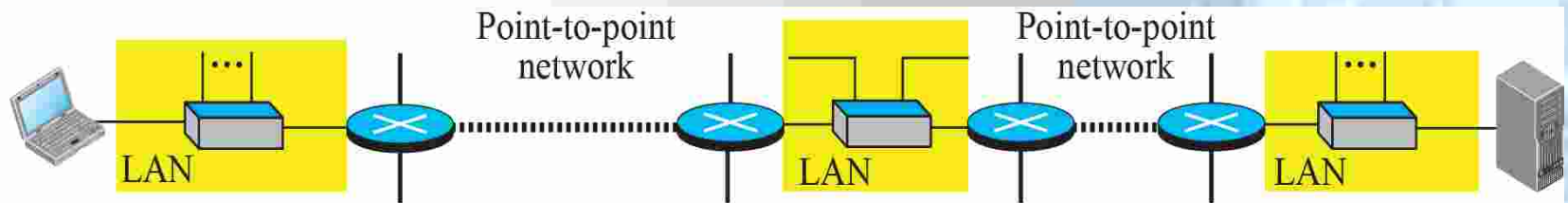


# Nodes and Links

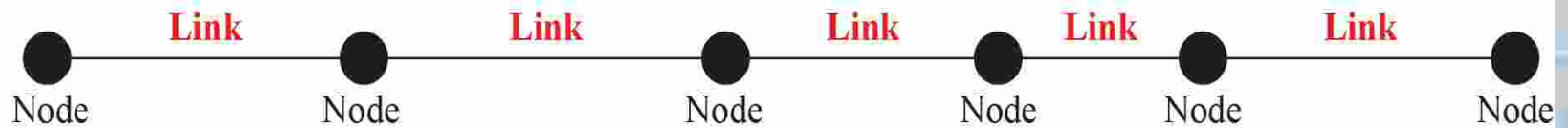
- Communication at the data-link layer is node-to-node
- A data unit from one point in the Internet needs to pass through many networks (LANs and WANs) to reach another point
- We refer to the two end hosts and the routers as nodes and the networks in between as links



# Nodes and Links



a. A small part of the Internet

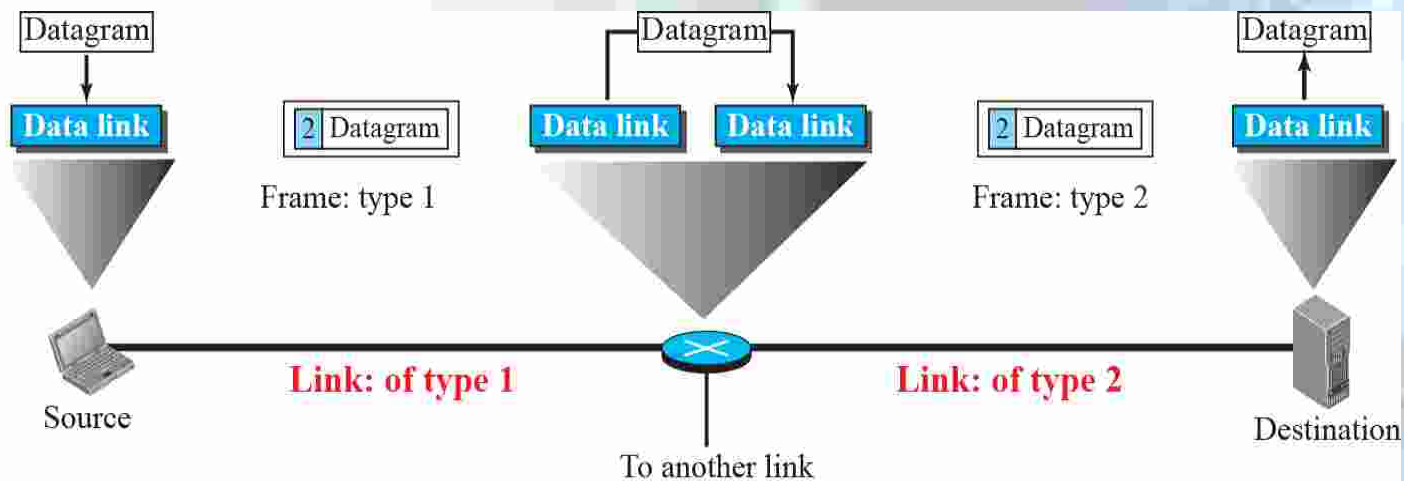


b. Nodes and links

# Services provided by Data-Link Layer

- Located between the physical and the network layers
- Provides services to Network Layer and receives services from Physical layer
- Framing
- Flow Control
- Error Control
- Congestion Control

# A Communication with only Three Nodes



# Services provided by Data-Link Layer

- **Data-Link layer provides services to Network Layer and receives services from Physical layer**
  - ✓ **Framing**
  - ✓ **Flow Control**
  - ✓ **Error Control**
  - ✓ **Congestion Control**

# Two Categories of Links

- Two nodes are physically connected by a transmission medium such as cable or air
- Data-link layer controls how the medium is used
  - ✓ Data-link layer can use whole capacity
  - ✓ Data-link layer can use only part of the capacity

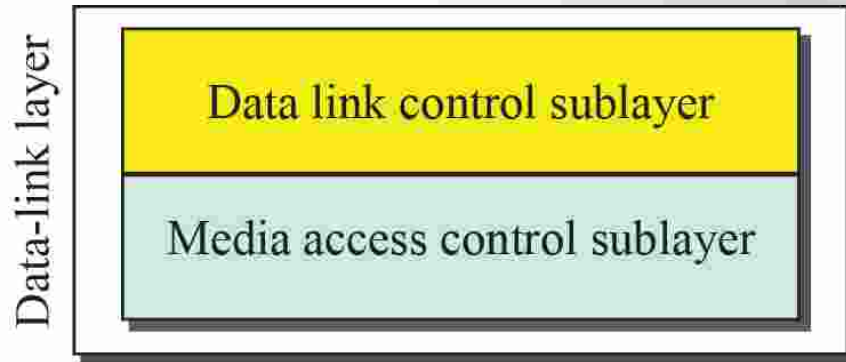
# Two Categories of Links

- We can have the following two types of links:
  - ✓ Point-to-point link or a
  - ✓ Broadcast link

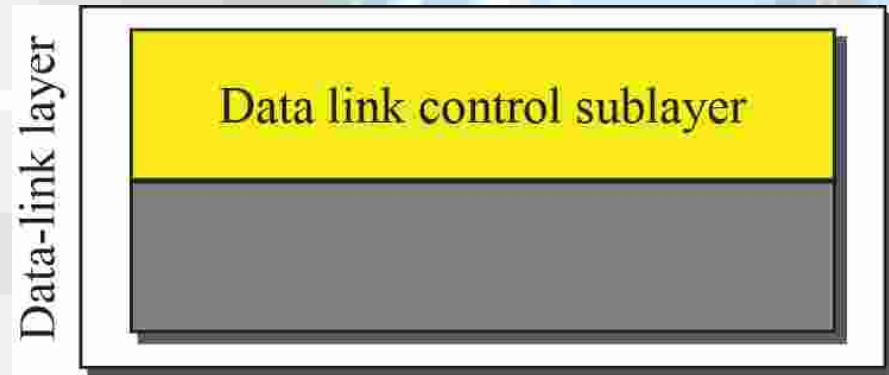
# Two Sublayers of Data-Link Layer

- We can divide the data-link layer into two sublayers:
  - Data Link Control (DLC)
  - Media Access Control (MAC)

# Dividing the data-link layer into two sublayers



a. Data-link layer of a broadcast link



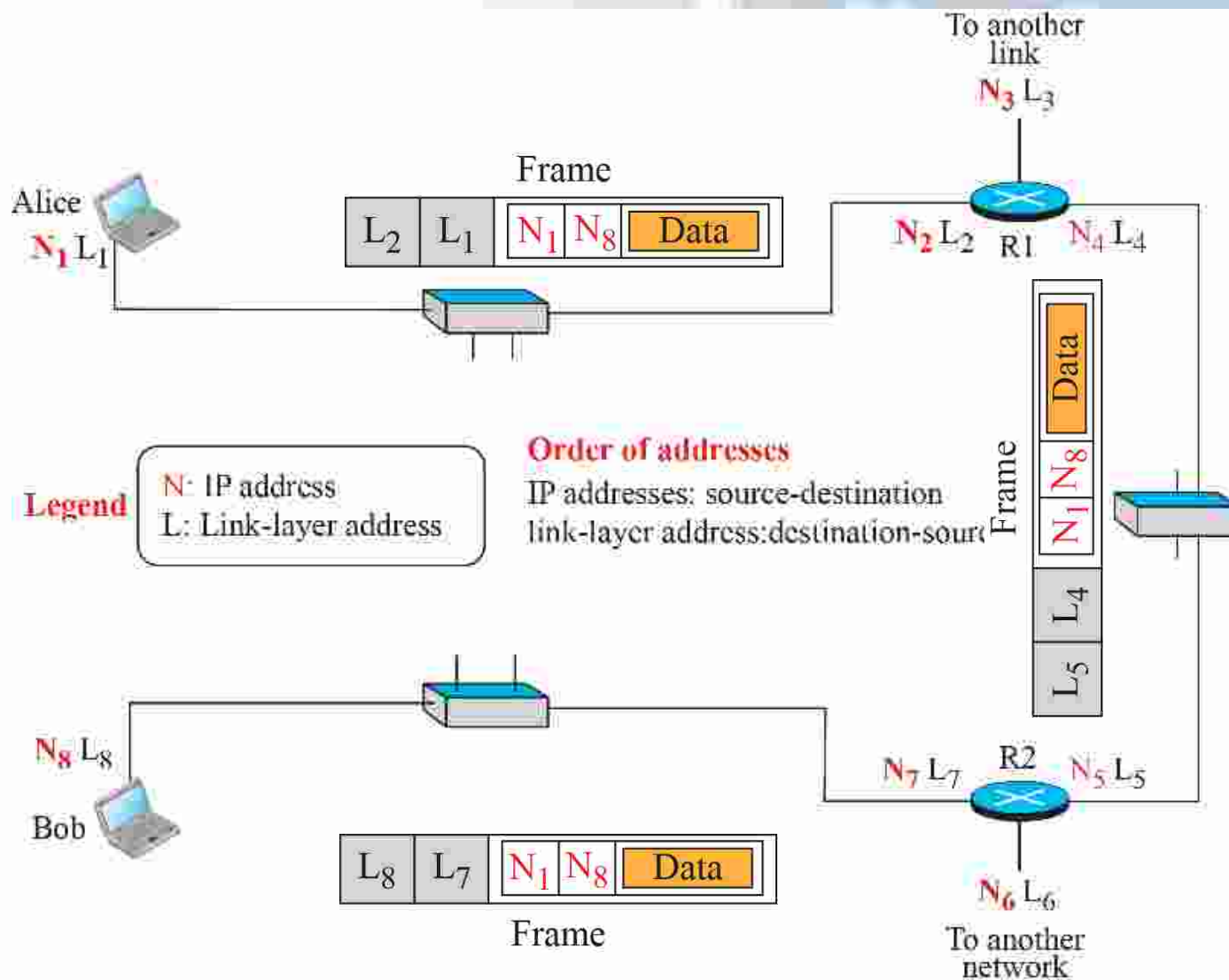
b. Data-link layer of a point-to-point link



# Why LINK-LAYER ADDRESSING?

- IP addresses are the identifiers at the network layer
- In Internet we cannot make a packet reach its destination using only IP addresses
- Source and destination IP addresses define the two ends but cannot define which links the packet will take

# IP addresses & Link-Layer Addresses



# Three Types of addresses

- Some link-layer protocols define three types of addresses:
- Unicast
- Multicast
- Broadcast

# Example

The unicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons; for example, the following is a link-layer address of a computer. The second digit needs to be an odd number.

**A3:34:45:11:92:F1**

# Example

The multicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons. The second digit, however, needs to be an even number in hexadecimal. The following shows a multicast address:

**A2:34:45:11:92:F1**

# Example

The broadcast link-layer addresses in the most common LAN, Ethernet, are 48 bits, all 1s, that are presented as 12 hexadecimal digits separated by colons. The following shows a broadcast address:

**FF:FF:FF:FF:FF:FF**

# Address Resolution Protocol (ARP)

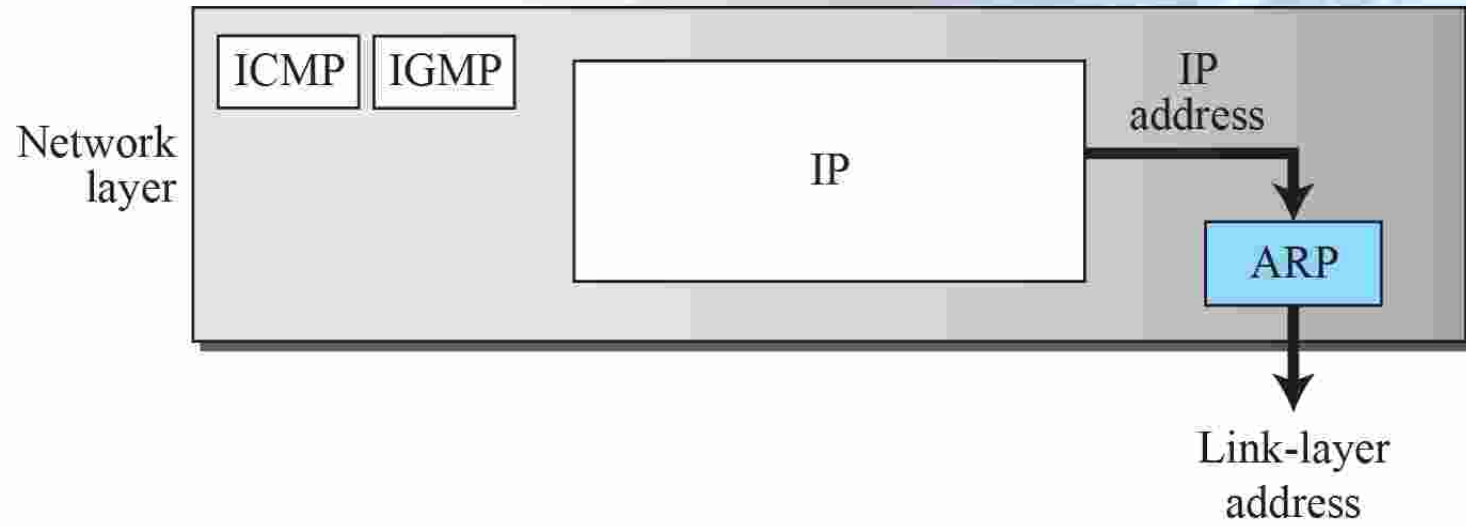
- Anytime a node has an IP packet to send to another node in a link, it has the IP address of the receiving node
- IP address of the next node is not helpful in moving a frame through a link; we need the link-layer address of the next node

# Address Resolution Protocol (ARP)

- We need Address Resolution Protocol (ARP)



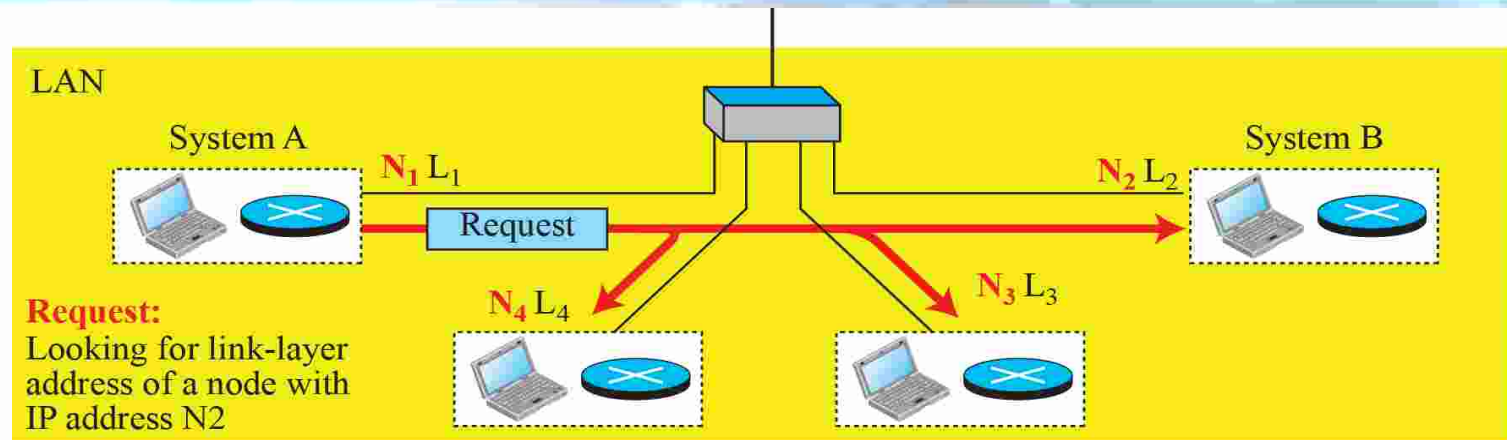
# Position of ARP in TCP/IP protocol suite



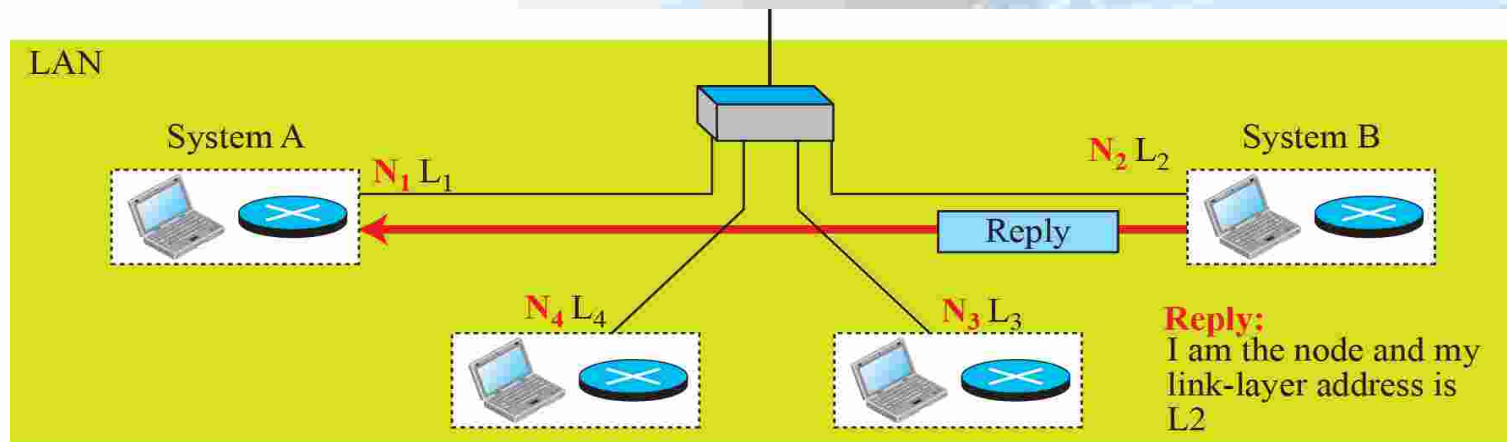
# Address Resolution Protocol (ARP)

- Anytime a node has an IP packet to send to another node in a link, it has the IP address of the receiving node
- IP address of the next node is not helpful in moving a frame through a link; we need the link-layer address of the next node

# ARP Operation



a. ARP request is broadcast



b. ARP reply is unicast

# Address Resolution Protocol (ARP)

- Anytime a node has an IP packet to send to another node in a link, it has the IP address of the receiving node
- IP address of the next node is not helpful in moving a frame through a link; we need the link-layer address of the next node

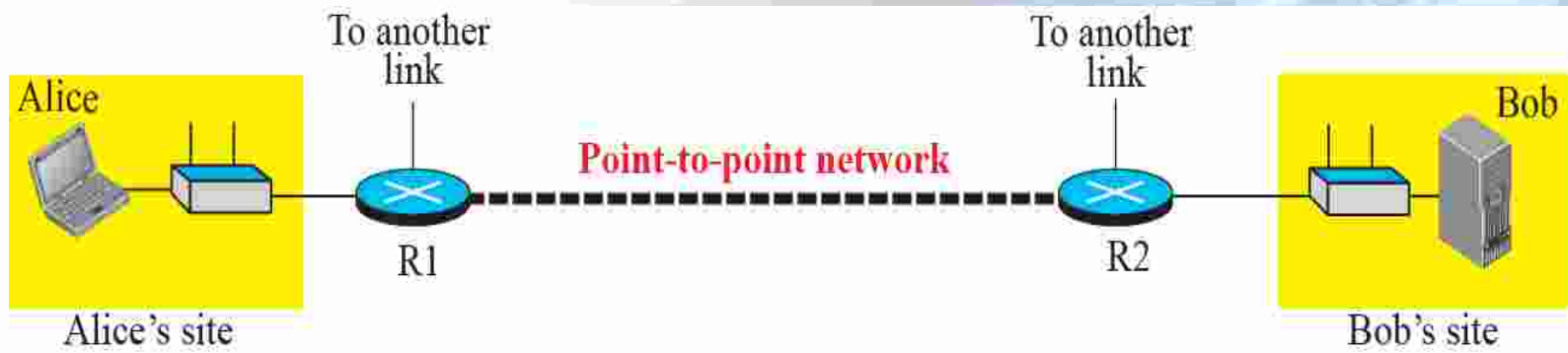
# ARP Packet

**Hardware:** LAN or WAN protocol

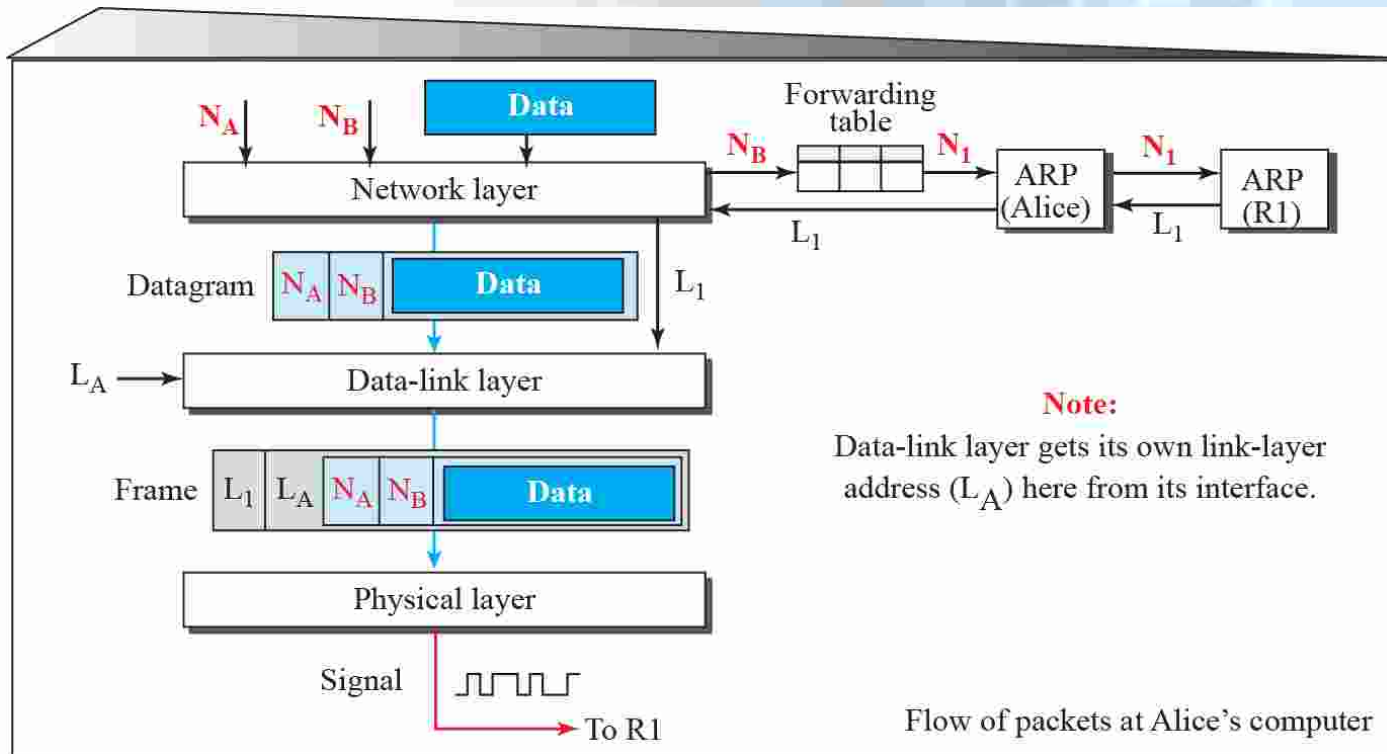
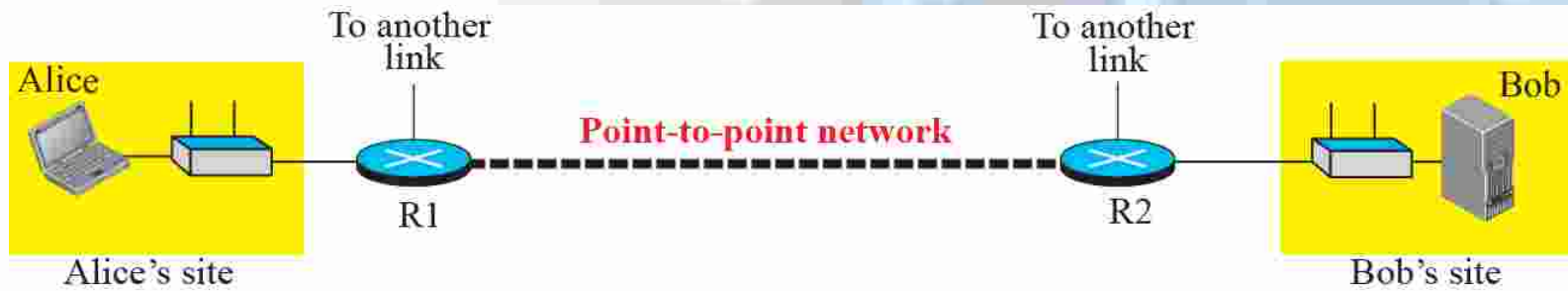
**Protocol:** Network-layer protocol

|                                                    |                 |                                        |    |
|----------------------------------------------------|-----------------|----------------------------------------|----|
| 0                                                  | 8               | 16                                     | 31 |
| Hardware Type                                      |                 | Protocol Type                          |    |
| Hardware length                                    | Protocol length | Operation<br><b>Request:1, Reply:2</b> |    |
| Source hardware address                            |                 |                                        |    |
| Source protocol address                            |                 |                                        |    |
| Destination hardware address<br>(Empty in request) |                 |                                        |    |
| Destination protocol address                       |                 |                                        |    |

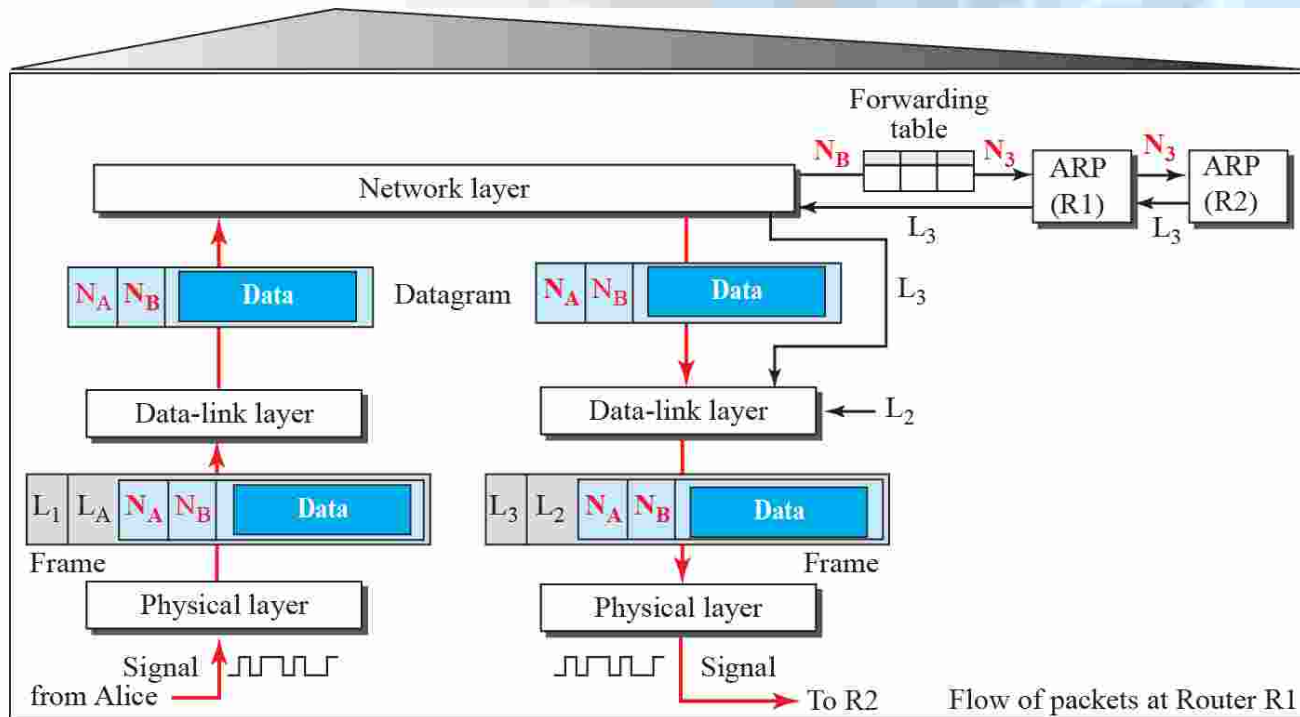
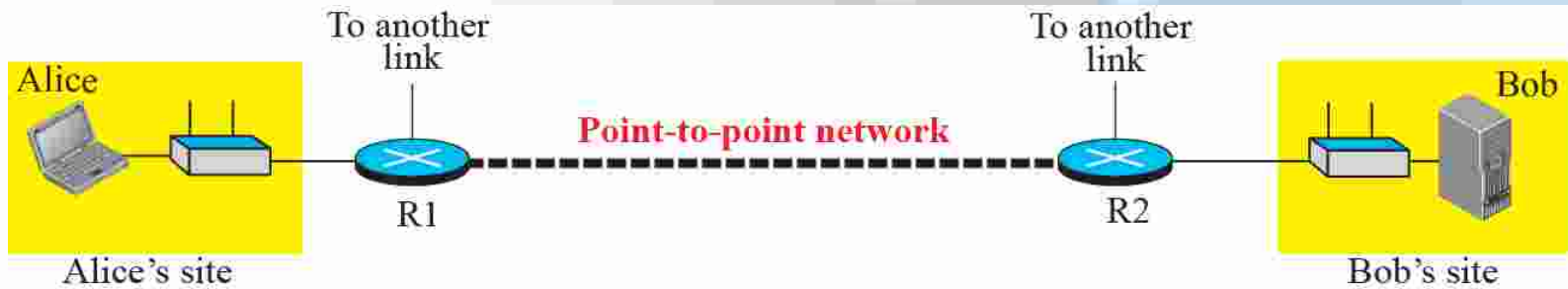
# The internet for our example



# Flow of packets at Alice site

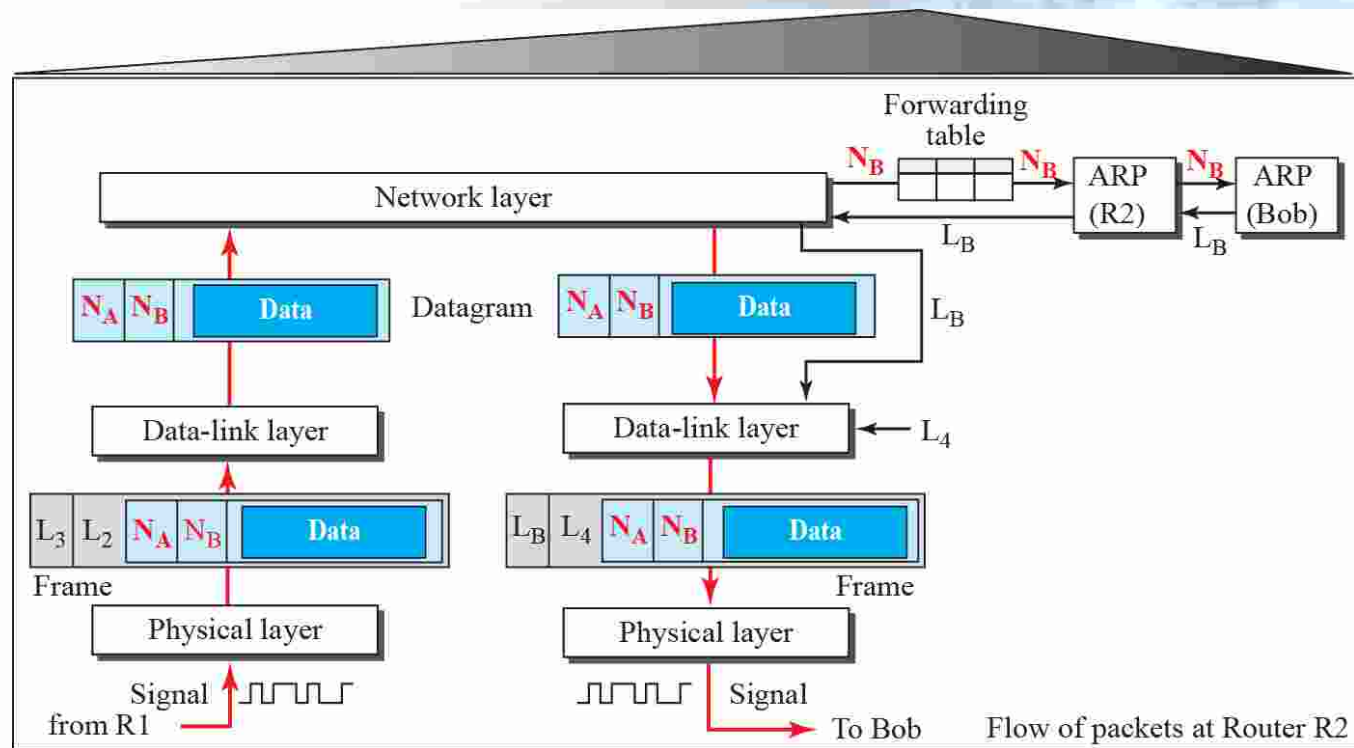
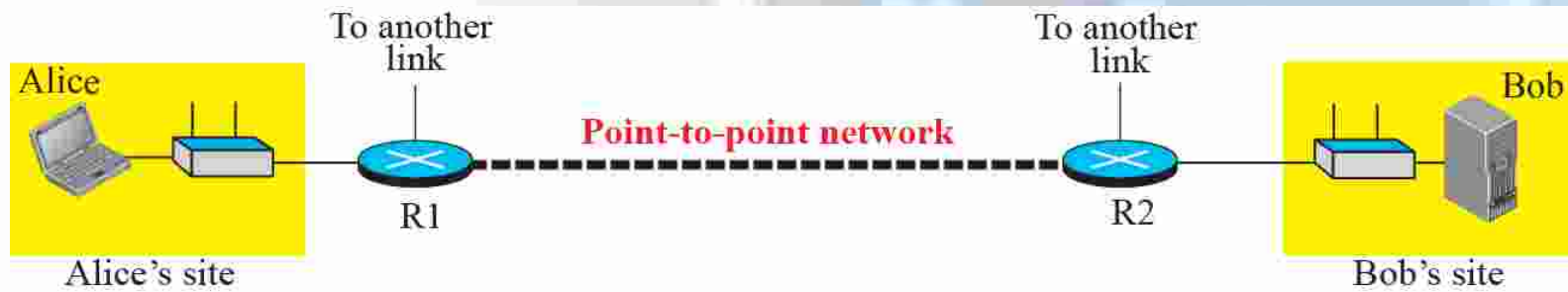


# Flow of activities at router R1

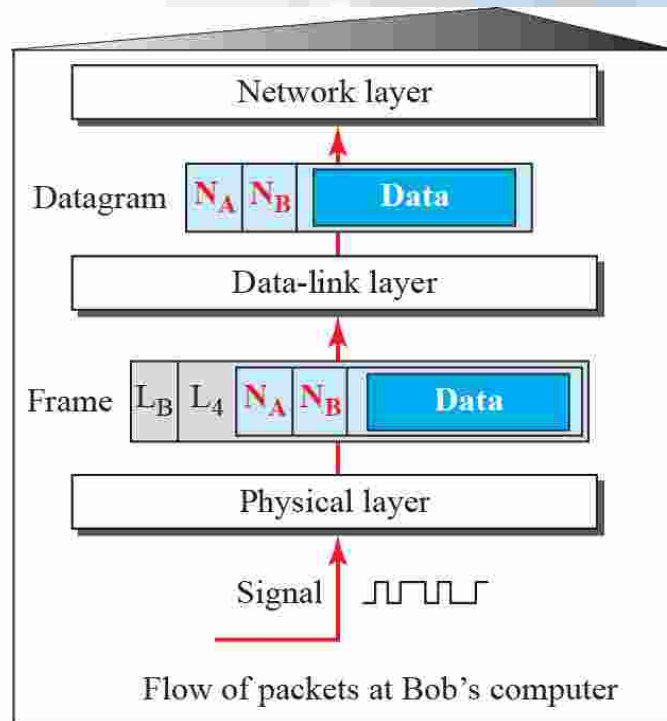
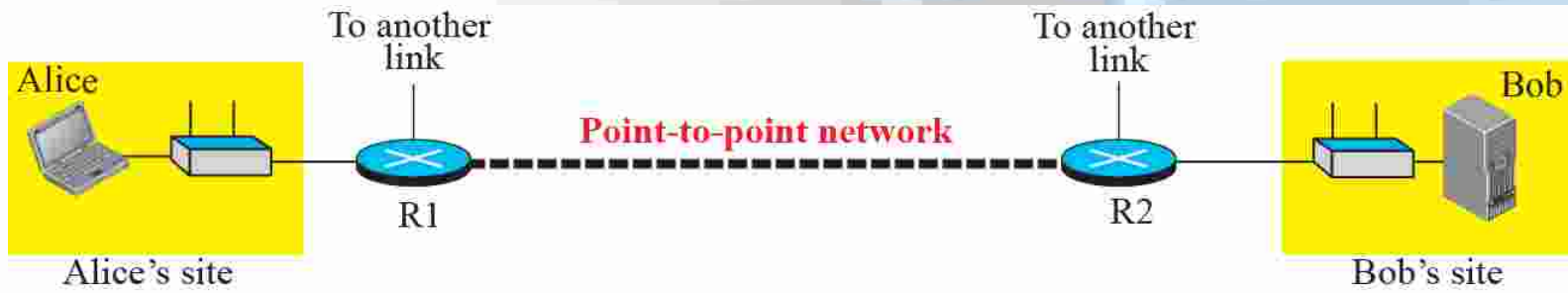




# Flow of activities at router R2



# Activities at Bob's site



# Types of Errors

- Data transmission suffers unpredictable changes because of interference
- The interference can change the shape of the signal
  - ✓ Single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1

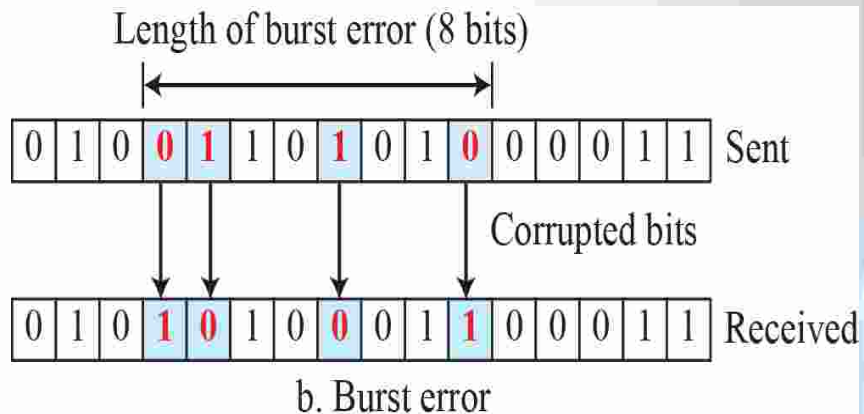
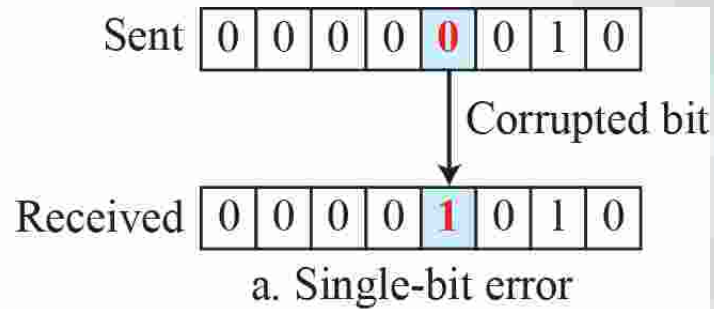
# Types of Errors

- ✓ Burst Error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1

# Types of Errors

- ✓ Single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1
- ✓ Burst Error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1

# Single-Bit and Burst Error



# Redundancy

- **Central concept in detecting or correcting errors is Redundancy**
- **To be able to detect or correct errors, we send some extra bits with our data**
- **The presence of these redundant bits allows the receiver to detect or correct corrupted bits**

# Detection versus Correction

- Correction is more difficult than the detection
- In error detection, we are only looking to see if any error has occurred (Yes or No)
- We are not interested in the number of corrupted bits in Detection
- Single-bit error is same as a Burst error



# Detection versus Correction

- In Error Correction, we need to know:
  - ✓ The exact number of bits that are corrupted and,
  - ✓ Their location in the message

# Coding

- Redundancy is achieved through various coding schemes
- Sender adds redundant bits through a process that creates a relationship between redundant bits and the actual data bits
- The receiver checks the relationships between the two sets of bits to detect errors

# Coding

- The ratio of redundant bits to data bits and the robustness of the process are important factors in any coding scheme

# Types of Coding Schemes

- Coding schemes can be divided into 2 broad categories:
  - ✓ Block Coding
  - ✓ Convolution Coding

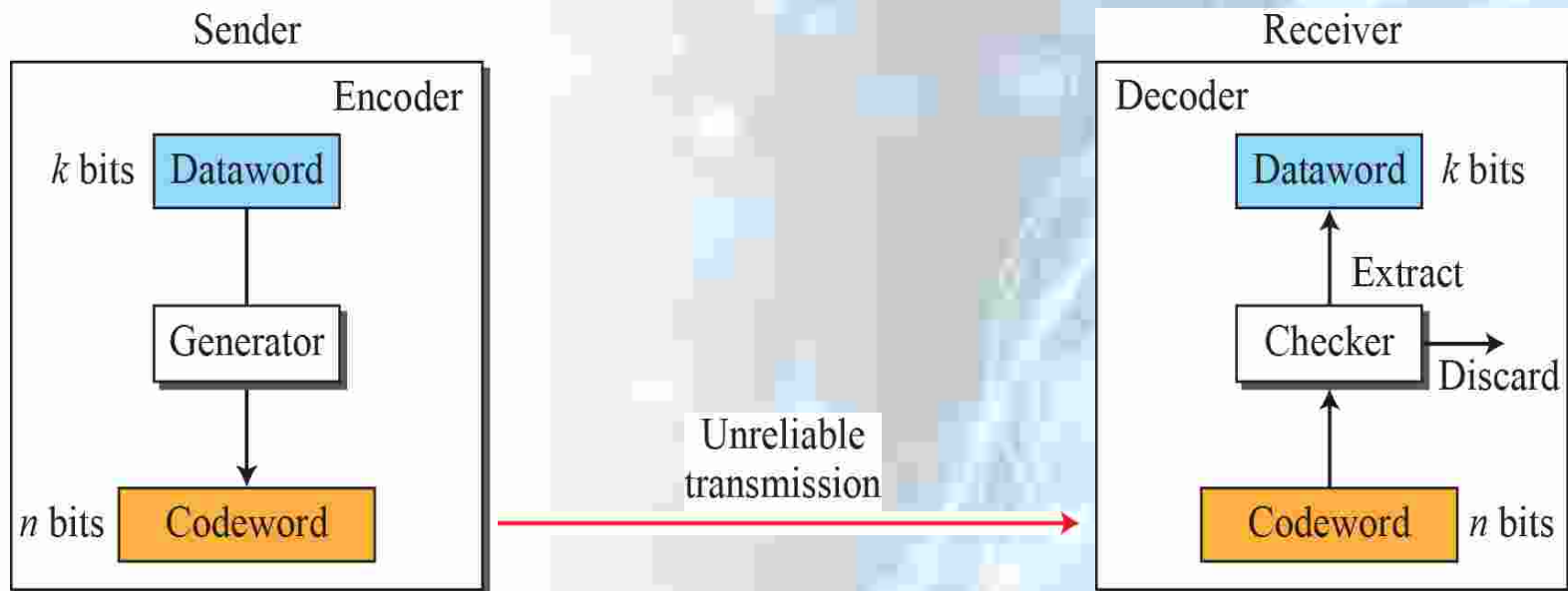
# Block Coding

- We divide our message into blocks, each of 'k' bits, called datawords
- We add 'r' redundant bits to each block to make the length ' $n = k + r$ '
- The resulting 'n-bit' blocks are called codewords

# BLOCK CODING in Error Detection

- If the following two conditions are met, the receiver can detect a change in the original codeword:
  - ✓ The receiver has (or can find) a list of valid codewords
  - ✓ The original codeword has changed to an invalid one

# Process of Error Detection in Block Coding



# Block Coding

- We divide our message into blocks, each of 'k' bits, called datawords
- We add 'r' redundant bits to each block to make the length ' $n = k + r$ '
- The resulting 'n-bit' blocks are called codewords



# Example

Let us assume that  $k = 2$  and  $n = 3$ . Table below shows the list of datawords and codewords. Later, we will see how to derive a cod word from a dataword.

| <i>Datawords</i> | <i>Codewords</i> | <i>Datawords</i> | <i>Codewords</i> |
|------------------|------------------|------------------|------------------|
| 00               | 000              | 10               | 101              |
| 01               | 011              | 11               | 110              |

# Hamming Distance

- Hamming Distance between two words of the same size is the number of differences between the corresponding bits
- Hamming Distance between two words  $x$  and  $y$  is  $d(x,y)$
- Hamming distance between received codeword and sent codeword is number of bits corrupted

# Example

Let us find the Hamming distance between two pairs of words.

1.  $d(000, 011)$
2.  $d(10101, 11110)$

# Minimum Hamming Distance

- Minimum Hamming Distance is smallest hamming distance between all possible pairs of codewords
- $d_{\min} = s + 1$   
where,  
 $s \rightarrow$  no. of detectable errors  
 $d_{\min} \rightarrow$  minimum hamming distance

## Example

A code scheme has a Hamming distance  $d_{\min} = 4$ . This code guarantees the detection of up to how many errors?

# Linear Block Codes

- Subset of Block Codes in which the exclusive OR of two valid codewords creates another valid codeword

# Example

The code below is a linear block code because the result of XORing any codeword with any other codeword is a valid codeword. For example, the XORing of the second and third codewords creates the fourth one.

| <i>Datawords</i> | <i>Codewords</i> | <i>Datawords</i> | <i>Codewords</i> |
|------------------|------------------|------------------|------------------|
| 00               | 000              | 10               | 101              |
| 01               | 011              | 11               | 110              |

The numbers of 1s in the nonzero codewords are 2, 2, and 2. So the minimum Hamming distance is  $d_{\min} = 2$ .

# Parity-Check Code

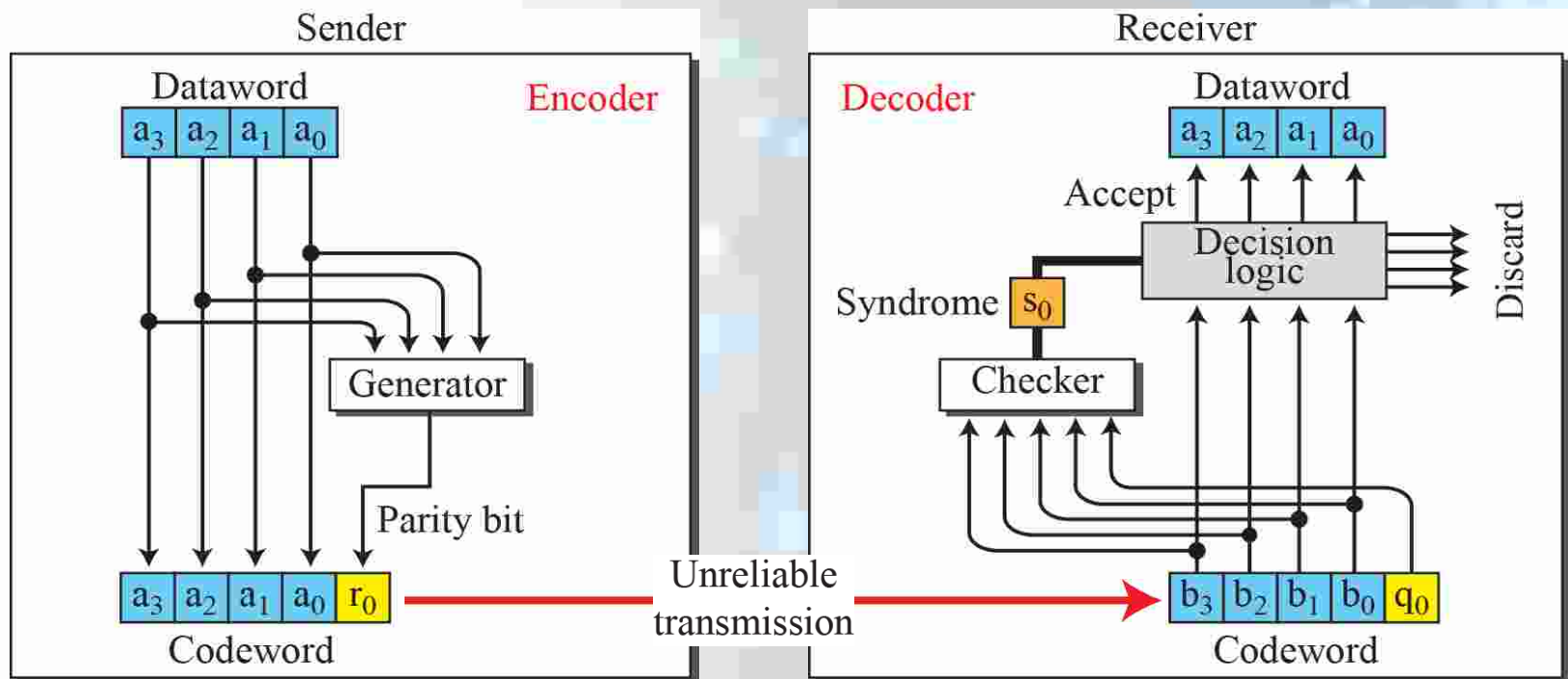
- Most common error-detecting code
- Linear block code ( $n=k+1$ )
- The extra parity bit is selected to make total number of 1s in codeword even



# Simple parity-check code C(5, 4)

| <i>Datawords</i> | <b>Codewords</b> | <i>Datawords</i> | <b>Codewords</b> |
|------------------|------------------|------------------|------------------|
| 0000             | <b>00000</b>     | 1000             | <b>10001</b>     |
| 0001             | <b>00011</b>     | 1001             | <b>10010</b>     |
| 0010             | <b>00101</b>     | 1010             | <b>10100</b>     |
| 0011             | <b>00110</b>     | 1011             | <b>10111</b>     |
| 0100             | <b>01001</b>     | 1100             | <b>11000</b>     |
| 0101             | <b>01010</b>     | 1101             | <b>11011</b>     |
| 0110             | <b>01100</b>     | 1110             | <b>11101</b>     |
| 0111             | <b>01111</b>     | 1111             | <b>11110</b>     |

# Encoder & decoder for simple parity-check code



# CYCLIC CODES

- Special linear block codes with one extra property
- If a codeword is cyclically shifted (rotated), the result is another codeword
- If 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword

# Cyclic Redundancy Check (CRC)

- **Subset of Cyclic Codes**
- **Cyclic redundancy check (CRC) is used in networks such as LANs and WANs**

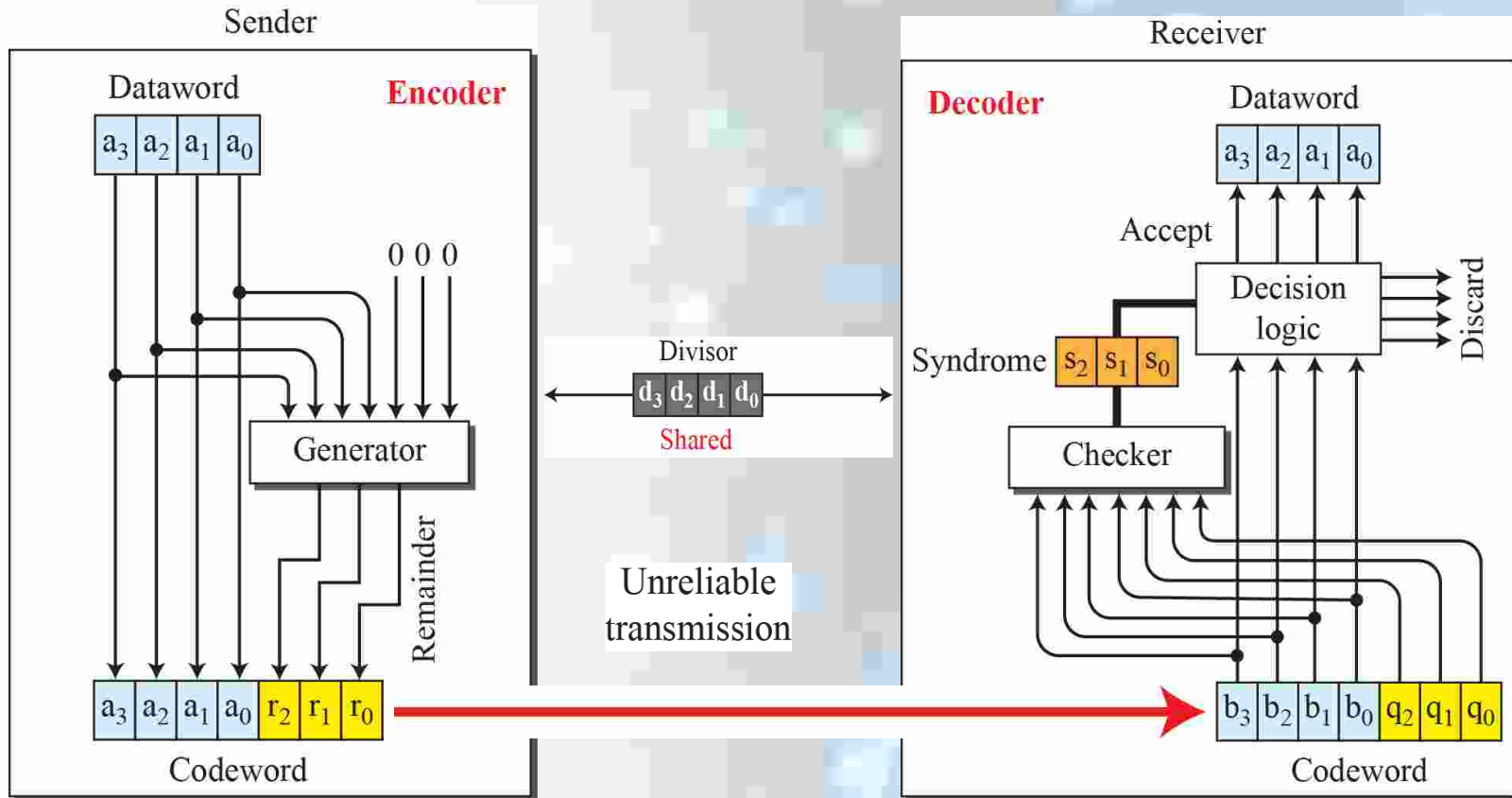
# A CRC code with C(7, 4)

| <i>Dataword</i> | <i>Codeword</i> | <i>Dataword</i> | <i>Codeword</i> |
|-----------------|-----------------|-----------------|-----------------|
| 0000            | 0000000         | 1000            | 1000101         |
| 0001            | 0001011         | 1001            | 1001110         |
| 0010            | 0010110         | 1010            | 1010011         |
| 0011            | 0011101         | 1011            | 1011000         |
| 0100            | 0100111         | 1100            | 1100010         |
| 0101            | 0101100         | 1101            | 1101001         |
| 0110            | 0110001         | 1110            | 1110100         |
| 0111            | 0111010         | 1111            | 1111111         |

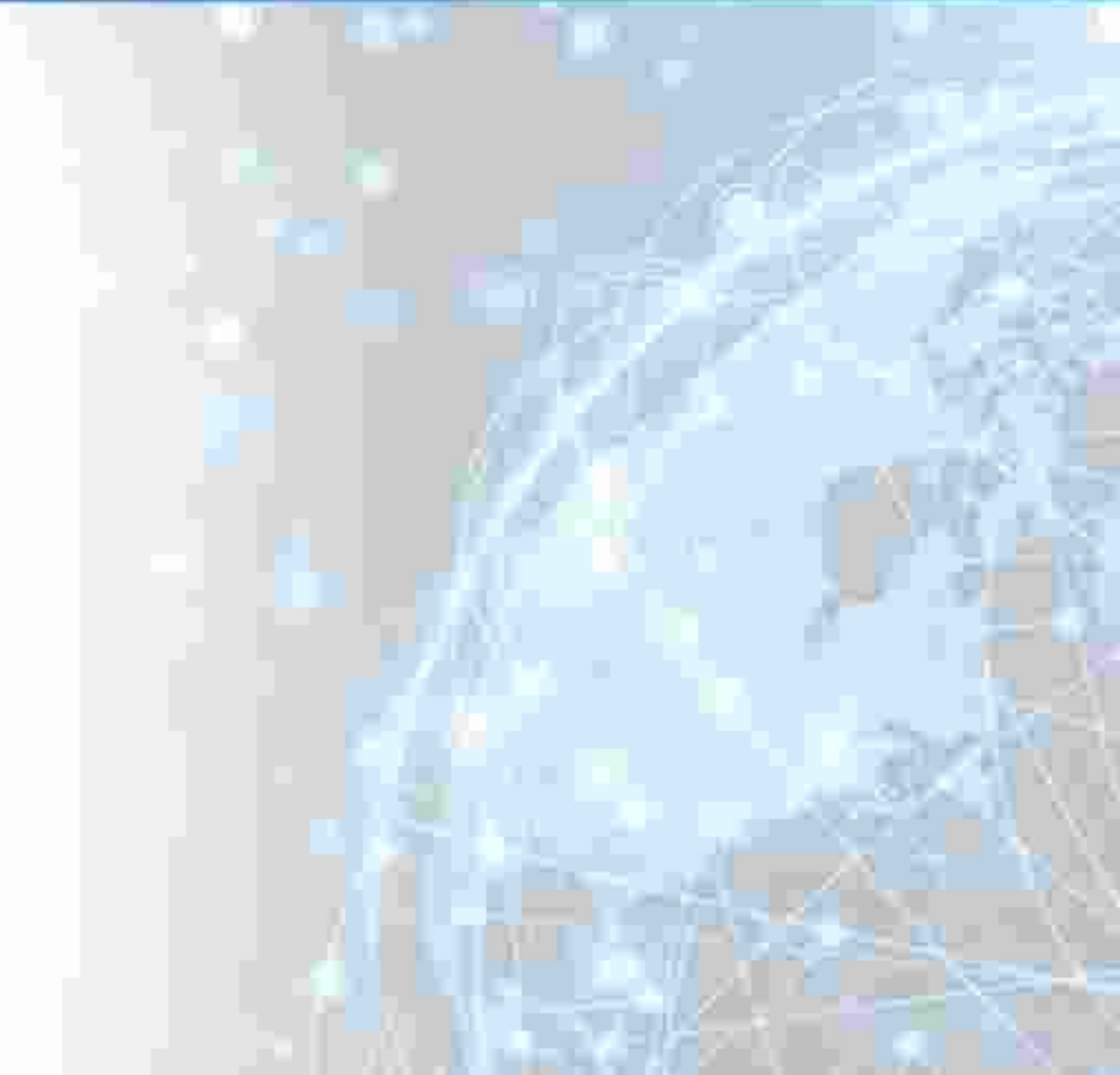
# Cyclic Redundancy Check (CRC)

- **Subset of Cyclic Codes**
- **Cyclic redundancy check (CRC) is used in networks such as LANs and WANs**

# CRC Encoder and Decoder



# Cyclic Code Analysis using Polynomials





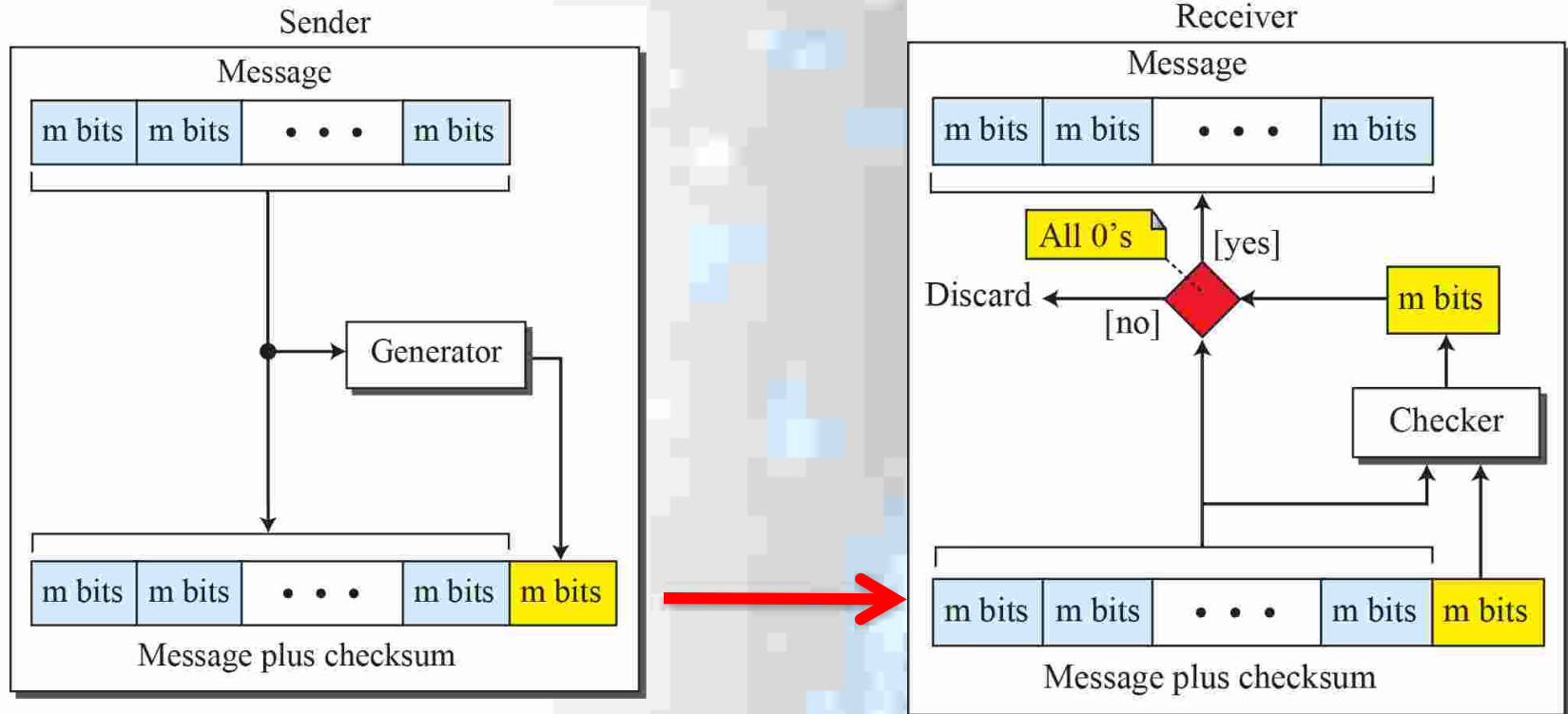
# Advantages of Cyclic Codes

- **Good performance in detection:**
  - **Single-bit errors**
  - **Double errors**
  - **Odd number of errors**
  - **Burst errors**
- **Easy Implementation**
- **Fast Implementation**

# CHECKSUM

- **Error-detection technique that can be applied to a message of any length**
- **Checksum mostly used at the network and transport layer rather than the data-link layer**

# Checksum



# Concept behind Checksum

- The idea of the traditional checksum is simple. We show this using a simple example

## Example

**Suppose the message is a list of five 4-bit numbers that we want to send to a destination. In addition to sending these numbers, we send the sum of the numbers. Set of numbers is (7, 11, 12, 0, 6)**

## Example

In the previous example, the decimal number 36 in binary is  $(100100)_2$ . To change it to a 4-bit number we add the extra leftmost bit to the right four bits as shown below

$$\begin{array}{l} (10)_2 + (0100)_2 = (0110)_2 \\ \longrightarrow (6)_{10} \end{array}$$

# Forward Error Correction

- Retransmission of corrupted and lost packets is not useful for real-time multimedia transmission
- We need to correct the error or reproduce the packet immediately
- Several techniques developed and are commonly called Forward Error Correction techniques

# Using Hamming Distance

- For error detection, we definitely need more distance
- It can be shown that to correct 't' errors, we need to have:
$$d_{\min} = 2t + 1$$
- If we want to correct 10 bits in a packet, we need to make the minimum hamming distance 21 bits
- A lot of redundant bits need to be sent with the data



# Using Hamming Distance

**If we want to correct 10 bits in a packet, we need to make the minimum hamming distance 21 bits**

# Using XOR

Another recommendation is to use the property of the exclusive OR operation as shown below.

$$R = P_1 \oplus P_2 \oplus \dots \oplus P_i \oplus \dots \oplus P_N$$

This means:

$$P_i = P_1 \oplus P_2 \oplus \dots \oplus R \oplus \dots \oplus P_N$$

# Using XOR

Another recommendation is to use the property of the exclusive OR operation as shown below.

$$R = P_1 \oplus P_2 \oplus \dots \oplus P_i \oplus \dots \oplus P_N$$

# Chunk Interleaving

- Another way to achieve FEC in multimedia is to allow some small chunks to be missing at the receiver
- We cannot afford to let all the chunks belonging to the same packet be missing; however, we can afford to let one chunk be missing in each packet

# Interleaving

|          |    |    |    |    |    |
|----------|----|----|----|----|----|
| Packet 1 | 05 | 04 | 03 | 02 | 01 |
| Packet 2 | 10 | 09 | 08 | 07 | 06 |
| Packet 3 | 15 | 14 | 13 | 12 | 11 |
| Packet 4 | 20 | 19 | 18 | 17 | 16 |
| Packet 5 | 25 | 24 | 23 | 22 | 21 |

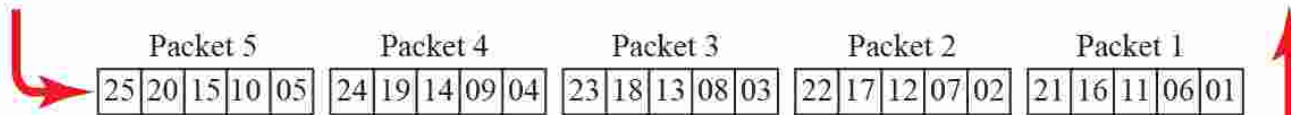
Sending  
column by column  
↓

a. Packet creation at sender

|          |    |    |  |    |    |
|----------|----|----|--|----|----|
| Packet 1 | 05 | 04 |  | 02 | 01 |
| Packet 2 | 10 | 09 |  | 07 | 06 |
| Packet 3 | 15 | 14 |  | 12 | 11 |
| Packet 4 | 20 | 19 |  | 17 | 16 |
| Packet 5 | 25 | 24 |  | 22 | 21 |

Receiving  
column by column  
↑

d. Packet recreation at receiver



b. Packets sent



c. Packets received

Packet 5      Packet 4      Packet 3      Packet 2      Packet 1

# Combining Hamming Distance & Interleaving

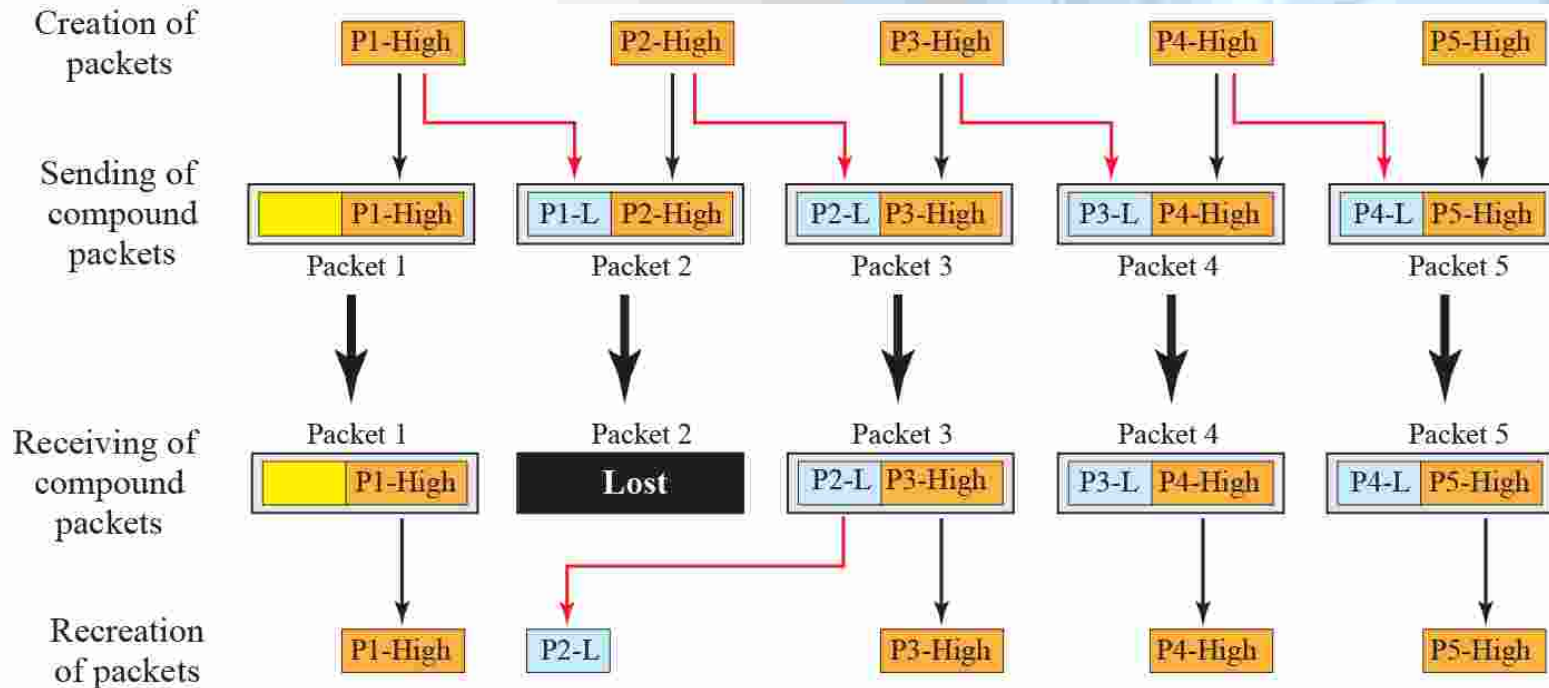
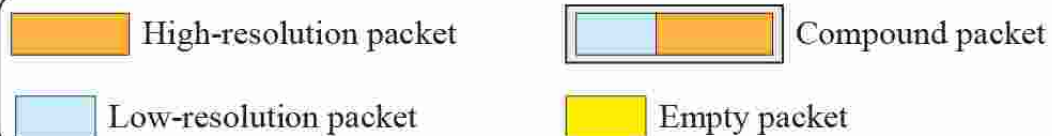
- Hamming distance and interleaving can be combined
- We can first create  $n$ -bit packets that can correct  $t$ -bit errors
- Then we interleave  $m$  rows and send the bits column by column
- Possible to correct burst errors up to  $m \times t$  bits of errors

# Compounding High & Low Resolution Packets

- Creation of a duplicate of each packet with a low-resolution redundancy and combine the redundant version with the next packet
- For example, we can create four low-resolution packets out of five high-resolution packets and send them

# Compounding High-and-Low resolution Packets

## Legend

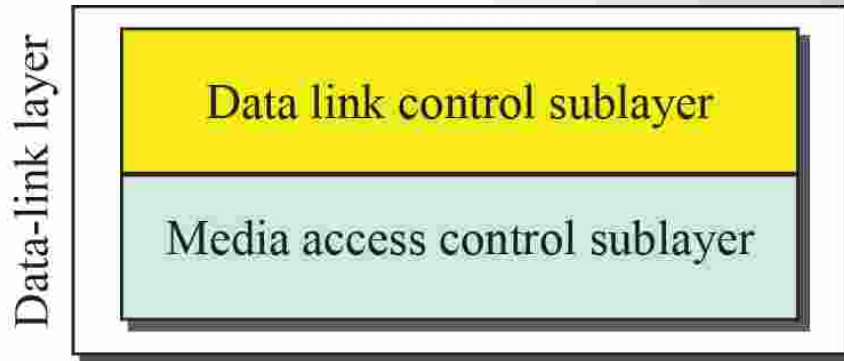




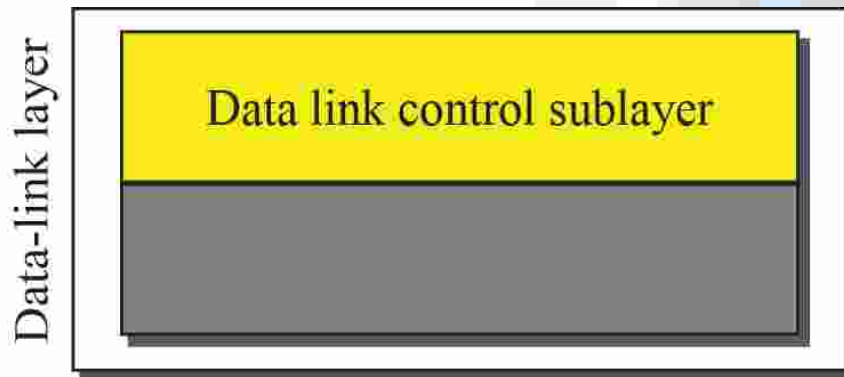
# Data Link Control (DLC) Services

- The data link control (DLC) deals with procedures for communication between two adjacent nodes no matter whether the link is dedicated or broadcast
- Data link control functions include framing, flow control and error control

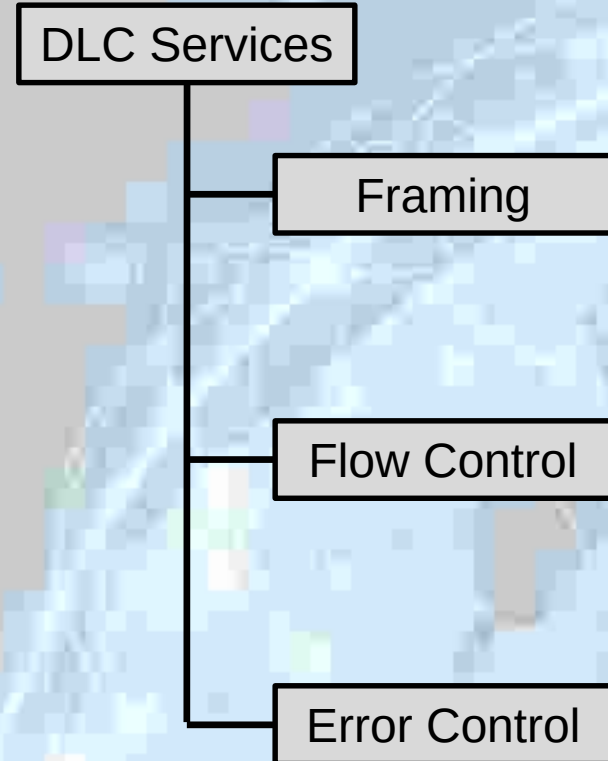
# DLC Services



a. Data-link layer of a broadcast link



b. Data-link layer of a point-to-point link



# Framing

- Data-Link layer needs to pack bits into frames, so that each frame is distinguishable from another
- Our postal system practices a type of framing
- Framing separates a message by adding a sender address and a destination address

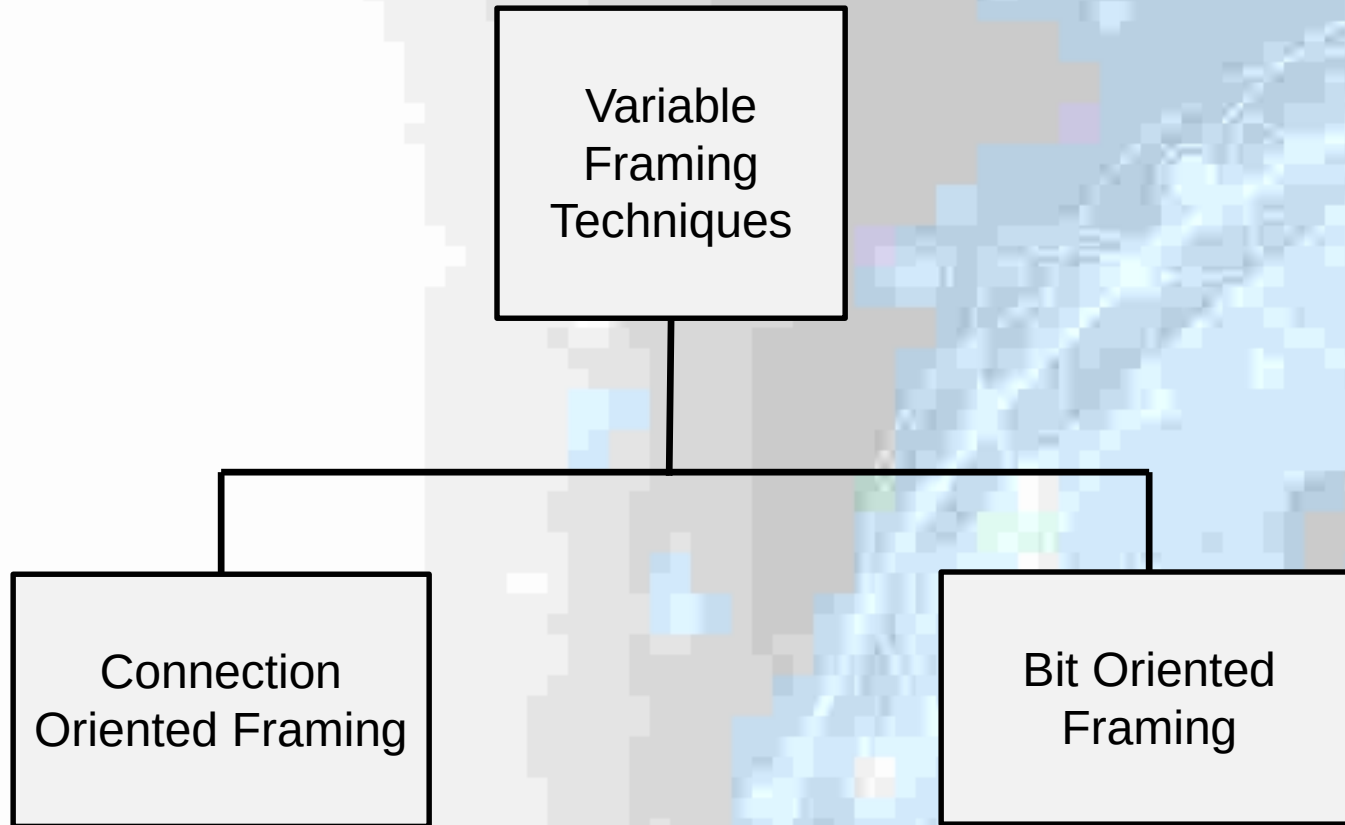
# Framing

- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt

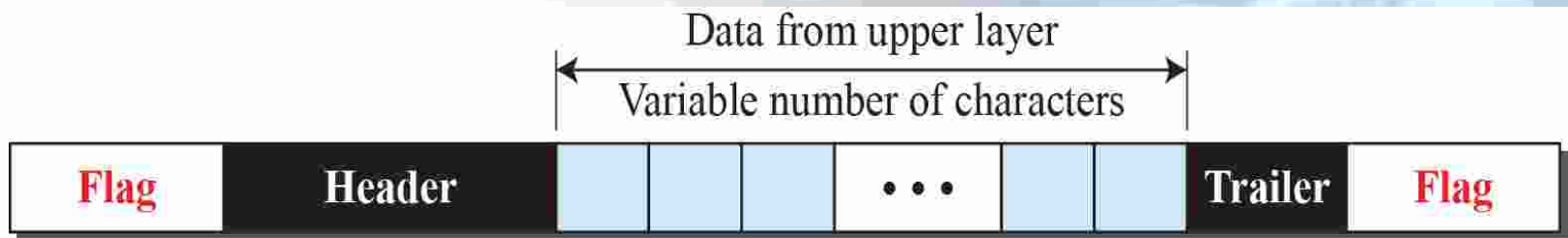
# Frame Size

- **Why not one BIG Frame?**
- **Frames can be of:**
  - ✓ **Fixed Size**
    - **Size acts as a boundary/delimiter**
  - ✓ **Variable Size**
    - **How to define Beginning and End of a Frame?**

# Variable Framing Techniques



# A Frame in a Character-Oriented Protocol



# Connection Oriented Framing

- Data to be carried are 8-bit characters



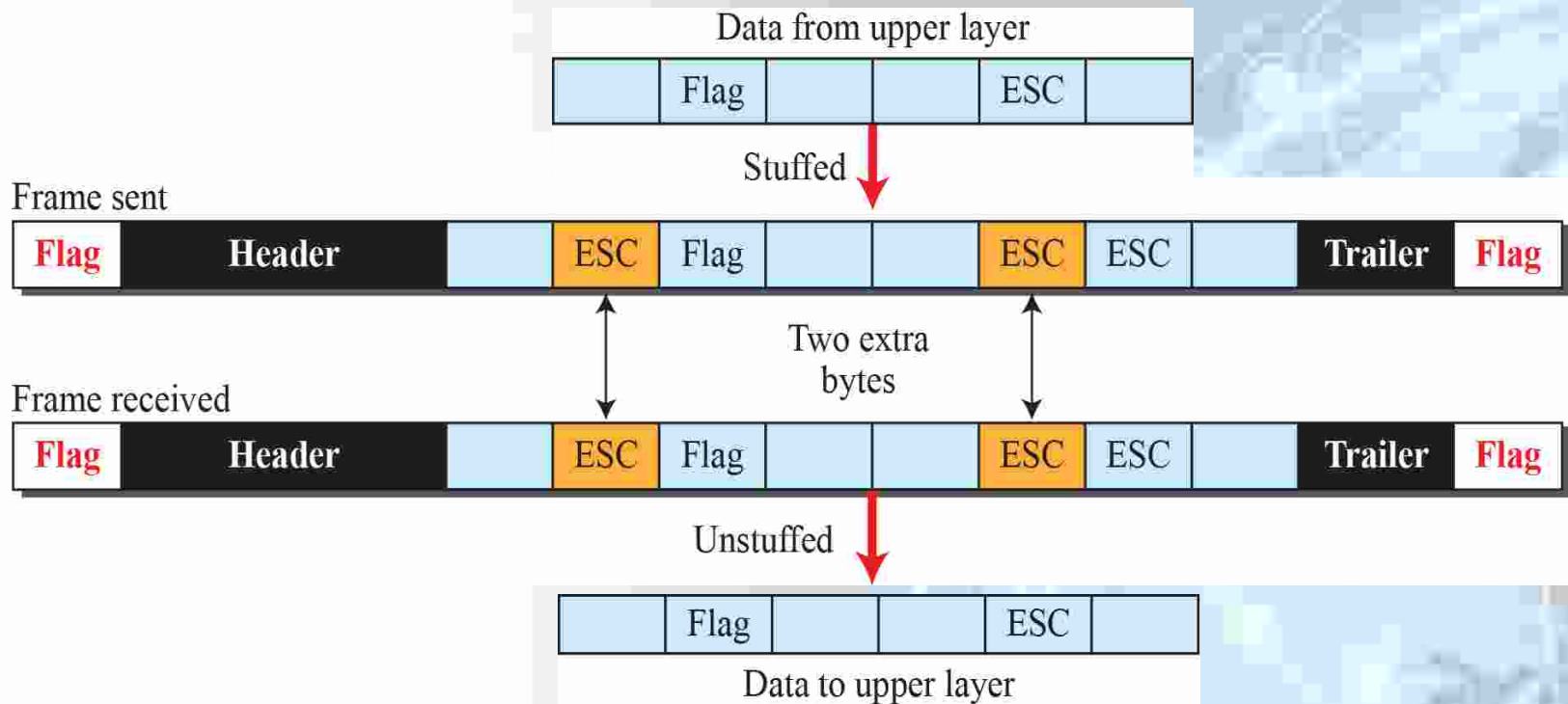
# Byte Stuffing in Connection-Oriented Framing

- **Connection-oriented Framing used text characters as flags**
- **Nowadays any character used for flag can also be a part of the data**
- **In order to avoid confusing the receiver, we use Byte Stuffing**

# Byte Stuffing in Connection-Oriented Framing

- **Several Issues:**
  - ✓ **One or more escape characters followed by a byte with same pattern as a flag?**
  - ✓ **Unicode (16/32 bit) vs. 8-bit characters**
- **Data is stuffed with a pre-defined Escape Character (byte) when there is a character with same pattern as a flag**

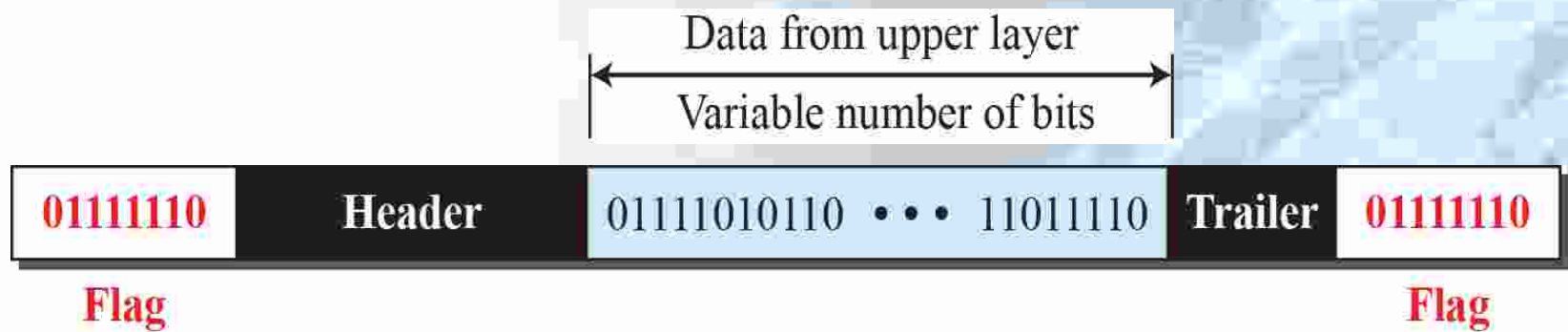
# Byte Stuffing and Unstuffing



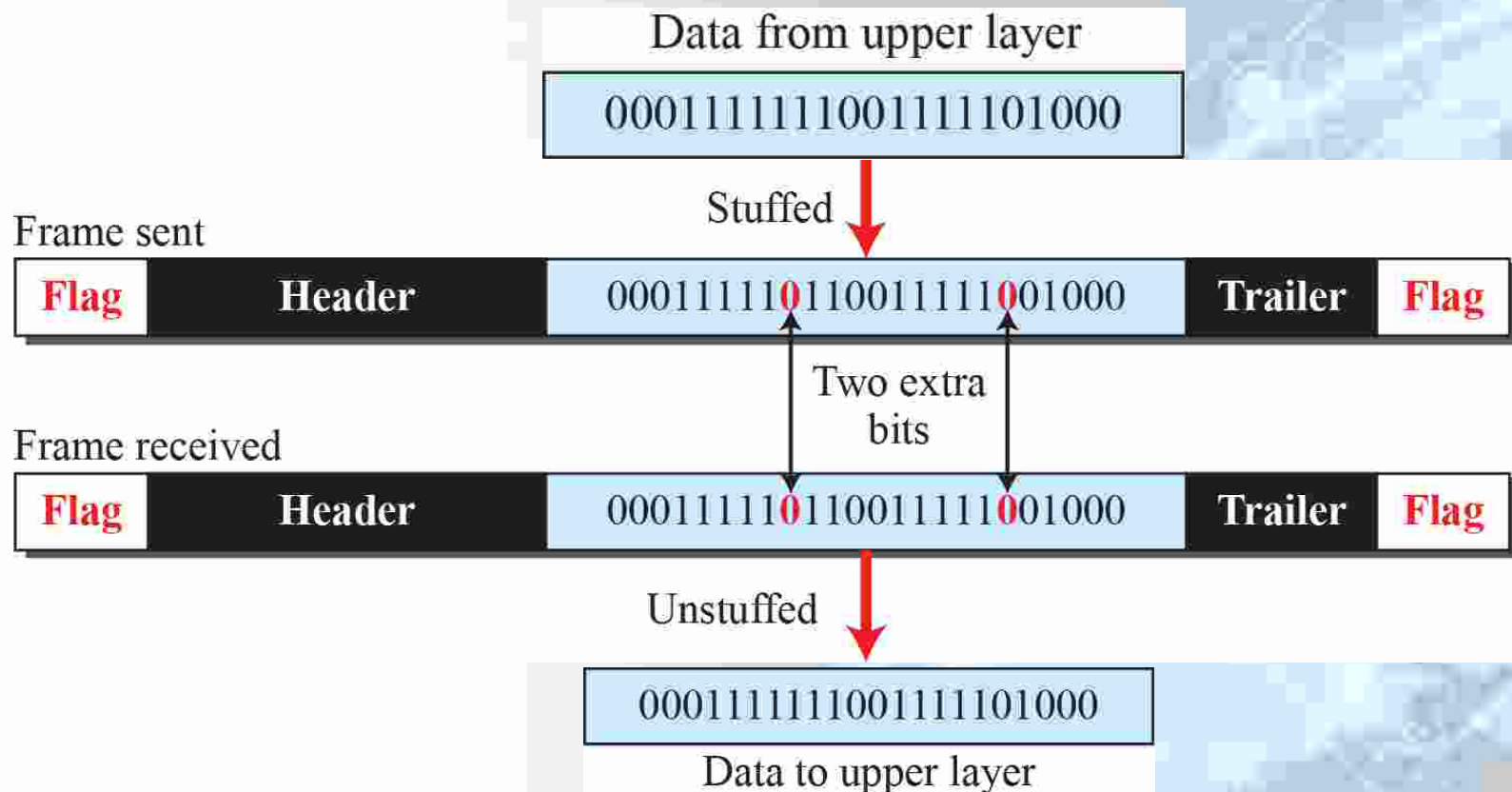
# Bit-Oriented Framing

- Data section of frame is a sequence of bits
- We need a delimiter to separate one frame from the other
- A special 8-bit pattern (01111110) to define beginning and end of a frame
- Same issue as Connection-oriented Framing

# A Frame in a Bit-Oriented Protocol



# Bit Stuffing and Unstuffing



# Flow and Error Control

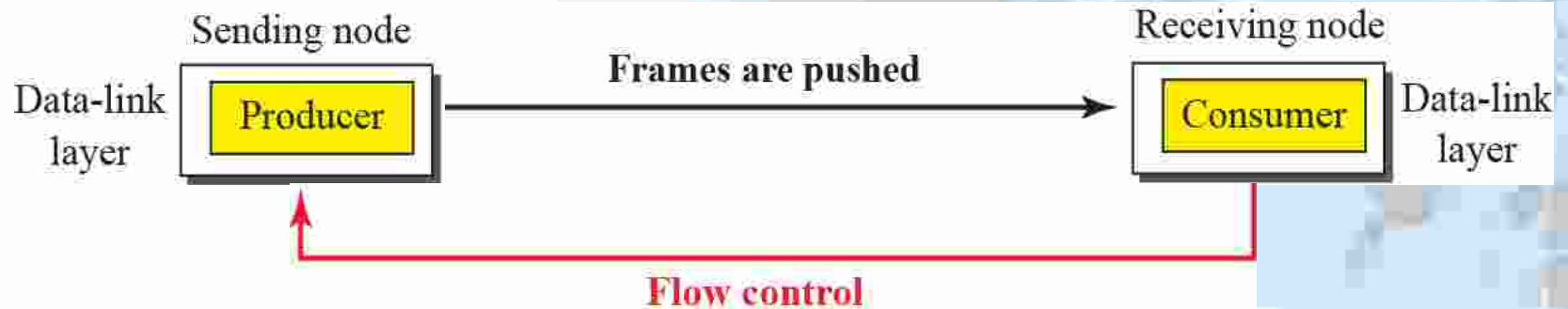
- One of the responsibilities of the data-link control sublayer is flow and error control at the data-link layer

# Flow Control

- **Balance between production and consumption rates**
- **If frames are produced faster than they are consumed at the receiving data link layer, the frames will be discarded**
- **Use of buffers; one at sending end and other at receiving end**



# Flow Control at the Data Link Layer



# Example

- Consumers need to communicate with the producers on two occasions:
  - ✓ When the buffer is full; &
  - ✓ When there are vacancies
- ✓ If the two parties use a buffer with only one slot, the communication can be easier

# Error Control

- Error Control at Data Link layer uses CRC in one of the two ways:
  - ✓ If a frame is corrupted, it is silently discarded and if it is good, it is delivered to network layer
  - ✓ If frame is corrupted, it is silently discarded and if it is good, an acknowledgement is sent to sender

# Connectionless and Connection-Oriented

- A DLC protocol can be either connectionless or connection-oriented
- **Connectionless:** No relationship between the frames
- **Connection-Oriented:** Frames are numbered and sent in order

# DATA-LINK LAYER PROTOCOLS

- Traditionally four protocols have been defined for the data-link layer to deal with flow and error control:
  - ✓ Simple Protocol
  - ✓ Stop-and-Wait Protocol
  - ✓ Go-Back-N Protocol
  - ✓ Selective-Repeat Protocol
- Last two protocols have almost disappeared completely

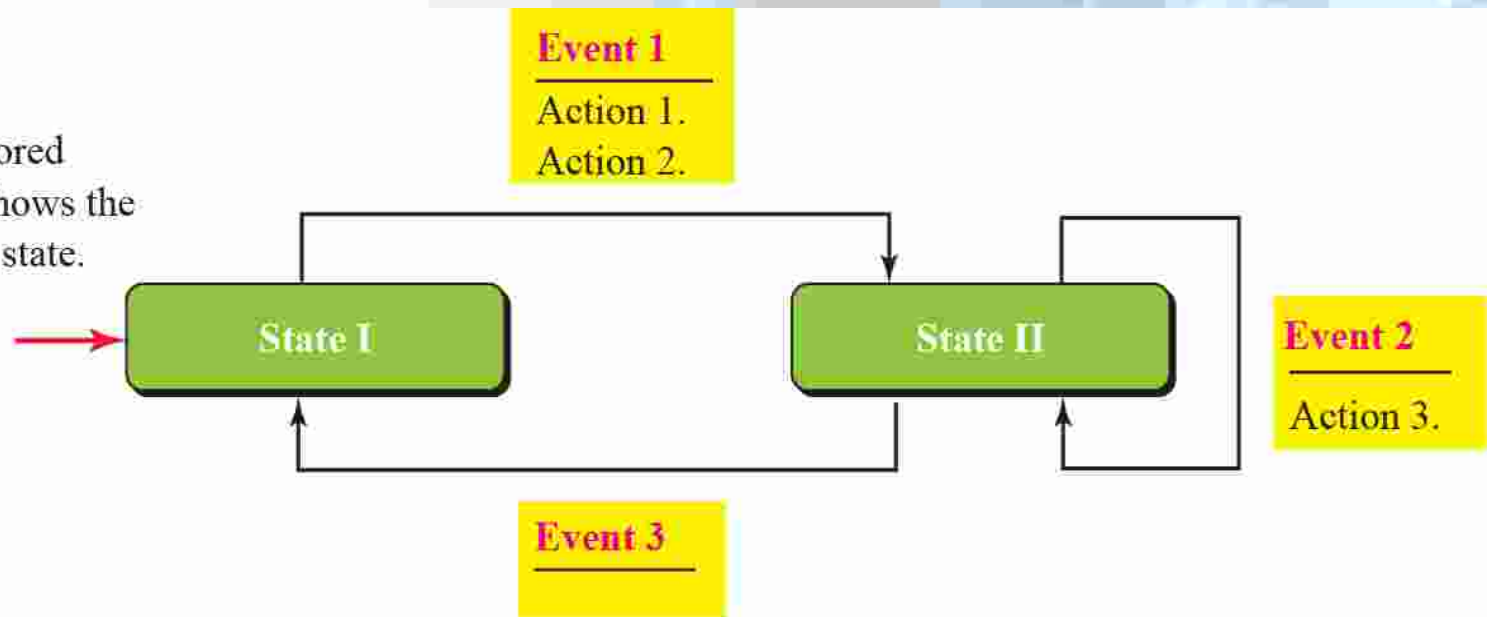
# Finite State Machine (FSM)

- A machine with a finite number of states
- Machines stays in one of the states until an event occurs
- Each event is associated with 2 reactions:
  - ✓ List of actions to be performed
  - ✓ Determining the next state

# Finite State Machine (FSM)

**Note:**

The colored arrow shows the starting state.



# DATA-LINK LAYER PROTOCOLS

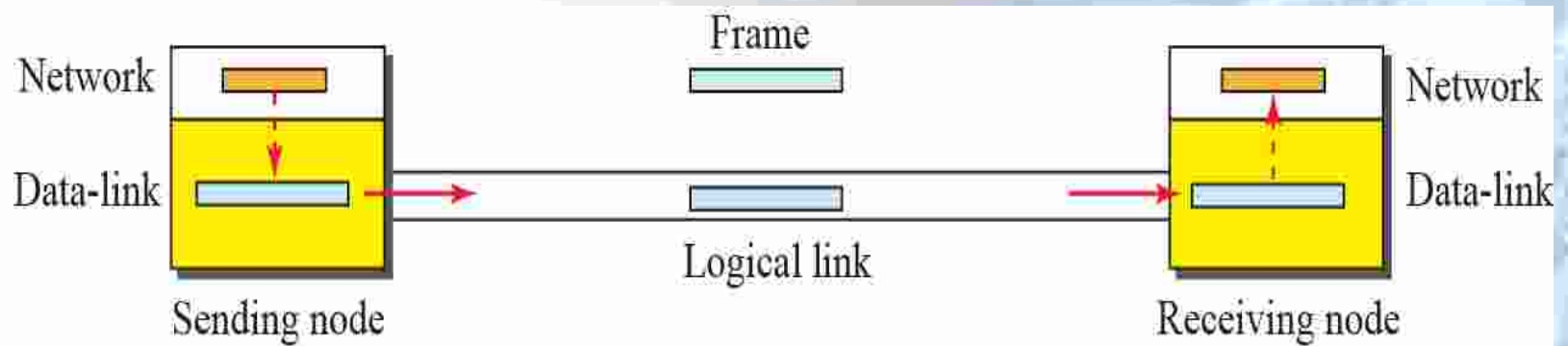
- Traditionally four protocols have been defined for the data-link layer to deal with flow and error control:
  - ✓ Simple Protocol
  - ✓ Stop-and-Wait Protocol
  - ✓ Go-Back-N Protocol
  - ✓ Selective-Repeat Protocol



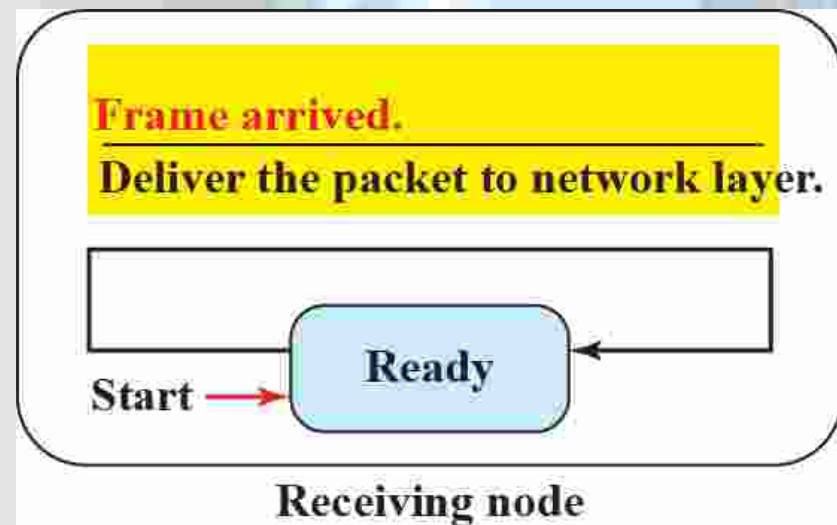
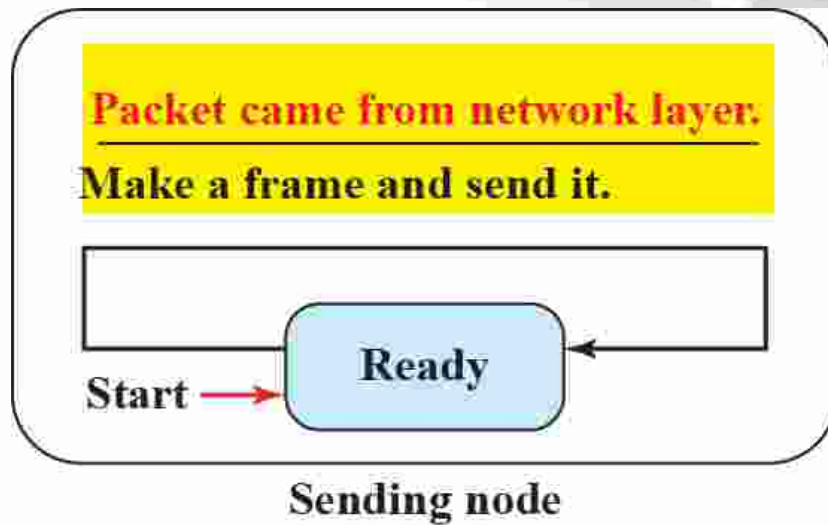
# Simple Protocol

- Simple protocol has neither flow nor error control
- Assumption: The receiver can immediately handle any frame it receives
- The receiver can never be overwhelmed with incoming frames

# Simple Protocol

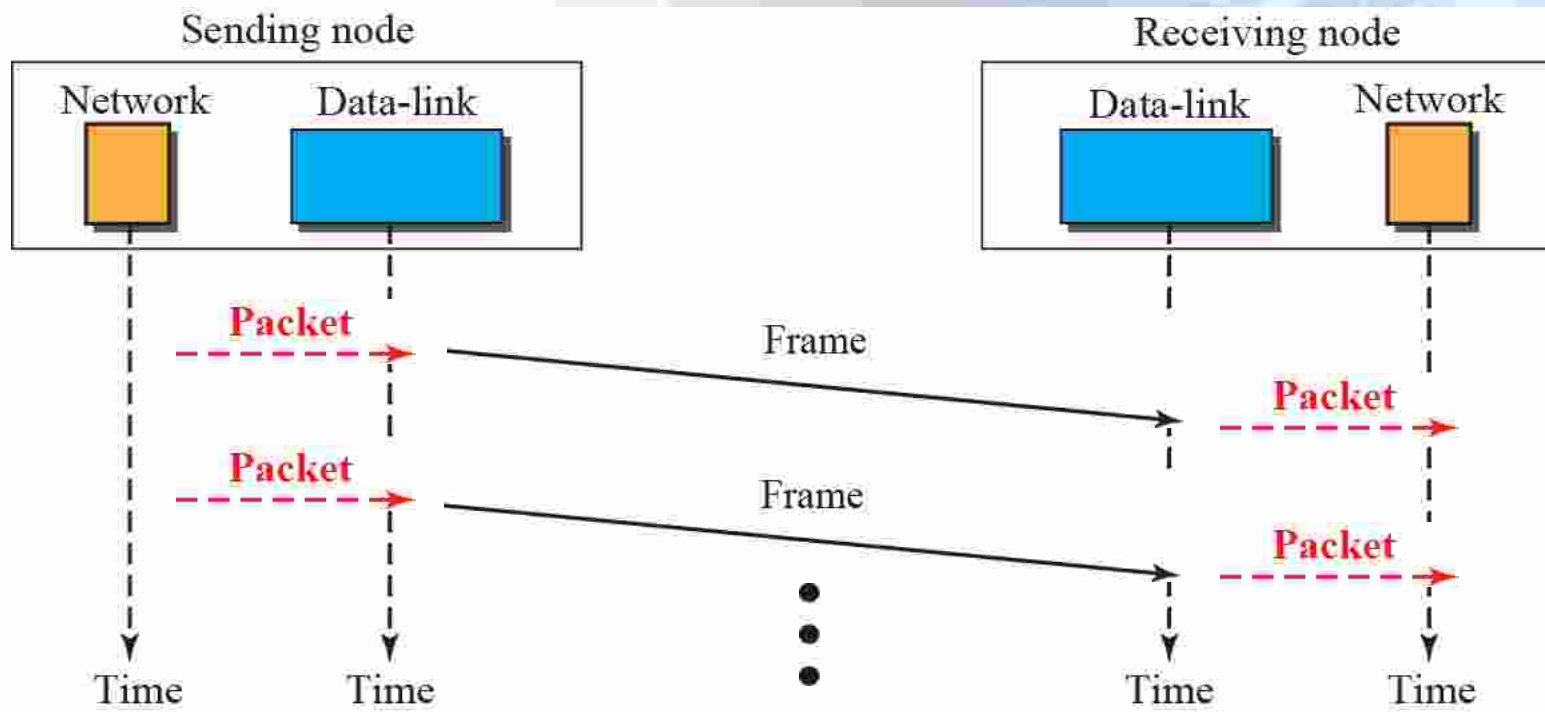


# FSM for Simple Protocol



# Example

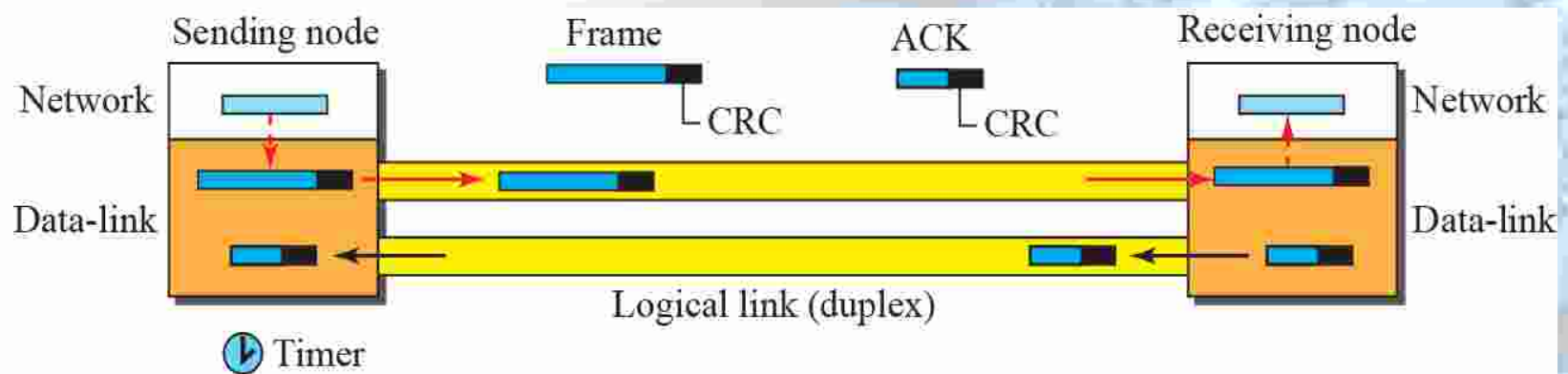
Here is an example of communication using this protocol. It is very simple. The sender sends frames one after another without even thinking about the receiver.



# Stop-and-Wait Protocol

- Stop-and-Wait protocol uses both flow and error control
- The sender sends one frame at a time and waits for an acknowledgment before sending the next one
- To detect corrupted frames, we add a CRC code

# Stop-and-Wait Protocol

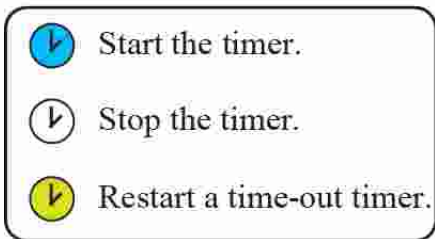


# Stop-and-Wait Protocol

- Stop-and-Wait protocol uses both flow and error control
- The sender sends one frame at a time and waits for an acknowledgment before sending the next one
- To detect corrupted frames, we add a CRC code

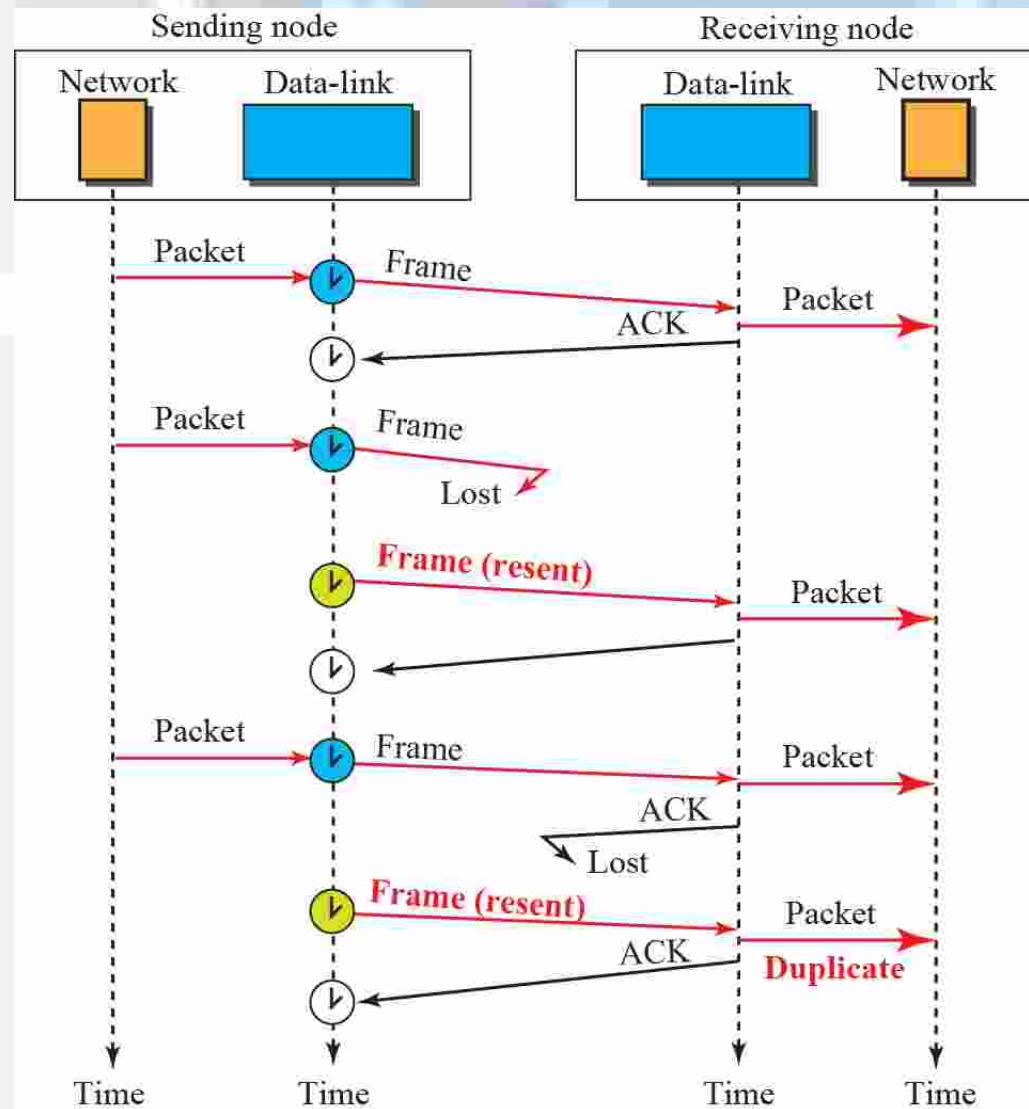
# Example

## Legend



## Notes:

A lost frame means either lost or corrupted.  
A lost ACK means either lost or corrupted.





# Example

## Legend

-  Start the timer.
-  Stop the timer.
-  Restart a time-out timer.

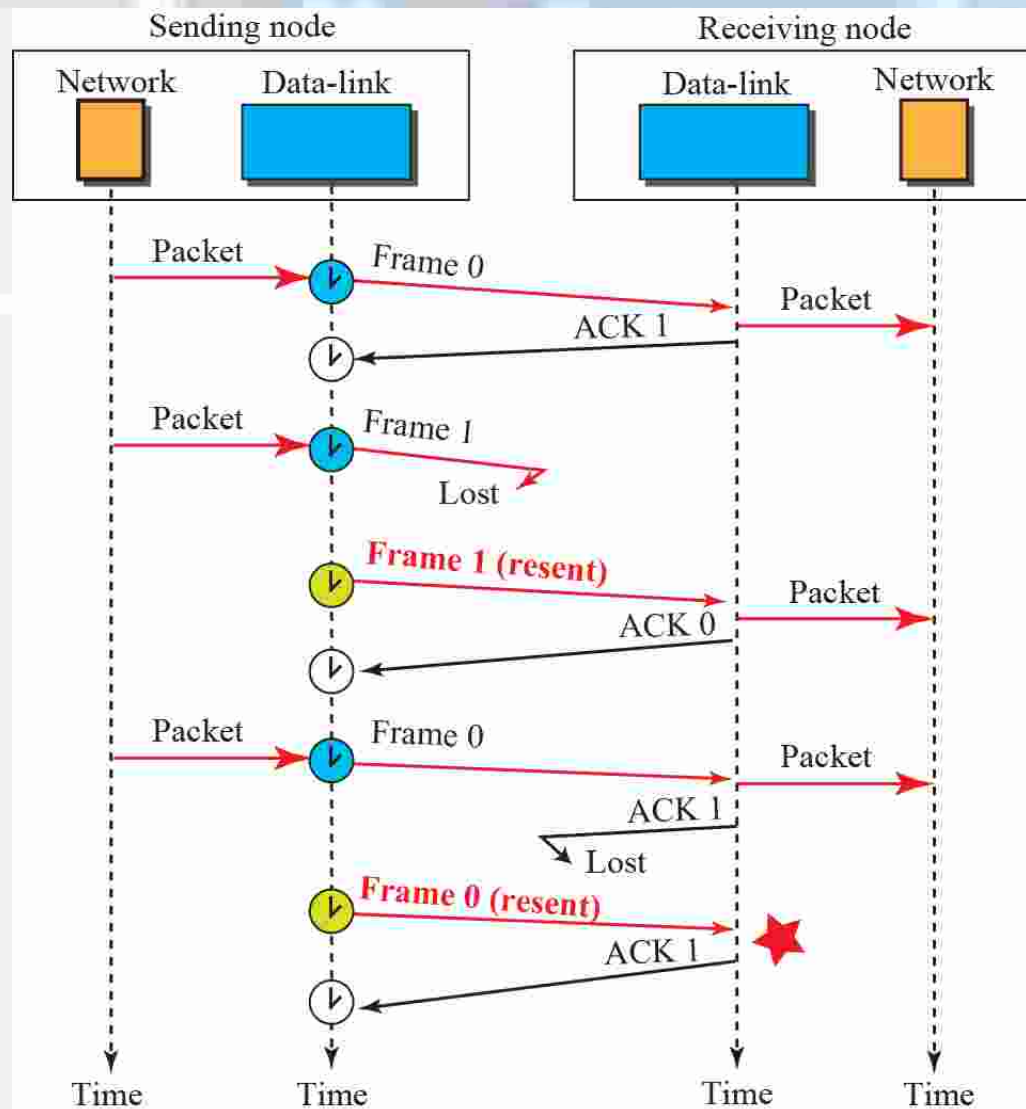
## Notes:

A lost frame means either lost or corrupted.

A lost ACK means either lost or corrupted.



Frame 0 is discarded because the receiver expects frame 1.



# Piggybacking

- Both Simple and Stop-and-wait protocols are designed for unidirectional communication
- Data flows in one direction and ACK travels in the other
- To make the system efficient, the data in one direction is piggybacked with the acknowledgment in the other direction

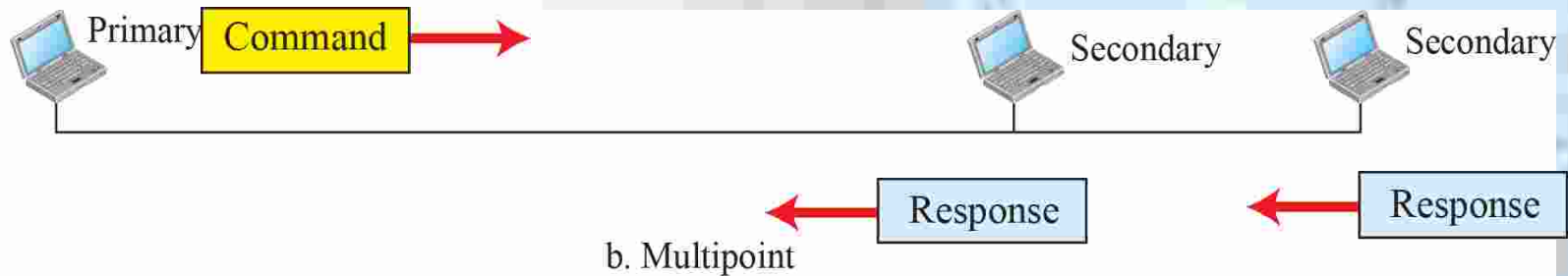
# High-level Data Link Control (HDLC)

- Bit -oriented protocol for communication over point-to-point and multipoint links
- It implements Stop-and-Wait protocol
- Most of the concepts defined in this protocol is the basis for other protocols such as PPP, Ethernet, or wireless LANs

# Configurations & Transfer Modes in HDLC

- HDLC provides two common transfer modes that can be used in different configurations:
  - ✓ Normal Response Mode (NRM) &
  - ✓ Asynchronous Balanced Mode (ABM)

# Normal Response Mode



# Asynchronous Balanced Mode

Combined



Command/response



Combined



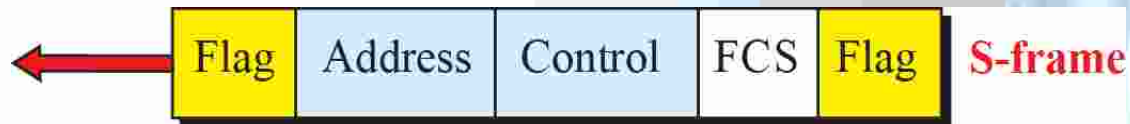
Command/response



# Framing

- HDLC defines three types of frames:
  - ✓ information frames (I-frames)
  - ✓ Supervisory frames (S-frames)
  - ✓ Unnumbered frames (U-frames)

# HDLC Frames





# Point-to-Point Protocol (PPP)

- Most common protocol for point-to-point access
- Millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP
- To control and manage the transfer of data, there is a need for a PPP at the data-link layer

# Services provided by PPP

The designers of PPP have included several services to make it suitable for a point-to-point protocol, but have ignored some traditional services to make it simple

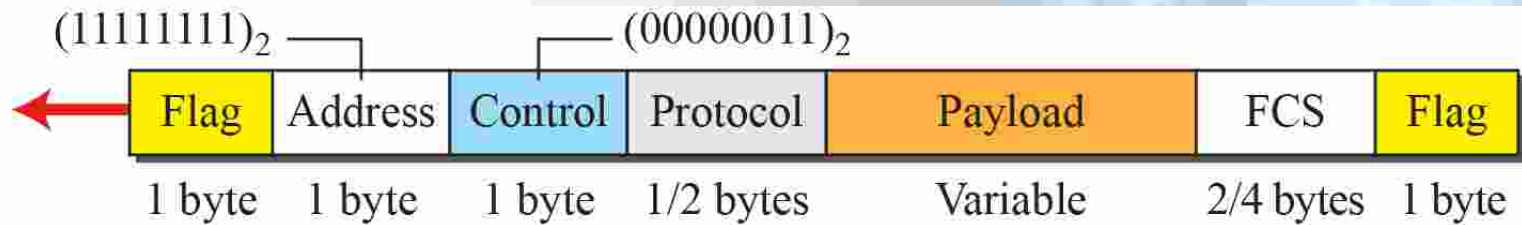
| Services Included                    | Services Not Included                         |
|--------------------------------------|-----------------------------------------------|
| Framing                              | Flow Control                                  |
| Link Establishment and Data Exchange | Error Correction (PPP has CRC detection only) |
| Authentication                       | No Sequence Numbering                         |
| Multilink PPP Address configuration  | Absence of sophisticated Addressing Mechanism |
| Network Address configuration        |                                               |

# Point-to-Point Protocol (PPP)

- Most common protocol for point-to-point access
- Millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP
- To control and manage the transfer of data, there is a need for a PPP at the data-link layer

# PPP Frame Format

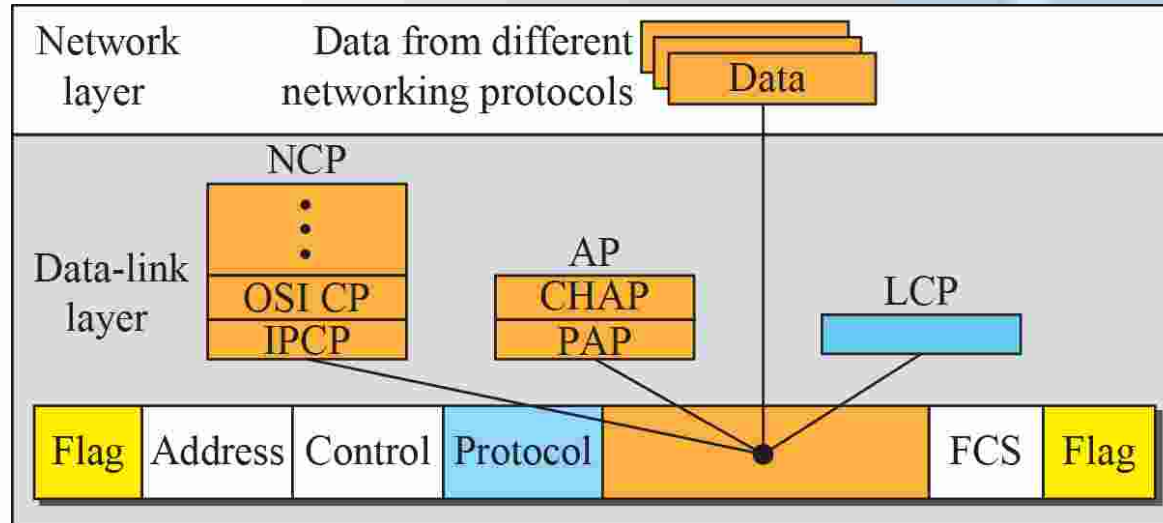
PPP uses a character-oriented (or byte-oriented) frame



# Multiplexing in PPP

- Although PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate and carry the network-layer data
- Three sets of protocols are:
  - Link Control Protocol (LCP)
  - Two Authentication Protocols (APs)
  - Several Network Control Protocols (NCPs)

# Multiplexing in PPP



## Legend

LCP : Link control protocol  
AP : Authentication protocol  
NCP: Network control protocol

## Protocol values:

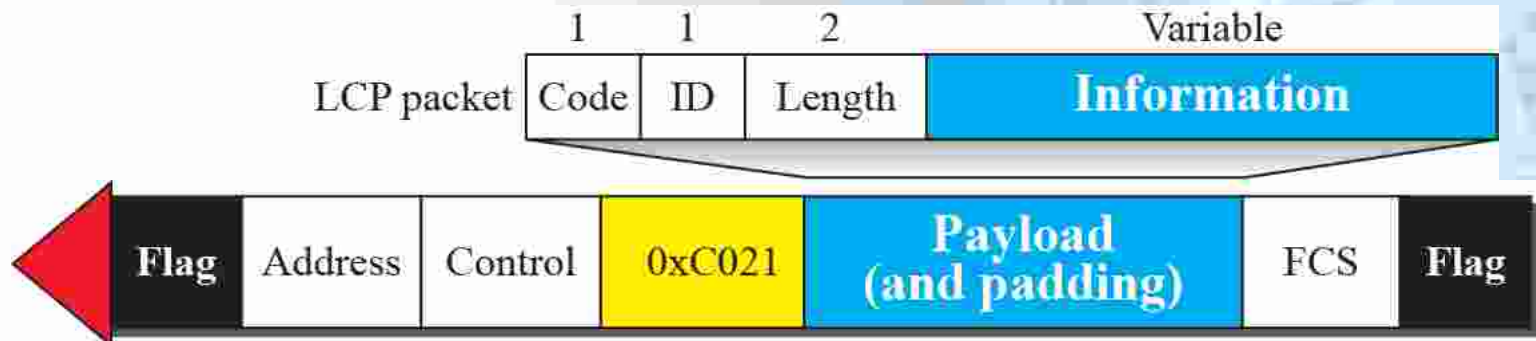
LCP : 0xC021

AP : 0xC023 and 0xC223

NCP: 0x8021 and ....

Data: 0x0021 and ....

# LCP Packet encapsulated in a Frame



# LCP Packets

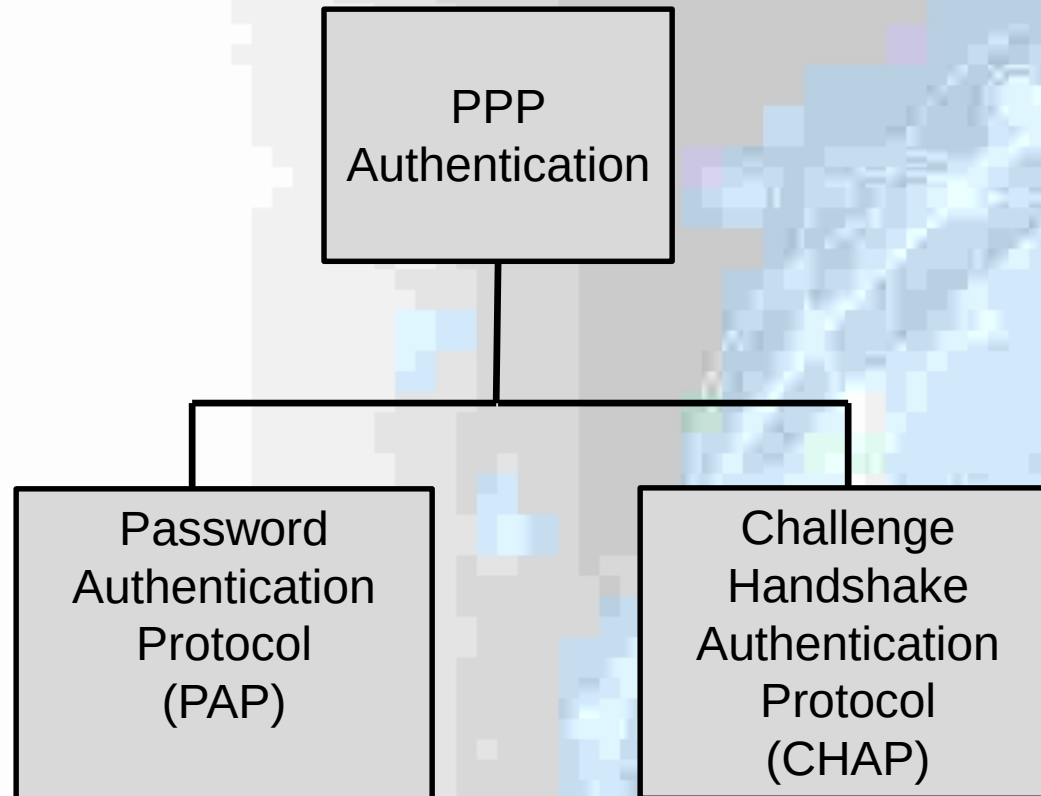
| Code | Packet Type       | Description                                                |
|------|-------------------|------------------------------------------------------------|
| 0x01 | Configure-request | Contains the list of proposed options and their values     |
| 0x02 | Configure-ack     | Accepts all options proposed                               |
| 0x03 | Configure-nak     | Announces that some options are not acceptable             |
| 0x04 | Configure-reject  | Announces that some options are not recognized             |
| 0x05 | Terminate-request | Request to shut down the line                              |
| 0x06 | Terminate-ack     | Accept the shutdown request                                |
| 0x07 | Code-reject       | Announces an unknown code                                  |
| 0x08 | Protocol-reject   | Announces an unknown protocol                              |
| 0x09 | Echo-request      | A type of hello message to check if the other end is alive |
| 0x0A | Echo-reply        | The response to the echo-request message                   |
| 0x0B | Discard-request   | A request to discard the packet                            |



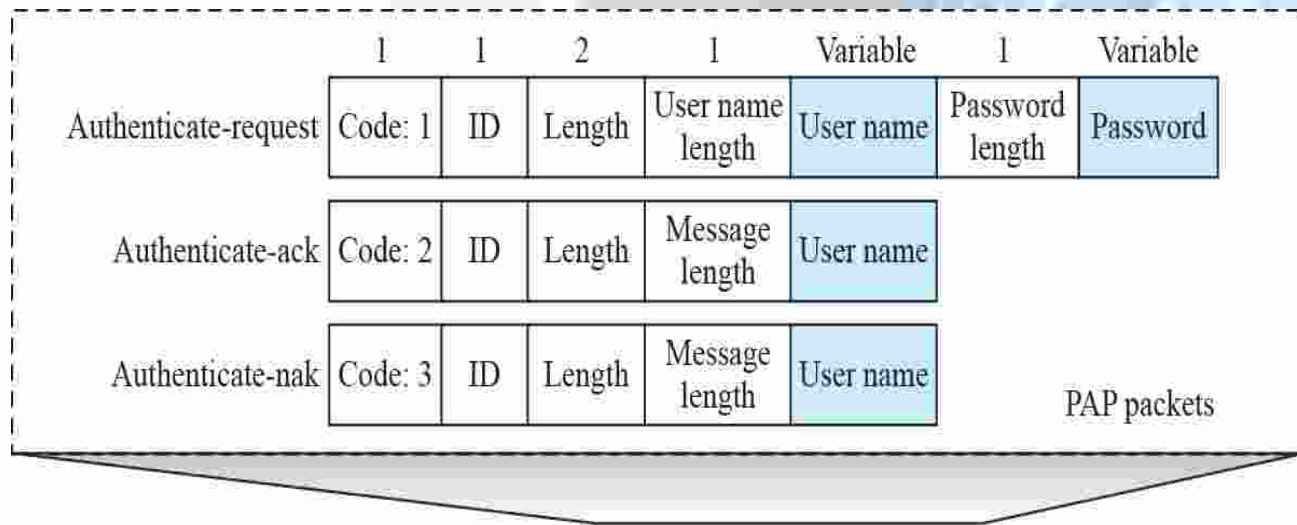
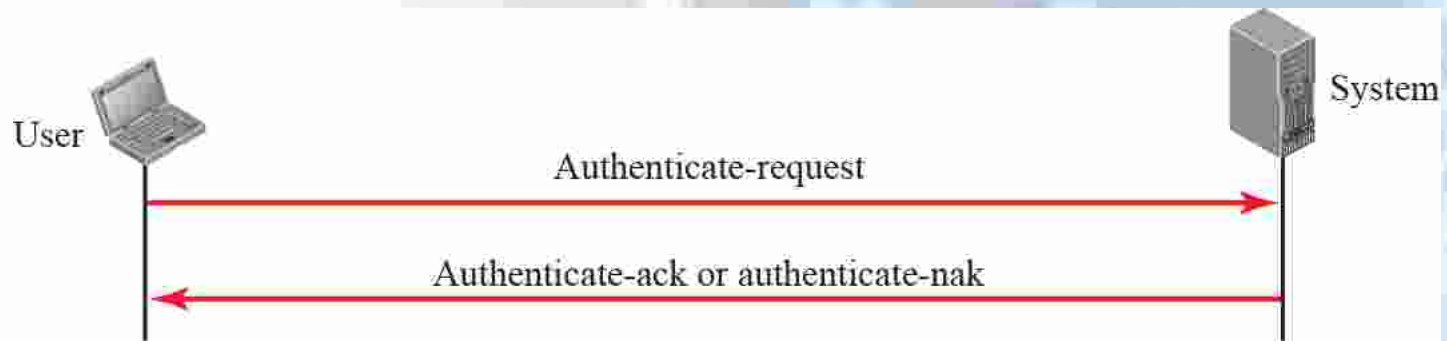
# Multiplexing in PPP

- Although PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate and carry the network-layer data
- Three sets of protocols are:
  - Link Control Protocol (LCP)
  - Two Authentication Protocols (APs)
  - Several Network Control Protocols (NCPs)

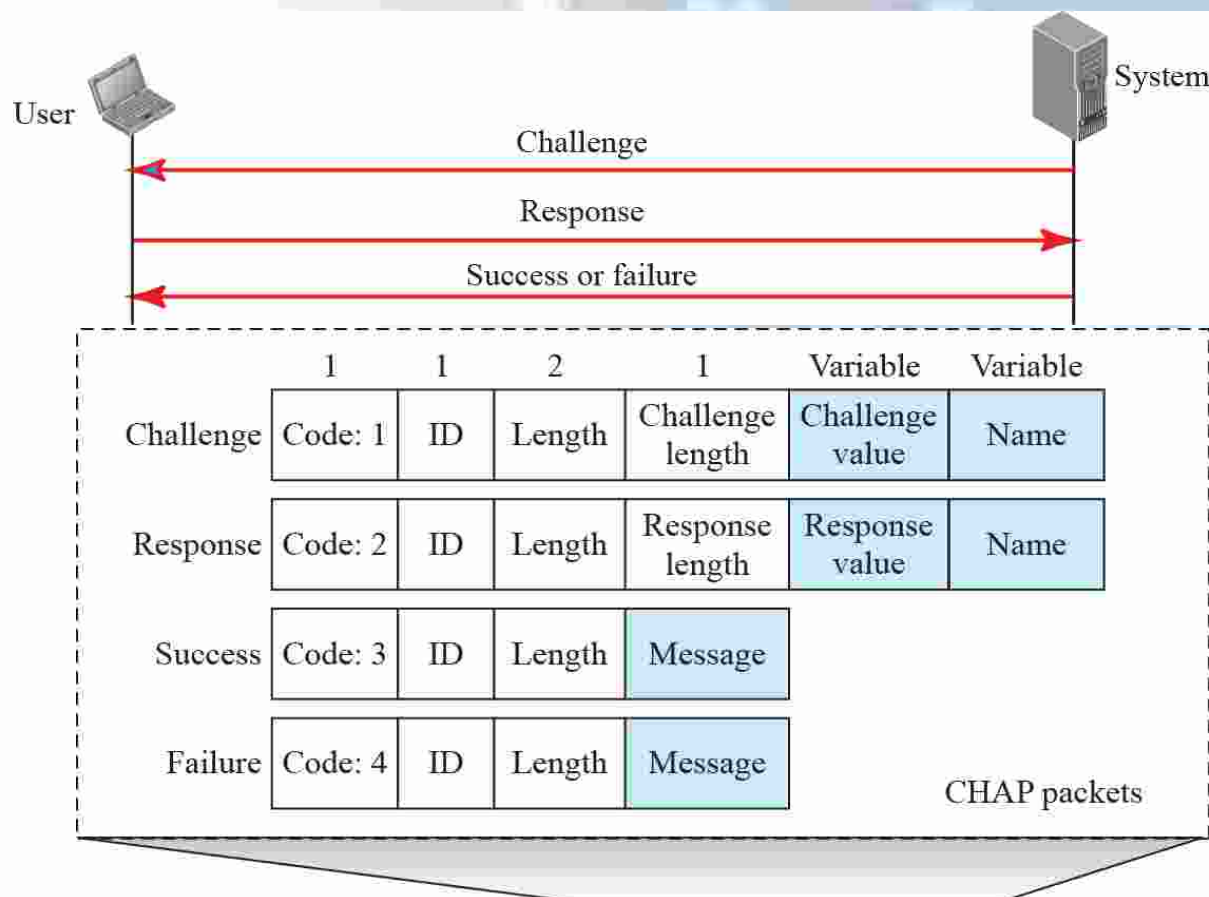
# Authentication Protocols in PPP



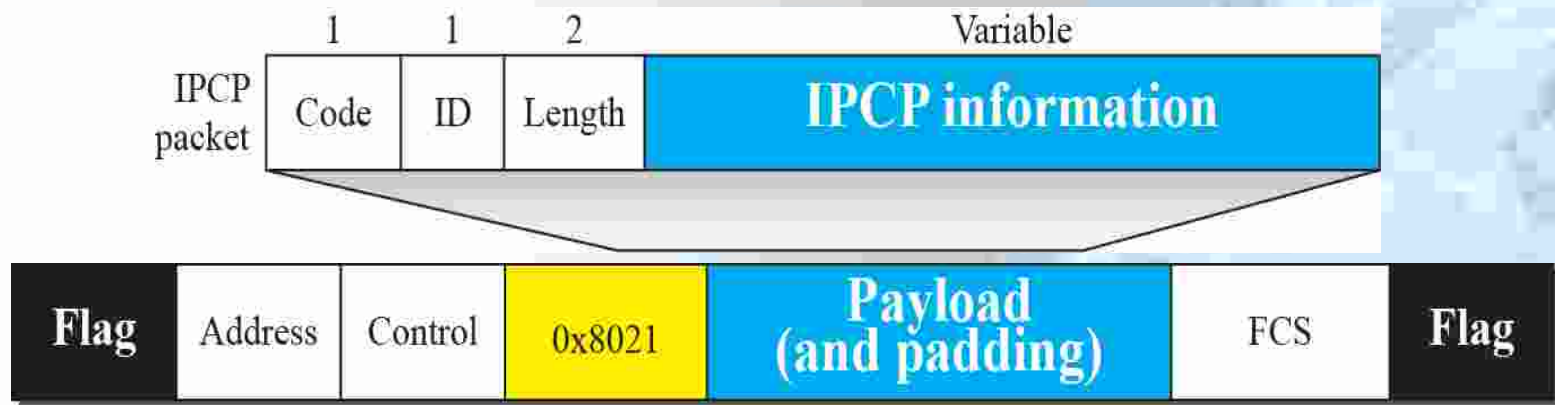
# PAP packets encapsulated in a PPP frame



# CHAP Packets encapsulated in a PPP frame



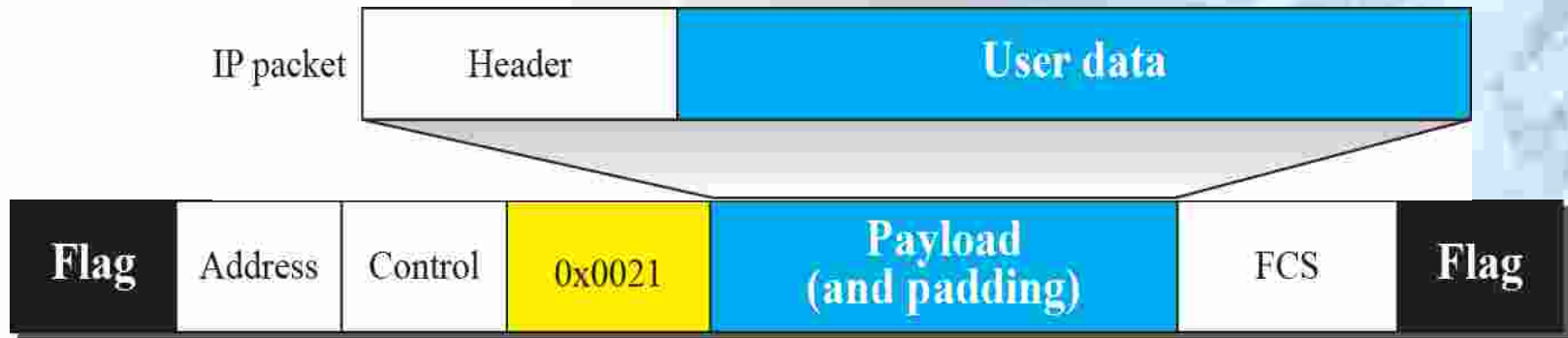
# Internet Protocol Control Protocol (IPCP)



# Code values for IPCP Packets

| <i>Code</i> | <i>IPCP Packet</i> |
|-------------|--------------------|
| 0x01        | Configure-request  |
| 0x02        | Configure-ack      |
| 0x03        | Configure-nak      |
| 0x04        | Configure-reject   |
| 0x05        | Terminate-request  |
| 0x06        | Terminate-ack      |
| 0x07        | Code-reject        |

# IP datagram encapsulated in a PPP frame

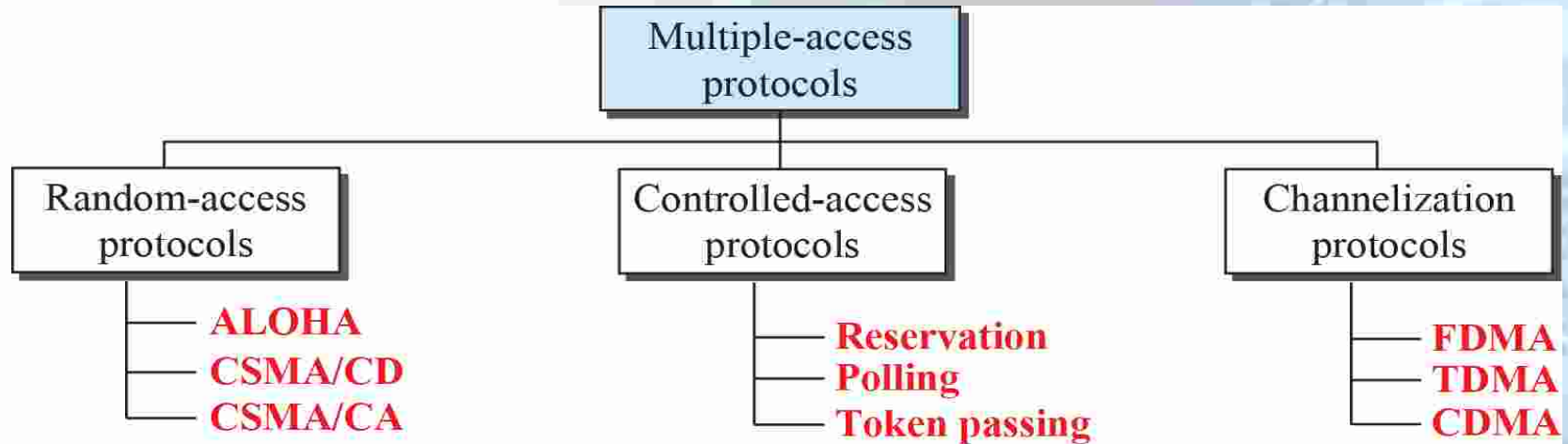


# Media Access Control (MAC) Sub-Layer

- When nodes use a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link
- Many protocols have been devised to handle access to a shared link
- All of these protocols belong to Media Access Control (MAC) sub-layer



# Taxonomy of Multiple-Access Protocols



# Random Access

- In random-access or contention no station is superior to the other and none is assigned control over the other
- Station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send
- This decision depends on the state of the medium (idle or busy)

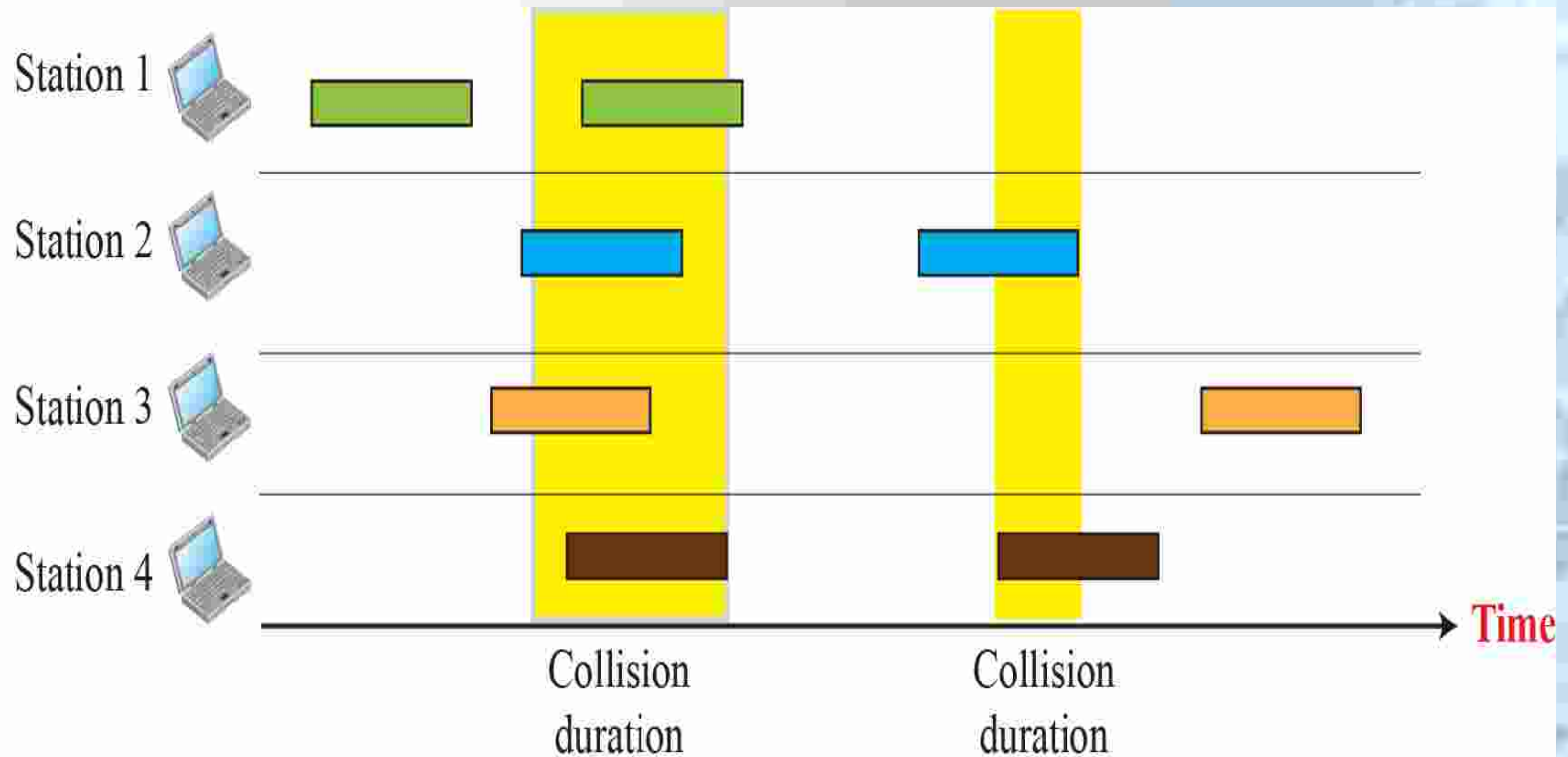
# ALOHA

- ALOHA, the earliest random access method, was developed in early 1970s
- Designed for a radio (wireless) LAN, but it can be used on any shared medium
- Potential collisions in this arrangement as the medium is shared between the stations

# ALOHA

- When a station sends data, another station may attempt to do so at the same time
- The data from the two stations collide and become garbled

# Frames in a pure ALOHA network



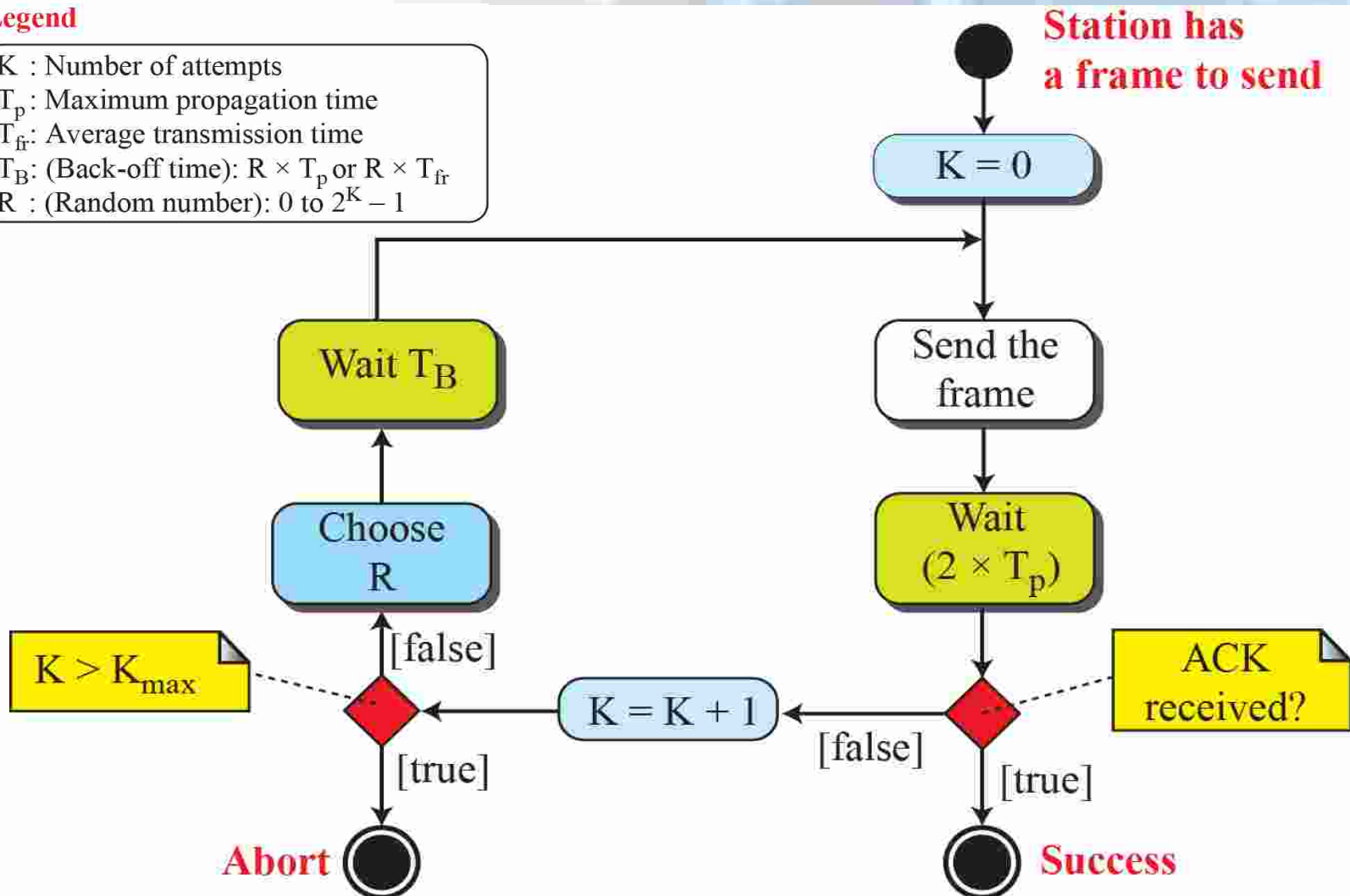
# ALOHA

- ALOHA, the earliest random access method, was developed in early 1970s
- Designed for a radio (wireless) LAN, but it can be used on any shared medium
- Potential collisions in this arrangement as the medium is shared between the stations

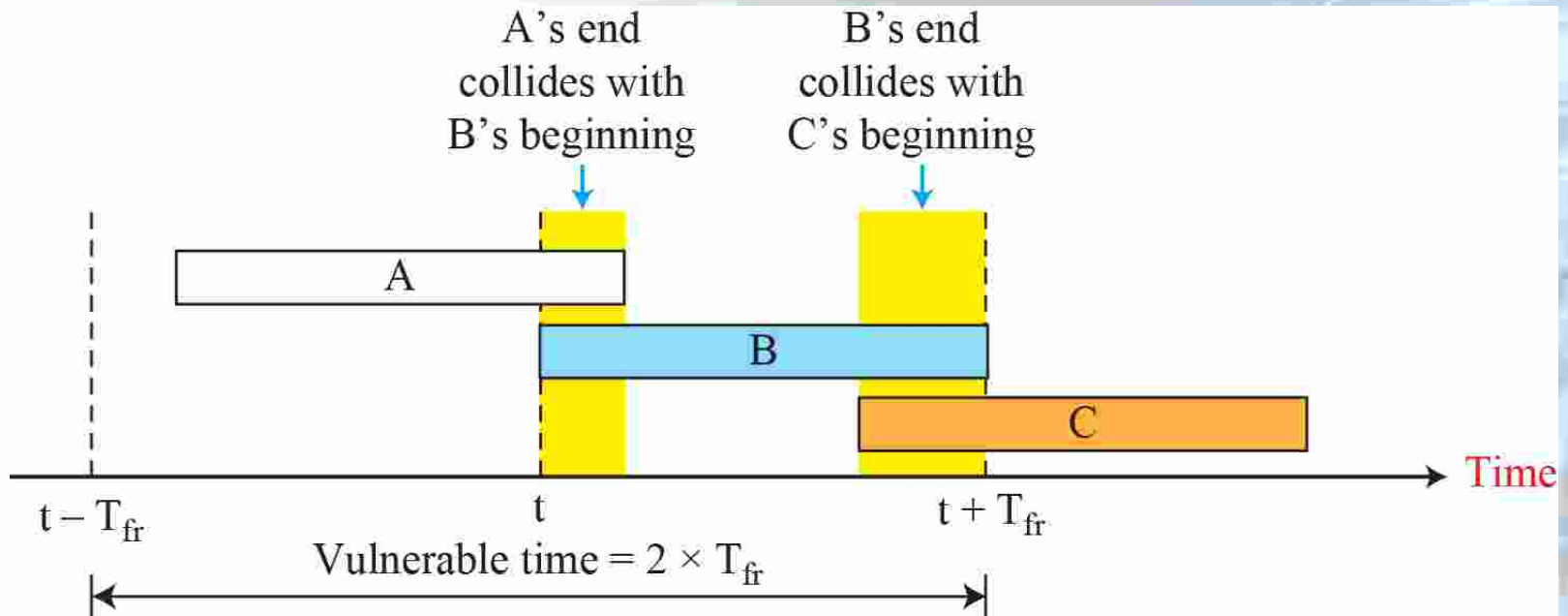
# Procedure for pure ALOHA protocol

## Legend

$K$  : Number of attempts  
 $T_p$  : Maximum propagation time  
 $T_{fr}$  : Average transmission time  
 $T_B$  : (Back-off time):  $R \times T_p$  or  $R \times T_{fr}$   
 $R$  : (Random number): 0 to  $2^K - 1$



# Vulnerable Time for pure ALOHA protocol

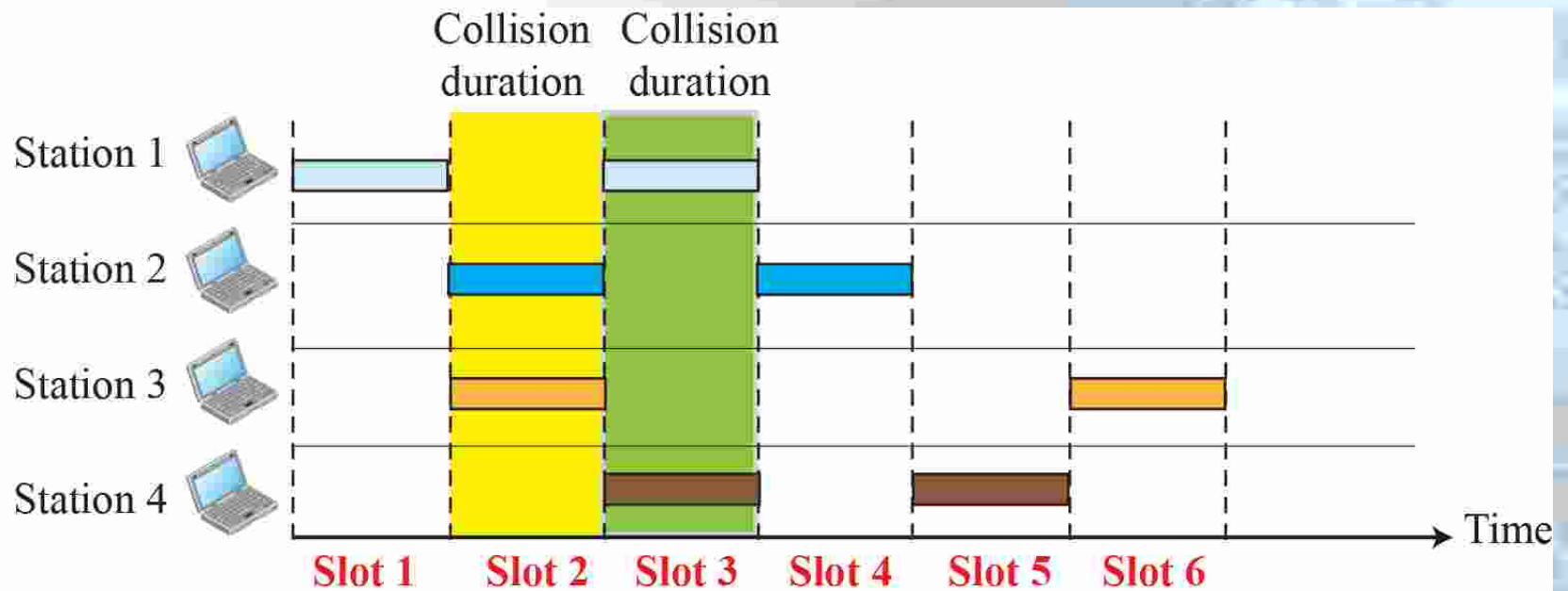




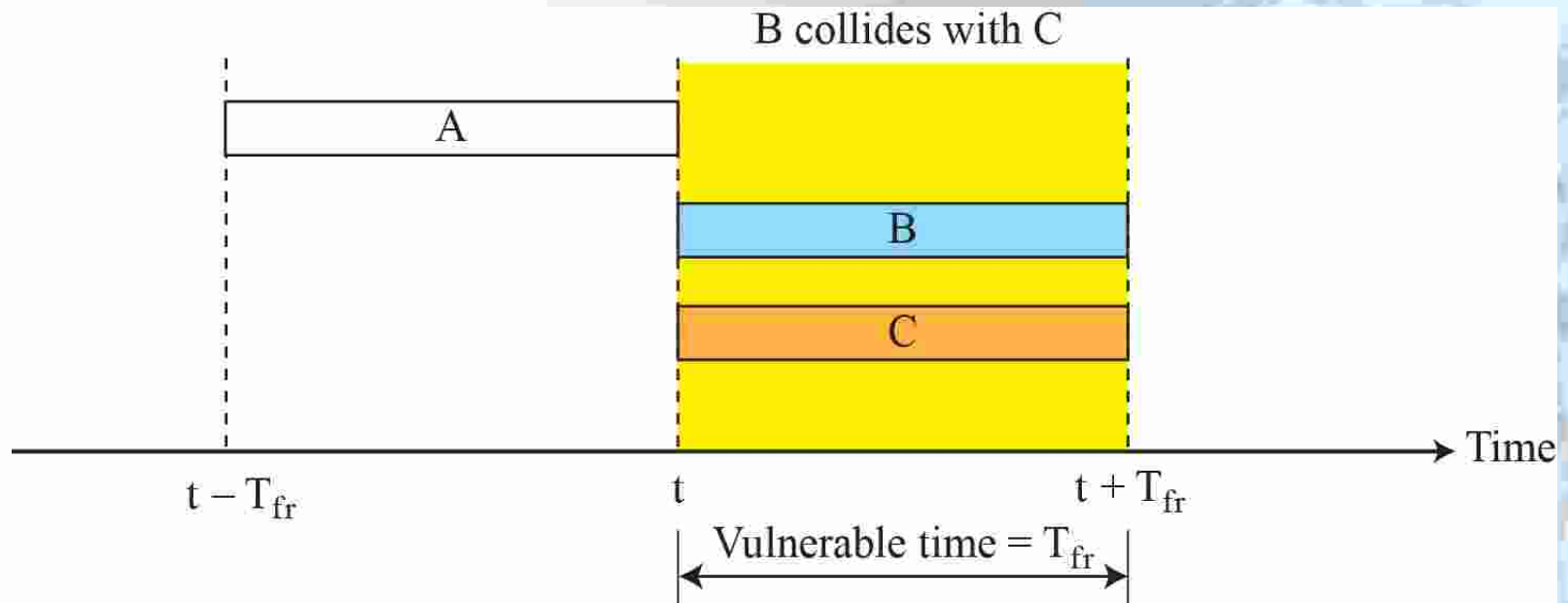
# Slotted ALOHA

- We divide time into slots of  $T_{fr}$  sec and force the station to send only at the beginning of the slot
- Invented to improve the efficiency of pure ALOHA
- If a station misses the time slot, it must wait until beginning of next time slot reducing vulnerable time to  $T_{fr}$  (vs.  $2 \times T_{fr}$  for pure ALOHA)

# Frames in a Slotted ALOHA Network



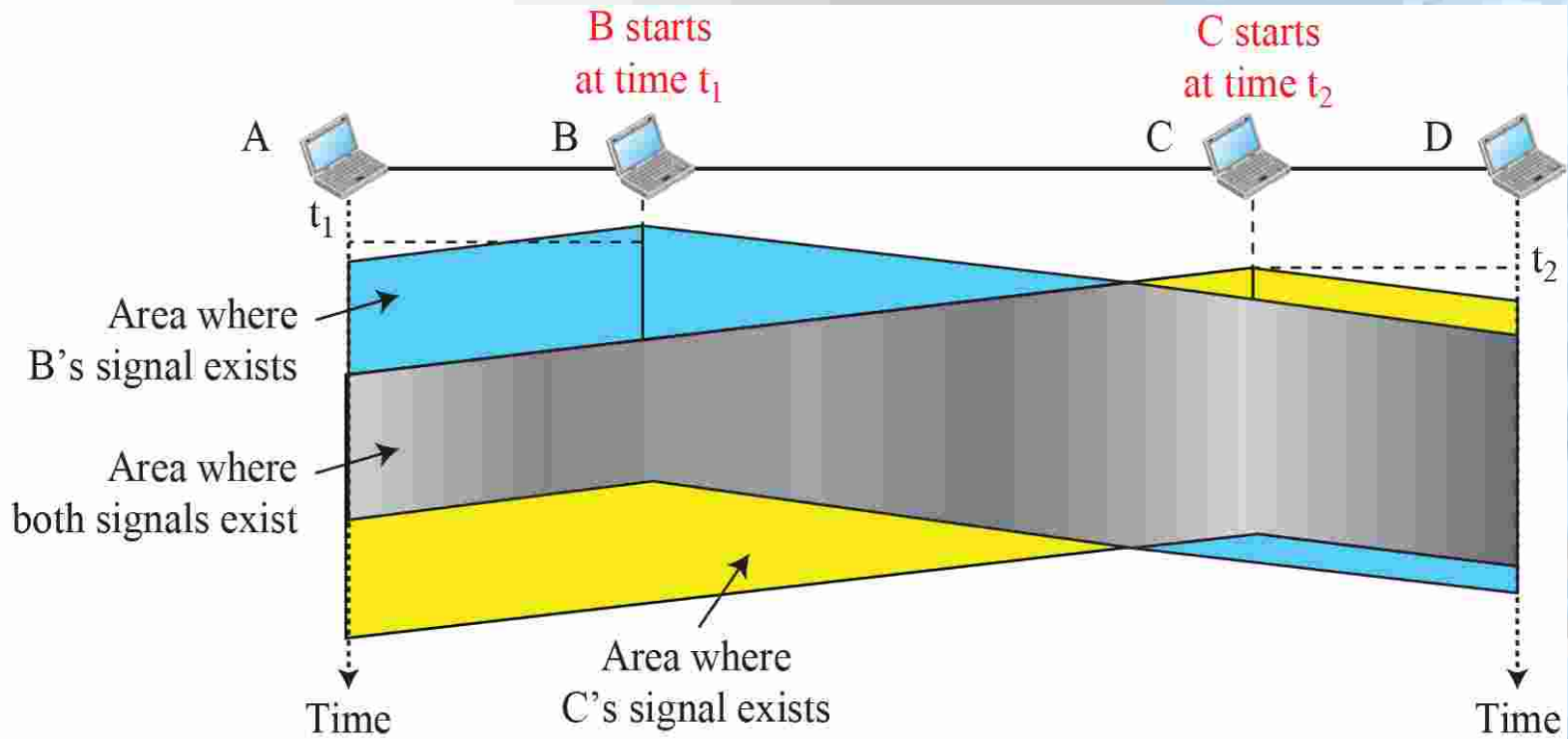
# Vulnerable Time for Slotted ALOHA



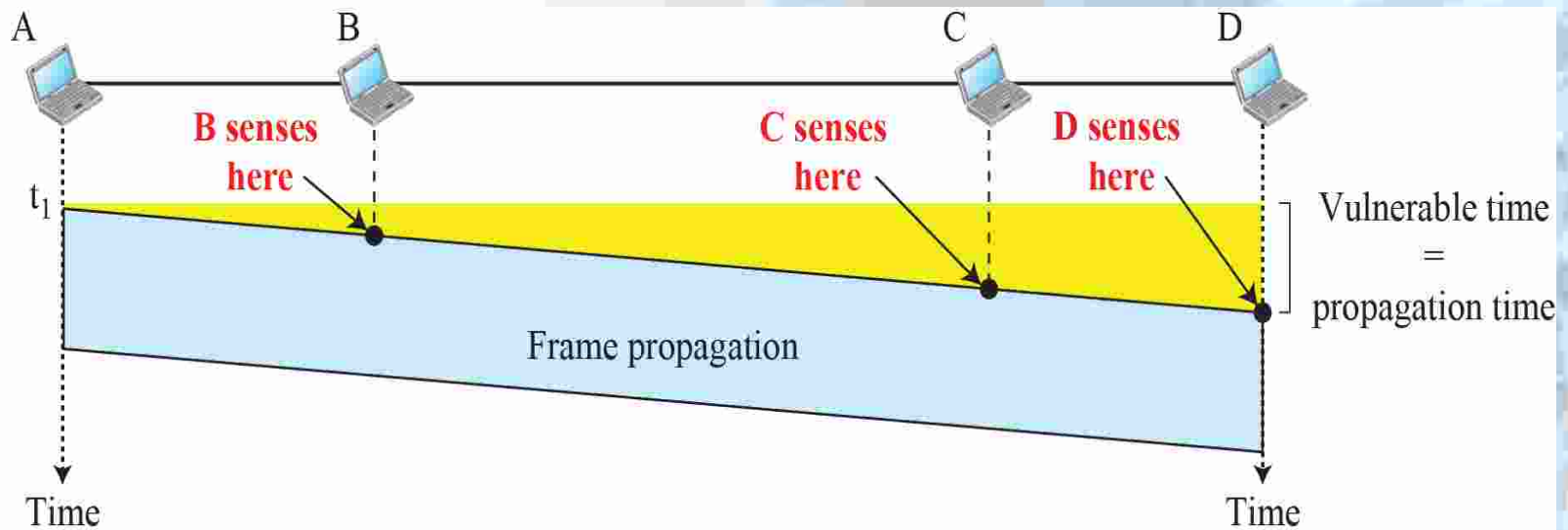
# Carrier Sense Multiple Access (CSMA)

- To minimize the chance of collision and, therefore, increase the performance, CSMA was developed
- The chance of collision is reduced as the station is required to sense/listen to the medium before sending data
- 'sense before transmit' or 'listen before talk'

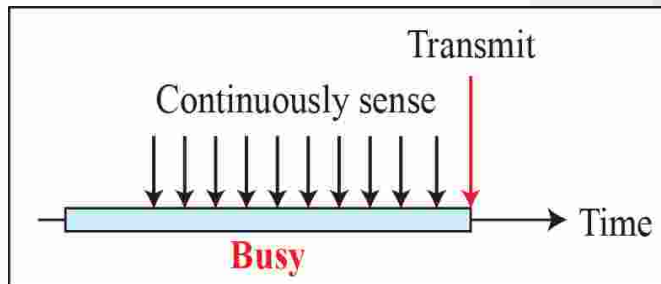
# Space/Time Model of a Collision in CSMA



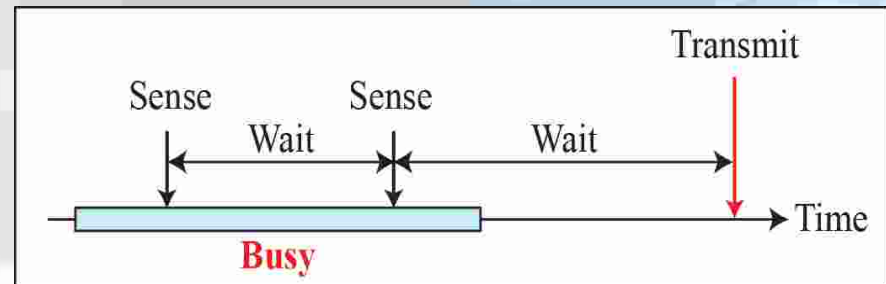
# Vulnerable Time in CSMA



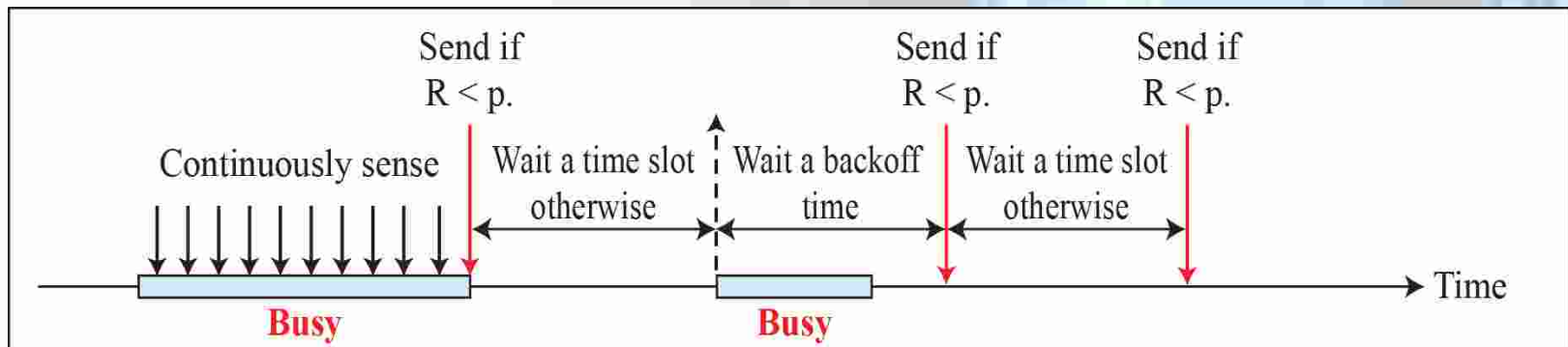
# Behavior of Three Persistence Methods



a. 1-persistent



b. Nonpersistent



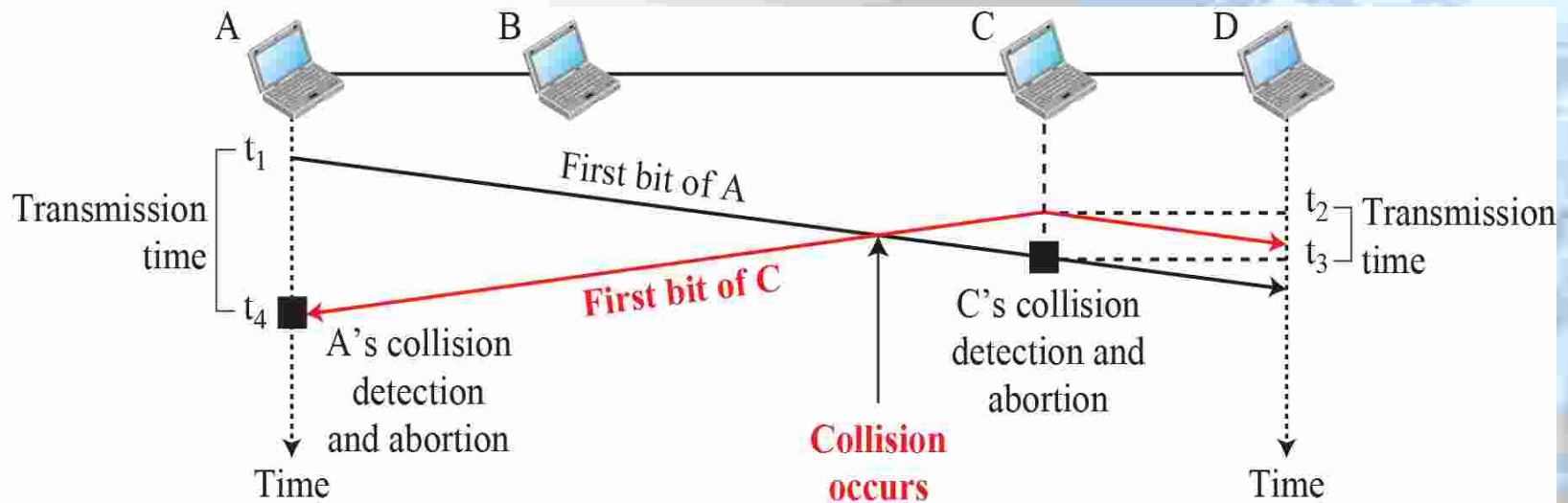
c.  $p$ -persistent

# Carrier Sense Multiple Access/Collision Detection

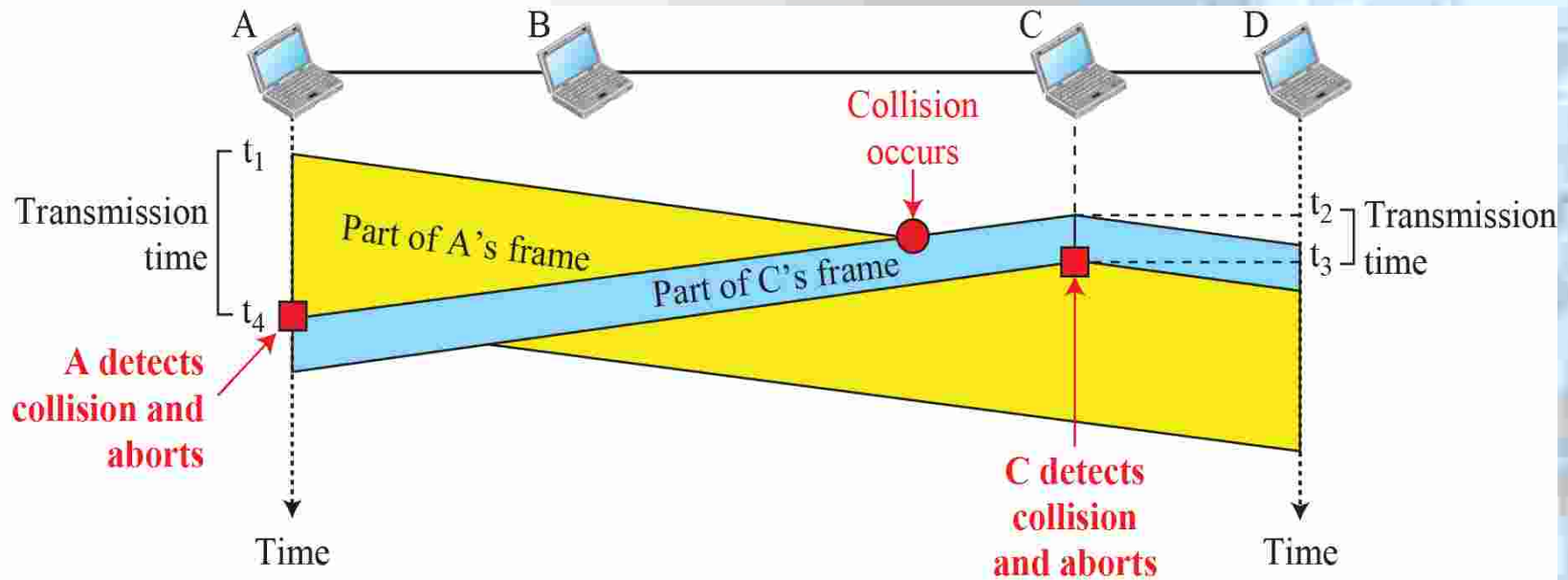
- CSMA method does not specify the procedure following a collision
- CSMA/CD augments the algorithm to handle the collision
- The station monitors the medium after it sends a frame to see if the transmission was successful. If there is a collision, the frame is sent again



# Collision of the First Bits in CSMA/CD



# Collision and Abortion in CSMA/CD



# Flow Diagram for the CSMA/CD

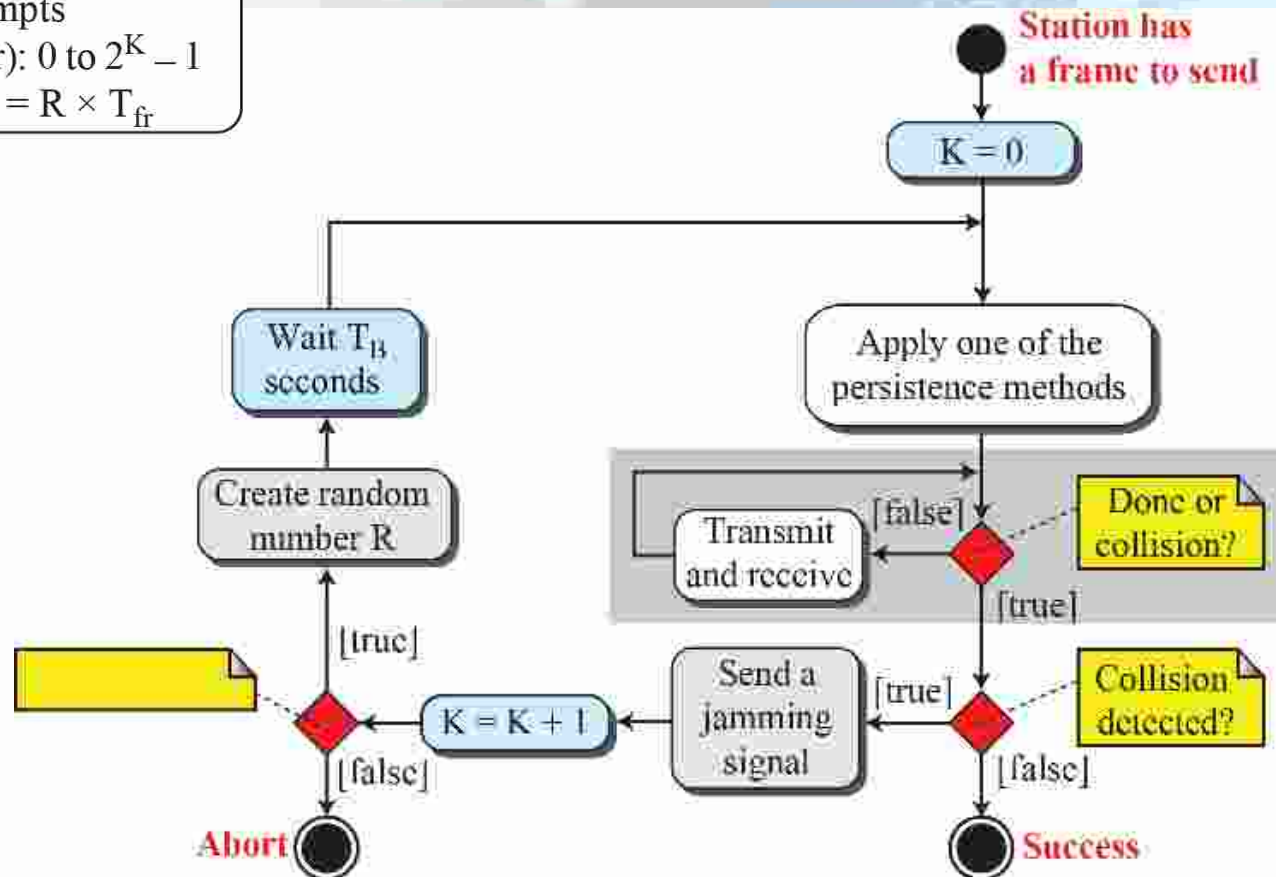
## Legend

$T_{fr}$ : Frame average transmission time

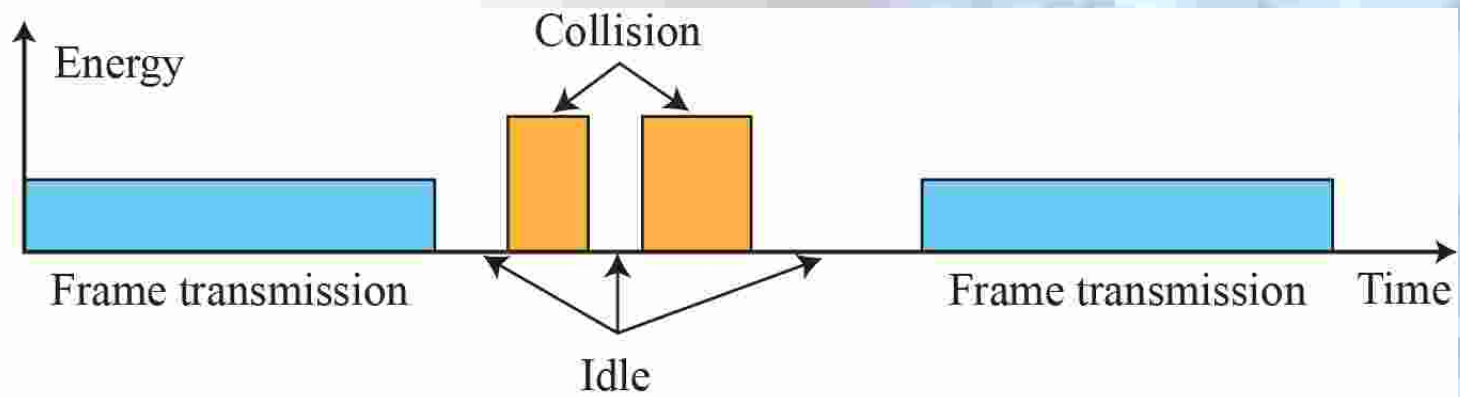
$K$  : Number of attempts

$R$  : (random number): 0 to  $2^K - 1$

$T_B$ : (Back-off time) =  $R \times T_{fr}$



# Energy Level During Transmission, Idleness and Collision



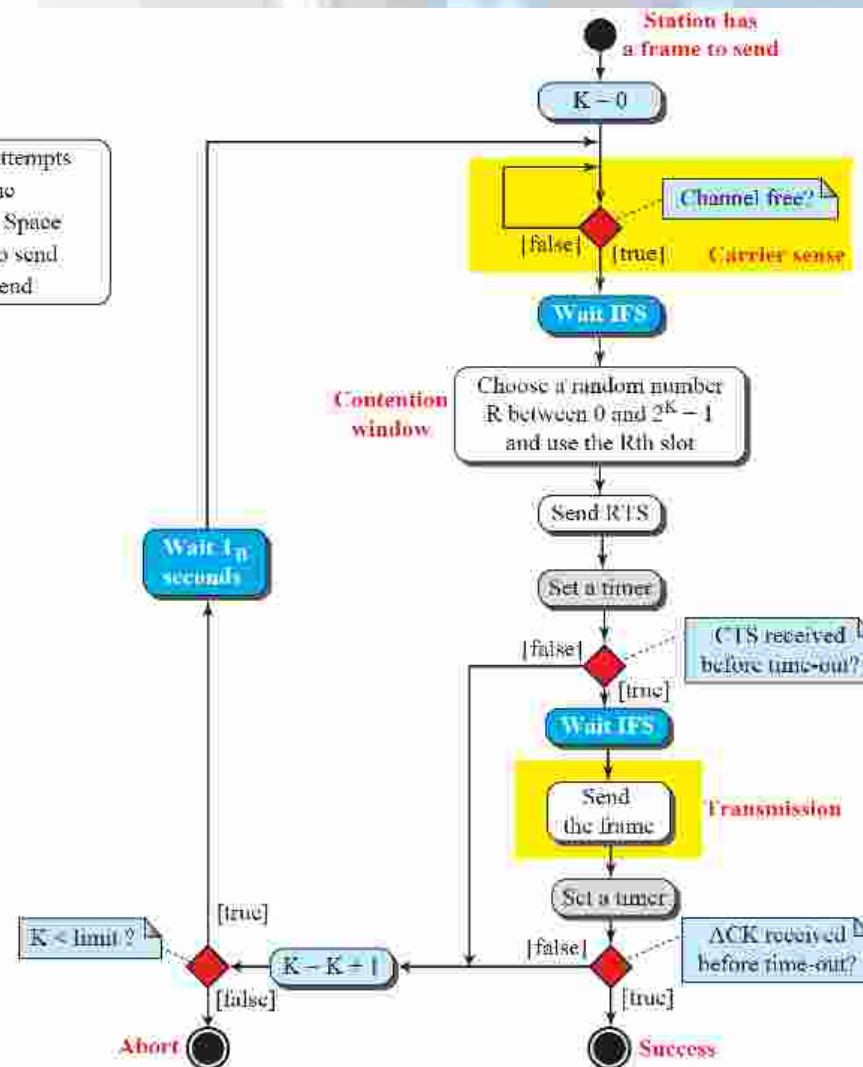
# Carrier Sense Multiple Access/Collision Avoidance

- CSMA/CA was invented for Wireless Networks
- Collisions are avoided through the use of three strategies:
  - ✓ The Interframe Space
  - ✓ The Contention Window
  - ✓ Acknowledgements

# Flow Diagram for CSMA/CA

## Legend

K: Number of attempts  
 $T_B$ : Backoff time  
 IFS: Interframe Space  
 RTS: Request to send  
 CTS: Clear to send



# Carrier Sense Multiple Access/Collision Avoidance

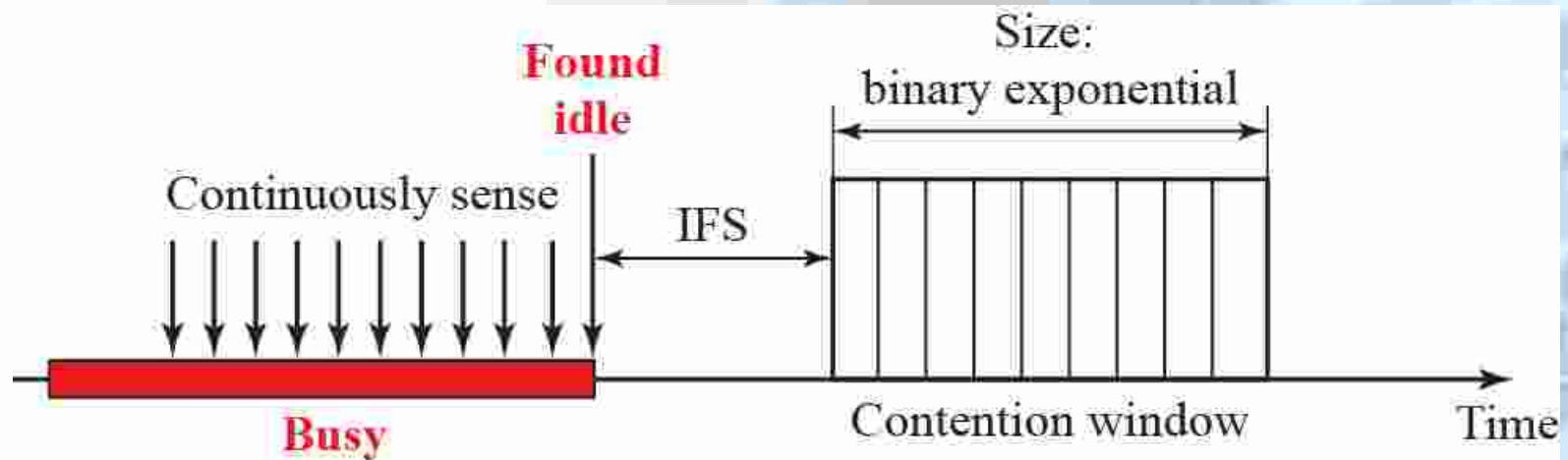
- CSMA/CA was invented for Wireless Networks
- Collisions are avoided through the use of three strategies:
  - ✓ The Interframe Space
  - ✓ The Contention Window
  - ✓ Acknowledgements

# CSMA/CA

- **Interframe Space (IFS):**  
Collisions are avoided by deferring transmission even if the channel is idle
- **Contention Window:**  
Amount of time divided into slots. Station chooses a random number of slots as its wait time (one slot first time and double each time system cannot detect an idle channel)



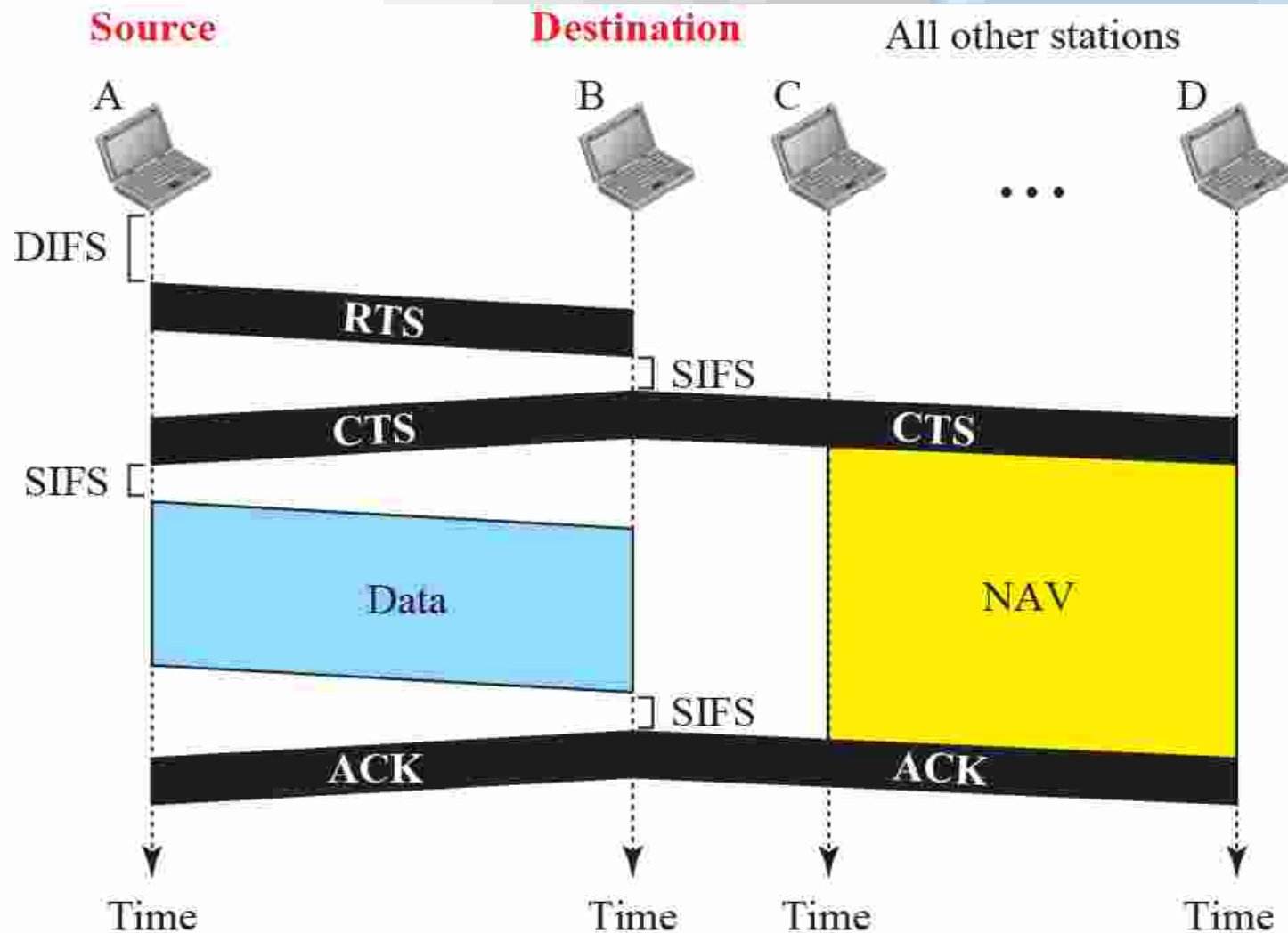
# Contention Window



# CSMA/CA

- **Acknowledgement:**  
Positive acknowledgement and time-out timer can help guarantee that the receiver has received the frame

# CSMA/CA and Network Allocation Vector (NAV)



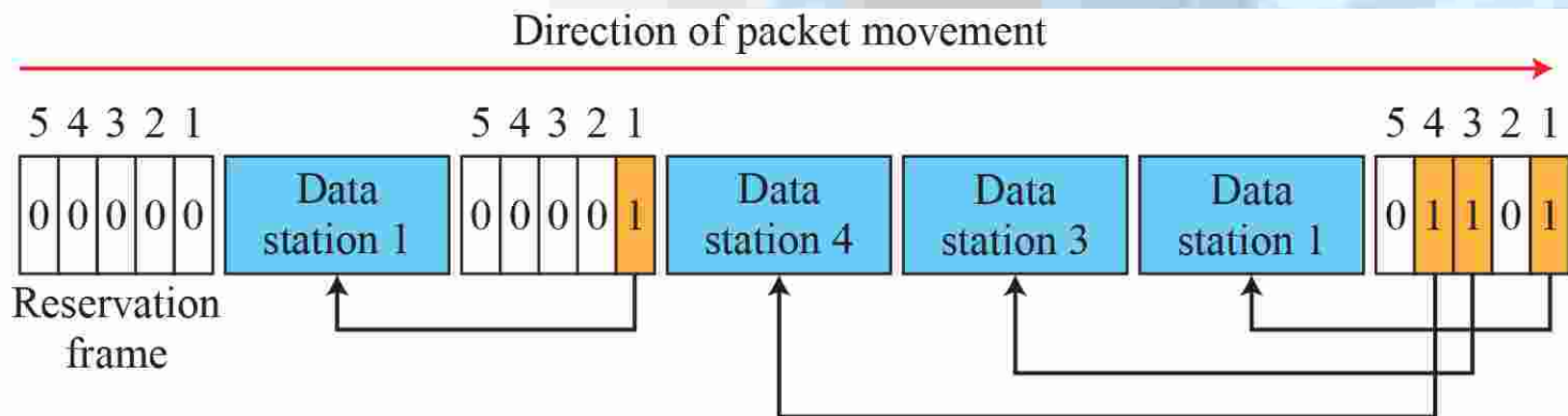
# CONTROLLED ACCESS

- The stations consult one another to find which station has the right to send
- A station cannot send unless authorized by other stations
- We discuss three controlled-access methods:
  - ✓ Reservation
  - ✓ Polling
  - ✓ Token Passing

# Reservation

- In the reservation method, a station needs to make a reservation before sending data
- Time is divided into intervals
- In each interval, a reservation frame precedes the data frames sent in that interval

# Reservation Access Method



# Polling

- Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations
- All data exchanges must be made through primary device even when the ultimate destination is a secondary device

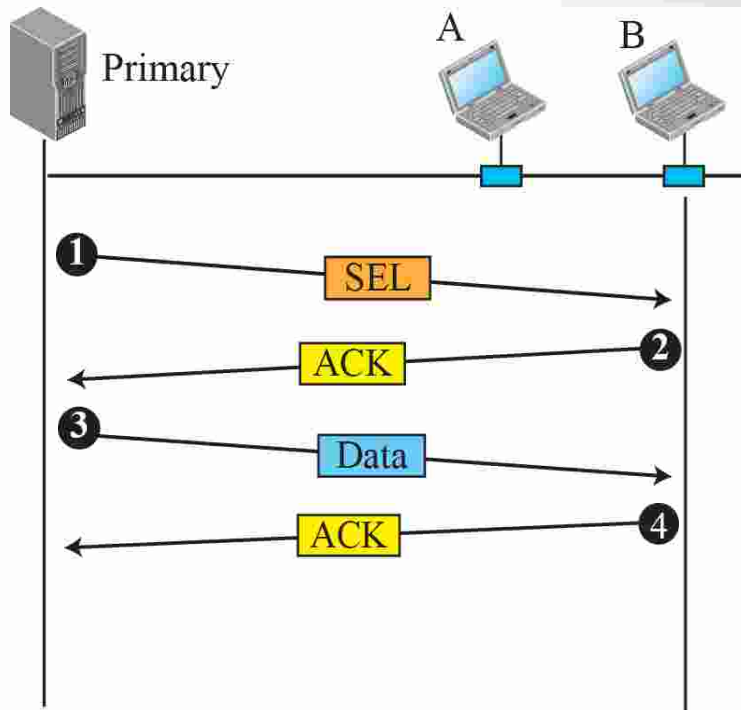
# Polling

- The primary device controls the link; the secondary devices follow its instructions

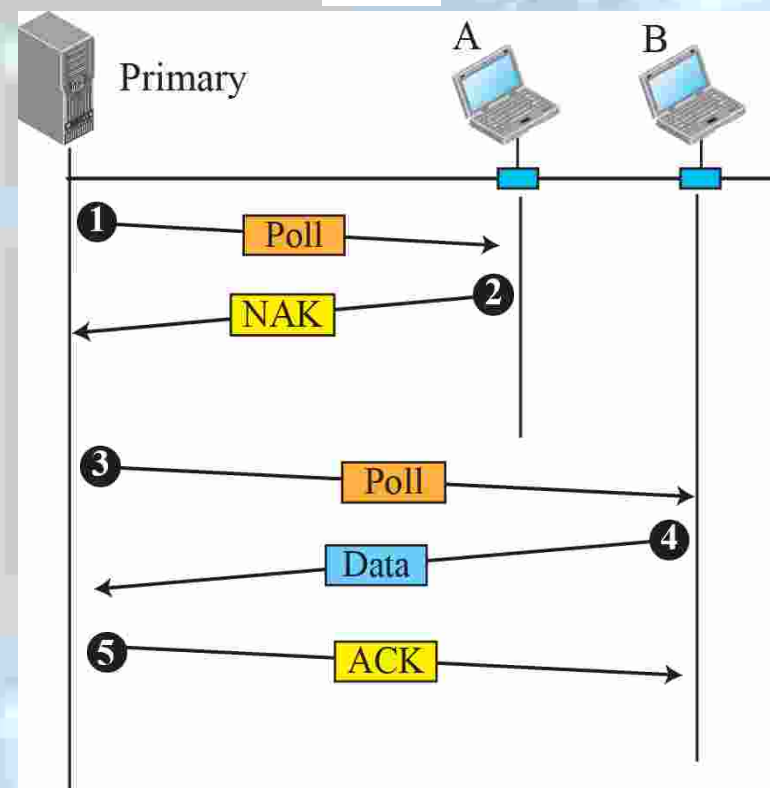


# Select and Poll Functions in Polling-Access Method

## Select



## Poll



# CONTROLLED ACCESS

- **Three controlled-access methods:**
  - ✓ **Reservation**
  - ✓ **Polling**
  - ✓ **Token Passing**

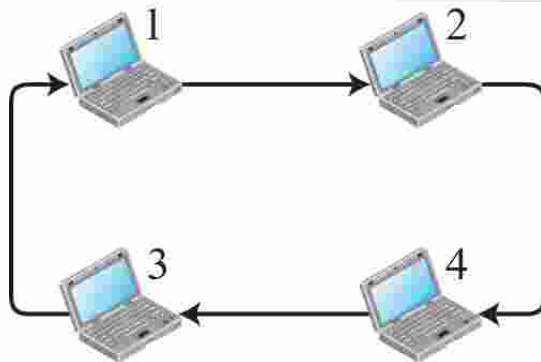
# Token Passing

- In the token-passing method, the stations in a network are organized in a logical ring
- For each station, there is a predecessor and a successor
- The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring

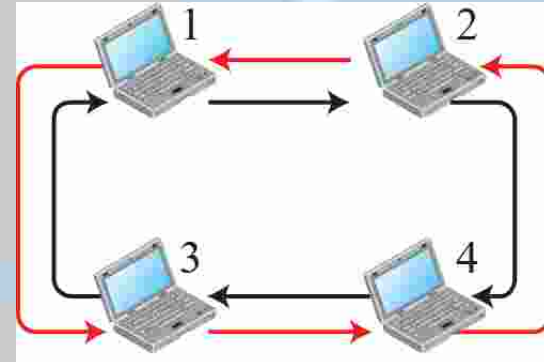
# Token Passing

- Special packet called **TOKEN** circulates through the ring
- Possession of **TOKEN** gives the station the right to send the data
- **TOKEN Management** is required to manage possession time, Token monitoring, priority assignment etc.

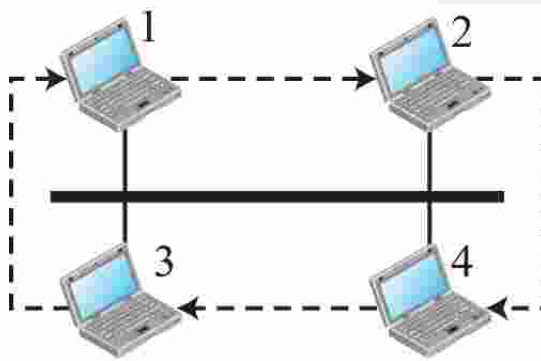
# Logical Ring & Physical Topology in Token-Passing Method



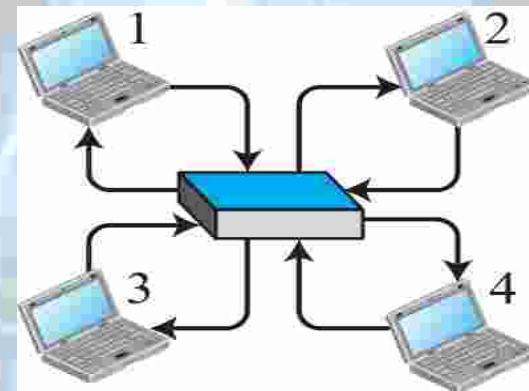
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

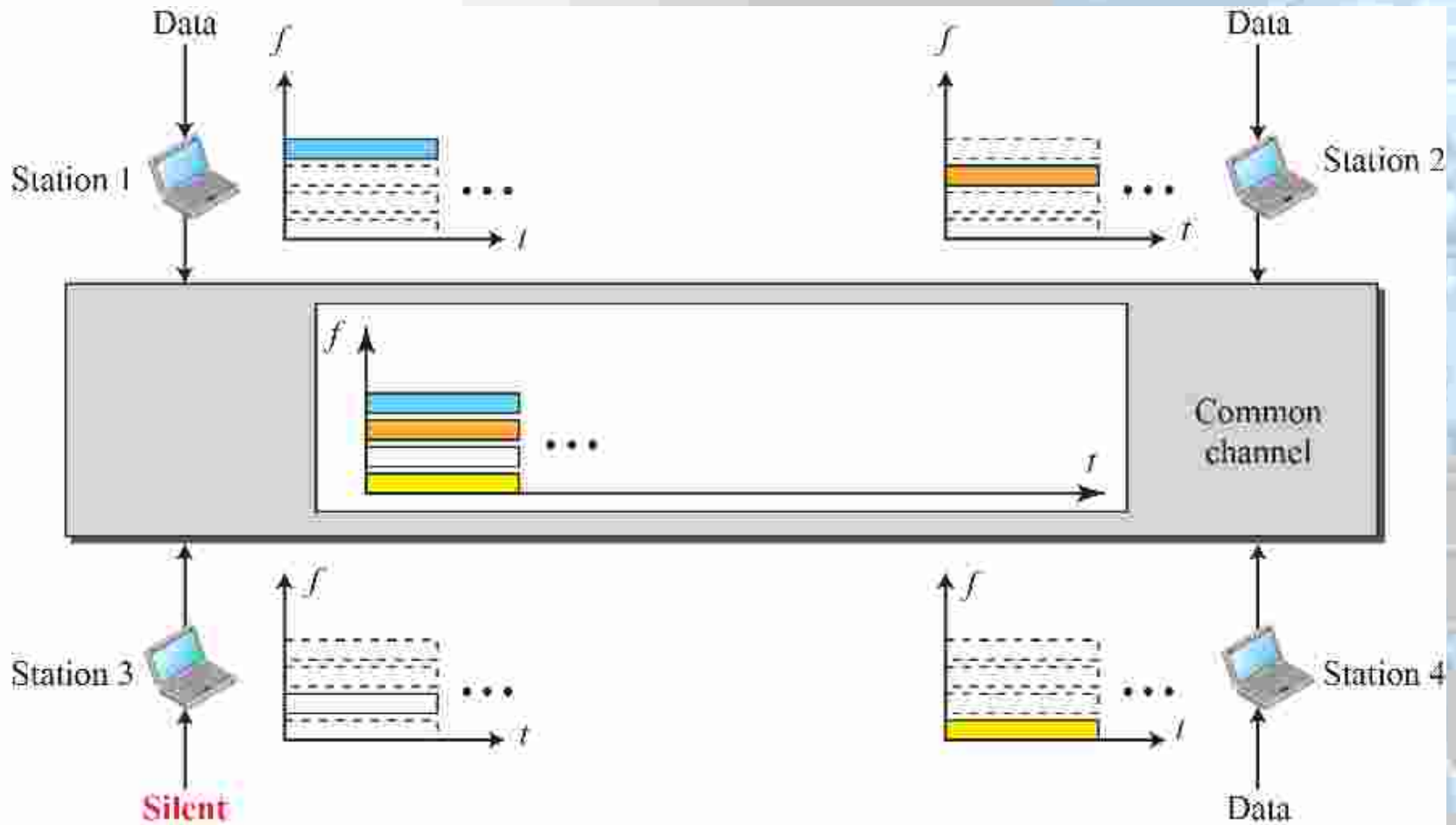
# **CHANNELIZATION (Channel Partition)**

- **The available bandwidth of a link is shared in time, frequency, or through code, among different stations**
- **We discuss three protocols:**
  - ✓ **Frequency Division Multiple Access (FDMA)**
  - ✓ **Time Division multiple Access (TDMA)**
  - ✓ **Code Division Multiple Access (CDMA)**

# Frequency-Division Multiple Access (FDMA)

- In FDMA, the available bandwidth is divided into frequency bands
- Each station is allocated a band to send its data i.e. each band is reserved for a specific station, and it belongs to the station all the time
- Each station also uses a bandpass filter to confine the transmitter frequencies

# Frequency-Division Multiple Access (FDMA)





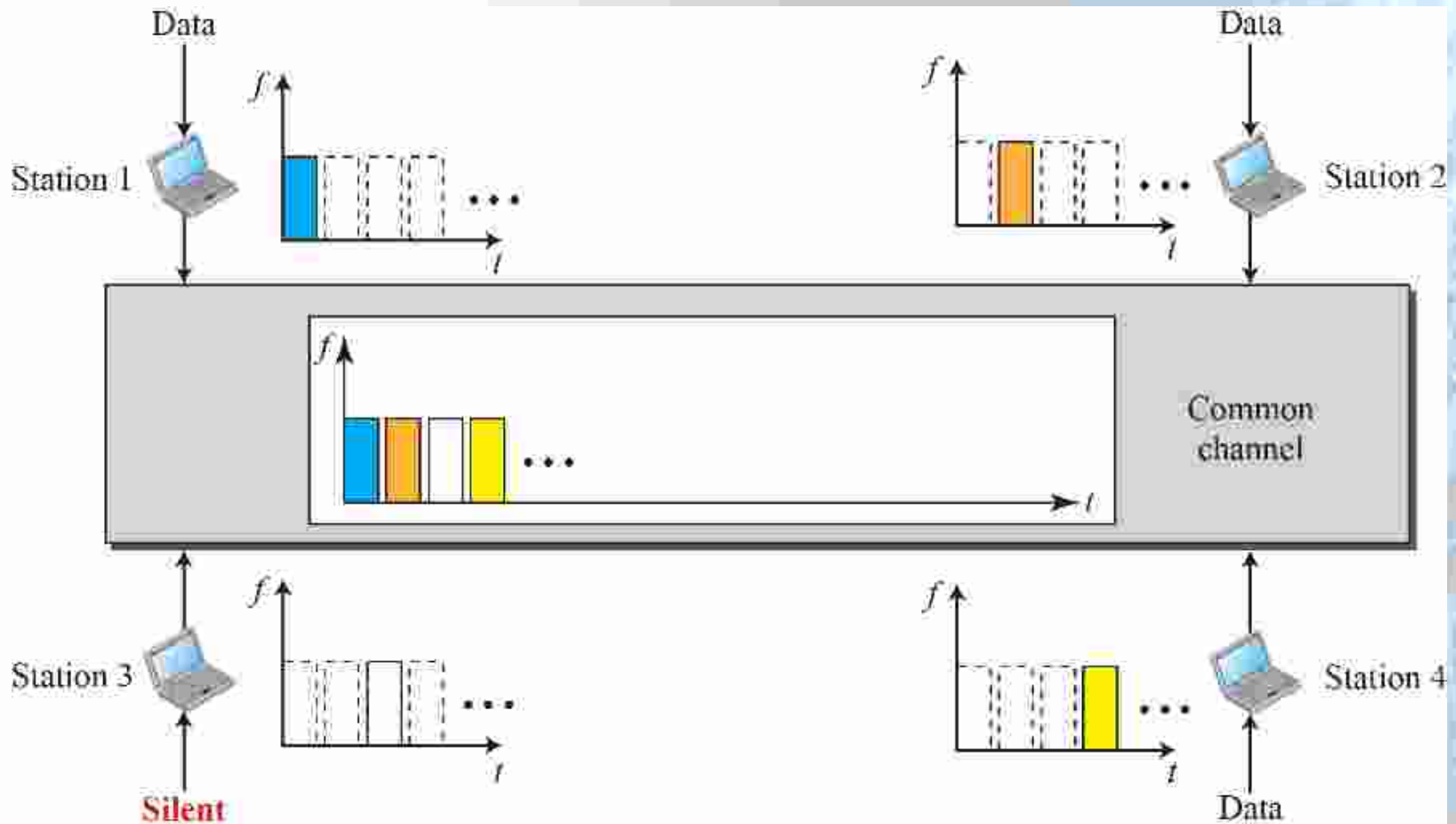
# CHANNELIZATION (Channel Partition)

- **Three protocols:**
  - ✓ **Frequency Division Multiple Access (FDMA)**
  - ✓ **Time Division Multiple Access (TDMA)**
  - ✓ **Code Division Multiple Access (CDMA)**

# TDMA

- Stations share the bandwidth of the channel in time
- Each station is allocated a time slot during which it can send data
- Each station transmits its data in its assigned time slot

# Time-Division Multiple Access (TDMA)



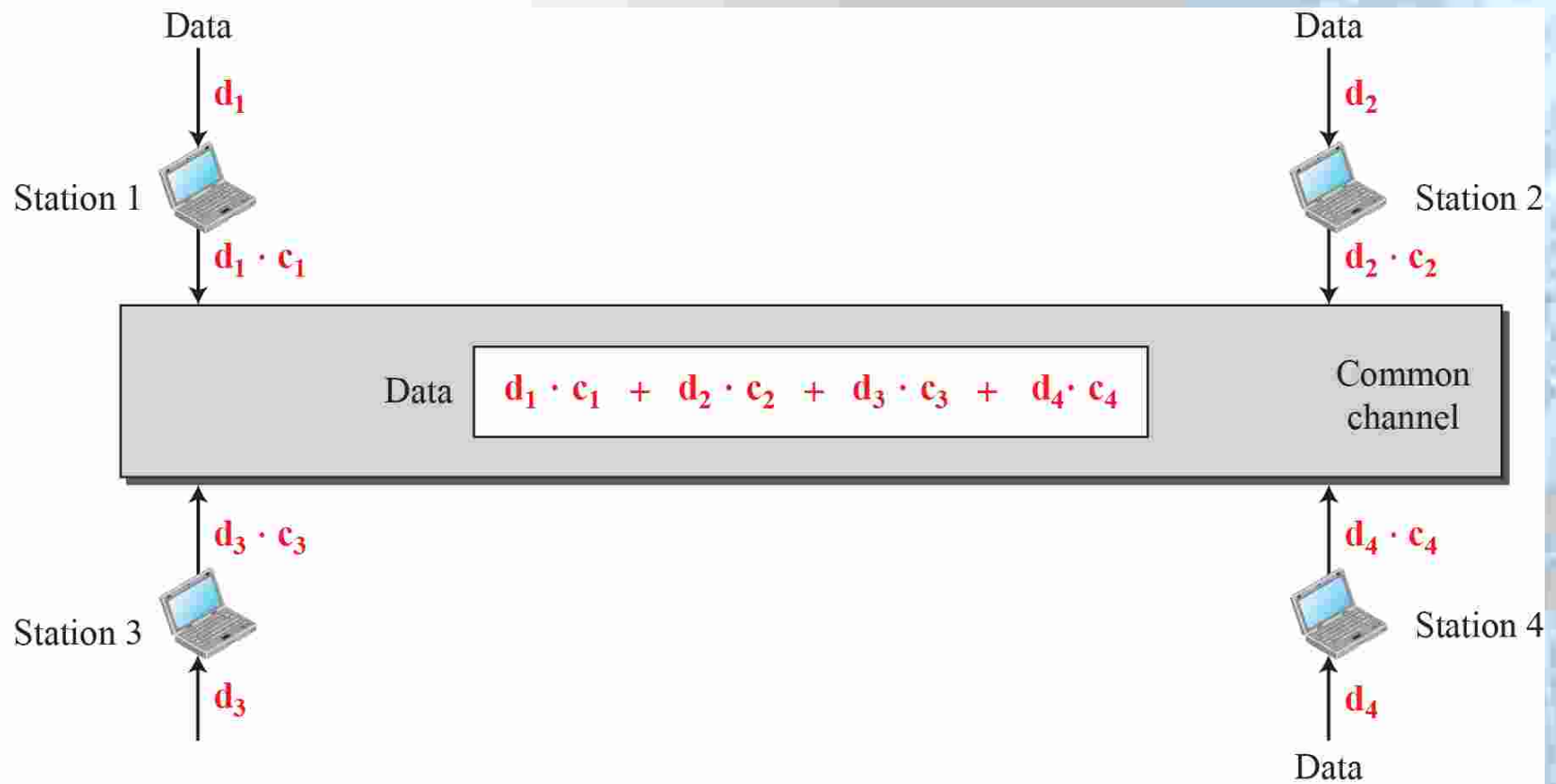
# CHANNELIZATION (Channel Partition)

- **Three protocols:**
  - ✓ **Frequency Division Multiple Access (FDMA)**
  - ✓ **Time Division Multiple Access (TDMA)**
  - ✓ **Code Division Multiple Access (CDMA)**

# Code Division Multiple Access (CDMA)

- CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link
- CDMA differs from TDMA in that all stations can send data simultaneously; there is no timesharing

# Simple idea of Communication with Code



# Data Representation in CDMA

Data bit 0  $\longrightarrow$  -1

Data bit 1  $\longrightarrow$  +1

Silence  $\longrightarrow$  0

# Ethernet Protocol

- Data-link layer and the physical layer are the territory of the local and wide area networks
- We can have wired or wireless networks



# IEEE Project 802

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable inter-communication among equipment from a variety of manufacturers
- Project 802 did not seek to replace any part of the OSI model or TCP/IP protocol suite

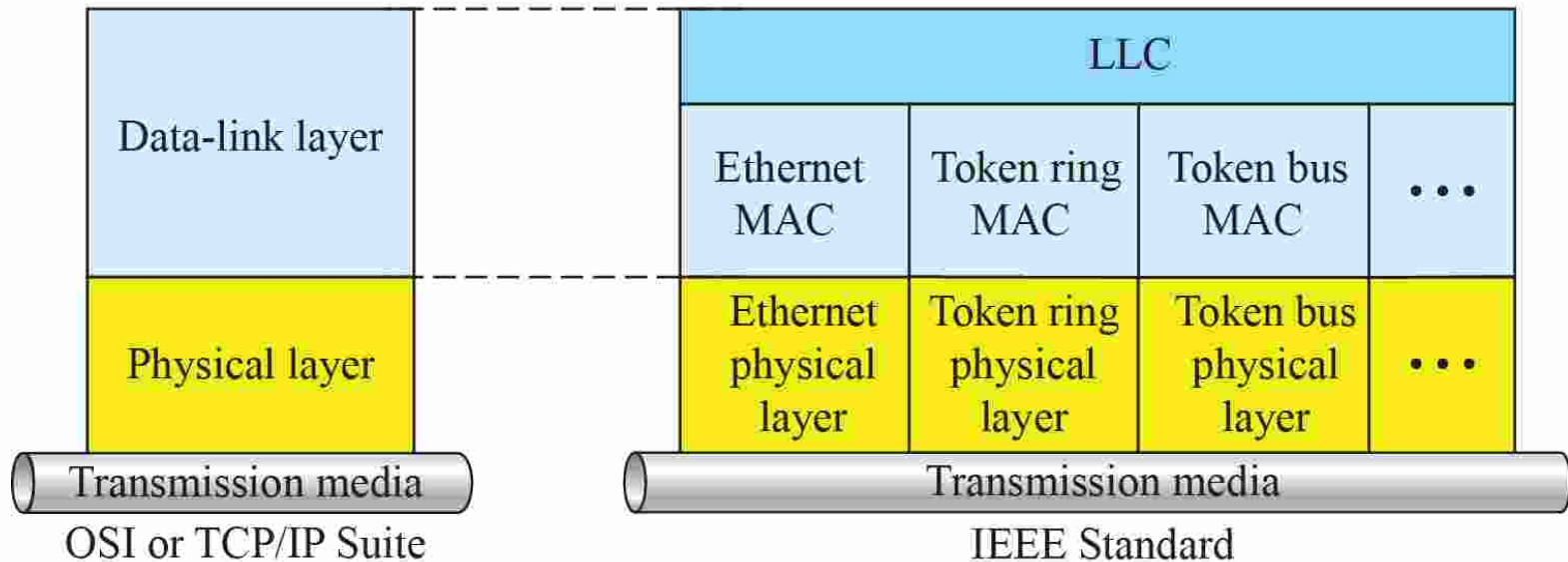
# IEEE Project 802

- A way of specifying functions of the physical layer and the data-link layer of major LAN protocols

# IEEE Standard for LANs

**LLC**: Logical link control

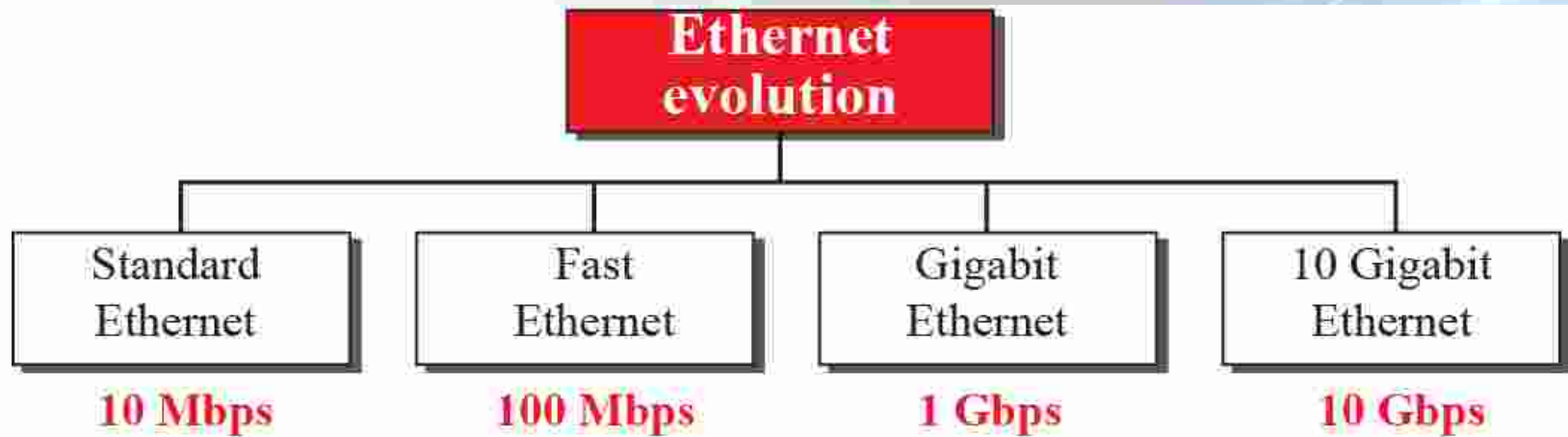
**MAC**: Media access control



# Ethernet Evolution

- The Ethernet LAN was developed in the 1970s
- Since then, it has gone through four generations:
  - ✓ Standard Ethernet (10 Mbps)
  - ✓ Fast Ethernet (100 Mbps)
  - ✓ Gigabit Ethernet (1 Gbps)
  - ✓ 10 Gigabit Ethernet (10 Gbps)

# Ethernet Evolution



# Standard Ethernet

- The original Ethernet technology with the data rate of 10 Mbps is called Standard Ethernet
- Most implementations have moved to later evolutions
- Still some features of the Standard Ethernet that have not changed during the evolution

# Connectionless & Unreliable Service

- Each frame is independent of other
- No connection establishment or tear down process
- The sender may overwhelm receiver with frames and frames are dropped
- If frame drops, sender will not know about it unless we are using TCP (Transport)

# Connectionless & Unreliable Service

- Ethernet is unreliable like IP and UDP
- If a frame is corrupted, receiver silently drops it
- Left to high level protocols to find out about it



# Standard Ethernet

- The original Ethernet technology with the data rate of 10 Mbps is called Standard Ethernet

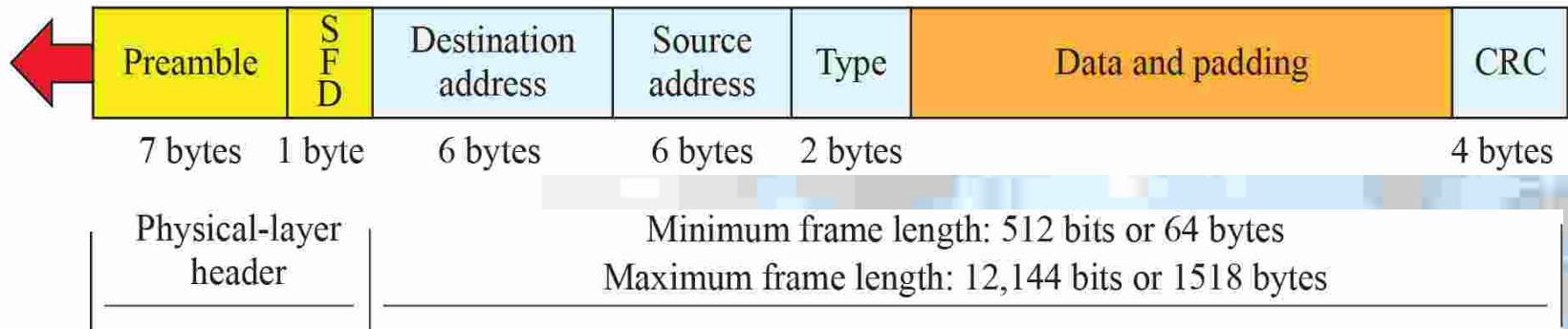
# Ethernet Frame Format

**Preamble:** 56 bits of alternating 1s and 0s

**SFD:** Start frame delimiter, flag (10101011)

Minimum payload length: 46 bytes

Maximum payload length: 1500 bytes



# Addressing in Standard Ethernet

- Each station on Ethernet has its own network interface card (NIC)
- The NIC fits inside the station and provides the station with a link-layer/physical address
- The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes

# Addressing

- For example, the following shows an Ethernet MAC address:

**4A:30:10:21:10:1A**

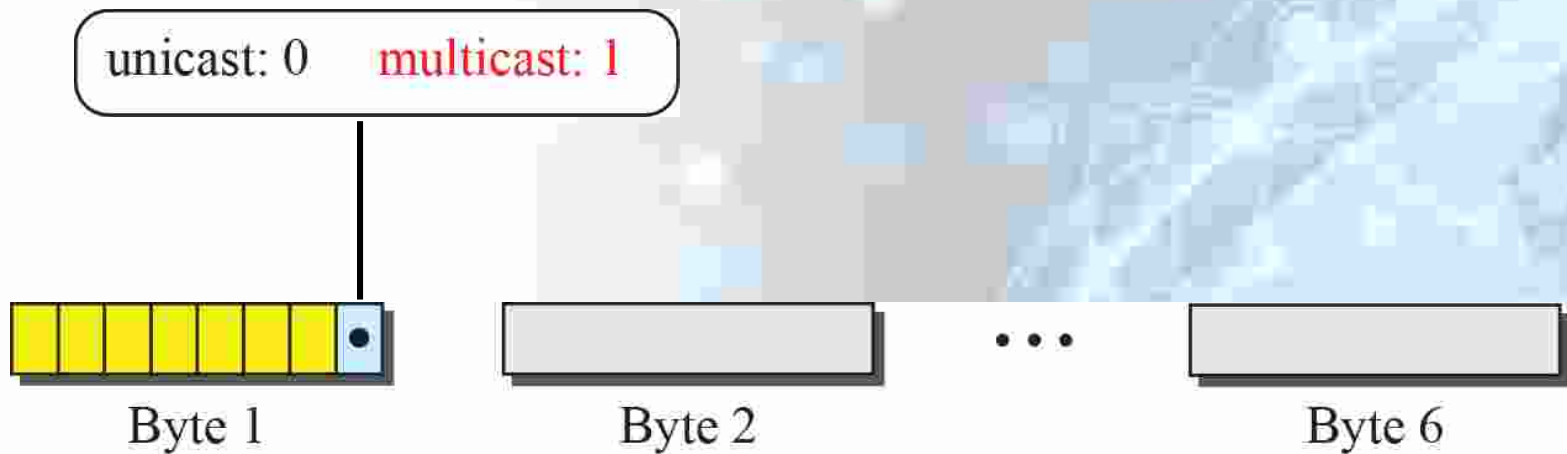
# Transmission of Address Bits

How the address 47:20:1B:2E:08:EE is sent out online.

The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

| Hexadecimal   | 47       | 20       | 1B       | 2E       | 08       | EE       |
|---------------|----------|----------|----------|----------|----------|----------|
| Binarys       | 01000111 | 00100000 | 00011011 | 00101110 | 00001000 | 11101110 |
| Transmitted ← | 11100010 | 00000100 | 11011000 | 01110100 | 00010000 | 01110111 |

# Unicast and Multicast Addresses



## Example 13.2

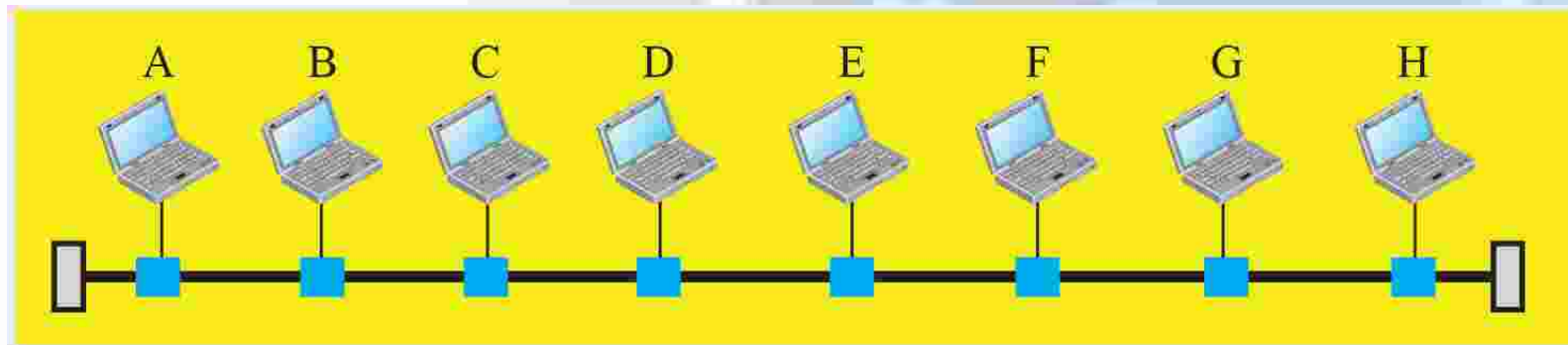
**Define the type of the following destination addresses:**

- a.** 4A:30:10:21:10:1A
- b.** 47:20:1B:2E:08:EE
- c.** FF:FF:FF:FF:FF:FF

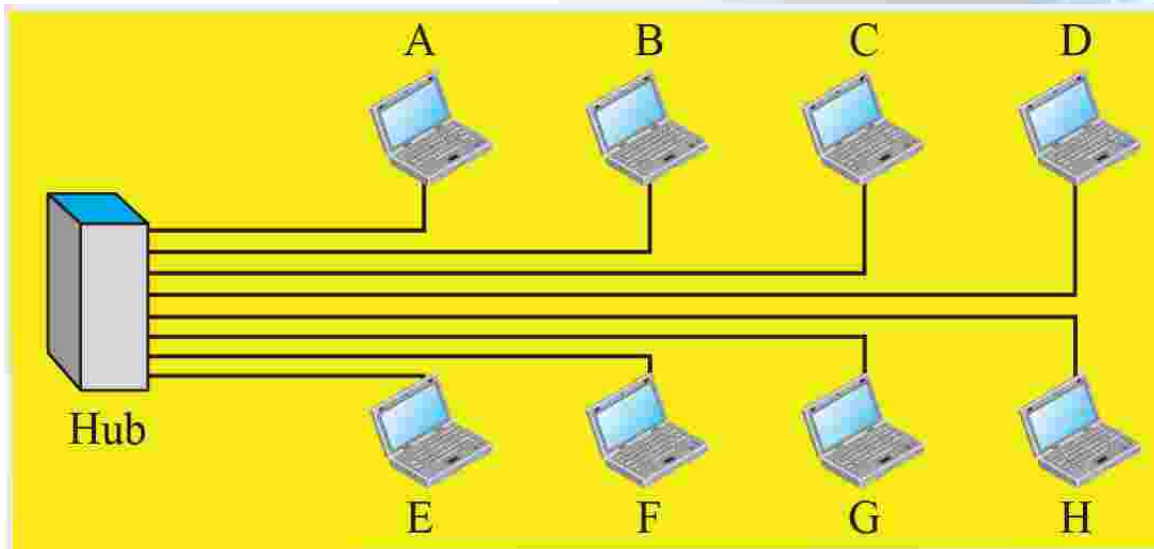
To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are Fs, the address is broadcast. Therefore, we have the following:

- a.** This is a unicast address because A in binary is 1010 (even).
- b.** This is a multicast address because 7 in binary is 0111 (odd).
- c.** This is a broadcast address because all digits are Fs in hexadecimal.

# Implementation of Standard Ethernet



a. A LAN with a bus topology using a coaxial cable



b. A LAN with a star topology using a hub

## Legend



A host (of any type)



A hub



A cable tap



A cable end



Coaxial cable



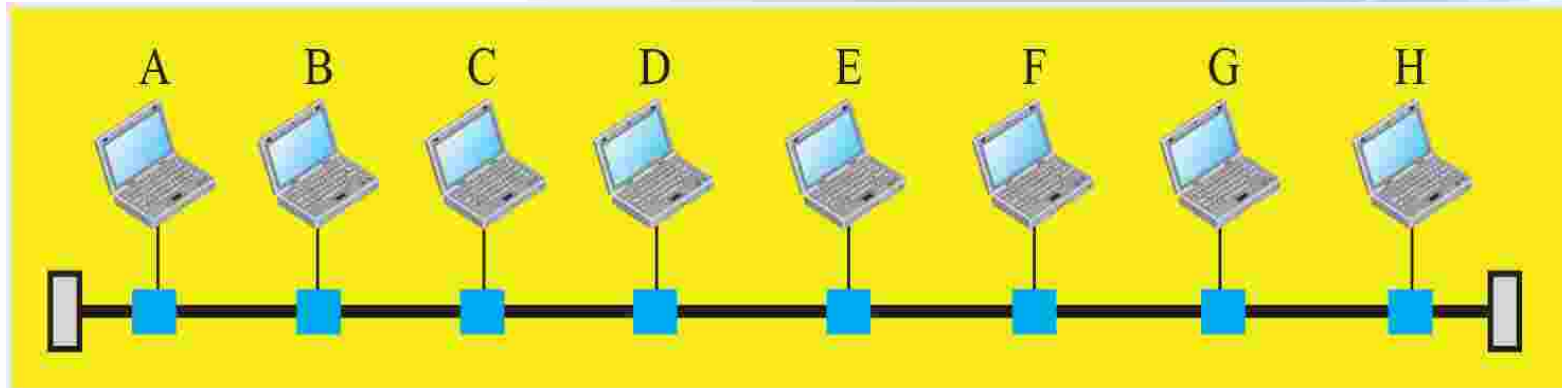
Twisted pair cable



# Access Method in Standard Ethernet

- Since the network that uses the standard Ethernet protocol is a broadcast network, we need to use an access method to control access to the sharing medium
- The standard Ethernet chose CSMA/CD with 1-Persistent Method

# Access Method in Standard Ethernet



a. A LAN with a bus topology using a coaxial cable

# Efficiency of Standard Ethernet

- The ratio of the time used by a station to send data to the time the medium is occupied by this station
- The practical efficiency of standard Ethernet has been measured to be:

$$\text{Efficiency} = 1/(1 + 6.4 \times a)$$

where  $a$  = number of frames that can fit on a medium

## Example

**In the Standard Ethernet with the transmission rate of 10 Mbps, we assume that the length of the medium is 2500 m and the size of the frame is 512 bits. The propagation speed of a signal in a cable is normally  $2 \times 10^8$  m/s.**

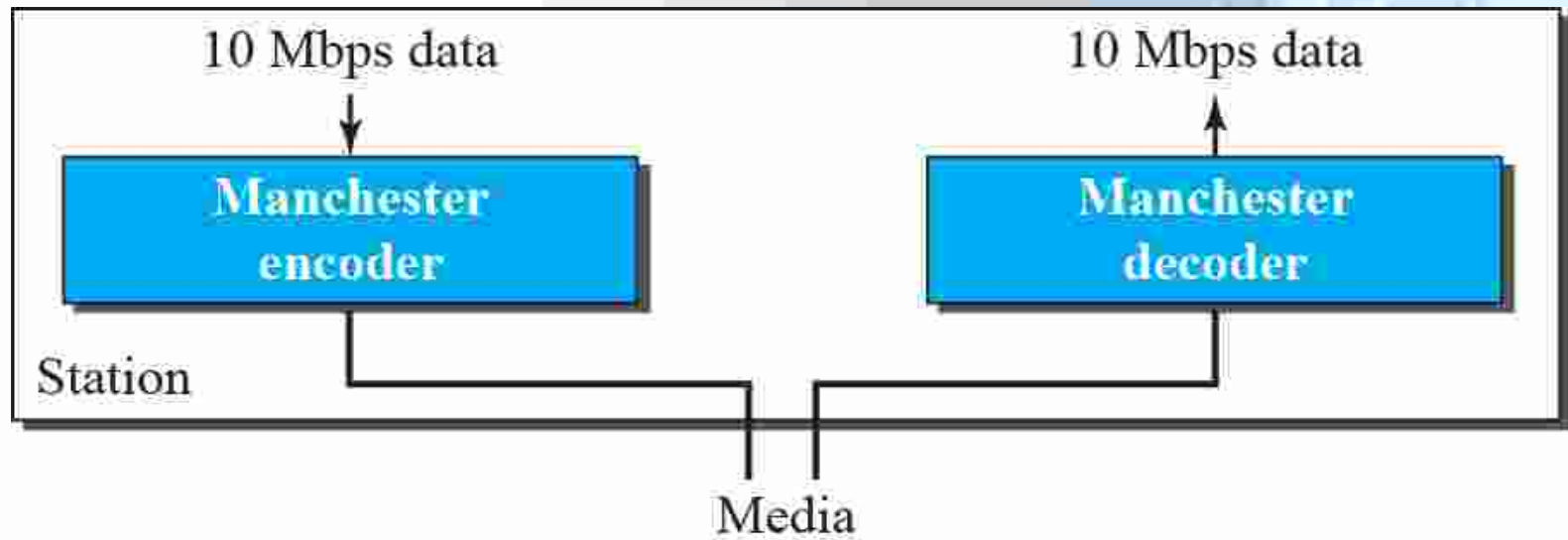
# Implementation of Standard Ethernet

- The Standard Ethernet defined several implementations, but only four of them became popular during the 1980s

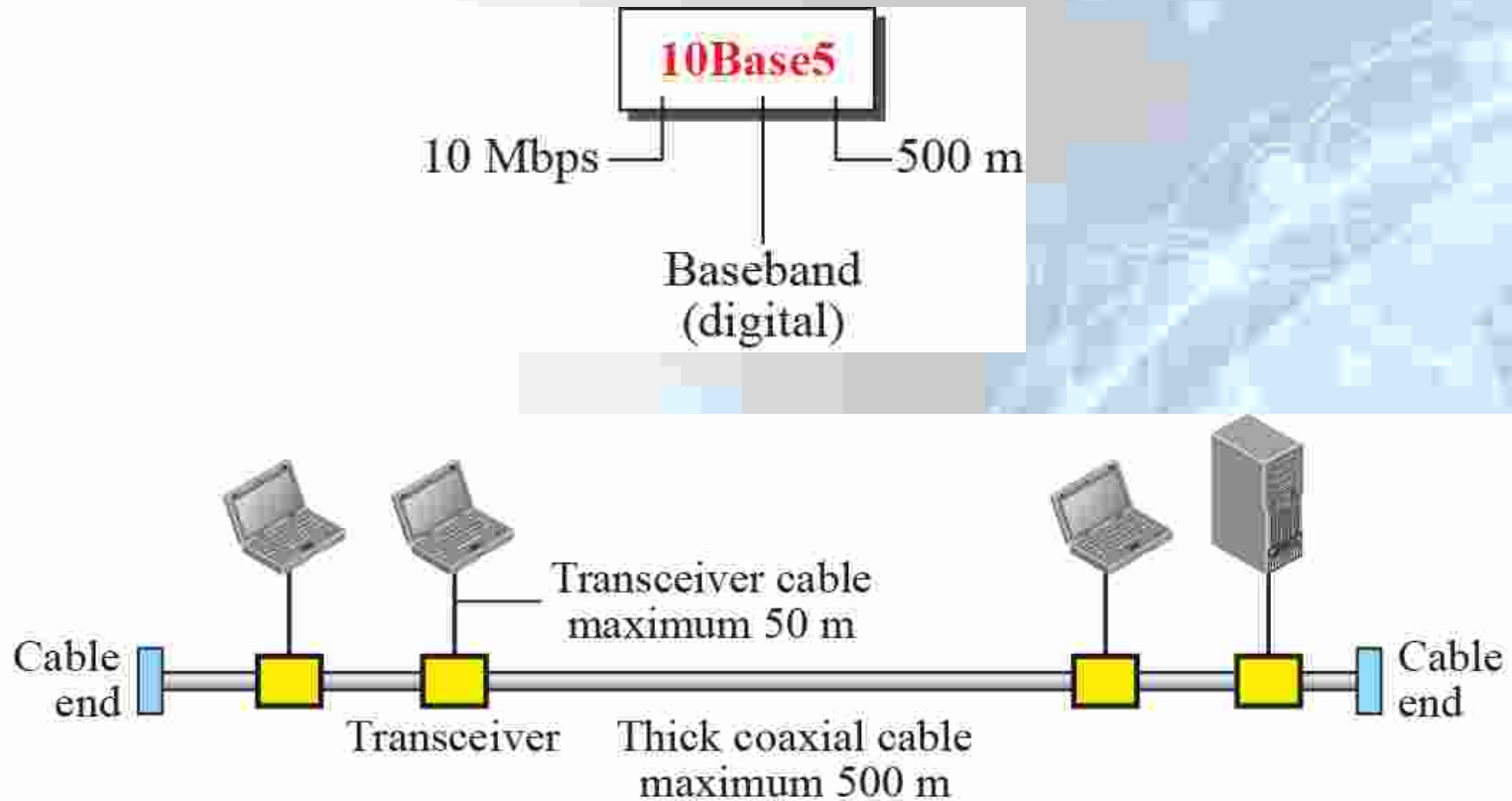
# Summary of Standard Ethernet implementations

| <i>Implementation</i> | <i>Medium</i> | <i>Medium Length</i> | <i>Encoding</i> |
|-----------------------|---------------|----------------------|-----------------|
| 10Base5               | Thick coax    | 500 m                | Manchester      |
| 10Base2               | Thin coax     | 185 m                | Manchester      |
| 10Base-T              | 2 UTP         | 100 m                | Manchester      |
| 10Base-F              | 2 Fiber       | 2000                 | Manchester      |

# Encoding in Standard Ethernet

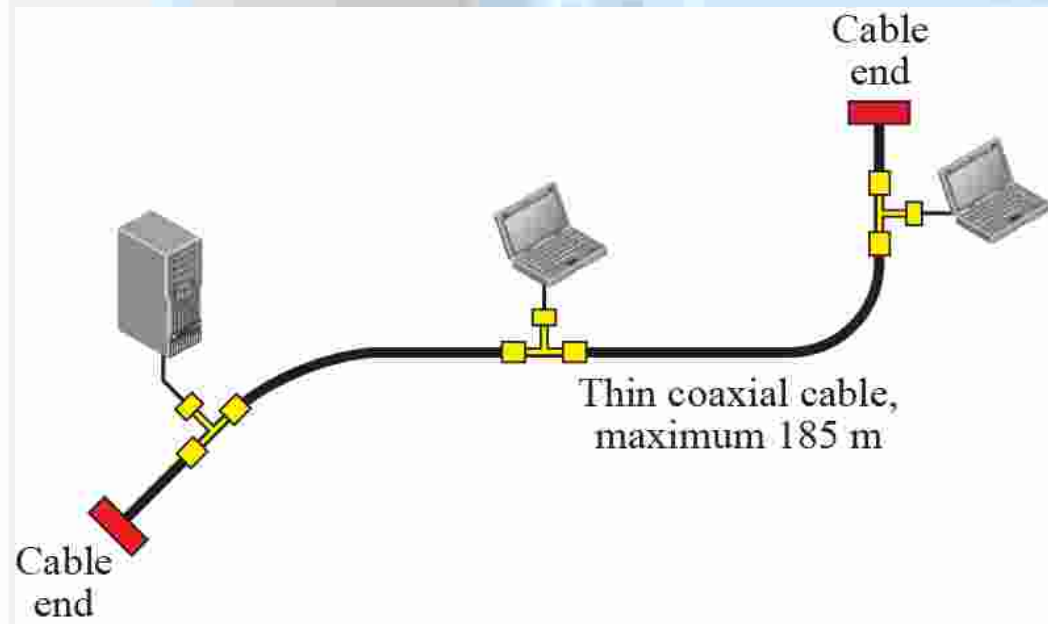
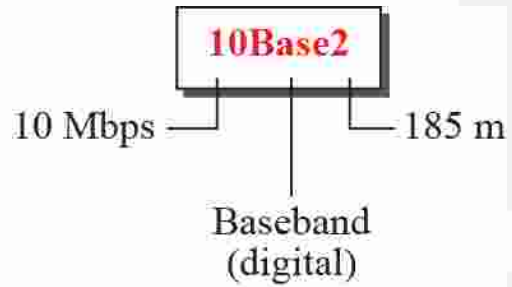


# 10Base5 implementation

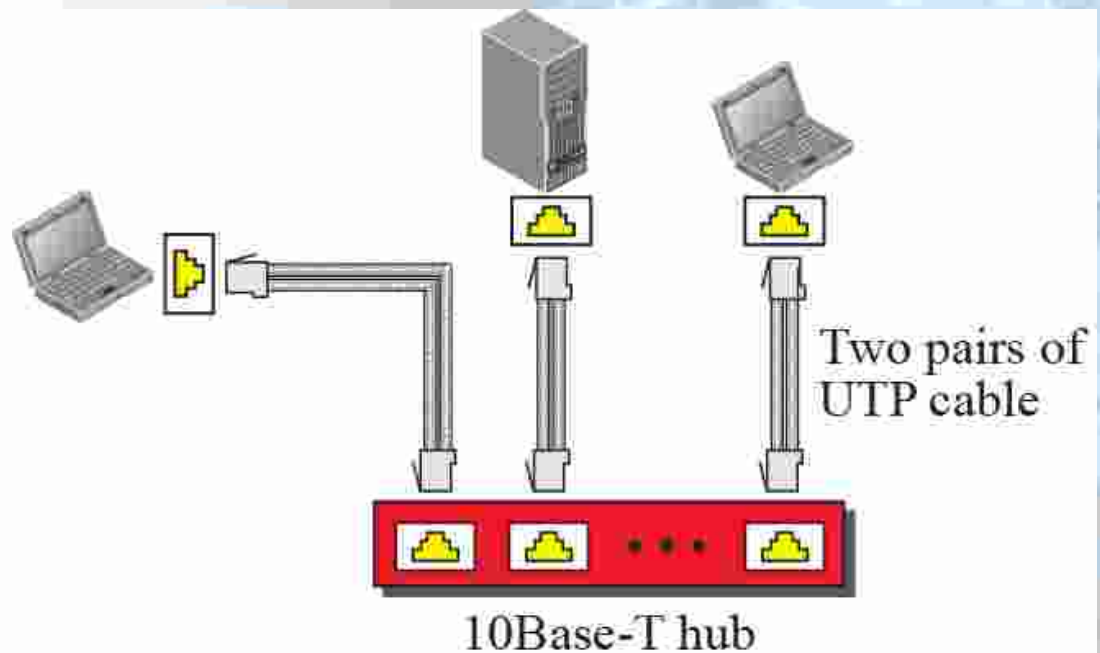
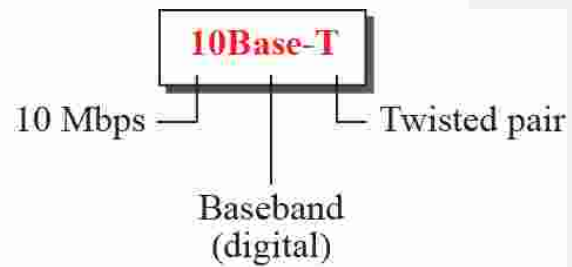




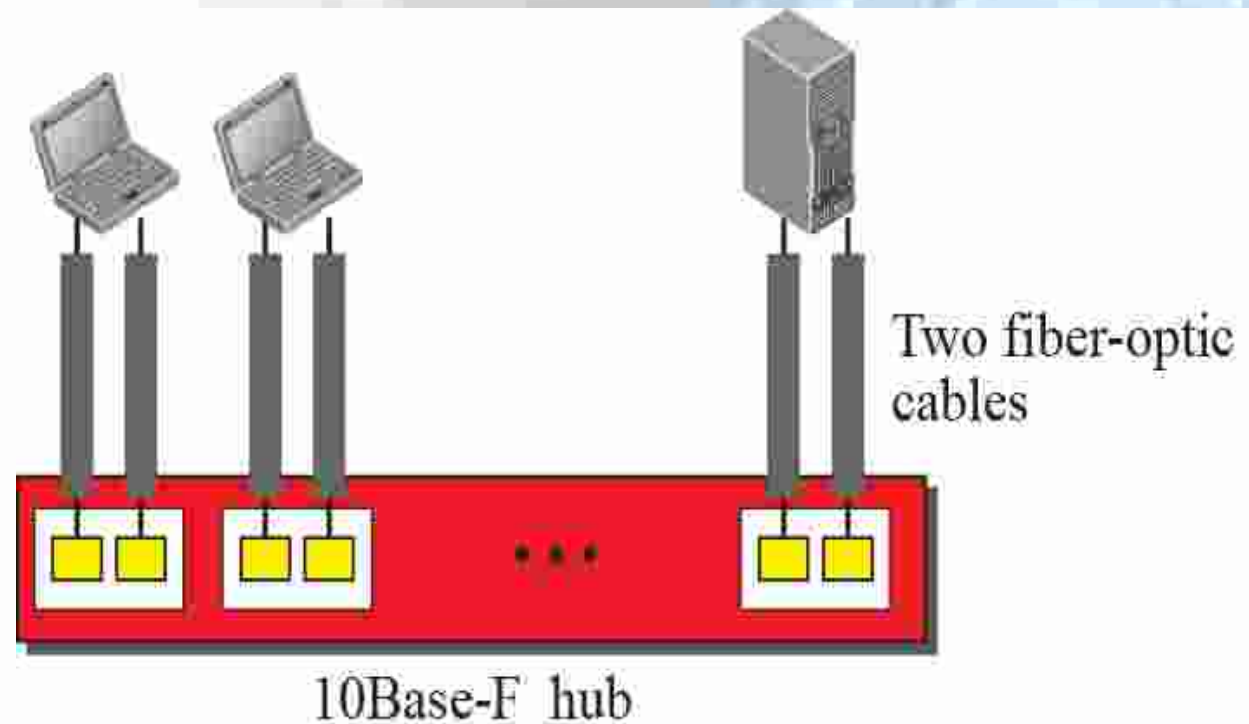
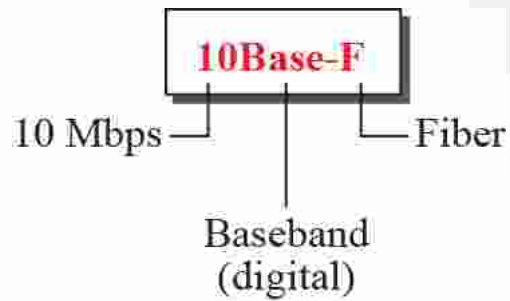
# 10Base2 implementation



# 10Base-T implementation



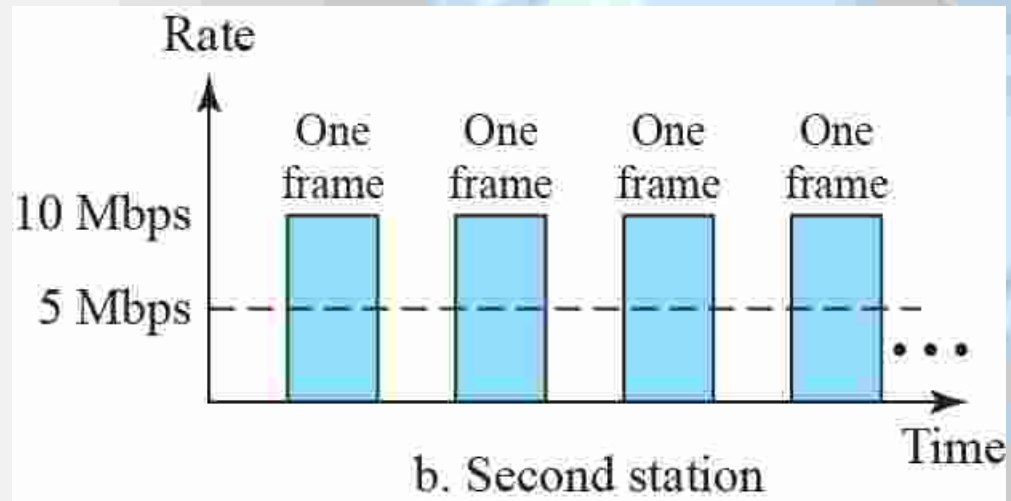
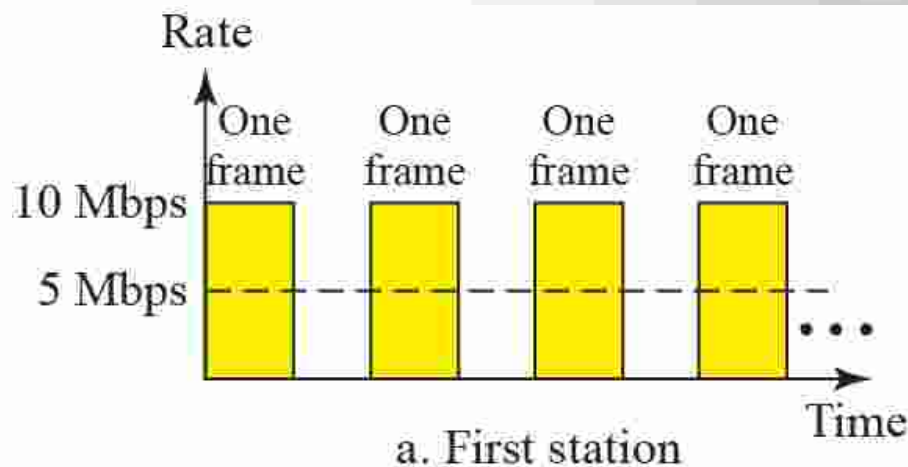
# 10Base-F implementation



# Changes in the Standard

- The changes that occurred to the 10-Mbps Standard Ethernet opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs
  - ✓ Bridged Ethernet
  - ✓ Switched Ethernet
  - ✓ Full-Duplex Ethernet

# Bridged Ethernet- Sharing Bandwidth



# A Network with and without Bridging



**a. Without bridging**

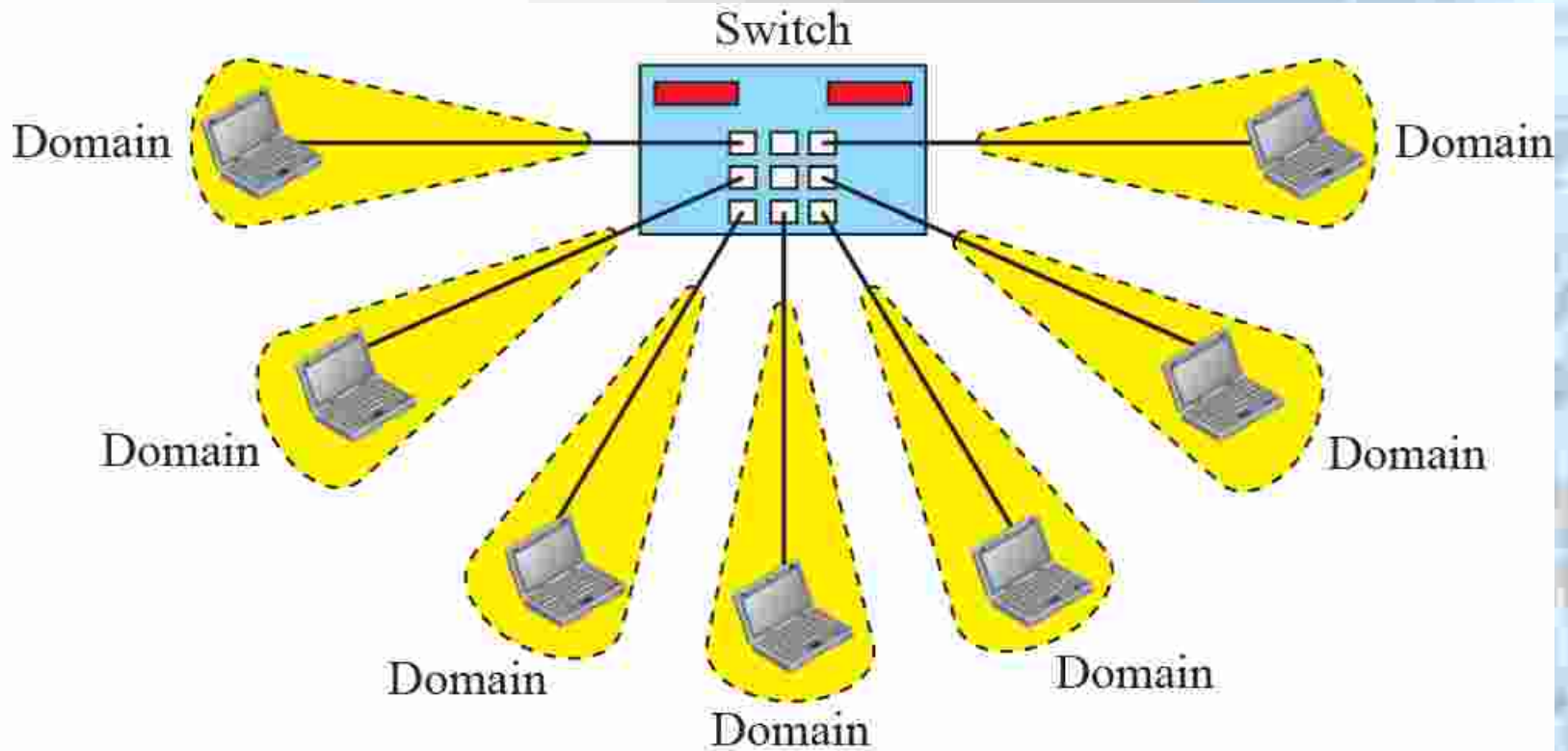


**b. With bridging**

# Changes in the Standard

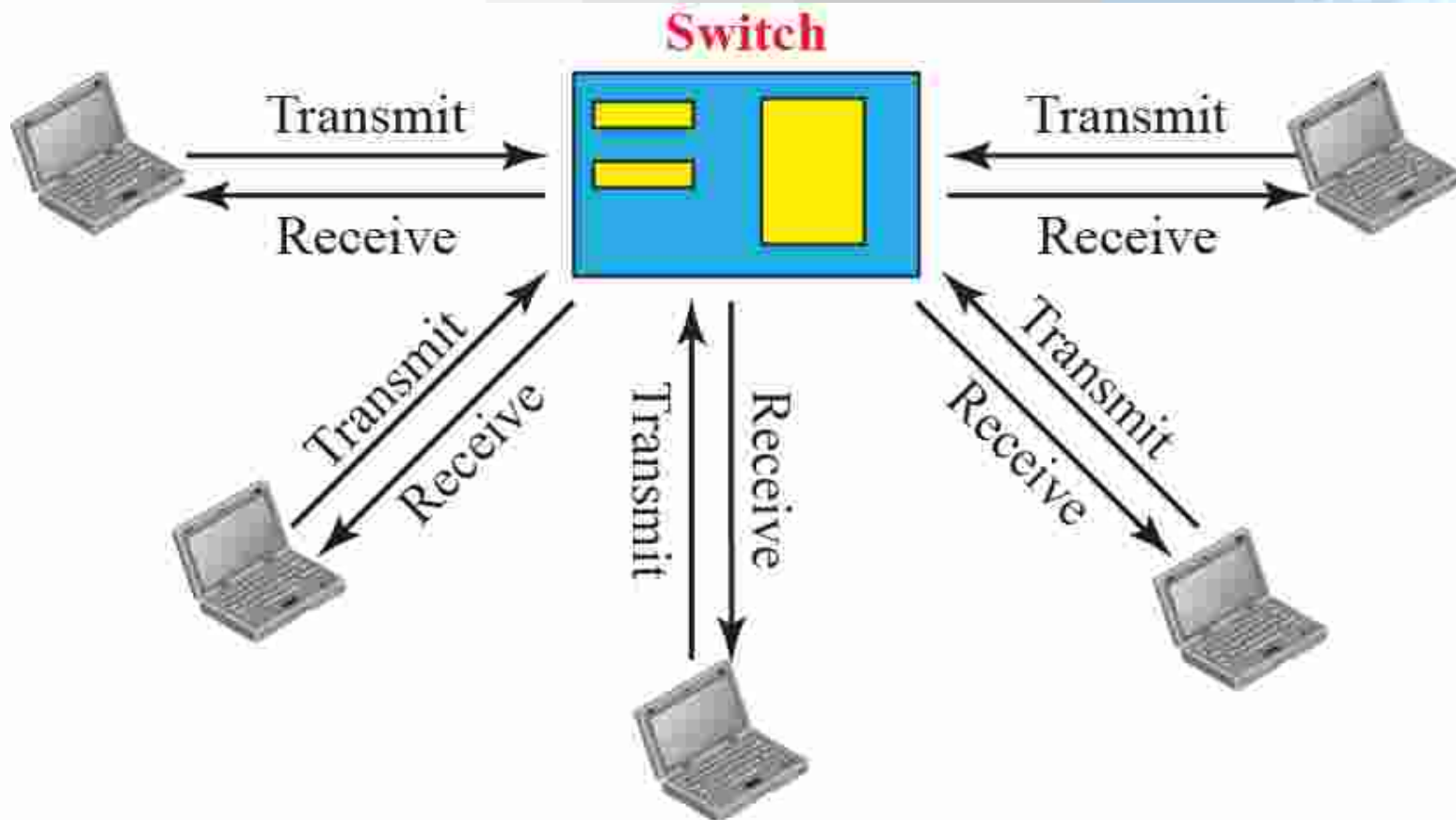
- The changes that occurred to the 10-Mbps Standard Ethernet opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs
  - ✓ Bridged Ethernet
  - ✓ Switched Ethernet
  - ✓ Full-Duplex Ethernet

# Switched Ethernet





# Full – Duplex Switched Ethernet



# Fast Ethernet

- In the 1990s, Ethernet made a big jump by increasing the transmission rate to 100 Mbps, and the new generation was called the Fast Ethernet
- To make it compatible with the Standard Ethernet, the MAC sublayer was left unchanged

# Fast Ethernet

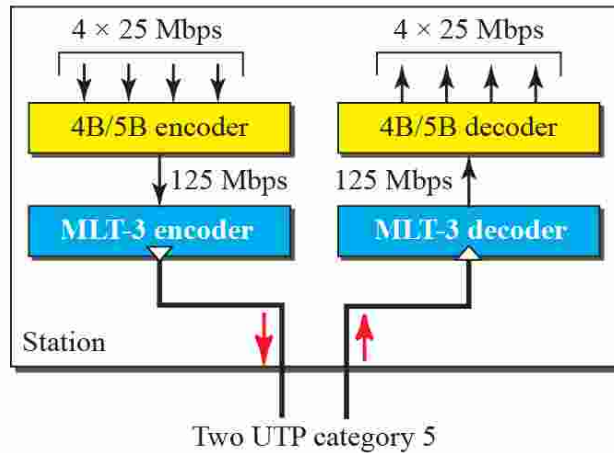
- But the features of the Standard Ethernet that depend on the transmission rate, had to be changed
- Goals of Fast Ethernet:
  - ✓ Upgrade data rate to 100Mbps
  - ✓ Make it compatible with Standard Ethernet
  - ✓ Keep same 48-bit address
  - ✓ Keep same frame format

# Physical Layer

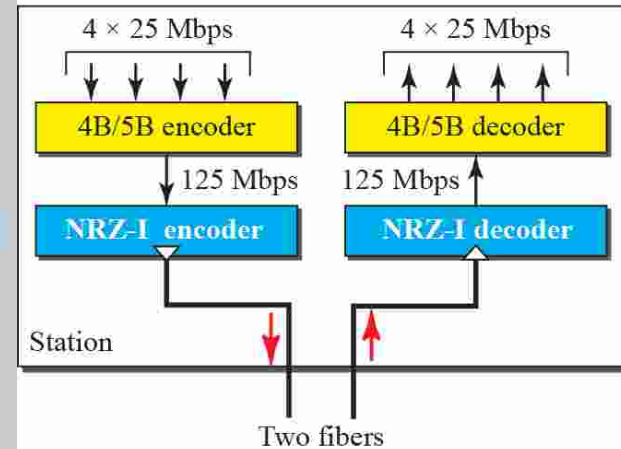
- To be able to handle a 100 Mbps data rate, several changes need to be made at the physical layer

# Encoding for Fast Ethernet

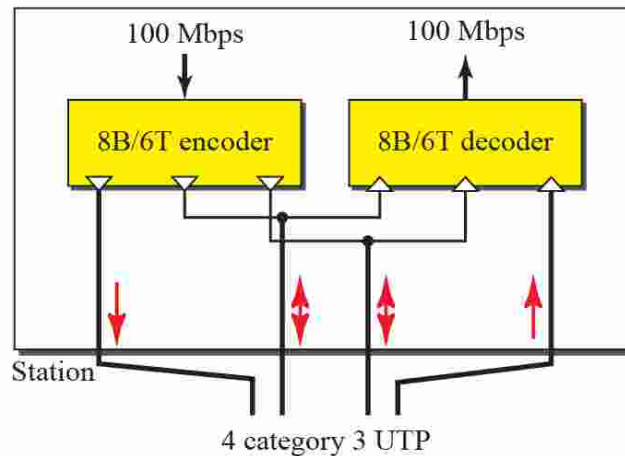
## 100Base-TX



## 100Base-FX



## 100Base-T4



# Implementation of Fast Ethernet implementations

| <i>Implementation</i> | <i>Medium</i> | <i>Medium Length</i> | <i>Wires</i> | <i>Encoding</i> |
|-----------------------|---------------|----------------------|--------------|-----------------|
| 100Base-TX            | STP           | 100 m                | 2            | 4B5B + MLT-3    |
| 100Base-FX            | Fiber         | 185 m                | 2            | 4B5B + NRZ-I    |
| 100Base-T4            | UTP           | 100 m                | 4            | Two 8B/6T       |

# Gigabit Ethernet

- Need for an even higher data rate resulted in the design of IEEE Standard 802.3z Gigabit Ethernet Protocol (1000 Mbps)

# Gigabit Ethernet

- The goals of the Gigabit Ethernet were:
  - ✓ Upgrade the data rate to 1 Gbps
  - ✓ Make it compatible with standard or Fast Ethernet
  - ✓ Use same 48 bit address
  - ✓ Use the same frame format
  - ✓ Keep same minimum and maximum frame lengths



# MAC Sub-layer

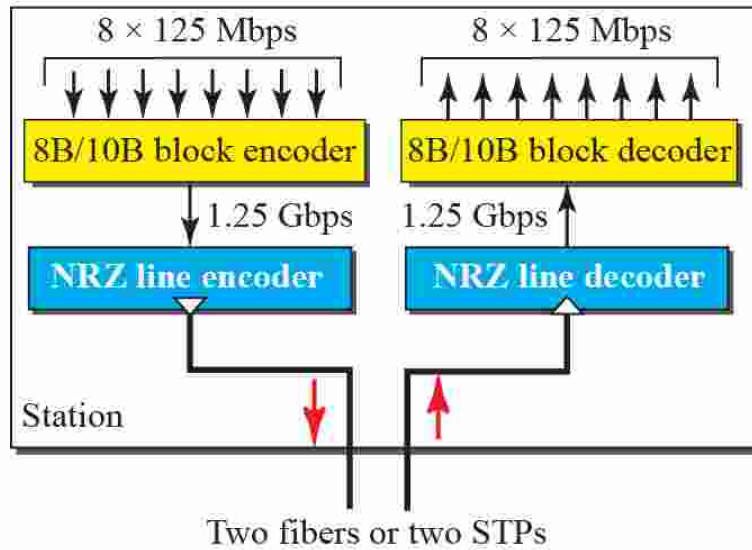
- A main consideration in the evolution of Ethernet was to keep the MAC sublayer untouched
- To achieve a data rate of 1 Gbps, this was no longer possible
- Gigabit Ethernet has two distinctive approaches for medium access:
  - ✓ Half-duplex
  - ✓ Full-duplex

# Physical Layer

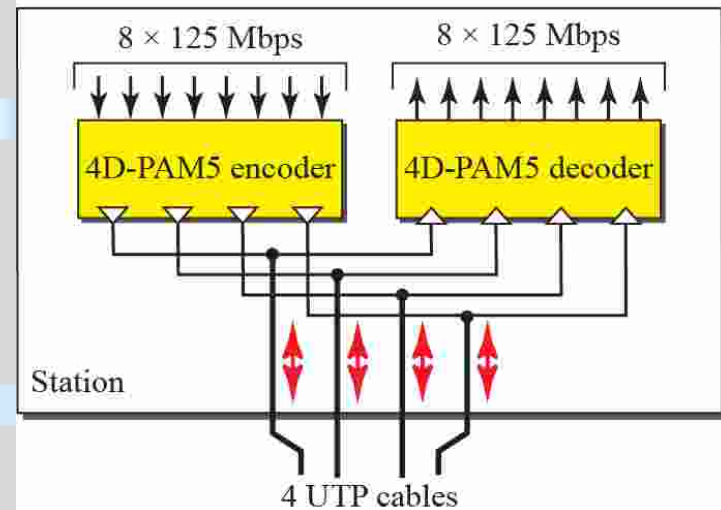
- The physical layer in Gigabit Ethernet is more complicated than that in Standard or Fast Ethernet
- We briefly discuss some features of this layer:

# Encoding in Gigabit Ethernet

## 1000Base-SX, 1000Base-LX, and 1000Base-CX



## 1000Base-T



# Summary of Gigabit Ethernet Implementations

| <i>Implementation</i> | <i>Medium</i> | <i>Medium Length</i> | <i>Wires</i> | <i>Encoding</i> |
|-----------------------|---------------|----------------------|--------------|-----------------|
| 1000Base-SX           | Fiber S-W     | 550 m                | 2            | 8B/10B + NRZ    |
| 1000Base-LX           | Fiber L-W     | 5000 m               | 2            | 8B/10B + NRZ    |
| 1000Base-CX           | STP           | 25 m                 | 2            | 8B/10B + NRZ    |
| 1000Base-T4           | UTP           | 100 m                | 4            | 4D-PAM5         |

# 10-gigabit Ethernet

- The idea is to extend the technology, the data rate, and the coverage distance so that the Ethernet can be used in LANs and MANs (metropolitan area network)
- The IEEE committee created 10 Gigabit Ethernet and called it Standard 802.3ae

# Implementation

- **10 Gigabit Ethernet operates only in full-duplex mode, which means there is no need for contention; CSMA/CD is not used in 10 Gigabit Ethernet**
- **Four implementations are most common:**

# Implementation

| <i>Implementation</i> | <i>Medium</i> | <i>Medium Length</i> | <i>Number of wires</i> | <i>Encoding</i> |
|-----------------------|---------------|----------------------|------------------------|-----------------|
| 10GBase-SR            | Fiber 850 nm  | 300 m                | 2                      | 64B66B          |
| 10GBase-LR            | Fiber 1310 nm | 10 Km                | 2                      | 64B66B          |
| 10GBase-EW            | Fiber 1350 nm | 40 Km                | 2                      | SONET           |
| 10GBase-X4            | Fiber 1310 nm | 300 m to 10 Km       | 2                      | 8B10B           |

# Other Wired Networks

- **Access Networks**
  - ✓ Networks that connect a small LAN to an ISP
- **Wide Area Networks**
  - ✓ Wired networks used to transfer data over long distances



# Telephone Network

- The telephone network had its beginnings in the late 1800s
- Plain Old Telephone System (POTS) was originally an analog system using analog signals to transmit voice
- With the advent of the computer era, the network, in the 1980s, began to carry data in addition to voice

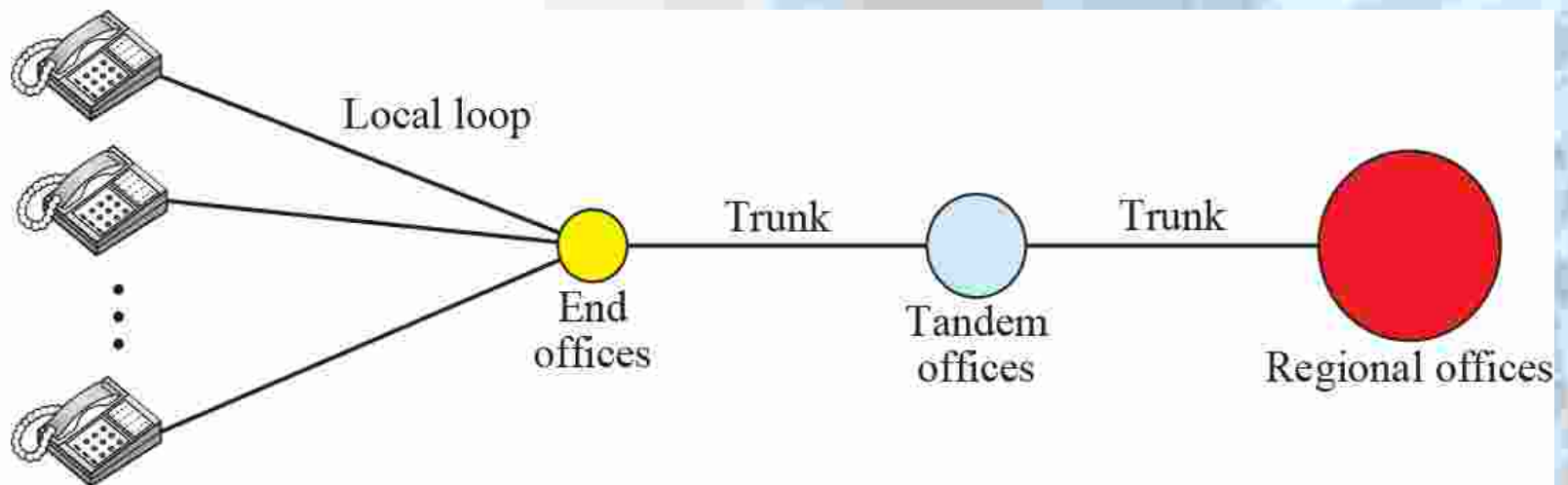
# Telephone Network

- During the last decade, the telephone network has undergone many technical changes and the network is now Digital as well as Analog

# Major Components

- The telephone network is made of three major components:
  - ✓ Local Loops
  - ✓ Trunks
  - ✓ Switching offices
- The telephone network has several levels of switching offices:
  - ✓ End offices
  - ✓ Tandem offices
  - ✓ Regional offices

# A Telephone System



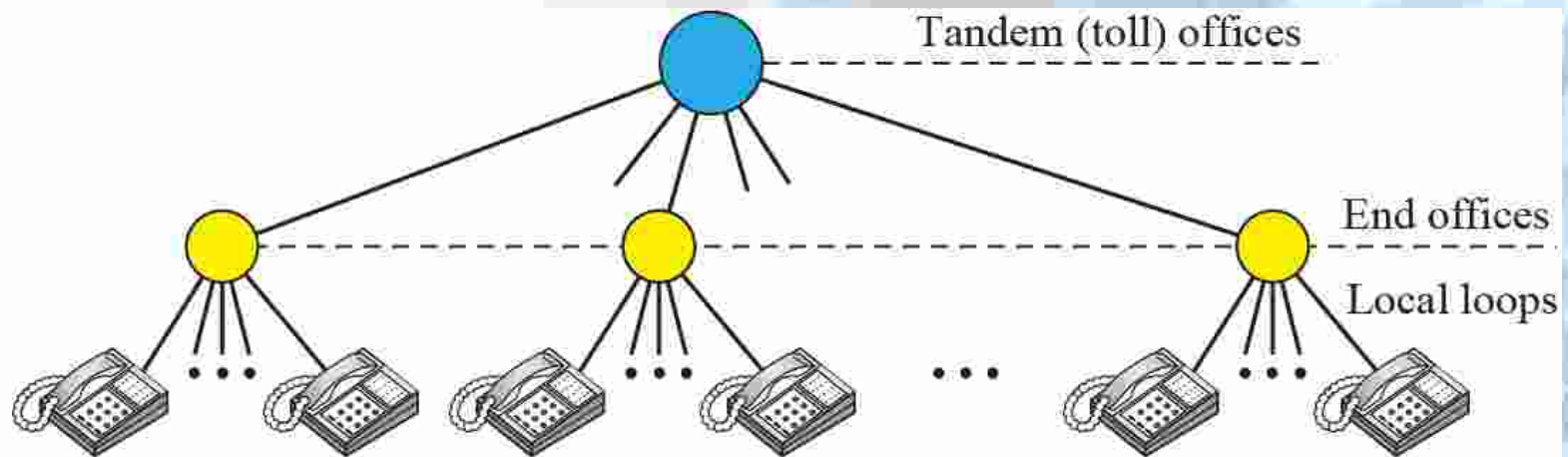
# Local-Access Transport Areas (LATAs)

- A LATA can be a small or large metropolitan area
- A small state may have a single LATA; a large state may have several LATAs
- A LATA boundary may overlap with state boundary; part of a LATA can be in one state, part in another state

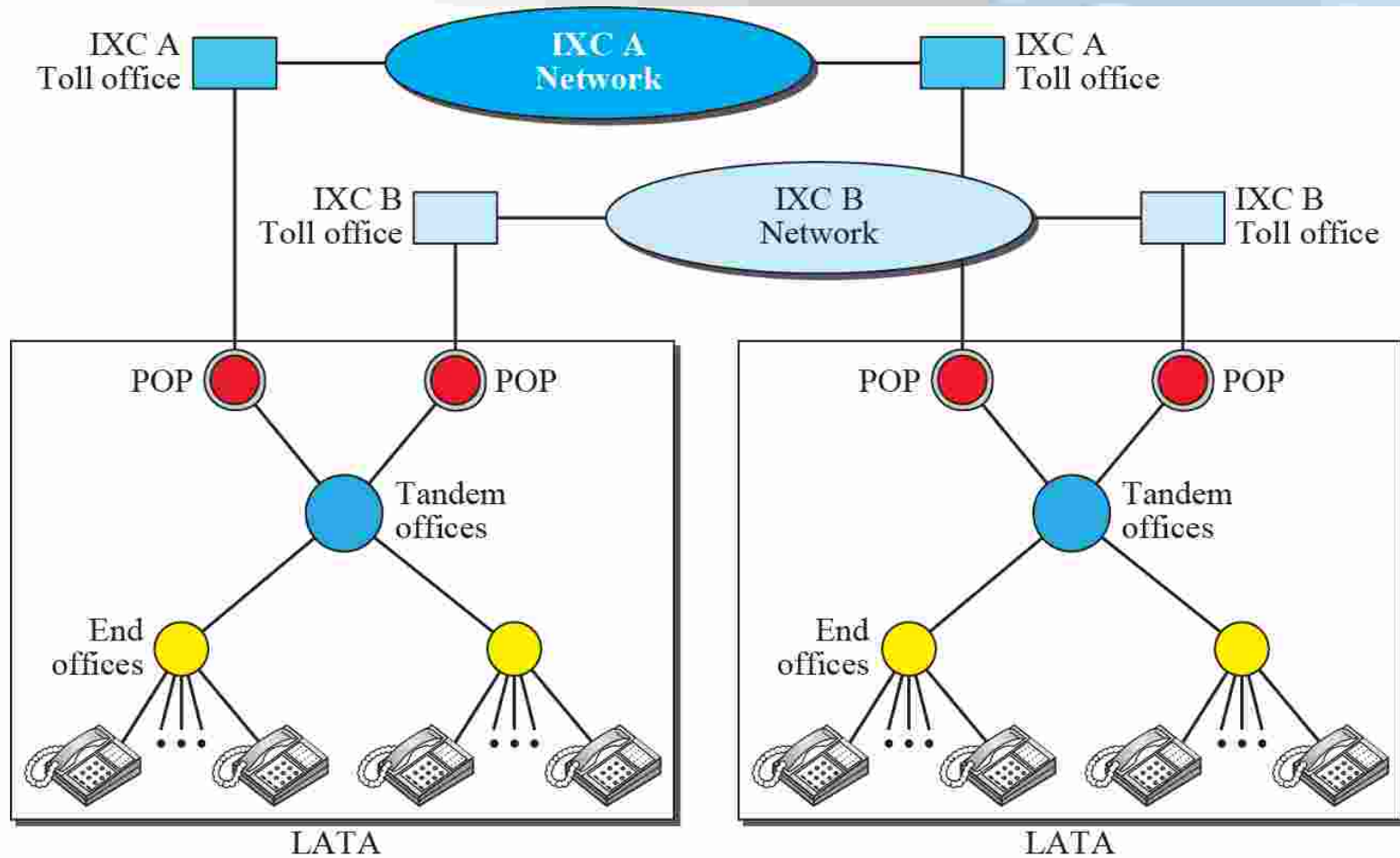
# Intra-LATA and Inter-LATA Services

- **Services offered by Telephone companies inside a LATA are called Intra-LATA services and between LATAs are called Inter-LATA services**
- **Carrier that handles Intra-LATA are called a Local Exchange Carrier (LEC) and the ones that handle Inter-LATA are called Interexchange Carriers (IXCs)**

# Switching Offices in a LATA



# Points of Presence (POPs)





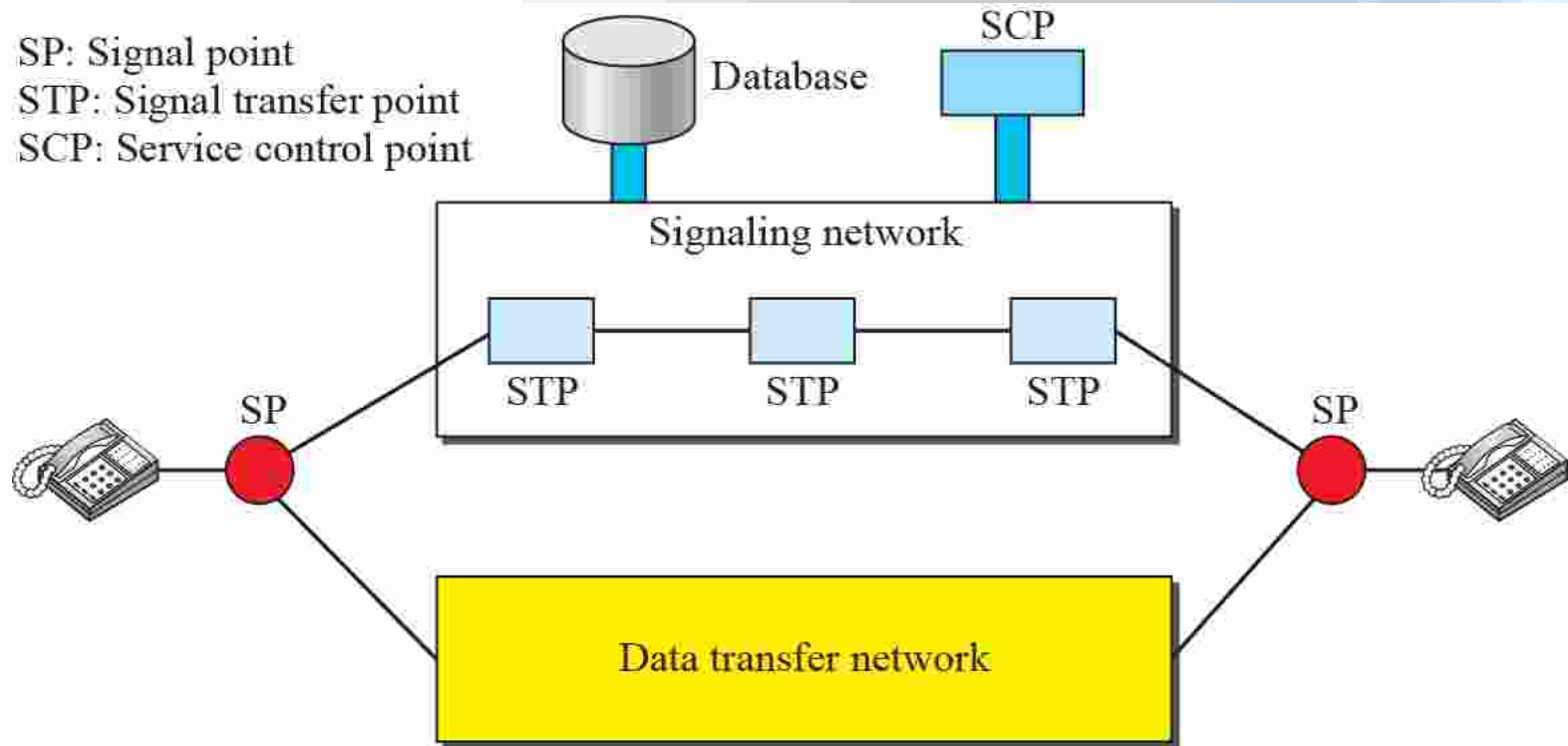
# Signaling

- The telephone network in the beginning, used a circuit-switched network with dedicated links to transfer voice communication
- The operator connected the two parties by using a wire with two plugs inserted into the corresponding two jacks
- Later, the signaling system became automatic

# Signaling

- Rotary telephones were invented that sent a digital signal defining each digit in a multi-digit telephone number
- As telephone networks evolved into a complex network, the functionality of the signaling system increased

# Data Transfer and Signaling Network



# Layers in SS7

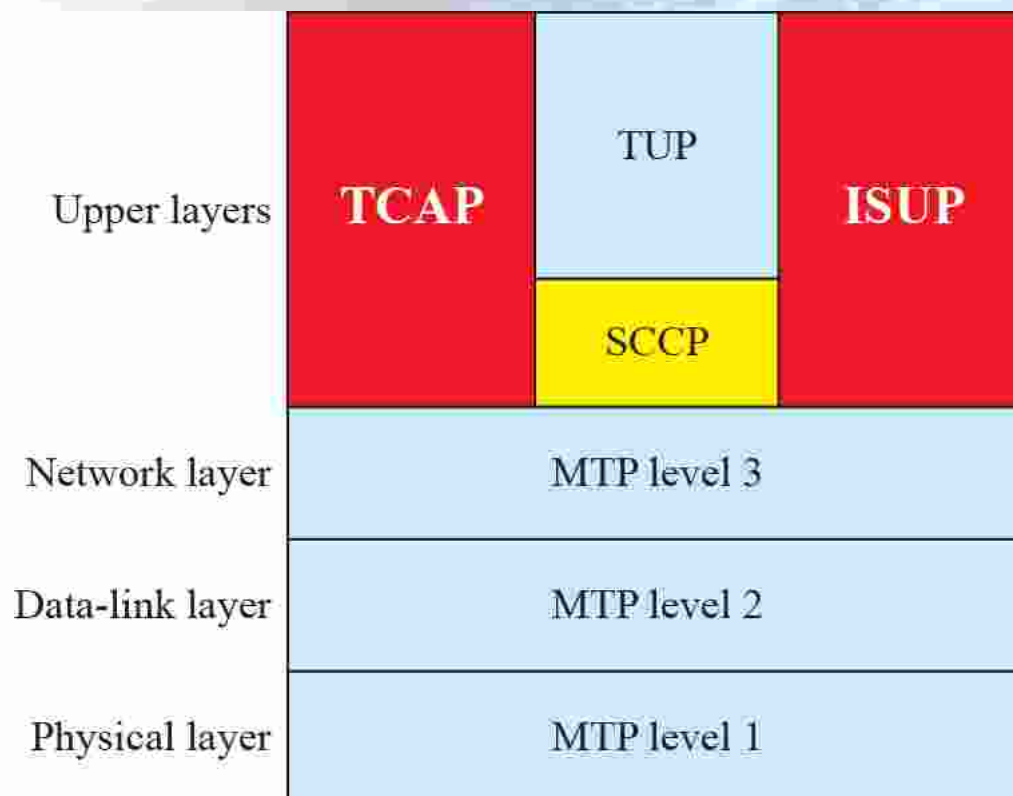
**MTP:** Message transfer part

**SCCP:** Signaling connection control point

**TCAP:** Transaction capabilities application port

**TUP:** Telephone user port

**ISUP:** ISDN user port



# Services

- Telephone companies provide two types of services:
  - ✓ Analog Services
    - Analog Switched Services
    - Analog Leased Services
  - ✓ Digital Services
    - Switched /56 Service
    - Digital Data Service

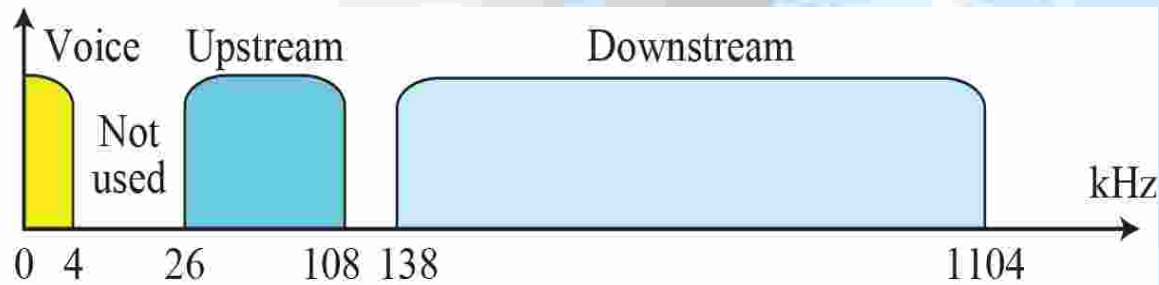
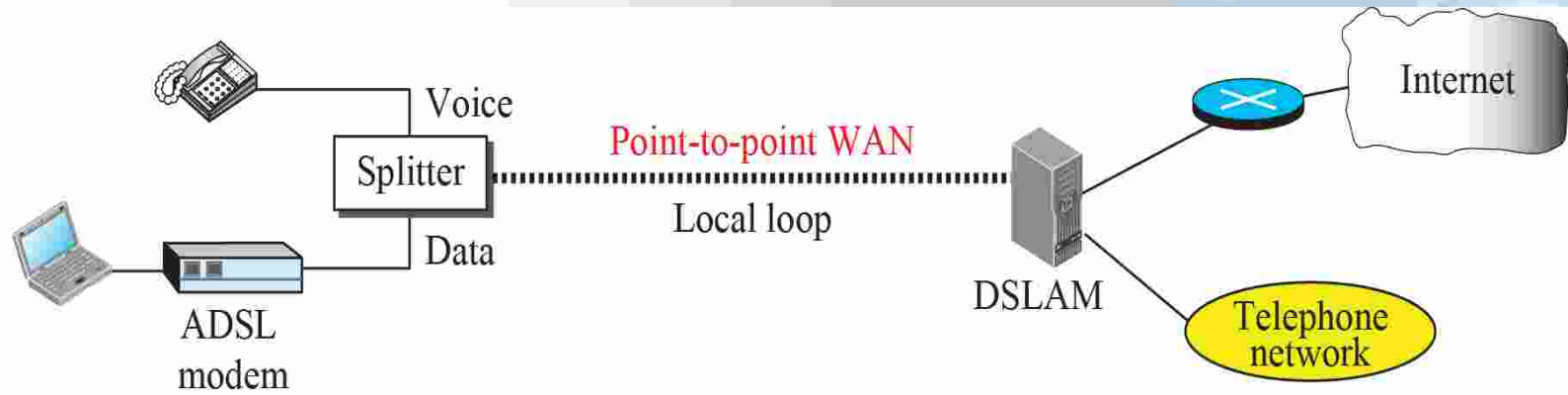
# Digital Subscriber Line (DSL)

- After traditional dial-up modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet
- DSL supports high-speed digital communication over the existing telephone

# Digital Subscriber Line (DSL)

- DSL technology is a set of technologies, each differing in the first letter (ADSL, VDSL, HDSL, and SDSL)

# ADSL Point-to-Point Network





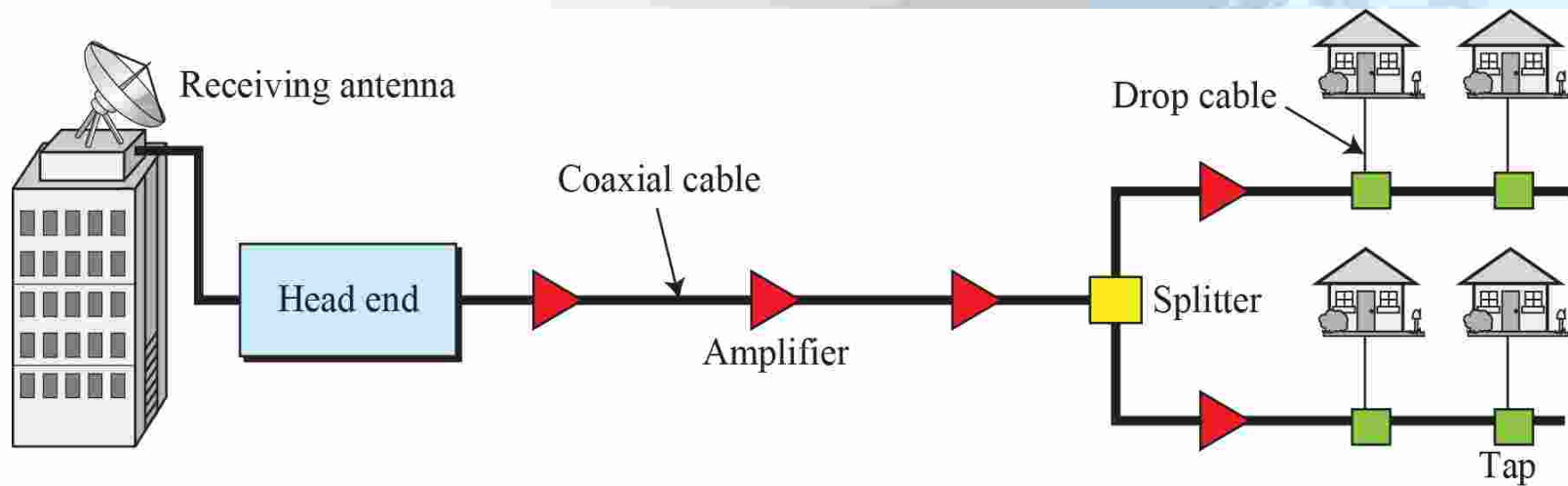
# Cable Network

- The Cable TV networks were initially created to provide remote subscribers access to TV programs
- Cable networks enabled access to remote broadcasting stations via microwave connections
- Cable TV also found a good ISP market by using some of the channels originally designed for video

# Traditional Cable Networks

- Cable TV started to distribute broadcast video signals to locations with poor or no reception in the late 1940s
- It was called community antenna television (CATV) because an antenna at the top of a tall hill or building received the signals from the TV stations

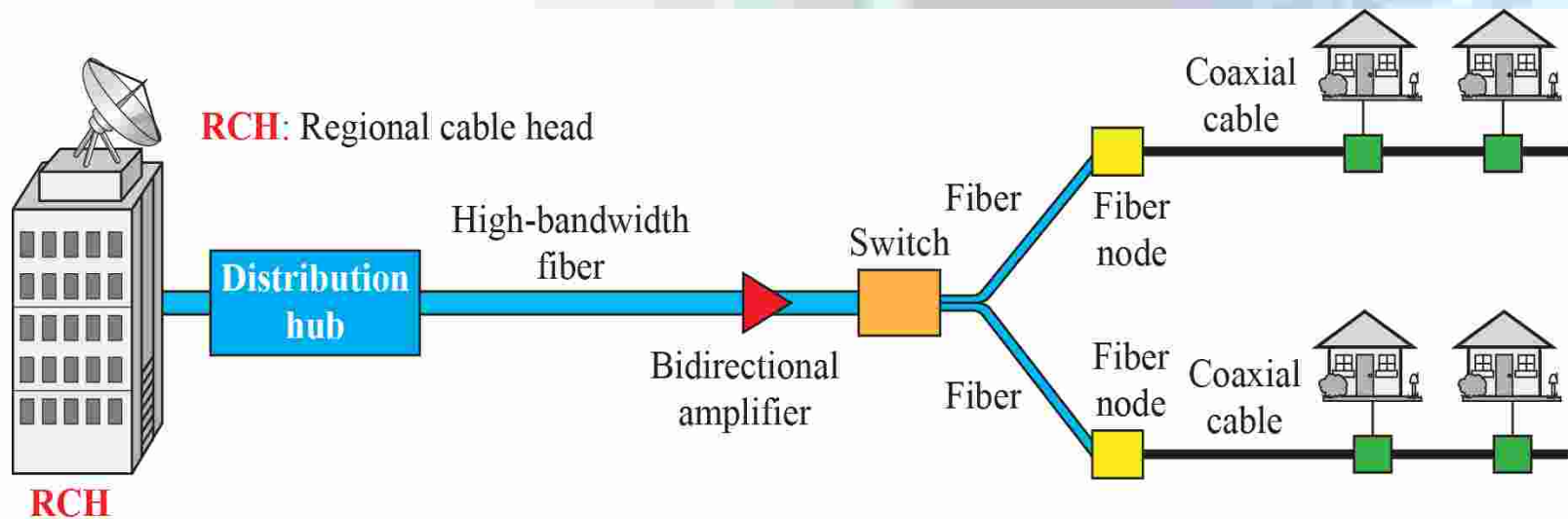
# Traditional Cable TV Network



# Hybrid Fiber Coaxial (HFC) Network

- Second generation of cable network is called a Hybrid Fiber-Coaxial (HFC) network
- The network uses a combination of fiber-optic and coaxial cable

# Hybrid Fiber-Coaxial (HFC) Network



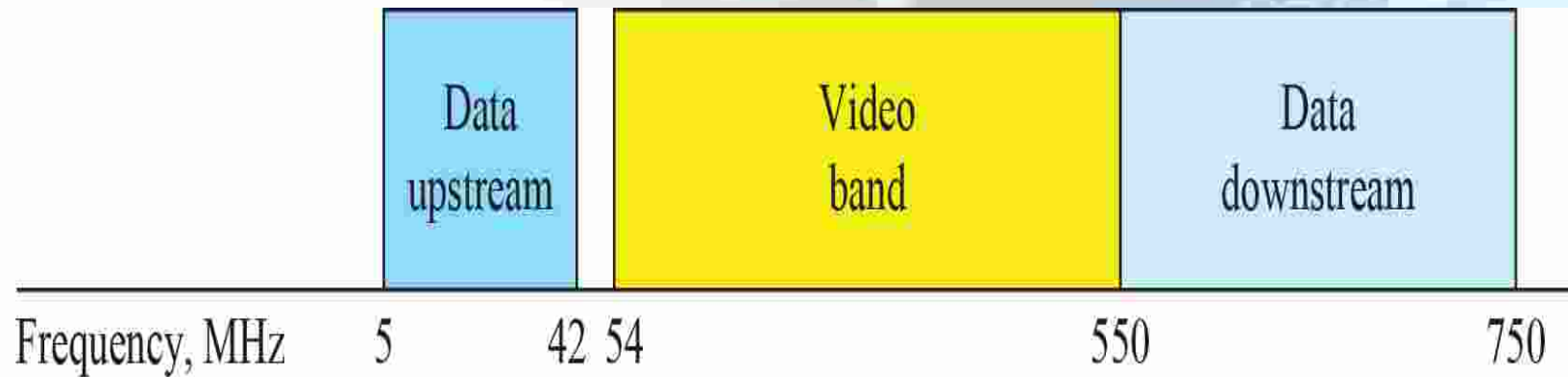
# Cable TV for Data Transfer

- Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer
- DSL technology provides high-data-rate connections for residential subscribers over the local loop BUT UTP is susceptible to Interference

# Cable TV for Data Transfer

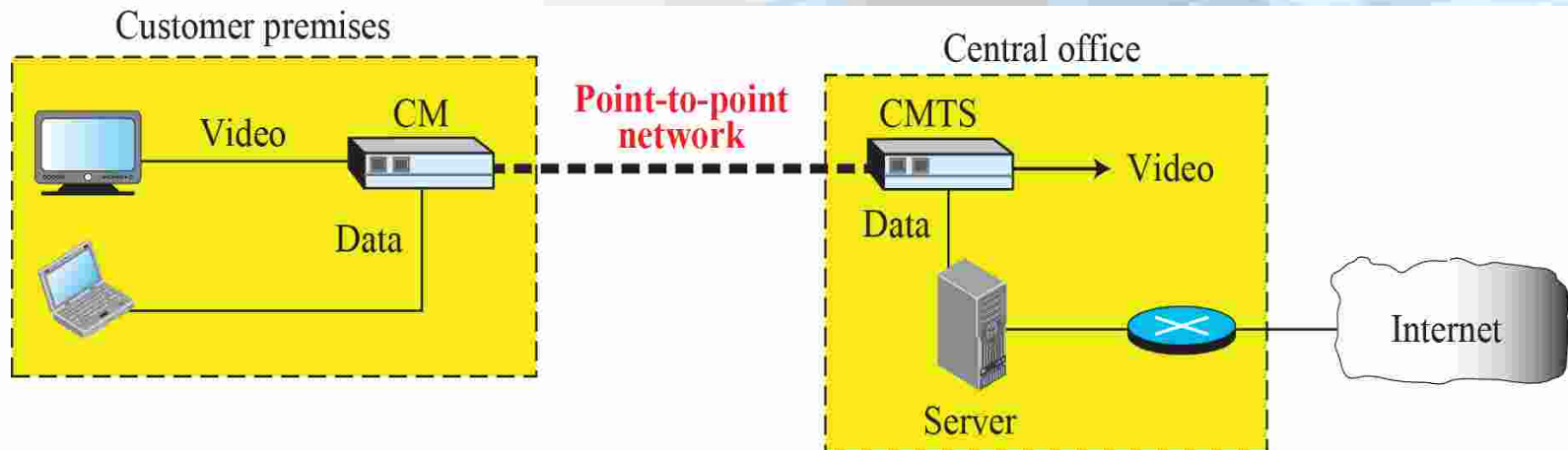
- This imposes an upper limit on the data rate. A solution is the use of the cable TV network

# Division of Coaxial Cable Band by CATV





# Cable Modem Transmission System (CMTS)



# Synchronous Optical Network (SONET)

- A wide area network (WAN) that is used as a transport network to carry loads from other WANs
- ITU–T standard called Synchronous Digital Hierarchy (SDH)
- Architecture of a SONET system consists of signals, devices, and connections

# SONET Architecture

- **Signals**
  - ✓ Synchronous Transport Signals (STS)
  - ✓ Optical Carriers (OCs)
  - ✓ Synchronous Transport Module (STM)
- **SONET Devcies**
  - ✓ STS Mux/Demux
  - ✓ Regenerators
  - ✓ Add-Drop Multiplexer and Terminals
- **Connections**
  - ✓ Section
  - ✓ Line
  - ✓ Path

# SONET Signals

| <i>STS</i> | <i>OC</i> | <i>Rate (Mbps)</i> | <i>STM</i>    |
|------------|-----------|--------------------|---------------|
| STS-1      | OC-1      | 51.840             |               |
| STS-3      | OC-3      | 155.520            | <b>STM-1</b>  |
| STS-9      | OC-9      | 466.560            | <b>STM-3</b>  |
| STS-12     | OC-12     | 622.080            | <b>STM-4</b>  |
| STS-18     | OC-18     | 933.120            | <b>STM-6</b>  |
| STS-24     | OC-24     | 1244.160           | <b>STM-8</b>  |
| STS-36     | OC-36     | 1866.230           | <b>STM-12</b> |
| STS-48     | OC-48     | 2488.320           | <b>STM-16</b> |
| STS-96     | OC-96     | 4976.640           | <b>STM-32</b> |
| STS-192    | OC-192    | 9953.280           | <b>STM-64</b> |

# SONET Architecture

- **Signals**
  - ✓ Synchronous Transport Signals (STS)
  - ✓ Optical Carriers (OCs)
  - ✓ Synchronous Transport Module (STM)
- **SONET Devcies**
  - ✓ STS Mux/Demux
  - ✓ Regenerators
  - ✓ Add-Drop Multiplexer and Terminals
- **Connections**
  - ✓ Section
  - ✓ Line
  - ✓ Path

# SONET Devices

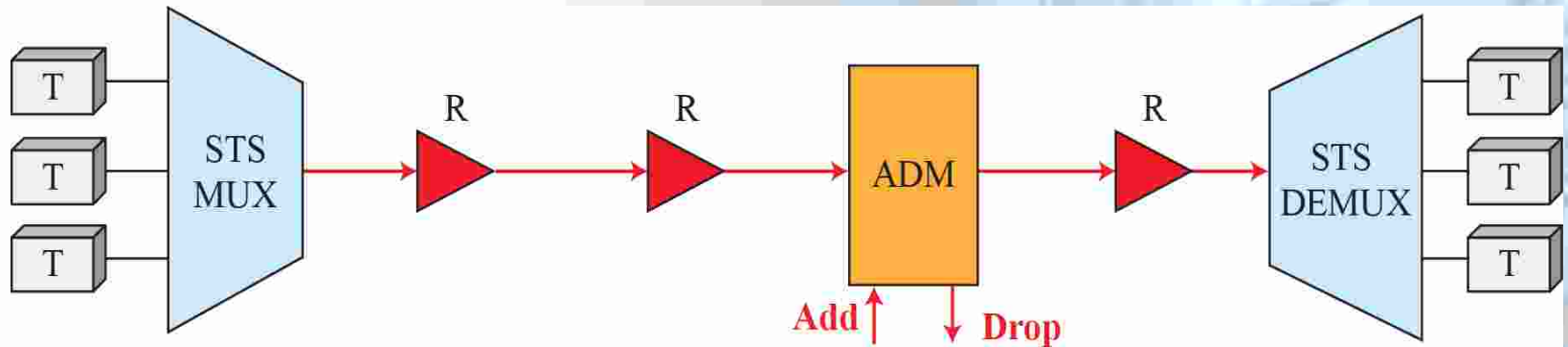
**ADM:** Add/drop multiplexer

**R:** Regenerator

**STS MUX:** Synchronous transport signal multiplexer

**T:** Terminal

**STS DEMUX:** Synchronous transport signal demultiplexer



# SONET Connections

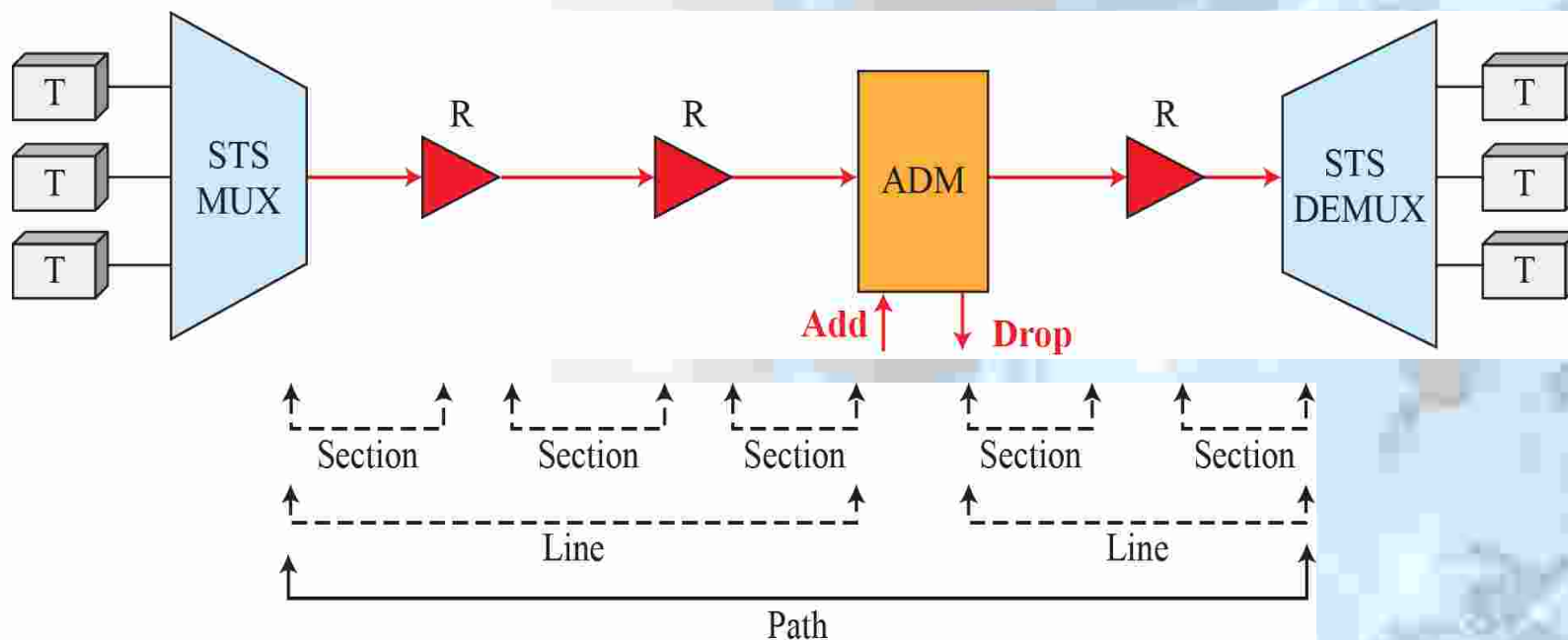
**ADM:** Add/drop multiplexer

**R:** Regenerator

**STS MUX:** Synchronous transport signal multiplexer

**T:** Terminal

**STS DEMUX:** Synchronous transport signal demultiplexer

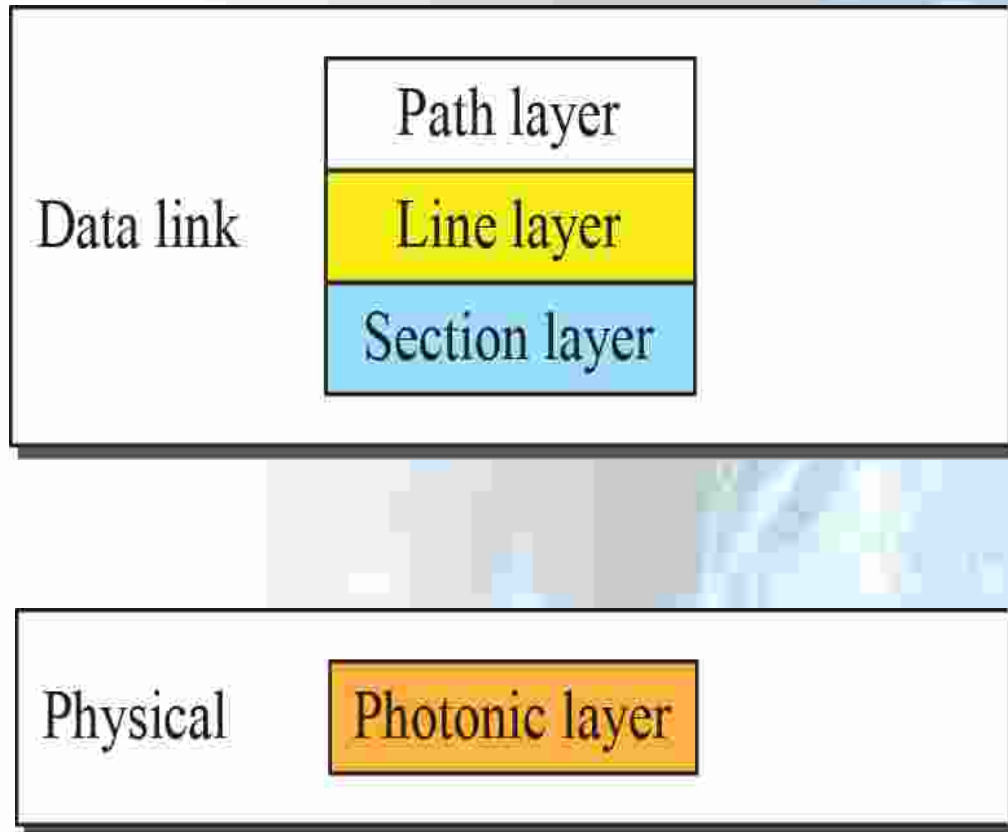


# SONET Layers

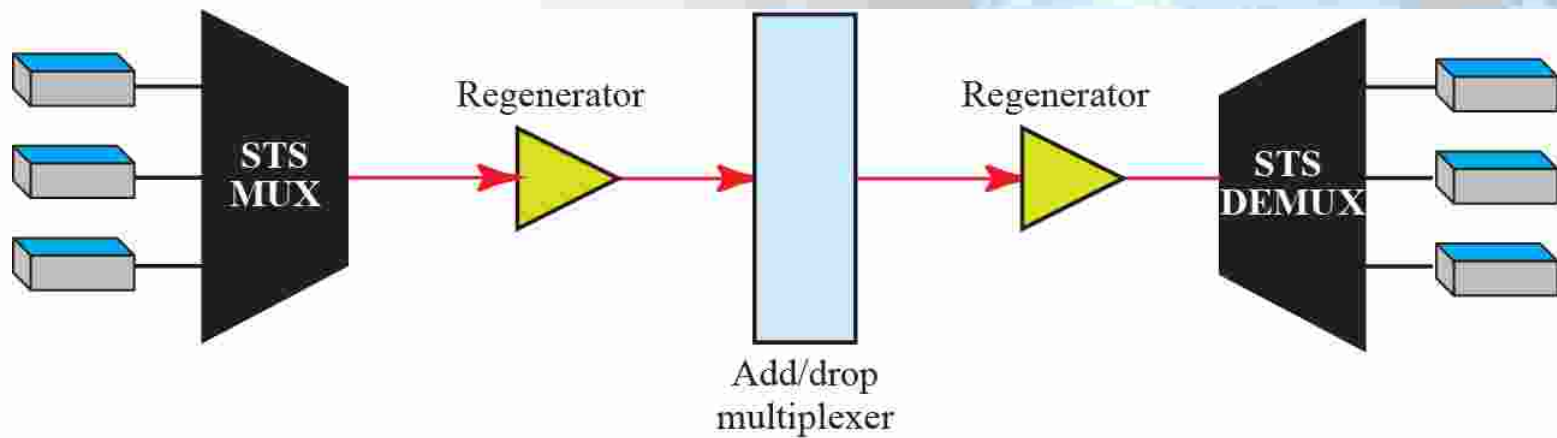
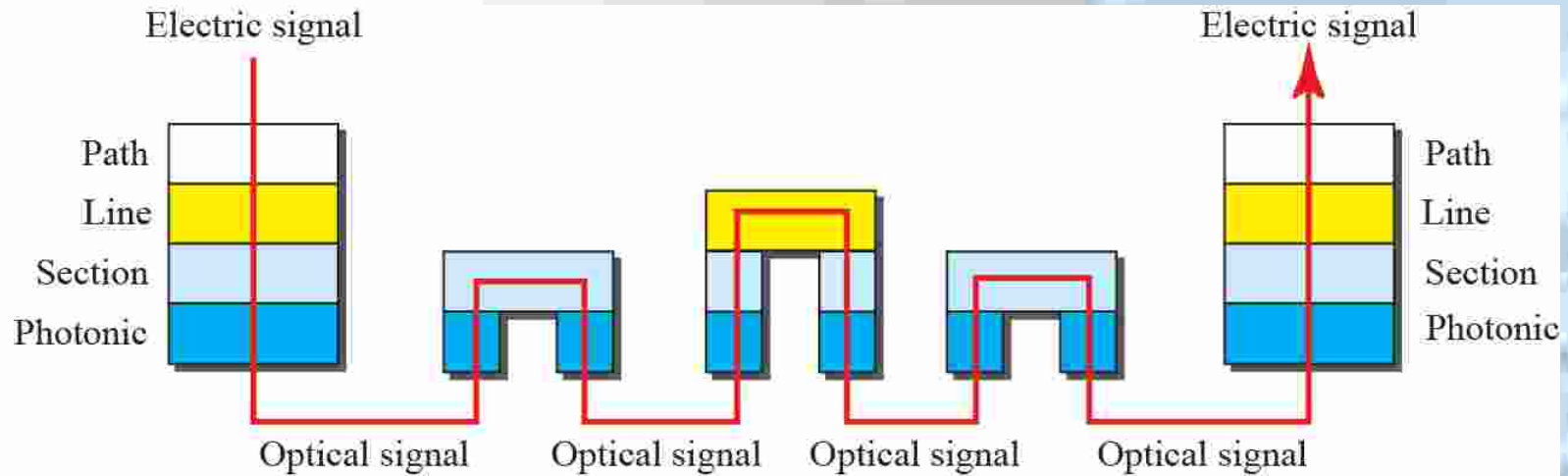
- The SONET standard includes four functional layers:
  - ✓ The Path Layer
  - ✓ The Line Layer
  - ✓ The Section Layer
  - ✓ The Photonic Layer
- The layers correspond to both the physical and the data-link layers



# SONET Layers



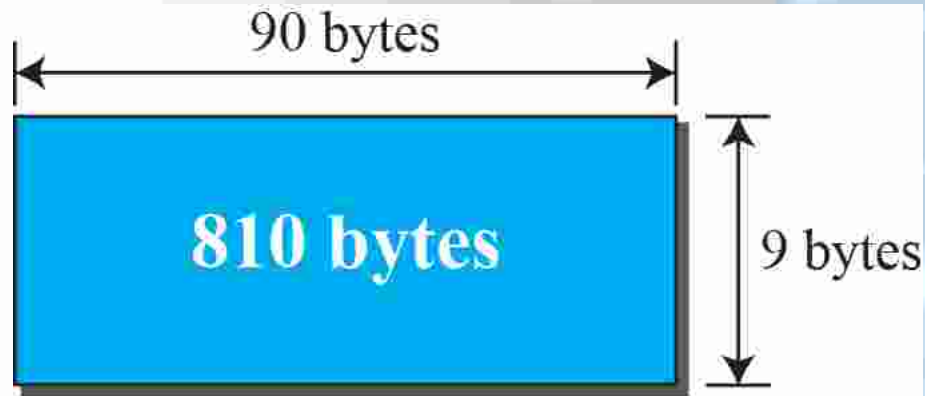
# Device-Layer Relationship in SONET



# SONET Frames

- Each synchronous transport signal STS-n is composed of 8000 frames
- Each frame is a two-dimensional matrix of bytes with 9 rows by  $90 \times n$  columns
- STS-1 frame is 9 rows by 90 columns (810 bytes), and an STS-3 is 9 rows by 270 columns (2430 bytes)

# An STS-1 and an STS-n Frame

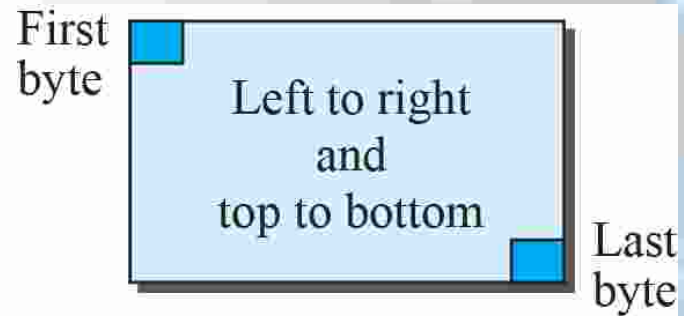


a. STS-1 frame

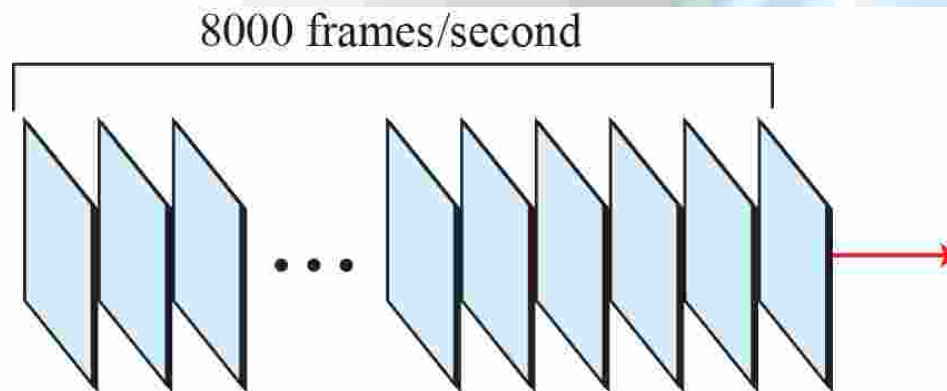


b. STS-*n* frame

# STS-1 Frames in Transition



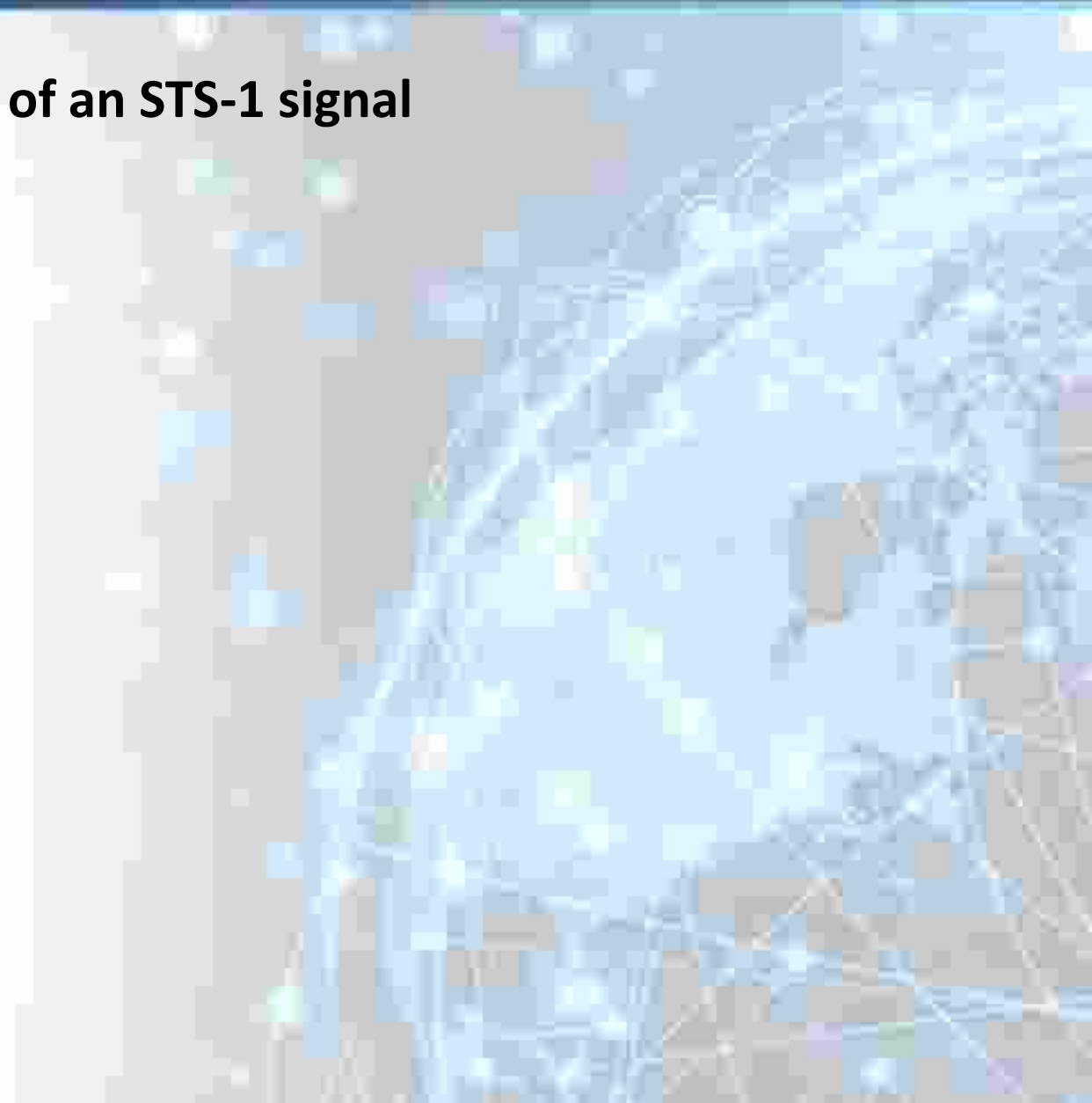
a. Byte transmission



b. Frame transmission

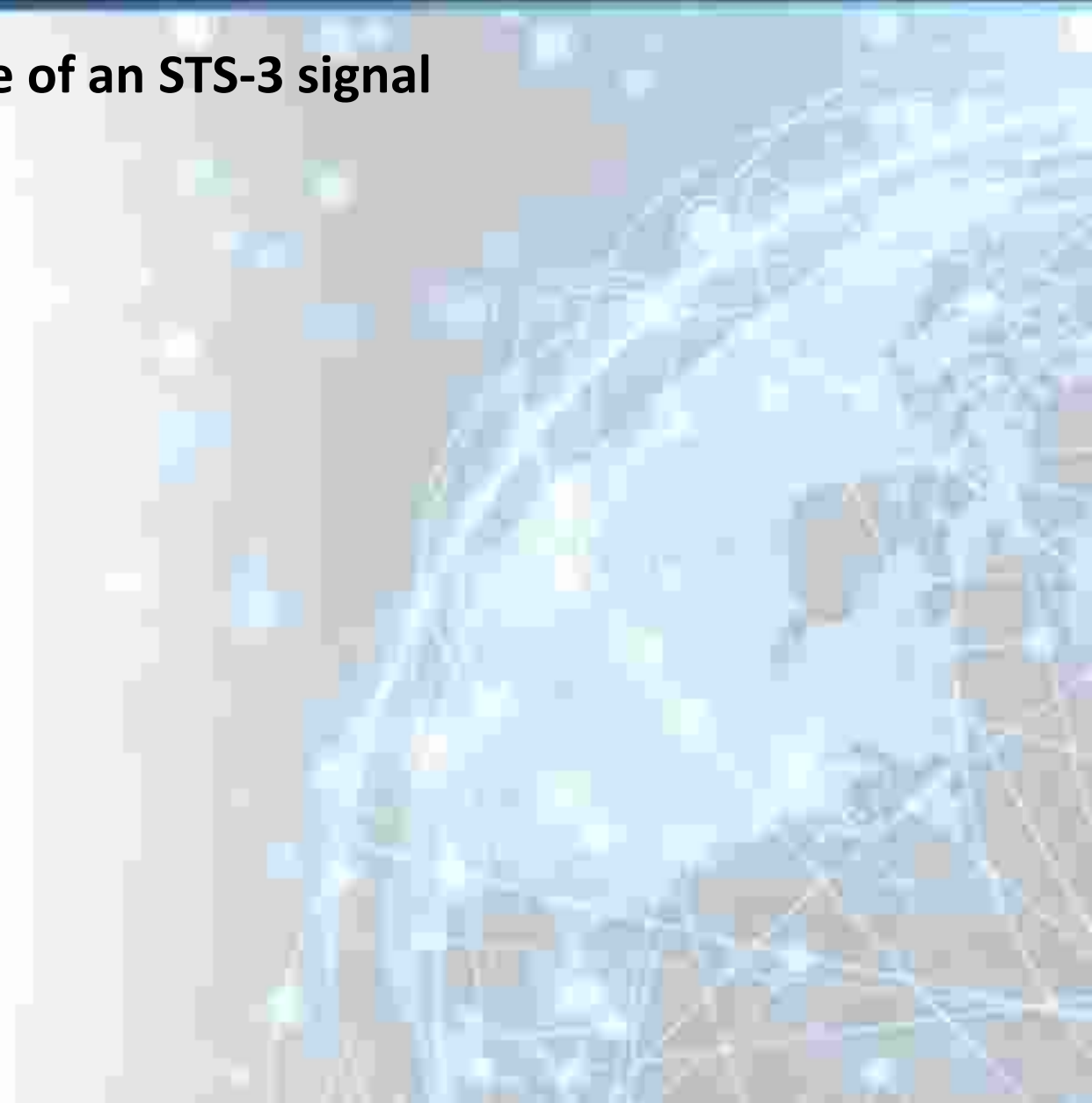
# Example

**Find the data rate of an STS-1 signal**

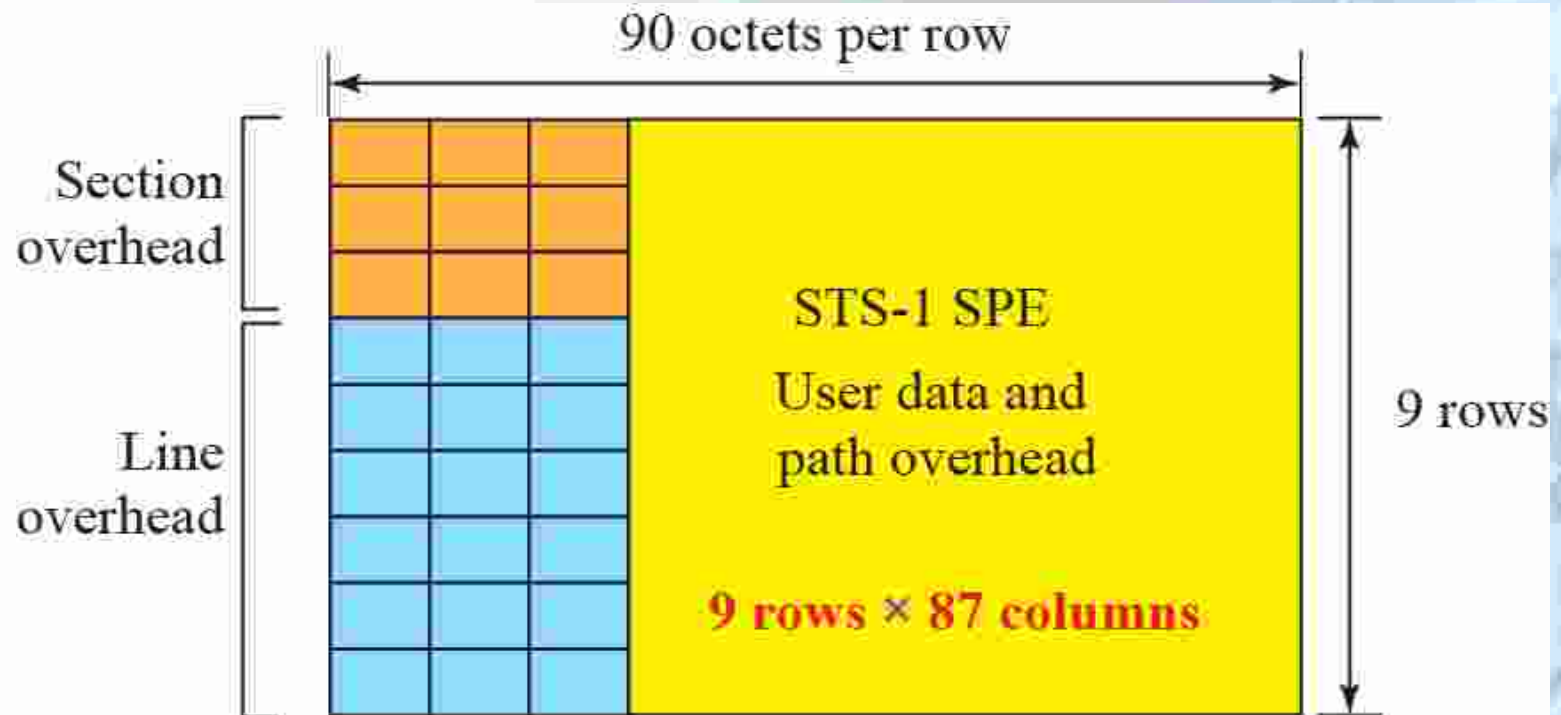


# Example

Find the data rate of an STS-3 signal



# STS-1 Frame Format





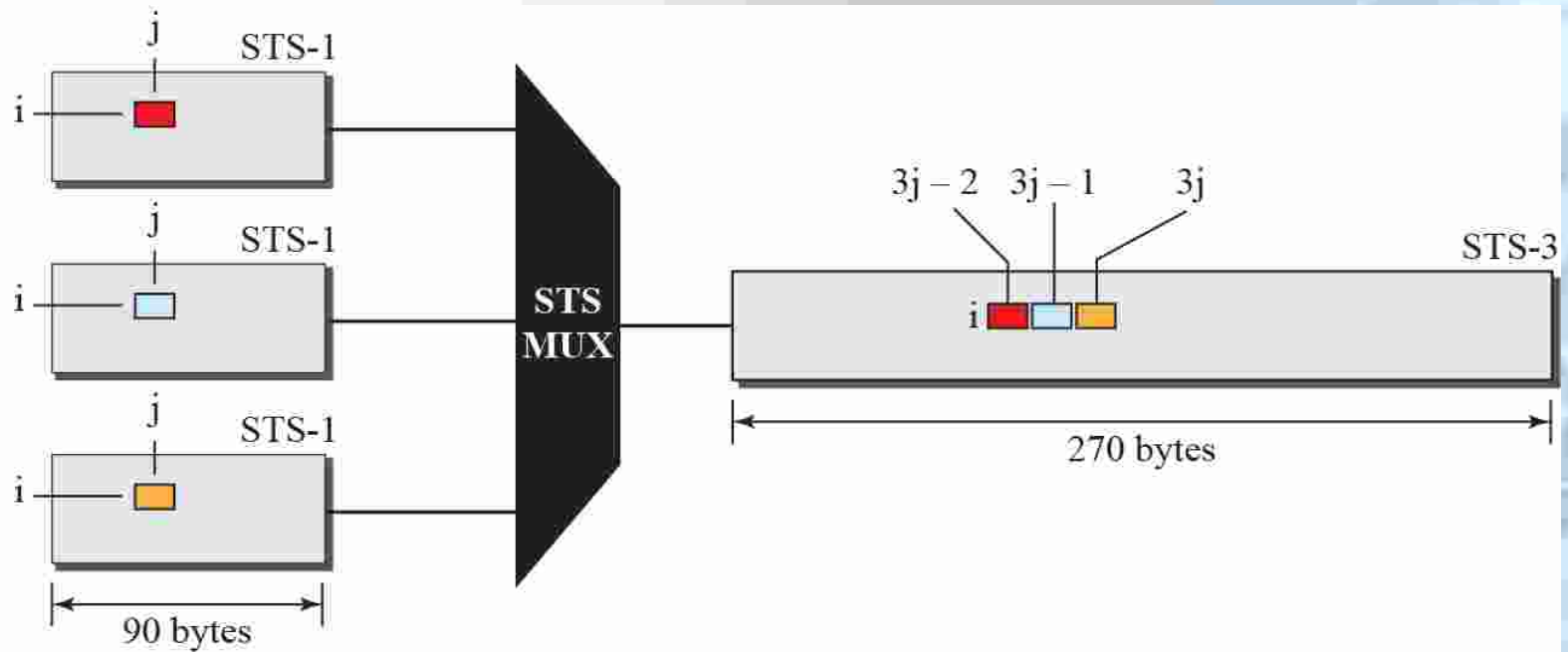
# STS Multiplexing

- In SONET, frames of lower rate can be synchronously time-division multiplexed into a higher-rate frame
- For example, three STS-1 signals (channels) can be combined into one STS-3 signal (channel), four STS-3s can be multiplexed into one STS-12, and so on

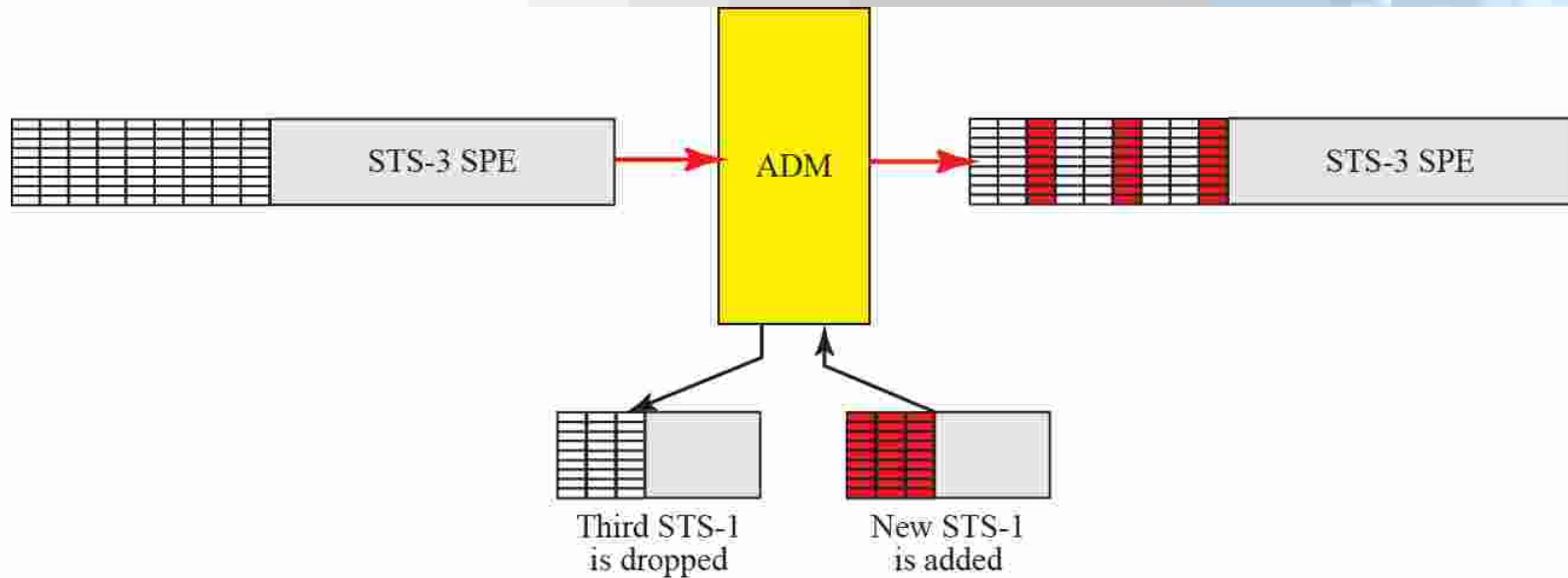
# STS Multiplexing/Demultiplexing



# Byte Interleaving



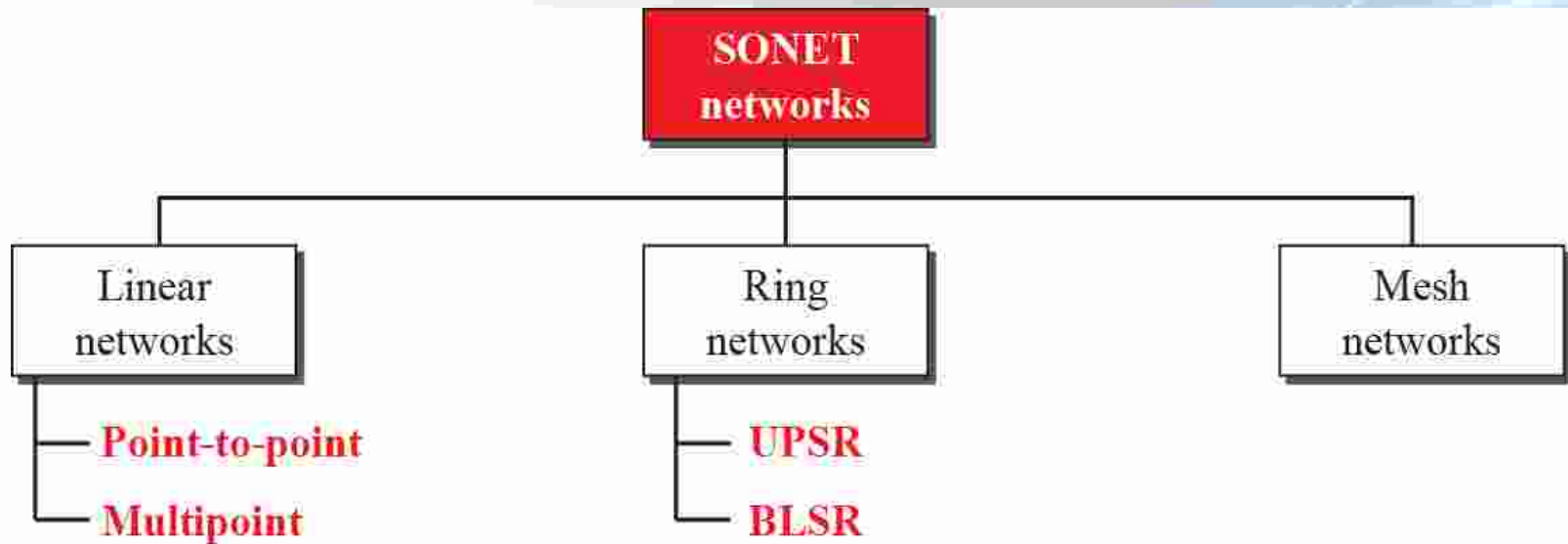
# Add/Drop Multiplexer



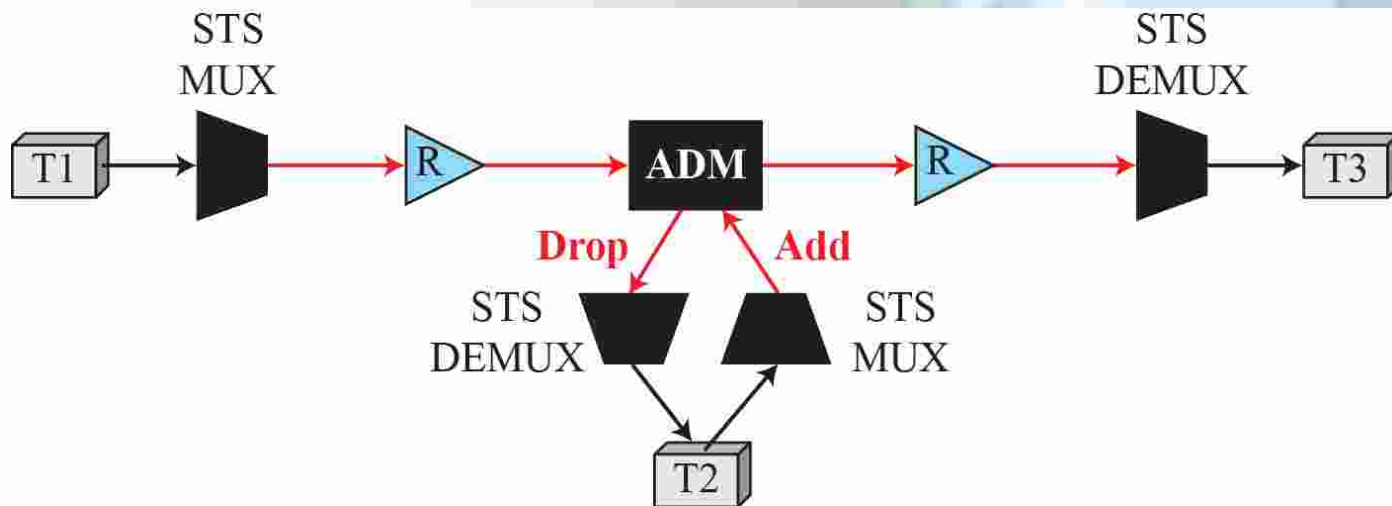
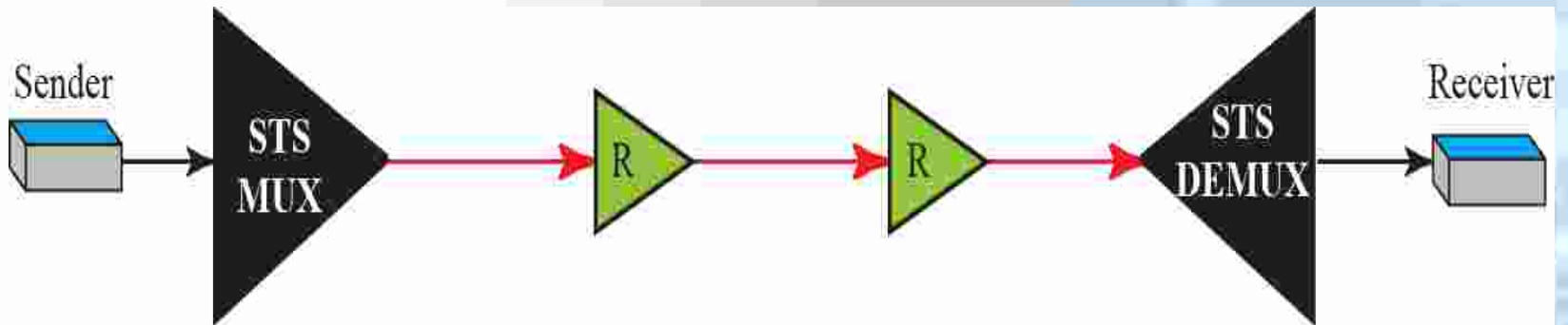
# SONET Networks

- **SONET network can be used as a high-speed backbone carrying loads from other networks such as ATM or IP**
- **We can roughly divide SONET networks into three categories:**
  - ✓ **Linear Networks**
  - ✓ **Ring Networks**
  - ✓ **Mesh networks**

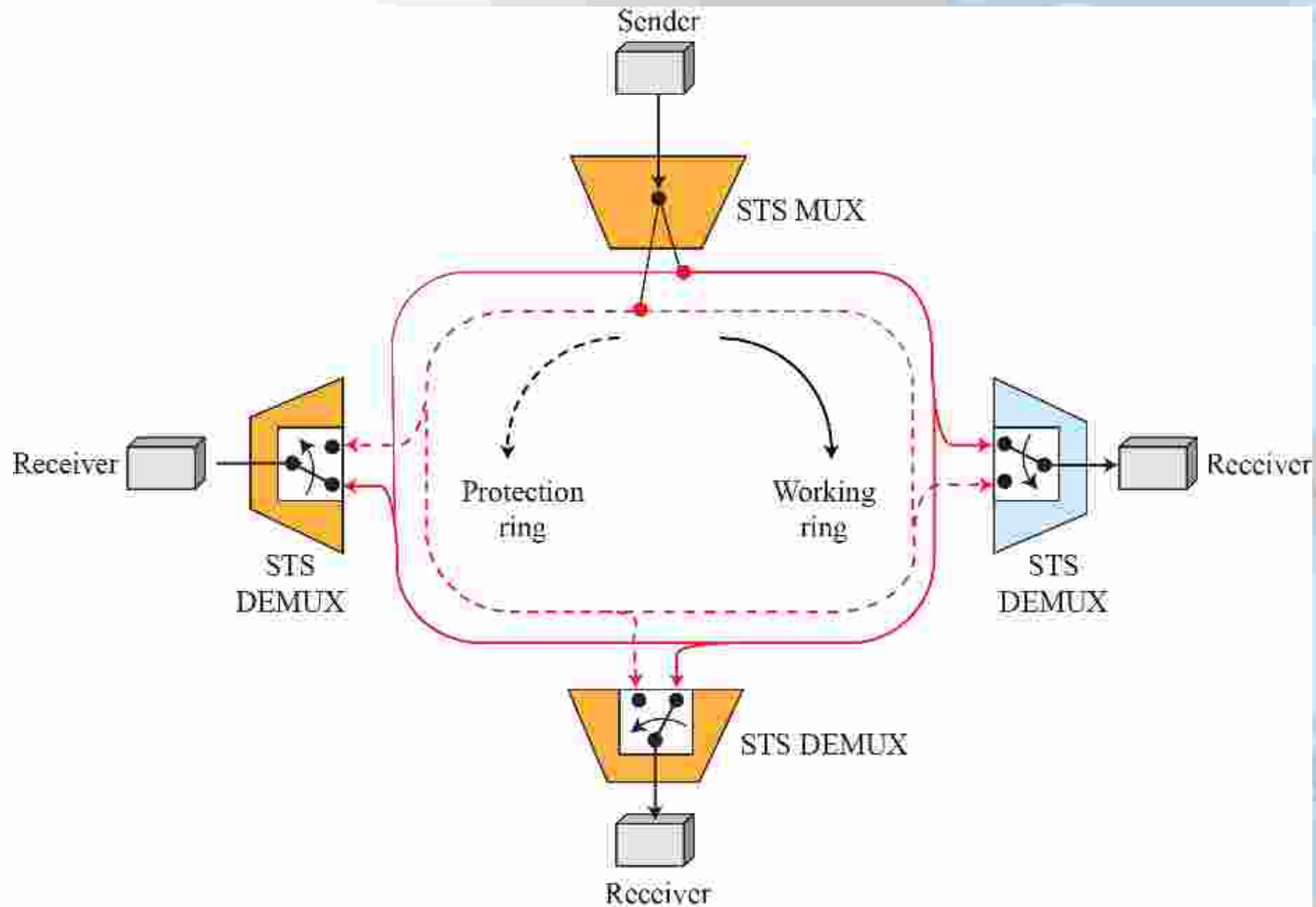
# Taxonomy of SONET Networks



# SONET Networks – Linear Networks

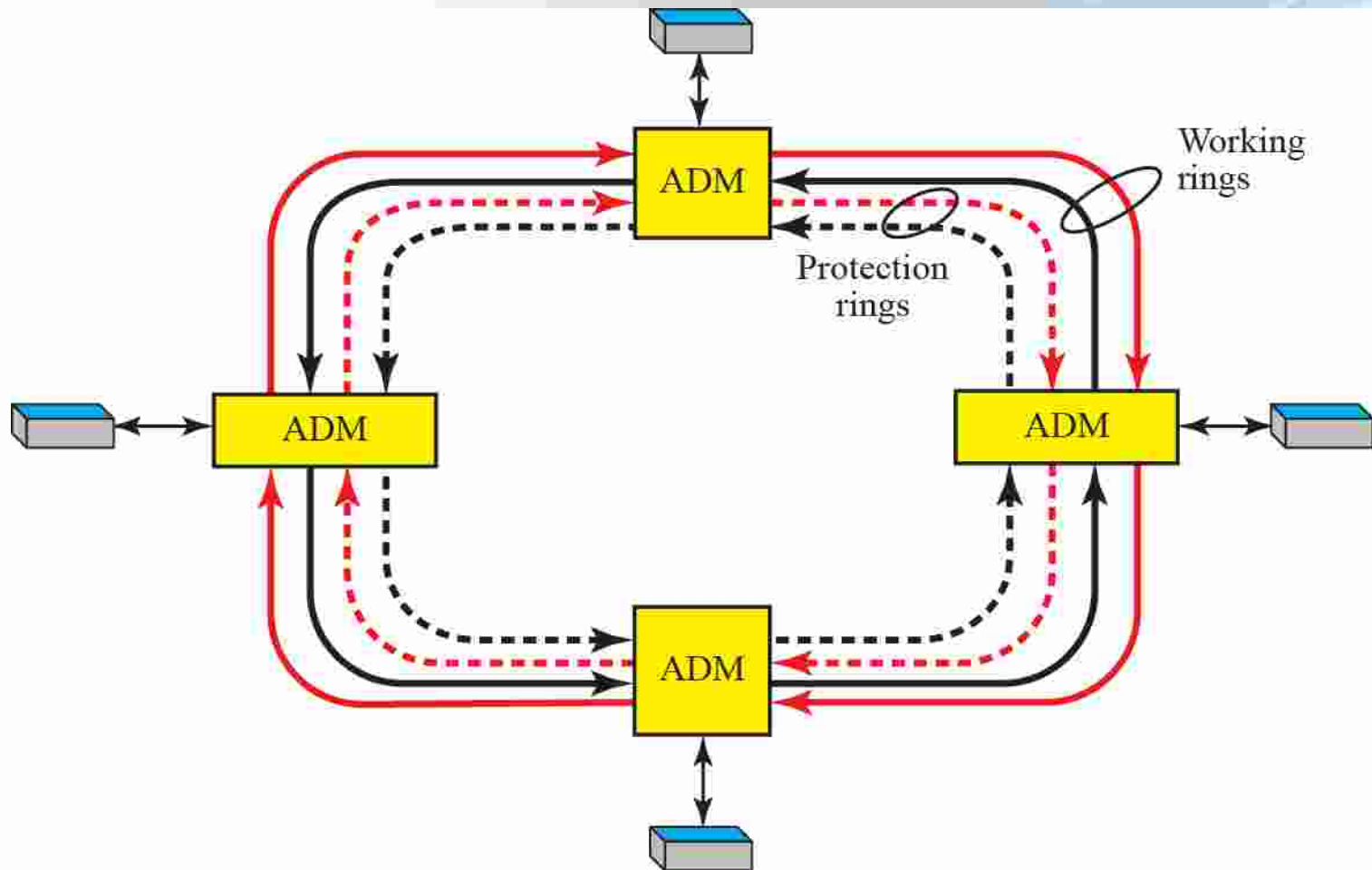


# SONET Networks – Ring Networks

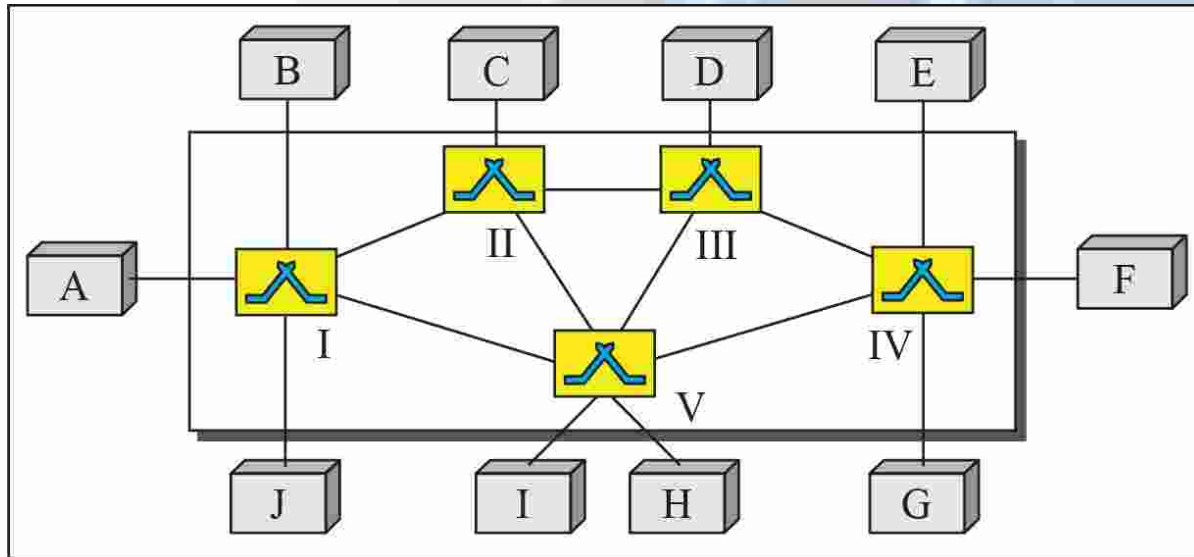




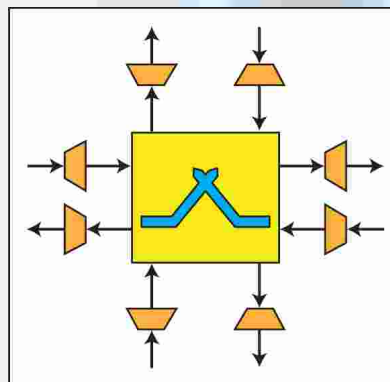
# SONET Networks – Ring Networks



# SONET Networks – Mesh Networks



a. SONET mesh network



b. Cross-connect switch

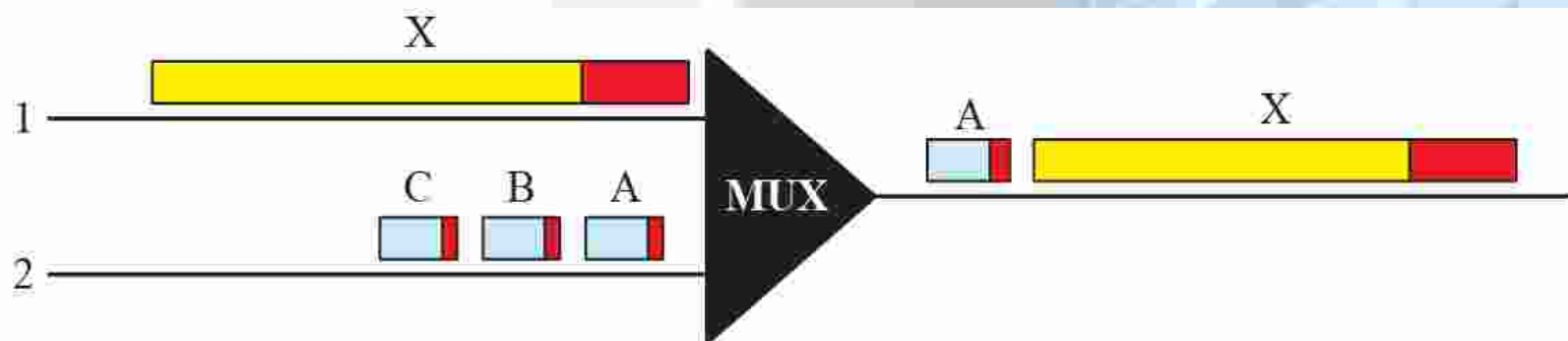
# ATM

- **Asynchronous Transfer Mode (ATM) is a switched wide area network based on the cell relay protocol designed by the ATM forum**
- **The combination of ATM and SONET will allow high-speed interconnection of networks**

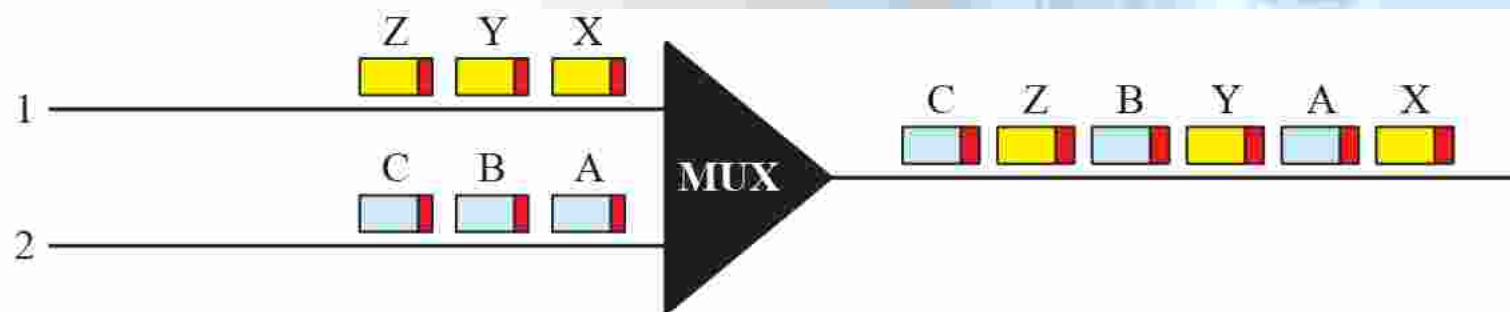
# Problems

- Some of the problems associated with existing systems are:
  - ✓ Frame Networks
  - ✓ Mixed Network Traffic
- Solution
  - ✓ Cell Networks
  - ✓ Asynchronous TDM

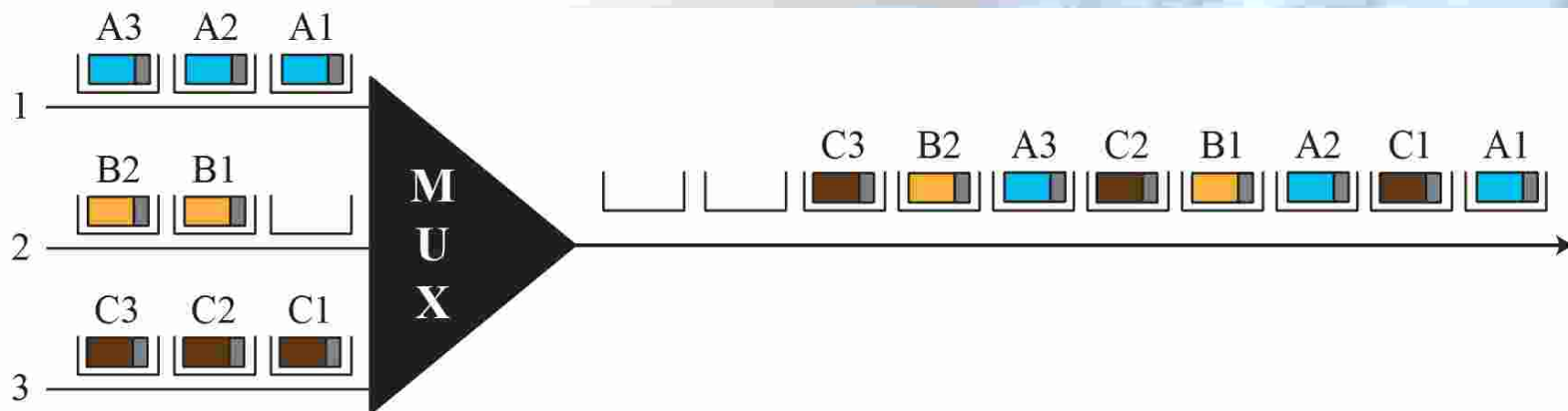
# Multiplexing using Different Frame Size



# Multiplexing using Cells



# ATM Multiplexing

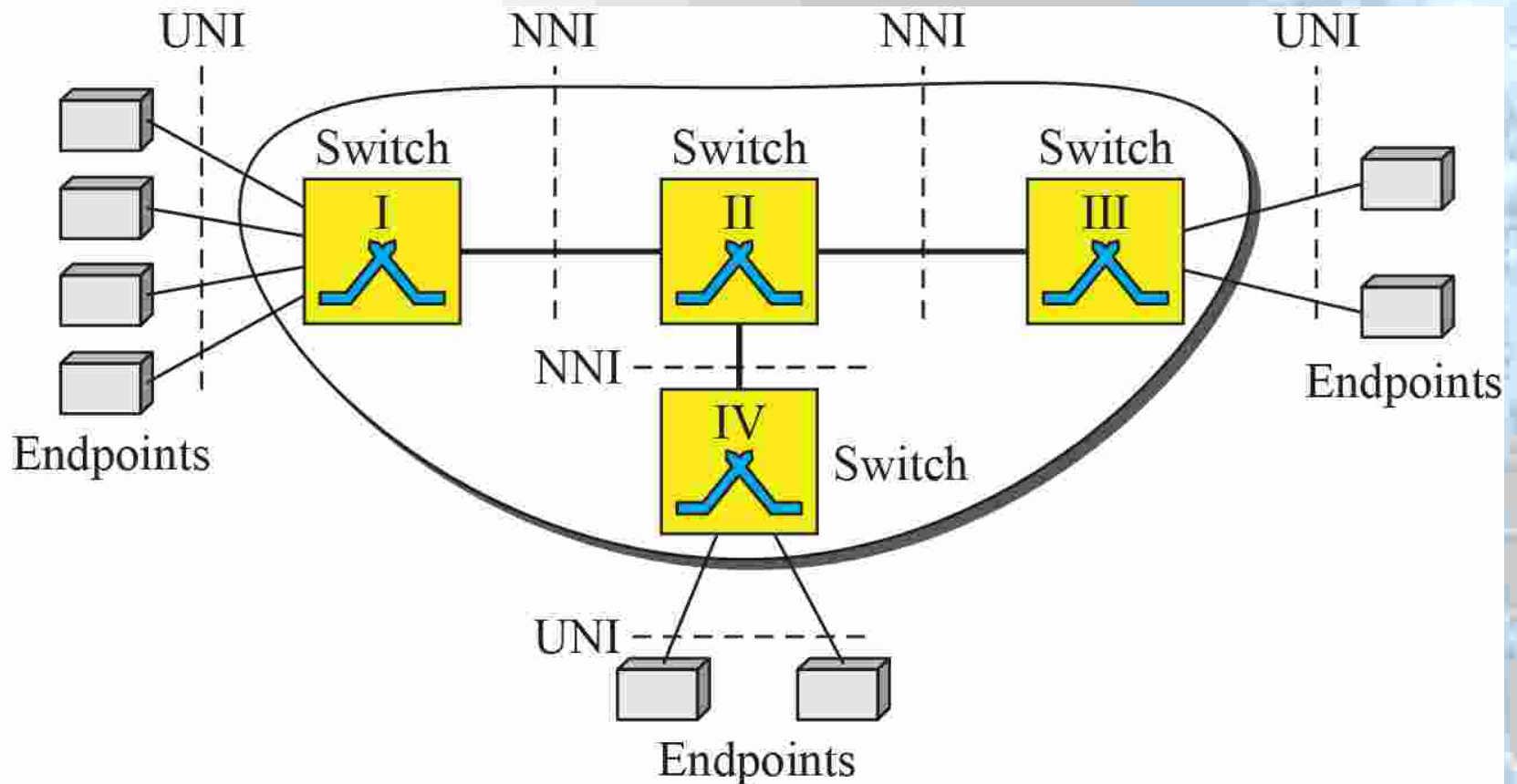


# Architecture

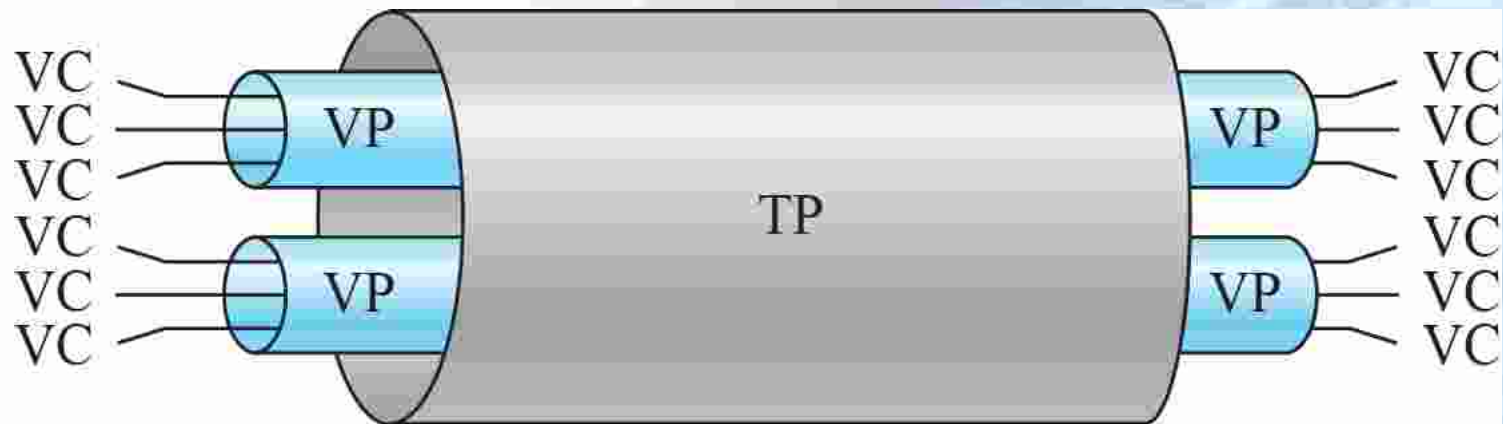
- **ATM is a cell-switched network**
- **The user access devices, called the endpoints, are connected through a user-to-network interface (UNI) to the switches inside the network**
- **The switches are connected through network-to-network interfaces (NNIs)**



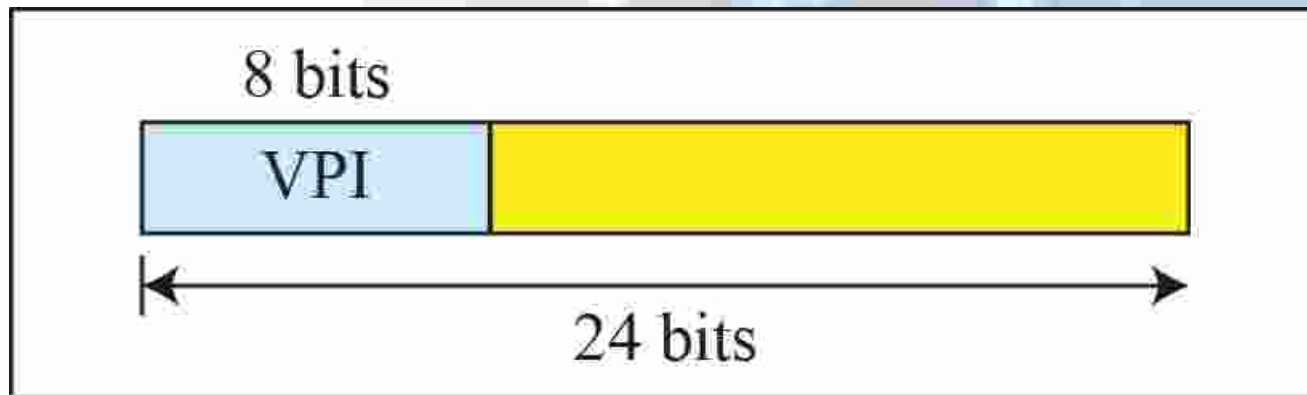
# Architecture of an ATM Network



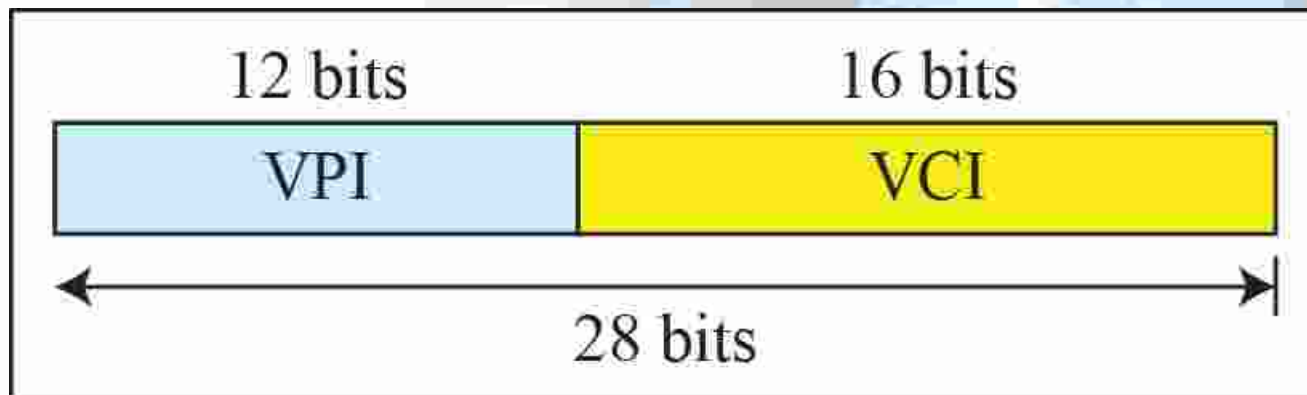
# TP, VPs, and VCs



# Virtual connection identifiers in UNIs & NNIs

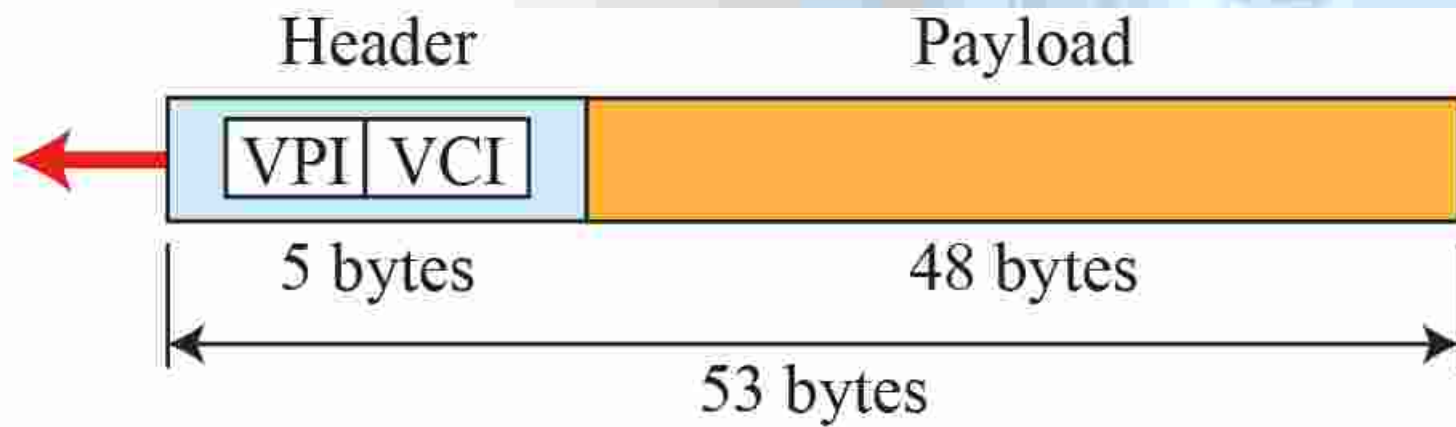


a. VPI and VCI in a UNI

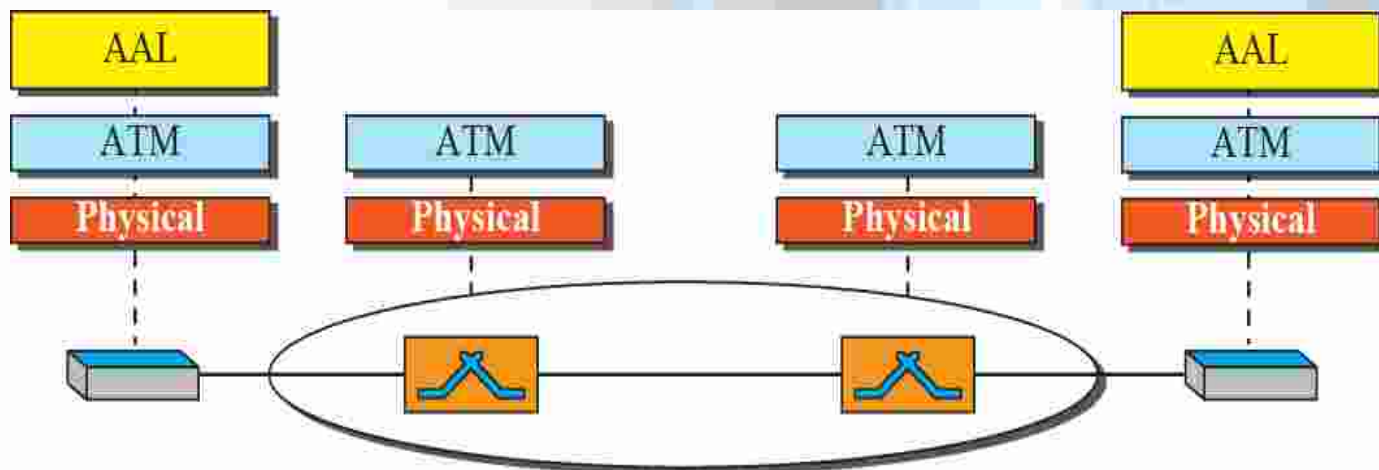
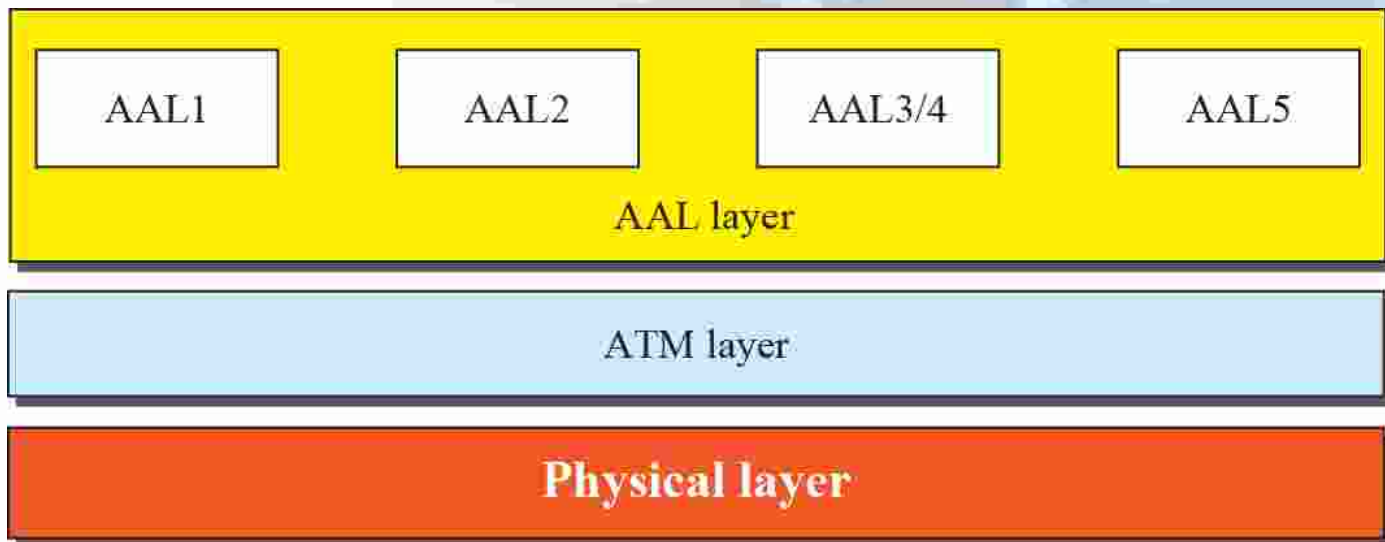


b. VPI and VCI in an NNI

# An ATM Cell



# ATM Layers



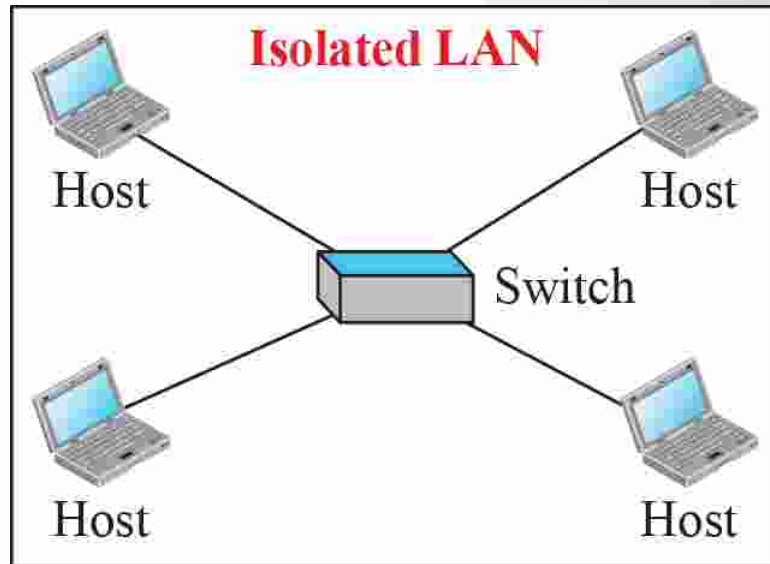
# Introduction

- **Wireless communication is one of the fastest-growing technologies**
- **The demand for connecting devices without the use of cables is increasing everywhere**
- **Wireless LANs can be found on college campuses, in office buildings, and in many public areas**

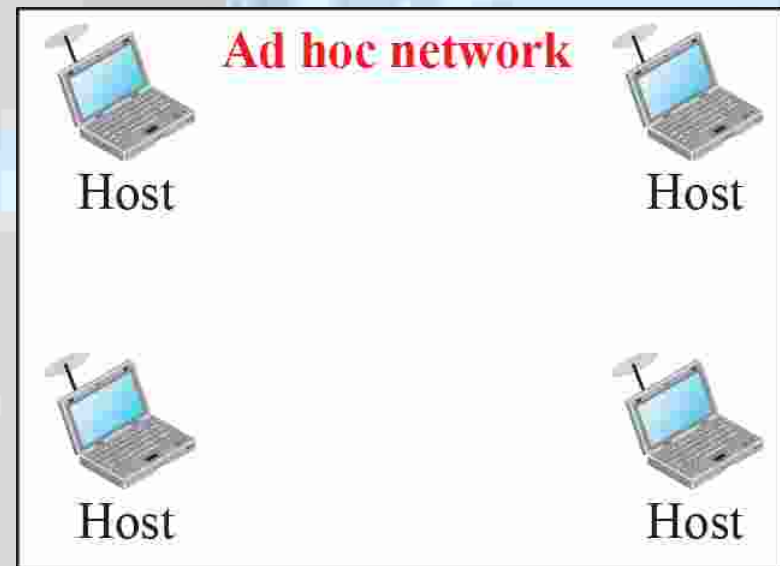
# Architectural Comparison

- **Architecture comparison of wired and wireless LANs**
  - ✓ **Medium**
  - ✓ **Hosts**
  - ✓ **Isolated LANs**
  - ✓ **Connection to other Networks**
  - ✓ **Moving between Environments**

# Isolated LANs: Wired versus Wireless



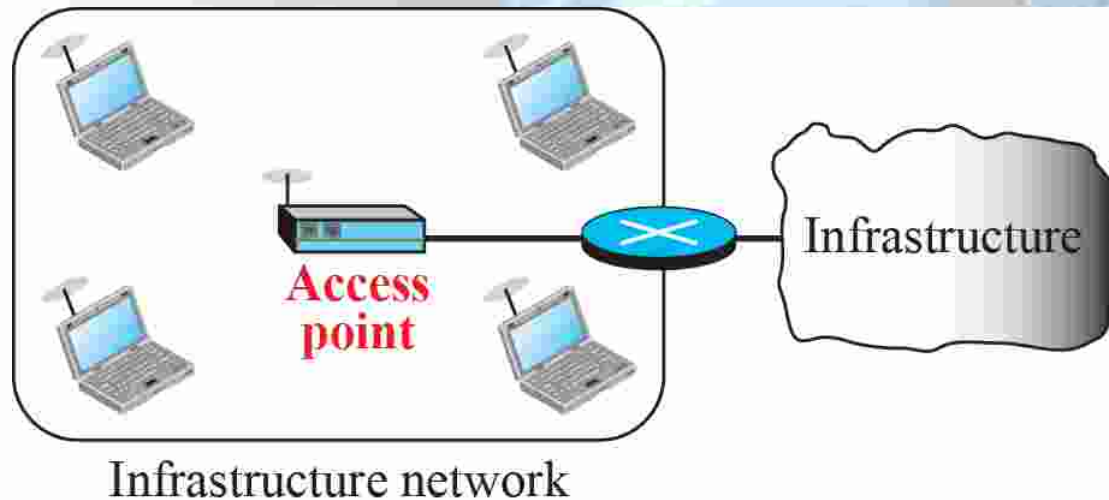
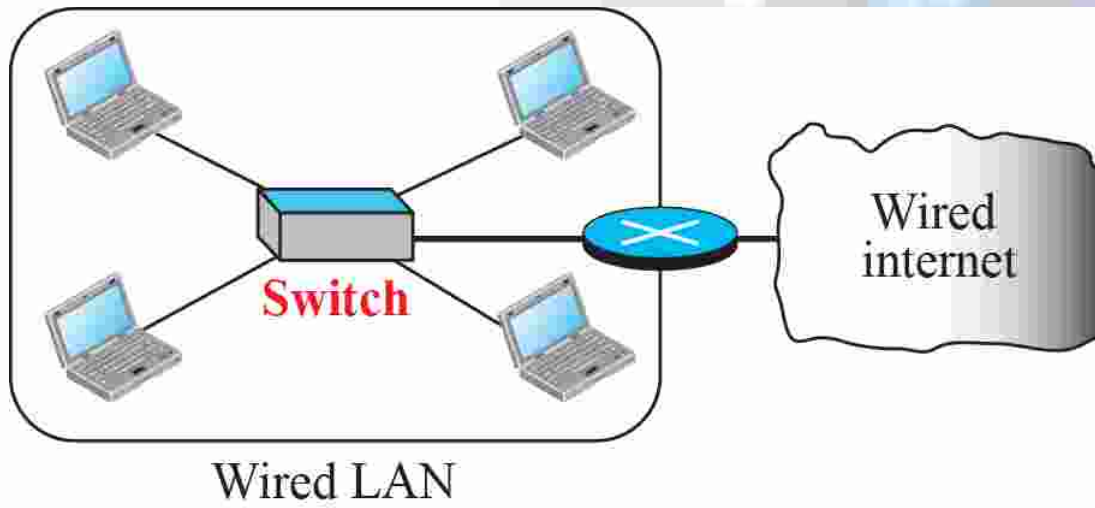
Wired



Wireless



# Connection of a Wired/Wireless LAN to other networks



# Characteristics of a Wireless LAN

- Several characteristics of wireless LANs either do not apply to wired LANs or the existence of these is negligible and can be ignored
  - ✓ Attenuation
  - ✓ Interference
  - ✓ Multipath Propagation
  - ✓ Error

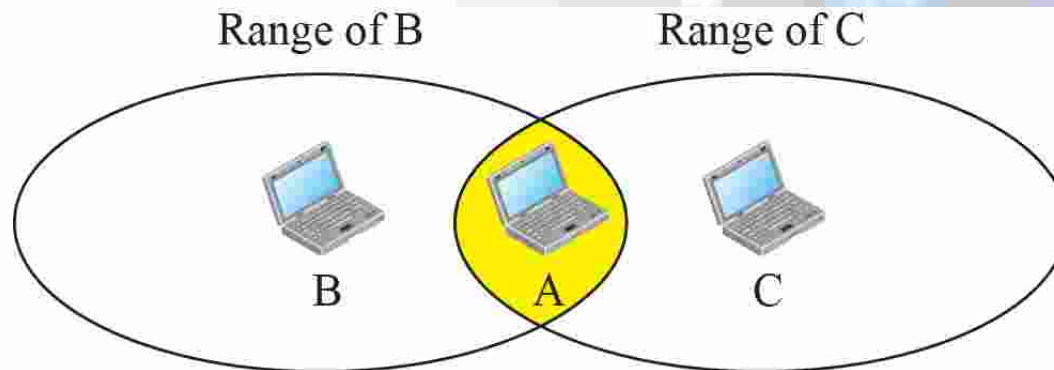
# Access Control

- Most important issue in a wireless LAN is how a wireless host can get access to the shared medium (air)
- CSMA/CD does not work in wireless LANs for three reasons:
  1. Wireless hosts don't have power to send and receive at the same time

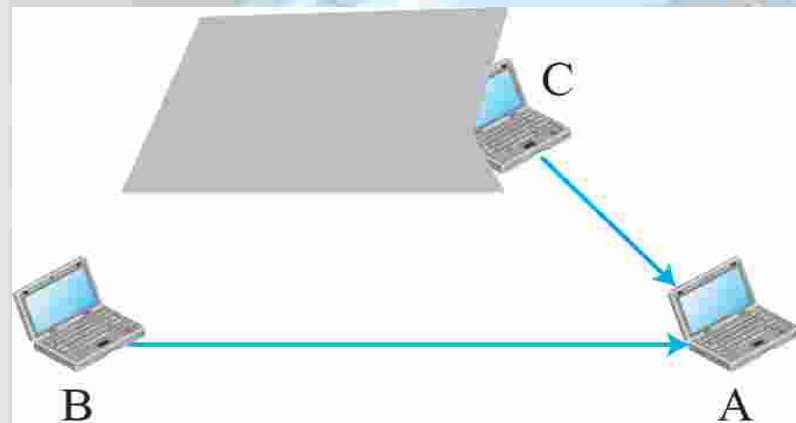
# Access Control

2. The hidden station problem prevents collision detection
3. The distance between stations can be large

# Hidden Station Problem



- a. Stations B and C are not in each other's range.



- b. Stations B and C are hidden from each other.

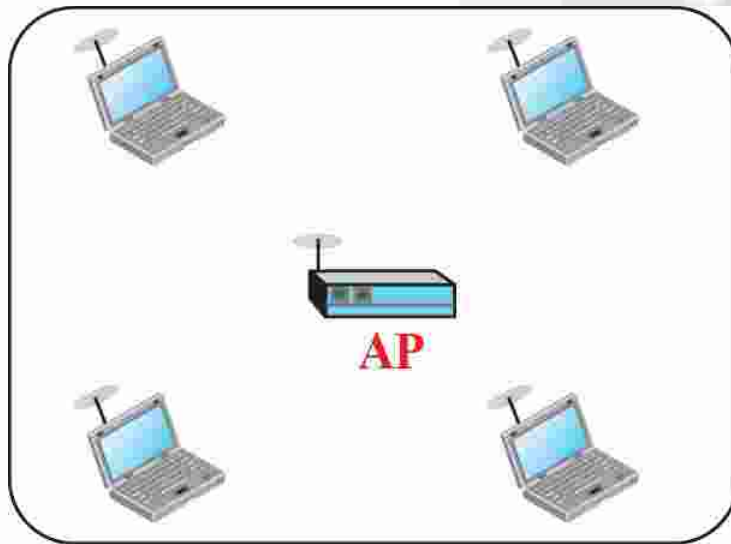
# IEEE 802.11 PROJECT

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers
- It is sometimes called Wireless Ethernet
- The term WiFi (short for wireless fidelity) as a synonym for wireless LAN (certified by WiFi alliance)

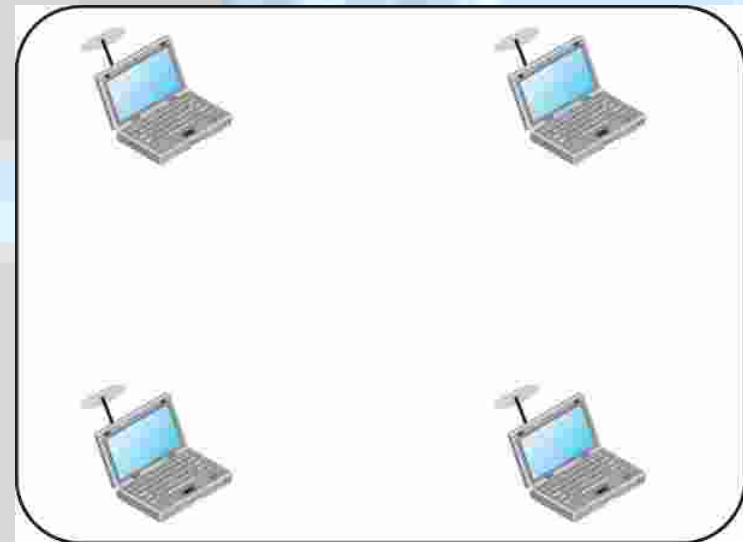
# Architecture

- The standard defines two kinds of services:
  - ✓ The basic service set (BSS); and
  - ✓ The Extended service set (ESS)

# Basic Service Sets (BSSs)



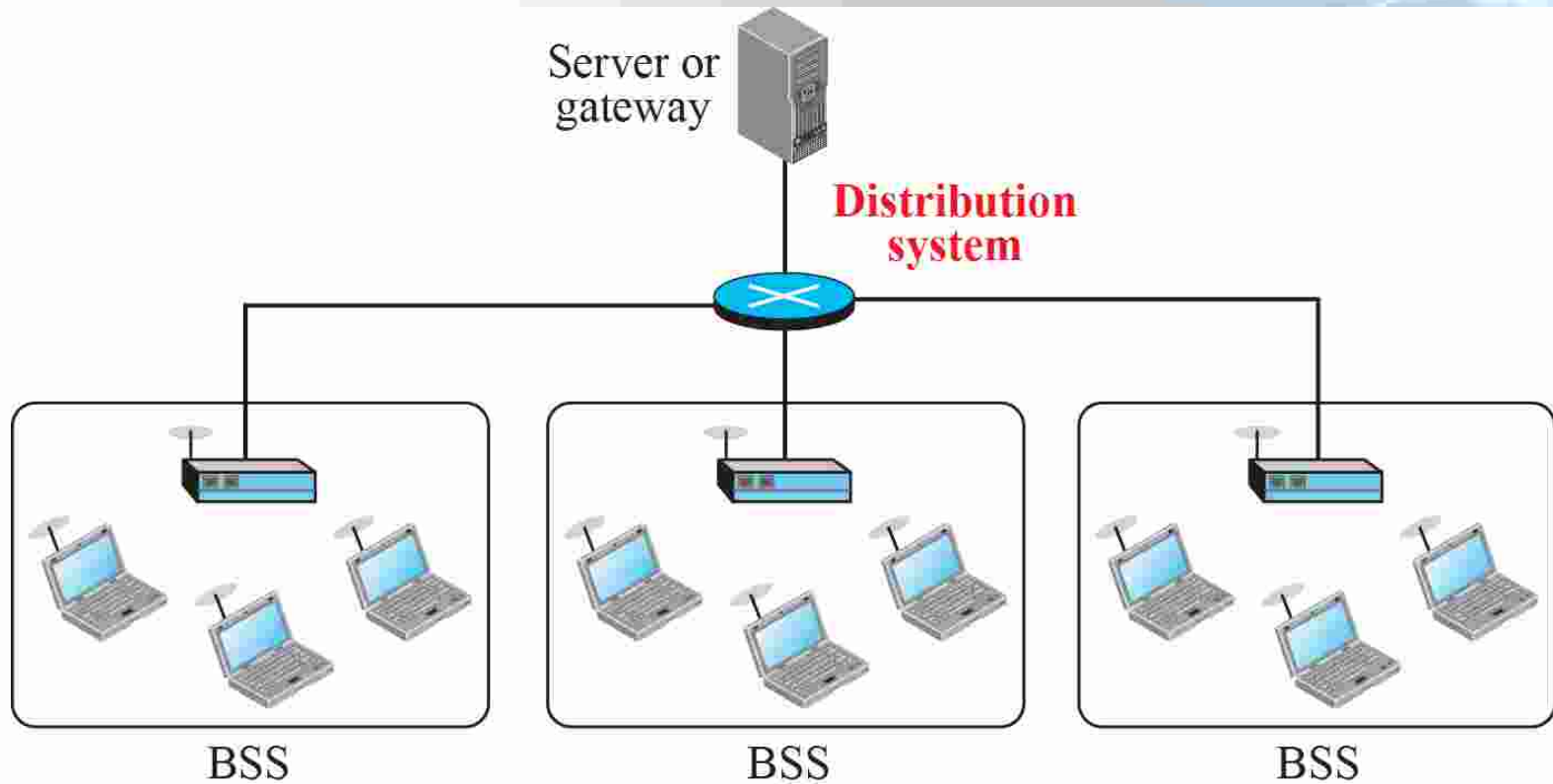
Infrastructure BSS



Ad hoc BSS



# Extended Service Set (ESS)



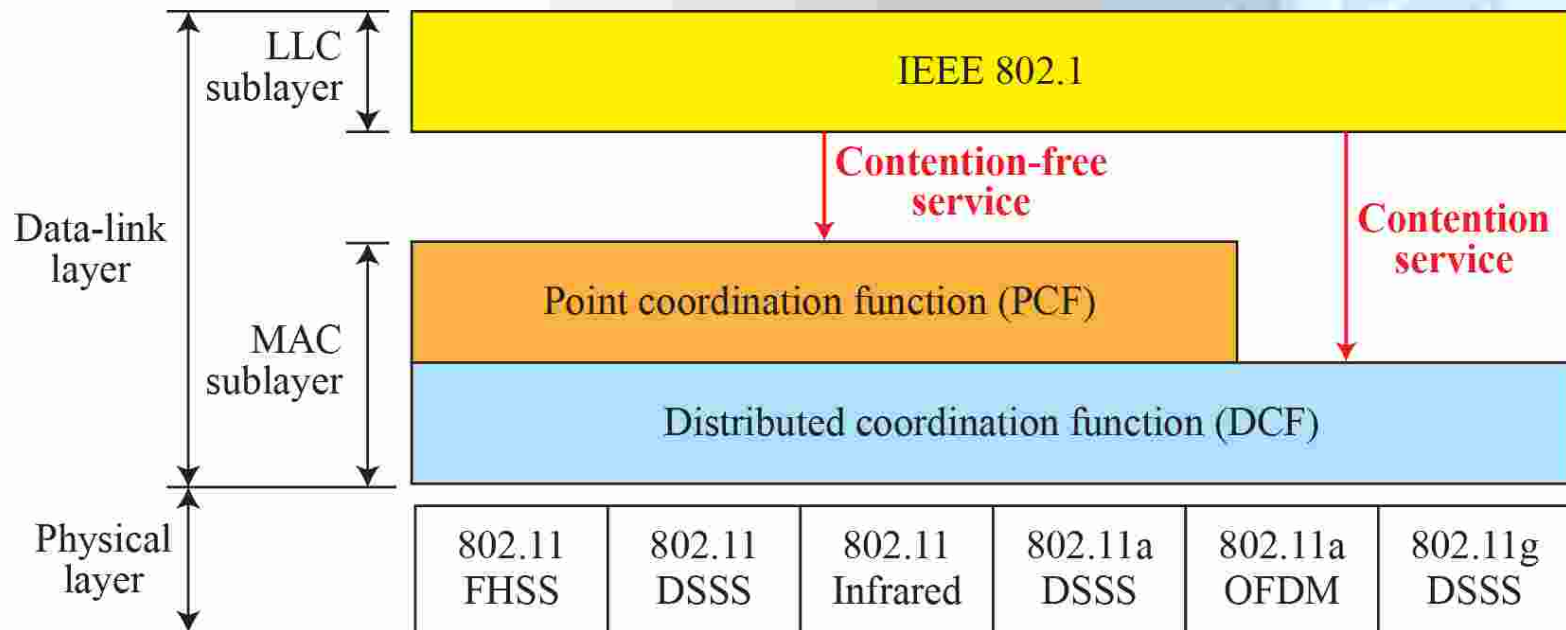
# Types of Stations

- **No-Transition Mobility**
- **BSS-Transition Mobility**
- **ESS-Transition Mobility**

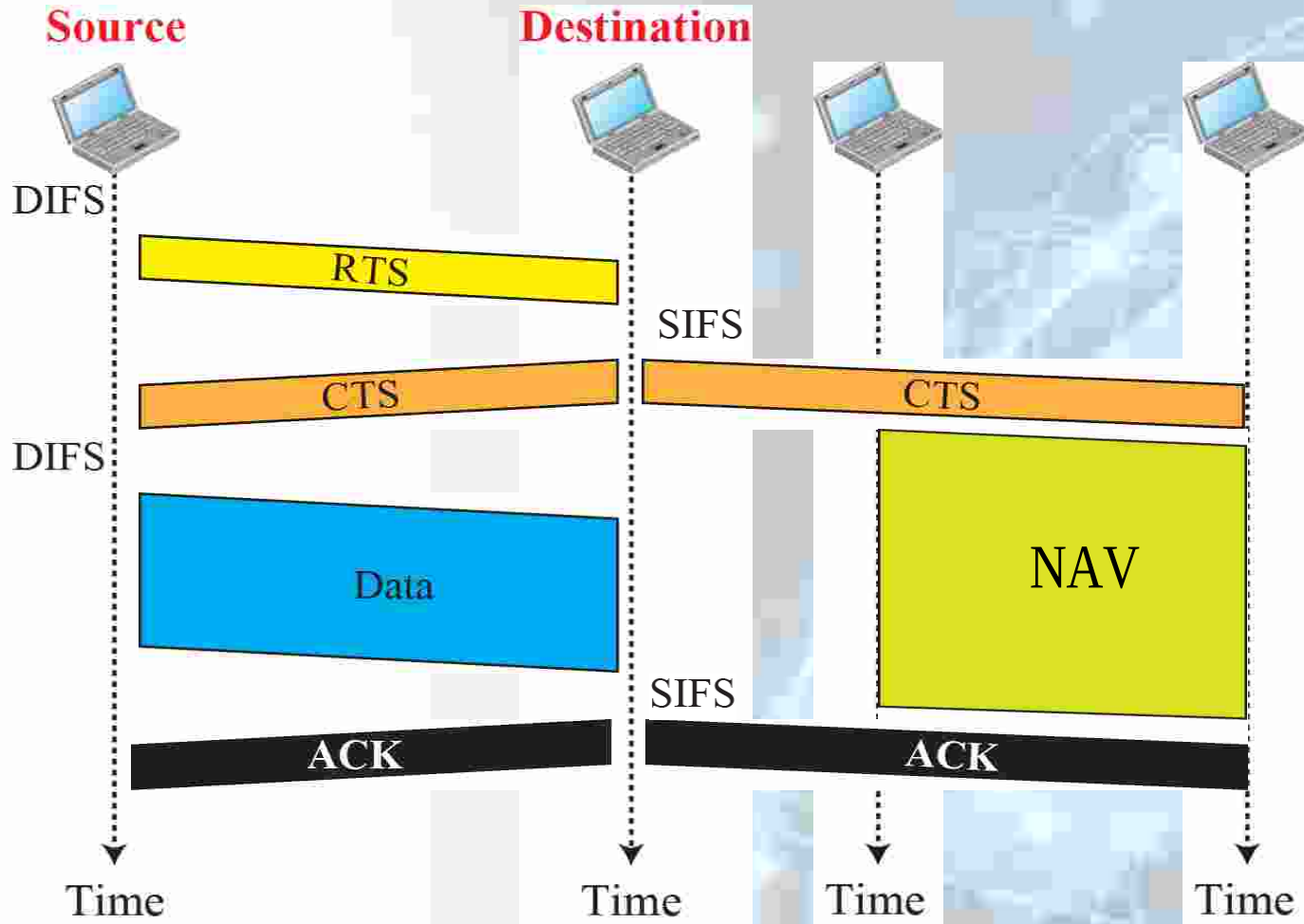
# MAC Sub-layer

- IEEE 802.11 defines two MAC sub-layers:
  - ✓ The Distributed Coordination Function (DCF) ; and
  - ✓ The Point Coordination Function (PCF)

# MAC Layers in IEEE 802.11 Standard



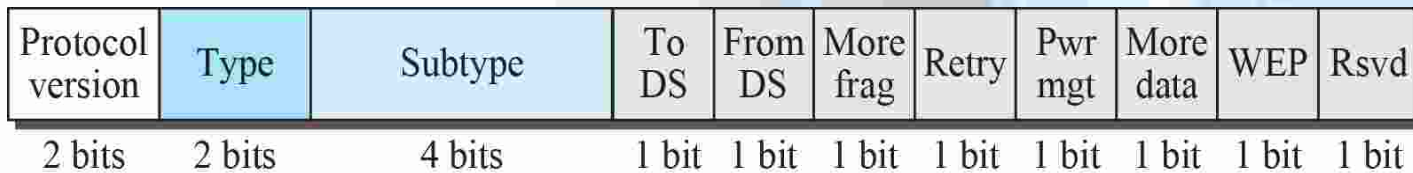
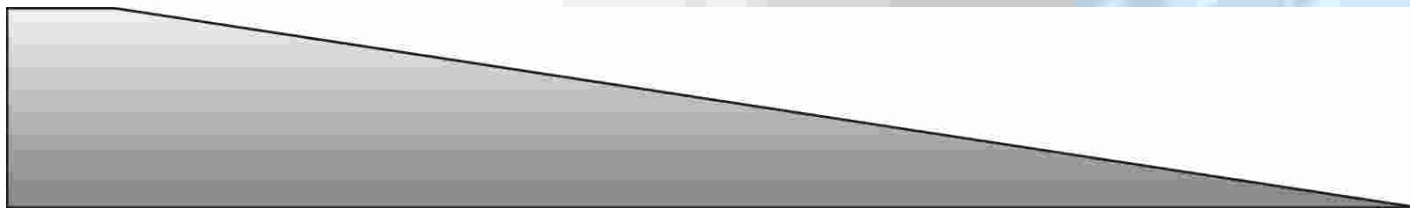
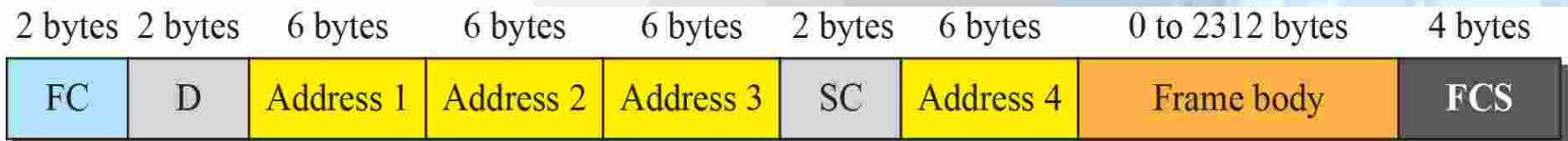
# CSMA/CA and NAV



# MAC Sub-layer

- IEEE 802.11 defines two MAC sub-layers:
  - ✓ The Distributed Coordination Function (DCF) ; and
  - ✓ The Point Coordination Function (PCF)

# Frame Format



# Subfields in FC field

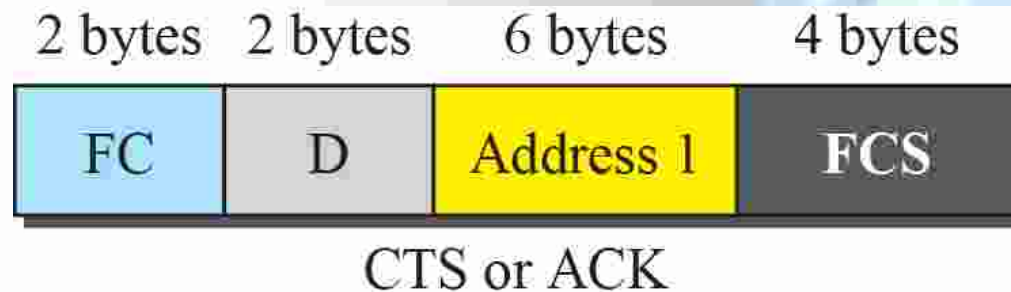
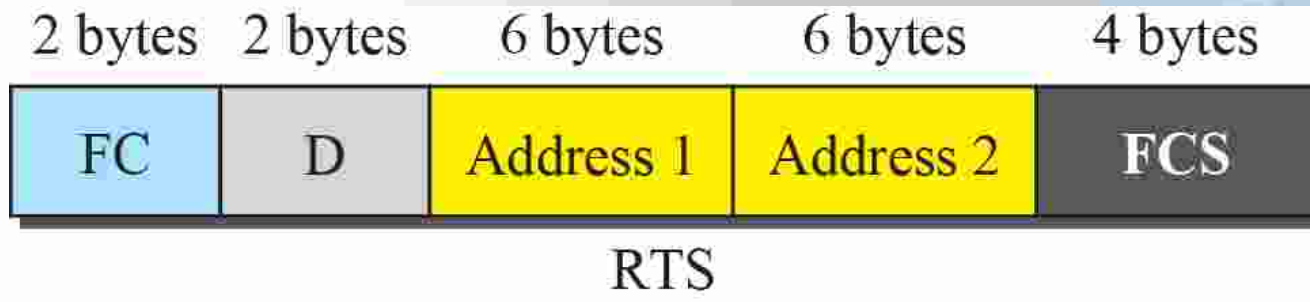
| <i>Field</i> | <i>Explanation</i>                                               |
|--------------|------------------------------------------------------------------|
| Version      | Current version is 0                                             |
| Type         | Type of information: management (00), control (01), or data (10) |
| Subtype      | Subtype of each type (see Table 6.2)                             |
| To DS        | Defined later                                                    |
| From DS      | Defined later                                                    |
| More flag    | When set to 1, means more fragments                              |
| Retry        | When set to 1, means retransmitted frame                         |
| Pwr mgt      | When set to 1, means station is in power management mode         |
| More data    | When set to 1, means station has more data to send               |
| WEP          | Wired equivalent privacy (encryption implemented)                |
| Rsvd         | Reserved                                                         |



# Frame Types

- **Management Frames**
- **Control Frames**
- **Data Frames**

# Control Frames



# Values of Subfields in Control Frames

| <i>Subtype</i> | <i>Meaning</i>        |
|----------------|-----------------------|
| 1011           | Request to send (RTS) |
| 1100           | Clear to send (CTS)   |
| 1101           | Acknowledgment (ACK)  |

# Physical Layer

- All physical implementations, except the infrared, operate in the industrial, scientific, and medical (ISM) band, which defines 3 unlicensed bands in 3 ranges:
  - ✓ 902–928 MHz
  - ✓ 2.400–4.835 GHz
  - ✓ 5.725–5.850 GHz

# Specifications

| <i>IEEE</i> | <i>Technique</i> | <i>Band</i>     | <i>Modulation</i> | <i>Rate (Mbps)</i> |
|-------------|------------------|-----------------|-------------------|--------------------|
| 802.11      | FHSS             | 2.400–4.835 GHz | FSK               | 1 and 2            |
|             | DSSS             | 2.400–4.835 GHz | PSK               | 1 and 2            |
|             | None             | Infrared        | PPM               | 1 and 2            |
| 802.11a     | OFDM             | 5.725–5.850 GHz | PSK or QAM        | 6 to 54            |
| 802.11b     | DSSS             | 2.400–4.835 GHz | PSK               | 5.5 and 11         |
| 802.11g     | OFDM             | 2.400–4.835 GHz | Different         | 22 and 54          |
| 802.11n     | OFDM             | 5.725–5.850 GHz | Different         | 600                |

# BLUETOOTH

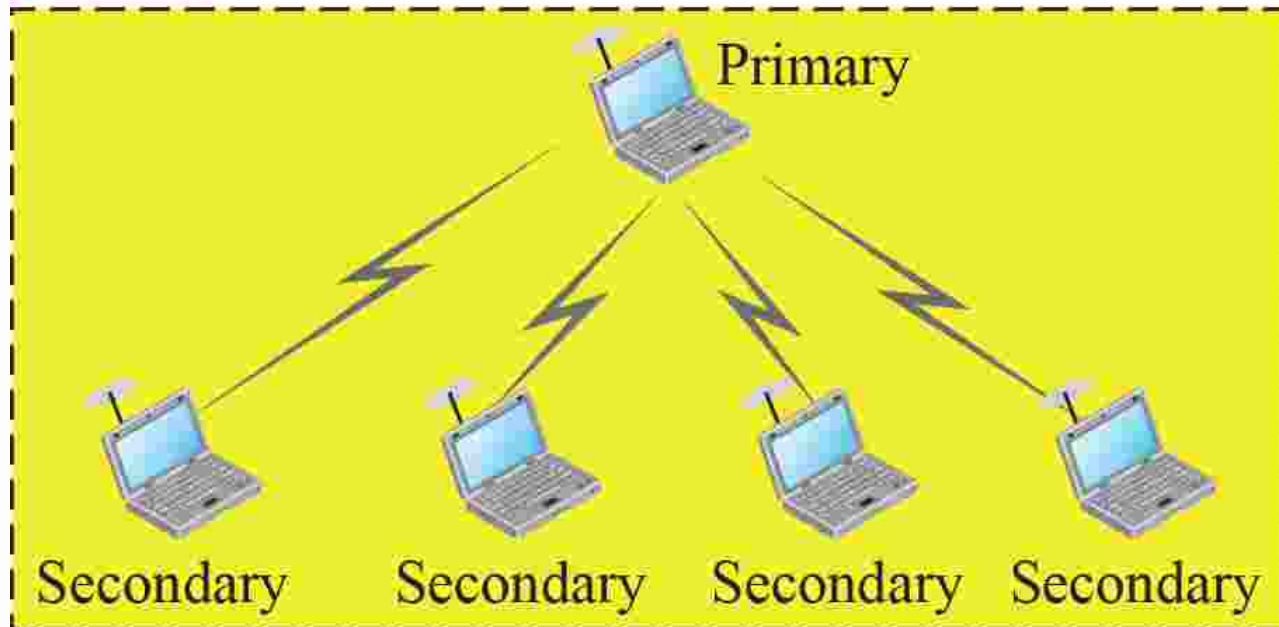
- Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other
- A Bluetooth LAN is an ad hoc network
- The devices, sometimes called gadgets, find each other and make a network called a Piconet

# Architecture

- Bluetooth defines two types of networks:
  - ✓ Piconet
  - ✓ Scatternet

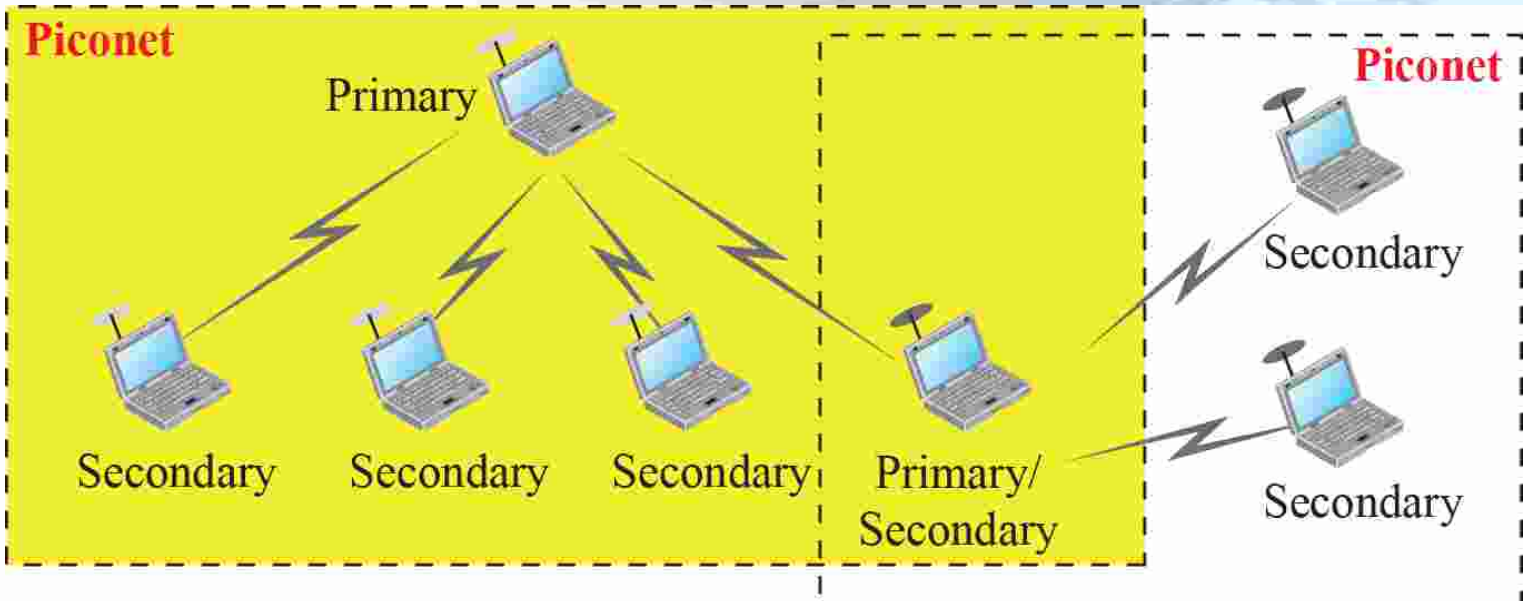
# Piconet

## Piconet





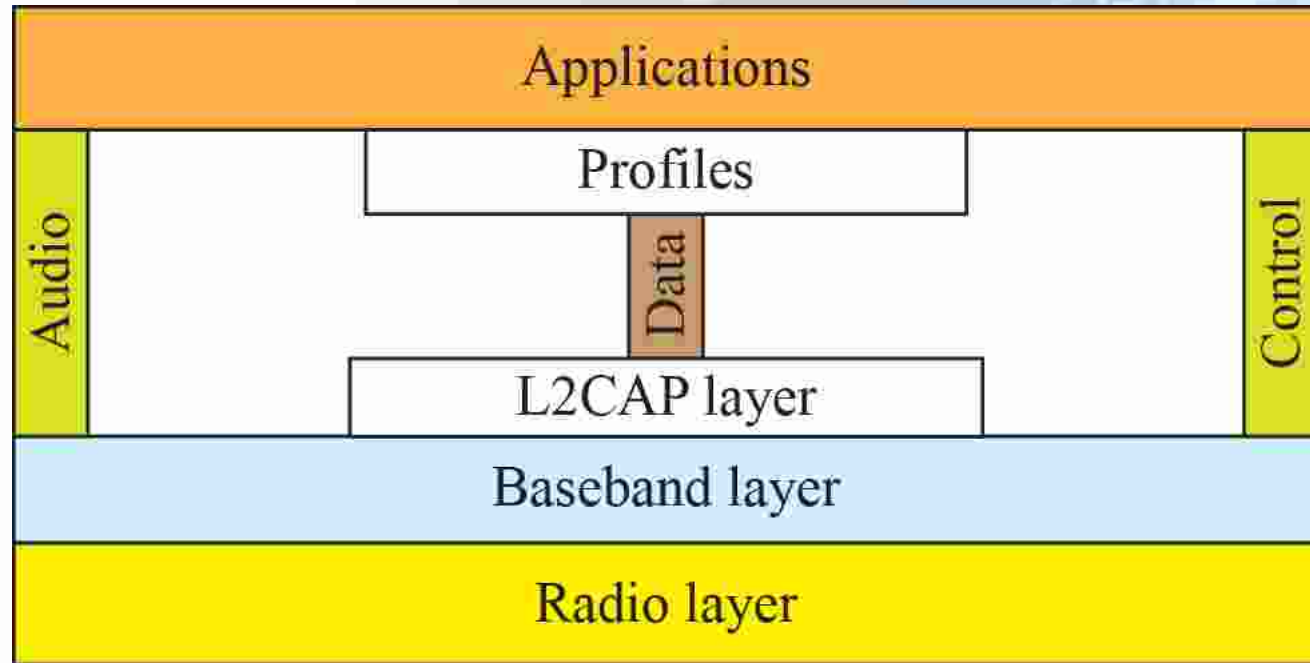
# Scatternet



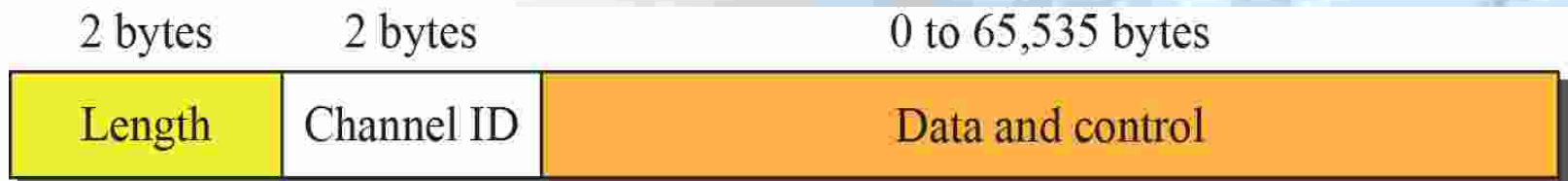
# Bluetooth Layers

- Bluetooth uses several layers that do not exactly match those of the Internet model we have defined in this book

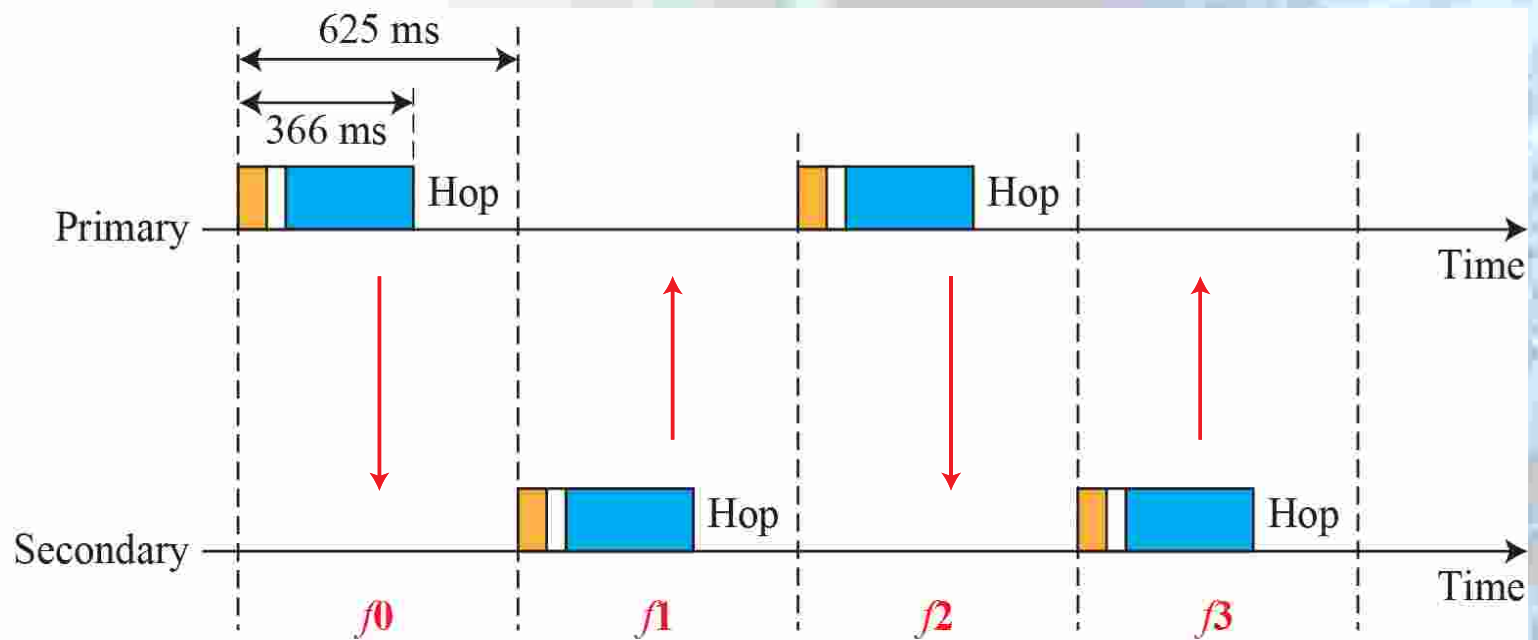
# Bluetooth Layers



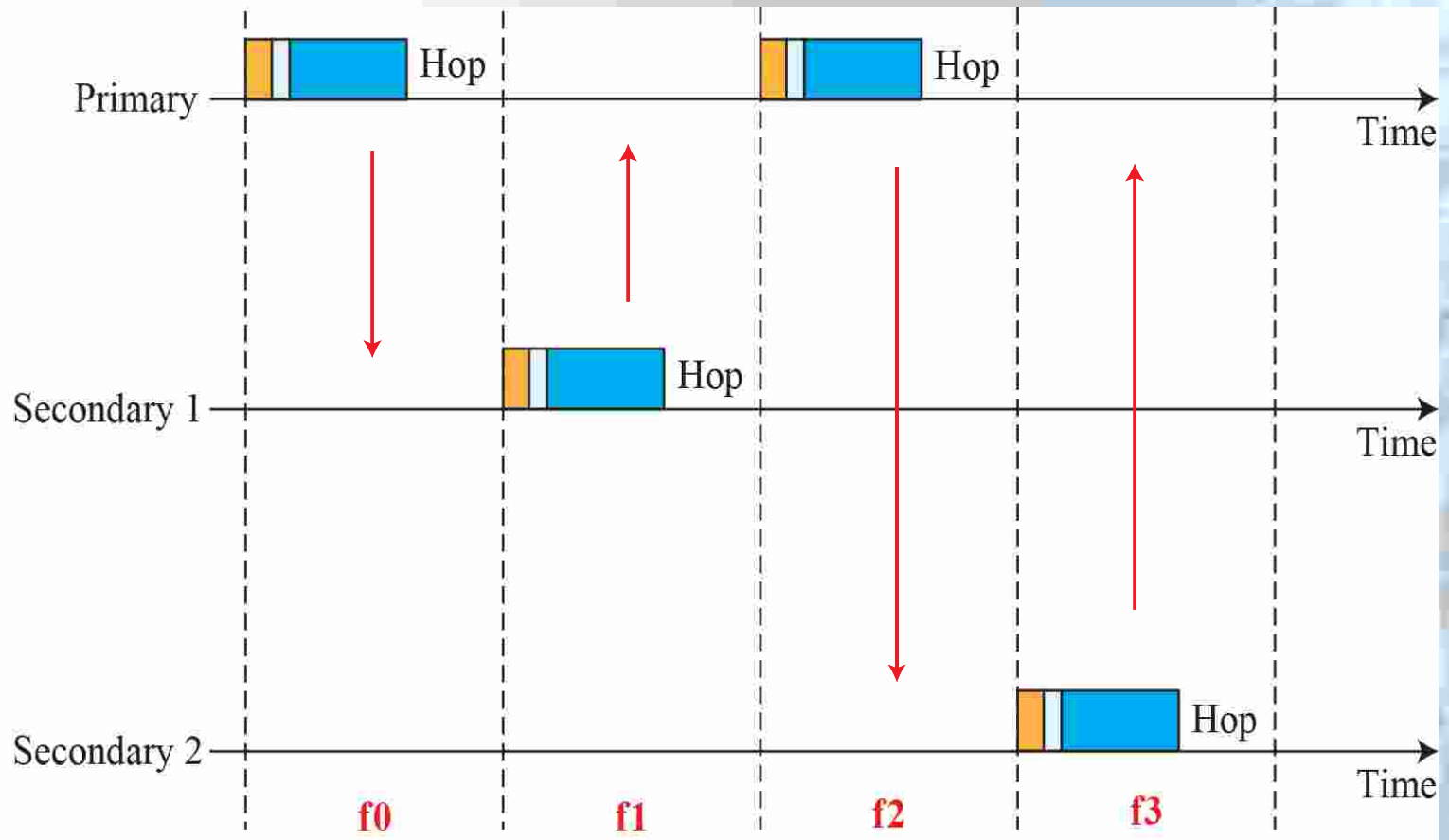
# L2CAP Data Packet Format



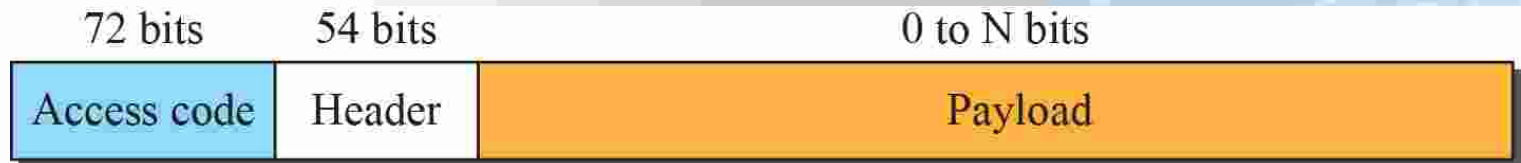
# Single-Secondary Communication



# Multiple-Secondary Communication



# Frame Format Types



3 bits    4 bits    1   1   1    8 bits

This 18-bit part is repeated 3 times.

**N = 240 for 1-slot frame**

**N = 1490 for 3-slot frame**

**N = 2740 for 5-slot frame**

# Bluetooth

- Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other
- A Bluetooth LAN is an ad hoc network
- The devices, sometimes called gadgets, find each other and make a network called a Piconet



# Bluetooth

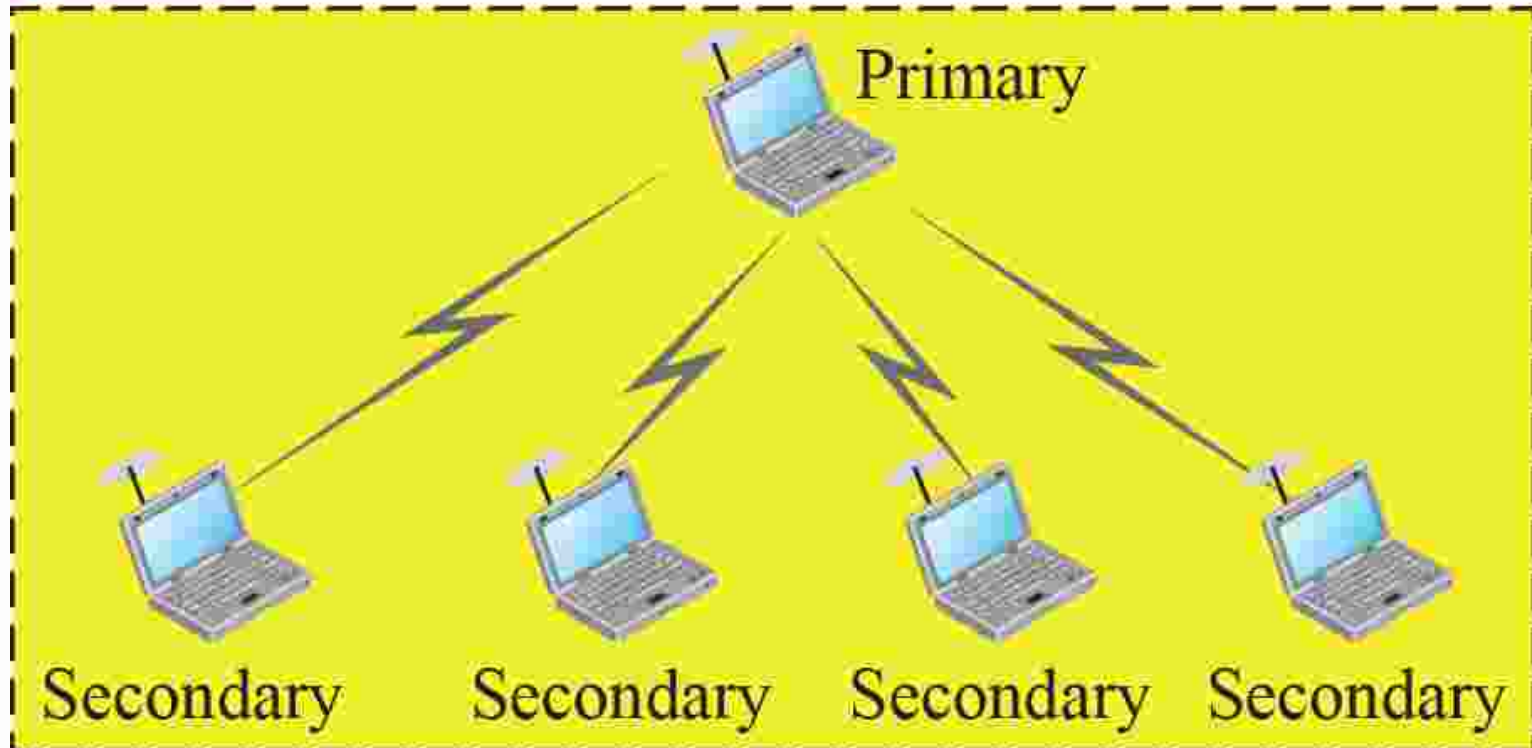
- Bluetooth technology is the implementation of a protocol defined by the IEEE 802.15 standard
- The standard defines a wireless Personal-Area Network (PAN) operable in an area the size of a room or a hall

# Architecture

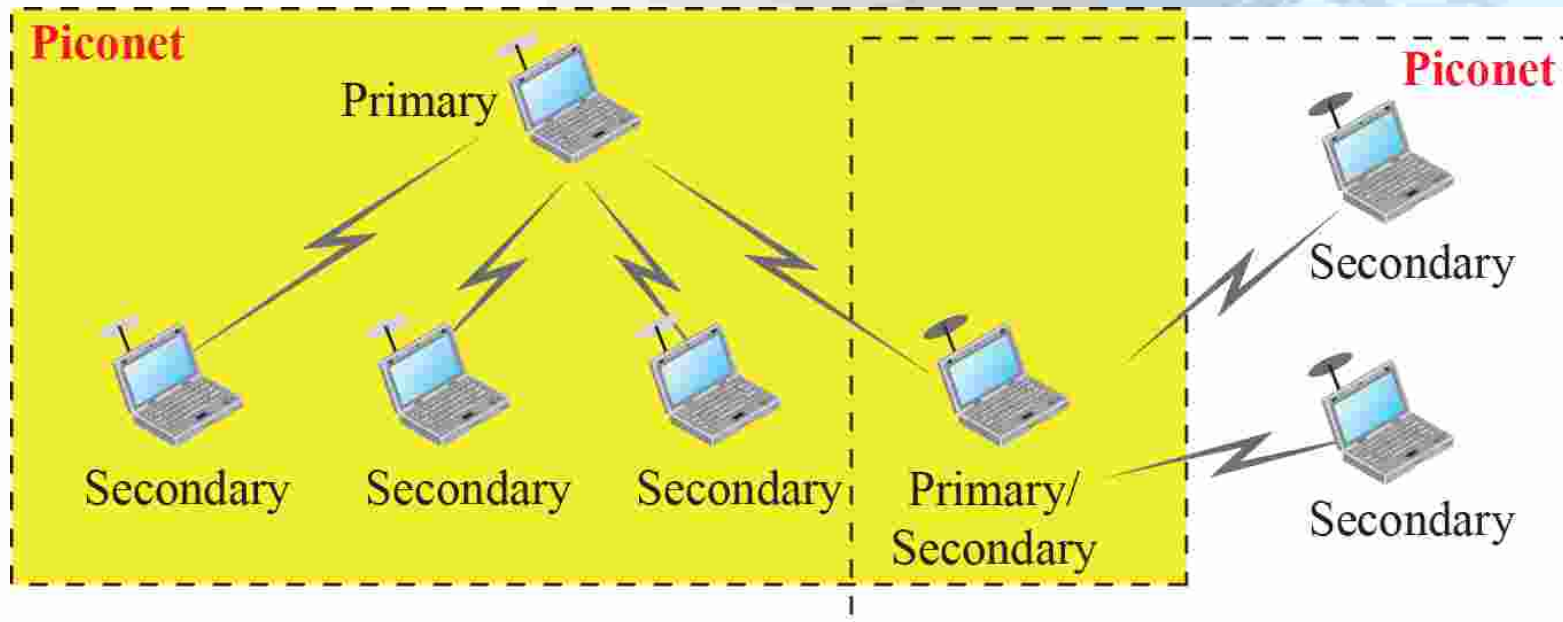
- Bluetooth defines two types of networks:
  - ✓ Piconet
  - ✓ Scatternet

# Piconet

## Piconet



# Scatternet



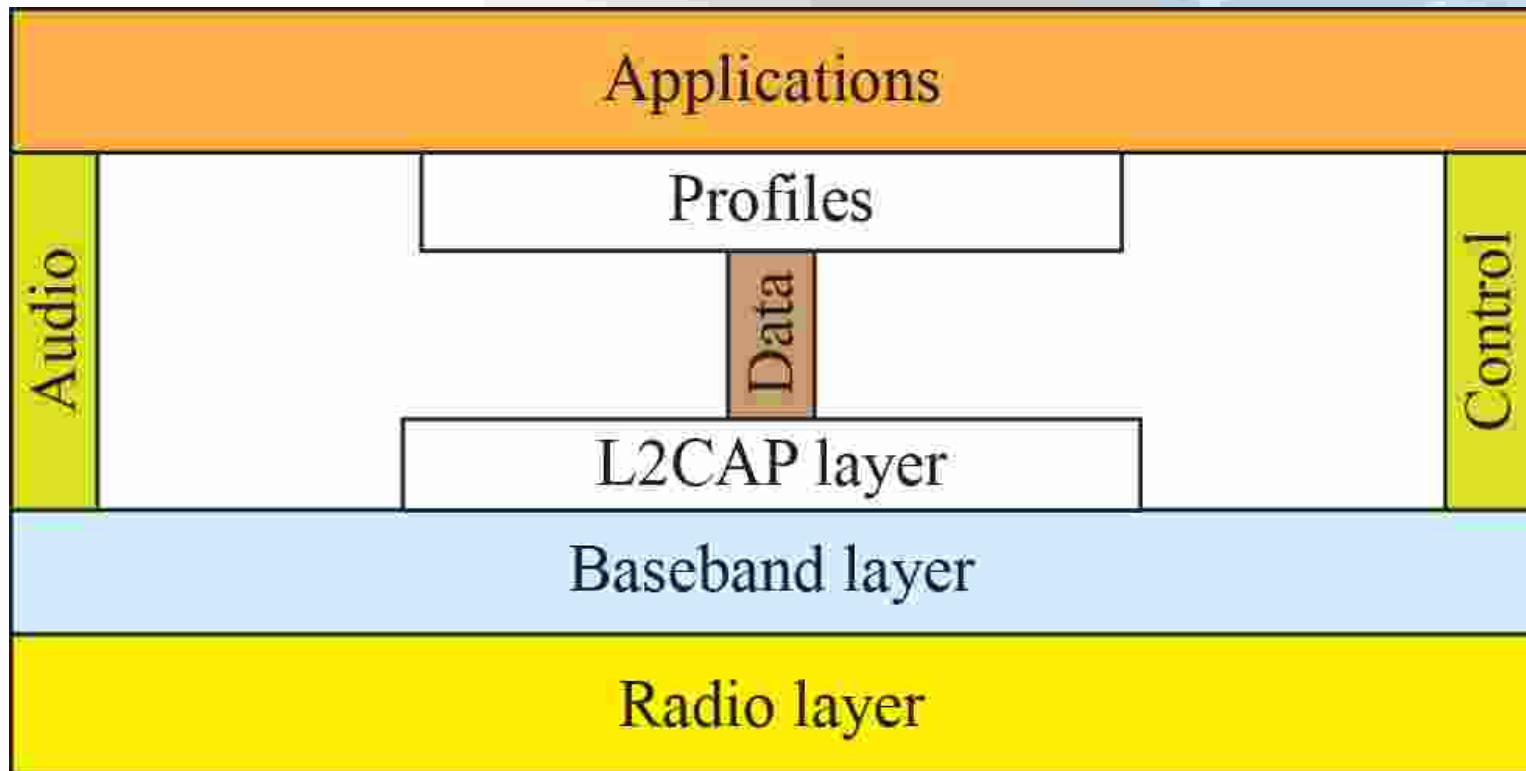
# Bluetooth Devices

- A Bluetooth device has a built-in short-range radio transmitter
- The current data rate is 1 Mbps with a 2.4-GHz bandwidth
- This means that there is a possibility of interference between the IEEE 802.11b wireless LANs and Bluetooth LANs

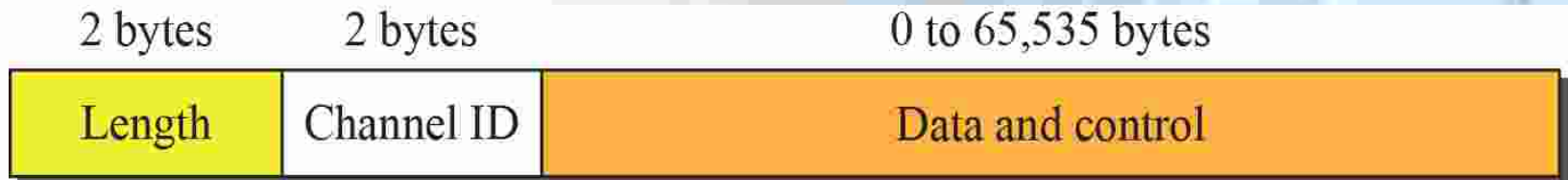
# Bluetooth Layers

- Bluetooth uses several layers that do not exactly match those of the Internet model we have defined already

# Bluetooth Layers

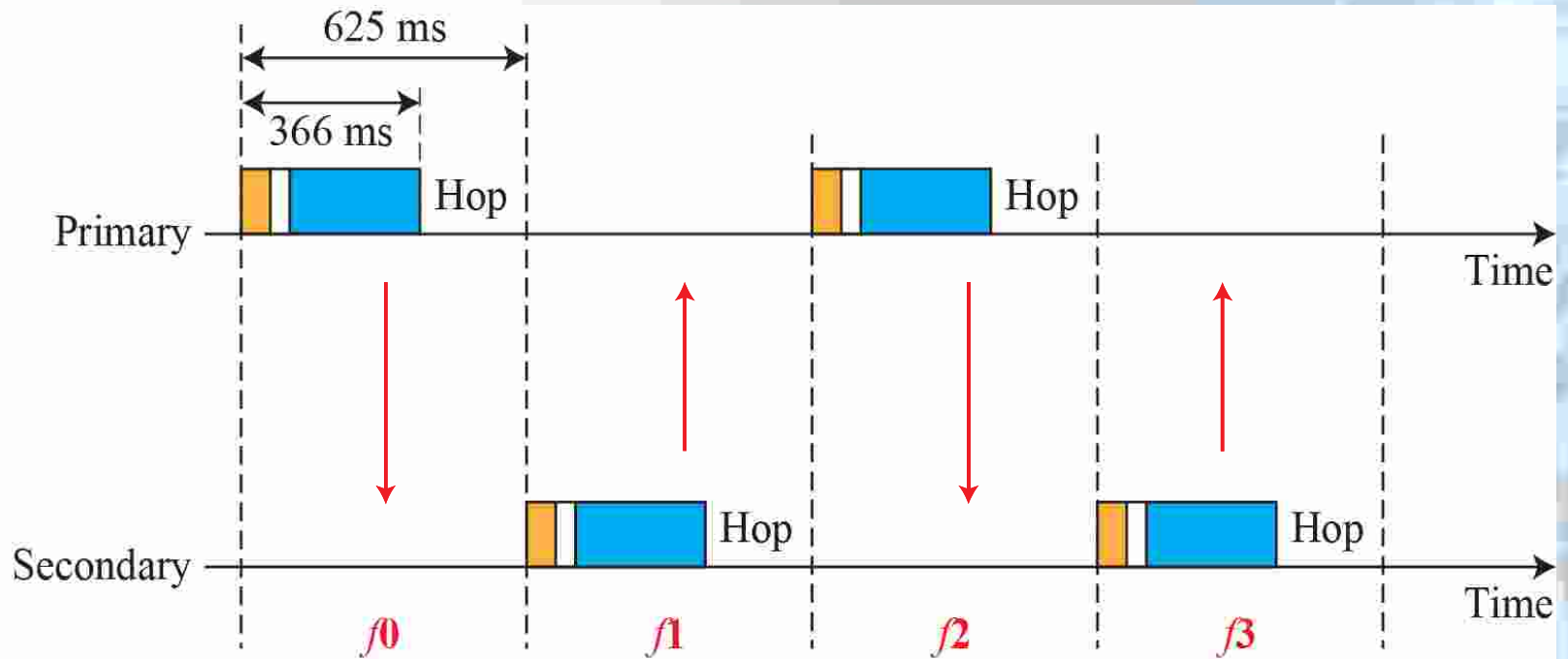


# L2CAP Data Packet Format

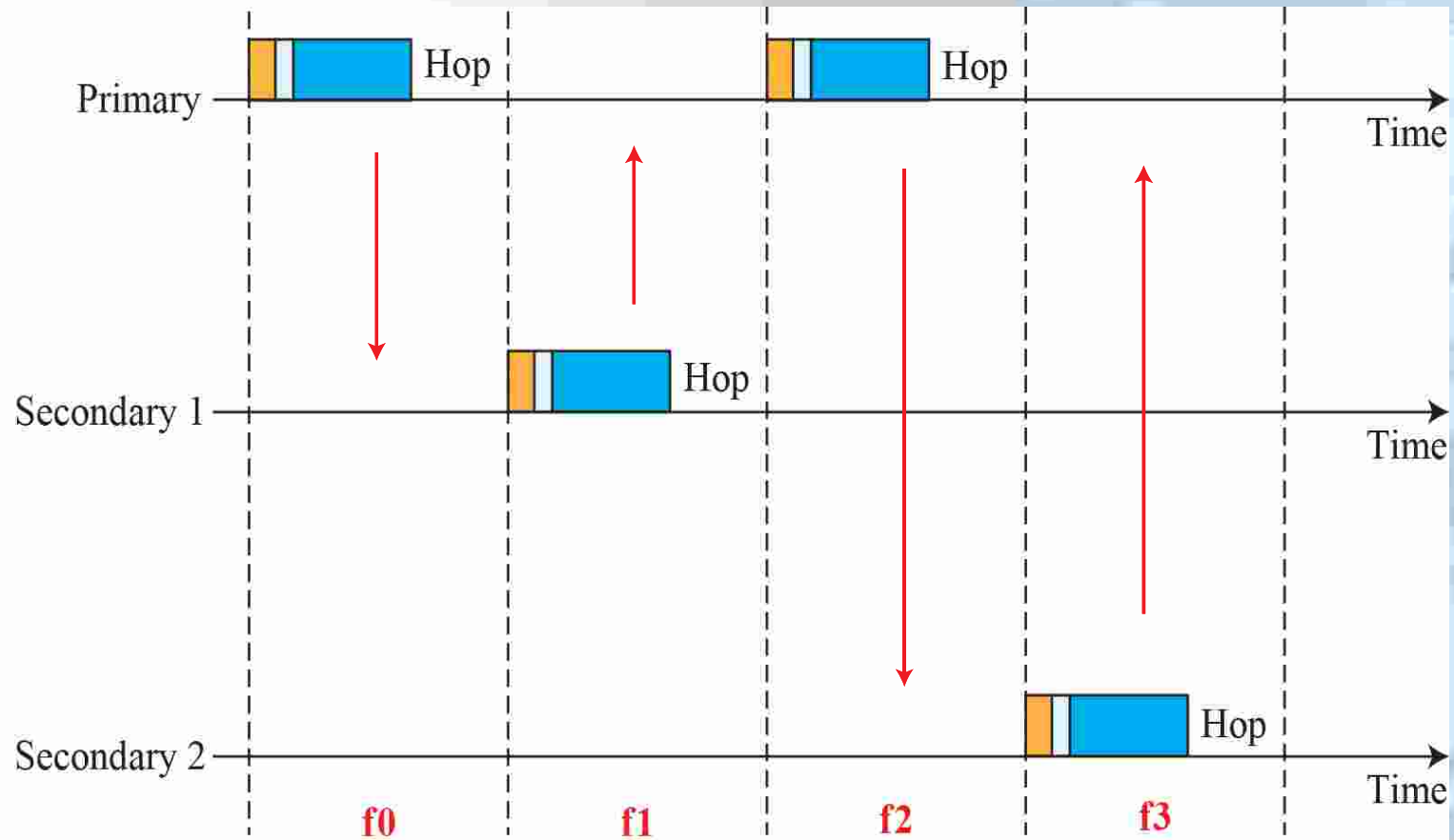




# Single-Secondary Communication



# Multiple-Secondary Communication



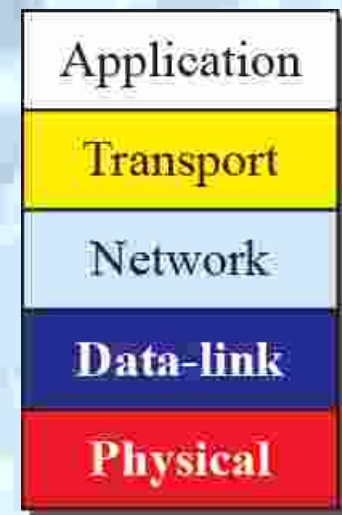
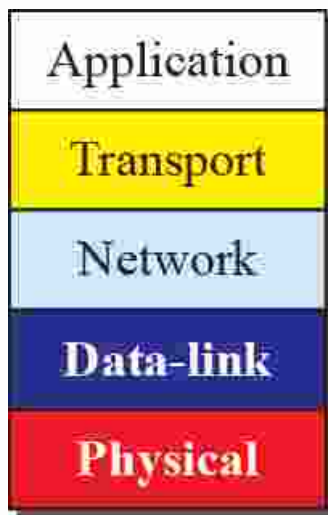
# Connecting Devices

- Hosts and networks do not normally operate in isolation
- Connecting devices connect hosts together to make a network or connect networks together to make an internet
- Connecting devices can operate in different layers of the Internet model

# Connecting Devices

- Three kinds of connecting devices:
  - ✓ Hubs
  - ✓ Link-layer switches
  - ✓ Routers

# Three Categories of Connecting Devices



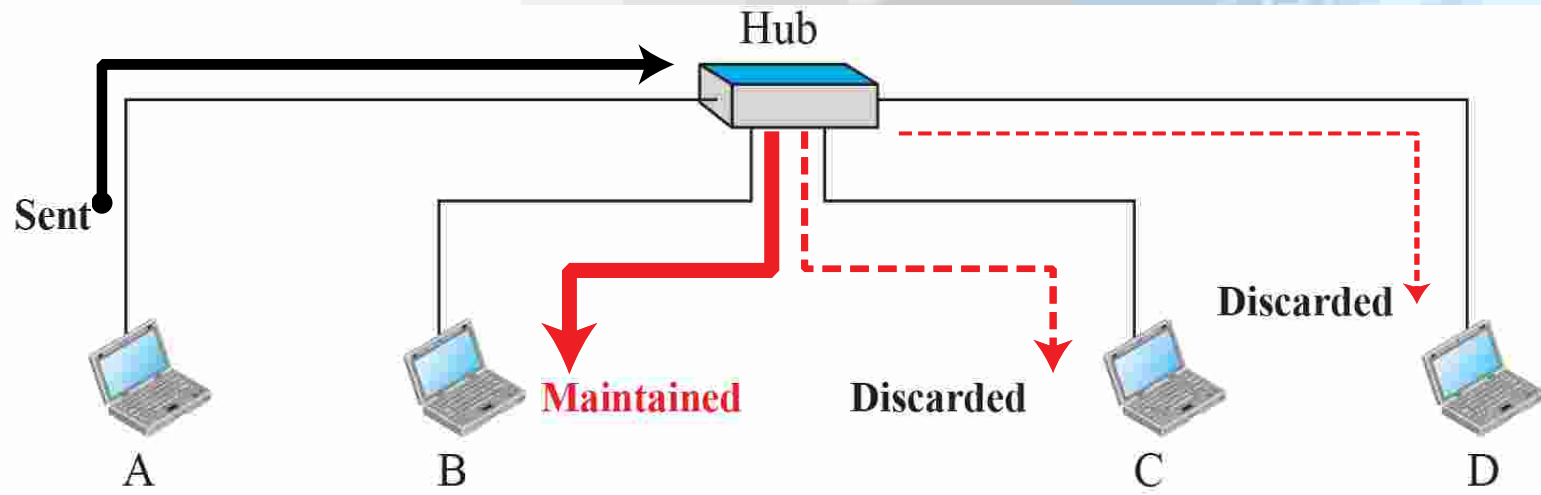
# Hubs

- Hub is a device that operates only in the physical layer
- Signals that carry information within a network can travel a fixed distance before attenuation impacts the data
- A hub (repeater) receives a signal and, before it becomes too weak or corrupted, regenerates it

# Hubs

- **Hub is a device that operates only in the physical layer**

# Hub





# Link-Layer Switches

- A link-layer switch (or switch) operates in both the physical and the data-link layers
- As a physical-layer device, it regenerates the signal it receives
- As a link-layer device, the link-layer switch can check the MAC addresses (source and destination) contained in the frame

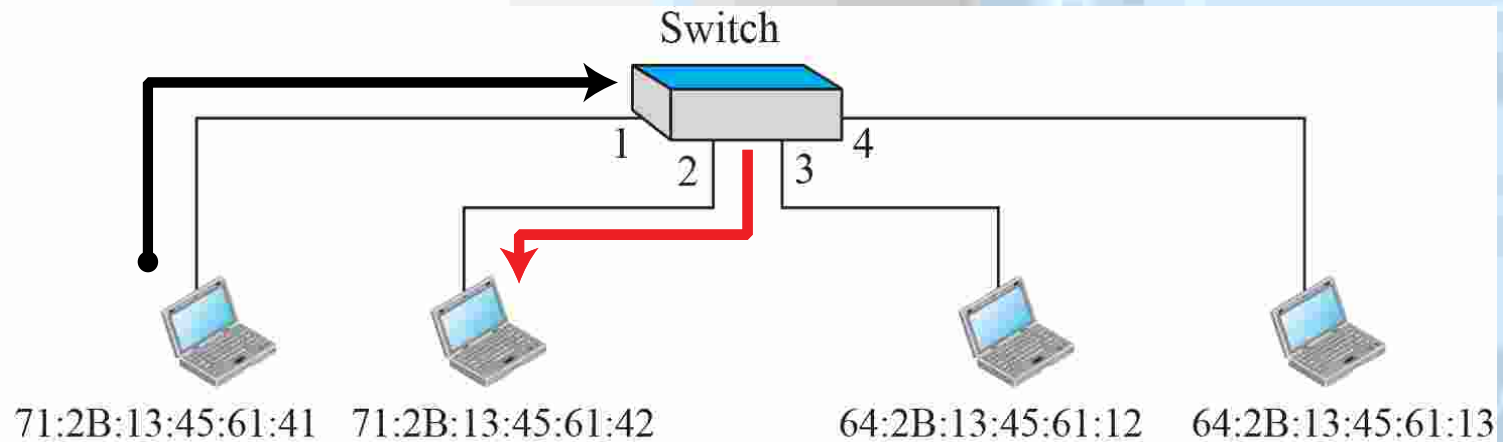
# Switch versus Hub

- Switch has the 'Filtering' capability
- Unlike hub, a switch can check the destination address of a frame and decide on outgoing port
- Switch eliminates collisions and does not require carrier sensing
- Switches connect heterogeneous devices

# Link-Layer Switch

Switching table

| Address           | Port |
|-------------------|------|
| 71:2B:13:45:61:41 | 1    |
| 71:2B:13:45:61:42 | 2    |
| 64:2B:13:45:61:12 | 3    |
| 64:2B:13:45:61:13 | 4    |



# Link-Layer Switches

- A link-layer switch (or switch) operates in both the physical and the data-link layers

# Learning Switch

| Address | Port |
|---------|------|
|---------|------|

a. Original

| Address           | Port |
|-------------------|------|
| 71:2B:13:45:61:41 | 1    |

b. After A sends a frame to D

| Address           | Port |
|-------------------|------|
| 71:2B:13:45:61:41 | 1    |
| 64:2B:13:45:61:13 | 4    |

c. After D sends a frame to B

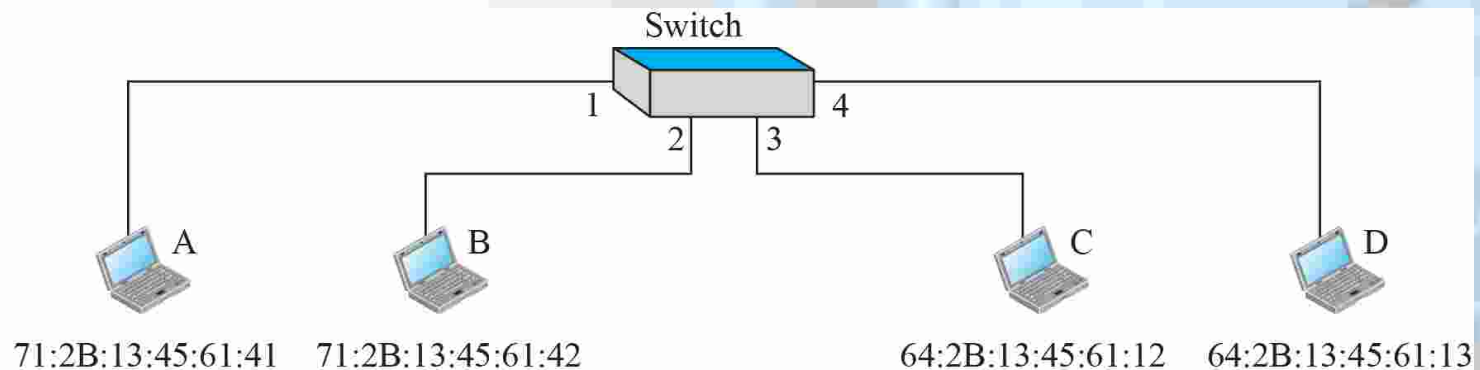
| Address           | Port |
|-------------------|------|
| 71:2B:13:45:61:41 | 1    |
| 64:2B:13:45:61:13 | 4    |
| 71:2B:13:45:61:42 | 2    |

d. After B sends a frame to A

| Address           | Port |
|-------------------|------|
| 71:2B:13:45:61:41 | 1    |
| 64:2B:13:45:61:13 | 4    |
| 71:2B:13:45:61:42 | 2    |
| 64:2B:13:45:61:12 | 3    |

e. After C sends a frame to D

## Gradual building of Table

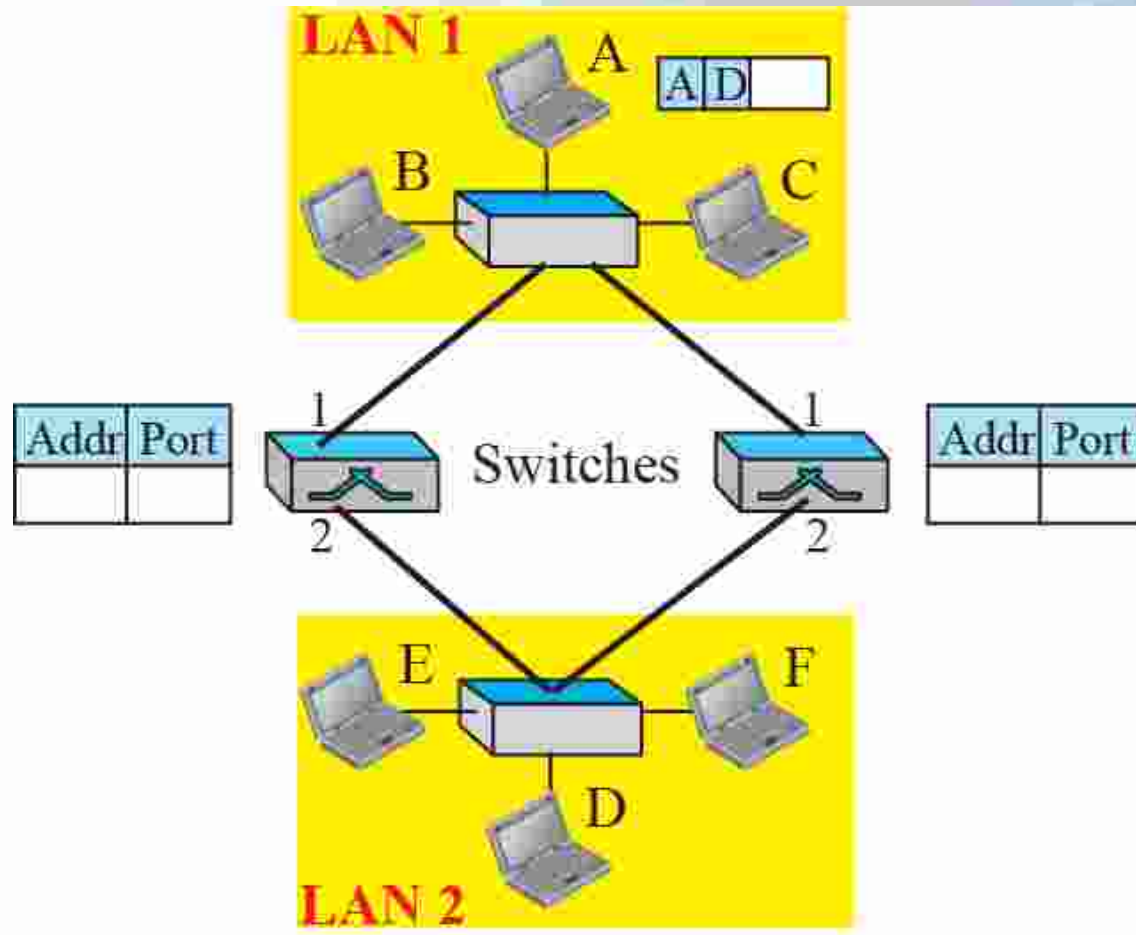


# Loop Problem in a Switch

- Redundant switches create Loops in the system
- Created when two or more broadcasting LANs are connected by more than one switch

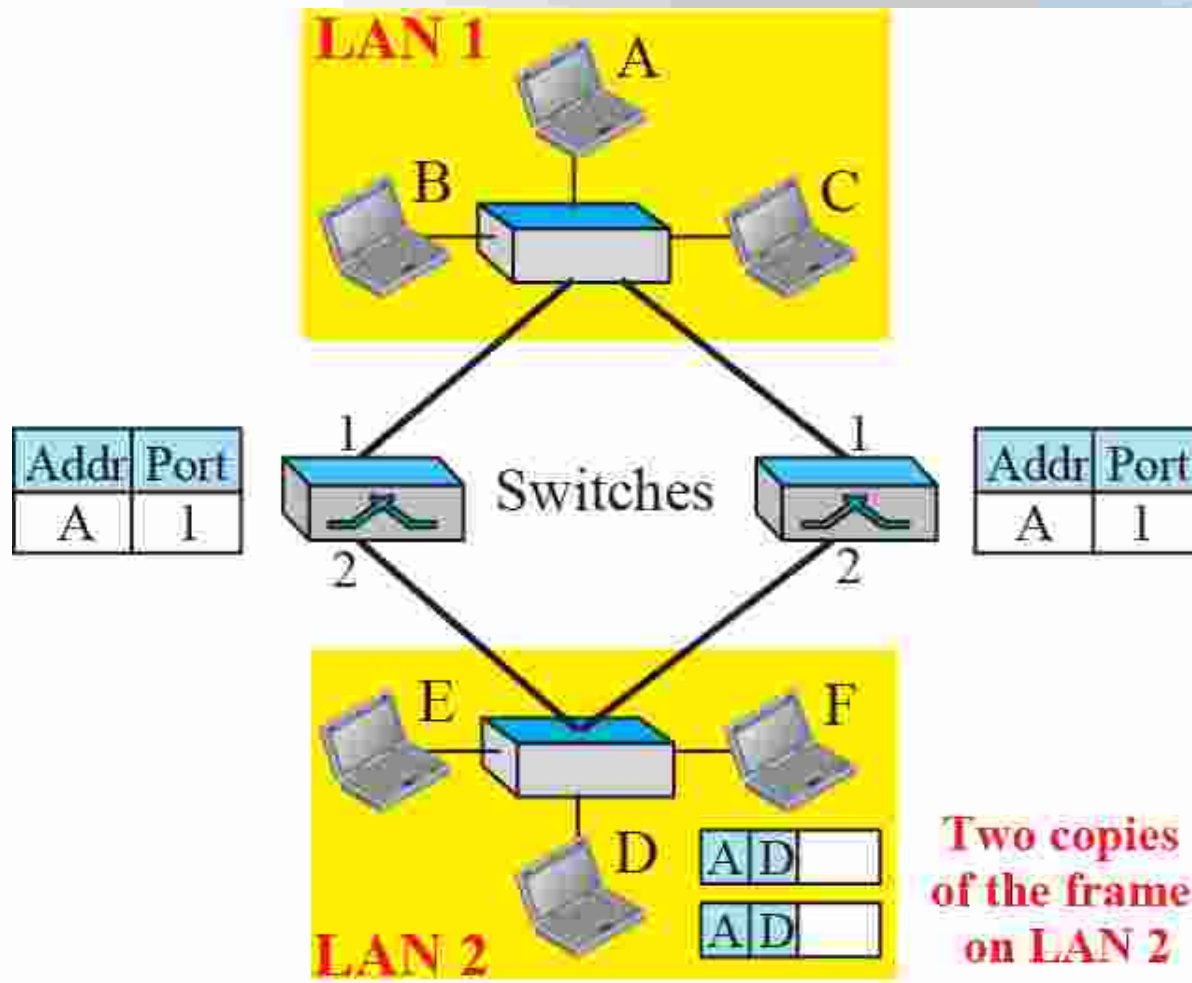
# Loop Problem in a Learning Switch (Part a)

a. Station A sends a frame to station D



# Loop Problem in a Learning Switch (Part b)

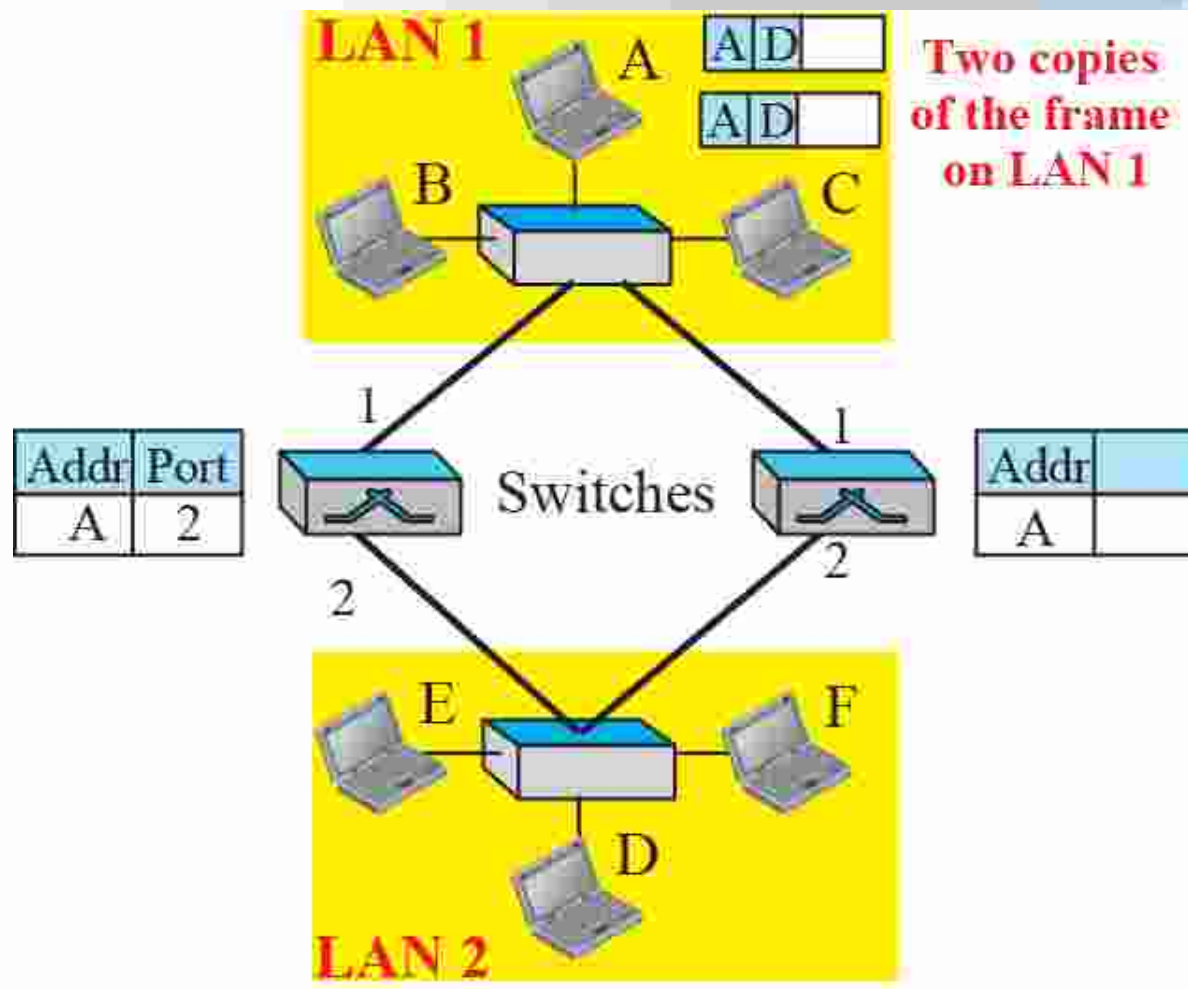
**b. Both switches forward the frame**





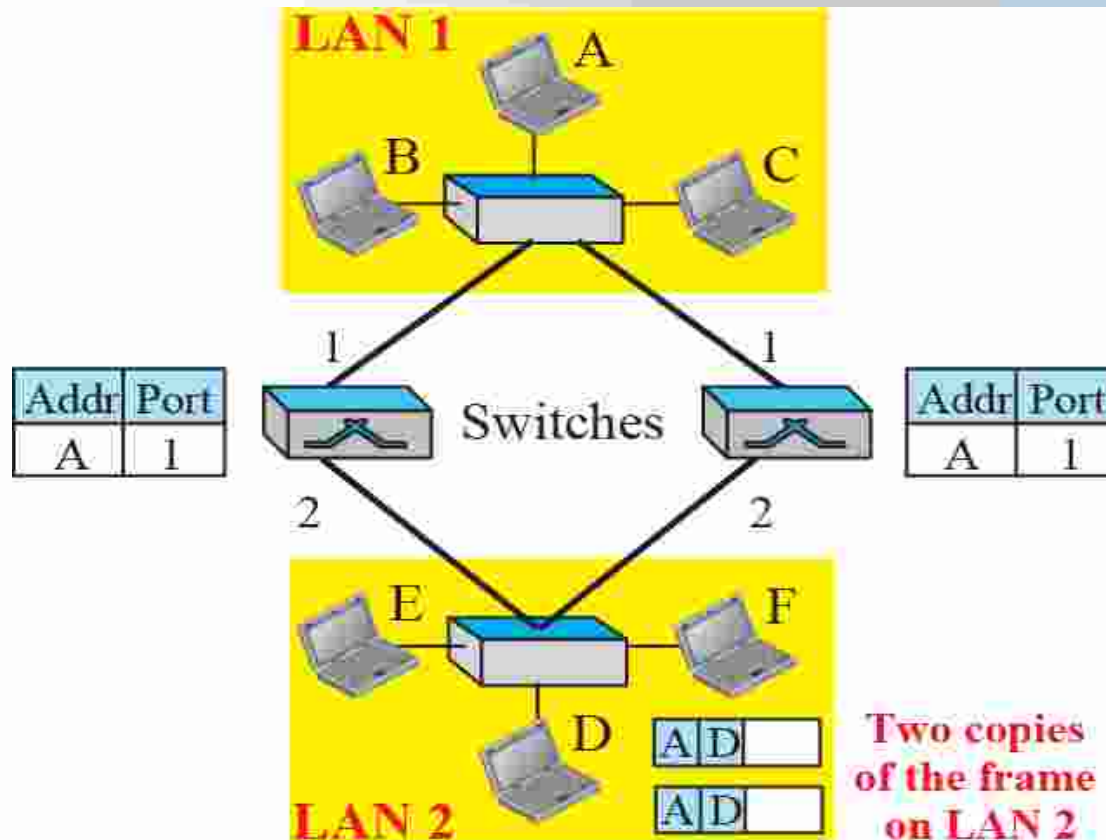
# Loop Problem in a Learning Switch (Part c)

## c. Both switches forward the frame



# Loop Problem in a Learning Switch (part d)

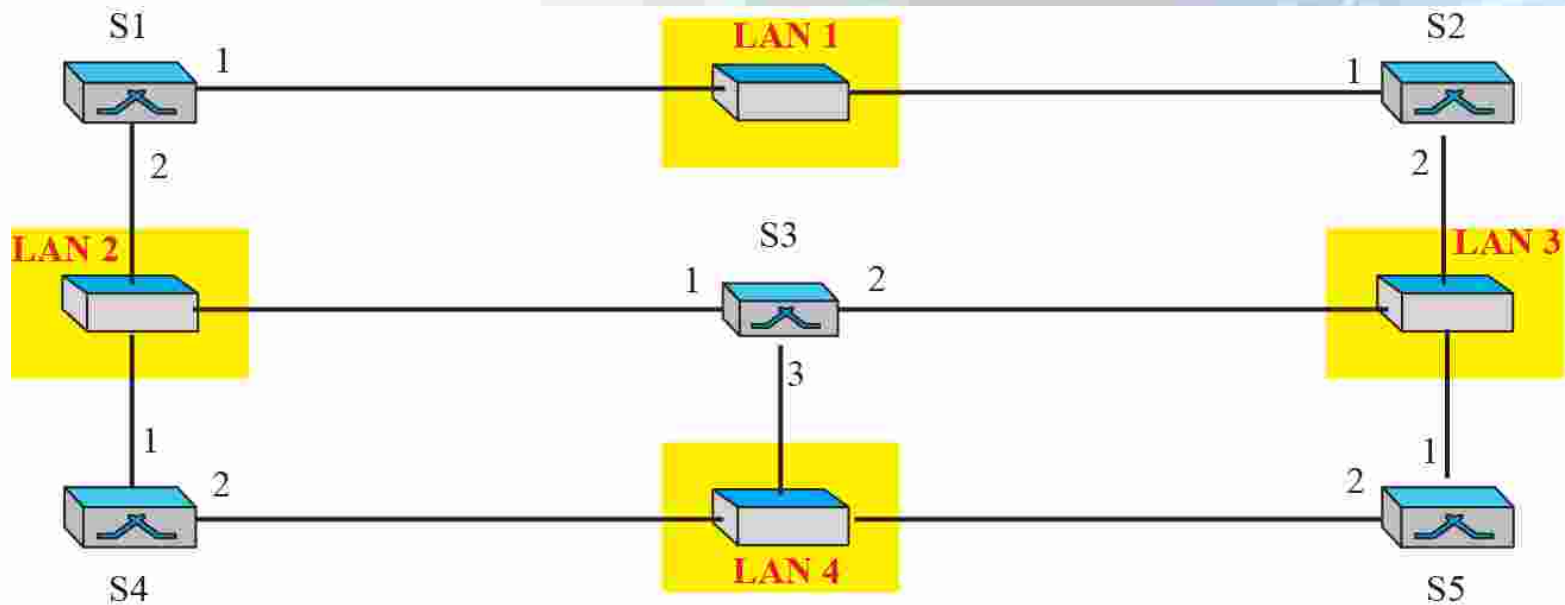
**d. Both switches forward the frame**



# Spanning Tree Algorithm

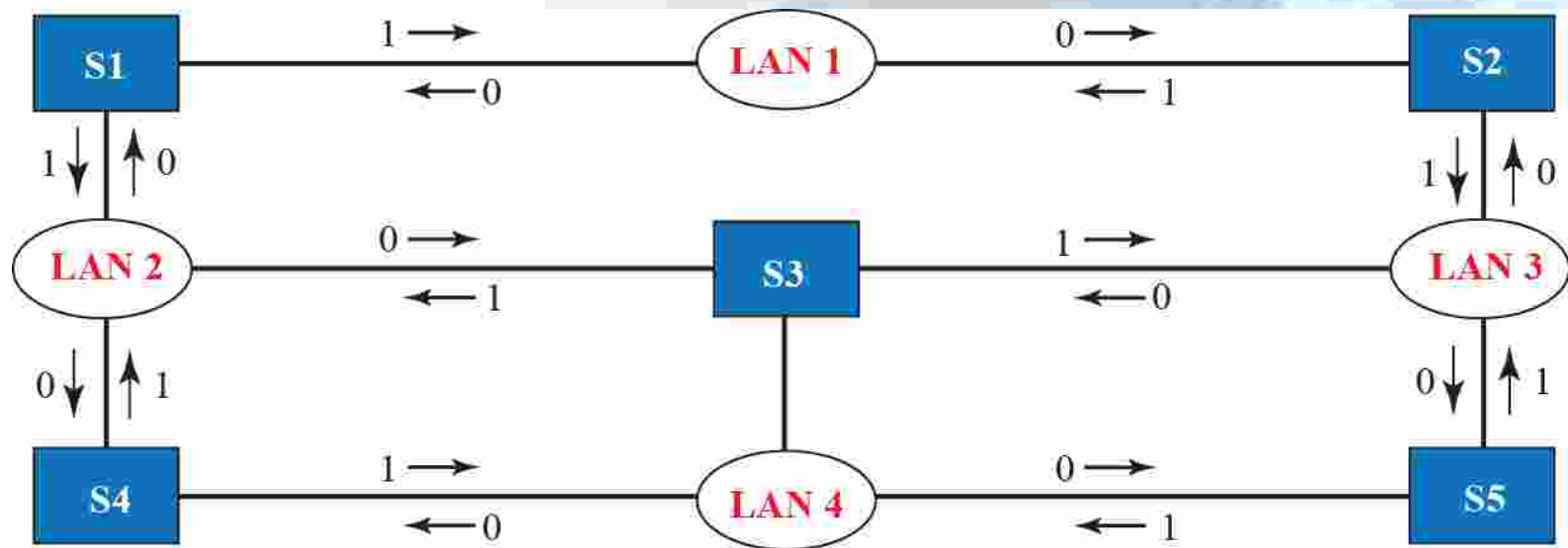
- In graph theory, Spanning Tree is a graph in which there is no loop
- In a switched LAN, this means creating a topology in which each LAN can be reached from any other LAN through one path only (no loop)
- To find the spanning tree, we assign a cost (metric) to each LAN link

# A system of Connected LANs and its Graph (Part a)



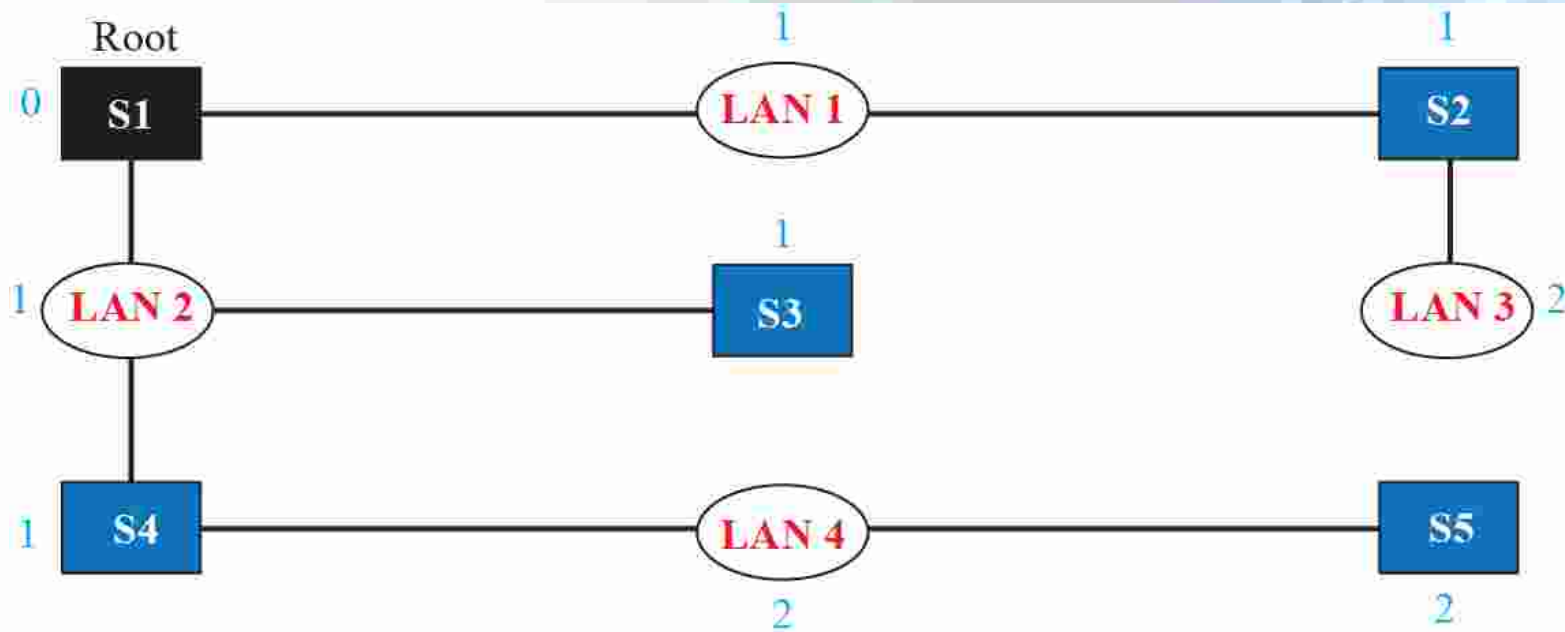
a. Actual system

# A System of Connected LANs and its Graph (Part b)



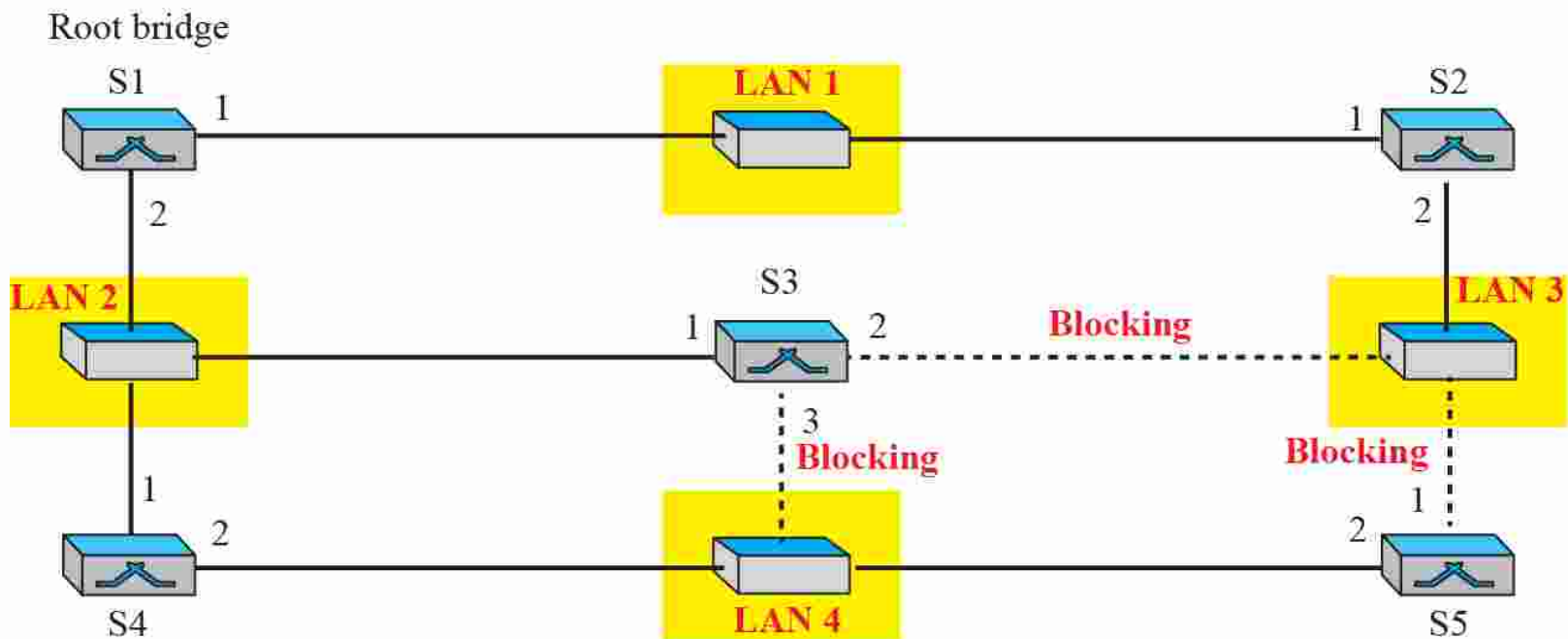
b. Graph representation with cost assigned to each arc

# Finding Shortest Path/Spanning Tree for a Switch



# Forwarding & Blocking Ports after using Spanning Tree

Ports 2 and 3 of bridge S3 are blocking ports (no frame is sent out of these ports).  
Port 1 of bridge S5 is also a blocking port (no frame is sent out of this port).



# Routers

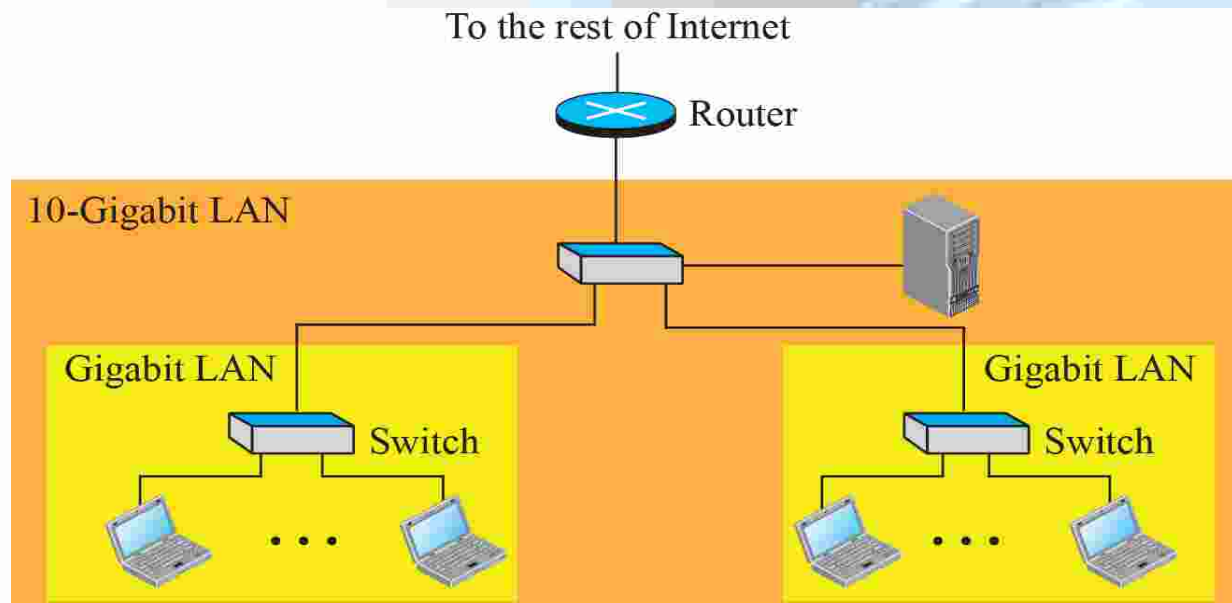
- We compare routers to two-layer switch and a hub
- A router is a three-layer device; it operates in the physical, data-link, and network layers



# Router vs. Switch

Three differences between a router and a repeater or a switch:

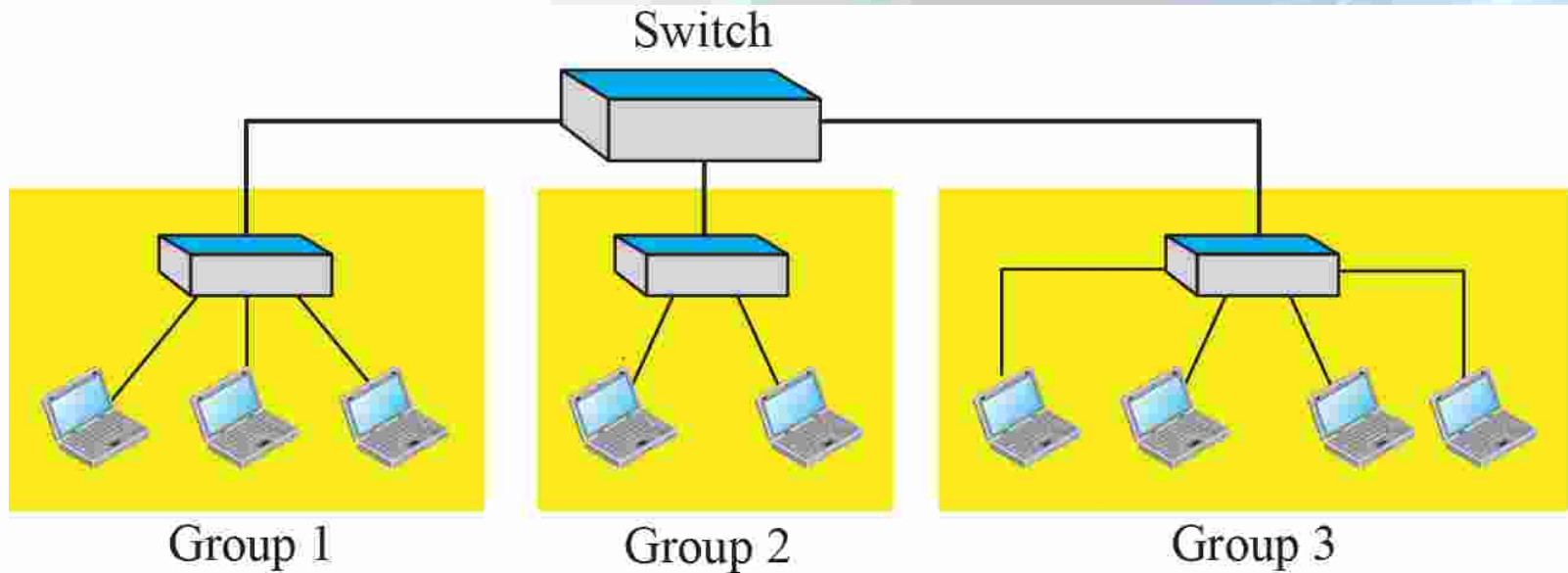
1. A router has a physical and logical (IP) address for each of its interfaces.
2. A router acts only on those packets in which the link-layer destination address matches the address of the interface at which the packet arrives.
3. A router changes the link-layer address of the packet (both source and destination) when it forwards the packet.



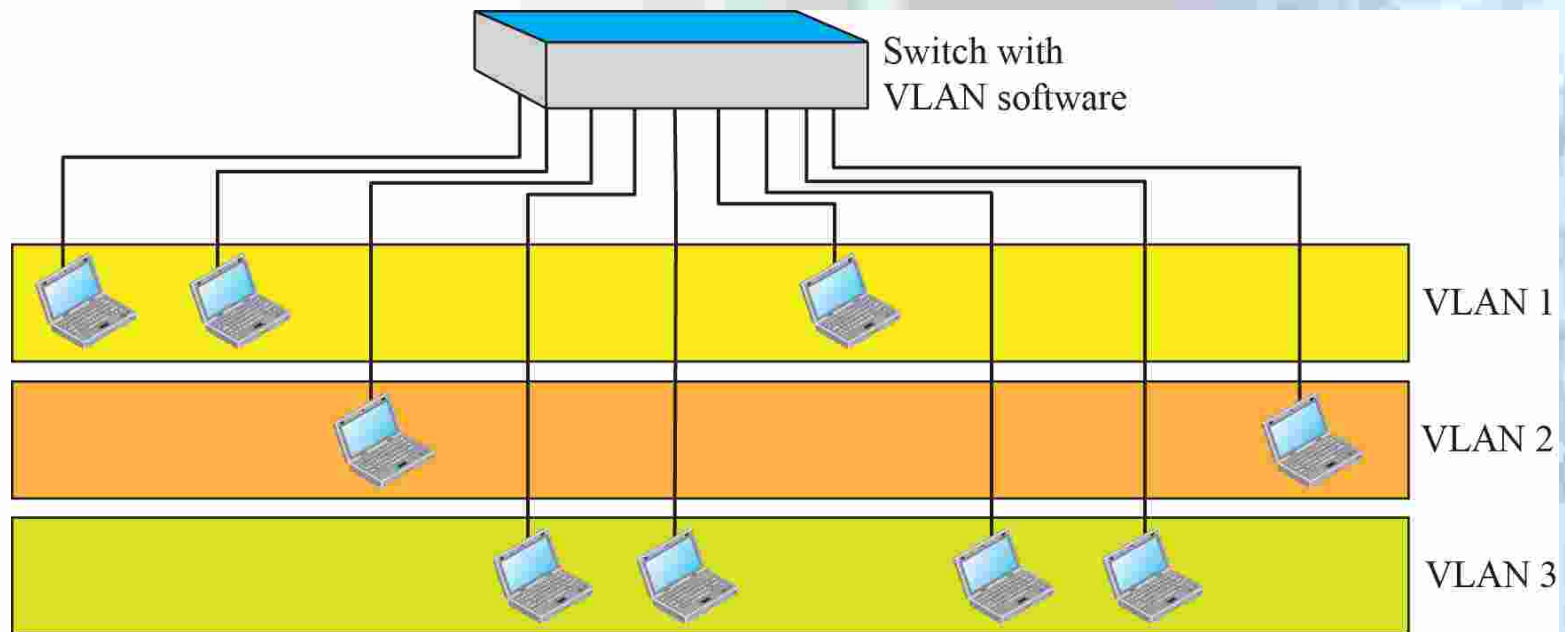
# VIRTUAL LANS (VLAN)

- A VLAN is a LAN configured by software, not by physical wiring
- A station is considered part of a LAN if it physically belongs to that LAN i.e. The criterion of membership is geographic
- Provides a virtual connection between two stations belonging to two different physical LANs

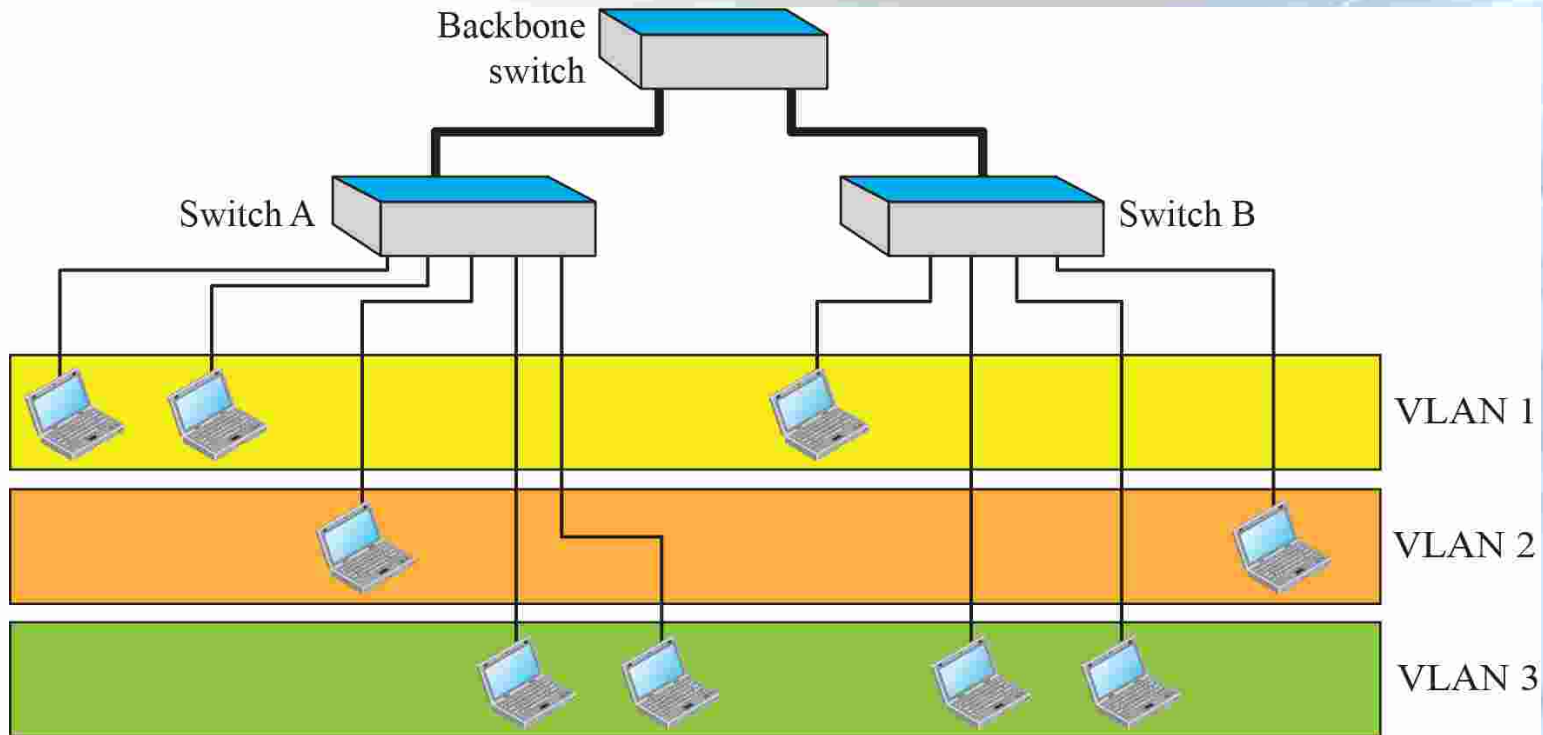
# A Switch Connecting three LANs



# A Switch using VLAN Software



# Two Switches in a Backbone using VLAN Software



# Membership of a VLAN

- What characteristic can be used to group stations in a VLAN?
- Vendors use different characteristics such as interface numbers, port numbers, MAC addresses, IP addresses, IP multicast addresses, or a combination of two or more of these

# Configuration of a VLAN

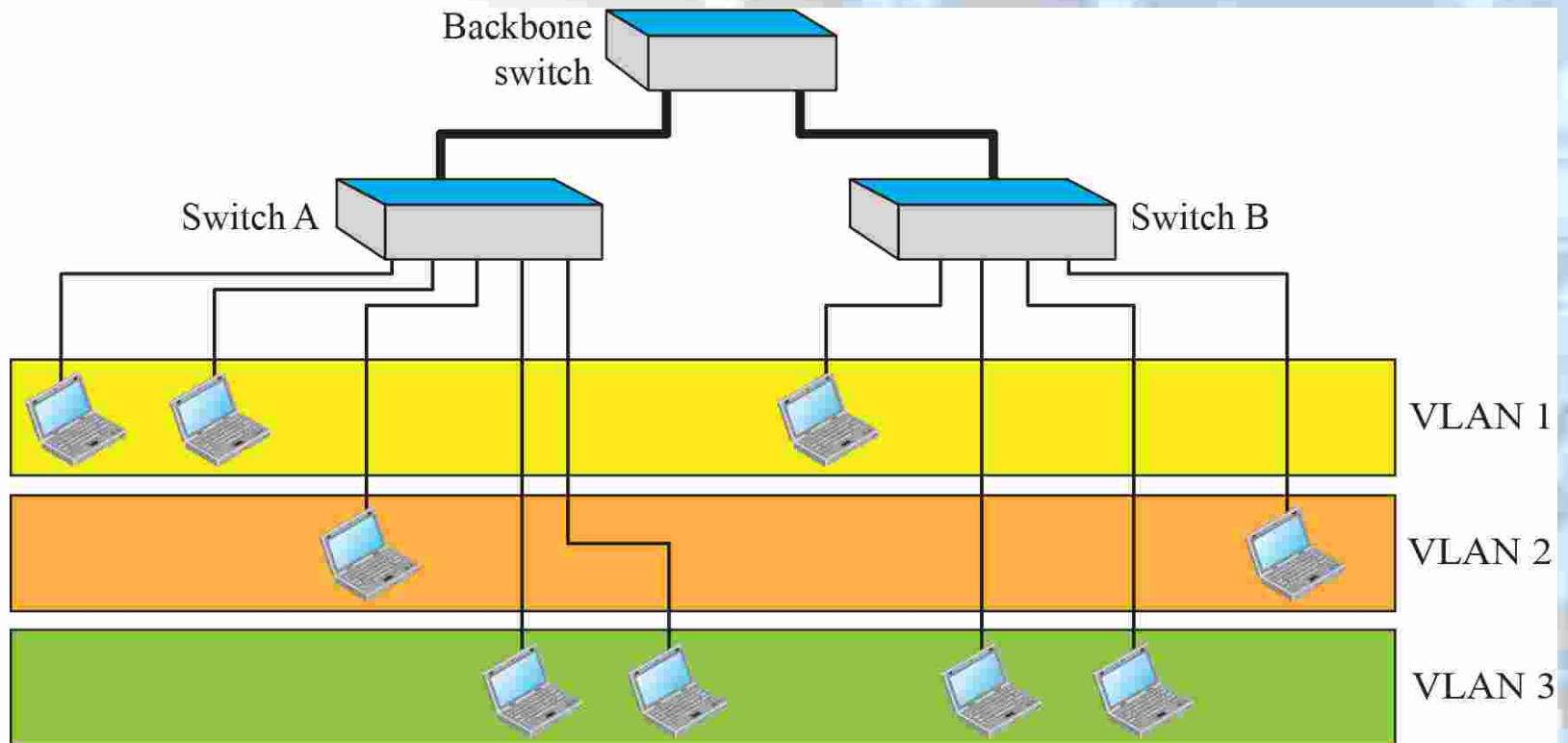
- How are the stations grouped into different VLANs?
- Stations are configured in one of three ways:
  - ✓ Manually
  - ✓ Semi-Automatically
  - ✓ Automatically

# Communication between Switches

- In a multi-switched backbone, each switch must know:
  - ✓ Which station belongs to which VLAN; and
  - ✓ The membership of stations connected to other switches



# Communication between Switches



***Switch A must know the membership status of stations connected to switch B, and switch B must know the same about switch A. Three methods have been devised for this purpose: table maintenance, frame tagging, and time-division multiplexing.***

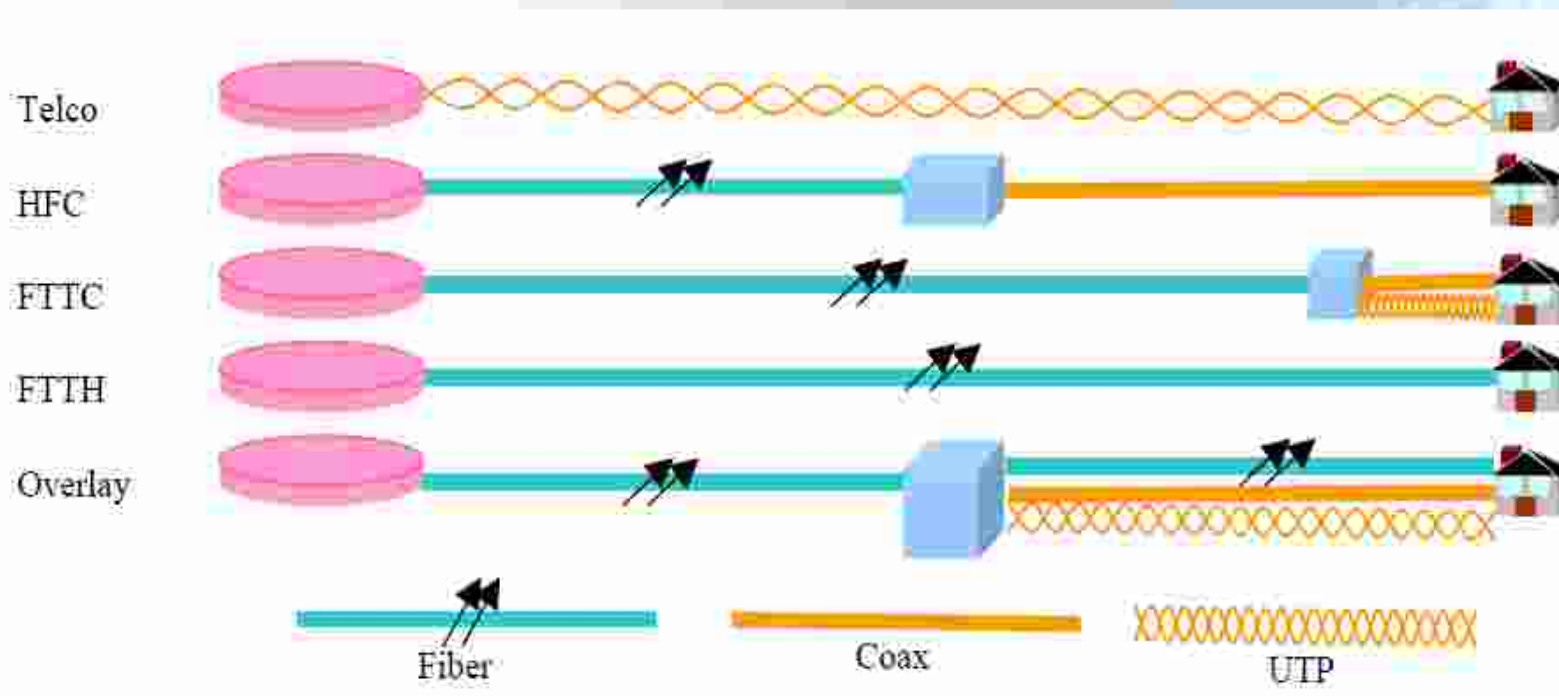
# Advantages of using VLANs

- **Cost and Time Reduction**
- **Creating virtual Workgroups**
- **Security**

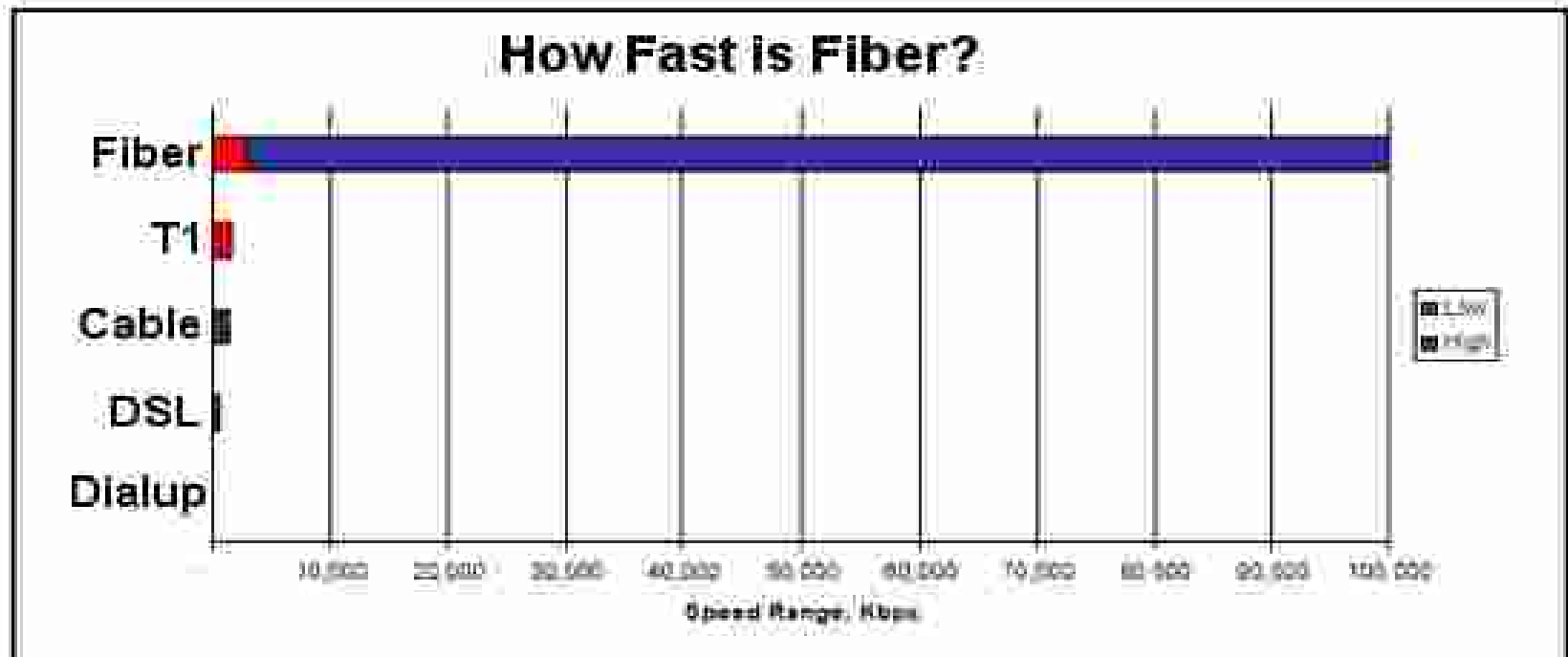
# Comparison of Modern Access Technologies

- Telco
- HFC
- FTTx

# Comparison of Modern Access Technologies



# Fiber – The Medium of the Future!

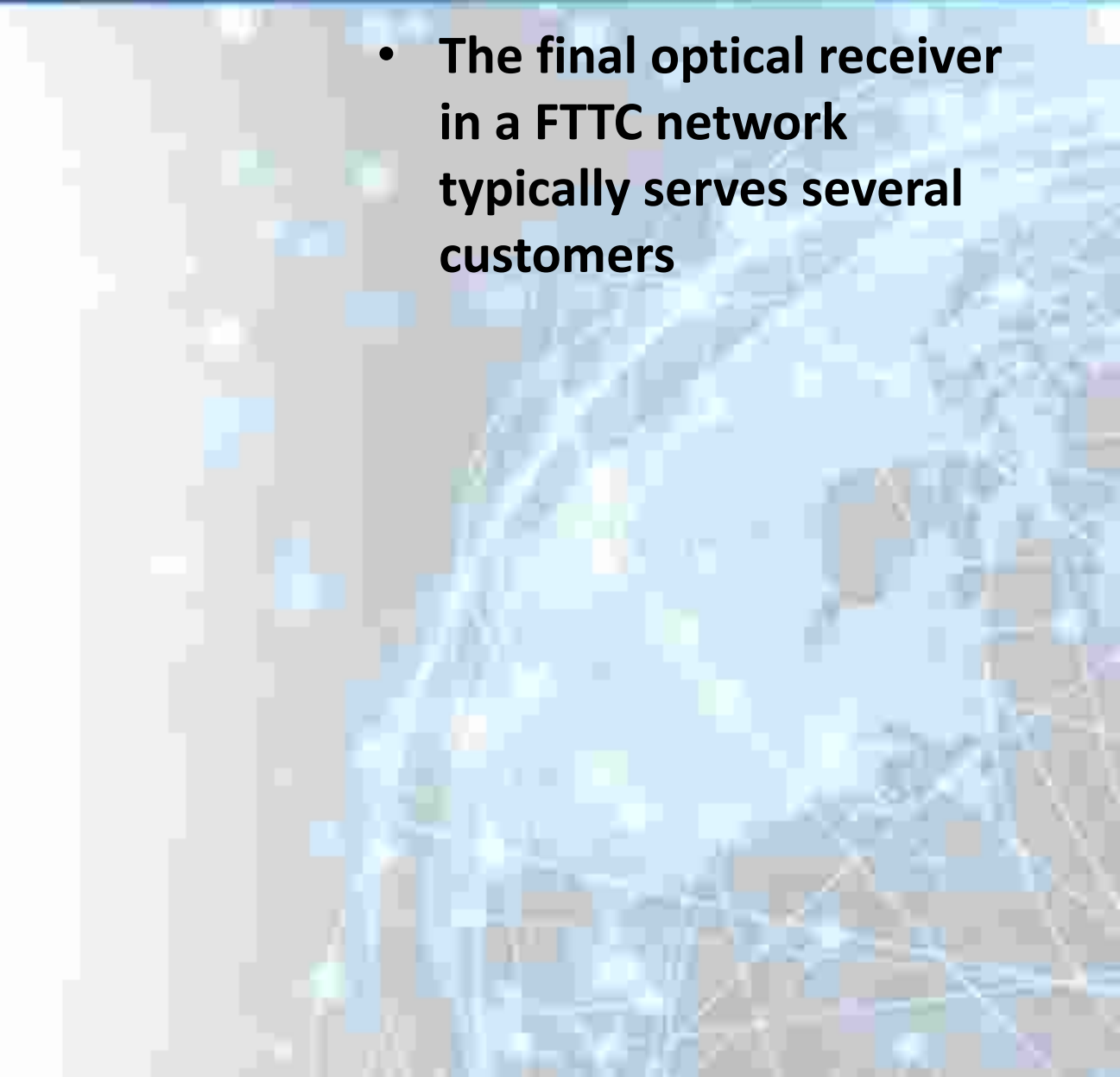


# Fiber To The Curb (FTTC)

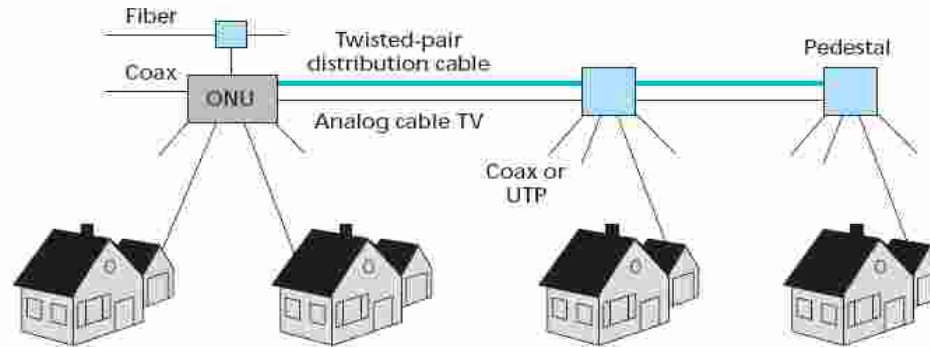
- An access network in which fiber is used for part, but not the entire link from the provider to the end-user
- An optical to electrical (O/E) conversion takes place somewhere near the end-user
- The terminal network segment of a FTTC network is usually twisted pair or coaxial cable

# Fiber To The Curb (FTTC)

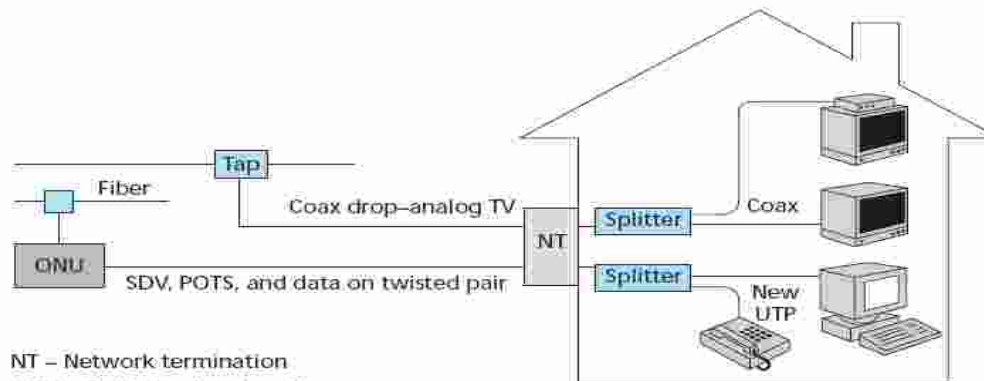
- The final optical receiver in a FTTC network typically serves several customers



# Fiber To The Curb (FTTC)



ONU – Optical network unit  
UTP – Unshielded twisted pair



NT – Network termination  
ONU – Optical network unit  
POTS – Plain old telephone service  
SDV – Switched digital video  
UTP – Unshielded twisted pair



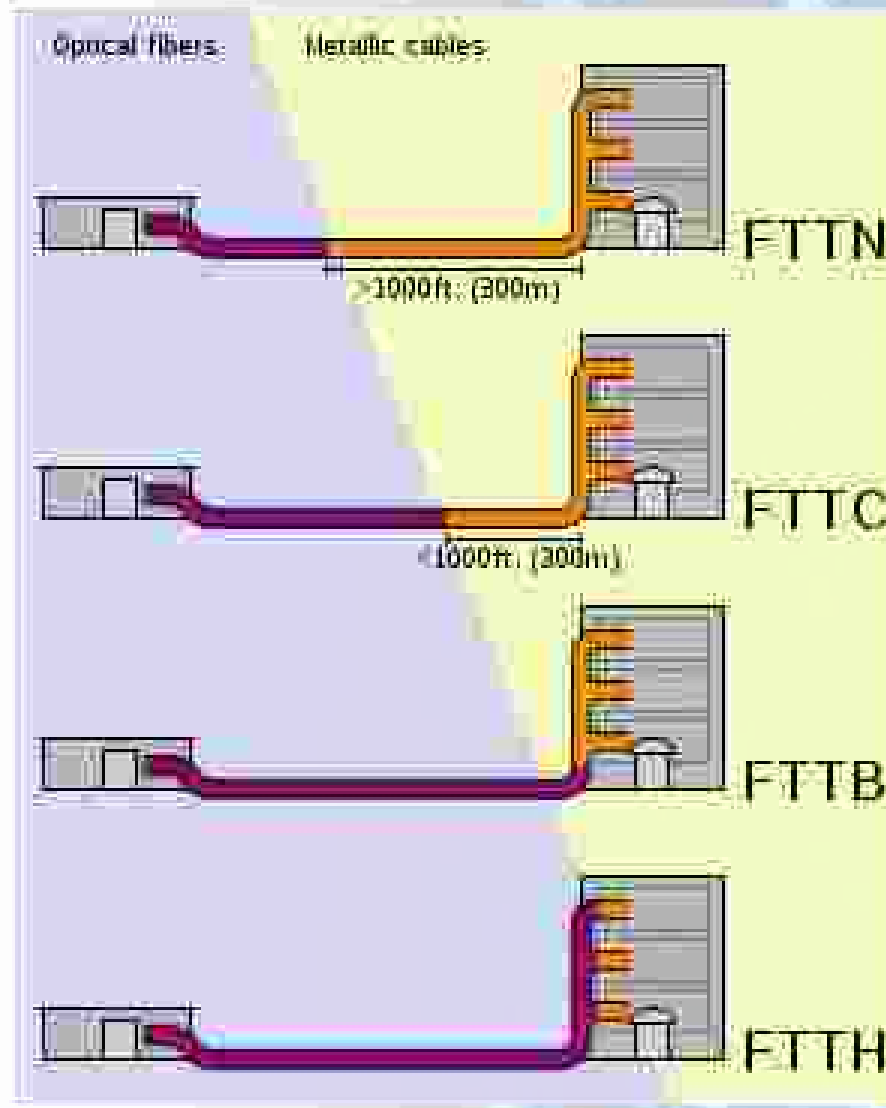
# Fiber To The Home (FTTH)

- Need: High-speed data, reliable voice and high-quality video
- Problems:
  - ✓ How to get high speed lines out to each customer?
  - ✓ How to future-proof the architecture?
- Solution: FTTH

# Fiber To The Home (FTTH)

- Fiber-to-the-home (FTTH) is the installation of optical fiber from a telephone switch directly into the subscriber's home
- It is one of the latest access technologies
- FTTH is also referred to as Fiber-to-the-Building (FTTB)

# Fiber To The Home (FTTH)



|                                                                                   |                                                                              |                           |                     |                    |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------|---------------------|--------------------|
|  | <b>CS601-Data Communication</b><br>Latest Solved MCQs from Final term Papers |                           |                     | <b>May 26,2011</b> |
| <b>Lectures 23-45</b>                                                             | <b>Mc100401285</b>                                                           | <b>moaaz.pk@gmail.com</b> | <b>Moaaz Siddiq</b> | <b>Latest Mcqs</b> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p align="center"><b>FINAL TERM EXAMINATION</b><br/><b>Spring 2011</b></p> <p><b>Question No: 1 ( Marks: 1 ) - Please choose</b><br/><u>The inversion of the level at 1 bit is called as _____</u><br/>▶ NRZ-L<br/>▶ <b>NRZ-I Page 73</b><br/>▶ RZ</p> <p><b>Question No: 2 ( Marks: 1 ) - Please choose</b><br/><u>In selective-reject ARQ, only the specific damaged or lost frame is _____.</u><br/>▶ <b>Retransmitted (Page 200)</b><br/>▶ Forwarded<br/>▶ Selected<br/>▶ Rejected</p> <p><b>Question No: 3 ( Marks: 1 ) - Please choose</b><br/><u>YMODEM has _____ Byte of data unit.</u><br/>▶ <b>1024 Page 205</b><br/>▶ 256<br/>▶ 128<br/>▶ 512</p> <p><b>Question No: 4 ( Marks: 1 ) - Please choose</b><br/><u>Which of the following sublayer, resolves the contention for the shared media</u><br/>▶ <b>MAC (Page 219)</b><br/>▶ LLC<br/>▶ Physical</p> <p><b>Question No: 5 ( Marks: 1 ) - Please choose</b><br/><u>Like 10 Base 5, 10 Base 2 is a _____ topology LAN</u><br/>▶ Ring<br/>▶ Mesh<br/>▶ Star<br/>▶ <b>Bus (Page 227)</b></p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Question No: 6 ( Marks: 1 ) - Please choose**

At the CRC generator, \_\_\_\_\_ added to the data unit after the division process

► **0s are (Page 176)**

- 1s are
- The polynomial
- The CRC remainder is

**Question No: 7 ( Marks: 1 ) - Please choose**

Flow control is needed to prevent \_\_\_\_\_.

- Bit errors
- Overflow of the sender buffer
- **Overflow of the receiver buffer (Page 46)**
- Collision between sender and receiver

**Question No: 8 ( Marks: 1 ) - Please choose**

In Y-MODEM Multiple files can be sent simultaneously

► **True (Page 205)**

- False

**Question No: 9 ( Marks: 1 ) - Please choose**

In 56K Modems Max Uploading speed is \_\_\_\_\_ bytes and downloading speed is \_\_\_\_\_ byte.

- 40K, 56K
- **33.6K, 56K (Page 117)**
- 56K, 33.6K
- None of given

**Question No: 10 ( Marks: 1 ) - Please choose**

If db is amplified then attenuation is -----

- **Positive (Page 142)**
- Negative
- Zero
- None of the above

**Question No: 11 ( Marks: 1 ) - Please choose**

Amplifiers are used to \_\_\_\_\_ the signal to heat.

- **Amplify (Page 142)**
- Rectify
- Testify
- Nullify

**Question No: 12 ( Marks: 1 ) - Please choose**

\_\_\_\_\_ requires the maximum number of I/O ports.

- Bus
- Star
- **Mesh (Page 29)**

- Ring

**Question No: 13 ( Marks: 1 ) - Please choose**

There are \_\_\_\_\_ basic categories of multiplexing.

- 3 (Page 148)
- 4
- 2
- 5

**Question No: 14 ( Marks: 1 ) - Please choose**

In \_\_\_\_\_ transmission, bits are transmitted over a single wire, one at a time.

- Asynchronous serial
- Synchronous serial
- Parallel (Page 98)
- Asynchronous & Synchronous serial

**Question No: 15 ( Marks: 1 ) - Please choose**

In \_\_\_\_\_ transmission, bits are transmitted over their own wires

- Asynchronous serial
- Synchronous serial
- Parallel
- [http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter4/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter4/mixed_quiz.html)
- Asynchronous & Synchronous serial

**Question No: 16 ( Marks: 1 ) - Please choose**

EIA 232 allows for a maximum bit rate of \_\_\_\_\_ Kbps.

- 40
- 30
- 20 (Page 104)
- 10

**Question No: 17 ( Marks: 1 ) - Please choose**

In CRC the quotient at the sender \_\_\_\_\_

- Becomes the dividend at the receiver
- Becomes the divisor at the receiver
- Is discarded

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

- Is the remainder

**Question No: 18 ( Marks: 1 ) - Please choose**

In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_ frame

- EOT (Page 189)
- EKT
- ENT
- ESP

**FINALTERM EXAMINATION**  
Spring 2010

**Question No: 1 ( Marks: 1 ) - Please choose one**

Encryption and encoding are the same terms.

- ▶ True
- ▶ **False (Page 52)**

**Question No: 2 ( Marks: 1 ) - Please choose one**

Traditional modems are wide spread now to a data rate of \_\_\_\_\_.

- ▶ **56 Kbps (Page 115)**
- ▶ 72 Kbps
- ▶ 42 Kbps
- ▶ 96 Kbps

**Question No: 3 ( Marks: 1 ) - Please choose one**

Bi phase encoding is a type of bipolar encoding in which we use two voltage levels.

- ▶ True
- ▶ **False (Page 73)**

**Question No: 4 ( Marks: 1 ) - Please choose one**

The \_\_\_\_\_ layer changes bits into electromagnetic signals.

- ▶ **physical** [http://www.ee.surrey.ac.uk/Projects/CAL/networks/Physical\\_Layer.htm](http://www.ee.surrey.ac.uk/Projects/CAL/networks/Physical_Layer.htm)
- ▶ data link
- ▶ transport
- ▶ none of given

**Question No: 5 ( Marks: 1 ) - Please choose one**

Which is not an element of protocol?

- ▶ Semantics
- ▶ Timing
- ▶ **Communication service module (Page 19)**

**Question No: 6 ( Marks: 1 ) - Please choose one**

A telephone network is an example of a \_\_\_\_\_ network.

- ▶ Packet-switched
- ▶ **Circuit-switched (Page 38)**

- ▶ Message-switched
- ▶ none of the given

**Question No: 7 ( Marks: 1 ) - Please choose one**

The bit rate always equals the baud rate in which type of signal?

- ▶ **FSK (Page 86)**
- ▶ QAM
- ▶ 4-PSK
- ▶ PSK

**Question No: 8 ( Marks: 1 ) - Please choose one**

Secondary hub in a tree must be a passive hub.

- ▶ True
- ▶ **False (Page 31)**

**Question No: 9 ( Marks: 1 ) - Please choose one**

In case of uploading at the switching station, data is converted to digital signal using \_\_\_\_\_.

- ▶ TCP
- ▶ **PCM (Page 116)**
- ▶ ICP
- ▶ TDM

**Question No: 10 ( Marks: 1 ) - Please choose one**

Category 5 UTP cable is used for data transmission of upto \_\_\_\_\_.

- ▶ **100 Mbps (Page 123)**
- ▶ 200 Mbps
- ▶ 250 Mbps
- ▶ 400 Mbps

**Question No: 11 ( Marks: 1 ) - Please choose one**

Which of the following primarily uses guided media?

- ▶ Cellular telephone system
- ▶ **Local telephone system (Page 120)**
- ▶ Satellite communications
- ▶ Radio broadcasting

**Question No: 12 ( Marks: 1 ) - Please choose one**

Ultra high-frequency waves always use \_\_\_\_\_ propagation.

- ▶ Ground
- ▶ Sky



► **Line of Sight (Page 136)**

► Space

**Question No: 13 ( Marks: 1 ) - Please choose one**

Distortion occurs in a \_\_\_\_\_ signal.

► Rectified

► **Composite (Page 141)**

► Amplified

► none of the given

**Question No: 14 ( Marks: 1 ) - Please choose one**

Which of the following are not used to measure the performance of TX Media.

► Throughput

► Propagation Speed

► Propagation Time

► **none of the given (Page 144)**

**Question No: 15 ( Marks: 1 ) - Please choose one**

A portion of the path that carries TX b/w a given pair of devices is known as \_\_\_\_\_.

► Node

► Bridge

► **Channel (Page 147)**

► Gateway

**Question No: 16 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ takes data from one high speed line and breaks it into portions.

► Multiplexing

► **Inverse multiplexing (Page 158)**

► Inverse subtraction

► Inverse addition

**Question No: 17 ( Marks: 1 ) - Please choose one**

The \_\_\_\_\_ layer is the layer closest to the transmission medium.

► **physical**

► data link

► network

► transport

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter2/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter2/multiple_choice_quiz.html)

**Question No: 18 ( Marks: 1 ) - Please choose one**

In primary-secondary communication \_\_\_\_\_ is always the initiator of a session

► **Primary (Page 189)**

- Secondary
- Sender
- Receiver

**Question No: 19 ( Marks: 1 ) - Please choose one**

YMODEM has \_\_\_\_\_ Byte of data unit.

► **1024 (Page 205)**

- 256
- 128
- 512

**Question No: 20 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ coordinates the amount of data that can be sent before receiving acknowledgment

► **flow control (Page 186)**

- error control
- data control

**Question No: 21 ( Marks: 1 ) - Please choose one**

If the ASCII character H is sent and the character I is received, what type of error is this?

► **Single-bit (Page 169)**

- Multiple-bit
- Burst
- Recoverable

**Question No: 22 ( Marks: 1 ) - Please choose one**

In \_\_\_\_\_ ARQ, if a NAK is received, only the specific damaged or lost frame is retransmitted.

- Stop-and-wait
- Go-Back-N
- **Selective reject (Page 200)**
- Stop-and-wait & Go-back-N

**Question No: 23 ( Marks: 1 ) - Please choose one**

Data link protocols can be divided into \_\_\_\_\_ sub-groups.

► **two (Page 202)**

- three
- four
- five

**Question No: 24 ( Marks: 1 ) - Please choose one**

If odd parity is used for ASCII error detection, the number of 0s per 8-bit symbol is \_\_\_\_\_

- ▶ Even
- ▶ **Odd (Page 173)**
- ▶ Indeterminate
- ▶ 42

**Question No: 25 ( Marks: 1 ) - Please choose one**

In Y-MODEM Multiple files can be sent simultaneously

- ▶ **True (Page 205)**
- ▶ False

**Question No: 26 ( Marks: 1 ) - Please choose one**

The PDU has no flag fields, no CRC, and no station address

- ▶ **TRUE (Page 220)**
- ▶ FALSE

**Question No: 27 ( Marks: 1 ) - Please choose one**

Check sum method is used for \_\_\_\_\_ layers.

- ▶ Physical
- ▶ Application
- ▶ Transport
- ▶ **Data link** <http://www.cse.iitk.ac.in/users/dheeraj/cs425/lec09.html>

**Question No: 28 ( Marks: 1 ) - Please choose one**

Repeater is an amplifier, not a regenerator.

- ▶ True
- ▶ **False (Page 240)**

**Question No: 29 ( Marks: 1 ) - Please choose one**

RARP stands for Reverse Address Resolution Protocol.

- ▶ **True (Page 244)**
- ▶ False

**Question No: 30 ( Marks: 1 ) - Please choose one**

The BNC-T connector is a T-shaped device with \_\_\_\_\_ ports

- ▶ **Three (Page 228)**
- ▶ Two
- ▶ Four
- ▶ Five

**FINAL TERM EXAMINATION**  
Spring 2010

**Question No: 1 ( Marks: 1 ) - Please choose one**

When data are transmitted from device A to device B, the header from A's layer 4 is read by B's \_\_\_\_\_ layer.

- ▶ physical
- ▶ **transport**
- ▶ application
- ▶ none of the given

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter2/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter2/multiple_choice_quiz.html)

**Question No: 2 ( Marks: 1 ) - Please choose one**

A periodic signal completes one cycle in 0.001 s. What is the frequency?

- ▶ 1 Hz
- ▶ 100 Hz
- ▶ **1 KHz (Page 61)**
- ▶ 1 MHz

**Question No: 3 ( Marks: 1 ) - Please choose one**

Zero crossing bandwidth is also called as equivalent noise bandwidth.

- ▶ True
- ▶ **False (Page 67)**

**Question No: 4 ( Marks: 1 ) - Please choose one**

Amplitude in ASK is more resistive to EMI and Noise.

- ▶ True
- ▶ **False (Page 87)**

**Question No: 5 ( Marks: 1 ) - Please choose one**

If FCC regulations are followed, the carrier frequencies of adjacent AM radio stations are \_\_\_\_\_ apart.

- ▶ 5 KHz
- ▶ **10 KHz (Page 95)**
- ▶ 200 KHz
- ▶ 530 KHz

**Question No: 6 ( Marks: 1 ) - Please choose one**

There are \_\_\_\_\_ types of serial transmission:

- ▶ 1
- ▶ 2 (Page 99)
- ▶ 3
- ▶ 4

**Question No: 7 ( Marks: 1 ) - Please choose one**

Synchronous transmission has \_\_\_\_\_.

- ▶ a start bit
- ▶ a stop bit
- ▶ gaps between bytes
- ▶ none of the given (page 99)

**Question No: 8 ( Marks: 1 ) - Please choose one**

Which of the following is an example of ITU-T modem standards?

- ▶ T-series
- ▶ X-series
- ▶ N-series
- ▶ V-series (Page 114)

**Question No: 9 ( Marks: 1 ) - Please choose one**

The maximum data rate in the uploading direction is still \_\_\_\_\_.

- ▶ 26.6 Kbps
- ▶ 33.6 Kbps (Page 117)
- ▶ 36.6 Kbps
- ▶ 46.6 Kbps

**Question No: 10 ( Marks: 1 ) - Please choose one**

Category 5 UTP cable is used for data transmission of upto \_\_\_\_\_.

- ▶ 100 Mbps (Page 123)
- ▶ 200 Mbps
- ▶ 250 Mbps
- ▶ 400 Mbps

**Question No: 11 ( Marks: 1 ) - Please choose one**

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- ▶ Conical
- ▶ Barrel (Page 131)
- ▶ Circular
- ▶ Rectangular

**Question No: 12 ( Marks: 1 ) - Please choose one**

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

- ▶ Ground
- ▶ Sky
- ▶ **Line of sight (Page 134)**
- ▶ Space

**Question No: 13 ( Marks: 1 ) - Please choose one**

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across \_\_\_\_\_ data link

- ▶ **Single (Page 147)**
- ▶ Multi
- ▶ Single and Multi
- ▶ none of the given

**Question No: 14 ( Marks: 1 ) - Please choose one**

Need for Addressing makes Asynchronous TDM inefficient for bit or byte \_\_\_\_\_.

- ▶ **Interleaving (Page 158)**
- ▶ Addition
- ▶ Subtraction
- ▶ None of the given

**Question No: 15 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ takes data from one high speed line and breaks it into portions.

- ▶ Multiplexing
- ▶ **Inverse multiplexing (Page 158)**
- ▶ Inverse subtraction
- ▶ Inverse addition

**Question No: 16 ( Marks: 1 ) - Please choose one**

A traditional telephone line has a bandwidth of \_\_\_\_\_.

- ▶ **3000Hz (Page 111)**
- ▶ 4000 Hz
- ▶ 2000 MHz
- ▶ 4000 MHz

**Question No: 17 ( Marks: 1 ) – Please choose one**

If the ASCII character H is sent and the character I is received, what type of error is this?

- ▶ **Single-bit (Page 169)**
- ▶ Multiple-bit

- ▶ Burst
- ▶ Recoverable

**Question No: 18 ( Marks: 1 ) - Please choose one**

At the CRC generator, \_\_\_\_\_ added to the data unit after the division process

- ▶ **0s are**
- ▶ 1s are
- ▶ The polynomial
- ▶ The CRC remainder is

[http://higherred.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://higherred.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 19 ( Marks: 1 ) - Please choose one**

Error control in the data link layer is based on \_\_\_\_\_

- ▶ **automatic repeat request (Page 196)**
- ▶ automatic repeat acknowledgment
- ▶ automatic send acknowledgment

**Question No: 20 ( Marks: 1 ) - Please choose one**

In line discipline the initiator first transmits a frame called an \_\_\_\_\_

- ▶ **Enquiry (Page 189)**
- ▶ Acknowledgment
- ▶ NAK
- ▶ Request

**Question No: 21 ( Marks: 1 ) - Please choose one**

Primary device uses \_\_\_\_\_ to receive transmission from the secondary devices.

- ▶ ACK
- ▶ ENQ
- ▶ **POLL (Page 191)**

**Question No: 22 ( Marks: 1 ) - Please choose one**

Flow control is needed to prevent \_\_\_\_\_.

- ▶ Bit errors
- ▶ Overflow of the sender buffer
- ▶ **Overflow of the receiver buffer (Page 46)**
- ▶ Collision between sender and receiver

**Question No: 23 ( Marks: 1 ) - Please choose one**

For stop-and-wait ARQ, for N data packets sent, \_\_\_\_\_ acknowledgments are needed.

- ▶ **N**

- ▶ 2N
- ▶ N-1
- ▶ N+1

[http://highereducation.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter11/mixed\\_quiz.html](http://highereducation.mcgraw-hill.com/sites/0072515848/student_view0/chapter11/mixed_quiz.html)

**Question No: 24 ( Marks: 1 ) - Please choose one**

In Y-MODEM Multiple files can be sent simultaneously

- ▶ **True (Page 205)**
- ▶ False

**Question No: 25 ( Marks: 1 ) - Please choose one**

HDLC is an acronym for \_\_\_\_\_.

- ▶ High-duplex line communication
- ▶ **High-level data link control (Page 210)**
- ▶ Half-duplex digital link combination
- ▶ Host double-level circuit

**Question No: 26 ( Marks: 1 ) - Please choose one**

Token Bus has no commercial application in data communications

- ▶ **True (Page 232)**
- ▶ False

**Question No: 27 ( Marks: 1 ) - Please choose one**

The BNC-T connector is a T-shaped device with \_\_\_\_\_ ports

- ▶ **Three (Page 228)**
- ▶ Two
- ▶ Four
- ▶ Five

**Question No: 28 ( Marks: 1 ) - Please choose one**

Each station in the Token Ring regenerates the frame.

- ▶ **True (Page 234)**
- ▶ False

**Question No: 29 ( Marks: 1 ) - Please choose one**

Repeater works on \_\_\_\_\_ layer.

- ▶ Data Link
- ▶ Physical
- ▶ **Network (224)**
- ▶ Application



**Question No: 30 ( Marks: 1 ) - Please choose one**

Trunks are transmission media such as \_\_\_\_\_ that handle the telephone to the nearest end office.

- ▶ Satellite links
- ▶ **Twisted-pair & Fiber-optic**
- ▶ Twisted-pair
- ▶ Fiber-optic

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter8/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter8/mixed_quiz.html)

**Spring 2010**  
**CS601- Data Communication**

**Question No: 1 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ representation of links that connect nodes is called as physical topology.

- ▶ geometrical
- ▶ logical
- ▶ **physical (page 44)**

**Question No: 2 ( Marks: 1 ) – Please choose one**

The internet model consists of \_\_\_\_\_ layers.

- ▶ three
- ▶ two
- ▶ **five**
- ▶ seven

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter2/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter2/multiple_choice_quiz.html)

**Question No: 3 ( Marks: 1 ) – Please choose one**

Encryption and encoding are the same terms.

- ▶ True
- ▶ **False (Page 52)**

**Question No: 4 ( Marks: 1 ) - Please choose one**

The amplitude of a digital signal depends upon the \_\_\_\_\_ to represent a bit.

- ▶ phase
- ▶ **voltage (Page 59)**
- ▶ wavelength

**Question No: 5 ( Marks: 1 ) - Please choose one**

The inversion of the level at 1 bit is called as \_\_\_\_\_

- ▶ NRZ-L

► **NRZ-I (Page 73)**

► RZ

**Question No: 6 ( Marks: 1 ) - Please choose one**

Modulation of an analog signal can be accomplished through changing the \_\_\_\_\_ of the carrier signal.

- amplitude
- frequency
- phase
- **all of the given**

**Question No: 7 ( Marks: 1 ) - Please choose one**

If FCC regulations are followed, the carrier frequencies of adjacent AM radio stations are \_\_\_\_\_ apart.

- 5 KHz
- **10 KHz (Page 95)**
- 200 KHz
- 530 KHz

**Question No: 8 ( Marks: 1 ) - Please choose one**

Category 5 UTP cable is used for data transmission of upto \_\_\_\_\_.

- **100 Mbps (Page 123)**
- 200 Mbps
- 250 Mbps
- 400 Mbps

**Question No: 9 ( Marks: 1 ) - Please choose one**

The RG number gives us information about \_\_\_\_\_.

- Twisted pairs
- **Coaxial cables (Page 125)**
- Optical fibers
- all of the given

**Question No: 10 ( Marks: 1 ) - Please choose one**

The inner core of an optical fiber is \_\_\_\_\_ in composition.

- **Glass plastic (Page 130)**
- Copper
- Bimetallic
- Liquid

**Question No: 11 ( Marks: 1 ) - Please choose one**

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- ▶ Conical
- ▶ **Barrel (Page 131)**
- ▶ Circular
- ▶ Rectangular

**Question No: 12 ( Marks: 1 ) - Please choose one**

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

- ▶ Ground
- ▶ Sky
- ▶ **Line of sight (Page 134)**
- ▶ Space

**Question No: 13 ( Marks: 1 ) - Please choose one**

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across \_\_\_\_\_ data link

- ▶ **Single (Page 147)**
- ▶ Multi
- ▶ Single and Multi
- ▶ none of the given

**Question No: 14 ( Marks: 1 ) - Please choose one**

A portion of the path that carries TX b/w a given pair of devices is known as \_\_\_\_\_.

- ▶ Node
- ▶ Bridge
- ▶ **Channel (Page 147)**
- ▶ Gateway

**Question No: 15 ( Marks: 1 ) - Please choose one**

Which error detection method involves polynomials?

- ▶ Checksum
- ▶ Two-dimensional parity check
- ▶ **CRC (Page 177)**
- ▶ Simple parity check

**Question No: 16 ( Marks: 1 ) - Please choose one**

If the ASCII character G is sent and the character D is received, what type of error is this?

- ▶ Single-bit
- ▶ Multiple-bit

► **Burst**

► Recoverable

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 17 ( Marks: 1 ) - Please choose one**

Which error detection method involves the use of parity bits?

► **Simple parity check & two dimensional parity check**

► CRC

► Two-dimensional parity check

► Simple parity check

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 18 ( Marks: 1 ) - Please choose one**

Which error detection method can detect a single-bit error?

► Simple parity check

► Two-dimensional parity check

► CRC

► **All of the given**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 19 ( Marks: 1 ) - Please choose one**

The Hamming code is a method of \_\_\_\_\_

► Error detection

► **Error correction (Page 181)**

► Error encapsulation

► Error detection & Error encapsulation

**Question No: 20 ( Marks: 1 ) - Please choose one**

Sliding window requires that data frames be transmitted \_\_\_\_\_

► **Sequentially (Page 199)**

► Frequently

► Synchronously

► Asynchronously

**Question No: 21 ( Marks: 1 ) - Please choose one**

In selective-reject ARQ, only the specific damaged or lost frame is \_\_\_\_\_.

► **Retransmitted (Page 200)**

► Forwarded

► Selected

► Rejected

**Question No: 22 ( Marks: 1 ) - Please choose one**

Which of the following sub layer, resolves the contention for the shared media

▶ **MAC (Page 219)**

- ▶ LLC
- ▶ Physical

**Question No: 23 ( Marks: 1 ) - Please choose one**

The PDU has no flag fields, no CRC, and no station address

▶ **TRUE (Page 220)**

- ▶ FALSE

**Question No: 24 ( Marks: 1 ) - Please choose one**

IEEE divides the base band category into \_\_\_\_\_ standards.

▶ **5 (Page 222)**

- ▶ 4
- ▶ 3
- ▶ 6

**Question No: 25 ( Marks: 1 ) - Please choose one**

Like 10 Base 5, 10 Base 2 is a \_\_\_\_\_ topology LAN

- ▶ Ring
- ▶ Mesh
- ▶ Star

▶ **Bus (Page 227)**

**Question No: 26 ( Marks: 1 ) - Please choose one**

Check sum method is used for \_\_\_\_\_ layers.

- ▶ Physical
- ▶ Application
- ▶ Transport

▶ **Data link** <http://www.cse.iitk.ac.in/users/dheeraj/cs425/lec09.html>

**Question No: 27 ( Marks: 1 ) - Please choose one**

Repeater works on \_\_\_\_\_ layer.

- ▶ Data Link
- ▶ Physical
- ▶ **Network (Page 224)**
- ▶ Application

**Question No: 28 ( Marks: 1 ) - Please choose one**

Trunks are transmission media such as \_\_\_\_\_ that handle the telephone to the nearest end office.

- ▶ Satellite links
- ▶ **Twisted-pair & Fiber-optic**
- ▶ Twisted-pair
- ▶ Fiber-optic

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter8/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter8/mixed_quiz.html)

**Question No: 29 ( Marks: 1 ) - Please choose one**

Which of the following \_\_\_\_\_ uses a series of filters to decompose multiplexed signal into its constituent signals?

- ▶ MUX
- ▶ **DEMUX (Page 150)**
- ▶ Switch
- ▶ Bridge

**Question No: 30 ( Marks: 1 ) - Please choose one**

In Fast Ethernet, data rate can be increased by \_\_\_\_\_ collisions.

- ▶ Increasing
- ▶ **Decreasing (Page 230)**
- ▶ Keeping Constant
- ▶ None of the given

**FINALTERM EXAMINATION  
Spring 2009**

**Question No: 1 ( Marks: 1 ) - Please choose one**

Which is not an element of protocol

- ▶ semantics
- ▶ timing
- ▶ **communication service module (Page 19)**

**Question No: 2 ( Marks: 1 ) - Please choose one**

Layers 5, 6 and 7 also called as network support layers.

- ▶ True
- ▶ **False (Page 42)**

**Question No: 3 ( Marks: 1 ) - Please choose one**

Fourier transform tells us that any digital signal can be decomposed into infinite number of periodic signals

- ▶ True
- ▶ **False (Page 42)**

**Question No: 4 ( Marks: 1 ) - Please choose one**

Time domain plot show changes in signal phase with respect to time.

- ▶ True
- ▶ **False (Page 63)**

**Question No: 5 ( Marks: 1 ) - Please choose one**

Analog to digital conversion is also termed as modulating an analog signal.

- ▶ True
- ▶ **False (Page 70)**

**Question No: 6 ( Marks: 1 ) - Please choose one**

DC component is also termed as Direct current component or a component with non-zero frequency.

- ▶ **False (Page 72)**
- ▶ True

**Question No: 7 ( Marks: 1 ) - Please choose one**

Manchester is a type of \_\_\_\_\_ encoding.

- ▶ **biphase (Page 75)**
- ▶ polar
- ▶ biphase & polar
- ▶ none of the given

**Question No: 8 ( Marks: 1 ) - Please choose one**

The inversion of the level at 1 bit is called as \_\_\_\_\_

- ▶ NRZ-L
- ▶ **NRZ-I (Page 73)**
- ▶ RZ

**Question No: 9 ( Marks: 1 ) - Please choose one**

PCM is the first process of PAM.

- ▶ True
- ▶ **False (Page 80)**

**Question No: 10 ( Marks: 1 ) - Please choose one**

In 4PSK each phase change represents \_\_\_\_\_ bits.

- ▶ 3
- ▶ 6
- ▶ **2 (Page 91)**
- ▶ 4

**Question No: 11 ( Marks: 1 ) - Please choose one**

In RS 422 Balanced mode two lines carry \_\_\_\_\_ signals which are not identical to each other.

► **Same (Page 109)**

- different
- digital
- analog

**Question No: 12 ( Marks: 1 ) - Please choose one**

A \_\_\_\_\_ converts an analog signal into a digital signal.

► **Demodulator (Page 110)**

- Modulator
- Digital-to-analog converter

**Question No: 13 ( Marks: 1 ) - Please choose one**

**Which of the following is an example of ITU-T modem standards:**

- T-series
- X-series
- N-series

► **V-series (Page 114)**

**Question No: 14 ( Marks: 1 ) - Please choose one**

**The maximum data rate in the uploading direction is still \_\_\_\_\_.**

- 26.6 Kbps
- **33.6 Kbps (Page 117)**
- 36.6 Kbps
- 46.6 Kbps

**Question No: 15 ( Marks: 1 ) - Please choose one**

**Which of the following primarily uses guided media**

- Cellular telephone system
- **Local telephone system (Page 120)**
- Satellite communication
- Radio broadcasting

**Question No: 16 ( Marks: 1 ) - Please choose one**

**When a beam of light travels through media of two different densities, if the angle of incidence is greater than the critical angle, \_\_\_\_\_ occurs.**

- Reflection
- **Refraction (Page 127)**
- Incidence
- Criticism

**Question No: 17 ( Marks: 1 ) - Please choose one**

**When we talk about unguided media, usually we are referring to \_\_\_\_\_.**

- Metallic wires
- Nonmetallic wires



► **The air (Page 132)**

- Water

**Question No: 18 ( Marks: 1 ) - Please choose one**

Optical fibers are defined by the ratio of the \_\_\_\_\_ of their core to the diameter of their cladding.

► **Diameter (Page 130)**

- Radius
- Length
- Width

**Question No: 19 ( Marks: 1 ) - Please choose one**

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- Conical

► **Barrel (Page 131)**

- Circular
- Rectangular

**Question No: 20 ( Marks: 1 ) - Please choose one**

Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.

- Four
- Three
- Two

► **Five (Page 132)**

**Question No: 21 ( Marks: 1 ) - Please choose one**

dB is \_\_\_\_\_ if a signal is amplified.

- Negative

► **Positive (Page 142)**

- Null
- Zero

**Question No: 22 (Marks: 1) - Please choose one**

A prism can deflect the light depending upon the angle of \_\_\_\_\_ and the frequency.

- Deviation

► **Incident (Page 151)**

- Refraction (Page 127)
- Reflection

**Question No: 23 ( Marks: 1 ) - Please choose one**

Asynchronous TDM is efficient only when the size of the time slot is kept relatively \_\_\_\_\_

► **Large (Page 158)**

- Small
- Medium
- None of the given

**Question No: 24 ( Marks: 1 ) - Please choose one**

The local loop has \_\_\_\_\_ cable that connects the subscriber telephone to the nearest end office.

▶ **Twisted-pair (Page 160)**

- ▶ Coaxial
- ▶ Fiber-optic
- ▶ None of the given

**Question No: 25 ( Marks: 1 ) - Please choose one**

Data from a computer are \_\_\_\_\_; the local loop handles \_\_\_\_\_ signals.

- ▶ Analog; analog
- ▶ Analog; digital
- ▶ Digital; digital

▶ **Digital; analog**

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter9/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter9/multiple_choice_quiz.html)

**Question No: 26 ( Marks: 1 ) - Please choose one**

Which error detection method uses ones complement arithmetic?

- ▶ Simple parity check
- ▶ Two-dimensional parity check
- ▶ CRC

▶ **Checksum (Page 179)**

**Question No: 27 ( Marks: 1 ) - Please choose one**

Flow control is needed to prevent \_\_\_\_\_

- ▶ Bit errors
- ▶ Overflow of the sender buffer
- ▶ **Overflow of the receiver buffer (Page 46)**
- ▶ Collision between sender and receiver

**Question No: 28 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ coordinates the amount of data that can be sent before receiving acknowledgment

▶ **flow control (Page 186)**

- ▶ error control
- ▶ data control

**Question No: 29 ( Marks: 1 ) - Please choose one**

Error control is both error \_\_\_\_\_ and error \_\_\_\_\_

▶ **detection; correction (Page 186)**

- ▶ detection; deletion
- ▶ detection; avoidance
- ▶ detection; forwarding

**Question No: 30 ( Marks: 1 ) - Please choose one**

Addressing is not needed in \_\_\_\_\_ configuration.

► **Point to Point (Page 190)**

- Multipoint
- Point to point and multipoint

**Question No: 31 ( Marks: 1 ) - Please choose one**

In a Go-Back-N ARQ, if the window size is 63, what is the range of sequence number?

► **0 to 63**

- 0 to 64
- 1 to 63
- 1 to 64

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter11/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter11/multiple_choice_quiz.html)

**Question No: 32 ( Marks: 1 ) - Please choose one**

A timer is set when \_\_\_\_\_ is sent out.

- A data frame
- **An ACK (Page 197)**
- A NAK

**Question No: 33 ( Marks: 1 ) - Please choose one**

Sliding window requires that data frames be transmitted \_\_\_\_\_

► **Sequentially (Page 199)**

- Frequently
- Synchronously
- Asynchronously

**Question No: 34 ( Marks: 1 ) - Please choose one**

In Y-MODEM Multiple files can be sent simultaneously

► **True (Page 205)**

- False

**Question No: 35 ( Marks: 1 ) - Please choose one**

BLAST stands for :

► **Blocked asynchronous transmission (Page 205)**

- Blocked synchronous transmission
- Barrel asynchronous transmission
- Below asynchronous transmission

**Question No: 36 ( Marks: 1 ) - Please choose one**

HDLG is an acronym for \_\_\_\_\_.

- High-duplex line communication
- **High-level data link control (Page 210)**
- Half-duplex digital link combination
- Host double-level circuit

**Question No: 37 ( Marks: 1 ) – Please choose one**

The HDLC \_\_\_\_\_ field defines the beginning and end of a frame.

► **Flag (Page 214)**

- Address
- Control
- FCS

**Question No: 38 ( Marks: 1 ) – Please choose one**  
\_\_\_\_\_ is the access protocol used by traditional Ethernet.

► **CSMA/CD (Page 222)**

- CSMA/CA
- Token Ring
- CSMA

**Question No: 39 ( Marks: 1 ) – Please choose one**  
Bridges can divide a large \_\_\_\_\_ into smaller segments

► **Network (Page 241)**

- Packet
- Frame
- Address

**Question No: 40 ( Marks: 1 ) – Please choose one**  
Like VRC, LRC and CRC, Checksum is also based on \_\_\_\_\_.

► **Redundancy (Page 179)**

- Decimal Division
- Encryption
- Encoding

**FINAL TERM EXAMINATION**  
Fall 2009

**Question No: 1 ( Marks: 1 ) – Please choose one**  
The information to be communicated in a data communications system is the

- Medium
- Protocol
- **Message (Page 7)**
- Transmission

**Question No: 2 ( Marks: 1 ) – Please choose one**  
A set of devices connected by communication links is called networking

- **True (Page 12)**
- False

**Question No: 3 ( Marks: 1 ) – Please choose one**

Internet with small “i” specifies the world wide Network the actual internet.

- ▶ True
- ▶ **False (Page 39)**

**Question No: 4 ( Marks: 1 ) – Please choose one**

Data chunk at data link layer is called \_\_\_\_\_

- ▶ frame
- ▶ **packet (Page 38)**
- ▶ datagram

**Question No: 5 ( Marks: 1 ) – Please choose one**

Data synchronization is a function related with \_\_\_\_\_

- ▶ **session layer (Page 51)**
- ▶ presentation layer
- ▶ transport

**Question No: 6 ( Marks: 1 ) – Please choose one**

When data are transmitted from device A to device B, the header from A’s layer 4 is read by B’s \_\_\_\_\_ layer.

- ▶ physical
- ▶ **transport**
- ▶ application
- ▶ none of the given

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter2/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter2/multiple_choice_quiz.html)

**Question No: 7 ( Marks: 1 ) – Please choose one**

Data must be converted into \_\_\_\_\_ before transmission.

- ▶ **signal (Page 56 and 84)**
- ▶ wave
- ▶ electric pulse

**Question No: 8 ( Marks: 1 ) – Please choose one**

PCM is the first process of PAM.

- ▶ True
- ▶ **False (Page 80)**

**Question No: 9 ( Marks: 1 ) – Please choose one**

In \_\_\_\_\_ transmission, bits are transmitted over a single wire, one at a time.

- ▶ Asynchronous serial
- ▶ Synchronous serial
- ▶ **Parallel (Page 98)**
- ▶ Asynchronous & Synchronous serial

**Question No: 10 ( Marks: 1 ) – Please choose one**

EIA 232 allows for a maximum bit rate of \_\_\_\_\_Kbps.

- ▶ 40
- ▶ 30
- ▶ **20 (Page 104)**
- ▶ 10

**Question No: 11 ( Marks: 1 ) – Please choose one**

When a beam of light travels through media of two different densities, if the angle of incidence is greater than the critical angle, \_\_\_\_\_occurs.

- ▶ Reflection
- ▶ **Refraction (Page 127)**
- ▶ Incidence
- ▶ Criticism

**Question No: 12 ( Marks: 1 ) – Please choose one**

A parabolic dish antenna is a(n) \_\_\_\_\_ antenna.

- ▶ Omnidirectional
- ▶ Bidirectional
- ▶ **Unidirectional (Page 138)**
- ▶ Horn

**Question No: 13 ( Marks: 1 ) – Please choose one**

Ultra high-frequency waves always use \_\_\_\_\_ propagation.

- ▶ Ground
- ▶ Sky
- ▶ **Line of Sight (Page 136)**
- ▶ Space

**Question No: 14 ( Marks: 1 ) – Please choose one**

Middle frequency waves having range 300 KHz-3 MHz always use \_\_\_\_\_ propagation.

- ▶ Ground
- ▶ Sky
- ▶ **Line of Sight**
- ▶ Space

<http://www.scribd.com/doc/22810820/Data-Communication-and-Networking> **Page 8 of this book.**

**Question No: 15 ( Marks: 1 ) – Please choose one**  
DB is \_\_\_\_\_ if a signal is attenuated.

- ▶ **Negative (Page 142)**
- ▶ Positive
- ▶ Null
- ▶ Zero

**Question No: 16 ( Marks: 1 ) – Please choose one**  
Which multiplexing technique involves signals composed of light beams?

- ▶ FDM
- ▶ TDM
- ▶ **WDM (Page 150)**
- ▶ none of the given

**Question No: 17 ( Marks: 1 ) – Please choose one**  
Multiplexing has long been used as an essential tool in the \_\_\_\_\_.

- ▶ Electronic industry
- ▶ **Telephone industry (Page 159)**
- ▶ Space science
- ▶ VLAN

**Question No: 18 ( Marks: 1 ) – Please choose one**  
A telephone network is an example of a \_\_\_\_\_ network.

- ▶ Packet-switched
- ▶ **Circuit-switched (Page 38)**
- ▶ Message-switched
- ▶ none of the given

**Question No: 19 ( Marks: 1 ) – Please choose one**  
The local loop has \_\_\_\_\_ cable that connects the subscriber telephone to the nearest end office.

- ▶ **Twisted-pair (Page 160)**
- ▶ Coaxial
- ▶ Fiber-optic
- ▶ None of the given

**Question No: 20 ( Marks: 1 ) – Please choose one**  
Which error detection method involves polynomials?

- ▶ Checksum
- ▶ Two-dimensional parity check
- ▶ **CRC (Page 177)**
- ▶ Simple parity check

**Question No: 21 ( Marks: 1 ) – Please choose one**

In CRC there is no error if the remainder at the receiver is \_\_\_\_\_

- ▶ Equal to the remainder at the sender
- ▶ **Zero (Page 176)**
- ▶ Nonzero
- ▶ The quotient at the sender

**Question No: 22 ( Marks: 1 ) – Please choose one**

In CRC the quotient at the sender \_\_\_\_\_

- ▶ Becomes the dividend at the receiver
- ▶ Becomes the divisor at the receiver
- ▶ **is discarded**
- ▶ is the remainder

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 23 ( Marks: 1 ) – Please choose one**

Which error detection method can detect a bust error?

- ▶ The parity check
- ▶ Two-dimensional parity check
- ▶ CRC
- ▶ **Two-dimensional parity check and CRC**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 24 ( Marks: 1 ) – Please choose one**

At the CRC checker, \_\_\_\_\_ means that the data unit is damaged

- ▶ A string of 0s
- ▶ A string of 1s
- ▶ A string of alternating 1s and 0s
- ▶ **A nonzero remainder**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 25 ( Marks: 1 ) – Please choose one**

Error control is both error \_\_\_\_\_ and error \_\_\_\_\_

- ▶ **detection; correction (Page 186)**
- ▶ detection; deletion
- ▶ detection; avoidance
- ▶ detection; forwarding

**Question No: 26 ( Marks: 1 ) – Please choose one**

In line discipline the initiator first transmits a frame called an \_\_\_\_\_



► **Enquiry (Page 189)**

- Acknowledgment
- NAK
- Request

**Question No: 27 ( Marks: 1 ) – Please choose one**

In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_ frame

► **EOT (Page 189)**

- EKT
- ENT
- ESP

**Question No: 28 ( Marks: 1 ) – Please choose one**

In primary-secondary communication \_\_\_\_\_ is always the initiator of a session

► **Primary (Page 189)**

- Secondary
- Sender
- Receiver

**Question No: 29 ( Marks: 1 ) – Please choose one**

Primary device uses \_\_\_\_\_ to receive transmission from the secondary devices.

- ACK
- ENQ
- **POLL (Page 191)**

**Question No: 30 ( Marks: 1 ) – Please choose one**

In a Go-Back-N ARQ, if the window size is 63, what is the range of sequence number?

- **0 to 63**
- 0 to 64
- 1 to 63
- 1 to 64

[http://highereducation.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter11/multiple\\_choice\\_quiz.html](http://highereducation.mcgraw-hill.com/sites/0072967757/student_view0/chapter11/multiple_choice_quiz.html)

**Question No: 31 ( Marks: 1 ) – Please choose one**

Sliding window requires that data frames be transmitted \_\_\_\_\_

- **Sequentially (Page 199)**
- Frequently
- Synchronously
- Asynchronously

**Question No: 32 ( Marks: 1 ) – Please choose one**

Data link protocols can be divided into \_\_\_\_\_ sub-groups.

▶ **two (Page 202)**

- ▶ three
- ▶ four
- ▶ five

**Question No: 33 ( Marks: 1 ) – Please choose one**

XMODEM is a \_\_\_\_\_ protocol designed for telephone-line communication b/w PCs.

▶ **file transfer (Page 203)**

- ▶ hardware
- ▶ software
- ▶ application exchange

**Question No: 34 ( Marks: 1 ) – Please choose one**

In Y-MODEM Multiple files can be sent simultaneously

▶ **True (Page 205)**

- ▶ False

**Question No: 35 ( Marks: 1 ) – Please choose one**

Which of the following combines features of the other two?

▶ **ZMODEM (Page 205)**

- ▶ YMODEM
- ▶ XMODEM
- ▶ None of given

**Question No: 36 ( Marks: 1 ) – Please choose one**

BLAST stands for :

▶ **Blocked asynchronous transmission (Page 205)**

- ▶ Blocked synchronous transmission
- ▶ Barrel asynchronous transmission
- ▶ Below asynchronous transmission

**Question No: 37 ( Marks: 1 ) – Please choose one**

Which one of the following uses full duplex operation with sliding window flow control.

▶ **BLAST (Page 205)**

- ▶ XMODEM
- ▶ YMODEM
- ▶ ZMODEM

**Question No: 38 ( Marks: 1 ) – Please choose one**

HDLC is an acronym for \_\_\_\_\_.

- ▶ High-duplex line communication
- ▶ **High-level data link control (Page 210)**
- ▶ Half-duplex digital link combination
- ▶ Host double-level circuit

**Question No: 39 ( Marks: 1 ) – Please choose one**

The DSAP and SSAP are addresses used by \_\_\_\_\_ to identify the protocol stacks.

- ▶ **LLC (Page 220)**
- ▶ MAC
- ▶ Network

**Question No: 40 ( Marks: 1 ) – Please choose one**

The PDU has no flag fields, no CRC, and no station address

- ▶ **TRUE (Page 220)**
- ▶ FALSE

**Question No: 41 ( Marks: 1 ) – Please choose one**

Which of the following is a bus topology LAN that uses base band signaling and has a max? segment length of 500 meters

- ▶ **10 Base5 (Page 223)**
- ▶ 10 Base2
- ▶ 100 Base2
- ▶ 100 Base5

**Question No: 42 ( Marks: 1 ) – Please choose one**

Token Bus has no commercial application in data communications

- ▶ **True (Page 232)**
- ▶ False

**Question No: 43 ( Marks: 1 ) – Please choose one**

The BNC-T connector is a T-shaped device with \_\_\_\_\_ ports

- ▶ **Three (Page 228)**
- ▶ Two
- ▶ Four
- ▶ Five

**Question No: 44 ( Marks: 1 ) – Please choose one**

\_\_\_\_\_ is the access protocol used by traditional Ethernet.

► **CSMA/CD (Page 222)**

- CSMA/CA
- Token Ring
- CSMA

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter13/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter13/mixed_quiz.html)

**Question No: 45 ( Marks: 1 ) – Please choose one**

When a collision is detected in a network using CSMA/CD. \_\_\_\_\_

- The frame is immediately resent
- **A jam signal is sent by the station**
- The backoff value is set to 0
- The backoff value is decremented by 1

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter13/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter13/mixed_quiz.html)

**Question No: 46 ( Marks: 1 ) – Please choose one**

Like VRC, LRC and CRC, Checksum is also based on \_\_\_\_\_.

► **Redundancy (Page 179)**

- Decimal Division
- Encryption
- Encoding

**Question No: 47 ( Marks: 1 ) – Please choose one**

Check sum method is used for \_\_\_\_\_ layers.

- Physical
- Application
- Transport

► **Data link** <http://www.cse.iitk.ac.in/users/dheeraj/cs425/lec09.html>

**Question No: 48 ( Marks: 1 ) – Please choose one**

Repeater works on \_\_\_\_\_ layer.

- Data Link
- Physical
- **Network (Page 224)**
- Application

**Question No: 49 ( Marks: 1 ) – Please choose one**

When systems share a point to point link, then protocols used are called \_\_\_\_\_ protocols.

- **Direct (Page 21)**
- Indirect

- ▶ Monolithic
- ▶ structured

**Question No: 50 ( Marks: 1 ) – Please choose one**  
\_\_\_\_\_ requires more bandwidth.

- ▶ **FSK** [http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter5/](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter5/)
- ▶ ASK
- ▶ PSK
- ▶ QAM

**Fall 2009**  
**CS601- Data Communication**

**Question No: 1 ( Marks: 1 ) – Please choose one**  
An unauthorized user is a network \_\_\_\_\_ issue.

- ▶ Performance
- ▶ Reliability
- ▶ **Security (Page 15)**
- ▶ All of the given

**Question No: 2 ( Marks: 1 ) – Please choose one**  
Which is not an element of protocol?

- ▶ semantics
- ▶ timing
- ▶ **communication service module (Page 19)**

**Question No: 3 ( Marks: 1 ) – Please choose one**  
\_\_\_\_\_ is a multipoint topology.

- ▶ Ring
- ▶ Mesh
- ▶ Tree
- ▶ **Bus (Page 31)**

**Question No: 4 ( Marks: 1 ) – Please choose one**  
Unidirectional traffic movement is overcome by dual ring technology.

- ▶ **True (Page 33)**
- ▶ False

**Question No: 5 ( Marks: 1 ) – Please choose one**  
Physical layer defines characteristics of interface between device and \_\_\_\_\_

- ▶ **transmission medium** [http://en.wikipedia.org/wiki/OSI\\_model](http://en.wikipedia.org/wiki/OSI_model)
- ▶ another device

- ▶ another peer physical layer at other side
- ▶ modem

**Question No: 6 ( Marks: 1 ) – Please choose one**

\_\_\_\_\_ layer deals with syntax and semantics of information exchange.

▶ **presentation (Page 51)**

- ▶ session
- ▶ application
- ▶ physical

**Question No: 7 ( Marks: 1 ) – Please choose one**

To allow access to network resources is the function of \_\_\_\_\_

▶ **application layer (Page 53)**

- ▶ physical layer
- ▶ network layer

**Question No: 8 ( Marks: 1 ) – Please choose one**

Time domain plot show changes in signal phase with respect to time.

- ▶ True
- ▶ **False (Page 63)**

**Question No: 9 ( Marks: 1 ) – Please choose one**

In 8QAM each signal shift or one baud represents \_\_\_\_\_.

- ▶ 4 bits
- ▶ 2 bits
- ▶ 5 bits
- ▶ **3 bits (Page 93)**

**Question No: 10 ( Marks: 1 ) – Please choose one**

Modulation of an analog signal can be accomplished through changing the \_\_\_\_\_ of the carrier signal.

- ▶ amplitude
- ▶ frequency
- ▶ phase
- ▶ **all of the given (Page 62)**

**Question No: 11 ( Marks: 1 ) – Please choose one**

EIA 449 provides much better functionality than EIA \_\_\_\_\_

- ▶ **232 (Page 110)**
- ▶ 223

- ▶ 262
- ▶ 222

**Question No: 12 ( Marks: 1 ) – Please choose one**

Which of the following is an example of ITU-T modem standards:

- ▶ T-series
- ▶ X-series
- ▶ N-series
- ▶ **V-series (Page 114)**

**Question No: 13 ( Marks: 1 ) – Please choose one**

Traditional modems are wide spread now to a data rate of \_\_\_\_\_.

- ▶ **56 Kbps (Page 115)**
- ▶ 72 Kbps
- ▶ 42 Kbps
- ▶ 96 Kbps

**Question No: 14 ( Marks: 1 ) – Please choose one**

In case of uploading at the switching station, data is converted to digital signal using \_\_\_\_\_.

- ▶ TCP
- ▶ **PCM (Page 116)**
- ▶ ICP
- ▶ TDM

**Question No: 15 ( Marks: 1 ) – Please choose one**

The RG number gives us information about \_\_\_\_\_.

- ▶ Twisted pairs
- ▶ **Coaxial cables (Page 125)**
- ▶ Optical fibers
- ▶ all of the given

**Question No: 16 ( Marks: 1 ) – Please choose one**

The \_\_\_\_\_ is an association that sponsors the use of infrared waves.

- ▶ **IrDA** [http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter7/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter7/mixed_quiz.html)
- ▶ EIA
- ▶ FCC
- ▶ PUD

**Question No: 17 ( Marks: 1 ) – Please choose one**

Optical fibers are defined by the ratio of the \_\_\_\_\_ of their core to the diameter of their cladding.

► **Diameter (Page 130)**

- Radius
- Length
- Width

**Question No: 18 ( Marks: 1 ) – Please choose one**

The section of EM spectrum defined as Radio Communication is divided into \_\_\_\_\_ ranges called BANDS.

► **8 (Page 132)**

- 10
- 5
- 6

**Question No: 19 ( Marks: 1 ) – Please choose one**

Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.

- Four
- Three
- Two

► **Five (Page 132)**

**Question No: 20 ( Marks: 1 ) – Please choose one**

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

- Ground
- Sky

► **Line of sight (Page 136)**

- Space

**Question No: 21 ( Marks: 1 ) – Please choose one**

In \_\_\_\_\_ propagation, low-frequency radio waves hug the earth.

- Ground
- Sky
- Line of Sight

► **Space (Page 133)**

**Question No: 22 ( Marks: 1 ) – Please choose one**

dB is \_\_\_\_\_ if a signal is amplified.

- Negative

► **Positive (Page 142)**

- Null
- Zero



**Question No: 23 ( Marks: 1 ) – Please choose one**

Distortion occurs in a \_\_\_\_\_ signal.

- ▶ Rectified
- ▶ **Composite (Page 141)**
- ▶ Amplified
- ▶ none of the given

**Question No: 24 ( Marks: 1 ) – Please choose one**

There are \_\_\_\_\_ basic categories of multiplexing.

- ▶ **3 (Page 148)**
- ▶ 4
- ▶ 2
- ▶ 5

**Question No: 25 ( Marks: 1 ) – Please choose one**

In bit \_\_\_\_\_, MUX adds extra bits to a device.

- ▶ **Stuffing (Page 156)**
- ▶ Adding
- ▶ Multiplication
- ▶ Exchanging

**Question No: 26 ( Marks: 1 ) – Please choose one**

The local loop has \_\_\_\_\_ cable that connects the subscriber telephone to the nearest end office.

- ▶ **Twisted-pair (Page 160)**
- ▶ Coaxial
- ▶ Fiber-optic
- ▶ None of the given

**Question No: 27 ( Marks: 1 ) – Please choose one**

FTTC stands for \_\_\_\_\_

- ▶ flexible to the curb
- ▶ **fiber to the curb (Page 166)**
- ▶ fiber to the cable
- ▶ fiber to the center

**Question No: 28 ( Marks: 1 ) – Please choose one**

If the ASCII character G is sent and the character D is received, what type of error is this?

- ▶ Single-bit
- ▶ Multiple-bit
- ▶ **Burst**

- ▶ Recoverable

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 29 ( Marks: 1 ) – Please choose one**

Which error detection method can detect a single-bit error?

- ▶ Simple parity check
- ▶ Two-dimensional parity check
- ▶ CRC

▶ **All of the given**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 30 ( Marks: 1 ) – Please choose one**

Flow control is needed to prevent \_\_\_\_\_

- ▶ Bit errors
- ▶ Overflow of the sender buffer
- ▶ **Overflow of the receiver buffer (Page 46)**
- ▶ Collision between sender and receiver

**Question No: 31 ( Marks: 1 ) – Please choose one**

In data link layer, communication requires at least \_\_\_\_\_ devices working together

- ▶ 3
- ▶ **2 (Page 185)**
- ▶ 4
- ▶ 5

**Question No: 32 ( Marks: 1 ) – Please choose one**

Data link control is composed of \_\_\_\_\_ important functions.

- ▶ 2
- ▶ **3 (Page 185)**
- ▶ 4
- ▶ 5

**Question No: 33 ( Marks: 1 ) – Please choose one**

\_\_\_\_\_ coordinates the amount of data that can be sent before receiving acknowledgment

- ▶ **flow control (Page 186)**
- ▶ error control
- ▶ data control

**Question No: 34 ( Marks: 1 ) – Please choose one**

Primary device uses \_\_\_\_\_ to receive transmission from the secondary devices.

- ▶ ACK
- ▶ ENQ
- ▶ **POLL (Page 191)**

**Question No: 35 (Marks: 1) – Please choose one**

In a Go-Back-N ARQ, if the window size is 63, what is the range of sequence number?

- ▶ **0 to 63**
- ▶ 0 to 64
- ▶ 1 to 63
- ▶ 1 to 64

[http://highereducation.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter11/multiple\\_choice\\_quiz.html](http://highereducation.mcgraw-hill.com/sites/0072967757/student_view0/chapter11/multiple_choice_quiz.html)

**Question No: 36 (Marks: 1) – Please choose one**

Data link protocols can be divided into \_\_\_\_\_ sub-groups.

- ▶ **two (Page 202)**
- ▶ three
- ▶ four
- ▶ five

**Question No: 37 (Marks: 1) – Please choose one**

XMODEM is a \_\_\_\_\_ protocol designed for telephone-line communication b/w PCs.

- ▶ **file transfer (Page 203)**
- ▶ hardware
- ▶ software
- ▶ application exchange

**Question No: 38 (Marks: 1) – Please choose one**

YMODEM uses ITU-T CRC-\_\_\_\_\_ for Error Checking

- ▶ **16 (Page 205)**
- ▶ 32
- ▶ 8
- ▶ 4

**Question No: 39 (Marks: 1) – Please choose one**

In Y-MODEM Multiple files can be sent simultaneously

- ▶ **True (Page 205)**
- ▶ False

**Question No: 40 (Marks: 1) – Please choose one**

HDLC is an acronym for \_\_\_\_\_.

- ▶ High-duplex line communication
- ▶ **High-level data link control (Page 210)**
- ▶ Half-duplex digital link combination
- ▶ Host double-level circuit

**Question No: 41 ( Marks: 1 ) - Please choose one**

What is present in all HDLC control fields?

- ▶ **P/F bit (Page 215)**
- ▶ N(R)
- ▶ N(S)
- ▶ Code bits

**Question No: 42 ( Marks: 1 ) - Please choose one**

Which of the following sublayer, resolves the contention for the shared media

- ▶ **MAC (Page 219)**
- ▶ LLC
- ▶ Physical

**Question No: 43 ( Marks: 1 ) - Please choose one**

Ethernet LANs can support data rates between \_\_\_\_\_

- ▶ **1 and 100 Mbps (Page 223)**
- ▶ 1 and 200 Mbps
- ▶ 1 and 500 Mbps
- ▶ 1 and 100 Gbps

**Question No: 44 ( Marks: 1 ) - Please choose one**

In FDDI, Token Passing is used as Access method.

- ▶ **True (Page 236)**
- ▶ False

**Question No: 45 ( Marks: 1 ) - Please choose one**

Bridges can divide a large \_\_\_\_\_ into smaller segments

- ▶ **Network (Page 241)**
- ▶ Packet
- ▶ Frame
- ▶ Address

**Question No: 46 ( Marks: 1 ) - Please choose one**

Like VRC, LRC and CRC, Checksum is also based on \_\_\_\_\_.

► **Redundancy (Page 179)**

- Decimal Division
- Encryption
- Encoding

**Question No: 47 ( Marks: 1 ) - Please choose one**

Check sum method is used for \_\_\_\_\_ layers.

- Physical
- Application
- Transport

► **Data link** <http://www.cse.iitk.ac.in/users/dheeraj/cs425/lec09.html>

**Question No: 48 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ Which of the following \_\_\_\_\_ uses a series of filters to decompose multiplexed signal into its constituent signals?

- Bridge
- MUX

► **DEMUX (Page 150)**

- Switch

**Question No: 49 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ We need \_\_\_\_\_ to decompose a composite signal into its components.

► **Fourier transform (Page 64)**

- nyquist theorem
- shannon capacity

**Question No: 50 ( Marks: 1 ) - Please choose one**

Data from computer is in \_\_\_\_\_ form and the local loop handles \_\_\_\_\_ signals.

- Analog; analog
- Analog; digital
- Digital; digital

► **Digital; analog**

[http://highered.mcgraw-hill.com/sites/0072967757/student\\_view0/chapter9/multiple\\_choice\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072967757/student_view0/chapter9/multiple_choice_quiz.html)

**FINALTERM EXAMINATION**  
Spring 2009

**Question No: 1 ( Marks: 1 ) - Please choose one**

There are how many factors on which the performance of a network depends?

- ▶ Three
- ▶ **Five (Page 14)**
- ▶ Four
- ▶ Two

**Question No: 2 (Marks: 1) - Please choose one**  
\_\_\_\_\_requires the maximum number of I/O ports.

- ▶ Bus
- ▶ Star
- ▶ **Mesh (Page 29)**
- ▶ Ring

**Question No: 3 (Marks: 1) - Please choose one**  
Headers are added at layers 1 and 7 of OSI model.

- ▶ True
- ▶ **False (Page 41)**

**Question No: 4 (Marks: 1) - Please choose one**  
Trailer is only added at \_\_\_\_\_layer of OSI model.

- ▶ **data link (Page 41)**
- ▶ physical
- ▶ network
- ▶ application

**Question No: 5 (Marks: 1) - Please choose one**  
A sine wave must be an analog signal.

- ▶ **True (Page 58)**
- ▶ False

**Question No: 6 (Marks: 1) - Please choose one**  
If there is \_\_\_\_\_in voltage then the frequency is infinite.

- ▶ **instantaneous change (Page 61)**
- ▶ continuous change
- ▶ no change

**Question No: 7 (Marks: 1) - Please choose one**  
The inversion of the level at 1 bit is called as \_\_\_\_\_

- ▶ NRZ-L
- ▶ **NRZ-I (Page 73)**

► RZ

**Question No: 8 ( Marks: 1 ) - Please choose one**

The last process in PCM is \_\_\_\_\_ digital data into digital signal.

► **encoding (Page 82)**

► decoding

► modulating

**Question No: 9 ( Marks: 1 ) - Please choose one**

Amplitude in ASK is more resistive to EMI and Noise.

► True

► **False (Page 87)**

**Question No: 10 ( Marks: 1 ) - Please choose one**

If FCC regulations are followed, the carrier frequencies of adjacent AM radio stations are \_\_\_\_\_ apart.

► 5 KHz

► **10 KHz (Page 95)**

► 200 KHz

► 530 KHz

**Question No: 11 ( Marks: 1 ) - Please choose one**

In \_\_\_\_\_ transmission, a start bit and a stop bit frame a character byte.

► **Asynchronous serial (Page 99)**

► Synchronous serial

► Parallel

► Asynchronous & Synchronous serial

**Question No: 12 ( Marks: 1 ) - Please choose one**

Synchronous transmissions have \_\_\_\_\_.

► a start bit

► a stop bit

► gaps between bytes

► **none of the given (Page 99)**

**Question No: 13 ( Marks: 1 ) - Please choose one**

Transmission media are usually categorized as \_\_\_\_\_

► Fixed or Unfixed

► **Guided or Unguided (Page 119)**

► Determinate or Indeterminate

- ▶ Metallic or Nonmetallic

**Question No: 14 ( Marks: 1 ) - Please choose one**

The RG number gives us information about \_\_\_\_\_.

- ▶ Twisted pairs
- ▶ **Coaxial cables (Page 125)**
- ▶ Optical fibers
- ▶ all of the given

**Question No: 15 ( Marks: 1 ) - Please choose one**

Radio wave and microwave frequencies range from \_\_\_\_\_.

- ▶ 3 to 300 KHz
- ▶ **300 KHz to 300GHz (Page 132)**
- ▶ 3 KHz to 300 GHz
- ▶ 3 KHz to 3000 GHz

**Question No: 16 ( Marks: 1 ) - Please choose one**

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- ▶ Conical
- ▶ **Barrel (Page 131)**
- ▶ Circular
- ▶ Rectangular

**Question No: 17 ( Marks: 1 ) - Please choose one**

In \_\_\_\_\_ method a signal can be directed in a straight from Antenna to antenna.

- ▶ **Line of sight (Page 134)**
- ▶ Ground propagation
- ▶ Sky propagation
- ▶ Microwaves

**Question No: 18 ( Marks: 1 ) - Please choose one**

Amplifiers are used to \_\_\_\_\_ the signal to heat.

- ▶ **Amplify (Page 142)**
- ▶ Rectify
- ▶ Testify
- ▶ Nullify

**Question No: 19 ( Marks: 1 ) - Please choose one**

There are \_\_\_\_\_ basic categories of multiplexing.

- ▶ **3 (Page 148)**



- ▶ 4
- ▶ 2
- ▶ 5

**Question No: 20 ( Marks: 1 ) - Please choose one**

If a T-1 carries 8000 frames, the data rate is \_\_\_\_\_

- ▶ 2.544 Mbps
- ▶ **1.544 Mbps (Page 165)**
- ▶ 1.544 Kbps
- ▶ 1.544 Gbps

**Question No: 21 ( Marks: 1 ) - Please choose one**

FTTC stands for \_\_\_\_\_

- ▶ flexible to the curb
- ▶ **fiber to the curb (Page 166)**
- ▶ fiber to the cable
- ▶ fiber to the center

**Question No: 22 ( Marks: 1 ) - Please choose one**

Optical signals are multiplexed using \_\_\_\_\_ at switching office to create wider BW optical signals

- ▶ **WDM (Page 167)**
- ▶ FDM
- ▶ TDM
- ▶ MUX

**Question No: 23 ( Marks: 1 ) - Please choose one**

Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?

- ▶ Simple parity check
- ▶ **Two-dimensional parity check**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

- ▶ CRC
- ▶ Checksum

**Question No: 24 ( Marks: 1 ) - Please choose one**

If the ASCII character G is sent and the character D is received, what type of error is this?

- ▶ Single-bit
- ▶ Multiple-bit
- ▶ **Burst**

- ▶ Recoverable

**Question No: 25 ( Marks: 1 ) - Please choose one**

Flow control is needed to prevent \_\_\_\_\_

- ▶ Bit errors
- ▶ Overflow of the sender buffer
- ▶ **Overflow of the receiver buffer. (Page 46)**
- ▶ Collision between sender and receiver

**Question No: 26 ( Marks: 1 ) - Please choose one**

In data link layer, communication requires at least \_\_\_\_\_ devices working together

- ▶ 3
- ▶ **2 (Page 185)**
- ▶ 4
- ▶ 5

**Question No: 27 ( Marks: 1 ) - Please choose one**

Data link control is composed of \_\_\_\_\_ important functions.

- ▶ 2
- ▶ **3 (Page 185)**
- ▶ 4
- ▶ 5

**Question No: 28 ( Marks: 1 ) - Please choose one**

Error control is both error \_\_\_\_\_ and error \_\_\_\_\_

- ▶ **detection; correction (Page 186)**
- ▶ detection; deletion
- ▶ detection; avoidance
- ▶ detection; forwarding

**Question No: 29 ( Marks: 1 ) - Please choose one**

ENQ/ACK stands for \_\_\_\_\_

- ▶ **Enquiry/ Acknowledgment (Page 187)**
- ▶ Enque/ Acknowledgment
- ▶ Enquist/ Acknowledgment
- ▶ none of the given

**Question No: 30 ( Marks: 1 ) - Please choose one**

In \_\_\_\_\_ ARQ, if a NAK is received, only the specific damaged or lost frame is retransmitted.

- ▶ Stop-and-wait
- ▶ Go-Back-N
- ▶ **Selective reject (Page 200)**
- ▶ Stop-and-wait & Go-back-N

**Question No: 31 ( Marks: 1 ) - Please choose one**

Sliding window requires that data frames be transmitted \_\_\_\_\_

- ▶ **Sequentially (Page 199)**
- ▶ Frequently
- ▶ Synchronously
- ▶ Asynchronously

**Question No: 32 ( Marks: 1 ) - Please choose one**

Which of the following combines features of the other two?

- ▶ **ZMODEM (Page 205)**
- ▶ YMODEM
- ▶ XMODEM
- ▶ None of given

**Question No: 33 ( Marks: 1 ) - Please choose one**

BLAST stands for :

- ▶ **Blocked asynchronous transmission (Page 205)**
- ▶ Blocked synchronous transmission
- ▶ Barrel asynchronous transmission
- ▶ Below asynchronous transmission

**Question No: 34 ( Marks: 1 ) - Please choose one**

HDLC is an acronym for \_\_\_\_\_.

- ▶ High-duplex line communication
- ▶ **High-level data link control (Page 210)**
- ▶ Half-duplex digital link combination
- ▶ Host double-level circuit

**Question No: 35 ( Marks: 1 ) – Please choose one**

The HDLC \_\_\_\_\_ field defines the beginning and end of a frame.

- ▶ **Flag (Page 214)**
- ▶ Address
- ▶ Control
- ▶ FCS

**Question No: 36 ( Marks: 1 ) – Please choose one**

The shortest frame in HDLC protocol is usually the \_\_\_\_\_ frame.

- ▶ Information
- ▶ **Supervisory (Page 213)**
- ▶ Management
- ▶ None of the given

**Question No: 37 ( Marks: 1 ) - Please choose one**

The PDU has no flag fields, no CRC, and no station address

- ▶ **TRUE (Page 220)**
- ▶ FALSE

**Question No: 38 ( Marks: 1 ) - Please choose one**

\_\_\_\_\_ is the access protocol used by traditional Ethernet.

- ▶ **CSMA/CD (Page 222)**
- ▶ CSMA/CA
- ▶ Token Ring
- ▶ CSMA

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No: 39 ( Marks: 1 ) - Please choose one**

Repeater is an amplifier, not a regenerator.

- ▶ True
- ▶ **False (Page 240)**

**Question No: 40 ( Marks: 1 ) - Please choose one**

Bridges can divide a large \_\_\_\_\_ into smaller segments

- ▶ **Network (Page 241)**
- ▶ Packet
- ▶ Frame
- ▶ Address

**CS601 Data Communication  
Final Term Examination – Spring 2006**

**Question No. 1 Marks : 2**

**Which error detection method uses ones complement arithmetic?**

- Simple parity check
- Two-dimensional parity check
- CRC
- **Checksum (Page 179)**

**Question No. 2 Marks : 2**

**In a time-domain plot, the vertical axis is the measure of \_\_\_\_\_.**

- **Amplitude (Page 59)**
- Frequency
- Phase
- Time

**Question No. 4 Marks : 2**

**Which topology requires a multipoint connection?**

- Mesh
- Star
- **Bus (Page 31)**
- Ring

**Question No. 8 Marks : 2**

**A timer is set when \_\_\_\_\_ is (are) sent out.**

- A data frame
- **An ACK (Page 197)**
- A NAK
- An ARQ

**Question No. 9 Marks : 2**

**If the data unit is 11111, the divisor 1010, and the remainder 110, what is the dividend at the receiver?**

- 111111011
- **111111110**
- 1010110
- 110111111

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

## **FINAL TERM EXAMINATION**

### **FALL 2006**

Question No: 1 ( Marks: 1 ) - Please choose one

**OSI standard is practically implemented in the form of internet.**

- ▶ TRUE
- ▶ FALSE

Question No: 2 ( Marks: 1 ) - Please choose one

**TCP/IP application layers combine the functions of OSI Application, Session and Transport layers.**

- ▶ **TRUE (Page 55)**
- ▶ FALSE

Question No: 3 ( Marks: 1 ) - Please choose one

**Unipolar use one voltage levels.**

▶ **TRUE (Page 71)**

▶ FALSE

Question No: 4 ( Marks: 1 ) - Please choose one

**The material of core and cladding in fiber optic cable is same.**

▶ TRUE

▶ **FALSE (Page 127)**

Question No: 5 ( Marks: 1 ) - Please choose one

**Burst error is easily corrected then bit error.**

▶ TRUE

▶ FALSE

Question No: 6 ( Marks: 1 ) - Please choose one

**Set of rules governing communication is known as**

▶ OSI model

▶ **Protocol (Page 08)**

▶ Medium

Question No: 7 ( Marks: 1 ) - Please choose one

**WDM(wave division multiplexing) deals with**

▶ electric waves

▶ **light waves (Page 150)**

▶ both

Question No: 8 ( Marks: 1 ) - Please choose one

**A Null modem is a**

▶ traditional modem

▶ **technique to connect two DTE (Page 106)**

▶ both

Question No: 9 ( Marks: 1 ) - Please choose one

**Router work on**

▶ data link layer

▶ physical layer

▶ **network layer (Page 48)**

Question No: 10 ( Marks: 1 ) - Please choose one

**Poll/Select is a technique related with**

▶ line control

▶ **line discipline (Page 187)**

- ▶ error control

## Final Term Examination – Spring 2005

**Question No. 1 Marks : 02**

**In cyclic redundancy checking, what is the CRC?**

- ▶ the divisor
- ▶ the quotient
- ▶ the dividend
- ▶ **the remainder**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No. 2 Marks : 02**

**In fiber optics the signal source is \_\_\_\_\_ waves.**

- ▶ **Light (Page 126)**
- ▶ Radio
- ▶ Very Low frequency
- ▶ Infrared

**Question No. 3 Marks : 02**

**Each device has a dedicated point – to – point line configuration only with the two devices on either side of it.**

- ▶ **Ring Topology (Page 33)**
- ▶ Star Topology
- ▶ Mesh Topology
- ▶ Tree Topology

**Question No. 6 Marks : 02**

**Measures the relative strengths of two signals**

- ▶ **Decibel (Page 141)**
- ▶ Bandwidth
- ▶ Wavelength
- ▶ Phase

**Question No. 7 Marks : 02**

**Which multiplexing technique shifts each signal to a different carrier frequency?**

- ▶ **FDM (Page 149)**

- ▶ Synchronous TDM
- ▶ Asynchronous TDM
- ▶ None of the above

**Question No. 8 Marks : 02**

The \_\_\_\_\_ of a signal is the width of the frequency spectrum.

- ▶ Amplitude
- ▶ **Bandwidth (Page 66)**
- ▶ Bit interval
- ▶ Phase

## Final Term Examination – Spring 2005

**Question No. 1**

In cyclic redundancy checking, what is the CRC?    Marks : 02

- ▶ the divisor
- ▶ the quotient
- ▶ the dividend
- ▶ **the remainder**

[http://highered.mcgraw-hill.com/sites/0072515848/student\\_view0/chapter10/mixed\\_quiz.html](http://highered.mcgraw-hill.com/sites/0072515848/student_view0/chapter10/mixed_quiz.html)

**Question No. 2**

In fiber optics the signal source is \_\_\_\_\_ waves.    Marks : 02

- ▶ **Light (Page 126)**
- ▶ Radio
- ▶ Infrared
- ▶ Very Low frequency

**Question No. 3 Marks : 02**

Each device has a dedicated point – to – point line configuration only with the two devices on either side of it.

- ▶ **Ring Topology (Page 33)**
- ▶ Mesh Topology
- ▶ Star Topology
- ▶ Tree Topology

**Question No. 6**

Measures the relative strengths of two signals    Marks : 02



► **Decibel (Page 141)**

- Bandwidth
- Phase
- Wavelength

**Question No. 7    Marks : 02**

**Which multiplexing technique shifts each signal to a different carrier frequency?**

► **FDM (Page 149)**

- Synchronous TDM
- Asynchronous TDM
- None of the above

**Question No. 8**

**The \_\_\_\_\_ of a signal is the width of the frequency spectrum.    Marks : 02**

- Amplitude
- **Bandwidth (Page 66)**
- Bit interval
- Phase



# CS601-Data Communication

(Solved MCQ's)

**LECTURE FROM**  
**(23 to 45)**



[Junaidfazal08@gmail.com](mailto:Junaidfazal08@gmail.com)  
[Bc190202640@vu.edu.pk](mailto:Bc190202640@vu.edu.pk)

FOR MORE VISIT  
VULMSHELP.COM

**JUNAID MALIK**  
**0304-1659294**

## AL-JUNAID TECH INSTITUTE



[www.vulmshelp.com](http://www.vulmshelp.com)



### Language Courses Training Available

I'm providing paid courses in different languages within 3 Months, Certificate will be awarded after completion.

- HTML
- CSS
- JAVASCRIPT
- BOOTSTRAPS
- JQUERY
- PHP MYSQL
- NODES.JS
- REACT JS

### LMS Handling Services

### LMS Activities Paid Task

Assignments 95% Results

Quizes 95% Results

GDB 95% Results

For CS619 Project Feel Free To Contact With Me

**Ph# 0304-1659294**  
**Email: [junaidfazal08@gmail.com](mailto:junaidfazal08@gmail.com)**

# **AL-JUNAID INSTITUTE GROUP**

**ALL answers are verified if found any mistake then Correct ACCORDINGLY**

## **Question No: 1**

Ethernet Address is of----- bytes

- 4
- **6**
- 8
- 12

## **Question No: 2**

In Pure ALOHA, the vulnerable time is -----the frame transmission time.

- Same as
- **Two times**
- Three times
- Four times

## **Question No: 3**

\_\_\_\_\_ was used as a medium in 10 Base-T

- Thick coaxial Cable
- Think coaxial Cable
- **Twisted Pair Cable**
- Two Mode For Cable

## **Question No: 4**

When data are transmitted from device A to device B, the header from A's layer 4 is read by B's \_\_\_\_\_ layer.

- **physical**
- transport
- application
- none of the given

## **Question No: 5**

In \_\_\_\_\_ encoding, we use three levels: positive, zero, and negative.

- polar
- nonpolar
- mmultilevel
- **bipolar**

## **Question No: 6**

A periodic signal completes one cycle in 0.001 s. What is the frequency?

- 1 Hz
- **100 Hz**
- 1 KHz
- 1 MHz

# AL-JUNAID TECH INSTITUTE

## Question No: 7

There are \_\_\_\_\_ basic modes for the propagation of light in fiber-optic cables.

- 3
- 4
- 5
- 2

## Question No: 8

A parabolic dish antenna is a(n) \_\_\_\_\_ antenna.

- Omnidirectional
- Bidirectional
- Horn
- Unidirectional

## Question No: 9

Zero crossing bandwidth is also called as equivalent noise bandwidth.

- True
- False

## Question No: 10

During the transmission, transmitting device produces a high frequency signal which acts as a basis for the transmission. This base signal is called \_\_\_\_\_.

- Analog signal
  - Digital signal
  - Modulated signal
  - Carrier signal
- Not Confirm

## Question No 11:

There are \_\_\_\_\_ types of serial transmission:

- 1
- 2
- 3
- 4

## Question No 12:

Synchronous transmission have \_\_\_\_\_.

- a start bit
- a stop bit
- gaps between bytes
- none of the given

## Question No 13

Using \_\_\_\_\_ encoding scheme in digital transmission, we represent 0 by zero voltage level and represent 1 by any positive voltage level.

- Polar
- AMI
- Bipolar
- Unipolar

## Question No 14

# AL-JUNAID TECH INSTITUTE

\_\_\_\_\_ is the formula for finding number of bits sent per level.

- Log L
- Log2
- Log2\*2
- Log2 L

## Question No 15

Which of the following is an example of ITU-T modem standards:?

- T-series
- X-series
- N-series
- V-series

## Question No 16

The maximum data rate in the uploading direction is still\_\_\_\_\_.

- 26.6 Kbps
- 33.6 Kbps
- 36.6 Kbps
- 46.6 Kbps

## Question No 16

In\_\_\_\_\_ each station is forced to send a frame only at beginning of time slot

- Pure ALOHA
- Slotted ALOHA
- Fragmented ALOHA
- None of the given

## Question 17

In\_\_\_\_\_ each station sends a frame whenever it has a frame to send

- Pure ALOHA
- Slotted ALOHA
- Fragmented ALOHA
- None of the given

## Question No 18

The data rate of a T-1 line is\_\_\_\_\_.

- 1.544 Gbps
- 2.544 Mbps
- 1.544 Kbps
- 1.544 Mbps

## Question 19

Circuit switched networks have\_\_\_\_\_ efficiency but\_\_\_\_\_ delay compared to the packet switched networks.

- high, minimal

# AL-JUNAID TECH INSTITUTE

- high, high
- low, minimal

## Question No 20

Category 5 UTP cable is used for data transmission of upto\_\_\_\_\_.

- 100 Mbps
- 200 Mbps
- 250 Mbps
- 400 Mbps

## Question No 21

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- Conical
- Barrel
- Circular
- Rectangular

## Question No 22

SONET stands for \_\_\_\_\_

- synchronous optical network
- synchronous operational network
- stream optical network
- shell operational network

## Question No 23

Fast Ethernet has a data rate of \_\_\_\_\_ Mbps.

- 10
- 100
- 1000
- 10,000

## Question No 24

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

- Ground
- Sky
- Line of sight
- Space

## Question No 25

In Bit-oriented framing, \_\_\_\_\_ bit pattern is used as a flag to define beginning and end of a frame.

- $(11111111)_2$
- $(10000001)_2$
- Single
- $(01111110)_2$

## Question No 26

Traditionally, \_\_\_\_\_ protocols have been defined for the data-link layer to deal with flow and error control.

- Two
- Three



# AL-JUNAID TECH INSTITUTE

- Five
- **Four**

Question No 27

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across \_\_\_\_\_ data link

- **Single**
- Multi
- Single and Multi
- none of the given

Question No 28

Need for Addressing makes Asynchronous TDM inefficient for bit or byte \_\_\_\_\_.

- Interleaving
- **Addition**
- Substraction
- None of the given

Question No 29

\_\_\_\_\_ takes data from one high speed line and breaks it into portions.

- ▶ **Multiplexing**
- ▶ Inverse multiplexing
- ▶ Inverse subtraction
- ▶ Inverse addition

Question No 30

For Carrier Sense Multiple Access/Collision Detection (CSMA/CD) we need a restriction on the \_\_\_\_\_

- Collision Size
- Signal Size
- **Frame Size**
- Station Size

Question No 31:

In Y-MODEM Multiple files can be sent simultaneously

- ▶ **True**
- ▶ False

Question No 32

At the data-link layer, \_\_\_\_\_ protocol is designed to handle both flow and error control, but communication is one frame at a time.

- Simple
- Go-Back-N
- Selective-Repeat
- **Stop-and-Wait**

Question No 33

# AL-JUNAID TECH INSTITUTE

The PPP protocol uses \_\_\_\_\_, which is responsible for establishing, maintaining, configuring, and terminating links.

- Internet Protocol Control Protocol (IPCP)
- Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocols (NCPs)
- **Link Control Protocol (LCP)**

Question No 34

Aloha, CSMA/CD and CSMA/CA are part of \_\_\_\_\_.

Controlled Access protocol

Channelization protocol

Mutual access protocol

**Random Access protocol**

Question No 35

Which of the following is an example of ITU-T modem standards:

- **T-series**
- X-series
- N-series
- V-series

Question No 36

The maximum data rate in the uploading direction is still \_\_\_\_\_.

- 26.6 Kbps
- **33.6 Kbps**
- 36.6 Kbps
- 46.6 Kbps

Question No 37

Category 5 UTP cable is used for data transmission of upto \_\_\_\_\_.

- **100 Mbps**
- 200 Mbps
- 250 Mbps
- 400 Mbps

Question No 38

ALOHA the earliest random access method was developed in early \_\_\_\_\_.

1950s

1960s

1980s

**1970s**

Question No 39

In block coding, we divide our message into blocks, each of \_\_\_\_\_ bits; and add \_\_\_\_\_ redundant bits to each block to make the codewords of \_\_\_\_\_ bit length.

m, r, c



# AL-JUNAID TECH INSTITUTE

n, k, r

n, r, k

k, r, n

## Question No 40

All of popular Fiber optic connectors are \_\_\_\_\_ shaped.

- Conical
- **Barrel**
- Circular
- Rectangular

## Question No 41

MAC stand for \_\_\_\_\_

- Media Access Controller
- Main Access Control
- Mandatory Access Control
- **Media Access Control**

## Question No 42

Which one of the following is a Random-Access protocol

- CDMA/CA
- **ALOHA OR CSMA/CD Both are correct**
- CDMA/CD
- TDMA

## Question No 43

The VLF and LF bands use \_\_\_\_\_ propagation for communications.

- Ground
- **Sky**
- Line of sight
- Space

## Question No 44

Multiplexing is the set of techniques that allows simultaneous TX of multiple signals across \_\_\_\_\_ data link

- **Single**
- Multi
- Single and Multi
- none of the given

## Question No 45

Need for Addressing makes Asynchronous TDM inefficient for bit or byte \_\_\_\_\_.

- Interleaving
- **Addition**
- Subtraction
- None of the given

## Question No 46

CDMA stands for \_\_\_\_\_

- Carrier Data Multiple Access
- **Code Division Multiple Access**
- Code Data Multiple Access
- Carrier Division Multiple Access

## Question No 47

# AL-JUNAID TECH INSTITUTE

In Ethernet, the source address field in the MAC frame is the \_\_\_\_\_ address.

- Sender's logical
- Previous station's physical
- Next destination's physical
- Sender's service port

Question No 48

\_\_\_\_\_ takes data from one high speed line and breaks it into portions.

- Multiplexing
- Inverse multiplexing
- Inverse subtraction
- Inverse addition

Question No 49

A traditional telephone line has a bandwidth of \_\_\_\_\_.

- ▶ 2000Hz
- ▶ 4000 Hz
- ▶ 2000 MHz
- ▶ 4000 MHz

Question No 50

To detect five errors, the Hamming distance between each pair of codewords should be at least \_\_\_\_\_.

- 4
- 5
- 11
- 6

Question No 51

For FEC using XOR operation, we divide a packet into N chunks, and need to send \_\_\_\_\_ chunks to the receiver side.

- N
- 2N
- $(N+1)/2$
- N+1

Question No 52

If the ASCII character H is sent and the character I is received, what type of error is this?

- Single-bit
- Multiple-bit
- Burst
- Recoverable

Question No 53

At the CRC generator, \_\_\_\_\_ added to the data unit after the division process

- 0s are
- 1s are
- The polynomial
- The CRC remainder is

Question No 54

Error control in the data link layer is based on \_\_\_\_\_

# AL-JUNAID TECH INSTITUTE

- ▶ automatic repeat request
- ▶ automatic repeat acknowledgment
- ▶ automatic send acknowledgment

Question No 55

In a technique called, compounding High-and-Low Resolution Packets, the low-resolution section in the first packet is \_\_\_\_\_.

- ▶ CRC Not Sure
- ▶ Parity code
- ▶ Empty

Question No 56

In line discipline the initiator first transmits a frame called an \_\_\_\_\_

- Enquiry
- Acknowledgment
- NAK
- Reques

Question No 57

Primary device uses \_\_\_\_\_ to receive transmission from the secondary devices.

- ACK Not Sure
- ENQ
- POLL

Question N0 58

Token Bus has no commercial application in data communications

- True
- False

Question No 59

HDLC is an acronym for \_\_\_\_\_.

- High-duplex line communication
- High-level data link control
- Half-duplex digital link combination
- Host double-level circuit

Question No 59

The BNC-T connector is a T-shaped device with \_\_\_\_\_ ports

- Three
- Two
- Four
- Five

Question No 60

Each station in the Token Ring regenerates the frame.

- True
- False

# AL-JUNAID TECH INSTITUTE

## Question No 61

Repeater works on \_\_\_\_\_ layer.

- Data Link
- Physical
- **Network**
- Application

## Question No 62

Trunks are transmission media such as \_\_\_\_\_ that handle the telephone to the nearest end office.

- Satellite links
- **Twisted-pair & Fiber-optic**
- Twisted-pair
- Fiber-optic

## Question No 63

At the data-link layer, \_\_\_\_\_ protocol is designed to handle both flow and error control, but communication is one frame at a time.

- Simple
- Go-Back-N
- Selective-Repeat
- **Stop-and-Wait**

## Question No 64

The PPP protocol uses \_\_\_\_\_, which is responsible for establishing, maintaining, configuring, and terminating links.

- Internet Protocol Control Protocol (IPCP)
- Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocols (NCPs)
- **Link Control Protocol (LCP)**

## Question No 65

The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?

- **Standard Ethernet**
- Fast Ethernet
- Gigabit Ethernet
- 10 Gigabit Ethernet

## Question No 66

Normally the value of Kmax in pure ALOHA is \_\_\_\_\_.

- 10
- **15**
- 25
- 30

## Question No 67

Which one of the following is not a Channelization Protocol?

- **CSMA**
- CDMA

# AL-JUNAID TECH INSTITUTE

➤ FDMA

➤ TDMA

Question No 68

In CSMA/CA \_\_\_\_\_ is the amount of time divided into slots.

➤ Contention Window

➤ Interframe Window

➤ Fragmented Window

➤ Collided Window

Question No 69

Aloha, CSMA/CD and CSMA/CA are part of \_\_\_\_\_.

Controlled Access protocol

Channelization protocol

Mutual access protocol

Random Access protocol

Question No 70

ALOHA the earliest random access method was developed in early \_\_\_\_\_.

• 1950s

• 1960s

• 1980s

• 1970s

Question No 71

For FEC using XOR operation, we divide a packet into N chunks, and need to send \_\_\_\_\_ chunks to the receiver side.

N

2N

$(N+1)/2$

N+1

Question No 71

The \_\_\_\_\_ layer is responsible for the creation and delivery of a frame to another node, along the link.

➤ Application

➤ Network

➤ Physical

➤ Data-link

Question No 72

IP addresses are the identifiers at the \_\_\_\_\_ layer.

Physical

Data-link

Transport

➤ Network

Question No 73

Each host or each interface of a router is assigned \_\_\_\_\_ address.

➤ A multicast

➤ A broadcast

➤ Both multicast or broadcast

# AL-JUNAID TECH INSTITUTE

## ➤ A unicast

Question No 74

The ARP protocol is one of the auxiliary protocols defined in the \_\_\_\_\_ layer, which maps an IP address to a logical-link address.

- Physical
- Data-link
- Transport
- Network**

Question No 75

Following are the examples of connecting devices EXCEPT \_\_\_\_\_.

- Link Layer Switches
- Hub
- Routers
- **Wire**

Question No 76

\_\_\_\_\_ eliminates collision and connect heterogeneous devices.

- Hub
- Repeater
- Bridge
- **Switch**

Question No 77

\_\_\_\_\_ creates looping problem in learning switch.

- Unicast
- Multicast
- None of the Given
- **Broadcast**

Question No 78

The term FTTC stands for \_\_\_\_\_.

Front Technology Terminal Connection

Free Technology Transmission Connection

Fiber Terminal Transmission Commission

**Fiber to the Curb**

Question No 79

Multiple-access Protocol is divided into \_\_\_\_\_.

- Two categories
- Four categories
- Five categories
- Three categories**

# AL-JUNAID TECH INSTITUTE

Question No 80

The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_ Ethernet.

- Fast
- Gigabit
- 10 Gigabit
- Standard**

Question No 81

CSMA/CD is not used in \_\_\_\_\_ Ethernet.

- Standard
- Fast
- Gigabit
- 10 Gigabit**

Question No 82

The amplitude of a digital signal depends upon the \_\_\_\_\_ to represent a bit.

- Phase
- Wavelength
- Bandwidth
- Voltage**

Question No 83

In IEEE 802.11, a station with \_\_\_\_\_ mobility can move from one BSS to another, but the movement is confined inside one ESS.

- None of the choices are correct
- No-transition
- ESS-transition**
- BSS-transition

Question No 84

In IEEE 802.11, communication between two stations in two different BSSs usually occurs via two \_\_\_\_\_.

- BSSs
- ESSs
- None of the given

**Aps**

Question No 89

A property of a signal where it is being received by receiver after reflection from different intermediate objects (Building, iron polls, walls) is called \_\_\_\_\_

- ☐ **Attenuation**
- ☐ Interference
- ☐ Error
- ☐ Multipath Propagation

Question No 90

\_\_\_\_\_ medium is used in wireless LANs for transportation of data.

Copper pair cable

# AL-JUNAID TECH INSTITUTE

Optical fiber cable

Coaxial cable

**Air**

Question No 91

In context of control signaling network, what does STP stand for \_\_\_\_\_

- Shielded twisted pair
- Signaling transmission point
- Signal transportation procedure
- **Signal transfer point**

Question No 92

POP stands for \_\_\_\_\_

- Presentation of points
- Points of points
- Points of packets
- **Points of Presence**

Question No 93

What does LATA stand for \_\_\_\_\_

- Local Area Transmission Arena
- Local-Access Transit Access
- Local-Areas Transport Areas
- **Local-Access Transport Areas**

Question No 94

Unipolar encoding scheme is not supported over the medium having \_\_\_\_\_.

- The capability of handling DC component
- Low speed
- High speed
- **No capability of handling DC component**

Question No 95

The amplitude of a digital signal depends upon the \_\_\_\_\_ to represent a bit.

- ▶ Phase
  - ▶ Wavelength
  - ▶ Bandwidth
  - ▶ **Voltage**

Question No 96

Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol

- Application
- Network
- Session
- **Transport**



# AL-JUNAID TECH INSTITUTE

Question No 97

In \_\_\_\_\_ each station is allotted a time slot during which it can send data

- NDMA
- CDMA
- TDMA
- FDMA

Question No 98

In \_\_\_\_\_ collisions are avoided by deferring transmission even if the channel is idle

- The Slotted Frame
- The Deployment Frame
- The Contention Window
- The Interframe Space

Question No 99

In the context of Ethernet Implementations, maximum medium length of 10 Base-F was \_\_\_\_\_ meters.

- 500
- 1000
- 2000
- 150

Question No 100

Collisions in CSMA/CA are avoided through \_\_\_\_\_

- The Interframe Space
- The Contention Window
- Acknowledgments
- All of the given

Question No 101

The extra bits added with the original data for error detection/correction are called

- ▶ Process IDs
- ▶ Redundant bits
- ▶ Fault bits
- ▶ Port Number

Question No 102

\_\_\_\_\_ are used to exchange session management and control between connected devices.

- ▶ U-frames
- ▶ N-frames
- ▶ S-frames
- ▶ I-frames

Question No 103

The message 1110 sent by source is received by a destination as 1011. This is \_\_\_\_\_ types of err.

- ▶ Hamming

# **AL-JUNAID TECH INSTITUTE**

- ▶ Single-Bit
- ▶ Uni-Bit
- ▶ **Brust**

Question No 104

In Block code we divide our message into blocks of K bits called as \_\_\_\_\_

- ▶ Password
- ▶ Code word
- ▶ **Data words**
- ▶ Cross words

Question No 105

DLC in Data Link Layer stand for \_\_\_\_\_

- ▶ Demand Link coordination
- ▶ Data Line Code
- ▶ **Data Link Control**
- ▶ Divide Line Communication

Question No 106

\_\_\_\_\_ is responsible for governing node to node Communication.

- ▶ **Data Link Layer**
- ▶ Session Layer
- ▶ Presentation Layer
- ▶ Application Layer

Question No 107

Go-Back-N protocol is one of the \_\_\_\_\_ Layer protocol, which is used to control the flow of data during the transmission

- ▶ Presentation
- ▶ Network
- ▶ **Data Link**
- ▶ Physical

Question No 108

Error correction is more \_\_\_\_\_ than the error detection.

- ▶ Useless
- ▶ Informal
- ▶ **Difficult**
- ▶ Easy

Question No 109

Stop-and-Wait protocol is used for \_\_\_\_\_

- ▶ Flow control Only
- ▶ Neither flow nor error control

# AL-JUNAID TECH INSTITUTE

- ▶ Both flow and error control

Question No 110

\_\_\_\_\_ is not a function of Data Link control

- ▶ Framing
- ▶ Error Control
- ▶ Flow Control
- ▶ Modulation

Question No 111

Payload filed of the PPP frame is of variables Size and it can carry maximum Data up to \_\_\_\_\_

- ▶ 10 Bytes
- ▶ 500 Bytes
- ▶ 100 Bytes
- ▶ 1500 Bytes

Question No 112

In Block Coding Scheme, number of Code word is always \_\_\_\_\_ data wods.

- ▶ Equal to
- ▶ same as
- ▶ Greater Then
- ▶ Less then

Question No 113

High-Level Data Link Control(HDLC) defines \_\_ types of frames to deal with the transfer modes.

- ▶ Three
- ▶ Two
- ▶ Four
- ▶ Six

Question No 114

Congestion control is a feature of \_\_\_\_\_ layer(S)

- ▶ Transport
- ▶ All Mentioned
- ▶ Network
- ▶ Data Link

Question No 115

Error detection and correction are the services provided by \_\_\_\_\_ Layer.

- ▶ Physical Layer
- ▶ Application Layer
- ▶ Data Link Layer

# AL-JUNAID TECH INSTITUTE

## ▶ Session Layer

Question No 116

IN Linear Block codes, \_\_ of two valid code words creates another valid code word.

- ▶ AND
- ▶ OR
- ▶ XOR
- ▶ NOR

Question No 117

Stop-and-Wait protocol is used for \_\_\_\_\_

- ▶ Bothe flow and error control
- ▶ Error control Only
- ▶ Flow control Only
- ▶ Nether flow nor error control

Question No 118

CHAP is an Authentication protocol, which uses \_\_\_\_\_ process to authenticate user information.

- ▶ Four way
- ▶ Three Way
- ▶ Two Way
- ▶ Five way

Question No 119

Data link layer is divided into \_\_\_\_\_ sub Layer

- ▶ Seven
- ▶ Two
- ▶ Five
- ▶ Three

Question No 120

If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error location.

- ▶ 16
- ▶ 2
- ▶ 8
- ▶ 4

Question No 121

\_\_\_\_\_ is not a function of Data Link Control.

- ▶ Framing
- ▶ Modulation
- ▶ Flow control
- ▶ Error control

# AL-JUNAID TECH INSTITUTE

Question No 122

The\_\_\_\_\_is the number of single elements sent per unit time.

- ▶ **Single rate**
- ▶ Period
- ▶ Bit rate
- ▶ Data rate

Question No 123

In ASK, both\_\_\_\_\_and\_\_\_\_\_remain constant.

- ▶ Amplitude, Frequency
- ▶ Amplitude, phase
- ▶ **Frequency, Phase**
- ▶ Amplitude, Samples

Question No 124

In an analog hierarchy to carry voice channels, a group can carry \_\_\_\_\_voice channels.

- ▶ 60
- ▶ 12
- ▶ **20**
- ▶ 10

Question No 125

Which of the following sublayer resolves the contention for the shared media.

- ▶ **MAC**
- ▶ LLC
- ▶ Physical
- ▶ None of the given.

Question No 126

What is the period of a Sine wave having frequency of 5 Hz?

- ▶ **0.2**
- ▶ 0.1
- ▶ 0.5
- ▶ 0.3

Question No 127

In CRC there is no error if the remainder at the receiver is \_\_\_\_\_.

- ▶ Equal to the remainder at the sender
- ▶ **Zero**
- ▶ Nonzero
- ▶ The quotient at the sender

Question No 128

Analog To Analog Conversion Methods include\_\_\_\_\_.

- ▶ **AM, PM and FM**
- ▶ AM, FSK and PM

# AL-JUNAID TECH INSTITUTE

- ▶ AM, PM and QAM
- ▶ None of the given

Question No 129

Signals travel through fiber optic cable are in the form of

- ▶ **Light**
- ▶ Bits
- ▶ Electromagnetic
- ▶ Bytes

Question No 130

If the Hamming distance between sent and received code word is \_\_\_\_\_, then it shows that received data is corrupted.

- ▶  **$\neq 0$**
- ▶  $= 0$
- ▶  $\neq 1$
- ▶  $= 1$

Question No 131

The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

- ▶ Duplicated
- ▶ Blocked
- ▶ Cleaned
- ▶ **Escaped**

Question No 132

\_\_\_\_\_ is normally used where speed is priority in datatransfer.

- ▶ **Serial data transmission**
- ▶ Hybrid data transmission
- ▶ Parallel data transmission
- ▶ Both parallel and serial data transmission

Question No 133

Baud rate is greater than or equal to the bit rate.

- ▶ True
- ▶ **False**

Question No 134

\_\_\_\_\_ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

- ▶ CHAP
- ▶ **PAP**
- ▶ LCP
- ▶ ICMP

Question No 135

In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.

- ▶ Same as

# AL-JUNAID TECH INSTITUTE

- ▶ **Two times**
- ▶ Three times
- ▶ Four times

Question No 136

A broadcast signal is received by the orbiting satellites which rebroadcasts \_.

- ▶ to the resender
- ▶ to the sender
- ▶ to the atmosphere
- ▶ **none of the given**

Question No 137

In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as \_\_\_\_\_.

- ▶ Encapsulation
- ▶ De-encapsulation
- ▶ **Packetizing**
- ▶ Framing

Question No 138

By using \_\_\_\_\_ system, change in the file contents during file transmission can be avoided.

- ▶ Routing
- ▶ Flow Control
- ▶ **Error Detection and Correction**
- ▶ Congestion Control

Question No 139

Which of the following consist of just one redundant bit per data unit?

- ▶ Two dimensional Parity check.
- ▶ CRC
- ▶ **Simple Parity Check**
- ▶ Checksum

Question No 140

If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise.

- ▶ Constant
- ▶ **More**
- ▶ Less
- ▶ Zero

Question No 141

The extra bits added with the original data for error detection/correction are called \_\_\_\_\_.

- ▶ Fault bits
- ▶ Port numbers
- ▶ **Redundant bits**
- ▶ Process IDs

# AL-JUNAID TECH INSTITUTE

Question No 142

Flow control is needed to prevent\_\_\_\_\_.

- ▶ Bit errors
- ▶ Overflow of the sender buffer
- ▶ **Overflow of the receiver buffer**
- ▶ Collision between sender and receiver

Question No 143

The inversion of the level represents a 1 bit in \_\_\_\_\_encoding

- ▶ **NRZ-I**
- ▶ NRZ-L
- ▶ RZ
- ▶ Manchester

Question No 144

The term\_\_\_\_\_means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.

- ▶ Packet Error
- ▶ Burst Error
- ▶ **Single Bit Error**
- ▶ Character Error

Question No 145

A sine wave is defined by\_\_\_\_\_characteristics.

- ▶ 2
- ▶ **3**
- ▶ 4
- ▶ 5

Question No 146

Quadrature Amplitude Modulation (QAM) is the combination of \_\_\_\_\_and\_\_\_\_\_.

- ▶ **FSK, PSK**
- ▶ PSK, FSK
- ▶ ASK, PSK
- ▶ ASK, FSK

Question No 147

If we need to correct a single error in an 8-bit data unit, we need to consider\_\_\_\_\_possible error locations.

- ▶ **16**
- ▶ 2
- ▶ 4
- ▶ 8

Question No 148

To calculate the data rate for noisy channel\_\_\_\_\_formula is used.

- ▶ **Shannon**
- ▶ Nyquist



# AL-JUNAID TECH INSTITUTE

- ▶ Propagation
- ▶ Greedy

Question No 149

GO-BACK-N protocol is one of the \_\_\_\_\_ layer protocols, which is used to control the flow of data during the transmission.

- ▶ Physical
- ▶ **Network**
- ▶ Presentation
- ▶ Data link

Question No 150

A \_\_\_\_\_ modem is used to connect two DTEs directly.

- ▶ **'NULL'**
- ▶ Cable
- ▶ DSL
- ▶ None of the given

Question No 151

Time division multiplexing is used in \_\_\_\_\_ systems.

- ▶ Analog
- ▶ Hybrid
- ▶ **Digital**
- ▶ Automated

Question No 152

In OSI model headers are added at layer \_\_\_\_\_.

- ▶ 1,2,3 only
- ▶ 6,5,7 only
- ▶ 5,4,3 only

**6,5,4,3,2 only**

Question No 153

Line Configuration refers to the way two or more devices attach to a \_\_\_\_\_.

- ▶ Path
- ▶ Circuit
- ▶ **Link**
- ▶ Router

Question No 154

A sine wave is \_\_\_\_\_.

- ▶ periodic and discrete
- ▶ aperiodic and discrete
- ▶ **periodic and continuous**
- ▶ aperiodic and continuous

Question No 155

The logical connection between the peer layers is \_\_\_\_\_ connection.

- ▶ Physical
- ▶ Direct
- ▶ Indirect

# AL-JUNAID TECH INSTITUTE

## ▶ Tengible

Question No 155

What is the frequency of a sine wave if it completes one cycle in 2 seconds?

▶ 0.5

▶ 0.6

▶ 0.3

▶ 0.2

Question No 156

In Synchronous TDM, Time slots are not same in number as input devices.

▶ True

▶ False

Question No 157

\_\_\_\_\_ category of coaxial cable is used for thin Ethernet.

▶ RG-58

▶ RG-59

▶ RG-1

▶ RG-47

Question No 157

Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.

▶ Two

▶ Three

▶ Four

▶ Five

Question No 158

BSC supports \_\_\_\_\_ transmission using \_\_\_\_\_ flow control.

▶ Full-duplex, Stop and wait ARQ

▶ Half-duplex, Selective reject ARQ

▶ Full-duplex, Selective reject ARQ

▶ Half-duplex, Stop and wait ARQ

Question No 159

In selective-reject ARQ, only the specific damaged or lost frame is \_\_\_\_\_

▶ Retransmitted

▶ Forwarded

▶ Selected

▶ Rejected

Question No 160

Error detection and correction are the services provided by \_\_\_\_\_ layer.

▶ Data link layer

▶ Application layer

▶ Physical layer

▶ Session layer

Question No 161

# AL-JUNAID TECH INSTITUTE

The \_\_\_\_\_ generates the data and passes it along with any control information to a \_\_\_\_.

- ▶ **DTE, DCE**
- ▶ DCE, DTE
- ▶ DTE, DTE
- ▶ DCE, DCE

Question No 162

Analog refers to something that is continuous in \_\_\_\_\_.

- ▶ **Time**
- ▶ Space
- ▶ Frequency
- ▶ None of the given

Question No 163

Time domain shows changes in \_\_\_\_\_.

- ▶ time w.r.t signal amplitude
- ▶ **signal amplitude w.r.t time**
- ▶ time
- ▶ All of the given

Question No 164

Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of \_\_\_\_.

- ▶ Local Area Network
- ▶ **Metropolitan Area Network**
- ▶ Wide Area Network
- ▶ Home Based Network

Question No 165

Which level of the TCP/IP reference model routes data/information across a network channel?

- ▶ Application Layer
- ▶ Data Link Layer
- ▶ Transport Layer
- ▶ **Network Layer**

Question No 166

There are \_\_\_\_\_ types of serial transmission.

- ▶ 3
- ▶ 4
- ▶ **2**
- ▶ None of the given

Question No 167

Latency is made of \_\_\_\_\_ components.

- ▶ **Two**
- ▶ Three
- ▶ Four
- ▶ Five

# AL-JUNAID TECH INSTITUTE

Question No 168

Like 10 Base 5, 10 Base 2 is a \_\_\_\_\_ topology LAN.

- ▶ Ring
- ▶ Mesh
- ▶ Star
- ▶ **Bus**

Question No 169

\_\_\_\_\_ is the rate of change with respect to time.

- a. Voltage
- b. Time
- c. Frequency
- d. **Amplitude**

Question No 169

Two or more computers connected so that they can communicate with each other and share information is called a \_\_\_\_\_.

- ▶ satellite
- ▶ protocol
- ▶ broadcast
- ▶ **network**

Question No 170

Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?

- ▶ Checksum
- ▶ Cyclic Redundancy Check (CRC)
- ▶ **Longitudinal Red Check (LRC)**
- ▶ Vertical Redundancy Check (VRC)

Question No 171

Central Hub in a Tree is an \_\_\_\_\_.

- ▶ **Active hub**
- ▶ passive hub

Question No 172

Time domain plot show changes in signal phase with respect to time.

- ▶ true
- ▶ **false**

Question No 172

\_\_\_\_\_ is an Authentication Protocol, which uses two-step process to authenticate user information?

- ▶ CHAP
- ▶ LCP
- ▶ **PAP**
- ▶ ICMP

Question No 173

CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to

# AL-JUNAID TECH INSTITUTE

authenticate user information.

- ▶ Two-way
- ▶ **Three-way**
- ▶ Four-way
- ▶ Five-way

Question No 174

No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between \_\_\_\_\_.

- ▶ Virtual LANs
- ▶ **Two adjacent nodes**
- ▶ Ethernet links
- ▶ Source and destination PCs

Question No 175

Encoding digital data into digital signals is called \_\_\_\_\_.

- ▶ Analog-to-Digital Conversion
- ▶ **Digital-to-Digital Conversion**
- ▶ Digital-to-Analog Conversion
- ▶ None of the given

Question No 176

\_\_\_\_\_ encoding is almost obsolete today

- ▶ Bipolar
- ▶ **Unipolar**
- ▶ Polar
- ▶ None of the given

Question No 177

The Internet is \_\_\_\_\_.

- ▶ software for sending e-mail around the world
- ▶ a government-owned agency that links computers
- ▶ **a global network of computers networks**
- ▶ a specialised form of local area network

Question No 178

PSK is susceptible to the noise degradation.

- ▶ True
- ▶ **False**

Question No 179

Low pass channel has \_\_\_\_\_ bandwidth between two stations.

- ▶ **Dedicated**
- ▶ Shared
- ▶ Multiplexed
- ▶ Infinite

Question No 180

In ASK, Noise usually affects the \_\_\_\_\_.

- ▶ **Amplitude**
- ▶ Phase

# AL-JUNAID TECH INSTITUTE

- ▶ Sample
- ▶ Frequency
- ▶ Infinite

Question No 181

The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

- ▶ Duplicated
- ▶ Blocked
- ▶ Cleaned
- ▶ **Escaped**

Question No 182

Which of the following is not a characteristic of a sine wave?

- ▶ Amplitude
- ▶ **Segmentation**
- ▶ Phase
- ▶ Frequency
- ▶ Infinite

Question No 183

In statistical time division multiplexing, number of slots in frame are less than \_\_\_\_\_.

- ▶ Output data rate
- ▶ **Input lines**
- ▶ Input frequency
- ▶ Multiplexed T lines
- ▶ Infinite

Question No 184

If a digital signal has "L" number of levels. \_\_\_\_\_ is the number of bits required to represent each level.

- ▶ **Log L**
- ▶ Log Base 2 L
- ▶ L Square
- ▶  $\log(2 \cdot L)$
- ▶ Infinite

Question No 185

\_\_\_\_\_ is responsible for governing node to node communication.

- ▶ Application Layer
- ▶ Session Layer
- ▶ **Data Link layer**
- ▶ Presentation Layer
- ▶ Infinite

Question No 186

The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?

- ▶ **Standard Ethernet**
- ▶ Fast Ethernet
- ▶ Gigabit Ethernet

# AL-JUNAID TECH INSTITUTE

- ▶ 10 Gigabit Ethernet
- ▶ Infinite

Question No 187

What is the period of a Sine wave having frequency of 2 Hz?

- 0.2
- 0.1
- 0.5**
- 0.3

Question No 188

In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_

- ▶ **EOT**
- ▶ EKT
- ▶ ENT
- ▶ ESP
- ▶ Infinite

Question No 189

\_\_\_\_\_ are used for Multi cas communication, such as radio and Television.

- ▶ Microwaves
- ▶ **Radio Waves**
- ▶ Light waves
- ▶ Infrared Waves

Question No 190

POTS stand for \_\_\_\_\_?

- ▶ **Plain Old Telephone System**
- ▶ Plain Old Telephone Services
- ▶ Plain Old Time Sharing
- ▶ Plain Old Telephone Set

Question No 191

In \_\_\_\_\_, a station monitors the medium after it sends a frame to see if the transmission was successful.

- ▶ CSMA
- ▶ None of Given
- ▶ **CSMA/CA**
- ▶ CSMA/CD

Question No 192

DSL stand For

- ▶ **Digital Subscribe Line**
- ▶ Data Subscribe Line
- ▶ Data Switched Line
- ▶ Digital Switched Line

Question No 193

In Ethernet frame, both destination and sender addresses are of length \_\_\_\_\_.

# **AL-JUNAID TECH INSTITUTE**

- ▶ 2 byte
- ▶ 4 byte
- ▶ 6 byte
- ▶ 5 byte

Question No 193

Data Link layer provides services to the \_\_\_\_\_ layer.

- ▶ Network
- ▶ Transport
- ▶ Application
- ▶ Session

Question No 194

\_\_\_\_\_ is/are un-reliable protocol(s).

- ▶ TCP
- ▶ IP and TCP
- ▶ IP and UDP
- ▶ UDP

Question No 195

When we represent the data in CDMA, if a station is idle then it sends \_\_\_\_\_.

- ▶ -1
- ▶ 1
- ▶ Infinity
- ▶ 0

Question No 196

In the context of Ethernet address, the source address is always a \_\_\_\_\_ address.

- ▶ Multicast
- ▶ Broadcast
- ▶ Unicast
- ▶ 128 Broad Cast

Question No 197

\_\_\_\_\_ is the process of converting digital data to digital signals.

- ▶ Line Coding
- ▶ Multicast
- ▶ Broadcast
- ▶ Unicast

Question No 198

In even parity check code, the value of syndrome is \_\_\_\_\_ if the number of 1s is even.

- ▶ 4
- ▶ 0
- ▶ 1
- ▶ 2

Question No 199

In \_\_\_\_\_ each band is reserved for a specific station, and it belongs to the station all the time.

- ▶ FDMA



# AL-JUNAID TECH INSTITUTE

- ▶ TDMA
- ▶ CSMA
- ▶ CDMA

Question No 200

MAC address is of\_\_\_\_\_.

- ▶ 48 bits or 64 bits
- ▶ 24 bits
- ▶ 36 bits
- ▶ 42 bits

Question No 201

Checksum is an error-detection technique that can be applied to a message of\_\_\_\_\_length.

- ▶ Exactly 8
- ▶ Exactly 4
- ▶ Exactly 32
- ▶ Any

Question No 202

A personal computer or workstation on an Ethernet network must have\_\_\_\_\_card.

- ▶ None of the Given
- ▶ TDI
- ▶ PCI
- ▶ NCI

Question No 203

One of the most common protocols for point-to-point access is\_\_\_\_\_.

- ▶ PPP
- ▶ IP
- ▶ UDP
- ▶ TCP

Question No 204

In order to find Hamming distance between two pair of words,\_\_\_\_\_logical operation is used.

- ▶ NAND
- ▶ XOR
- ▶ AND
- ▶ OR

Question No 205

In order to resolve link layer address of the next node, the first generated ARP request is always in\_\_\_\_\_mode.

- ▶ Multicast
- ▶ Unicast
- ▶ Multiplex
- ▶ Broadcast

Question No 206

\_\_\_\_\_transmission technology (based on Ethernet) provides a data rate of 1 billion bits per

# AL-JUNAID TECH INSTITUTE

second.

- ▶ Gigabit Ethernet
- ▶ Standard Ethernet
- ▶ 10BASE-T Ethernet
- ▶ Fast Ethernet

Question No 207

The message 1110 sent by a source is received by a destination as 1011. This is \_\_\_\_\_ type of error.

- ▶ Hamming
- ▶ Single-Bit
- ▶ Uni-Bit
- ▶ Burst

Question No 208

a3:34:45:11:92:F1 is a \_\_\_\_\_ link layer address.

- ▶ Unicast OR Data Both are correct
- ▶ Broadcast
- ▶ Multicast
- ▶ Multiplexer

Question No 209

In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

- ▶ Port Forwarding
- ▶ Reservation
- ▶ Polling
- ▶ Token Passing

Question No 210

\_\_\_\_\_ was used as the medium in 10 Base-T.

- ▶ Twisted Cable
- ▶ Thick Coaxial Cable
- ▶ Thin Coaxial Cable
- ▶ Two Mode Fiber Cable

**CS601 Finalterm Mcqs**

**Updated handouts 2022**

**#Final-Term**

**ORANGE MONKEY TEAM**

**Important NOTE:-**

Maybe toppers are successful in school or college carrier but backbenchers are successful in their life because toppers run according to books while backbenchers don't follow rules and regulation that is why backbenchers are successful. For life, there are no books and another reason why toppers can't succeed because they fear failure and don't try anything new. Backbenchers don't fear failure as they have faced the failure in their school or college carrier so that is the reason they try new things and at last gain success.

As Abdul Kalam nicely quoted, "The best brains of the nation may be found on the last benches of the classroom"

**Translation into Urdu:**

---

**ORANGE MONKEY TEAM**

ہوسکتا ہے کہ ٹاپرز اسکول یا کالج کیریئر میں کامیاب ہوں لیکن بیک بینچر اپنی زندگی میں کامیاب ہوتے ہیں کیونکہ ٹاپرز کتابوں کے مطابق چلتے ہیں جب کہ بیک بینچر قواعد و ضوابط پر عمل نہیں کرتے اسی وجہ سے بیک بینچر کامیاب ہوتے ہیں۔ زندگی کے لیے، کوئی کتابیں نہیں ہیں اور ایک اور وجہ ہے کہ ٹاپرز کامیاب نہیں ہو پاتے کیونکہ وہ ناکامی سے ڈرتے ہیں اور کچھ نیا کرنے کی کوشش نہیں کرتے۔ بیک بینچر ناکامی سے نہیں ڈرتے کیونکہ انہوں نے اپنے اسکول یا کالج کیریئر میں ناکامی کا سامنا کیا ہے یہی وجہ ہے کہ وہ نئی چیزیں آزما رہے ہیں اور آخر کار کامیابی حاصل کرتے ہیں۔

جیسا کہ عبدالکلام نے اچھی طرح سے نقل کیا ہے، "قوم کے بہترین سماج کلاموں روم کے آخری بینچوں پر مل سکتے ہیں۔"





1. In \_\_\_\_\_ each station is allocated a time slot during which it can send data  
TDMA.....confirm
2. PAP and CHAP are used by \_\_\_\_\_ as Authentication Protocols.  
PPP.....confirm
3. In each character protocol each character of frame is encoded using \_\_\_\_\_ coding system .  
Bit-oriented
4. IN MAC, when nodes use a multi-point or broadcast link, we need a \_\_\_\_\_ to coordinate access to the link.  
Multiple-access protocol/.....confirm
5. The original Ethernet technology with the data rate of 10 MBps is called \_\_\_\_\_?  
Standard Ethernet....confirm
6. \_\_\_\_\_ uses fiber-optic cable

10base-F....confirm

7. One of the most common protocols for point-to-point access is \_\_\_\_\_.

PPP.....confirm

8. \_\_\_\_\_ is an Authentication Protocol, which uses two-step process to authenticate user information.

PAP.....confirm

9. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise

More.....confirm

10. For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the

Collision size....confirm

11. Stop-and-Wait protocol is used for \_\_\_\_\_.

Both flow and error control.....confirm

12. In hexadecimal notation of Ethernet address, each byte is separated by \_\_\_\_\_ symbol.

Colon....confirm

13. In a linear block code, the \_\_\_\_\_ of any two valid codewords creates another valid codeword.

- 14.

XOR.....confirm

15. In the \_\_\_\_\_ method, the stations in a network are organized in a logical ring.

Token-passing....confirm

16. Which one of the following is not a Channelization Protocol?

**CSMA.....confirm**

17. In order to find the location of two errors in an eight bit stream, we have to see \_\_\_\_\_ different combinations.

**28....confirm**

18. If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error locations

**16.....confirm**

19. In \_\_\_\_\_ each station sends a frame whenever it has a frame to send.

**Slotted ALoHA.....confirm**

20. \_\_\_\_\_ is the process of converting digital data to digital signals.

**Line coding.....confirm**

21. In case of higher data rate, the number of impacted bits will be \_\_\_\_\_.

**Higher**

22. In \_ protocol, the sender sends one frame at a time and wants for an acknowledgment before sending the next one.

**Stop-and-wait**

23. In block coding we divide our message into blocks of K bits called as \_\_\_\_\_.

**Data words....confirm**

24. CDMA stands for \_\_\_\_\_.

**Code division Multiple Access.....confirm**

25. \_\_\_\_\_ is not a function of data link control.

**Modulation.....confirm**

26. CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to authenticate user information.

**Three-way.....confirm from net**



27. In the context of Ethernet address, the source address is always a \_\_\_\_\_ address.

Unicast...confirm

28. In the context of Ethernet address, the destination address is always a \_\_\_\_\_ address.

Unicast, multicast, broadcast....confirm

29. \_\_\_\_\_ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

Gigabit Ethernet....confirm

30. Fast Ethernet has a data rate of \_\_\_\_\_ Mbps.

100.....confirm

31. In random access method, when two or more than two nodes transmit at the same time when

All the nodes in collision survives

32. Data link layer is divided into \_\_\_\_\_ sub layers

Two.....confirm

33. If the hamming distance between sent and received code word is \_\_\_\_\_ then it shows that received data is corrupted.

(Is not equal to)0....confirm

34. In order to find hamming distance between two pair of words, \_\_\_\_\_ logical operation is used.

XOR....confirm

35. Ethernet address is of \_\_\_\_\_ bytes

6....confirm

36. In \_\_\_\_\_ a station monitors the medium after it sends a frame to see if the transmission was successful.

CSMA/CD.....confirm



37. For carrier sense Multiple Access/Collision Detection (CSMA/CD), we need a restriction on the \_\_\_\_\_.

Frame size...confirm

38. Ethernet address is of \_\_\_\_\_ bytes

6

39. What was the main drawback in the aloha random access method

Redundancy in data

40. In telephone, network POP stands for

Point-of presence

41. Possession of token gives the station the right to \_\_\_\_\_.

Send the data

42. In Ethernet, the source address field in the MAC frame is the \_ address

Previous station physical

43. Normally the value of Kmax in pure aloha is

15

44. \_\_\_\_\_ uses thin coaxial cable.

10 base 5

45. MAC stands for \_\_\_\_\_

Media access control

46. In the context of Ethernet implementations maximum medium length of 10 base-F

2000

47. Transmission technology based on Ethernet provides a data rate of 1 billion bits per second.

### Standard Ethernet

48. Works with topologies in which one device is designated as a primary station and the other devices are secondary stations

None of the given

49. MAC address is of \_\_\_\_\_.

48 bits

50. The type of aloha which improves the efficiency of pure aloha is.

Slotted aloha

51. A personal computer or workstation on an Ethernet network must have \_\_\_\_\_ card

NIC

52. In which controlled access method all data exchange must be made through primary device even when the ultimate destination is a secondary device

Polling

53. In \_\_\_\_\_ each band is reserved for a specific station and it belongs to the station all the time

FDMA

54. In pure aloha the vulnerable time is \_\_\_\_\_ the frame transmission time

Two times

55. DSL stands for \_\_\_\_\_.

Digital subscriber line

56. In the context of Ethernet address, the source address is always a \_\_\_\_\_ address

Multicast

57. The original Ethernet technology with the data rate of 10 Mbps is called

Standard ethernet

58. Which of the following is a correct controlled access method

All options

59. When we represent the data in CDMA if a station is idle then it sends

0

60. In telephone, network POP stands for \_\_\_\_\_.

Point of presence

61. What was the main drawback in the aloha random access method

Redundancy data rate

62. \_\_\_\_\_ transmission technology based on Ethernet provides a data rate of 1 billion bits per second

Gigabit Ethernet

63. In which controlled access method all data exchanges must be made through a primary device even when the ultimate destination is a secondary device.

Polling

64. In telephone, network POP stands for

Point of presence

65. Mac address is of \_\_\_\_\_

48 bits

66. In \_\_\_\_\_ collision are avoided by deferring transmission even if the channel is idle

The interframe space

67. POTS stand for \_\_\_\_\_?

Plain old telephone system.....confirm

68. Normally the value of  $K_{max}$  in pure aloha is

15

69. In Ethernet frame both destination and sender addresses are of length  
6 byte

70. In \_\_\_\_\_ each band is reserved for a specific station and it belongs to the station all the time  
FDMA

71. The original Ethernet technology with a data rate of 10MBps is called  
Standard Ethernet

72. The \_\_\_\_\_ method has the highest chance of collision because two or more stations may find the line idle and send their frames immediately  
1 persistent method

73. IN allocated a time slot during which it can send data each station is  
TDMA

74. In which controlled access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device  
Polling

75. CDMA stands for  
Code data multiply access

76. Works with topologies in which one device is designated as a primary station and the other device are secondary stations  
Polling

77. In Ethernet, the source address field in the MAC frame is the address  
Next destination physical

1. VCI address is \_\_\_\_\_ address in virtual circuit approach.

1. Private

2. Variable

3. Local

4. Global

2. Traditionally MANs have been implemented using one of the 2 technologies, Circuit Switching and Packet Switching

1. true

2. false

3. In the \_\_\_\_\_ method, the stations in a network are organized in a logical ring.

1. Token-passing

2. Polling

3. Reservation

4. None of the given

4. Like 10 Base 5, 10 Base 2 is a \_\_\_\_\_ topology LAN.

1. Ring

2. Mesh

3. Star

4. Bus

5. In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.

1. Same as

2. Three times

**3. Two times**

4. Four times

**6. The transmission medium that carries the message is referred to as the \_\_\_\_\_.**

1. send and receive device

**2. communication channel**

3. protocol

4. gateways

**7. Analog signals are refers to be \_\_\_\_\_**

**1. Continuous**

2. Discrete

3. Periodic

4. Aperiodic

**8. There are \_\_\_\_\_ types of serial transmission.**

1. 3

2. 4

**3. 2**

4. None of the given

9. To calculate the data rate for noisy channel \_\_\_\_\_ formula is used.

1. Shannon

2. Nyquist

3. Propagation

4. Greedy

10. There are three types of Hubs that exist. Which of the following options correctly describes these three?

1. Passive, dormant, special

2. Active , dormant , passive

3. Passive, Active, Turbo

4. Passive, Active, Intelligent

11. \_\_\_\_\_ is the rate of change with respect to time.

1. Voltage

2. Time

3. Frequency

4. Amplitude

12. To control and manage the transfer of data, a protocol called Point to Point Protocol (PPP) is used at the \_\_\_\_\_ layer.

1. Physical

2. Network

3. Presentation

4. Data link

**13. What is the frequency of a sine wave if it completes one cycle in 2 seconds?**

1. 0.5

2. 0.6

3. 0.3

4. 0.2

**14. \_\_\_\_\_ relatively measures the strength of two signals.**

1. Signal rate

2. Bit rate

3. Decibel

4. Pulse Rate

**15. Central Hub in a Tree is an \_\_\_\_\_.**

1. Active hub

2. passive hub

**16. At the CRC generator, \_\_\_\_\_ added to the data unit before the divisional process.**

1. 0s are

2. 1s are

3. a polynomial is

4. a CRC remainder is



17. A broadcast signal is received by the orbiting satellites which rebroadcasts \_\_\_\_\_.

1. to the resender
2. to the sender
3. to the atmosphere
4. none of the given

18. The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

1. Duplicated
2. Blocked
3. Cleaned
4. Escaped

19. In \_\_\_\_\_, a signal is directed straight from antenna to antenna.

1. Line of sight propagation
2. Ground propagation
3. Space propagation
4. Tropospheric propagation

20. Which of the following consist of just one redundant bit per data unit?

1. Two dimensional Parity check.

2. CRC

**3. Simple Parity Check**

4. Checksum

**21. The term \_\_\_\_\_ means that only one bit of a given data unit is changed from 1 to 0 or from 0 to 1.**

1. Packet Error

2. Burst Error

**3. Single Bit Error**

4. Character Error

**22. Flow control is needed to prevent \_\_\_\_\_.**

1. Bit errors

2. Overflow of the sender buffer

**3. Overflow of the receiver buffer**

4. Collision between sender and receiver

**23. Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol.**

1. Transport

2. Application

3. Session

**4. Network**

24. A fiber -optic cable transmit signals in the form of \_\_\_\_\_.

1. Light

2. Sound

3. Wave

4. None of the given

25. Possession of TOKEN gives the station the right to \_\_\_\_\_.

1. Modify the data

2. Send the data

3. Delete the data

4. Receive the data

26. The inversion of the level at 1 bit is called as \_\_\_\_\_.

1. NRZ-L

2. NRZ-I

3. RZ

4. None of the given.

27. What was the main draw back in ALOHA Random-Access method?

1. Redundancy in Data

2. Potential Collision

3. Inconsistent Data rate

4. All of the given

28. Data is transmitted using light through a \_\_\_\_\_ cable.

1. twisted pair

2. fiber-optic

3. coaxial

4. microwave

29. In \_\_\_\_\_, each station sends a frame whenever it has a frame to send.

1. Slotted ALOHA

2. Fragmented ALOHA

3. Pure ALOHA

4. None of the given

30. In OSI model headers are added at layer \_\_\_\_\_.

1. 1,2,3 only

2. 6,5,7 only

3. 5,4,3 only

4. 6,5,4,3,2 only

31. Is there any difference between Data communication and telecommunication are network?

1. Yes

2. No

32. \_\_\_\_\_ is an Authentication Protocol, which is used by the PPP to authenticate passwords.

1. CHAP

**2. PAP**

3. LCP

4. ICMP

**33. Which level of the TCP/IP reference model routes data/information across a network channel?**

1. Application Layer

2. Data Link Layer

3. Transport Layer

**4. Network Layer**

**34. In ASK, both \_\_\_\_\_ and \_\_\_\_\_ remain constant.**

1. Amplitude, Frequency

2. Amplitude, phase

**3. Frequency, Phase**

4. Amplitude, Samples

**35. Which error detection method consists of a parity bit for each unit as well as an entire data unit of parity bits?**

1. Checksum

2. Cyclic Redundancy Check (CRC)

**3. Longitudinal Red Check (LRC)**

4. Vertical Redundancy Check (VRC)

36. Baud rate is greater than or equal to the bit rate.

1. True

2. False

37. The most common type of connector used by coaxial cable is \_\_\_\_\_.

1. BNC

2. RJ-45

3. RJ-11

4. RJ-57

38. In even parity check code, the value of syndrome is \_\_\_\_\_ if the number of 1s is even.

1. 1

2. 2

3. 0

4. 4

39. What is the period of a Sine wave having frequency of 10 Hz?

1. 0.2

2. 0.1

3. 0.5

4. 0.3

40. In \_\_\_\_\_, both peak amplitude and frequency remains constant as the phase changes.

1. PSK

2. QAM

3. Both PSK & QAM

4. None of the given

41. There are \_\_\_\_\_ types of guided media.

1. 2

2. 3

3. 4

4. 5

42. \_\_\_\_\_ is responsible for governing node to node communication.

1. Application Layer

2. Session Layer

3. Data Link layer

4. Presentation Layer

43. TROPOSHERE is the layer of atmosphere above the troposphere but below space

1. True

2. False

44. Terminal, microprocessor, computer, printer or any other device that generates or consumes digital data is known as \_\_\_\_\_

1. DCE

2. DTE

3. DCE & DTE

4. None of the given

45. Normally the value of Kmax in pure ALOHA is \_\_\_\_\_.

1. 25

2. 10

3. 15

4. 30

46. Sampling means measuring \_\_\_\_\_ of signal at equal intervals

1. Amplitudes

2. Frequencies

3. Phases

4. None of the given

47. A sine wave is \_\_\_\_\_.

1. periodic and discrete

2. aperiodic and discrete

3. periodic and continuous

4. aperiodic and continuous

48. The inversion of the level represents a 1 bit in \_\_\_\_\_ encoding



1. NRZ-I

2. NRZ-L

3. RZ

4. Manchester

49. To make sure that source does not overwhelm destination by sending data faster than it can be handled and processed is called?

1. Addressing & Routing

2. Exchange Management

3. Flow Control

4. Recovery

50. Time domain plot show changes in signal phase with respect to time.

1. true

2. false

51. In which controlled-access method all data exchanges must be made through primary device even when the ultimate destination is a secondary device?

1. Polling

2. Reservation

3. Token Passing

4. Port Forwarding

52. If a digital signal has "L" number of levels. \_\_\_\_\_ is the number of bits required to represent each level.

1. Log L
2. Log Base 2 L
3. L Square
4. Log(2\*L)

53. \_\_\_\_\_ category of coaxial cable is used for Cable TV.

1. RG-58
2. RG-59
3. RG-11
4. RG-47

54. The level of signal is always positive in NRZ encoding.

1. true
2. false

55. Radio is an example of \_\_\_\_\_ signal conversion.

1. Analog to analog
2. Analog to digital
3. Digital to Digital
4. Digital to analog

56. Analog To Analog Conversion Methods include \_\_\_\_\_.

1. AM, PM and FM

- 2. AM, FSK and PM
- 3. AM, PM and QAM
- 4. None of the given

**57. The original Ethernet technology with the data rate of 10 Mbps is called \_\_\_\_\_?**

- 1. Standard Ethernet**
- 2. Fast Ethernet
- 3. Gigabit Ethernet
- 4. 10 Gigabit Ethernet

**58. Which of the following is not a characteristic of a sine wave?**

- 1. Amplitude
- 2. Segmentation**
- 3. Phase
- 4. Frequency

**59. By using \_\_\_\_\_ system, change in the file contents during file transmission can be avoided.**

- 1. Routing
- 2. Flow Control
- 3. Error Detection and Correction**
- 4. Congestion Control

**60. In order to find the location of two errors in an eight bit stream, we have to see \_\_\_\_\_ different combinations.**

1. 28

2. 16

3. 8

4. 4

61. In CSMA/CA, \_\_\_\_\_ is the amount of time divided into slots.

1. Interframe Window

2. Contention Window

3. Fragmented Window

4. Collided Window

62. \_\_\_\_\_ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

1. Frame sequencing

2. Acknowledgement numbering

3. Counter and Timer reset

4. Frame sequencing and Acknowledgement numbering

63. Which one of the following is a correct-access method?

1. Polling

2. Token Passing

3. Reservation

4. All of the given

**64. In the encoding where positive to negative transition is one and vice versa is zero.**

1. Manchester
2. Differential Manchester
3. All of the given
4. None of given

**65. If the window size is 63 in go-back N ARQ then what is the range of sequence number?**

1. 0 to 63
2. 0 to 64
3. 1 to 63
4. 1 to 64

**66. Following characteristics fully describe \_\_\_\_\_,**  
Amplitude  
Period/Frequency  
Phase

1. Sine Waves
2. Digital Signals
3. Aperiodic Signals
4. None of the given

**67. Digitizing an Analog Signal is called \_\_\_\_\_.**

1. Analog-to-Digital Conversion
2. Digital-to-Digital Conversion

3. Digital-to-Analog Conversion

4. None of the given

**68. At the CRC checker, \_\_\_\_\_ means that the data unit is damaged.**

1. A string of 0s

2. A string of 1s

3. A string of alternating 1s and 0s

**4. A nonzero remainder**

**69. Which one of the following is a Random-Access protocol?**

1. TDMA

**2. CDMA/CD**

3. ALOHA

4. CDMA/CA

**70. Human voice is example of \_\_\_\_\_ signal.**

1. Digital

**2. Analog**

3. Discrete

4. None of the given

**71. In statistical time division multiplexing, number of slots in frame are less than \_\_\_\_\_.**

1. Output data rate

2. Input lines

3. Input frequency

4. Multiplexed T lines

72. A \_\_\_\_\_ modem is used to connect two DTEs directly.

1. 'NULL

2. Cable

3. DSL

4. None of the given

73. If we need to correct a single error in an 8-bit data unit, we need to consider \_\_\_\_\_ possible error locations.

1. 16

2. 2

3. 4

4. 8

74. Asynchronous TDM is efficient only when the size of the time slot is kept relatively \_\_\_\_\_.

1. Large

2. Small

3. Medium

4. Zero

75. The \_\_\_\_\_ protocol uses both flow and error control.

1. TCP/IP

2. HDLC

3. ICMP

**4. Stop-and-Wait**

76. In \_\_\_\_\_, each station is allocated a time slot during which it can send data.

1. CDMA

2. NDMA

**3. TDMA**

4. FDMA

77. The extra bits added with the original data for error detection/correction are called \_\_\_\_\_.

1. Fault bits

2. Port numbers

**3. Redundant bits**

4. Process IDs

78. \_\_\_\_\_ is correct formula to determine the total number of ports needed for one system connected in a mesh network.

1.  $N+1$

2.  $N-1$

**3.  $N(N+1)$**

4.  $N \times N$



79. \_\_\_\_\_ is sometimes called the bit rate.

1. signal rate
2. modulation rate
3. Data rate
4. pulse rate

80. Collisions in CSMA/CA are avoided through \_\_\_\_\_.

1. The Interframe Space
2. The Contention Window
3. Acknowledgements
4. All of the given

81. \_\_\_\_\_ category of coaxial cable is used for thin Ethernet.

1. RG-58
2. RG-59
3. RG-1
4. RG-47

82. Which one is not the function of data link layer?

1. Line discipline
2. Flow control
3. Error control
4. Network control

83. Session layer is responsible for \_\_\_\_\_.

1. Reassembly of data
2. Maintaining the connection

84. \_\_\_\_\_ mode of serial transmission guarantees fixed rate of data.

1. Synchronous
2. Asynchronous
3. Isochronous
4. Metasochronous

85. Data Link layer provides services to the \_\_\_\_\_ layer.

1. Application
2. Transport
3. Network
4. Session

86. The \_\_\_\_\_ is the number of signal elements sent per unit time.

1. Data rate
2. Signal rate
3. Bit rate
4. Period

87. The logical connection between the peer layers is \_\_\_\_\_ connection.

1. Physical

**2. Direct**

3. Indirect

4. Tangible

**88. What is the period of a Sine wave having frequency of 5 Hz?**

**1. 0.2**

2. 0.1

3. 0.5

4. 0.3

**89. A sine wave is defined by \_\_\_\_\_ characteristics.**

1. 2

**2. 3**

3. 4

4. 5

**90. If duration of noise is decreased during a transmission over noisy channel, fixed data rate of the channel will cause \_\_\_\_\_ number of bits being impacted by the noise.**

1. Constant

**2. More**

3. Less

4. Zero

91. S-frames in High-Level Data Link Control (HDLC) are only used to transfer \_\_\_\_\_ information.

- 1. User data
- 2. Control
- 3. Redundant
- 4. Original

92. The \_\_\_\_\_ generates the data and passes it along with any control information to a \_\_\_\_\_.

- 1. DTE, DCE
- 2. DCE, DTE
- 3. DTE, DTE
- 4. DCE, DCE

93. The type of ALOHA which improves the efficiency of Pure ALOHA is \_\_\_\_\_.

- 1. Slotted ALOHA
- 2. Upper ALOHA
- 3. Integrated ALOHA
- 4. Lower ALOHA

94. What is the period of a Sine wave having frequency of 2 Hz?

- 1. 0.2
- 2. 0.1
- 3. 0.5

4. 0.3

**95. WDM stands for \_\_\_\_\_.**

1. Wideband De-Modulation

2. Worst Data Manipulation

**3. Wavelength Division Multiplexing**

4. None of the given

**96. Latency is made of \_\_\_\_\_ components.**

**1. Two**

2. Three

3. Four

4. Five

**97. The logical connection between the peer layers is \_\_\_\_\_ connection.**

1. Physical

2. Direct

3. Indirect

**4. Tengible**

**98. In line discipline after the data transmission, the sending system finishes with an \_\_\_\_\_ frame.**

**1. EOT**

2. EKT

3. ENT

4. ESP

99. \_\_\_\_\_ of a signal is the collection of all the component frequencies it contains.

1. Time domain

2. Frequency domain

3. Frequency spectrum

4. None of the given

100. CHAP is an Authentication Protocol, which uses \_\_\_\_\_ process to authenticate user information.

1. Two-way

2. Three-way

3. Four-way

4. Five-way

101. RZ stands for \_\_\_\_\_.

1. Return to Z

2. Retire a Zero

3. Return of Zero

4. None of the given

1. In \_\_\_\_\_ each band is reserved for a specific station, and it belongs to the station all the time.

1. FDMA

2. CDMA

3. TDMA

4. CSMA

**2. In TCP/IP protocol suite, the process of adding header at each layer on sending side is known as \_\_\_\_\_.**

1. Encapsulation

2. De-encapsulation

**3. Packetizing**

4. Framing

**3. In Block coding we divide our message into blocks of k bits called as \_\_\_\_\_.**

1. Code words

**2. Data words**

3. Passwords

4. Cross words

**4. In circuit switched networks we have low efficiency but minimal \_\_\_\_\_.**

1. Delay

**2. Speed**

3. Throughput

4. Errors

**5. Error correction is more \_\_\_\_\_ than the error detection.**

1. Easy

2. Useless

3. Informal

4. Difficult

6. AM signal requires \_\_\_\_\_ the BW of original signal

1. Twice

2. Half

3. Same

4. Three Times

7. Asynchronous transmission is \_\_\_\_\_.

1. Slow

2. Fast

3. Costly

4. None effective

8. The Internet is \_\_\_\_\_.

1. software for sending e-mail around the world

2. a government-owned agency that links computers

3. a global network of computers networks

4. a specialised form of local area network

9. Time division multiplexing is used in \_\_\_\_\_ systems.



1. Analog

2. Hybrid

**3. Digital**

4. Automated

**10. Radio wave transmission utilizes \_\_\_\_\_ different types of propagation.**

1. Two

2. Three

3. Four

**4. Five**

**11. Checksum is an error-detection technique that can be applied to a message of \_\_\_\_\_ length.**

1. Exactly 4

2. Exactly 8

3. Exactly 32

**4. Any**

**12. At the Application layer, object/information is in the form of \_\_\_\_\_.**

1. Message

**2. Packet**

3. Segment

4. Frame

13. POST stand for \_\_\_\_\_?

1. Plain Old Telephone Set
2. Plain Old Time Sharing
3. Plain Old Telephone System
4. Plain Old Telephone Service

14. Which of the following statement is correct?

1. Digital signals are less prone to Noise
2. Digital signals are highly prone to Noise
3. Analog Signals are less prone to Noise
4. Digital signals are not affected by Noise

15. In selective-reject ARQ, only the specific damaged or lost frame is \_\_\_\_\_.

1. Retransmitted
2. Forwarded
3. Selected
4. Rejected

16. In ASK correct formula for calculating the bandwidth is as \_\_\_\_\_.

1.  $B=(1+d)S$
2.  $B=(1*d)S$
3.  $B=(d-1)S$

4.  $B=(d-5)S$

17. Encoding digital data into digital signals is called \_\_\_\_\_.

1. Analog-to-Digital Conversion

2. Digital-to-Digital Conversion

3. Digital-to-Analog Conversion

4. None of the given

18. If the Hamming distance between sent and received code word is \_\_\_\_\_, then it shows that received data is corrupted.

1.  $\neq 0$

2.  $= 0$

3.  $\neq 1$

4.  $= 1$

19. Quadrature Amplitude Modulation (QAM) is the combination of \_\_\_\_\_ and \_\_\_\_\_.

1. FSK, PSK

2. PSK, FSK

3. ASK, PSK

4. ASK, FSK

20. PSK is susceptible to the noise degradation.

1. True

2. False

21. \_\_\_\_\_ works with topologies in which one device is designated as a primary station and the other devices are secondary stations.

1. Reservation
- 2. Polling**
3. Token
4. None of the given

22. No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between \_\_\_\_\_.

1. Virtual LANs
- 2. Two adjacent nodes**
3. Ethernet links
4. Source and destination PCs

23. CDMA stands for \_\_\_\_\_.

1. Carrier Division Multiple Access
2. Carrier Data Multiple Access
3. Code Data Multiple Access
- 4. Code Division Multiple Access**

24. If users must take turns using the link, then it is called Spatial Sharing.

1. true
- 2. false**

25. **GO-BACK-N** protocol is one of the \_\_\_\_\_ layer protocols, which is used to control the flow of data during the transmission.

1. Physical
- 2. Network**
3. Presentation
4. Data link

26. In \_\_\_\_\_ a dedicated communication path is established between two stations through the nodes of the network

- 1. Circuit switching**
2. Packet switching

27. Signals travel through fiber optic cable are in the form of \_\_\_\_\_.

- 1. Light**
2. Bits
3. Electromagnetic
4. Bytes

28. \_\_\_\_\_ contains a repeater.

- 1. Active hub**
2. Passive hub

29. In frequency shift keying, \_\_\_\_\_ remain(s) constant.

1. Frequency
- 2. Amplitude**

3. Both Amplitude and Phase

**4. Both Phase and Frequency**

**30. \_\_\_\_\_ is/are un-reliable protocol(s).**

1. IP and TCP

2. UDP

**3. IP and UDP**

4. TCP

**31. \_\_\_\_\_ is not a function of Data Link Control.**

1. Framing

**2. Modulation**

3. Error Control

4. Flow Control

**32. Each computer on the Internet has a unique numeric address called a(n) \_\_\_\_\_.**

1. domain address

2. protocol address

**3. IP address**

4. Web address

**33. dB is negative if a signal is \_\_\_\_\_.**

**1. Attenuated**

2. Distorted

3. Amplified

4. Noisy

**34. We can have \_\_\_\_\_ different groups with a 4 bit block.**

1. 4

2. 8

3. 16

4. 32

**35. Line Configuration refers to the way two or more devices attach to a \_\_\_\_\_.**

1. Path

2. Circuit

3. Link

4. Router

**36. Error detection and correction are the services provided by \_\_\_\_\_ layer.**

1. Data link layer

2. Application layer

3. Physical layer

4. Session layer

**37. Which of the following sublayer resolves the contention for the shared media.**

1. MAC

- 2. LLC
- 3. Physical
- 4. None of the given.

**38. There are \_\_\_\_\_ basic categories of multiplexing.**

- 1. 2
- 2. 3
- 3. 7
- 4. 5

**39. Low pass channel has \_\_\_\_\_ bandwidth between two stations.**

- 1. Dedicated
- 2. Shared
- 3. Multiplexed
- 4. Infinite

**40. BSC supports \_\_\_\_\_ transmission using \_\_\_\_\_ flow control.**

- 1. Full-duplex, Stop and wait ARQ
- 2. Half-duplex, Selective reject ARQ
- 3. Full-duplex, Selective reject ARQ
- 4. Half-duplex, Stop and wait ARQ

**41. Quantization is a method of assigning \_\_\_\_\_ values in a specific range to sampled instances.**



1. Negative

2. Positive

3. Binary

4. Integral

42. \_\_\_\_\_ encoding is almost obsolete today

1. Bipolar

2. Unipolar

3. Polar

4. None of the given

43. YMODEM has \_\_\_\_\_ Byte of data unit.

1. 1024

2. 256

3. 128

4. 512

44. Like Data link layer, \_\_\_\_\_ is also responsible for Flow control.

1. Transport Layer

2. Session Layer

45. In CRC there is no error if the remainder at the receiver is \_\_\_\_\_.

1. Equal to the remainder at the sender

**2. Zero**

3. Nonzero

4. The quotient at the sender

**46. \_\_\_\_\_ is normally used where speed is priority in data transfer.**

**1. Serial data transmission**

2. Hybrid data transmission

3. Parallel data transmission

4. Both parallel and serial data transmission

**47. In Block coding scheme, number of code words is always \_\_\_\_\_ data words.**

1. Less than

**2. Equal to**

3. Greater than

4. Same as

**48. A portion of the path that carries TX between a given pair of devices is known as \_\_\_\_\_.**

1. Direct Point

2. Bridge

**3. Channel**

4. Access Point

49. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 200, 400, 600 and 800?

1. 100

2. 500

3. 300

4. 600

50. \_\_\_\_\_ is the process of converting digital data to digital signals.

1. CSMA/CD

2. CSMA/CA

3. Line coding

4. Segmentation

51. \_\_\_\_\_ transmission technology (based on Ethernet) provides a data rate of 1 billion bits per second.

1. Fast Ethernet

2. Standard Ethernet

3. Gigabit Ethernet

4. 10BASE-T Ethernet

52. Congestion Control is a feature of layer(s).

1. Data Link

2. Network

3. Transport

**4. All mentioned**

**53. In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time.**

- 1. Same as
- 2. Two times**
- 3. Three times
- 4. Four times

**54. What is the Bandwidth of a periodic signal if it is decomposed into 4 sine waves with frequencies 300, 600, 900 and 1200?**

- 1. 100
- 2. 900**
- 3. 300
- 4. 600

**55. \_\_\_\_\_ signal can take infinite levels of intensity over time.**

- 1. Digital
- 2. Discrete
- 3. Analog**
- 4. Logical

**56. Mr. Asif while sitting in Lahore is talking with his friend in Dubai through Skype is an example of \_\_\_\_\_.**

- 1. Local Area Network
- 2. Metropolitan Area Network**

3. Wide Area Network

4. Home Based Network

**57. Which layer is concerned with the syntax and semantics of info exchange between two systems?**

1. Physical

2. Application

3. Session

**4. Presentation**

**58. Time domain shows changes in \_\_\_\_\_.**

1. time w.r.t signal amplitude

**2. signal amplitude w.r.t time**

3. time

4. All of the given

**59. In ASK, Noise usually affects the \_\_\_\_\_.**

**1. Amplitude**

2. Phase

3. Sample

4. Frequency

**60. In analog transmission, the base signal is called the \_\_\_\_\_.**

**1. Carrier Signal**

2. Analog signal

3. Digital signal

4. Modulated signal

**61. Which of the following is most affected by noise?**

1. PSK

**2. ASK**

3. FSK

4. QAM

**62. In Synchronous TDM, Time slots are not same in number as input devices.**

1. True

**2. False**

**63. The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.**

1. Duplicated

2. Blocked

3. Cleaned

**4. Escaped**

**64. Guard bands are used in \_\_\_\_\_ multiplexing technique, to avoid overlapping of frequency bands assigned to each user.**

1. PDM

2. CSMA

3. TDM

#### 4. FDM

65. DLC in Data Link Layer stands for \_\_\_\_\_.

1. Divide Line Communication
2. Data Line Code
3. Data Link Control
4. Demand Link Coordination

66. In Pulse Code Modulation (PCM), the sampling is dependent on \_\_\_\_\_.

1. Time
2. Amplitude
3. Frequency
4. Signal Rate

67. Digital signals are referred to be \_\_\_\_\_.

1. Continuous
2. Discrete
3. All of the given
4. None of the given

68. Two or more computers connected so that they can communicate with each other and share information is called a \_\_\_\_\_.

1. satellite
2. protocol

3. broadcast

4. network

69. Which error detection method uses one's complement arithmetic.

1. Cyclic Redundancy Check (CRC)

2. Longitudinal Red Check (LRC)

3. Checksum

4. Simple Parity Check

70. WDM stands for \_\_\_\_\_.

1. Wave Division Multiplexing

2. Wrong Division Multiplexing

3. Well Division Multiplexing

4. West Division Multiplexing

71. Analog refers to something that is continuous in \_\_\_\_\_.

1. Time

2. Space

3. Frequency

4. None of the given

72. What is the frequency of a sine wave if it completes one cycle in 5 seconds?

1. 0.5



2. 0.6

3. 0.3

4. 0.2

**73. In an analog hierarchy to carry voice channels, a group can carry \_\_\_\_\_ voice channels.**

1. 60

2. 12

3. 20

4. 10

**74. STP is more expensive than UTP and is less susceptible to noise.**

1. True

2. False

**75. Switch is a network device which operates on the \_\_\_\_\_ layer of the TCP/IP protocol suite.**

1. Application

2. Transport

3. Presentation

4. Data link

**76. \_\_\_\_\_ are used to exchange session management and control information between connected devices.**

1. I-frames

2. U-frames

3. S-frames

4. N-frames

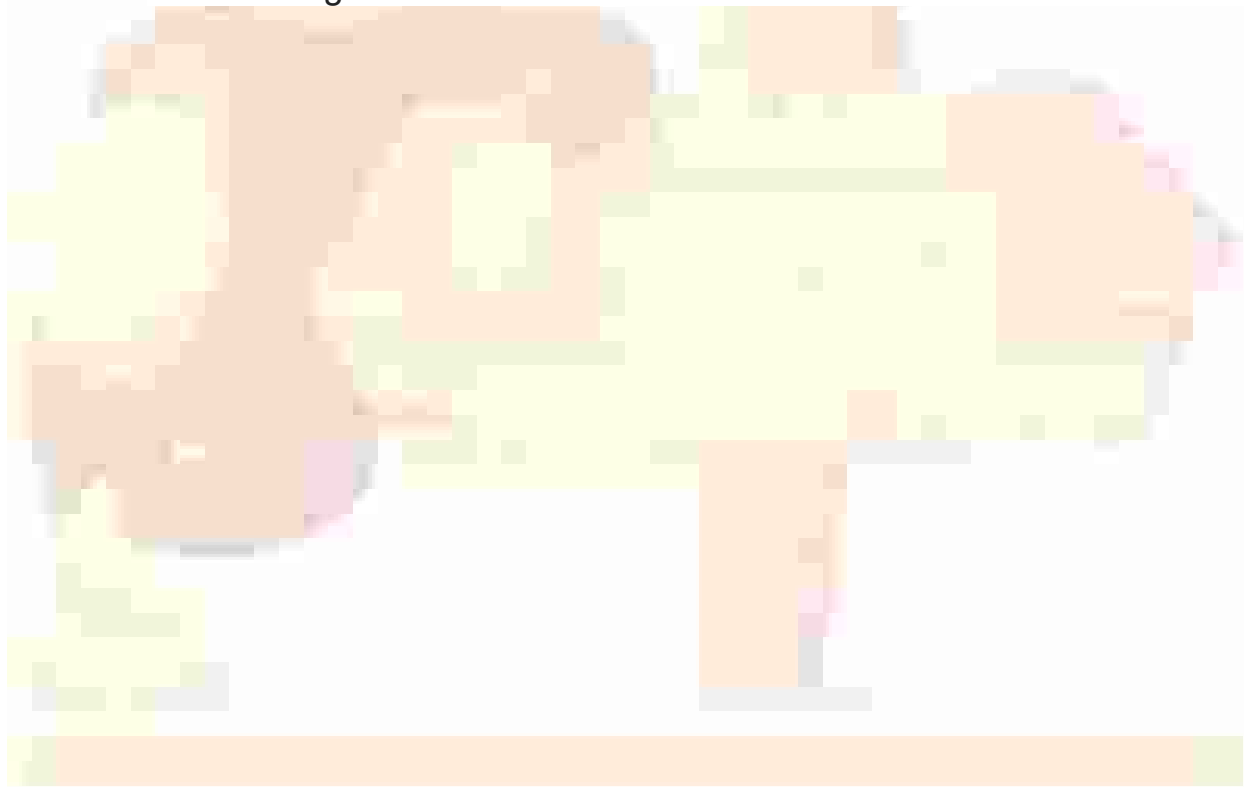
77. The message 1110 sent by a source is received by a destination as 1011. This is \_\_\_\_\_ type of error.

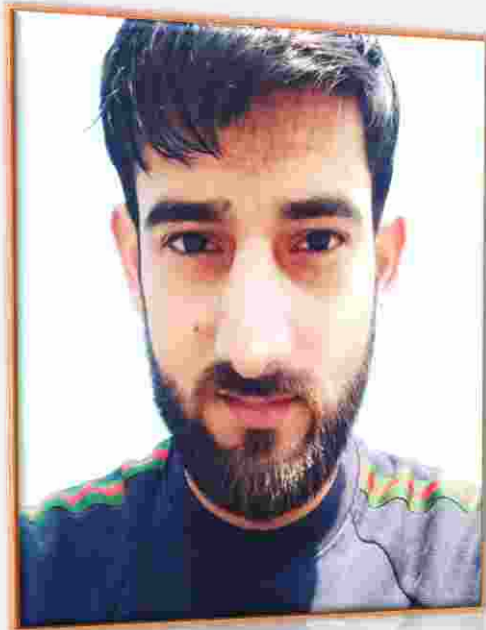
1. Burst

2. Single-Bit

3. Uni-Bit

4. Hamming





# CS601 QUIZ(2)

Topics 151 to 200

## RIZ MUGHAL SQA ENGINEER:

I'm providing 100% correct quiz solution.

You can visit my YouTube channel for more quiz solution, also final year project including project assignments, and viva.

## YOUTUBE:

<https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists>

## FACEBOOK:

<https://www.facebook.com/groups/923887914750307>

CS601: Quiz No. 2

Question # 1 of 10 ( Start time: 08:17:18 AM, 24 February 2021 )

Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol.

Select the correct option:

|                                  |             |
|----------------------------------|-------------|
| <input type="radio"/>            | Application |
| <input type="radio"/>            | Network     |
| <input type="radio"/>            | Session     |
| <input checked="" type="radio"/> | Transport   |

RIZ MUGHAL

## Question # 2 of 10 ( Start time: 08:17:35 AM, 24 February 2021 )

In MAC, when nodes use a multi-point or broadcast link, we need a \_\_\_\_\_ to coordinate access to the link.

Select the correct option

|                                  |                          |
|----------------------------------|--------------------------|
| <input checked="" type="radio"/> | Multiple-access Protocol |
| <input type="radio"/>            | Data encrypt Protocol    |
| <input type="radio"/>            | Telnet Protocol          |
| <input type="radio"/>            | Application Protocol     |

RIZ MUGHAL

Question # 3 of 10 ( Start time: 08:17:47 AM, 24 February 2021 )

\_\_\_\_\_ is an Authentication Protocol, which uses two-step process to authenticate user information

Select the correct option

|                                  |      |
|----------------------------------|------|
| <input type="radio"/>            | CHAP |
| <input type="radio"/>            | LCP  |
| <input checked="" type="radio"/> | PAP  |
| <input type="radio"/>            | ICMP |

CS601: Quiz No. 2

Question # 4 of 10 ( Start time: 08:17:58 AM, 24 February 2021 )

\_\_\_\_\_ uses fiber-optic cable.

Select the correct option

- |                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | 10Base5  |
| <input type="radio"/>            | 10Base2  |
| <input type="radio"/>            | 10Base-T |
| <input checked="" type="radio"/> | 10Base-F |

Question # 5 of 10 ( Start time: 08:18:09 AM, 24 February 2021 )

In \_\_\_\_\_ each band is reserved for a specific station, and it belongs to the station all the time.

Select the correct option.

|                                     |      |
|-------------------------------------|------|
| <input checked="" type="checkbox"/> | FDMA |
| <input type="checkbox"/>            | TDMA |
| <input type="checkbox"/>            | CDMA |
| <input type="checkbox"/>            | CSMA |



Question # 6 of 10 ( Start time: 08:18:20 AM, 24 February 2021 )

In character oriented protocol, each character of frame is encoded using \_\_\_\_\_ coding system.

Select the correct option

|                                     |             |
|-------------------------------------|-------------|
| <input checked="" type="checkbox"/> | ASCII       |
| <input type="checkbox"/>            | NRZ         |
| <input type="checkbox"/>            | Unicode     |
| <input type="checkbox"/>            | Bipolar AMI |

## Question # 7 of 10 ( Start time: 08:18:31 AM, 24 February 2021 )

The flag in PPP is a byte that needs to be \_\_\_\_\_ whenever it appears in the data section of the frame.

Select the correct option

- |                                  |            |
|----------------------------------|------------|
| <input type="radio"/>            | Duplicated |
| <input type="radio"/>            | Blocked    |
| <input type="radio"/>            | Cleaned    |
| <input checked="" type="radio"/> | Escaped    |
- RIZ MUGHAL
- R

Question # 8 of 10 ( Start time: 08:18:44 AM, 24 February 2021 )

Stop-and-Wait protocol is used for \_\_\_\_\_

Select the correct option

|                                  |                                |
|----------------------------------|--------------------------------|
| <input type="radio"/>            | Flow control only              |
| <input type="radio"/>            | Error control only             |
| <input checked="" type="radio"/> | Both flow and error control    |
| <input type="radio"/>            | Neither flow nor error control |

CS601: Quiz No: 2

Question # 9 of 10 ( Start time: 08:18:58 AM, 24 February 2021 )

PAP and CHAP are used by \_\_\_\_\_ as Authentication Protocols.

Select the correct option

|                                  |         |
|----------------------------------|---------|
| <input type="radio"/>            | SMTP    |
| <input type="radio"/>            | TCP/ IP |
| <input checked="" type="radio"/> | PPP     |
| <input type="radio"/>            | ICMP    |

Question # 10 of 10 ( Start time: 08:19:10 AM, 24 February 2021 )

In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time

Select the correct option

- |                                  |             |
|----------------------------------|-------------|
| <input type="radio"/>            | Same as     |
| <input checked="" type="radio"/> | Two times   |
| <input type="radio"/>            | Three times |
| <input type="radio"/>            | Four times  |

2<sup>nd</sup> account

CS601 Quiz No. 2

Question # 1 of 10 ( Start time: 08:22:51 AM, 24 February 2021 )

\_\_\_\_\_ uses thin coaxial cable.

Select the correct option

- |                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | 10Base5  |
| <input checked="" type="radio"/> | 10Base2  |
| <input type="radio"/>            | 10Base-T |
| <input type="radio"/>            | 10Base-F |
- RIZ MUGHAL

CS601: Quiz No. 2

Question # 2 of 10 ( Start time: 08:23:12 AM, 24 February 2021 )

\_\_\_\_\_ was used as the medium in 10 Base-T.

Select the correct option

- |                                  |                      |
|----------------------------------|----------------------|
| <input type="radio"/>            | Thick Coaxial Cable  |
| <input type="radio"/>            | Thin Coaxial Cable   |
| <input checked="" type="radio"/> | Twisted Pair Cable   |
| <input type="radio"/>            | Two Mode Fiber Cable |



CS601: Quiz No. 2

Question # 3 of 10 ( **Start time: 08:23:23 AM, 24 February 2021** )

The \_\_\_\_\_ protocol uses both flow and error control.

Select the correct option:

|                                  |               |
|----------------------------------|---------------|
| <input type="radio"/>            | TCP/IP        |
| <input type="radio"/>            | HDLC          |
| <input type="radio"/>            | ICMP          |
| <input checked="" type="radio"/> | Stop-and-Wait |



CS601:Quiz No. 2

Question # 4 of 10 ( ~~Start time: 08:23:36 AM, 24 February 2021~~ )

POTS stand for \_\_\_\_\_ ?

Select the correct option :

- |                                  |                             |
|----------------------------------|-----------------------------|
| <input type="radio"/>            | Plain Old Telephone Set     |
| <input checked="" type="radio"/> | Plain Old Telephone Service |
| <input type="radio"/>            | Plain Old Time Sharing      |
| <input type="radio"/>            | Plain Old Telephone System  |
- RIZ MUGHAL

CS601:Quiz No. 2

Question # 5 of 10 ( Start time: 08:23:49 AM, 24 February 2021 )

Transmission Control Protocol is a(n) \_\_\_\_\_ layer protocol:

Select the correct option

|                                  |             |
|----------------------------------|-------------|
| <input type="radio"/>            | Application |
| <input type="radio"/>            | Network     |
| <input type="radio"/>            | Session     |
| <input checked="" type="radio"/> | Transport   |

Question # 6 of 10 ( Start time: 08:24:03 AM, 24 February 2021 )

In Pure ALOHA, the vulnerable time is \_\_\_\_\_ the frame transmission time

Select the correct option

- |                                  |             |
|----------------------------------|-------------|
| <input type="radio"/>            | Same as     |
| <input checked="" type="radio"/> | Two times   |
| <input type="radio"/>            | Three times |
| <input type="radio"/>            | Four times  |

Question # 7 of 10 ( Start time: 08:24:14 AM, 24 February 2021 )

In character oriented protocol, each character of frame is encoded using \_\_\_\_\_ coding system.

Select the correct option

|                                     |             |
|-------------------------------------|-------------|
| <input checked="" type="checkbox"/> | ASCII       |
| <input type="checkbox"/>            | NRZ         |
| <input type="checkbox"/>            | Unicode     |
| <input type="checkbox"/>            | Bipolar AMI |

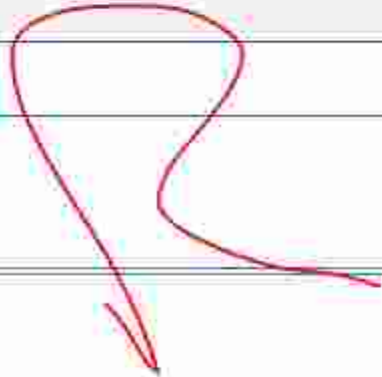
Question # 8 of 10 ( Start time: 08:24:27 AM, 24 February 2021 )

A personal computer or workstation on an Ethernet network must have \_\_\_\_\_ card.

Select the correct option

|                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | TDI               |
| <input checked="" type="radio"/> | NIC               |
| <input type="radio"/>            | PCI               |
| <input type="radio"/>            | None of the given |

RIZ MUGHAL



CS601: Quiz No. 2

Question # 9 of 10 ( Start time: 08:24:39 AM, 24 February 2021 )

Fast Ethernet has a data rate of \_\_\_\_\_ Mbps.

Select the correct option

|                                  |        |
|----------------------------------|--------|
| <input type="radio"/>            | 10     |
| <input checked="" type="radio"/> | 100    |
| <input type="radio"/>            | 1000   |
| <input type="radio"/>            | 10,000 |

Question # 10 of 10 ( Start time: 08:24:52 AM, 24 February 2021 )

DSL stands for \_\_\_\_\_

Select the correct option

- |                                  |                         |
|----------------------------------|-------------------------|
| <input checked="" type="radio"/> | digital subscriber line |
| <input type="radio"/>            | digital switched line   |
| <input type="radio"/>            | data subscriber line    |
| <input type="radio"/>            | data switched line      |

Thank you for watching 😊

Share with your fellows

[rizwanqadeer848@gmail.com](mailto:rizwanqadeer848@gmail.com)





## CS601-Data Communication

Latest Solved subjective from Final term Papers

May 08,2011

Mc100401285

moaaz.pk@gmail.com

mc100401285@Gmail.com

Latest subjectives

### FINAL TERM EXAMINATION SPRING 2011

**What is the difference between guided and unguided media?**

**Answer:** [Click here for detail](#)

Guided Media are those media that provide a conduit from one device to another. Guided Transmission Media uses a "cabling" system that guides the data signals along a specific path while Unguided Transmission Media consists of a means for the data signals to travel but nothing to guide them along a specific path. It passes through a vacuum; it is independent of a physical pathway.

**Write Commercial advantage and characteristics of token bus**

**Answer:** (Page 232)

- Other LANs are not suitable for this purpose
- Token Bus has no commercial application in data communications
- Token Ring allows each station to send one frame per turn
- Access method: Token passing

**What is the difference between FDM and TDM**

**Answer:** [Click here for detail](#)

- 1) FDM-Frequency division multiplexing where as TDM mean Time division Multiplexing.
- 2) In FDM spectrum is divided into frequency whereas in TDM divided into time slot.
- 3) FDM is used in 1st generation analog system whereas TDM is used in 2nd generation analog system.

**Write the types transmission noise**

**Answer:** (Page 143)

**Thermal Noise:** Due to random originally sent by TX

**Induced Noise:** Comes from sources like Motors and Appliances

**Crosstalk:** Effect of one wire on another

**Impulse Noise:** Spike (A signal with high energy in a very short period of time power lines, lightning etc.

**What is power bandwidth**

**Answer:** [Click here for detail](#)

The **power bandwidth** of an amplifier is sometimes taken as the frequency range (or, rarely, the upper frequency limit) for which the rated power output of an amplifier can be maintained (without excessive distortion) to at least *half* of the full rated power.

OR

# Muhammad Moaaz Siddiq

**Power Bandwidth**

**Answer:** (Page 67)

Frequency band in which 99% of the total power resides.

**What does the CRC generator append to data unit? [2]**

**Answer:** (Page 175)

Appending it to the end of the data must make the resulting bit sequence exactly divisible by the divisor

**How much bandwidth for modem is required in case of FSK? [2]**

**Answer:**

BW required for FSK is equal to the Baud rate of the signal plus the frequency shift. Because of the limitations of voice-grade telephone lines, these modems are restricted to a bandwidth of about 3 kHz

**What is even parity generator in VRC error detection mechanism? [2]**

**Answer:** (Page 172)

Even parity generator counts the 1's and appends the parity bit (1) to the end.

**What is the difference between angle of incident and angle of reflection? [2]**

**Answer:** (Page 126)

The difference between them is that Angle of refraction passes from less dense to denser medium whereas angle of incidence passes from more dense to less dense medium.

**What is daisy chaining in 1Base 5 star Lan? [2]**

**Answer:** (Page 229)

Slower speed in star lan can be increase by the use of DAISY CHAINING.

**What is the responsibility of Application layer? [3]**

**Answer:** Page 53

Enables the user either human or software to access the network

It provides user interface and support for the services such as Electronic mail, Remote File access and Transfer, Shared Database Management and other services

**What is critical angle?**

**Answer:** (Page 127)

We have a beam of light moving from a more dense to a less dense medium. We gradually increase the angle of incidence measured from vertical axis. As angle of incidence increases, so does the angle of refraction.

The angle at which refracted line lies on the horizontal axis is called **Critical Angle**

**Hamming code-Redundancy bit (5 marks Q)**

**Answer:** (Page 181)

**Redundancy Bits (r)**

- r must be able to indicate at least  $m+r+1$  states
- $m+r+1$  states must be discoverable by r bits
- Therefore,  $2r \geq m+r+1$

**Muhammad Moaaz Siddiq**

- If  $m=7$ ,  $r=4$  as  $24 \geq 7+4+1$

### **Tree topology advantages (3 or 5 marks)**

**Answer: (Page 31)**

- Because of Secondary Hub, More devices can be attached to a Central Hub and therefore increase the distance a signal can travel
- Enables Differentiated Services: Allows to prioritize communication, e.g. computers attached to one secondary hub can be given priority over others
- Therefore, TIME SENSITIVE data will not have to wait for access to the network
- Rest of the advantages are almost the same as STAR

### **Q #41: Whether VRC error detection method is used for single bit error or burst error. (2)**

**Answer: (Page 173)**

VRC can detect all single bit errors

Can also detect Burst errors as long as the total number of bits changed is ODD

### **Q # 42: Which modem was first developed commercially in 1970? (2)**

**Answer: (Page 114)**

Bell modems

- First commercial modems by Bell Telephone Co.
- Developed in early 1970s

### **Q # 45: Consider a major telecom company using RZ encoding for its signals**

**conversion. What will be the major problem faced by using such type of encoding? (2)**

**Answer: (Page 75)**

Any time, data contains long strings of 1's or 0's, Rx can lose its timing.

The only problem with RZ encoding is that it requires two signal changes to encode one bit and therefore occupies more BANDWIDTH

### **Q # 47: Geosynchronous Satellite? (3)**

**Answer: (Page 139)**

- Line of sight propagation requires the sending and receiving antennas must be locked into each other
- To ensure continuous communication, satellites must move with the same speed as earth. So that they seem fixed w.r.t earth
- These satellites are called Geosynchronous Satellites

**Muhammad Moaaz Siddiq**

## FINAL TERM EXAMINATION 2011

### Question#1

**What are the Asynchronous protocols in data communication layer? .....Marks (10)**

**Answer: (Page 206)**

Asynchronous protocols Treat each character in a Bit stream independently. Employed mainly in Modems.

Transmission does not require timing coordination; Timing is done by using extra bits

### Different Asynchronous Protocols

#### XMODEM

- The first field is a One Byte start of header (SOH) field
- The second field is a two-byte Header.
  - The first header byte , the Sequence number carries the Frame number
  - The second header byte is used to check the validity of the sequence number
- The fixed data field holds 128 bytes of data
- The last field CRC checks for errors in the data field only

#### YMODEM

YMODEM is similar to X-MODEM with only the following major differences:

- 1024-Byte data unit
- Two CANs to abort Transmission
- ITU-T CRC-16 for Error Checking
- Multiple files can be sent simultaneously

#### ZMODEM :

Newer Protocol

Combines features of XMODEM and YMODEM.

#### BLAST

- Blocked Asynchronous Transmission
- More powerful than XMODEM
- Full Duplex
- Sliding Window Flow Control
- Allows transfer of Data and Binary Files

#### KERMIT

- Designed at Columbia University
- Most Widely used Asynchronous Protocol
- File Transfer protocol is similar in operation to XMODEM, with sender waiting for an NAK before it starts TX
- Kermit allows the transmission of control characters as Text

**Muhammad Moaaz Siddiq**

**Question#2****What is Frequency division multiplexing ?.....Marks (5)****Answer: (Page 149)****Frequency division multiplexing (FDM)**

- An analog technique that can be applied when BW of the link is greater than the combined BW of the signals to be TX
- Signals generated by each sending device modulate different carrier frequencies
- These modulated signals are then combined into a single Composite signal that can be transported by the link
- Carrier frequencies are separated by enough BW to accommodate the modulated signal
- These BW ranges are the channels through which the various signals travel

**Question#4****What is stop and wait ARQ in error control ?....Marks (3)****Answer: Page 197**

Stop-and-Wait is an extended form of flow control to include retransmission of data in case of Lost or Damaged frames.

There are four main features added in it.

1. Sending device keeps a copy of the last frame transmitted until it receives the ACK for that frame.
2. Both data and ACK frames are numbered 0 and 1 alternately.
3. A data 0 frame is acknowledged by a ACK 1 frame indicating that the receiver has received data 0 and is now expecting data 1 .
4. For retransmission to work, 4 features are added to the basic flow control mechanism.

**Question#5****What is Interleaving ?.....Marks (3)****Answer: Page 153**

Synchronous TDM is considered as a very fast rotating switch. When this switch opens in front of a device, the device has the opportunity to send a specific amount of data on to the path.

The switch moves from device to device at a constant rate and in a fixed order. This process is called INTERLEAVING. Interleaving can be done by BITS, BYTES or by any other DATA UNIT

**Question#6****What is DSU in terms of digital services?.....Marks (3)****Answer: Page 163**

DSU (Digital service unit) changes the rate of digital data created by the subscriber's device to 56 Kbps and encodes it in the format used by service provider. It is used in dialing process and is more expensive than MODEM. But it has better speed, better quality and less susceptibility to noise.

**Question#7****Which architecture of Ethernet developed by ITU\_T and****ANSI?..... Marks (2)****Answer: 236**

FDDI (Fiber Distributed Data Interface) architecture of Ethernet developed by ITU\_T and ANSI.

# Muhammad Moaaz Siddiq

**Question#8**

**What is a spike in noise term?.....Marks (2)**

**Answer:143**

Spike is a signal with high energy in a very short period of time that comes from power lines, lightening etc,

**Question#9**

**Answer: Page 172**

**What is even parity generator in VRC error detection mechanism?.....Marks (2)**

Even parity generator counts the 1's and appends the parity bit (1) to the end.

**Question#10**

**Compare line discipline methods ENQ/ACK and Poll/ Select?**

**Answer: Page 188-189**

=>ENQ/ACK coordinates which device may start a transmission and whether or not the intended recipient is ready and enabled.

=> Using ENQ/ACK, a session can be initiated by either station on a link as long as both are of equal rank.

=> In both half-duplex and full-duplex transmission, the initiating device establishes the session.

=> In half duplex, the initiator then sends its data while the responder waits. The responder may take over the link when the initiator is finished or has requested a response.

=> In full duplex, both devices can transmit simultaneously once the session has been established.

POLL/SELECT:

=> The poll/select method of line discipline works with topologies where one device is designated as a primary station and the other devices are secondary stations.

=> Multipoint systems must coordinate several nodes, not just two.

=> The primary device controls the link and the secondary device follow its instruction

It is up to the primary to determine which device is allowed to use the channel data given time

**Why addressing is required in Poll / Select method and not required in ENQ/ACK method?3**

**Answer: (Page 190)**

Addressing is required in Poll / Select method as it is a not point-to-point configuration, For the primary device in a multipoint topology to be able to identify and communicate with a specific secondary device, there must be some addressing, while ENQ/ACK method is a point-to-point method and for point-to-point configuration, there is no need for addressing.

**What do you know about ITU-T Modems?3**

**Answer: (Page 114)**

ITU-T modem :

V-series: Today's most popular modem standards

Bell modem compatible:

V.21/22/23/26/27/29

**Muhammad Moaaz Siddiq**



**Following abbreviations stands for what?3**

**Answer: (Page 224)**

ARP..... (Address Resolution Protocol)

RARP..... (Reverse Address Resolution Protocol)

ICMP ..... (Internet Control Message Protocol)

**Write names of Link Access Protocols developed by ITU-T?3**

**Answer: (Page 211)**

LAPs: LAPB, LAPD, LAPM, LAPZ etc. all based on HDLC

**Write the names of different types of noise in the medium?3**

**Answer: (Page 144)**

Thermal Noise

Induced Noise

Crosstalk

Impulse Noise

**Write down some disadvantages of star topology.3**

**Answer: (Page 30)**

Although Cabling required is far less than Mesh

Still each node must be connected to a Hub , so Cabling is still much more

**What are the two major classes of synchronous protocols at data link layer?2**

**Answer: (Page 206)**

Character – Oriented Protocols

Bit – Oriented Protocols

**Whether Hamming code is the technique used for error detection or error correction?2**

**Answer: (Page 181)**

Hamming code is the technique used for error correction

**Define Multiplexing? What is its advantage?2**

**Answer: (Page 147)**

Set of techniques that allows the simultaneous transmission of multiple signals across a single data link

It allows multiple users to share total capacity of a Transmission Medium.

**What is the purpose of dual ring?2**

**Answer: (Page 34)**

Unidirectional traffic movement is overcome by dual ring technology.

**Which modem was first developed commercially in 1970?2**

**Answer: (Page 114)**

Bell modems

**Write any two functions of physical layer?2**

**Answer: (Page 45)**

**Muhammad Moaaz Siddiq**

It defines characteristics of Interface between device and transmission Medium  
It also defines the type of transmission medium  
Physical Layer is also concerned with Line Configuration

## **FINAL TERM EXAMINATION 2011**

**Which one has more overhead, a repeater or a bridge? Explain your answer. [3]**

**Answer:**

A bridge has more overhead than a repeater. A bridge processes the packet at two layers; a repeater processes a frame at only one layer. A bridge needs to search a table and find the forwarding port as well as to regenerate the signal; a repeater only regenerates the signal. In other words, a bridge is also a repeater (and more); a repeater is not a bridge

**Define high frequency [HF] and super high frequency [SHF], which devices uses these frequencies [3]**

**Answer: [Page 135 and 136](#)**

High frequency.

HF uses ionospheric propagation. These frequencies move into the ionosphere where the density difference reflects them back on earth.

It is used for Citizen's Band Radio, International Broadcasting, Military Communication, Telephone, Telegraph and Fax

Super high frequency.

SHF waves are TX using mostly line-of-sight and some Space propagation.

It is used for Terrestrial and Satellite Microwave and Radar Communication devices.

**Write all steps of checksum method. [3]**

**Answer: [\(Page 179\)](#)**

- o The sender subdivides data units into equal segments of 'n' bits(16 bits)
- o These segments are added together using one's complement
- o The total (sum) is then complemented and appended to the end of the original data unit as redundancy bits called CHECKSUM
- o The extended data unit is transmitted across the network
- o The receiver subdivides data unit as above and adds all segments together and complement the result
- o If the intended data unit is intact, total value found by adding the data segments and the checksum field should be zero
- o If the result is not zero, the packet contains an error & the receiver rejects it

**Differentiate internet and the Internet? [3]**

**Answer: [\(Page 240\)](#)**

**INTERNET**

- o *An internet* is a generic term used to mean an interconnection of individual networks
- o To create an internet, we need networking devices called routers and gateways
- o An internet is different from the Internet
- o Internet is the name of a specific worldwide network

# **Muhammad Moaaz Siddiq**



**What is the differences in between bit oriented and character oriented protocols [5]**

**Answer:** (Page 206)

### **Character – Oriented Protocols**

- Also called **Byte- Oriented Protocol**
- These protocols interpret a transmission frame or packet as a succession of characters, each usually composed of one byte
- All control information is in the form of an existing character encoding system

### **Bit – Oriented Protocols**

- Character –Oriented Protocols are not as efficient as bit – oriented protocols and are seldom used
- They are easy to comprehend and employ the same logic as bit-oriented protocols
- Their study will provide the basis for studying the other data link layer protocols
- IBN's BSC is the best known character oriented protocol

## **FINAL TERM EXAMINATION 2011**

### **Question#1**

**What is the formula to calculate the number of redundancy bits required to correct a bit error in a given number of data bits? [2]**

**Answer:** [Click here for detail](#)

Messages(frames) consist of m data (message) bits, yielding an  $n=(m+r)$ -bit codeword.

### **Question#2**

**What is R G rating of coaxial cable?**

**Answer:-** (Page 126)

Different coaxial cable designs are categorized by their Radio government ( RG ) ratings

Each cable defined by RG rating is adapted for a specialized function:

RG-8

- Used in Thick Ethernet

RG-9

- Used in Thick Ethernet

RG-11

- Used in Thick Ethernet

RG-58

- Used in Thin Ethernet

RG-59

- Used for TV

### **Question#3**

**What are the advantages of thin Ethernet?**

**Answer:** (Page 228)

The **advantages** of thin Ethernet are:

1. reduced cost
2. ease of installation

Because the cable is lighter weight and more flexible than that used in Thick net

# **Muhammad Moaaz Siddiq**

#### Question#4

What is the difference between a unicast, multicast, and broadcast address? [3]

**Answer:** [Click here for detail](#)

**Broadcast:** transmitting a packet that will be received by every device on the network

**Unicast:** the sending of information packets to a single destination

**Multicast:** delivery of information to a group of destinations.

#### Question#5

T lines are designed for Digital data how they can be used for Analog Transmission?

**Answer:** [\(Page 166\)](#)

- o DS-1 requires 8 Kbps of overhead
- o To understand this overhead, let's examine format of a 24-voice channel frame
- o Frame used on T-1 line is usually 193 bits divided into 24 slots of 8 bits each + 1 bit for synchronization ( $24 \times 8 + 1 = 193$ )
- o 24 segments are interleaved in one frame
- o If a T-1 carries 8000 frames, the data rate is 1.544 Mbps ( $193 \times 8000 = 1.544$  Mbps) which is capacity of the line

#### Question#6

What are the three types of Guided Media?

**Answer:** [\(Page 120\)](#)

1. Coaxial cable
2. Twisted-pair cable
3. Fiber optic cable.

#### Question#7

Why do we need Inverse Multiplexing? [5]

**Answer:** [\(Page 159\)](#)

- An organization wants to send data, voice and video each of which requires a different data rate
- To send voice it needs 64Kbps,
- To send data, it needs 128 Kbps link
- To send video it may need 1.544 Mbps link
- It can lease a 1.544 Mbps line from a common carrier and only use it fully for sometime
- Or it can lease several separate channels of lower data rates
- Voice can be sent over any of these channels
- Data & Video can be broken into smaller portions using Inverse Multiplexing and TX

#### Question#8

Describe method of checksum briefly?

**Answer:** [\(Page 180\)](#)

- o The sender subdivides data units into equal segments of 'n' bits (16 bits)
- o These segments are added together using one's complement
- o The total (sum) is then complemented and appended to the end of the original data unit as redundancy bits called CHECKSUM
- o The extended data unit is transmitted across the network
- o The receiver subdivides data unit as above and adds all segments together and complement the result
- o If the intended data unit is intact, total value found by adding the data segments and the checksum field should be zero
- o If the result is not zero, the packet contains an error & the receiver rejects it

# Muhammad Moaaz Siddiq

**Question#9**

**Explain Asynchronous Time Division Multiplexing in detail? Also discuss its advantages over synchronous TDM?**

**Answer:**

Asynchronous time-division multiplexing (ATDM) is a method of sending information that resembles normal TDM, except that time slots are allocated as needed dynamically rather than pre-assigned to specific transmitters. ATDM is more intelligent and has better bandwidth efficiency than TDM. asynchronous time-division multiplexing comprising receive circuits (CRI/i) supplying cells received via input links, transmit circuits (CTI/j) transmitting retransmitted cells on output links, a buffer memory (MT) storing the received cells and delivering the cells to be retransmitted and a buffer memory addressing device (SMT) including a write address source (SAE) and a read address source (fsl/j).

Advantages asynchronous TDM:

In asynchronous TDM, the timeslots are not fixed. They are assigned dynamically as needed.

The objective would be to switch from one user to another user whenever the one user is idle, and to asynchronously time multiplex the data. With such an arrangement, each user would be granted access to the channel only when he has a message to transmit. This is known as an Asynchronous Time Division Multiplexing System (ATDM). A segment of a typical ATDM data stream is shown in Figure 2. The crucial attributes of such a multiplexing technique are:

1. An address is required for each transmitted message, and
2. Buffering is required to handle the random message arrivals.

## FINAL TERM EXAMINATION 2011

**Question No: 31 ( Marks: 2 )**

What is hybrid topology?

**Answer:**

Hybrid topology is a kind of topology, In which Several topologies combined in a larger topology

**Question No: 32 ( Marks: 2 )**

What is combined station of HDLC?

**Answer: (Page 211)**

A combined station is one of a set of connected peer devices programmed to behave either as a primary or as a secondary depending on the nature and the direction of the transmission.

**Question No: 33 ( Marks: 2 )**

What kind of error is undetectable by the checksum?

**Answer: (Page 180)**

Error is invisible if a bit inversion is balanced by an opposite bit inversion in the corresponding digit of another segment

**Question No: 34 ( Marks: 2 )**

What's the name of the telephone service in which there is no need of dialing?

**Answer: (Page 161)**

# Muhammad Moaaz Siddiq

In Analog Leased Service there is no need of dialing

**Question No: 35 ( Marks: 3 )**

How Bit Rate & Baud rate are related?

**Answer: (Page 85)**

Bit rate equals the baud rate times the no. of bits represented by each signal unit.

The baud rate equals the bit rate divided by the no. of bits represented by each signal shift.

Bit rate is always greater than or equal to Baud rate

**Question No: 36 ( Marks: 3 )**

Following abbreviations stands for what?

**Answer: (Page 244)**

1. ARP: Address Resolution Protocol
2. RARP: Reverse Address Resolution Protocol
3. ICMP: Internet Control Message Protocol

**Question No: 37 ( Marks: 3 )**

Differentiate between Polling and Selecting.

**Answer: (Page 189)**

If the primary wants to receive data, it asks the second-ary if they have anything to send, This is called Polling

If the primary wants to send data, it tells the target secondary to get ready to receive, This function is called Selecting.

**Question No: 38 ( Marks: 3 )**

T lines are designed for Digital data how they can be used for Analog Transmission?

**Answer: repeat**

**Question No: 39 ( Marks: 5 )**

What is the difference between character oriented and bit oriented protocols?

**Answer: repeat**

**Question No: 40 ( Marks: 5 )**

Write down the function of Primary-Secondary communication in line discipline.

**Answer: (Page 189)**

1. Poll method works with topologies where one device is designed as a Primary station and the other devices are Secondary stations
2. The primary device controls the link and the secondary device follow its instruction
3. It is up to the primary to determine which device is allowed to use the channel at a given time.
4. The primary therefore is always the initiator of the a session
5. Whenever a multipoint link consists of a primary device and multiple secondary devices using a single TX line , all exchanges must be made through the primary device even when the ultimate destination is a secondary device

**Muhammad Moaaz Siddiq**

## FINAL TERM EXAMINATION 2010

**Question No: 31 ( Marks: 2 )**

**What kind of error is undetectable by the checksum? [2]**

**Answer: Page 180 (repeated)**

**Question No: 32 ( Marks: 2 )**

**What are properties of signals?**

**Answer: (Page 17)**

Capable of being propagated over TX. Medium ,Interpretable as data at the Receiver

**Question No: 33 ( Marks: 2 )**

**Whether in Asynchronous or Synchronous TDM, addressing is used?**

**Answer: (Page 158)**

Addressing is used only in Asynchronous TDM.

**Question No: 34 ( Marks: 2 )**

**What is the basic purpose of Router?**

**Answer: [Click here for detail](#)**

**Basic purpose of Router:**

A router is a device that extracts the destination of a packet it receives, selects the best path to that destination, and forwards data packets to the next device along this path. They connect networks together; a LAN to a WAN for example, to access the Internet.

**Question No: 35 ( Marks: 3 )**

**Why we need a Null Modem?**

**Answer: (Page 106)**

A null modem provide DTA –DTE interface w/o DCEs

**Question No: 36 ( Marks: 3 )**

**Count LRC for the following bits?**

**10011010 10100101 1101 0110**

**Question No: 37 ( Marks: 3 )**

**What are the categories of multiplexing?**

**Answer: Page 148**

There are three catogaries of multiplxing

FDM

TDM

Have a two other catagories

Synchronous and asyrounce

WDM

**Question No: 38 ( Marks: 3 )**

**What are the three purposes of control frames at data link layer?**

**Answer: (Page 209)**

Control frames serve 3 purposes:

# Muhammad Moaaz Siddiq

Establishing Connections

Maintaining Flow and Error Control during Data Transmission

Terminating Connection

**Question No: 39 ( Marks: 5 )**

**Compare line discipline methods ENQ/ACK and Poll/ Select?**

**Answer: (Page 188-189)**

=>ENQ/ACK coordinates which device may start a transmission and whether or not the intended recipient is ready and enabled.

=> Using ENQ/ACK, a session can be initiated by either station on a link as long as both are of equal rank.

=> In both half-duplex and full-duplex transmission, the initiating device establishes the session.

=> In half duplex, the initiator then sends its data while the responder waits. The responder may take over the link when the initiator is finished or has requested a response.

=> In full duplex, both devices can transmit simultaneously once the session has been established.

**POLL/SELECT:**

=> The poll/select method of line discipline works with topologies where one device is designated as a primary station and the other devices are secondary stations.

=> Multipoint systems must coordinate several nodes, not just two.

=> The primary device controls the link and the secondary device follow its instruction

It is up to the primary to determine which device is allowed to use the channel data given time

**Question No: 40 ( Marks: 5 )**

**What is the difference between character oriented and bit oriented protocols?**

**Answer: repeat**

## **FINAL TERM EXAMINATION 2010**

**Question No: 31 ( Marks: 2 )**

**What are the advantages of a multipoint connection over a point-to-point connection?**

**Answer:**

Point-to-point connection is limited to two devices, where else more than two devices share a single link in multipoint connection. Multipoint connection can be used for fail-over and reliability.

**Question No: 32 ( Marks: 2 )**

**What's the name of the telephone service in which there is no need of dialing.**

**Answer: (Page 164)**

DSS (digital data service) is the telephone service in which there is no need of dialing.

**Question No: 33 ( Marks: 2 )**

**Which type of frames is present in BSC frames?**

**Answer: (Page 206)**

# **Muhammad Moaaz Siddiq**



There are two types of frames that are present in BSC.

1. Control Frames and
2. Data Frames

**Question No: 34 ( Marks: 2 )**

**What methods of line discipline are used for peer to peer and primary secondary communication?**

**Answer: (Page 187)**

Line discipline is done in two ways:

1. ENQ/ACK (Enquiry Acknowledgement)

This is used for peer to peer communication.

2. Poll/ Select

This method is used for primary secondary communication.

**Question No: 35 ( Marks: 3 )**

**How does the checksum checker know that the received data unit is undamaged? [3]**

**Answer: (Page 179)**

**Checksum Checker or generator:**

The sender subdivides data units into equal segments of 'n' bits(16 bits)

1. These segments are added together using one's complement.
2. The total (sum) is then complemented and appended to the end of the original data unit as redundancy bits called CHECKSUM.
3. The extended data unit is transmitted across the network.
4. The receiver subdivides data unit and adds all segments together and complement the result.
5. If the intended data unit is intact, total value found by adding the data segments and the checksum field should be zero.
6. If the result is not zero, the packet contains an error & the receiver rejects it.

**Question No: 36 ( Marks: 3 )**

**Which one has more overhead, a repeater or a bridge? Explain your answer. [3]**

**Answer:**

A bridge has more overhead than a repeater. A bridge processes the packet at two layers; a repeater processes a frame at only one layer. A bridge needs to search a table and find the forwarding port as well as to regenerate the signal; a repeater only regenerates the signal. In other words, a bridge is also a repeater (and more); a repeater is not a bridge.

**Question No: 37 ( Marks: 3 )**

**Write down disadvantages of Ring Topology.**

**Answer: (Page 33)**

**Disadvantages of Ring Topology**

- 🕒 Unidirectional Traffic
- 🔌 A break in a ring I.e. a disabled station can disable the entire network

**Muhammad Moaaz Siddiq**

- ⌚ Can be solved by using:
- ⌚ Dual Ring or
  - ⌚ A switch capable of closing off the Break

**Question No: 38 ( Marks: 3 )**

**How parity bits are counted in VRC error detection method technique in case of odd parity generator?**

**Question No: 39 ( Marks: 5 )**

**How are lost acknowledgment and a lost frame handled at the sender site? [5]**

**Answer:**

A lost or damaged frame is handled in the same way by the receiver; when the receiver receives a damaged frame, it discards it, which essentially means the frame is lost. The receiver remains silent about a lost frame and keeps its value of R.

**Question No: 40 ( Marks: 5 )**

**Explain Protocol Data Unit (PDU)?**

**Answer: (Page 221)**

Protocol Data Unit (PDU)

The data unit in the LLC level is called the Protocol Data unit (PDU)

The PDU contains 4 fields familiar from HDLC:

- A destination service access point (DSAP)
- A source service access point (SSAP)
- A control field
- An Information field





# CS601-Data Communication

(Solved Subjective)  
**LECTURE FROM**  
**(23 to 45)**



[Junaidfazal08@gmail.com](mailto:Junaidfazal08@gmail.com)  
[Bc190202640@vu.edu.pk](mailto:Bc190202640@vu.edu.pk)

FOR MORE VISIT  
VULMSHELP.COME

**JUNAID MALIK**  
**0304-1659294**

## AL-JUNAID TECH INSTITUTE



[www.vulmshelp.com](http://www.vulmshelp.com)



### Language Courses Training Available

I'm providing paid courses in different languages within 3 Months, Certificate will be awarded after completion.

- HTML
- CSS
- JAVASCRIPT
- BOOTSTRAPS
- IQUERY
- PHP MYSQL
- NODES.JS
- REACT JS

### LMS Handling Services

#### LMS Activities Paid Task

**Assignments 95% Results**

**Quizes 95% Results**

**GDB 95% Results**

For CS619 Project Feel Free To Contact With Me

Ph# 0304-1659294  
Email: junaidfazal08@gmail.com

# **AL-JUNAID INSTITUTE GROUP**

**FOR MCQs UPDATED CS601  
VISIT MY SITE**

<https://vulmshelp.com/vu-final-term-past-papers-2021/>

## **Question No: 1**

Define Three Types of Cast

Data is transported over a network by three simple methods i.e. Unicast, Broadcast, and

Multicast. So let's begin to summarize the difference between these three:

- ❖ Unicast: from one source to one destination i.e. One-to-One
- ❖ Broadcast: from one source to all possible destinations i.e. One-to-All
- ❖ Multicast: from one source to multiple destinations stating an interest in receiving the traffic i.e. One-to-Many

## **Question No: 2**

RJ Stands for?

RJ stands for Registered Jack in Data Communication which a connector used for UTP cable.

## **Question No: 3**

Write Full Form of Channelization Protocol

Ans:

- ❖ DMA (Frequency Division Multiple Access)
- ❖ TDMA (Time-Division Multiple Access)
- ❖ CDMA (Code Division Multiple Access)

# **AL-JUNAID INSTITUTE GROUP**

## **Question No: 4**

What is ARP how does it works?

Ans:

ARP stands for Address Resolution Protocol. When you try to ping an IP address on your local network, say 10.12.2.1, if there is a value cached, ARP is not used. If the IP address is not found in the ARP table, the system will then send a broadcast packet to the network using the ARP protocol to ask "who has 10.12.2.1". All systems received broadcast but only intended system response with MAC address.

## **Question No: 6**

Difference between Hub, Switch and Router

### **Hub**

Hub is commonly used to connect segments of a LAN (Local Area Network).

A hub contains multiple ports.

When a packet arrives at one port, it is copied to the other ports so that all segments of the LAN can see all packets. Hub acts as a common connection point for devices in a network.

### **Switch**

A switch operates at the data link layer (layer 2) and sometimes the network layer (layer 3) of the OSI (Open Systems

Interconnection) Reference Model and therefore support any packet protocol.

LANs that use switches to join segments are called switched LANs or, in the case of Ethernet networks, switched Ethernet LANs.

In networks, the switch is

the device that filters and forwards packets between LAN segments.

See more information on Network

Switch and Selection Suggestions.

### **Router**

A router is connected to at least two networks, commonly two

LANs or WANs (Wide Area Networks) or a

# **AL-JUNAID INSTITUTE GROUP**

LAN and its ISP.s (Internet Service Provider's) network.

The router is generally located at gateways,

the places where two or more networks connect. Using headers and forwarding tables, router determines

the best path to forward the packets. In addition, router uses protocols such as ICMP (Internet Control Message Protocol) to communicate with each other and configures the best route between

any two hosts. In a word, router forwards data packets along with networks.

## **Question No: 7**

Abbreviation of the following

- ❖ DLC (Data Link Control)
- ❖ FSM (Finite State Machine)
- ❖ POTS (Plain old telephone service)
- ❖ FTTC (Fiber to the Curb)

## **Question No: 8**

HDLC Frame

Ans:

Types of HDLC Frames

- ❖ There are three types of HDLC frames. The type of frame is determined by the control field of the frame –
- ❖ I-frame – I-frames or Information frames carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.
- ❖ S-frame – S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.
- ❖ U-frame – U-frames or Un-numbered frames are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11.

## **Question No: 9**

Difference between data word and code word

Ans:

- ❖ We divide our message into blocks, each of 'k' bits, called data words

# **AL-JUNAID INSTITUTE GROUP**

- ❖ We add 'r' redundant bits to each block to make the length ' $n = k + r$ '
- ❖ The resulting 'n-bit' blocks are called code words

## **Question No: 10**

### **SERVICES PROVIDED BY DALA LINK**

Ans:

- ❖ Framing is a point-to-point connection between two computers or devices consists of a wire in which data is transmitted as a stream of bits. Source & Destination Address
- ❖ Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- ❖ Error control in data link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission.

## **Question No: 11**

### **Variable Framing Techniques**

- ❖ **Character-oriented framing**  
In character - oriented framing, data is transmitted as a sequence of bytes, from an 8-bit coding system like ASCII.
- ❖ **Bit-oriented framing**  
In bit-oriented framing, data is transmitted as a sequence of bits that can be interpreted in the upper layers both as text as well as multimedia data.

## **Question No: 12**

### **Configurations & Transfer Modes in HDLC**

Ans:

- ❖ HDLC supports two types of transfer modes, normal response mode and asynchronous balanced mode.
- ❖ Normal Response Mode (NRM) – Here, two types of stations are there, a primary station that send commands and secondary station that can respond to received commands. It is used for both point - to - point and multipoint communications.
- ❖ Asynchronous Balanced Mode (ABM) – Here, the configuration is balanced, i.e. each station can both send commands and respond to commands. It is used for only point - to - point communications.



# **AL-JUNAID INSTITUTE GROUP**

## **Question No: 13**

What is CONTROLLED ACCESS and its types?

Ans:

- ❖ The stations consult one another to find which station has the right to send
- ❖ A station cannot send unless authorized by other stations
- ❖ Three controlled-access methods:
- ❖ Reservation
- ❖ Polling
- ❖ Token Passing

## **Question No: 14**

Stop and Wait Flow Control

Ans:

- ❖ This protocol involves the following transitions –
- ❖ The sender sends a frame and waits for acknowledgment.
- ❖ Once the receiver receives the frame, it sends an acknowledgment frame back to the sender.
- ❖ On receiving the acknowledgment frame, the sender understands that the receiver is ready to accept the next frame. So it sends the next frame in queue.

## **Question No: 15**

Which form of signal represents binary 1 and 0 by positive and negative voltage level?

**Answer**

In Bipolar encoding scheme, **zero level represents binary 0**, and **binary 1 is represented by alternating positive and negative voltages**. If the first 1 bit is represented by positive amplitude, then the second 1 bit is represented by negative voltage, third 1 bit is represented by the positive amplitude and so on. The signal represents binary 1 and 0 by positive level then the negative level is signal represents binary 0 and 1 is negative level. the Baud Rate and minimum Bandwidth for FSK signal transmitted at 3000 bps by assuming that transmission medium is in half duplex mode and carrier must be separated by 4000 Hz. Using AMI encoding scheme in digital transmission, we represent 0 by zero voltage level and represent 1 by any positive voltage level.

# **AL-JUNAID INSTITUTE GROUP**

## **Question No: 16**

Distinguish between Slotted aloha and Pure aloha.

### **Answer**

In **pure ALOHA**, whenever a station has data to send it transmits it without waiting whereas, in **slotted ALOHA** a user waits till the next time slot begins to transmit the data. In **pure ALOHA** the time is continuous whereas, in **Slotted ALOHA** the time is discrete and divided into slots. Jun 23, 2016

Description. **Aloha** in computer networks is an access control protocol. Versions of **Aloha Protocol- Pure Aloha and Slotted Aloha**. **Pure Aloha** in networking allows the stations to transmit the data at any time. **Slotted Aloha** Protocol allows the stations to transmit data only at the beginning of the time slot.

ALOHA the earliest random access method was developed in early in 1960s the distinguish between Slotted aloha and Pure aloha in the data communication its work different before the 1960 people never know the aloha after the 1960s people were know the slotted aloha and pure aloha.

## **Question No: 17**

If we want to be able to detect 4-bit errors, what should be the minimum Hamming distance? Answer with the help of formula.

### **Answer**

Valid code words:  $2^8 = 256$  b. Invalid code words:  $2^{10} - 2^8 = 768$  Q10-6.

## **Question No: 18**

If we want to be able to detect two-bit errors what should be the minimum Hamming distance?

Ans:

**Hamming distance** =  $s+1$  Since  $s = 2$  **Hamming distance** =  $2+1 = 3$  Q10-8.

# **AL-JUNAID INSTITUTE GROUP**

## **Question No: 19**

What is the minimum Hamming distance?

Ans:

The **minimum distance** between any two vertices is the **Hamming distance** between the two binary strings.

## **Question No: 20**

Normally a telephone network is divided into several levels of switching offices, mention their names.

### **Answer**

The telephone network has several levels of switching offices:

- ❖ End offices
- ❖ Tandem offices
- ❖ Regional offices

## **Question No: 21**

If four computers are connected with wireless NIC in a LAN architecture like isolated LAN in wired network. What do we call such networks?

### **Answer**

If four computers are connected with wireless NIC in a LAN architecture like isolated LAN in wired network. Then we call this type of networks Multicast because into the timing the four computer are connected with wireless NIC in a local area network so that we called it's a multicast network

## **Question No: 22**

Find the Baud Rate and minimum Bandwidth for FSK signal transmitted at 3000 bps by assuming that transmission medium is in half duplex mode and carrier must be separated by 4000 Hz. Write all necessary steps, formulas a

### **Answer**

The Baud Rate and minimum Bandwidth for FSK signal transmitted at 3000 bps by assuming that transmission medium is in half duplex mode and carrier must be



# **AL-JUNAID INSTITUTE GROUP**

separated by 4000 Hz During the transmission, transmitting device produces a high frequency signal which acts as a basis for the transmission. This base signal is called Analog signal

## **Question No: 23**

Illustrate the two strategies used to avoid collision.

### **Answer**

additional energy. This is not useful for effective collision detection.

We need to avoid collisions on wireless networks because they cannot be detected. Carrier sense multiple access with collision avoidance (*CSMA/CA*) was invented for this

network. Collisions are avoided through the use of CSMA/CA's three strategies: the interframe

space, the contention window, and acknowledgments, as shown in Figure 12.16.

Figure 12.16 *Timing in CSMA/CA*

-Found Size:

idle 11Jillielxlp1obneinntiaalry

## **Question No: 24**

Show how the address 47:20:1B:2E:08:EE is sent out online by filling the spaces in figure given blow.

- Hexadecimal
- Binary
- Transmitted

### **Answer**

Show how the address 47:20:1B:2E:08:EE is sent out online by filling the spaces in figure given blow.

Hexadecimal      47:20:1B:2E:08:EE

Binary              47      20      1B      2E      08      EE

# **AL-JUNAID INSTITUTE GROUP**

Transmitted 47: 20: 1B : 2E: 08: EE

At the data-link layer, Simple protocol is designed to handle both flow and error control, but communication is one frame at a time. During the transmission, transmitting device produces a high frequency signal which acts as a basis for the transmission. This base signal is called Analog signal

## **Question No: 25**

A communication system is encoding 2-bit data words into 3-bit code words for error detection using even-parity encoding.

- I. If data word is '01', what will be the code word for transmission?
- II. If communication system suffers a 3-bit burst error and all the bits are damaged; what will be the code word received at the other end?
- III. With the effect of burst error; will this code word be accepted or rejected by the receiver? Mention the reason.

## **Answer**

A communication system is encoding 2-bit data words into 3-bit code words for error detection using even-parity encoding. If data word is '01', They will be the code word for transmission is <01>. If communication system suffers a 3-bit burst error and all the bits are damaged; They will be the code word received at the other end is 6-bit. With the effect of burst error; will this code word be accepted or rejected by the receiver? The reason of the code word is all the process will be hidden system for FEC using XOR operation, we divide a packet into N chunks, and need to send N chunks to the receiver side's a technique called, compounding High-and-Low Resolution Packets, the low-resolution section in the first packet is Checksum.

## **Question No: 26**

State the reasons for the Looping problem in a switch along with appropriate solutions based on logical reasoning.

## **Answer**

**Loop Problem** Transparent bridges work fine as long as there are no redundant

# **AL-JUNAID INSTITUTE GROUP**

bridges in the system. Systems administrators, however, like to have redundant bridges (more than one bridge between a pair of LANs) to make the system more reliable. If a bridge fails, another bridge takes over until the failed one is repaired or replaced. Redundancy can create loops in the system, which is very undesirable.

**Switching loop:** A **switching loop** or **bridge loop** occurs in computer networks when there is more than one Layer 2 (OSI model) path between two endpoints (e.g. multiple connections between two network **switches** or two ports on the same **switch** connected to each other).

## **Question No: 27**

Name the three wave propagation methods for communication through unguided media.

### **Answer**

The three wave propagation methods for communication through unguided media.

- ❖ cell phone
- ❖ mail
- ❖ internet

## **Question No: 28**

Enlist three channelization protocols.

### **Answer**

Enlist three channelization protocols.

- ❖ Aloha, CSMA/CD and CSMA/CA
- ❖ Reservation, Polling and Token
- ❖ FDMA, TDMA, and CDMA

## **Question No: 29**

If we want to be able to detect 4-bit errors, what should be the minimum Hamming distance? Answer with the help of formula.

### **Answer**

The Hamming distance between two pair of words (101101, 001011) is

# **AL-JUNAID INSTITUTE GROUP**

❖ 101101

❖ 001011

The answer is hamming distance is 3

The PAP is a simple authentication procedure with a twostep process, which uses Two type(s) of packet(s) for authentication. In CRC error detection, all the data words are added using 1's complement to calculate a sum, which is then complemented and transmitted with the data words. The Network layer is responsible for the creation and delivery of a frame to another node, along the link.

## **Question No: 30**

Give the names for the following two carriers in telephone system under LATA;

What is the name of carrier that handles Intra-LATA?

What is the name of carrier that handles Inter-LATA?

## **Answer**

The names for the following two carriers in telephone system under LATA;

LOCAL ACCESS TRANSPORT AREA

the name of carrier that handles Intra-LATA is the external access transport area

the name of carrier that handles Inter-LATA internal access transport area

## **Question No: 31**

Give the name of the characteristic in which signal gets degraded during its propagation and full strength of signal is not received at receiver.

## **Answer**

The name of the characteristic in which signal gets degraded during its propagation and full strength of signal is not received at receiver. is a signal where it is being received by receiver after reflection from different intermediate objects (Building, iron polls, walls) is called Interference

## **Question No: 32**

Two stations (M and N) are connected through a circuit switched network. If station M wants to transfer some data to station N; a dedicated link is used to

# **AL-JUNAID INSTITUTE GROUP**

transmit data. Write the names of five stages involved in this process for successful communication.

## **Answer**

Two stations (M and N) are connected through a circuit switched network. The station M wants to transfer some data to station N and dedicated link is used to transmit data to five stages involved in this process for successful communication.

- ❖ Radio
- ❖ mobile LAN
- ❖ WAN
- ❖ INTERNET
- ❖ FAX
- ❖ MESSAGE

## **Question No: 33**

Name and describe any **two** controlled-access methods.

## **Answer**

- ❖ LAN
- ❖ WAN

WAN, we search all the data into the wan when we use the browser and we find some data into the internet world we find easily everything just open and search the problem that is why we used the WAN

LAN in the lane when the router connects with the wire into the LAN then the system is start to working and we use the LAN

## **Question No: 34**

A communication system is encoding 2-bit data words into 3-bit code words for error detection using even-parity encoding.

- I.If data word is '01', what will be the code word for transmission?
- II.If communication system suffers a 3-bit burst error and all the bits are damaged; what will be the code word received at the other end?
- III. With the effect of burst error; will this code word be accepted or rejected by the receiver? Mention the reason.

# **AL-JUNAID INSTITUTE GROUP**

## **Answer**

A communication system is encoding 2-bit data words into 3-bit code words for error detection using even-parity encoding.

## **Question No: 35**

If data word is '01', what will be the code word for transmission?

The code word for transmission is 0. If communication system suffers a 3-bit burst error and all the bits are damaged; what will be the code word received at the other end? The code word received at the other end is error with the effect of burst error; will this code word be accepted or rejected by the receiver? The code word is rejected by the receiver because the code word is affected by burst error. That's why

## **Question No: 35**

Telephone system works in multiple small metropolitan area which are called LATA, answer the questions given below;

- Which name is given to services that are offered by telephone companies inside a LATA?
- And, what name shall be given to the services that are offered between LATAs?

## **Answer**

Telephone system works in multiple small metropolitan area which are called LATA, answer the questions given below; Which name is given to services that are offered by telephone companies inside a LATA? The name of the giving the companies is telecom local access transport area. And, the name shall be given to the services that are offered between

- ❖ LATAs is the Aloha, CSMA/CD and CSMA/CA
- ❖ Reservation, Polling and Token
- ❖ FDMA, TDMA, and CDMA

## **Question No: 36**

How does CSMA/CD technique get impacted in wireless LAN by the following factors?



# **AL-JUNAID INSTITUTE GROUP**

- ❖ Power at a host
- ❖ Hidden station

## **Answer**

How does CSMA/CD technique get impacted in wireless LAN by the following factors?

### **Power at a host**

The power at the host is the router openly in the room and the area covered the signal in the area and the user use easily some of the person use the wire to the router. The PAP is a simple authentication procedure with a two-step process, which uses two type(s) of packet(s) for authentication.

Cyclic codes are special linear block codes in which, rotation of a code word results in another code word.

### **Hidden station**

The hidden station is using the LAN it is difficult to use the LAN because the power of the station is hidden and the proper working but if the problem creates then it was difficult to use. In CRC error detection, all the data words are added using 1's complement to calculate a sum, which is then complemented and transmitted with the data words. The Network layer is responsible for the creation and delivery of a frame to another node, along the link.

## **Question No: 37**

What are the categories of multiplexing?

Ans:

There are three categories of multiplexing

- ❖ FDM
- ❖ TDM
- ❖ WDM

Have two other categories

- ❖ Synchronous
- ❖ Asynchronous

## **Question No: 38**

# **AL-JUNAID INSTITUTE GROUP**

What kind of error is undetectable by the checksum?

Ans:

Error detection method used by the higher layers like VRC, LRC, CRC, checksum is also based on the concept of redundancy

**Question No: 39**

**What are the basic purpose of routers?**

**Routers connect computers and other devices to the Internet.** A router acts as a dispatcher, choosing the best route for your information to travel. It connects your business to the world, protects information from security threats, and can even decide which computers get priority over others.

**Question No: 40**

**Why we need a Null Modem?**

Ans:

Null modem is a communication method to directly connect two DTEs (computer, terminal, printer, etc.) using an RS-232 serial cable. With a null modem connection, the transmit and receive lines are cross-linked. Depending on the purpose, sometimes also one or more handshake lines are cross-linked.

**Question No: 41**

What are properties of signals?

Ans:

**Question No: 42**

Coaxial cable designs are categorized by their Radio government (RG) ratings; you are required to match the following RG ratings with their respective uses.

| Category | Use            |
|----------|----------------|
| RG-11    | Used for TV    |
| RG-59    | Thin Ethernet  |
| RG-58    | Thick Ethernet |



# **AL-JUNAID INSTITUTE GROUP**

Ans:

| Category | Use            |
|----------|----------------|
| RG-11    | Thick Ethernet |
| RG-59    | Used for TV    |
| RG-58    | Thin Ethernet  |

## **Question No: 43**

Give Advantages of using VLAN on layer 3 switches.

Ans:

- ❖ VLANs enable logical grouping of end-stations that are physically dispersed on a network.
- ❖ VLANs reduce the need to have routers deployed on a network to contain broadcast traffic.
- ❖ Flooding of a packet is limited to the switch ports that belong to a VLAN.
- ❖ Confinement of broadcast domains on a network significantly reduces traffic
- ❖ Cost and Time Reduction
- ❖ Creating virtual Workgroups
- ❖ Security

## **Question No: 44**

Name and describe any **two** controlled-access methods.

**Answer**

The three popular controlled-access methods are as follows.

- ❖ Reservation: In the reservation method, a station needs to make a reservation before sending data.
- ❖ Polling: Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations. ...
- ❖ Token Passing: In the token-passing method, the stations in a network are organized in a logical ring

## **Question No: 45**

Consider a scenario where five computers are connected to each other in a wired LAN through switch. How can this wired LAN be converted to a wireless LAN architecture?

Ans:

According to given scenario, Wireless LANs can be replaced this. Wireless LANs are those Local Area Networks that use high frequency radio waves

# **AL-JUNAID INSTITUTE GROUP**

instead of cables for connecting the devices in LAN. Users connected by WLANs can move around within the area of network coverage. Most WLANs are based upon the standard IEEE 802.11 or Wi-Fi.

## **IEEE 802.11 Architecture**

The components of an IEEE 802.11 architecture are as follows

1. Stations (STA) – Stations comprise all devices and equipment's that are connected to the wireless LAN. A station can be of two types:
  - a. Wireless Access Points (WAP) – WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.
  - b. Client. – Clients are workstations, computers, laptops, printers, smartphones, etc.

Each station has a wireless network interface controller.

2. Basic Service Set (BSS) – A basic service set is a group of stations communicating at physical layer level. BSS can be of two categories depending upon mode of operation.